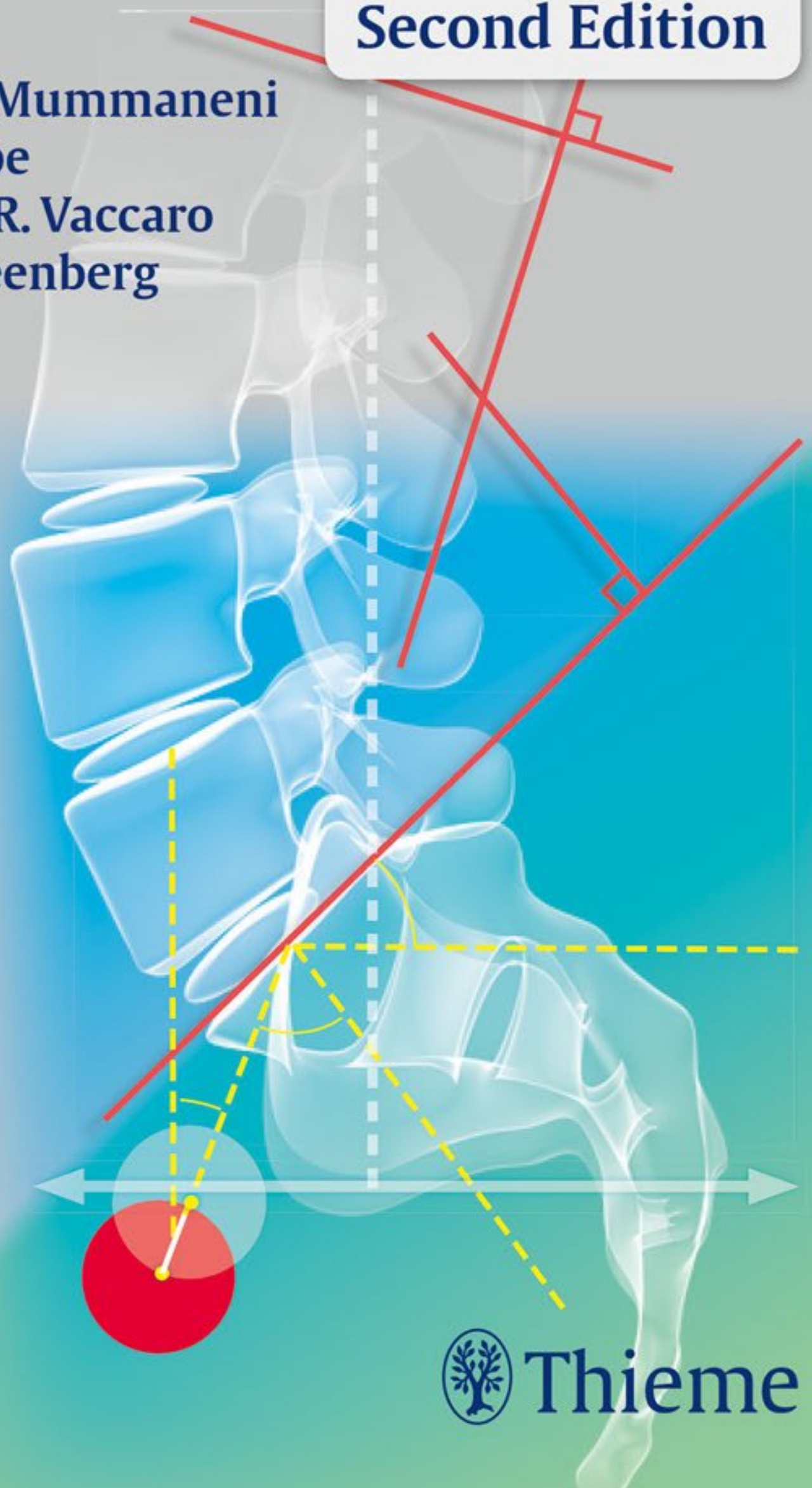


Handbook of Spine Surgery

Second Edition

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Praveen V. Mummaneni
Juan S. Uribe
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Mark S. Greenberg



 **Thieme**

Handbook of Spine Surgery

Second Edition

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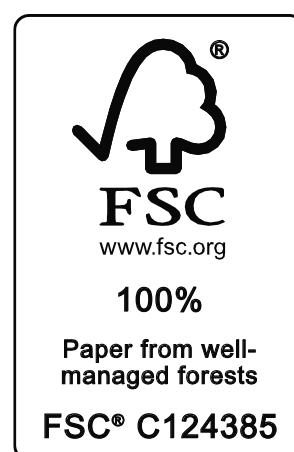
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To my parents, Abdulwahab and Hana, to my wife, Gabriela,
to all my mentors of spine surgery, especially Juan Uribe and Ziya Gokaslan,
and to the residents and fellows who give us the reason to do all this.

AAB

For Valli, Nikhita, Nkihil, and Neel for all their love and support.
For the fellows and residents whom I have had the pleasure to help teach.

PVM

I would like to dedicate this book to the birth of my child,
Christian John Vaccaro, and to my beautiful wife, Lauren. Lauren's
gift of caring and love has only added to the joy of our family life.

ARV

To my family.

MSG

To my wife, Catalina, my son, Sebastian, my daughter, Camila, my parents,
Carlos Santiago and Maria Cecilia, and my parents-in-law, Ivan and Maria
Cecilia, for their love and unconditional support.

JSU

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Foreword

In this exciting second edition of the *Handbook of Spine Surgery*, editors Ali A. Baaj, Praveen V. Mummaneni, Juan S. Uribe, Alexander R. Vaccaro, and Mark S. Greenberg have configured a virtual step-by-step “cookbook” of all the common and uncommon surgical spine procedures utilized around the globe, with chapters authored by highly recognized spinal surgeons. This task has been expertly accomplished by the overall organization of the chapters into sections: Anatomy, Clinical Spine Surgery, Spinal Pathology, and Surgical Techniques. Specific topics are thoroughly covered within and across sections, such as “Thoracolumbar Trauma” (Chapter 21 in the Spinal Pathology section), “Freehand Thoracic Pedicle Screw Placement Technique” (Chapter 47 in the Surgical Techniques section), and specific surgical procedures, such as “Sacrectomy” (Chapter 69 in the Surgical Techniques section). A common template is followed in which each chapter logically proceeds from Key Points to Indications, Techniques, Complications, Postoperative Care, Outcomes, Surgical Pearls, Common Clinical Questions and Answers (as is seen in current CME formats), and Key References. This structure is very effective and ideal for quick and easy review the evening before, or even the morning of, conferences and surgical procedures. Trainees still trying to master these topics, as well as more senior surgeons needing a quick refresher, will benefit. The organizational style is direct and compact but thorough, appropriately illustrated when helpful, and perfectly suited to a busy clinical schedule where essential information and details need to be gleaned in rapid fashion, whether in print, on a phone, tablet, or computer. The organizational consistency of the chapters also aids in quick dissemination and retention, which is critical to our training environment.

I can envision this text being extremely useful to a wide range of individuals: from a third-year medical student performing his or

her initial surgical rotation while assigned to the spine service, to a spine physician assistant helping out during surgery, to a spinal surgery fellow gaining technical confidence during the most important year of fellowship training. Those residents, fellows, and attendings utilizing this text will probably find the greatest “bang for their buck” in the pearls and tips, which are written to optimize safety and maximize efficiency during various specific spinal surgery procedures. I also found the Q and A sections a nice refresher for keeping up on the myriad of spine surgery techniques that have rapidly advanced over the past decade, as spine surgery itself continues down a subspecialization pathway distinct from orthopedic surgery and neurosurgery. In that regard, this book would also serve as an excellent review for those studying for board certification or recertification for the American Board of Orthopaedic Surgery or the American Board of Neurological Surgery exams.

I congratulate the editors and contributing authors for this important piece of work. Any “handbook” should, by definition, be able to concisely provide essential details of a condition and various remedies and solutions—as well as obstacles encountered along the way—to best educate the intended audience. I am confident that the *Handbook of Spine Surgery* has more than accomplished these objectives and will have an enduring and important impact on those fortunate enough to benefit from the assembled information provided.

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Foreword

The second edition of the *Handbook of Spine Surgery*, edited by Dr. Ali A. Baaj, Dr. Praveen V. Mummaneni, Dr. Juan S. Uribe, Dr. Alexander R. Vaccaro, and Dr. Mark S. Greenberg, is an outstanding distillation of the current knowledge of the evaluation and surgical management of spinal disorders. The book is divided into four well-organized sections: a detailed synopsis of spinal anatomy, an evaluation of the techniques used for clinical evaluation of spinal surgery, a detailed description of spinal surgical pathology, and a step-by-step description of surgical techniques. In addition, there are appendices on patient positioning, selected spinal orthoses, and commonly used outcome scales.

The *Handbook of Spine Surgery* particularly excels as a quick reference for almost any type of spinal surgical condition or surgical procedure, from occipitocervical fusion to spinopelvic fixation. Each of the chapters is written by a world-renowned physician. The surgical chapters are authored by experts from the field of orthopaedic or neurosurgical spinal surgery; every chapter has introductory key points and detailed descriptions of indications for each of the surgical procedures. A step-by-step description

of instrumentation techniques, frequently asked questions, and surgical pearls are provided by experts in the field.

This book will be an invaluable resource for orthopaedic and neurosurgical residents in training, spinal surgical fellows, and practicing orthopaedic surgeons and neurosurgeons who deal with spinal pathology. The book covers an exhaustive set of topics in a succinct manner, but sufficient information is provided to serve as a surgical atlas. I feel that this will become the standard resource for surgeons in training, early in practice, those preparing for board examinations, and experienced spine surgeons needing a refresher. I am positive that in its electronic form this book will be used as a quick reference on a daily basis by most practicing spine surgeons. I congratulate the editors on this outstanding resource and look forward to having it available to use in my own practice.

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Preface

We are delighted to present the second edition of the *Handbook of Spine Surgery*. As with the previous edition, our goal is to provide a comprehensive yet portable and compact text that distills the basic principles of contemporary spine surgery. We have once again been fortunate to receive contributions from dozens of reputable surgeons representing acclaimed orthopedic and neurosurgery programs.

Whereas the highly popular format of the first edition is once again adopted, we have significantly enhanced the text with several new chapters addressing topics like pediatric scoliosis and adult deformity principles. We have also expanded the spinal trauma

section to include dedicated chapters on cervical, thoracolumbar, and sacropelvic injuries. Surgical pearls and board-style questions at the end of each chapter emphasize the salient points of each topic.

We are confident that this text will continue to be an excellent resource for surgeons and trainees alike as we all strive to improve spine education and training and the clinical care of our patients.

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Anatomy

1 Embryology of the Spine

Eric Momin, Jared Fridley, and Andrew Jea

1.1 Key Points

- Vertebral body development: Gastrulation gives rise to paraxial mesoderm, which forms somites, which differentiate into sclerotomes, which surround neural tissue then resegment, and ossify into vertebral bodies.
- Vertebral body shape is determined by several homeobox (*Hox*) genes.¹
- Posterior element development: Cells adjacent to the neural tube form vertebral arches (or neural arches).
- Epiblast cells migrate to form the primitive groove, which in turn forms the notochord.
- The anterior neuropore closes on day 25 and the posterior neuropore on day 27.
- Neuroblasts form the mantle layer; the ventral portion forms the basal plates (motor), and the dorsal portion forms the alar plates (sensory).
- The caudal portion of the tube undergoes retrogressive differentiation and relative ascension of the conus.

1.2 Bony Development

- **Vertebral body:**
 - Somatogenesis: Paraxial mesoderm gives rise to 42 to 44 somites. Somites give rise to ventromedial sclerotomes and dorsolateral dermomyotomes. In week 4, cells of the sclerotomes move to surround the spinal cord and notochord.²
 - Resegmentation: Each sclerotome is at first separated by mesenchyme. Sclerotomes undergo resegmentation, which occurs when the caudal half of each sclerotome separates and fuses with the cephalic half of the next sclerotome.
- **Posterior elements:** Cells adjacent to the neural tube form vertebral arches (or neural arches) that give rise to the posterior elements.
- **Disk:** Cells from the caudal portion of the sclerotome form the annulus fibrosus. Notochord remnants form the nucleus pulposus.
- **Ossification:** An ossification center is a cartilaginous “model” that is ossified into bone.
 - Three primary ossification centers for each vertebrae: One for the vertebral body and one for each half of the vertebral arches. Five

secondary ossification centers for subaxial vertebrae: One for the superior and inferior endplates of the body, one for the spinous process, and one at the tip of each transverse process.

- C2 develops from five primary ossification centers: Two for the body of the dens, one for the vertebral body, and one for each neural arch.³ The tip of the dens represents a secondary ossification center.
- *Hox* genes: The shape of vertebral bodies is regulated by *Hox* genes that code for transcription factors.¹
- **Spinal curves:** Thoracic and sacral curves are present during the fetal period. Cervical lordosis develops when the child learns to hold up the head. Lumbar lordosis develops with walking.

1.3 Neural Development

- **Primitive pit:** The bilaminar disk consists of epiblast and hypoblast layers. Some epiblast accumulates at each side of the dorsal midline to form the primitive streak and, subsequently, the primitive groove. At the rostral edge of the primitive groove is a pit, the primitive node.
- **Genesis of notochord:** After gastrulation occurs, epiblast migrates rostrally from the primitive node, which is called the notochordal process. (This phenomenon can be compared to pushing one's finger into an inflated balloon.) The tube is formed exactly between the ectoderm and endoderm, and it divides the mesoderm. Thus, it is bordered laterally by mesoderm, superiorly by ectoderm, and inferiorly by endoderm.
- **Induction of neural plate:** At 3 weeks' gestation, the edges of the neural plate begin to elevate to form neural folds that begin to fuse in the cervical region, forming the neural tube (**Fig. 1.1**).²
 - Anterior neuropore closes on the 25th day.
 - Posterior neuropore closes on the 27th day.

Neural crest cells detach from the neural folds and migrate to form glia, arachnoid, pia, melanocytes, chondrocytes, chromaffin cells, osteocytes, Schwann cells, and enteric ganglia.

- **Mantle layer:** Neuroblasts form a mantle layer around the neuroepithelial layer that forms the gray matter of the spinal cord (**Fig. 1.2**).²
 - The ventral mantle layer forms the basal plates (motor horn), and the dorsal mantle layer forms the alar plates (sensory horn). The boundary between the plates is called sulcus limitans.
 - At the thoracic (T1–T12) and upper lumbar (L1–L2) region, the intermediate horn contains sympathetic neurons of the autonomic nervous system.

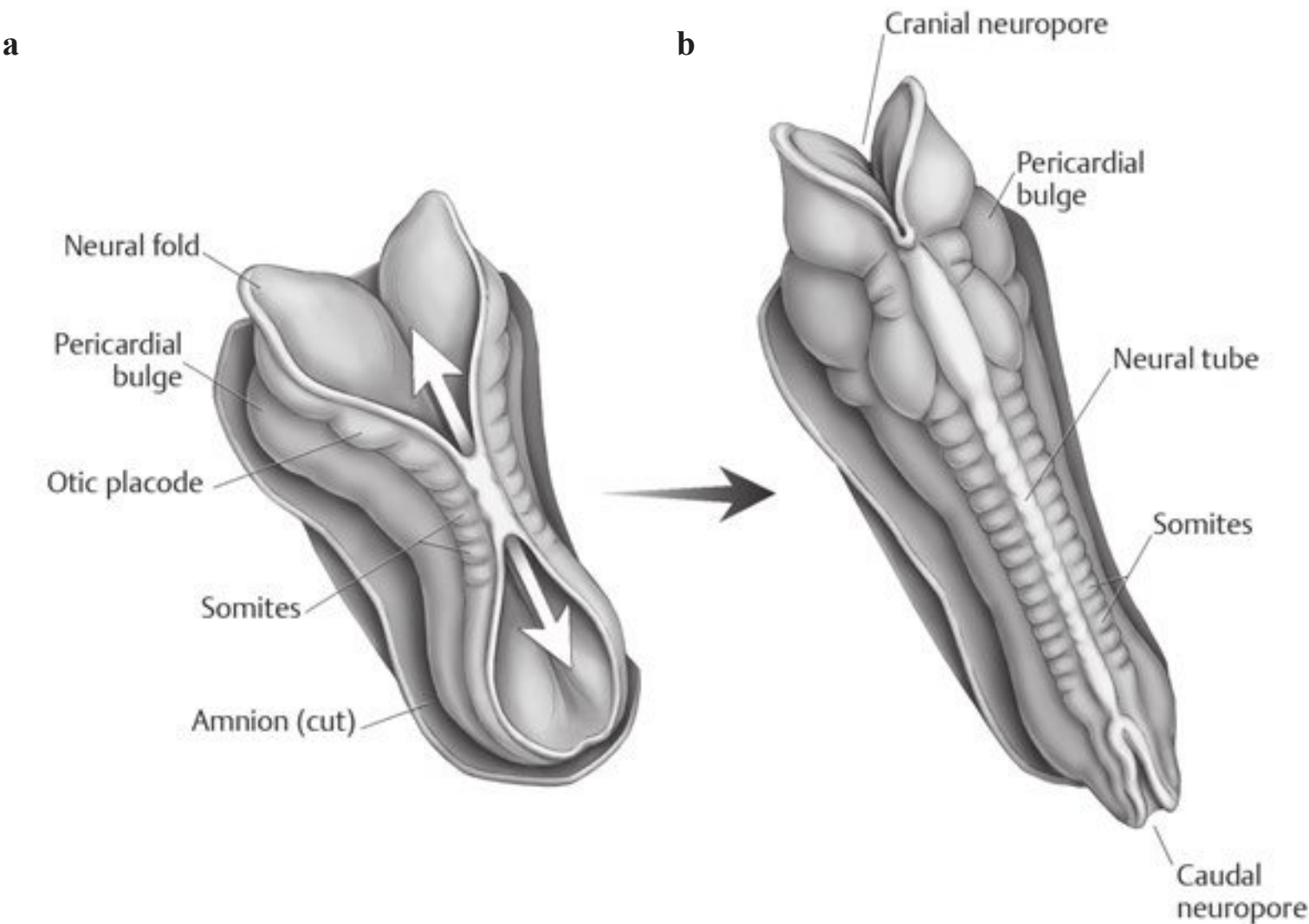


Fig. 1.1 (a) Dorsal view of the human embryo during the third week of gestation. Note the somites on each side of the neural tube as it begins to fuse in the cervical region. (b) The fused neural tube then continues to close both rostrally and caudally.

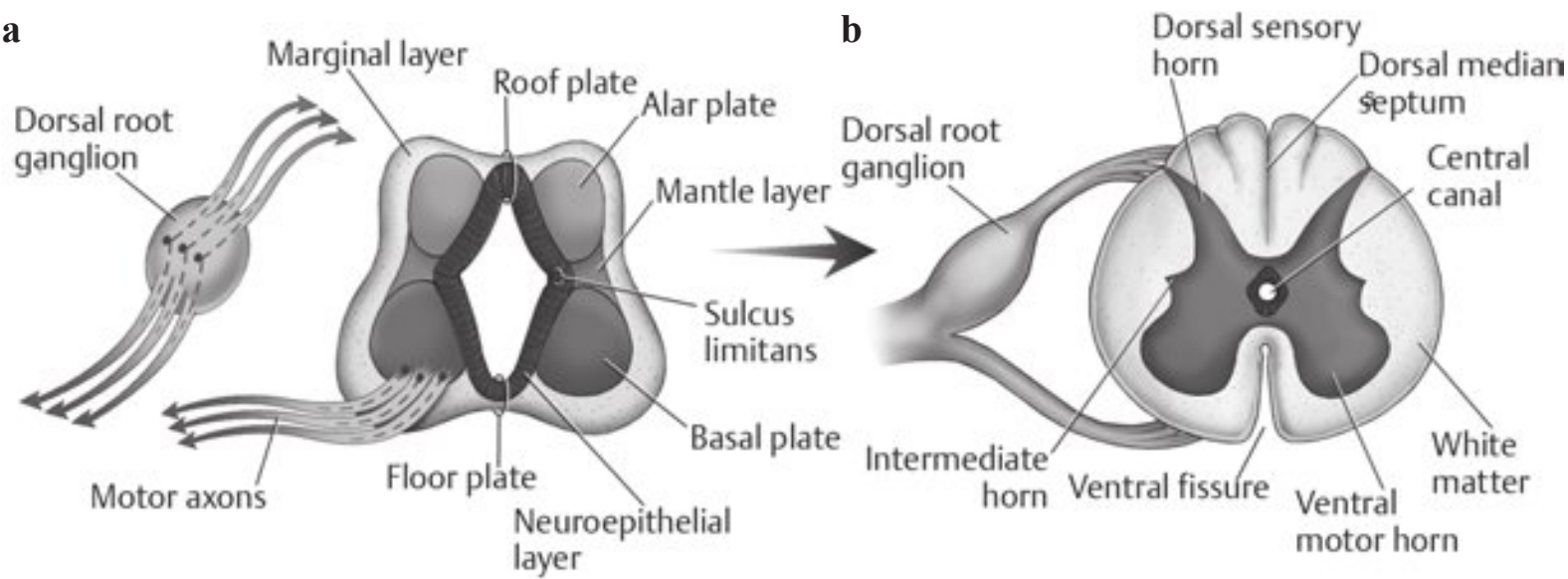


Fig. 1.2 (a) Cross-section of the developing spinal cord demonstrates how the migrating neuroblasts from the neuroepithelial layer form dorsal and ventral mantle layers. These ultimately become the gray matter of the spinal cord. (b) In addition, note the development of the dorsal root ganglion, as well as the outward growth of the motor axons.

- **Marginal layer:** The marginal layer contains nerve fibers from neuroblasts in the mantle layer that ultimately form the white matter of the spinal cord.
- The caudal tube forms during canalization (days 28–42).
- From day 43 to day 48, the ventriculus terminalis (a cystic structure at the caudal neural tube end) undergoes retrogressive differentiation, which is completed postnatally at 2 months.²
 - This results in relative ascension of the conus to its final level of L1/L2 and formation of the cauda equina and filum terminale (**Fig. 1.3**).

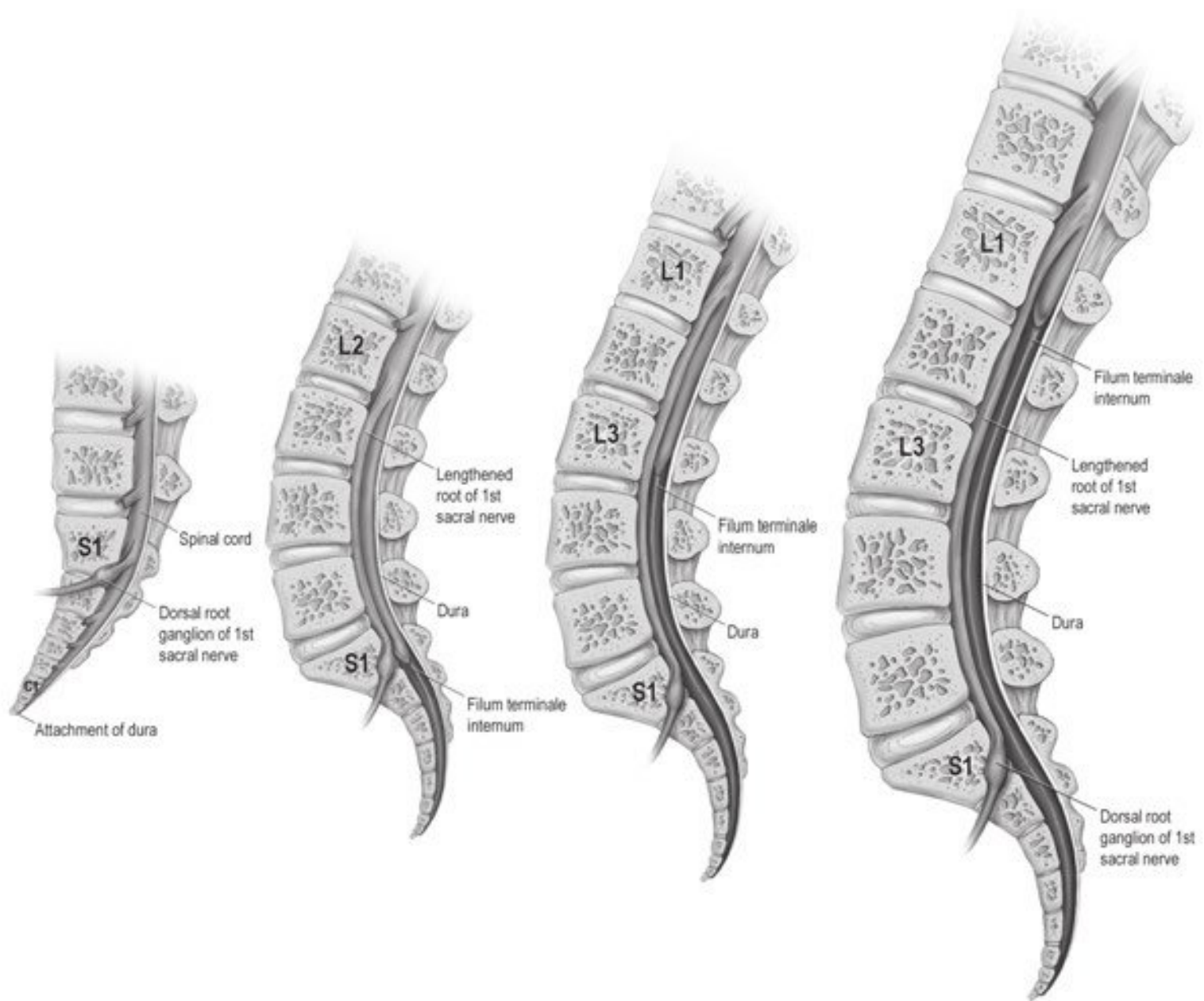


Fig. 1.3 The relative ascension of the conus and formation of the filum terminale via retrogressive differentiation.

1.4 Clinical Correlates

- The conus ascends to L1 to L2 by 2 months of age and may be lower in a neonate (keep in mind for lumbar punctures).
- Complete fusion of the ossification centers of C2 does not occur until age 12 (synchondroses between ossification centers can be mistaken for fractures).
- Incomplete closure of
 - Anterior neural *tube* (22 days) → occipital encephalocele
 - Anterior neuropore (24 days) → anencephaly
 - Anterior neuropore + anterior neurotube → craniorachischisis (brain and upper spinal cord remain open)
 - Posterior neuropore (26 days) → spina bifida/spinal dysraphism below L1/L2
 - Neural tube (defect of secondary neurulation, 28 to 35 days) → spinal dysraphism above L1/L2
 - Ventriculus terminalis → terminal myelocystocele (cyst lined with ependymal cells, communicates with the central canal)

Common Clinical Questions

1. Which structures border the notochordal process?
2. Why is the dorsal midline devoid of vascular structures?
3. Why do most mammalian species have seven cervical vertebrae?

Answers to Common Clinical Questions

1. Mesoderm laterally, ectoderm superiorly, endoderm inferiorly.
2. Mesoderm does not cross the dorsal midline in development, so the vascular structures derived from mesoderm are not present on the dorsal midline.
3. The *Hox* genes are highly conserved between species.

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2 Craniovertebral Junction

Jonathan Hobbs and Edwin Ramos

2.1 Key Points

- The craniovertebral junction (CVJ) is composed of the occiput (O), occipital condyles, atlas (C1), and axis (C2) and represents the transition between the cranium and mobile cervical spine.
- CVJ is composed of osseous structures articulated with synovial joints, muscles, ligaments and membranes.
- The principal motion segment of the O–C1 joint is flexion extension; the C1 to C2 motion segment is the most flexible of the cervical spine in respect to axial rotation.
- The unique ligament and membrane configuration of the CVJ provide stability and permit movement without compromising the traversing neural elements (**Fig. 2.1**).

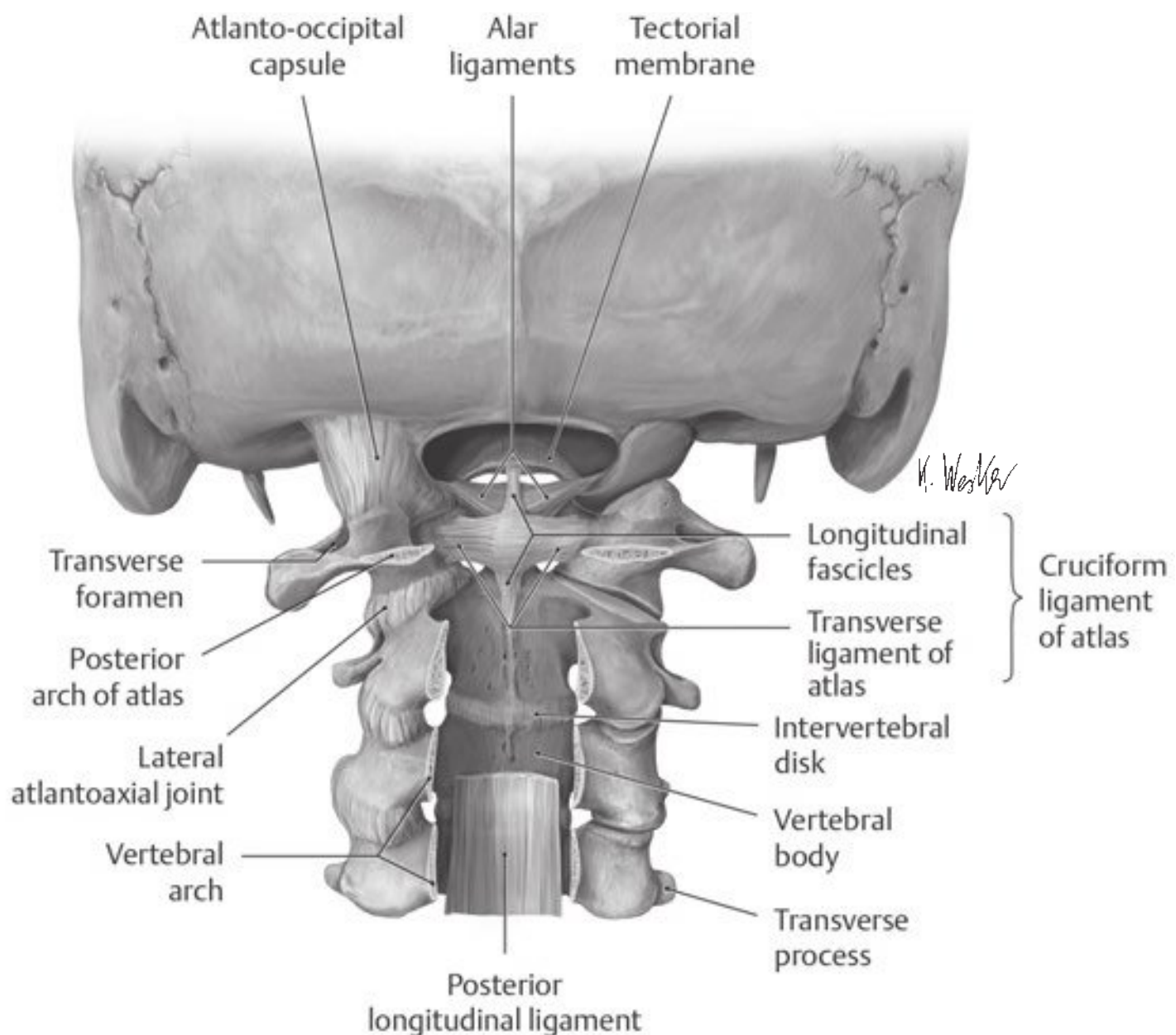


Fig. 2.1 CVJ ligamentous structures. (From Thieme Atlas of Anatomy, General Anatomy and Musculoskeletal System, © Thieme 2005, illustration by Karl Wesker.)

2.2 Bony Anatomy

- The CVJ consists of the base of the occiput, the atlas (C1), and the axis (C2).
- The boundaries of the foramen magnum are the basion anteriorly, the opisthion posteriorly, and the occipital condyles inferolaterally.
- The atlas (C1) has no vertebral body or spinous process. It is comprised of an anterior arch, a posterior arch, and two lateral masses. The superior facets are concave and accommodate the convex occipital condyles, allowing for flexion-extension motion segments (**Fig. 2.2**).
- The C1 anterior tubercle (C1 “button”) is the attachment site of the anterior longitudinal ligament (ALL) and the longus colli muscle.

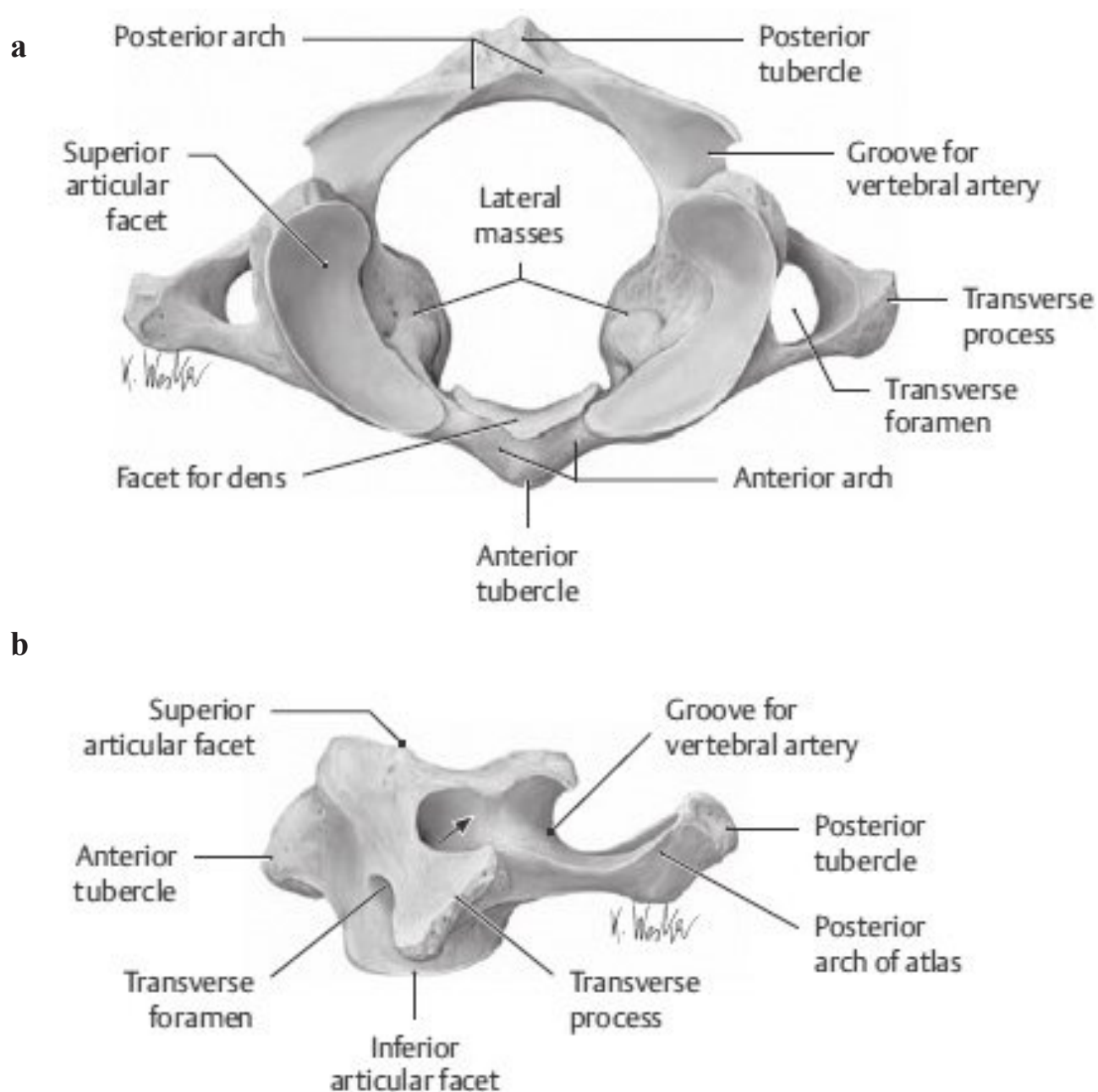


Fig. 2.2 (a) Superior and (b) lateral view of the atlas. (From Thieme Atlas of Anatomy, General Anatomy and Musculoskeletal System, © Thieme 2005, illustration by Karl Wesker.)

- The vertebral artery and C1 nerve run along the superior lateral groove on C1 (sulcus arteriosus). In less than 15% of the population, the groove is roofed, forming the arcuate foramen.
- The axis (C2) consists of the body, odontoid process, articulating surfaces, pedicles, pars interarticularis (note that pars and pedicles are distinct anatomical landmarks), lamina, and large, bifid spinous process (**Fig. 2.3**).¹
- The C2 odontoid (Gr. “tooth”) process projects superiorly and has multiple, overlapping ligamentous attachments to C1 and the occiput.

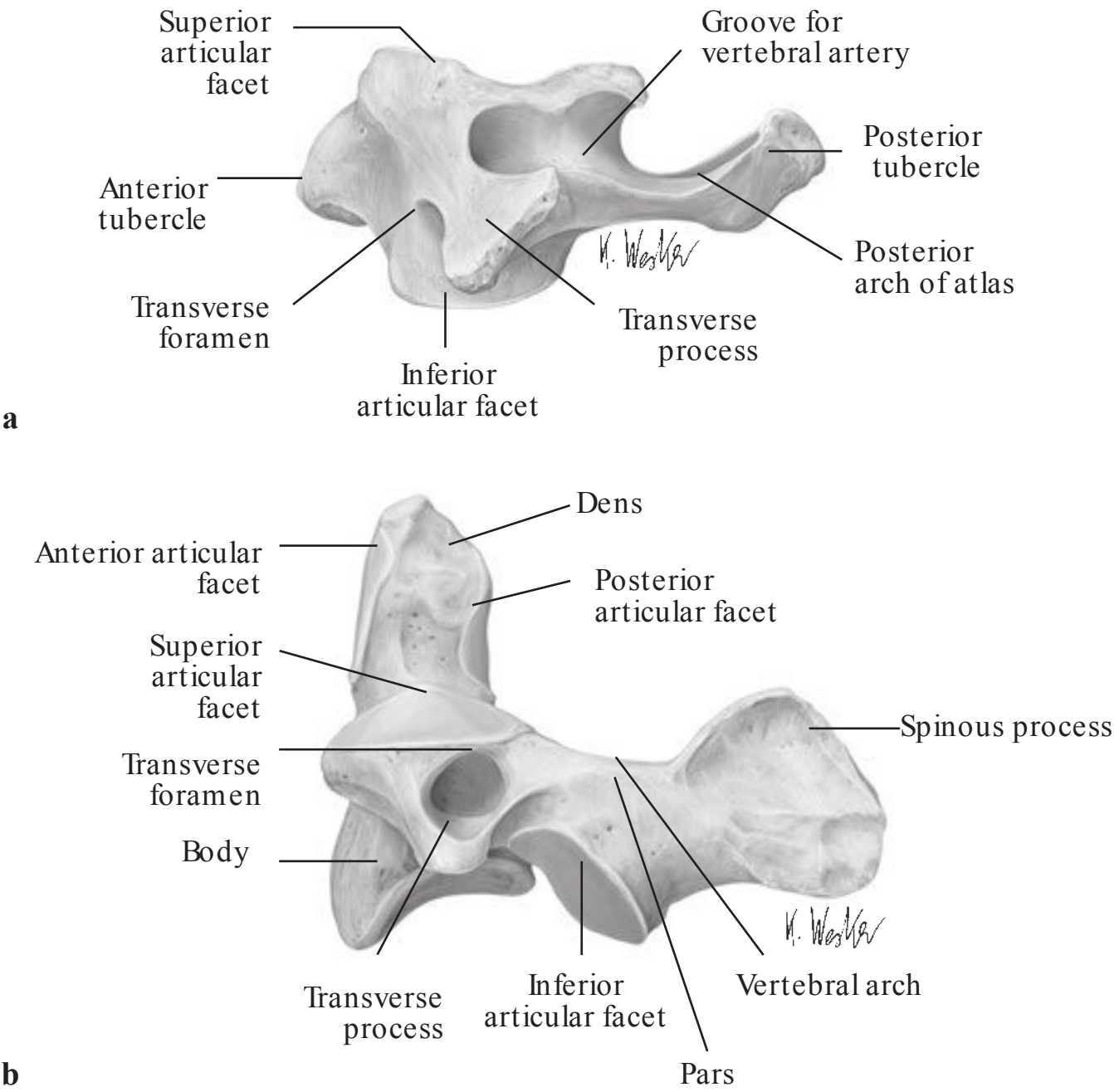


Fig. 2.3 (a) Superior and (b) lateral view of the axis. (From Thieme Atlas of Anatomy, General Anatomy and Musculoskeletal System, © Thieme 2005, illustrations by Karl Wesker.)

2.3 Neural Anatomy

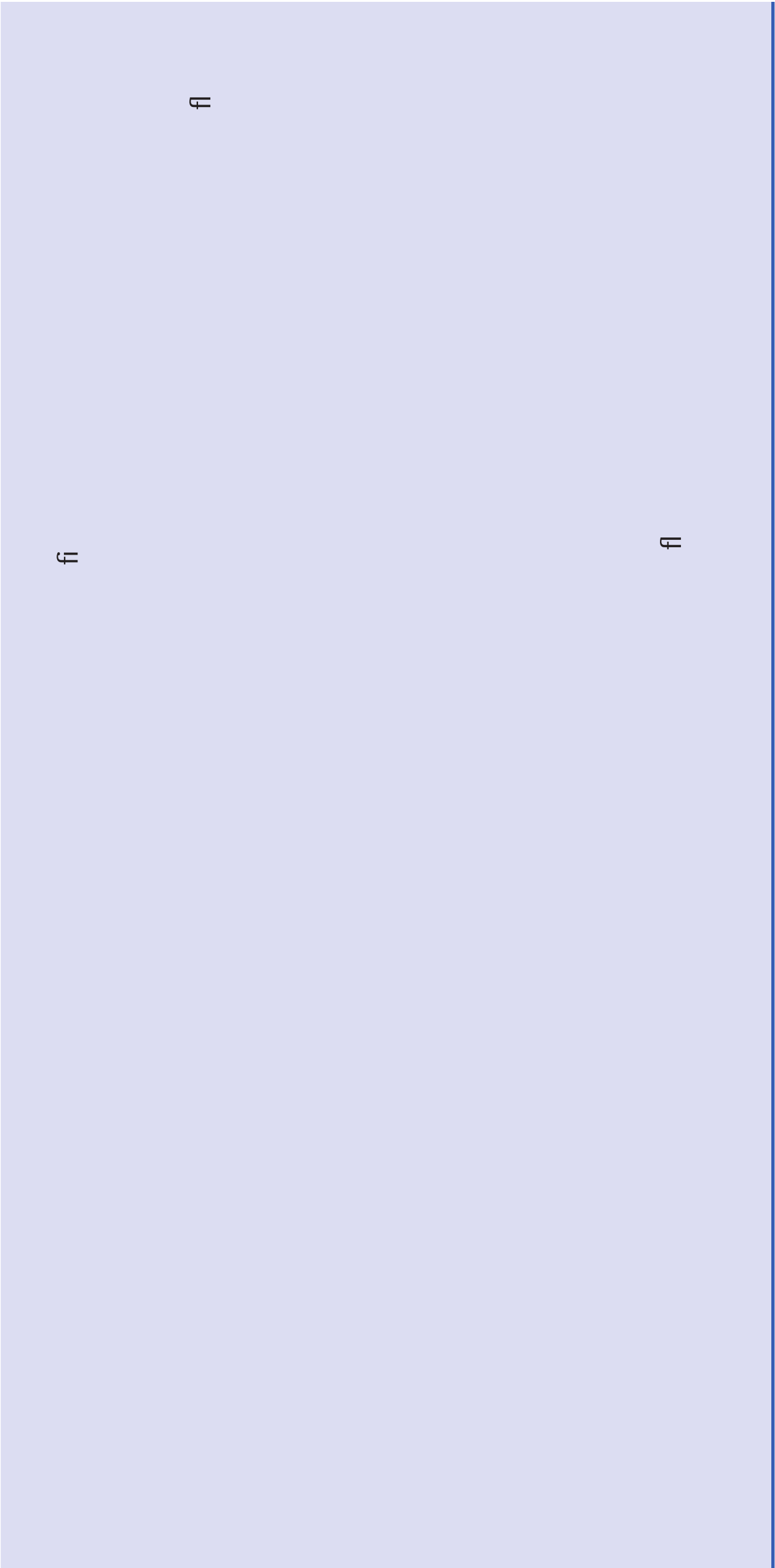
- Cervical nerve roots exit above their corresponding level (e.g., the C2 nerve root exits above the C2 pedicle).
- C1 nerve root: The posterior division (suboccipital nerve) is more prominent than the anterior division. It innervates suboccipital muscles and occasionally branches to the lesser/greater occipital nerve.
- C2 nerve root: Posterior, medial (greater occipital nerve), and lateral divisions innervate suboccipital muscles and scalp from occiput to vertex.
- The lesser occipital nerve is formed by dorsal divisions of C2 and C3.

2.4 Vascular Anatomy

- The vertebral artery (VA) leaves the C2 transverse foramen (becoming V3), takes a 45° lateral projection, and ascends (vertical portion of V3) into the C1 transverse foramen.
- The VA then courses medially (horizontal portion of V3) along the C1 sulcus arteriosus and then anteriorly through the atlantooccipital membrane, where it becomes intradural (beginning of V4 segment).
- Blood is supplied to the CVJ primarily through branches of the vertebral and occipital arteries.
- The base of the dens of C2 receives blood supply from vertebral artery branches (posterior circulation); the top is supplied by the apical branch of the hypoglossal artery (anterior circulation).
- Lymphatic drainage of the CVJ is through retropharyngeal and deep cervical nodes (Grisel's syndrome: CVJ instability with concomitant retropharyngeal inflammation/infection).

2.5 Ligamentous and Muscular Anatomy (Table 2.1)

- Suboccipital muscles and the CVJ (**Fig. 2.4**)
 - Superior oblique: C1 transverse process laterally to occiput medially
 - Inferior oblique: C1 transverse process laterally to spinous process of C2 medially
 - Rectus capitis posterior major: Spinous process of C2 up to base of occiput
 - Rectus capitis posterior minor: Posterior tubercle of C1 up to base of occiput
 - Semispinalis capitis: Transverse processes of cervical vertebrae to nuchal ligament and occipital bone; superficial to suboccipital muscles



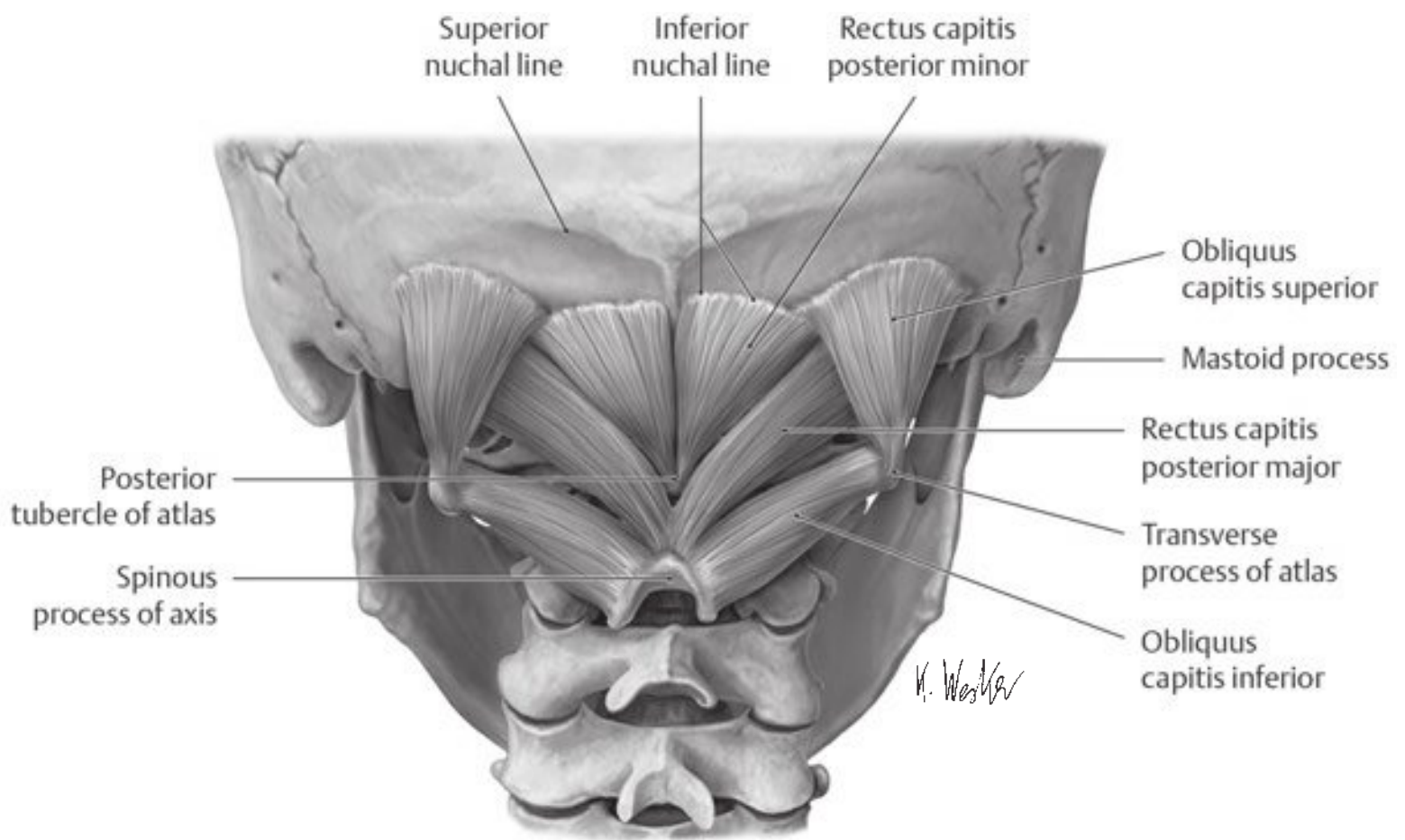


Fig. 2.4 Muscles of the suboccipital triangle. (From Thieme Atlas of Anatomy, General Anatomy and Musculoskeletal System, © Thieme 2005, illustration by Karl Wesker.)

- Longissimus capitis: Similar to semispinalis, but runs and attaches more laterally to the occiput

2.6 Surgical Pearls

- C1 nerve root, if present, can be sacrificed; though we don't routinely perform this, C2 nerve root may also be sacrificed (with minimal risk of occipital neuralgia).²
- Integrity of cruciate ligament must be considered before undertaking any CVJ stabilization procedure.
- Venous plexus around C2 ganglion may cause considerable bleeding, which should not be mistaken for VA injury.
- Thin-cut CT of the CVJ and/or CT angiography (CTA) should be obtained before C1/C2 fixation to verify route and patency of the vertebral arteries, as well as dimensions of the pars interarticularis and/or pedicle.

Common Clinical Questions

1. What is the source of the blood supply to the odontoid process of the axis?
2. What forms the continuation of the posterior longitudinal ligament (PLL) at the CVJ?

3. What persistent carotid–vertebrobasilar anastomoses may be encountered in the CVJ?
-

Answers to Common Clinical Questions

1. Superior part: Apical branch of hypoglossal artery; base: Branches of the vertebral artery.
 2. Tectorial membrane.
 3. Proatlanto intersegmental artery and persistent hypoglossal artery.
-

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3 Cervical Spine

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3.1 Key Points

- The subaxial cervical spine includes C3 to C7.
- The cervical spine, like the lumbar spine, normally demonstrates a lordotic curvature.
- Posterior instrumentation often uses lateral mass screws because the pedicles are narrower than in the thoracic and lumbar spine, increasing the risk of neurovascular injury.
- The size and volume of lateral masses decrease from the upper to lower subaxial cervical spine.
- The primary blood supply to the cervical spine is via the anterior spinal artery (ASA) and two posterior spinal arteries (PSA).

3.2 Bony Anatomy

- Radiographic landmarks:¹ C3 is at the hyoid, C4 is at the thyroid cartilage, and C6 is at the cricoid cartilage.
- Palpable landmark: The anterior tubercle of the C6 transverse process (Chassaignac tubercle) is palpable.
- In the coronal plane, uncovertebral joints are noted at the anterolateral aspect of the vertebral body (**Fig. 3.1**).
- The spinal canal is triangular and has a greater lateral than anteroposterior (AP) dimension.
- Lateral masses of the subaxial spine are composed of the superior and inferior articulating processes (the facet) (**Fig. 3.2**).
- A lateral mass begins lateral to where the lamina and pedicle meet.
- C7 is a transitional vertebra: The lateral mass is thinner, and the pedicle wider, than in C3 to C6.
- The AP diameter of the spinal canal decreases caudally:² 17 mm at C3, 15 mm at C7.
- The width of vertebral body is usually 17 to 20 mm.
- The normal lordotic curvature of the cervical spine is 16 to 25°.³
- Cervical disk herniation occurs most frequently at C5/C6 and C6/C7.
- Biomechanical studies show maximal flexion–extension at C4/C5 and C5/C6 and maximal lateral bending at C2/C3, C3/C4, and C4/C5.
- The least mobile segment is C7/T1.

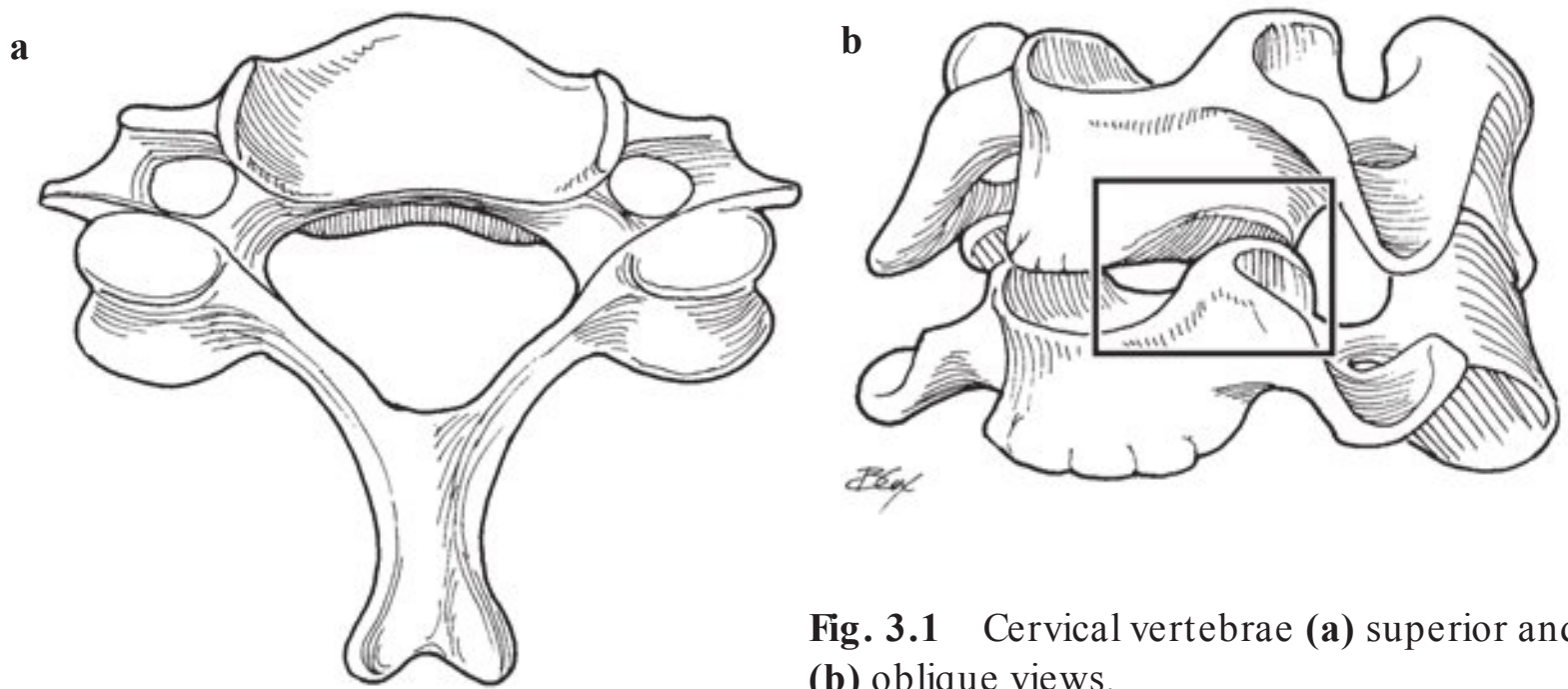


Fig. 3.1 Cervical vertebrae (a) superior and (b) oblique views.

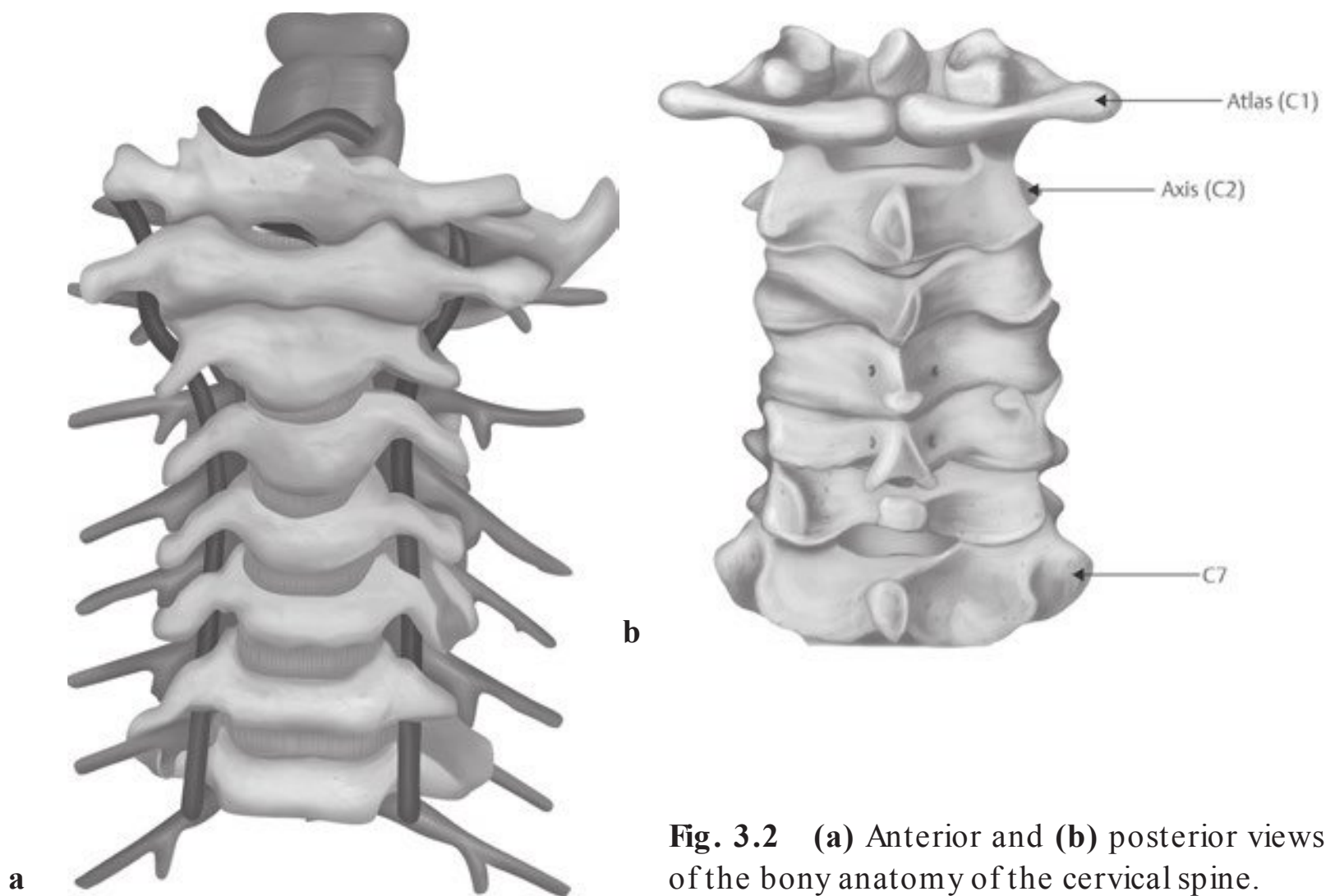


Fig. 3.2 (a) Anterior and (b) posterior views of the bony anatomy of the cervical spine.

3.3 Neural Anatomy

- C3 to C7 nerve roots exit above their corresponding level (e.g., C7 exits above the C7 pedicle).
- Cervical disk herniation is most frequently posterior–lateral and compresses the nerve associated with that level (e.g., C5–C6 herniation impinges on the C6 nerve).

- The C8 nerve root exits above the T1 pedicle.
- The cervical spinal cord enlarges caudally and reaches a maximal cross-sectional area at C6.
- Nerve roots enter the intervertebral foramina laterally, occupy approximately a third of the foramina, and are covered by epidural fat and venous plexus above.
- Nerve roots exit the spine at a point anterolateral to the superior joint facet.
- The cervical plexus is formed by the anterior rami of C1 to C4.
- The phrenic nerve is formed by the anterior rami of C3 to C5 and innervates the diaphragm
- The brachial plexus is formed by the anterior rami of C5 to T1.
- The cervical plexus gives rise to (1) the ansa cervicalis (supplies a branch to the hypoglossal nerve and innervates the strap muscles, except for the thyrohyoid), (2) phrenic nerve (C3 to C5, but mainly C4), and (3) cutaneous nerves of the posterior head and neck.
- Three major sympathetic ganglia rest outside the cervical column, behind the carotid sheath: (1) The superior cervical ganglia lies along the sympathetic trunk from C1 to C4 and carries the sympathetics for the head and neck traversing along the carotid artery. (2) The middle cervical ganglia is located at C5 to C6 (also targeting the head and neck). (3) The stellate ganglia extends from C7 to T1.

3.4 Vascular Anatomy

- Vertebral arteries usually originate from the subclavian artery and ascend between the anterior scalene and longus colli muscles.⁴
- Vertebral arteries enter the spine at the transverse foramen of C6 (occasionally at C7).
- Vertebral artery segments: V1 (preforaminal), origin to transverse foramen entrance; V2 (foraminal), C6 to C2; V3, C2 to dura; V4, intradural segment to basilar artery.
- The transverse foramen is lateral to the vertebral body and anterior to the nerve root groove.
- At C3 to C5, the lateralmost aspect of the transverse foramen is often anteromedial to the midpoint of the lateral mass.

- At C6 to C7, a portion of the transverse foramen is often anterior to the midpoint of the lateral mass.
- Blood supply to spinal cord includes the anterior spinal artery, the two posterior spinal arteries, and the segmental medullary arteries.
- The anterior spinal artery originates from the vertebral arteries just prior to the formation of the basilar artery at the pontomedullary junction.
- The anterior spinal artery is found in the ventral median fissure.
- The posterior spinal arteries most commonly originate from the vertebral arteries but can also arise from posterior inferior cerebellar arteries (PICA).
- The posterior spinal arteries are found in the posterolateral sulci of the spinal cord.
- Venous drainage of the spinal cord: Three anterior and three posterior longitudinally running veins.
- Spinal cord is surrounded by an anterior and a posterior venous plexus.
- The anterior venous plexus is most pronounced medial to the pedicles.

3.5 Ligamentous and Muscular Anatomy

- The anterior longitudinal ligament covers anterior vertebral bodies and resists hyperextension.
- The posterior longitudinal ligament covers posterior vertebral bodies and resists hyperflexion.
- The interspinous and supraspinous ligaments run between adjacent spinous processes and form the ligamentum nuchae.
- The ligamentum nuchae makes up the midline avascular plane.
- The carotid triangle of the neck is an important surgical landmark for anterior approaches, formed laterally by sternocleidomastoid, superiorly by dorsal portion of the digastric, and anteriorly by omohyoid.
- The carotid triangle contains the carotid sheath (common carotid artery, internal jugular vein, and vagus nerve). The artery is anteromedial, the vein is anterolateral, and the nerve runs posteriorly between the artery and vein.
- After bifurcation of the common carotid, the external branch exits the sheath, whereas the internal brain continues within.
- Longus colli muscles lie anterolateral to vertebral bodies and are elevated during anterior spinal procedures (**Fig. 3.3**).

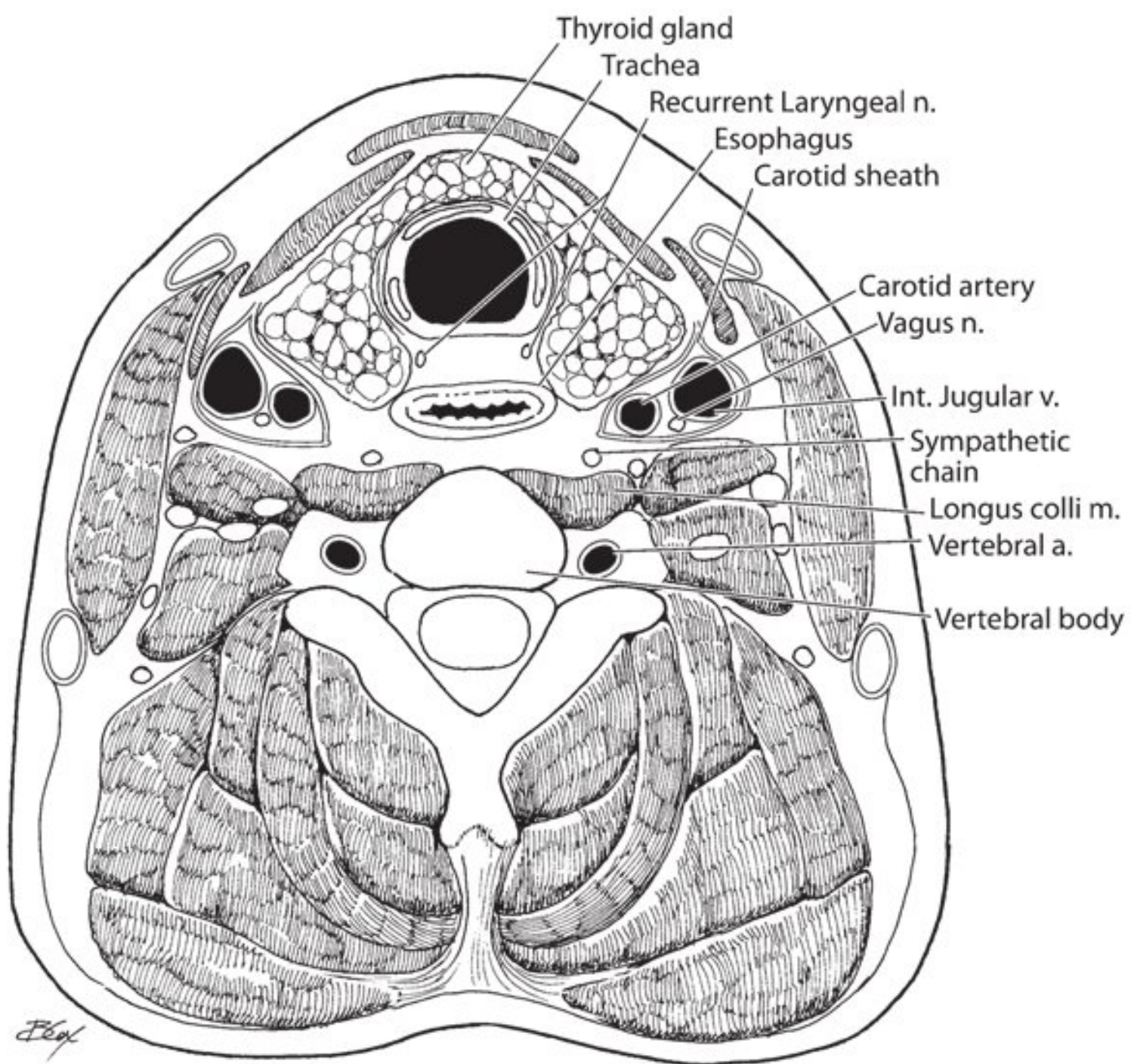


Fig. 3.3 Cross-section of the neck and spine at the C5 level.

3.6 Surgical Pearls

- Uncovertebral joints provide several surgical landmarks: They define the lateral borders for corpectomy or discectomy, and they define the midline for cervical plate placement.
- The Magerl technique for lateral mass screw placement is used to avoid injuring the vertebral artery or nerve root. The drill is placed 1 mm medial to the midpoint of the lateral mass and is angled 25° laterally and 30° superiorly.

- During anterior procedures, instruments are most safely inserted into the lateral aspect of the canal.
- During posterior cervical procedures, the patient can be placed in a slight reverse Trendelenburg position to reduce venous engorgement.
- C7 is a transitional-level vertebra. For a posterior fusion involving C7, some surgeons advocate extending the fusion to T1 to reduce adjacent-level disease.

Common Clinical Questions

1. At what level does cervical disk herniation most often occur?
2. What is the normal curvature of the cervical spine?
3. Which arteries provide the majority of blood circulation to the spinal cord?

Answers to Common Clinical Questions

1. C5/C6 and C6/7C.
2. The cervical spine normally has a lordotic curvature between 16 and 25°.
3. The spinal cord is supplied by the anterior spinal artery (from the vertebral arteries), two posterior spinal arteries (from either the vertebral arteries or PICA), and segmental medullary arteries.

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4 Thoracic Spine

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4.1 Key Points

- Twelve rib-bearing vertebral segments make up the normal thoracic spine; anatomical variants include 11 and 13 rib-bearing vertebral segments.
- Motion in the thoracic spine is limited by its osteoligamentous relationship with the rib cage.
- Average thoracic kyphosis ranges from 10 to 40°. The apex of kyphosis typically occurs at T7.¹
- Vascular supply in the thoracic spinal cord is tenuous and at great risk for ischemia during and after surgical procedures.

4.2 Bony Anatomy (Fig. 4.1 and Fig. 4.2)²

- The vertebral bodies of the thoracic spine increase in size from the upper to the lower thoracic spine.³

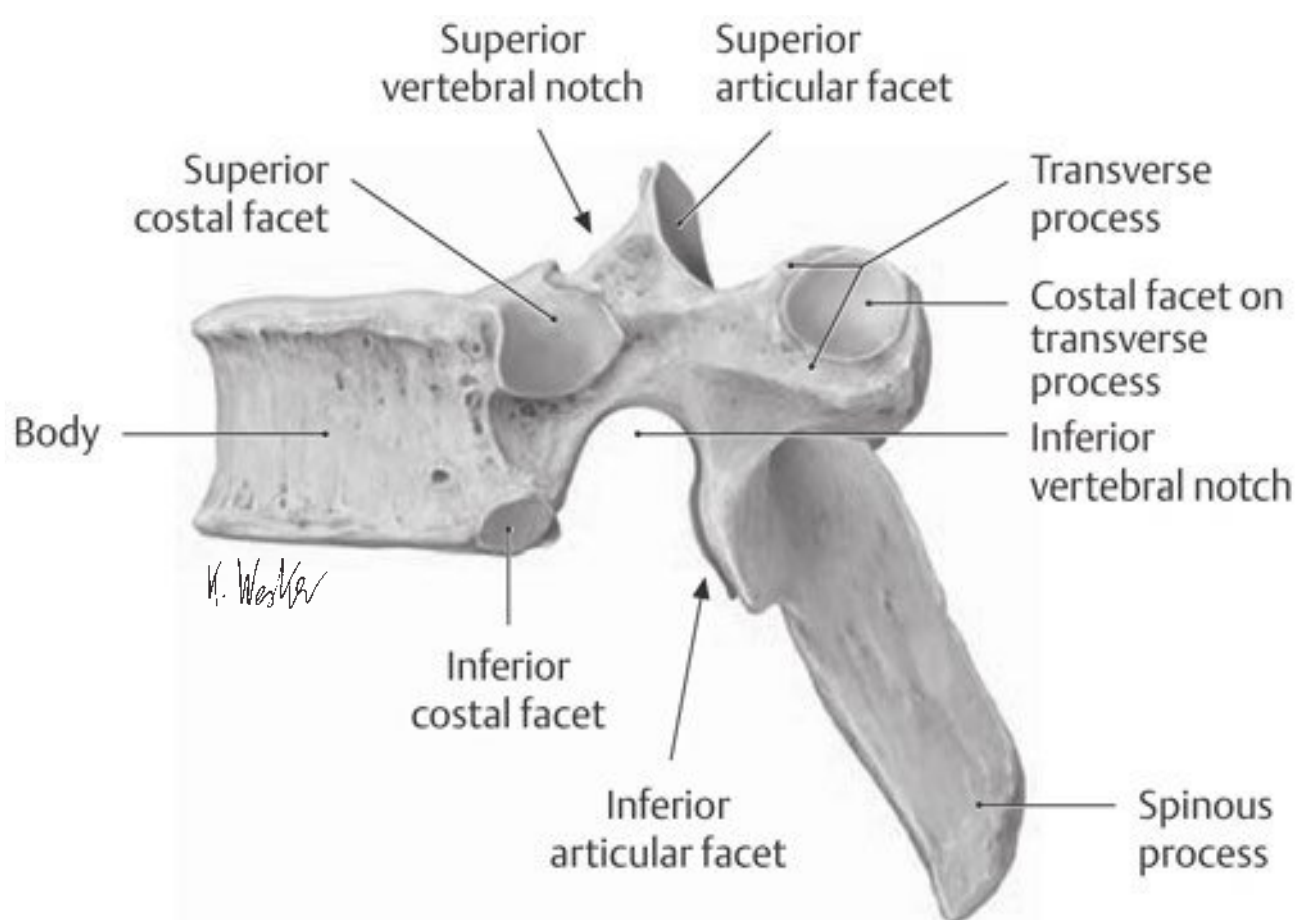


Fig. 4.1 Lateral view of thoracic vertebra. (From Thieme Atlas of Anatomy, General Anatomy and Musculoskeletal System, © Thieme 2005, illustration by Karl Wesker.)

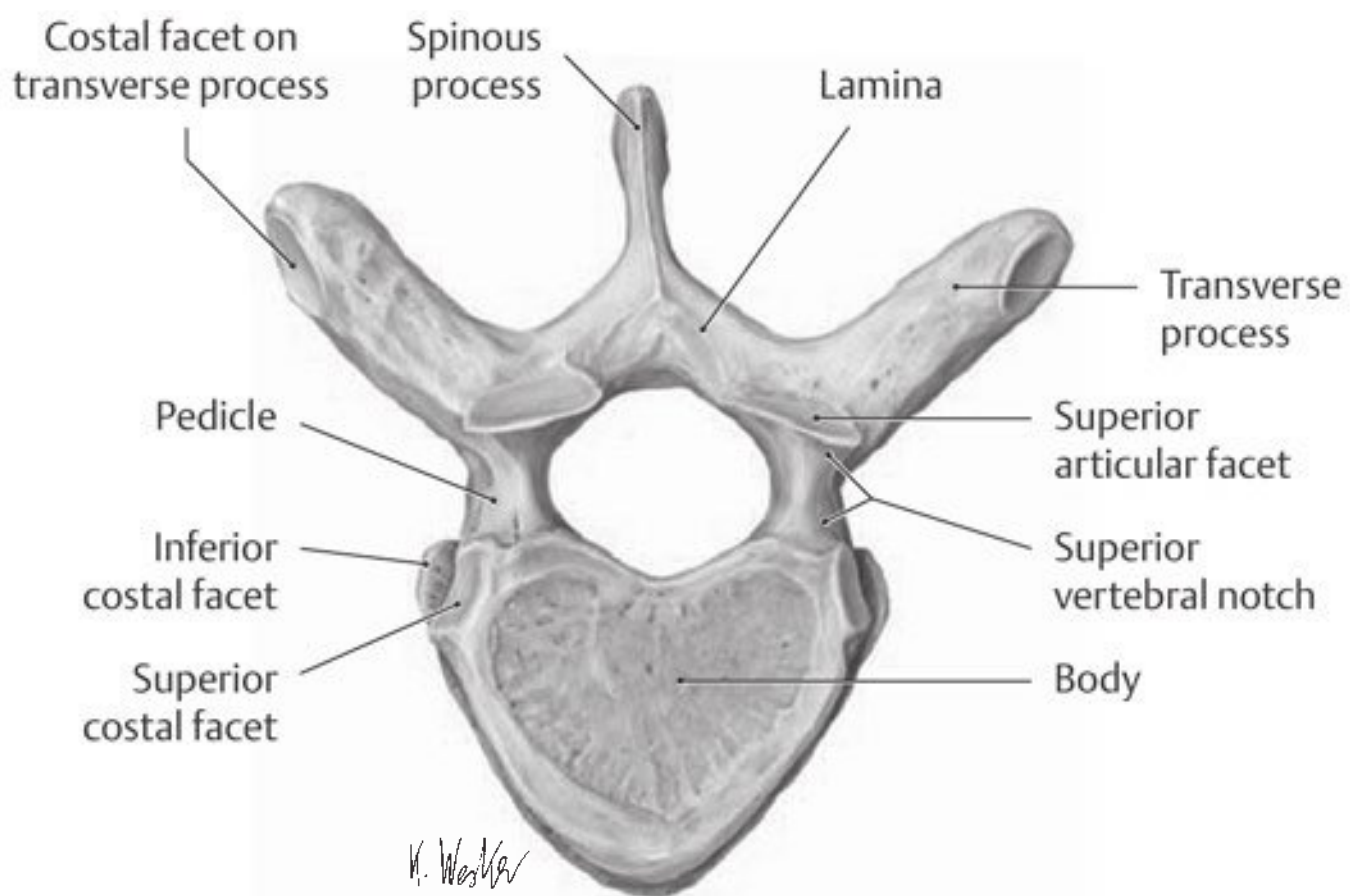


Fig. 4.2 Superior view of thoracic vertebra. (From Thieme Atlas of Anatomy, General Anatomy and Musculoskeletal System, © Thieme 2005, illustration by Karl Wesker.)

- Thoracic vertebral bodies are wedge-shaped, with shorter anterior aspects than posterior aspects. This results in a smooth kyphotic curvature of the thoracic spine.^{1,3}
- The thoracic facets from T1 to T10 are oriented in a coronal plane compared with the more axially oriented cervical facets and the more sagittally oriented T11, T12, and lumbar facets, providing stability in flexion and extension.^{1,3}
- The superior articular facet is anterior and inferior to the inferior articular facet of the level above.
- Pedicle height increases from the upper to lower thoracic spine.³
- Pedicle width decreases from the T1 to T5 to T6, then gradually increases from T5 to T6 to T12.^{1,3}
- Thoracic pedicles join the thoracic vertebral bodies at an acute angle with the superior aspect of the pedicle at the level of the disk space above the vertebral body and the inferior aspect of the pedicle at the midpoint of the vertebral body. The transverse pedicle angle decreases from T1 to T12.³
- The diameter of the thoracic spinal canal ranges from 20.3 mm at T1 to 15.9 mm at T5 to 22.3 mm at T12.⁴
- The vertebral bodies from T2 to T8 or T9 articulate with two ribs at bilateral demifacets, two superiorly and two inferiorly. The thoracic costotransverse facets from T1 to T10 are located on the transverse process and articulate with the same-level rib (e.g., T7 rib articulates with T6 and T7 vertebral bodies at the demifacets and the T7 transverse facet).

- The T1, T10, T11, and T12 (and sometimes T9) ribs have a full facet for articulation with the corresponding vertebra.
- The 11th and 12th ribs do not articulate with the transverse processes of the vertebral body at the corresponding level.

4.3 Neural Anatomy

- Nerve roots in the thoracic spine exit below the pedicle of the corresponding vertebral level (e.g., T1 nerve root exits below the T1 pedicle).
- Nerves of the thoracic spine innervate the trunk and abdomen. The T4 dermatome is at the nipple line, the T6 dermatome at the base of the sternum, and the T10 dermatome at the level of the umbilicus.
- Preganglionic sympathetic neurons are located in the intermediolateral cell columns from T1 to L2 or L3 and project axons to the sympathetic chain ganglia located on each side of the thoracolumbar vertebrae.
- The spinal cord typically runs through the entire length of the thoracic spinal canal, usually terminating in the conus medullaris at the L1 or L2 level.

4.4 Vascular Anatomy

- The anterior spinal artery runs in the anterior median fissure of the spinal cord and arises from vertebral and radicular arteries.
- The paired posterior spinal arteries run lateral to the posterior intermediate sulci of the spinal cord and usually arise from the posterior inferior cerebellar arteries and radicular arteries.
- Six to eight radicular arteries supplement the arterial supply of the thoracic spinal cord and arise from the ascending cervical artery, the deep cervical artery, the intercostal arteries, the lumbar arteries, and the sacral arteries.
- The artery of Adamkiewicz provides the main arterial supply to the spinal cord from the T8 level to the conus. It enters the spinal canal between the T9 and L2 levels in 85% of people and arises from the left side in 80% of people. It is the largest radicular artery.^{1,3}
- If arterial supply to the thoracic cord from the aorta is interrupted, collateral flow can come from the internal thoracic and lateral thoracic arteries, which anastomose with the posterior intercostal arteries.

4.5 Ligamentous and Muscular Anatomy (Table 4.1)

- Muscles of the thoracic spine
 - Superficial layers
 - Trapezius, rhomboid and latissimus dorsi

Table 4.1 Principal thoracic ligaments: Their attachments and modes of action

Ligament	Attachments	Action
Anterior longitudinal	Ventral vertebral body and annulus	Limits extension and distraction Thickest in thoracic spine
Posterior longitudinal	Dorsal vertebral body and annulus	Limits flexion and distraction Thickest in thoracic spine
Ligamentum flavum	Lamina of adjacent vertebrae	Limits flexion
Interspinous	Spinous processes of adjacent vertebra	Limits flexion and distraction
Supraspinous	Tips of spinous processes and thoracolumbar fascia	Limits flexion and distraction
Facet capsule	Superior and inferior facets of adjacent vertebra	Supports facet joint stability
Radiate	Rib and adjacent vertebral disk space and vertebral bodies anteriorly	Stabilizes rib attachments Contributes to limitation of thoracic flexion, extension, lateral bending, and axial rotation
Superior costotransverse	The rostral rib neck and the transverse process of the vertebra one level above	
Lateral costotransverse	The apex of the transverse process to the adjacent costal tubercle	

- Intermediate layers
 - Splenius cervicis, serratus posterior superior and inferior, and erector spinae (spinalis, longissimus, iliocostalis)
- Deep layers
 - Rotatores longus and brevis, levatores costarum longus and brevis, multifidus, semispinalis thoracis, and external intercostal muscles

4.6 Surgical Pearls

- Laminectomy of the thoracic spine disrupts the posterior tension band and can increase the risk of progressive kyphosis.
- Avoid injury to the artery of Adamkiewicz to prevent spinal cord infarction.
- Spinal cord insult from laminectomy for anterior thoracic pathology is often a result of vascular ischemia from cord retraction.
- Lateral, posterolateral, and anterior approaches in the thoracic spine have been developed to minimize risk of cord ischemia resulting from retraction of the thoracic spinal cord.
- Preoperative CT scan with fiducial markers can minimize the possibility of performing a wrong-level surgery in the thoracic spine and help in planning thoracic pedicle screw placement while avoiding neural injury.
- To prevent early instrumentation failures and progressive deformity, avoid ending spinal constructs at the thoracic kyphotic apex and the thoracolumbar junction.

Common Clinical Questions

1. True, or false? The pedicles on the convexity of a scoliotic curve tend to be larger than the pedicles on the concavity of the curve.
2. True, or false? Posterior laminectomy is the safest approach for midline ventral lesions in the thoracic spine.
3. True, or false? The artery of Adamkiewicz typically arises above the T8 level on the right.

Answers to Common Clinical Questions

1. True. Pedicles on the convexity are usually larger.
 2. False. Posterior laminectomy alone is not a safe option for midline ventral lesions in the thoracic spine.
 3. False. The artery of Adamkiewicz typically arises at the T9 to L2 levels. It enters the spinal canal from the left in 80% of people.
-

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5 Lumbar Spine

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and Ali Bydon

5.1 Key Points

- The lumbar spine (composed of vertebrae L1–L5) is a region of substantial loading and mobility, and it has an associated high incidence of trauma and age-related degeneration.
- The three-column model of the spine, developed to prognosticate the stability of thoracolumbar fractures, provides a framework to categorize the relevant clinical anatomy of the lumbar spine (**Fig. 5.1**):¹
 - Anterior column: Anterior half of the disk and vertebral body, as well as the anterior longitudinal ligament

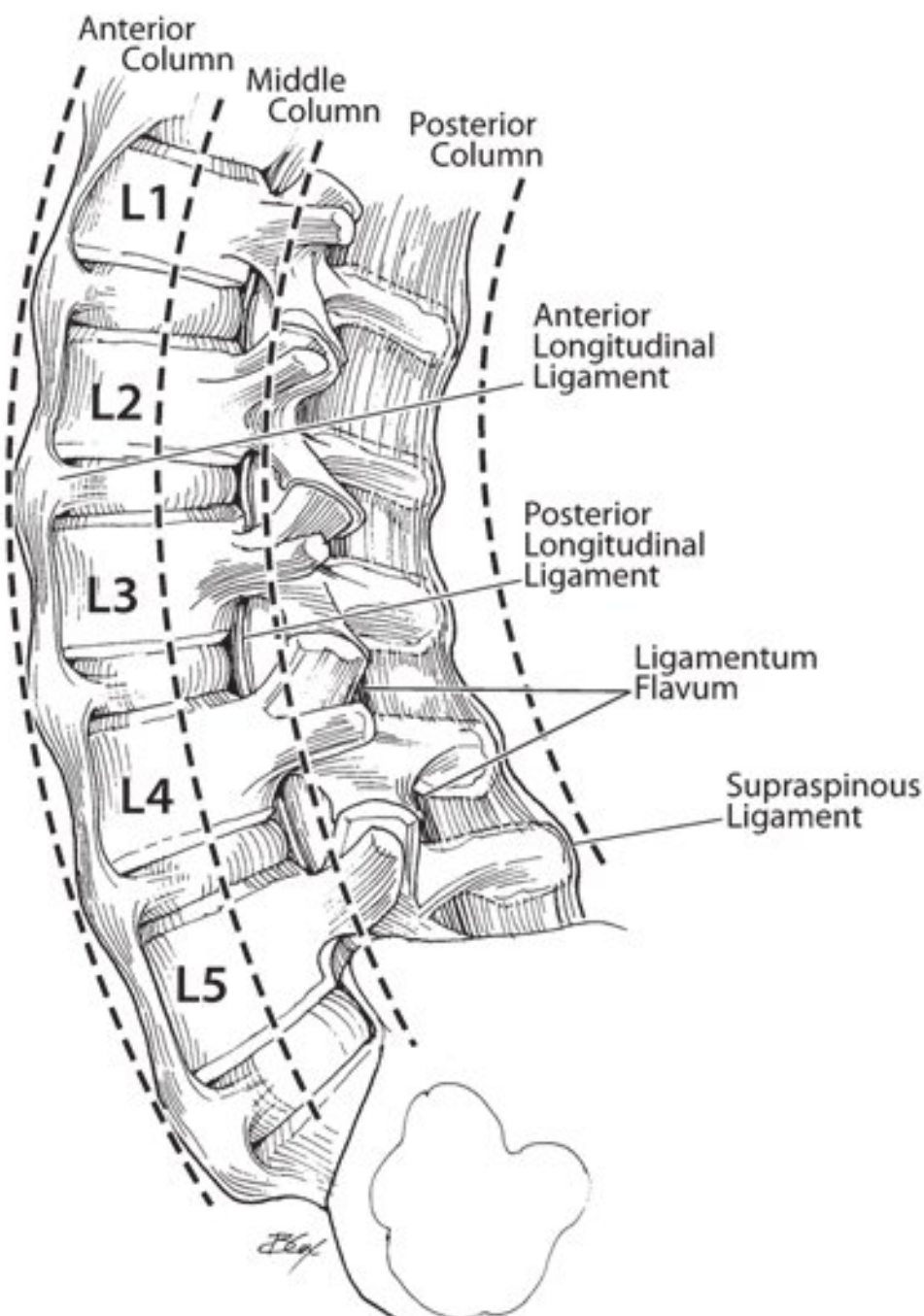


Fig. 5.1 Sagittal illustration of the lumbar spine demonstrating the three-column theory.

- Middle column: Posterior half of the disk and vertebral body, as well as the posterior longitudinal ligament
- Posterior column: Posterior bony arch, facet joint and its capsule, supraspinous and interspinous ligaments, and ligamentum flavum

5.2 Bony Anatomy (Table 5.1, Table 5.2)

- The vertebral body’s main biomechanical function is to support compressive loads caused by body weight and muscular forces.²
- The intervertebral disk lies between two adjacent vertebral bodies, and it is an avascular structure. The outer layer of the disk is called the annulus fibrosus and consists of 15 to 26 concentric layers of fibrous cartilage.² The nucleus pulposus lies in the center of the disk; primarily gelatinous,

Table 5.1 Bony borders of the vertebral column

Bone	Column	Description	Major pathology
Vertebral body	Anterior and middle	Lumbar vertebral bodies are the largest in the spine Major weight-bearing component Anterior portion of the vertebral column	Compression or burst in trauma, osteoporosis, or tumor
Pedicle	Posterior	Short and strong Arise from the upper and posterolateral vertebral body, forming the bilateral anterolateral portions of the vertebral column Frequently, transpedicular instrumentation is placed for stabilization or fusion	May fracture in trauma, osteoporosis, or tumor
Lamina	Posterior	Short, broad plates Form the bilateral posterolateral borders of the vertebral column May be removed in decompressive surgery	May fracture in trauma, osteoporosis, or tumor
Spinous process	Posterior	At the meeting point of the two laminae Forms the posterior border of the vertebral column	

Table 5.2 Borders of the intervertebral foramen

Anterior	Posterior borders of the adjacent vertebral bodies and disks
Superior	Inferior border of the pedicle of the superior vertebrae
Posterior	Pars interarticularis and facet joint
Inferior	Superior border of the pedicle of the inferior vertebrae

its function is to absorb compressive forces. Posterolateral herniations of the disk can compress an individual nerve root; central herniations can compress the entire cauda equina.¹

- In healthy disks, compressive axial loads increase the hydrostatic pressure within the nucleus, which is resisted by tensile stresses in the outer annulus fibrosus; “pressure within the nucleus is approximately 1.5 times the externally applied load per unit disk area.”²
- Bending and torsion forces are resisted by the fiber bundles of the outer annulus.²
- At the level of the pedicle, the transverse process arises. The processes of the first three lumbar vertebrae are long and slender, and those of the fourth and fifth are more pyramidal. In posterolateral lumbar fusion procedures, the bone graft is often placed in the lateral gutter—the area lateral to the facets where the transverse processes lie.
- The pars interarticularis, also referred to as the isthmus, is a thin bone of the posterior arch of the lumbar vertebrae where the lamina and the inferior articular process join the pedicle and superior articular process. A fracture in the pars interarticularis is referred to as spondylolysis, which can be found in 5 to 6% of the population; this may predispose the individual to the development of isthmic spondylolisthesis.¹
- The facet joint (zygapophysial joint) is composed medially of the inferior articulating process of the superior vertebrae and laterally of the superior articulating process of the inferior vertebrae. Surrounding the facet joint is an articular capsule. Hypertrophy of the facet joint and its capsule can contribute to both spinal canal and foraminal stenosis.³ Additionally, the facet joints may give rise to spinal synovial cysts, which in turn may also cause spinal stenosis.
- A motion segment, or *functional spinal unit*, comprises two adjacent vertebrae, the intervertebral disk, and the surrounding soft tissues.²

5.3 Neural Anatomy

- The spinal cord ends at the conus medullaris, most frequently at the level of the L1 or L2 vertebral body. Inferiorly, the roots descend within the thecal sac as the cauda equina before they exit individually (**Fig. 5.2** and **Fig. 5.3**).⁴
- The exiting nerve roots leave the vertebral column through the intervertebral foramen, closer to the superior pedicle (for example, the L4 nerve root exits right below the L4 pedicle). Far lateral disk herniations can compress the nerve root at this point (leading patients to present with radiculopathy); in such cases, the intervertebral disk actually compresses the nerve root of the superior level (for example, an L4–L5 disk compresses the L4 root). Far lateral herniation is much less common than posterolateral herniation of the disk.
- Facet hypertrophy can also lead to foraminal stenosis, which also causes radiculopathy.¹

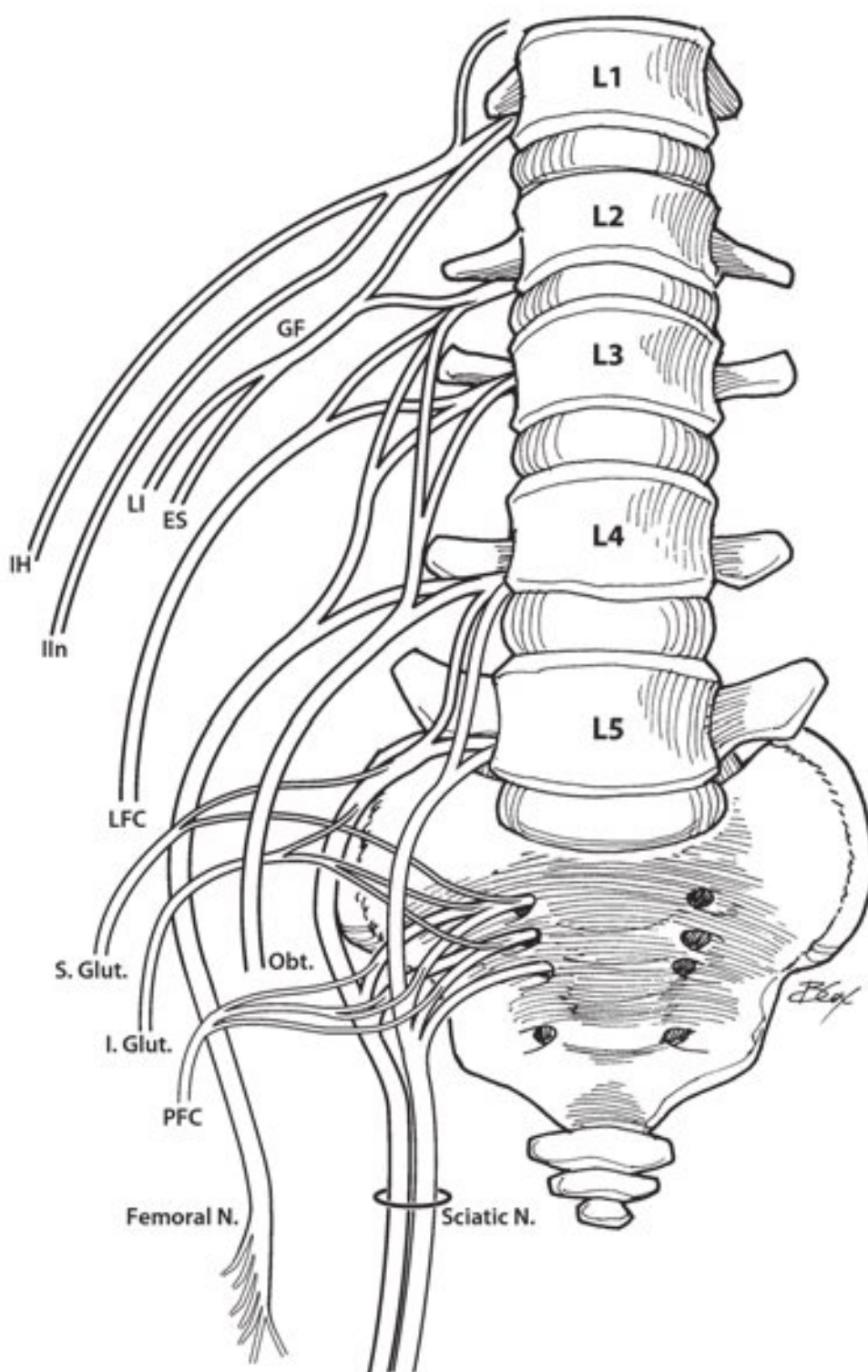


Fig. 5.2 Anterior view of the lumbar plexus.

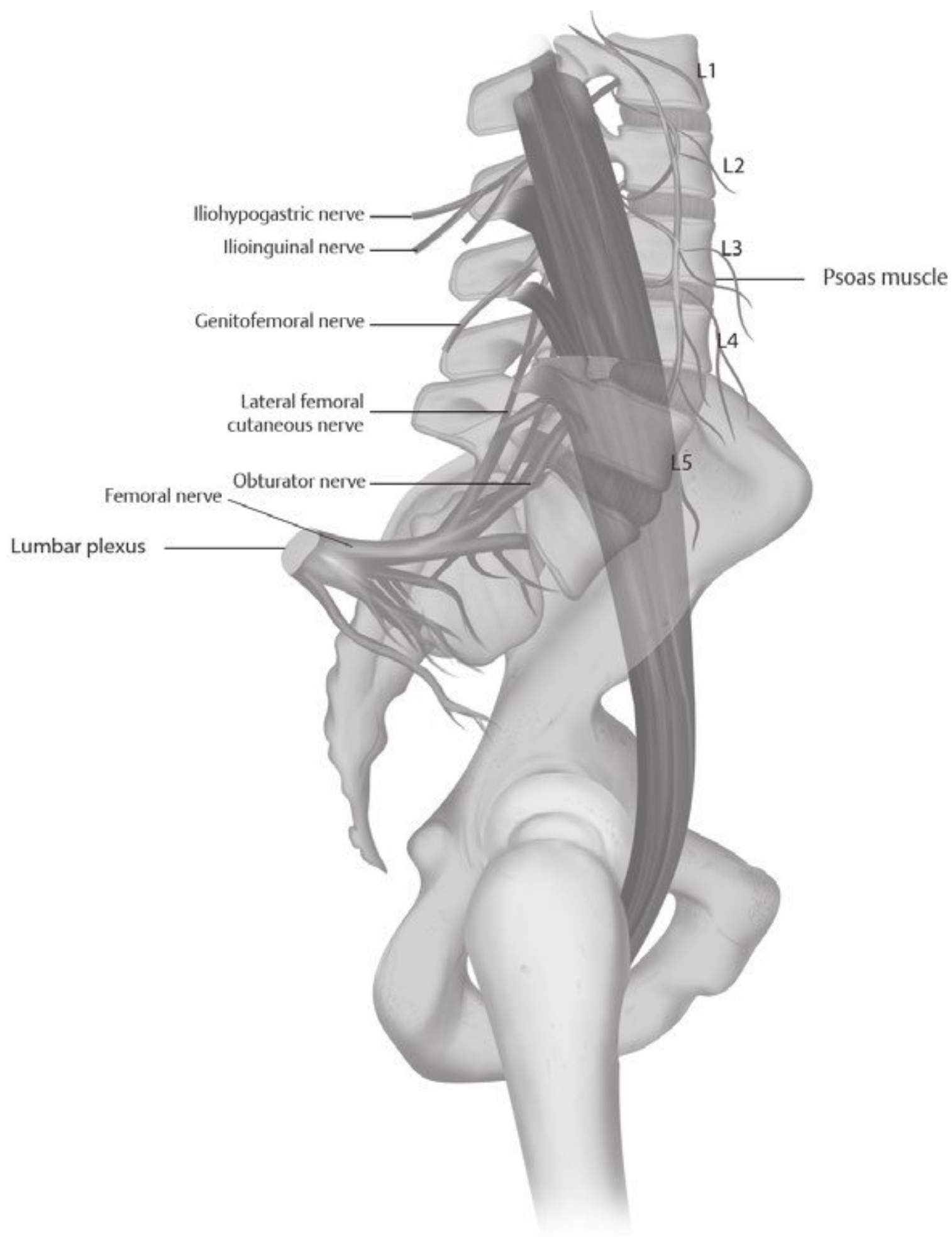


Fig. 5.3 Lateral view of the lumbar plexus.

5.4 Vascular Anatomy

- Segmental arteries arise primarily from the lumbar arteries, which in turn divide into anterior and posterior radicular arteries, which enter the intervertebral foramen along with the nerve roots. The arterial feeders of the spinal cord include branches of these radicular arteries as well as the

segmental medullary arteries, which also come off of the segmental spinal arteries, with one artery coursing anteriorly and two coursing posteriorly. The cauda equina is supplied by branches of the lumbar, iliolumbar, lateral, and median sacral arteries.⁴

- Two different plexuses of veins, one external and the other internal, extend along the vertebral column. The anterior external plexus lies in front of the vertebral bodies, and the posterior external plexus lies around the posterior arch of the vertebral column. The internal vertebral plexus is a network of veins in the epidural space within the vertebral canal that are in communication with the basivertebral plexus; the veins tunnel through the cancellous tissue of the vertebral bodies. Intradural venous drainage of the spinal cord involves retrograde anterior and posterior central veins.⁴
- Approaching the lumbar spine from an anterior approach, the great vessels (the aorta and the inferior vena cava) lie directly anterior to the vertebral bodies, with the bifurcation of the vessels at the level of the L4–L5 disk space or at the level of L5. The L5–S1 disk can generally be accessed inferior to the bifurcation; to reach the L4–L5 disk, these vessels must be retracted laterally.¹

5.5 Ligamentous and Muscular Anatomy (Table 5.3)

- Exposure of the lumbar spine from a posterior approach requires traversing the thoracolumbar fascia. In the lumbar region, the thoracolumbar fascia is thick and is attached to the spinous processes and the supraspinous ligament. It then extends laterally, covering the erector spinae muscles.³
- Deep in the thoracolumbar fascia lie the erector spinae muscles, which arise from the sacrum, the spinous processes of the lumbar and thoracic vertebrae, and the iliac crest. The erector spinae muscles include the iliocostalis, longissimus, and spinalis muscles, which are important in flexion, extension, and lateral rotation of the vertebral column. Deep in the erector spinae muscles are the multifidus and rotatores muscles.³

5.6 Surgical Pearls

- During transpedicular instrumentation placement, the landmarks used for cannulation of the pedicles are the meeting point of the pars interarticularis, the superior articulating process, and the transverse process. The small ridge of bone at that level is called the mammillary process.

Table 5.3 Ligaments of the lumbar spine

Ligament	Column	Attachment	Major pathology
Anterior longitudinal ligament	Anterior	Anterior margins of the vertebral bodies and intervertebral disks	May be disrupted in fracture–dislocation traumatic injuries and in spondylolisthesis
Posterior longitudinal ligament	Middle	Posterior margins of the vertebral bodies and intervertebral disks	May be disrupted in fracture–dislocation or seatbelt trauma and in spondylolisthesis
Ligamentum flavum	Posterior	Anterior border of adjacent laminae and spinous processes	May be hypertrophied in spinal stenosis
Supraspinal ligament	Posterior	Posterior borders of spinous processes	
Interspinous process ligaments	Posterior	Inferior border of superior spinous process and superior border of inferior spinous process	

- For lumbar disk herniation, after a laminotomy is performed, feel for the inferior pedicle (L5 in an L4–L5 disk herniation). Palpate its medial wall using a Woodson dissector. Immediately inferior is the nerve root (L5). With a nerve hook, retract the root medially and dissect superiorly until you feel a bulge (the herniated disk fragment).

Common Clinical Questions

1. Compression of any given nerve root in the neural foramen most often results from hypertrophy of which articulating process?
2. An L4–L5 far lateral herniated disk will compress which nerve root?
3. A midline herniated disk at the level of L5/S1 is more likely to compress which nerve root?

Answers to Common Clinical Questions

1. The superior articulating process of the lower vertebra.
 2. The exiting L4 root (not the traversing L5 root).
 3. The S1 nerve root.
-

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6 Sacral–Iliac Spine

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6.1 Key Points

- The sacrum is the structural link that distributes load from the lumbar spine to the pelvis through the sacroiliac joints (and vice versa).
- The bulbocavernosus reflex involves the S2 to S4 sacral nerves, and its absence or presence is of prognostic significance in spinal cord trauma.¹
- Because of the location of the lumbosacral plexus in relation to the sacrum, sacral fractures have a high incidence of neurological injury (up to 25%).¹

6.2 Bony Anatomy

- Normal adult spine consists of five fused sacral vertebrae that form the wedge-shaped sacrum and four fused coccygeal vertebrae that form the coccyx (the skeletal remnants of a tail).
- The sacrum has four paired sacral foramina, a sacral canal, a sacral promontory (anterior projection of the S1 body), and a sacral hiatus (clinically useful for caudal epidural anesthesia) (**Fig. 6.1** and **Fig. 6.2**).²

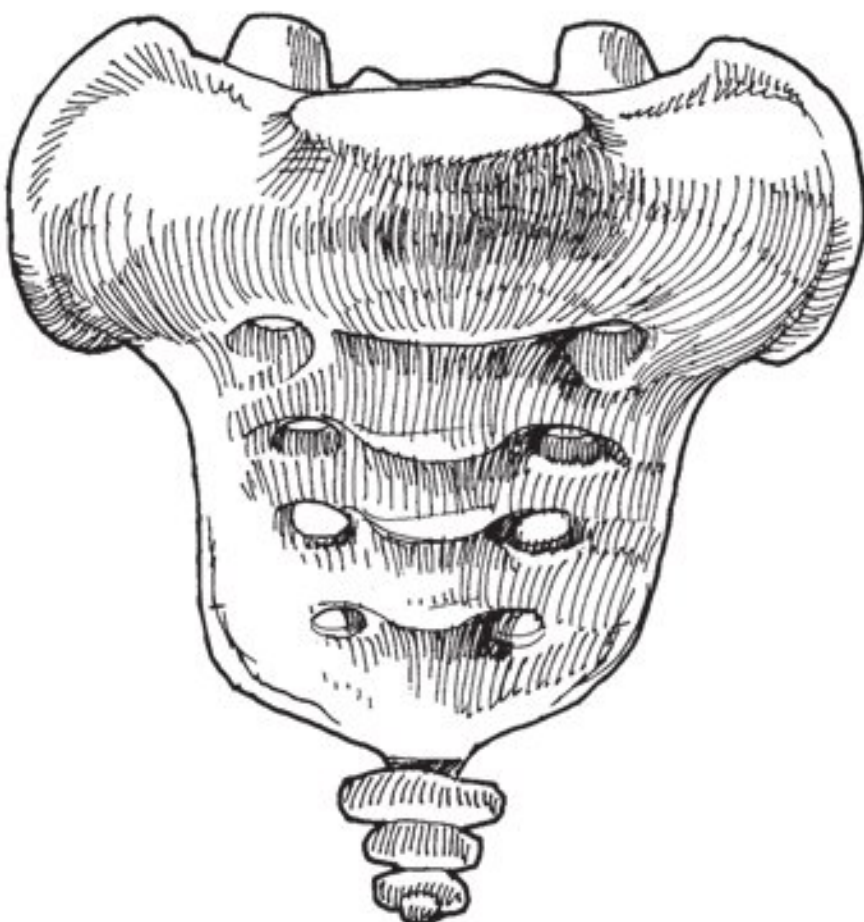


Fig. 6.1 Anterior view of the bony sacrum.

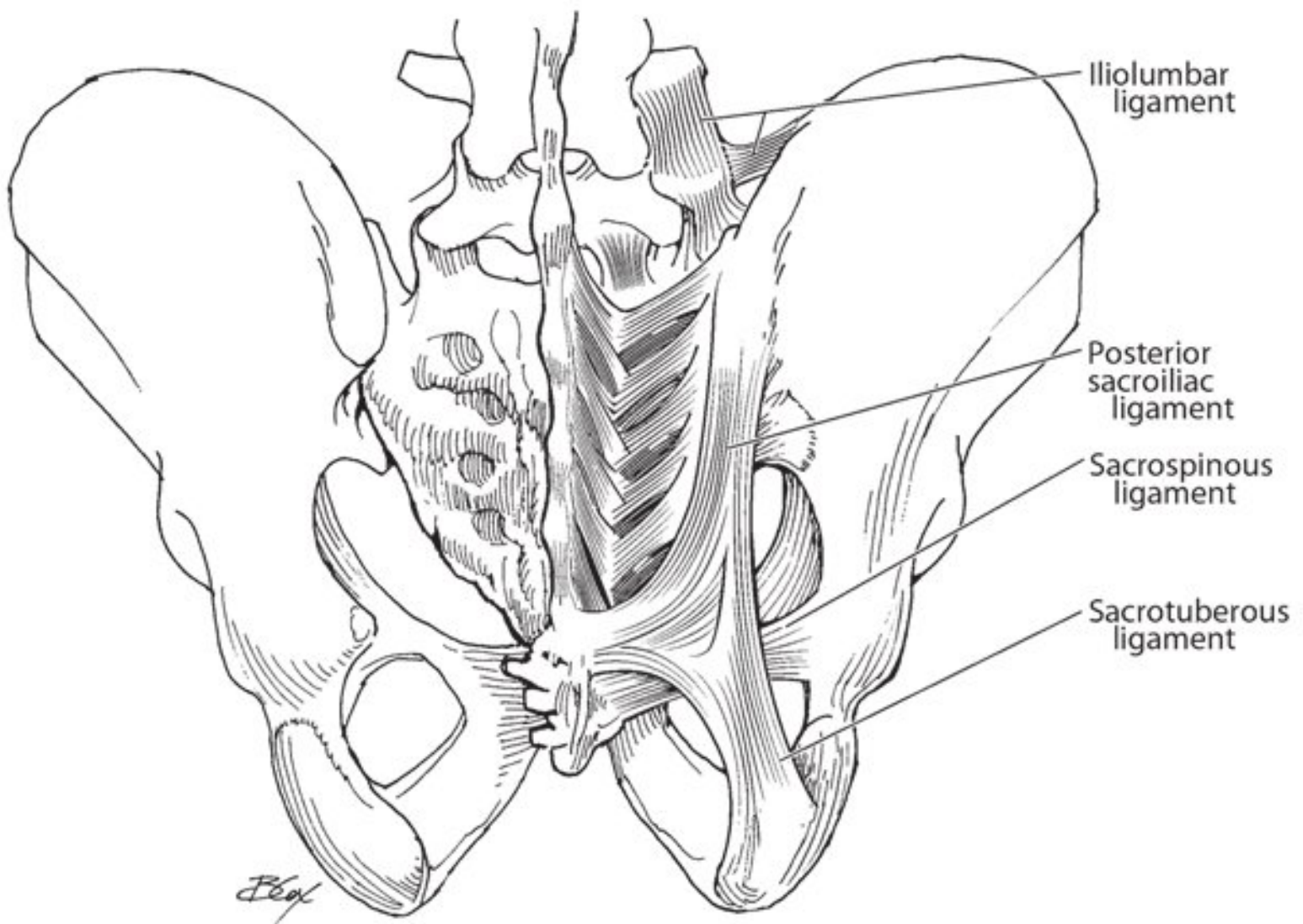


Fig. 6.2 Posterior view of bony sacrum.

- The sacroiliac (SI) spine has many palpable bony landmarks, including the sacral cornu and the iliac crest. The posterior superior iliac spine may be difficult to palpate but is readily identifiable by the permanent skin dimples above the buttocks.
 - An imaginary line connecting the dimples passes through the S2 spinous process and the middle of the SI joint in the anterior–posterior plane.³
 - An imaginary line connecting the highest points of the iliac crest passes through the L4–L5 intervertebral disk space.
- The sacrum has multiple points of articulation:
 - The S1 body articulates with the L5 body via the L5/S1 disk to form the lumbosacral angle, which varies from 130 to 160°.³
 - The inferior facet of L5 articulates with the superior facet of S1 and acts as a buttress to resist anterior translation.⁴
 - The apex of the sacrum articulates with the coccyx.
 - The lateral aspects of the sacrum articulate with the two coxal (innominate) bones to form the SI joint (a true synovial diarthrodial gliding joint with limited motion). Only the anterior 25% is synovial in nature (the rest is ligamentous attachment).³

6.3 Neural Anatomy

- Ventral and dorsal branches of the sacral nerves exit the four pair of anterior and posterior sacral foramina, respectively; the anterior foramina are larger in caliber than their posterior counterparts.⁴
 - Dermatomes supplied by sacral nerves (**Fig. 6.3**)
- The sacral canal contains the nerve roots of the cauda equine (**Fig. 6.4**).
- The pelvic splanchnic nerves are composed of parasympathetic fibers derived from S2 to S4 and supply autonomic innervation to various abdominal and pelvic viscera.¹

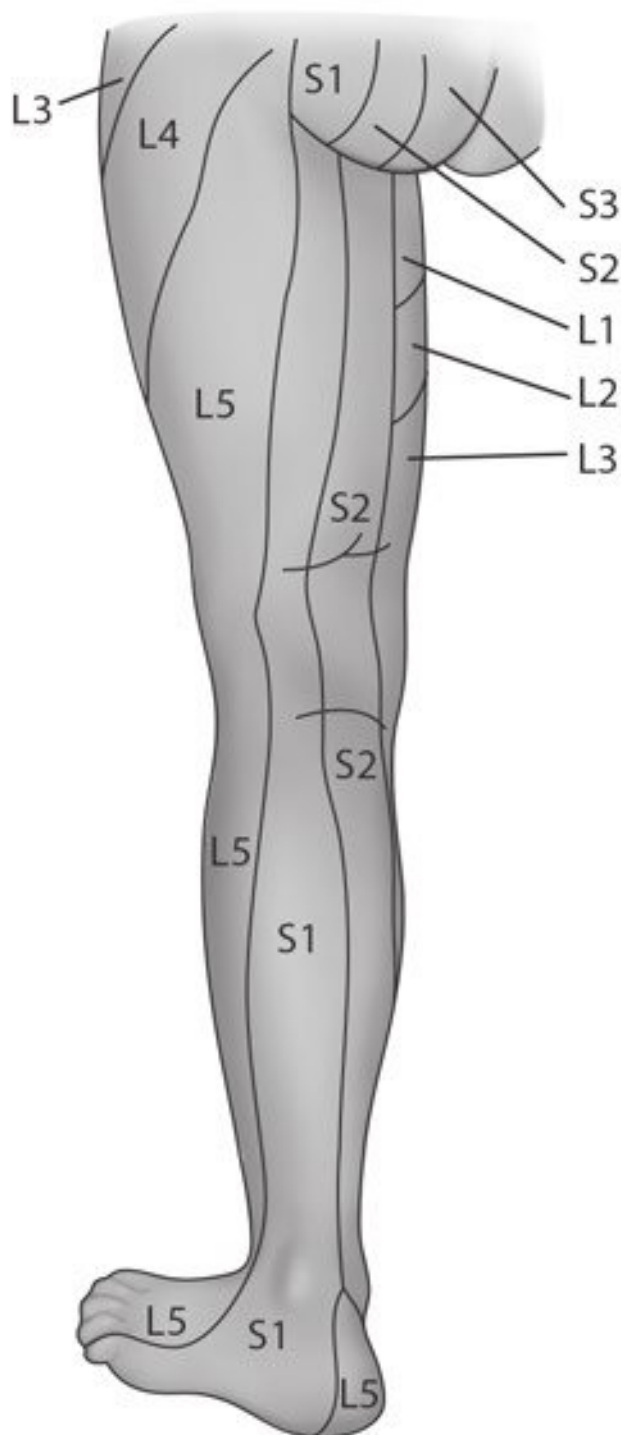


Fig. 6.3 Dermatomal map of sacral nerve roots.

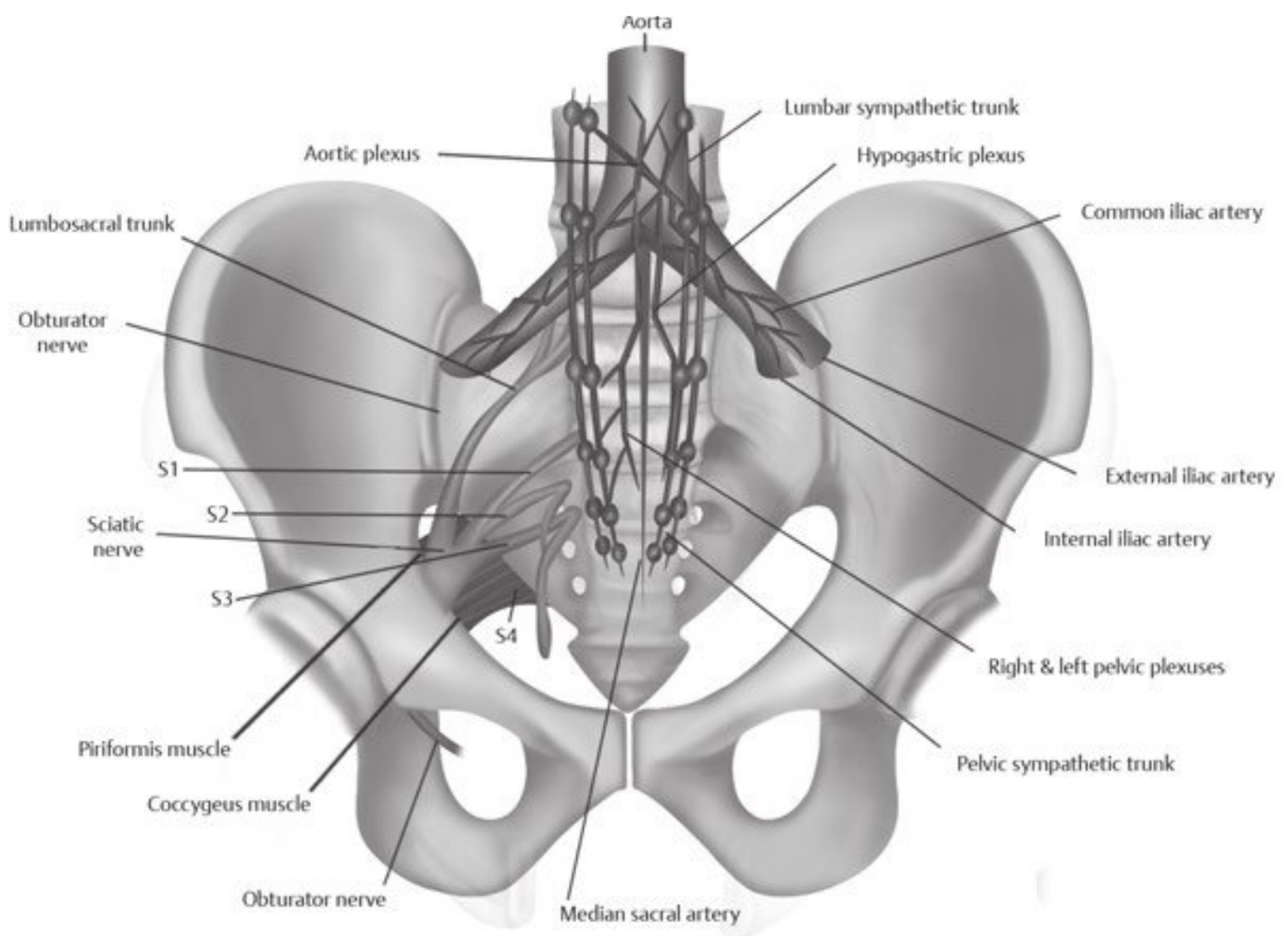


Fig. 6.4 Ventral view of the sacrum with exiting sacral nerve roots.

- The lumbosacral plexus is composed of the ventral rami from T12 to S3 and lies posterior to the psoas muscle.³
- The following are major nerves that arise from this plexus:³
 - Sciatic nerve: Composed of the ventral rami from L4 to S3 and has an anterior preaxial tibial division and a postaxial peroneal division; most commonly, it exits the pelvis via the greater sciatic foramen inferior to the piriformis muscle
 - Pudendal nerve: Composed of the anterior divisions of the ventral rami of S2 to S4; supplies the perineum and external genitalia
 - Superior gluteal nerve: Composed of the posterior divisions of the ventral rami of L4 to S1 to supply the gluteus medius, gluteus minimus, and tensor fascia lata
 - Inferior gluteal nerve: Composed of the posterior divisions of the ventral rami of L5 to S2; courses with the inferior gluteal artery to supply the gluteus maximus

6.4 Vascular Anatomy

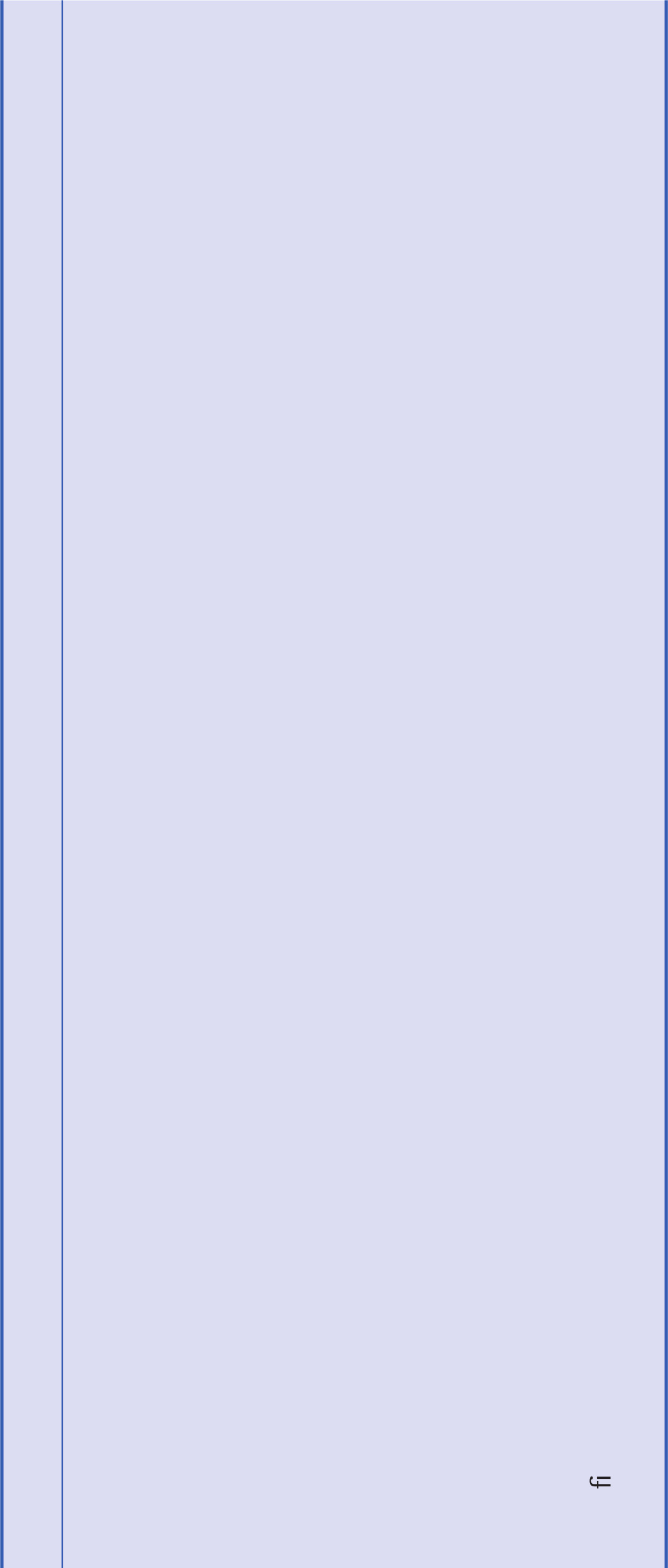
- The abdominal aorta begins at the aortic hiatus in the diaphragm at the level of T12 and bifurcates into the common iliac arteries at the level of L4.⁴
- The common iliac veins unite at the level of L5 to form the inferior vena cava.
- Common iliac artery bifurcates anterior to the SI joint to form the internal iliac artery and descends posteriorly into the greater sciatic foramen to supply the pelvis, buttocks, medial thigh, and perineum.³
 - The internal iliac vein sits between the SI joint and the internal iliac artery.
- The median sacral artery and vein is an unpaired vessel that originates from the posterior aspect of the abdominal aorta to supply the lower lumbar vertebrae, sacrum, and coccyx.³
- The lateral sacral artery paired and branches from the internal iliac artery and courses through the anterior sacral foramina to supply the spinal meninges.³

6.5 Ligamentous and Muscular Anatomy

- Iliolumbar ligament attaches to the transverse process of L5 with the ilium.
 - Unstable vertical shear fractures can avulse off the L5 transverse process.¹
- The anterior SI, posterior SI, and interosseous ligaments suspend the sacrum between the ilia.³
 - The posterior SI and interosseous ligaments are thicker and limit motion.
- The sacrotuberous ligament attaches the sacrum to the ischial tuberosity; the sacrospinous ligament attaches the sacrum to the ischial spine.
 - They function to limit upward movement of the caudal portion of the sacrum and delineate the greater and lesser sciatic foramen.³
- See **Table 6.1** for major muscles of the sacroiliac spine.

6.6 Surgical Pearls

- Transitional vertebra can lead to incorrect surgical localization.
- Aggressive exposure of sacral ala for posterolateral fusion can endanger L5 nerve root.
- Low L5 to S1 disk may not be accessible via an anterior approach as a result of pubic symphysis obstruction.
- Anterior dissection of the superior hypogastric sympathetic plexus can result in retrograde ejaculation in males, with rates ranging from 0.42 to 5.9%¹



Common Clinical Question

1. Of the listed nerves, which is the most susceptible to injury during an anterior approach to the SI joint?
 - A. S1 nerve root
 - B. Femoral
 - C. Ilioinguinal
 - D. L5 nerve root
 - E. Genitofemoral
-

Answer to Common Clinical Question

1. D. The L5 nerve root runs just anterior to the SI joint and is susceptible to injury not only in open procedures, but also with percutaneous sacroiliac screw placement. It courses 2 to 3 mm medial to the SI joint.
-

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II

Clinical Spine Surgery

7 Physical Examination

Mark S. Greenberg

7.1 Key Points

- There are four main components to the spinal exam: Motor, sensory, reflex and mechanical. This is in addition to the general exam, which includes observation for cutaneous and nail changes, deformity, pain behaviors, and so forth.
- Although components of a general survey nature should always be included, the physical exam is tailored to specific situations based on the history, the region of suspected involvement, and abnormal findings during the survey exam.
- No protocol can cover every contingency, and the exam must be individualized based on patient specific factors. The order of the exam must also be tailored to the situation.

7.2 Main Components of the Spinal-Related Exam

- Motor (strength, coordination, spasticity, and muscle bulk/tone, including atrophy/fasciculations)
 - Strength evaluation is usually graded using the Royal Medical Research Council of Great Britain (MRC) scale¹ shown in **Table 7.1**.
 - An overview of survey muscle groups to examine is shown in **Table 7.2**.
 - Further motor testing as indicated. It may include tibialis posterior and gluteus medius (in cases of foot drop to distinguish radiculopathy from peripheral neuropathy), first lumbrical (median nerve), and abductor digiti minimi (ulnar nerve).
- Sensation (pinprick, light touch, proprioception, temperature)
 - Pinprick testing. Dermatomes and sensory distributions of peripheral nerves are shown in **Fig. 7.1**.¹
 - Proprioception (posterior column function) is assessed by testing joint position sense in the second toe on each foot and/or vibratory sense with a low-frequency (128 Hz) tuning fork applied to bony prominences in the ankles.
 - Temperature sense may be crudely assessed by touching the cool metal of a reflex hammer handle to the skin.

Table 7.1 Muscle grading (modified Medical Research Council system)

Grade	Strength
0	No contraction (total paralysis)
1	Flicker or trace contraction (palpable or visible)
2	Active movement through full ROM against gravity
3	Active movement against resistance
4	Active movement against resistance (subdivisions: 4–, slight resistance; 4, moderate resistance; 4+, strong resistance)
5	Normal strength (against full resistance)
NT	Not testable

*Abbreviation: ROM, range of motion.

Table 7.2 Survey muscle groups

Muscle	Major root innervation	Peripheral nerve	Action to test
Upper extremity			
Deltoid	C5, C6	Axillary	Abduct shoulder over 90° above horizontal
Biceps brachii	C5, C6	Musculocutaneous	Flexion at elbow with forearm supinated
Brachioradialis	C5, C6	Radial	Flex elbow with thumb pointing up
Extensor carpi radialis	C5, C6	Radial	Wrist extension
Triceps brachii	C7, C8		Extension at elbow
Flexor digitorum profundus I and II	C7, C8, T1	Anterior interosseus	Flex distal phalanges digits 2 and 3

Table 7.2 (Continued)

Muscle	Major root innervation	Peripheral nerve	Action to test
Lower extremity			
Iliopsoas	L2, L3	Femoral + L1, 2, 3 roots	Thigh flexion
Quadriceps femoris	L3, L4	Femoral	Knee extension
Biceps femoris	L5, S1, S2	Sciatic	Knee flexion
Tibialis anterior	L4, L5	Deep peroneal	Ankle dorsiflexion
Extensor hallucis longus	L4, L5	Deep peroneal	Great toes extension
Gastrocnemius	S1, S2	Tibial	Ankle plantarflexion

- Reflexes (muscle stretch reflexes, pathologic reflexes, cutaneous reflexes, sacral reflexes, and priapism)
 - Muscle stretch reflexes are usually graded as shown in **Table 7.3**.¹ The muscle stretch reflex involved is shown in **Table 7.4**.
 - Babinski sign and Hoffmann’s sign and testing for ankle clonus: Part of assessment for long tract signs should be made in all patients to disclose unsuspected cervical or thoracic myelopathy or other causes of upper motor neuron deficit.
 - Further reflex testing as indicated includes abdominal cutaneous reflexes, anal wink, and bulbocavernosus. Absence of bulbocavernosus reflex in the setting of acute spinal cord injury may indicate the presence of spinal shock.
 - Priapism may indicate spinal cord injury.
- Mechanical factors: Includes observation for cutaneous changes, pain behaviors, muscle bulk/tone, and provocative maneuvers.
 - Gait and station: Casual gait is assessed in all patients to check balance, weakness that compromises gait, and pain manifestations. Use tandem gait or Romberg’s test to further assess balance and posterior column (proprioceptive) function. The ability to raise up on the toes tests gastrocnemius (S1) more reliably than does manual muscle testing.
 - Cervical spine²
 - Cervical range of motion and specific levels of tenderness should be documented to distinguish myofascial pain from bone pain. Specific

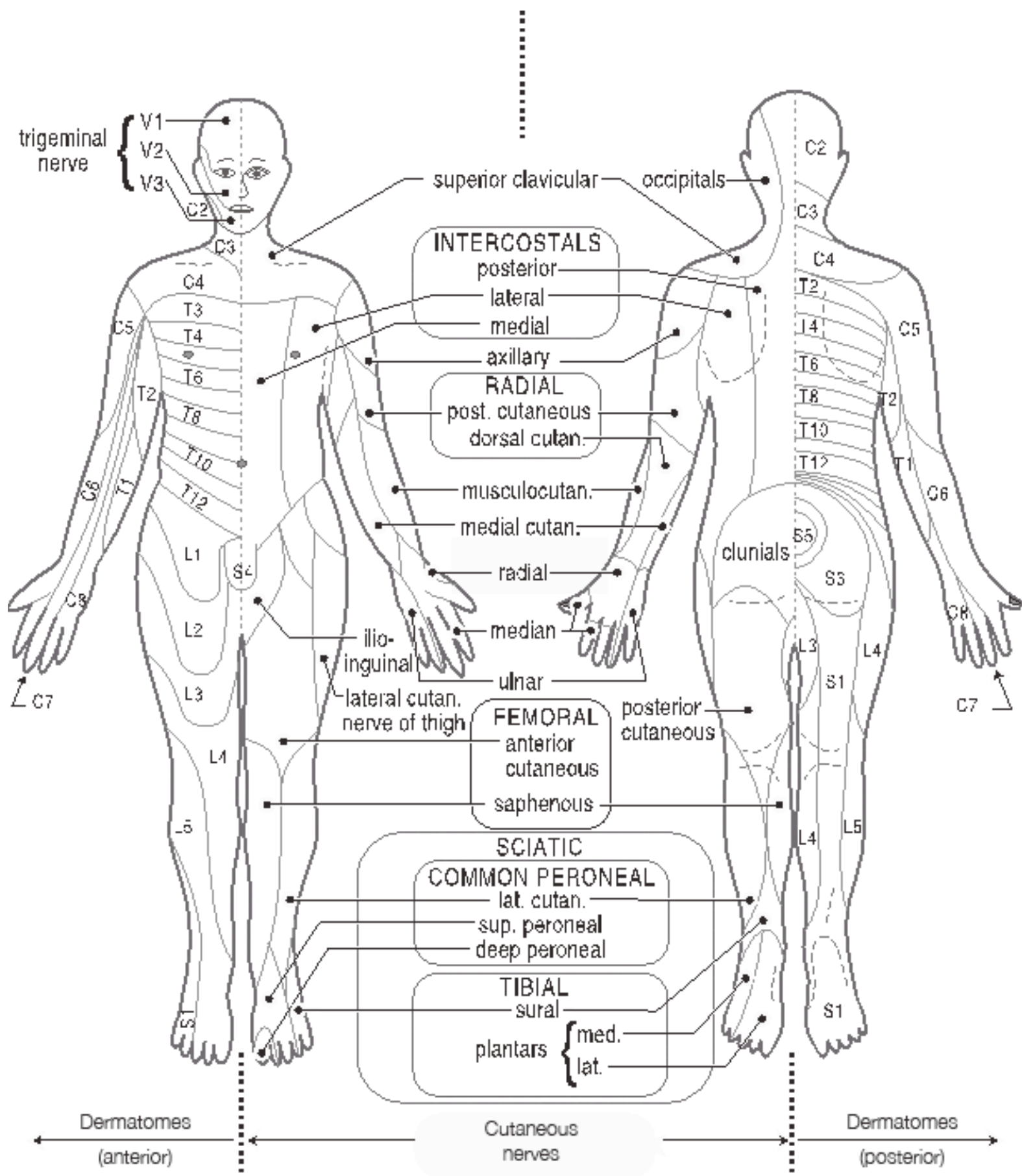


Fig. 7.1 Dermatomes and sensory distributions of peripheral nerves.

facet pain diagram patterns can be reviewed with the patient to identify specific problem levels.

- Spurling’s maneuver (axial loading on the vertex of the head with rotation to one side, then repeating with rotation to the other side) may reproduce cervical nerve root symptoms in a patient who has herniated cervical disk or foraminal stenosis.
- Shoulder pathology can often mimic cervical spine pathology. Tenderness of the acromioclavicular joint to palpation or a positive empty-can test (patient extends arms in front, spread ~ 30°

laterally from primary position with the thumbs pointing down, and resists while the examiner pushes down on the arms; stresses supraspinatus muscle and tendon) suggests primary shoulder pathology as a cause of shoulder pain.

Table 7.3 Muscle stretch reflex (deep tendon reflex) grading scale

Grade	Definition
0	No contraction (total paralysis)
0.5+	Elicitable only with reinforcement*
1+	Low normal
2+	Normal
3+	More brisk than normal (hyperreflexic)
4+	Hyperreflexic with clonus
5+	Sustained clonus

Note: *In the lower extremities, reinforcement consists of having the patient hook the tips of the fingers of the left hand into the tips of the hooked fingers to the right hand and pulling (Jendrassik maneuver). Reinforcement in the upper extremities consists of having the patient clench the teeth.

Table 7.4 Muscle stretch reflexes

Nerve root involved	Corresponding muscle stretch reflex
C5	Deltoid and pectoralis*
C6	Biceps and brachioradialis
C7	Triceps
C8	Finger flexor*
L4	Patellar (knee jerk)
L5	Medial hamstrings*
S1	Achilles (ankle jerk)

Note: *This reflex is not widely used and may be difficult to elicit.

- Lumbar spine²
 - Lumbar range of motion and specific levels of tenderness should be documented to distinguish myofascial from bone pain.
 - Nerve root tension signs: “Pull” on the nerve root that can reproduce pain in situations in which the nerve root is compressed.
 - Laségue’s sign, or straight-leg-raising (SLR): With the patient supine, raise one leg at a time, keeping the knee extended. Classically positive findings: Pain or paresthesias in the distribution of the involved nerve root (not just back pain) at $< 60^\circ$ elevation. In response, the patient may lift the hip on the involved side off the exam table. Positive in 83% of cases of nerve root compression. More sensitive for L5 or S1 compression than in upper lumbar roots.
 - Femoral stretch test or reverse straight-leg-raising: With the patient prone, flex the knee on one side at a time. More likely to be positive than SLR with upper lumbar nerve root compression (L2, L3, or L4).
- Hip and sacral pain: Important to distinguish between hip-mediated pathology and low back pathology
 - Palpation over spinous processes, paraspinal muscles, greater trochanters (to assess for greater trochanteric bursitis) and sacroiliac (SI) joints (“Fortin finger test” suggests sacroilitis when the patient points to SI joint as the primary locus of pain). Palpation over greater trochanters: Pain suggests trochanteric bursitis.
 - FABERE, or Patrick’s test (acronym: Flexion, Abduction, External Rotation), of the hip with the patient supine: Back pain suggests musculoskeletal lower back pain or sacroiliac pain; groin or hip pain suggests hip pathology.
 - The shear test can help distinguish sacral pain. Patient prone examiner applies pressure to the sacrum while simultaneously applying a traction force caudal with the corresponding limb. A test is positive if it reproduces the patient’s typical pain.
 - FADIR (Flexion, Adduction, Internal Rotation) can help distinguish piriformis syndrome of the hip with the patient supine. Positive response: Pain reproduction centering half the distance between the S3 foramen and the ipsilateral greater trochanter.
- Vascular considerations: Palpation for pedal pulses to rule out arterial insufficiency; inspection and palpation for pedal edema suggesting venous insufficiency, chronic heart failure.

7.3 Clinical Pearls

- Babinski sign or a positive Hoffmann’s sign: If there is no known etiology in a given patient, further investigation is required (to rule out cord compression or brain involvement).

- Cervical radiculopathy does not cause pain with shoulder abduction.
- Painless weakness in the LE almost never results from lumbar nerve root compression. Think of diabetic neuropathy, cervical spondylotic myelopathy, motor neuron disease, and the like.

Common Clinical Questions

1. A 55-year-old male presents to your clinic with progressive lower-extremity weakness and difficulty ambulating. He brings with him an MRI that shows grade 1 spondylolisthesis L5–S1 with severe central canal stenosis. Flexion-extension lumbar spine X-rays show no instability. His exam shows diffuse weakness of the LEs, diffuse reduction of pinprick sensation in the LEs, reduced Achilles reflexes, and bilateral upgoing toes. Options that could be considered at this point include the following:
 - I. Decompressive laminectomy L5–S1 with attempt at reduction with bilateral pedicle screw/rod instrumentation
 - II. Lateral interbody fusion L5–S1 with lateral vertebral body plating
 - III. Anterior lumbar interbody fusion with percutaneous L5–S1 pedicle screw/rod instrumentation
 - IV. MRI of the thoracic and cervical spine

Which are the appropriate options to choose at this time?

 - A. I and III
 - B. II and III
 - C. I, II, and III
 - D. IV
2. A patient presents with a 2-month history of left upper extremity pain radiating to the thumb and index finger that has not responded to conservative therapy. He has a positive Spurling's sign with the head turned to the left. Strength is normal. Reflexes are normal except for a reduction in the left biceps. An MRI of the cervical spine shows disk degeneration with protrusion into the left neural foramen at both C5 to C6 and C6 to C7. Flexion-extension cervical spine X-rays are without instability. After providing informed consent, he indicates his wish to proceed with surgical treatment. Appropriate surgical options include the following, *except* which?
 - A. Anterior cervical discectomy and fusion (ACDF) C5–C6
 - B. ACDF C5–C6 and C6–C7
 - C. ACDF C6–C7
 - D. Cervical disk arthroplasty C5–C6

3. A 60-year-old male has diffuse weakness of the UEs with reduced reflexes and has hyperreflexia in the LEs with bilateral upgoing toes. Etiologies that could be considered include the following:

- I. Cervical spinal stenosis
- II. Left C5–C6 foraminal herniated cervical disk
- III. Motor neuron disease (amyotrophic lateral sclerosis)
- IV. Coincident severe lumbar and cervical spinal stenosis

Which of the foregoing are appropriate diagnoses for this patient?

- A. I and III
 - B. II and IV
 - C. I, II, and III
 - D. IV
-

Answers to Common Clinical Questions

- 1. D. Upgoing toes and diffuse LE weakness and hypalgesia are never caused by compromise at L5 to S1. MRI to look for cord compression above the lumbar spine is appropriate.
 - 2. C. The sensory changes and reflex changes indicate that the C6 nerve root is involved, which implicates the disk at C5–C6. Some surgeons would also treat the disk at C6–C7 at the same time, for it may deteriorate rapidly after fusion of C5–C6; and some would perform disk arthroplasty at C5–C6 to try to shield the disk at C6–C7 from some of the forces that would occur with C5–C6 fusion. Operating on C6–C7 alone is inappropriate, for this is currently not a symptomatic level.
 - 3. A. Both cervical spinal stenosis and amyotrophic lateral sclerosis can produce lower motor neuron findings in the UEs and lower motor neuron findings in the LEs. A unilateral foraminal herniated cervical disk will not cause this. In coincidental lumbar and spinal stenosis, the lumbar stenosis generally masks the hyperreflexia in the lower extremities.
-

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8 Spinal Imaging

David Minges and Joon Y. Lee

8.1 Key Points

- A complete history and physical examination should be undertaken first to determine the best imaging modality of choice to help with the clinical picture.
- Spinal imaging should always be interpreted in conjunction with the clinical presentation and ongoing differential diagnosis.
- Multiple different imaging modalities can help confirm the diagnosis and guide treatment recommendations.

8.2 Description

- Plain radiographs
 - Universally available, inexpensive, and convenient.
 - Single planar imaging.
 - First modality of choice for evaluating degenerative disorders, trauma, neoplasm, infection, and deformity.
 - Unable to directly visualize nerve roots or cord compression.
 - Limited ability to determine the clinical severity of cervical and lumbar degenerative disk disease.
 - Dynamic radiographs can assess instability in the setting of spondylolisthesis.
 - First-line evaluation in the postoperative patient to assess the hardware and fusion mass for possible pseudoarthrosis.
 - Use of dynamic flexion-extension X-rays is preferred over static X-rays to assess for pseudoarthrosis.
- Myelography
 - Indirect, rather than direct, assessment of extradural compression.
 - Intrathecal water soluble contrast mixes with cerebrospinal fluid through C1 to C2 or lumbar puncture.
 - Diagnosis can be inferred based on changes in the contour of the contrast-filled thecal sac and nerve root sleeves.
 - Extradural impression on the dye column and filling defects of the nerve root sleeve demonstrate compression.
 - Water soluble agents are available and provide excellent contrast enhancement.

- Low rates of side effects (e.g., iohexol, iopamidol)
- Less toxic; absorption makes removal unnecessary
- Newer nonionic water soluble agents have milder side effects; severe side effects such as hallucinations, confusion, and seizures rarely occur.
- CT myelography can further improve visualization of foraminal and lateral recess stenosis with reconstructed images.
- Advantageous in patients who have implantable devices (e.g., pacemaker); as there can be significant metal artifact with MRI in those with stainless steel implants.
- Disadvantages include lack of diagnostic specificity, uncertainty of compressive lesion, and invasiveness.
- CT
 - Optimizes bony detail visualization.
 - High sensitivity in detecting fractures (**Fig. 8.1**).
 - Reconstruction views are useful to evaluate suspected postoperative pseudoarthrosis.
 - Direct visualization of potential bony neural compression.
 - For optimal bony detail, multiple thin cuts (1.5 to 3 mm) can be obtained.



Fig. 8.1 Sagittal CT image of the thoracolumbar spine demonstrating an L1 bony chance fracture.

- Limited visualization of neural structures, as well as intrathecal and soft tissue pathology, in comparison to MRI.
- Disadvantages include radiation exposure, streak artifact in cervical spine, effects of partial volume averaging, and time involved in performing multiple thin sections over multiple intervertebral disks and vertebral bodies.
- New multirow detector technology allows for rapid 3-mm image slices with isotropic voxels and no loss in spatial resolution.
- Metal artifact can lead to hollow projections causing “sunburst” or streak.
- MRI¹
 - Modality of choice for most spinal pathology.
 - Enhanced, noninvasive imaging of soft tissue pathology, including hematoma, infection, spinal cord injury, and tumor, as well as identification of intramedullary, intradural, and extradural compressive lesions.
 - High sensitivity in identifying ligamentous injury (anterior and posterior longitudinal ligaments, posterior ligamentous complex [PLC]).
 - Requires no radiation exposure, generally well tolerated, and uses pulsed radiofrequency (RF).
 - T1-weighted images:
 - MRI findings: Adipose, high signal; cortical bone, low signal; free water, low signal
 - T2-weighted images:
 - MRI findings: Adipose, low signal; cortical bone, low signal; free water, high signal
 - Gadolinium-based IV contrast medium can help distinguish recurrent disk herniation (avascular, no enhancement) from postoperative scar (vascular, enhancement) on T1-weighted images.
 - Contrast can be useful in detecting tumor and infection:
 - Tumor: Intervertebral disk is often spared but usually involves the entire vertebral body; may show multiple noncontiguous vertebrae with metastatic disease, indicating skip lesions (highest sensitivity with STIR sequences)
 - Infection (diskitis, vertebral osteomyelitis, epidural abscess): Increased signal on T2, decreased signal on T1, enhancement with gadolinium (**Fig. 8.2**)
 - Spinal cord injury:
 - MRI can quantify degree of spinal cord compression and injury
 - Can distinguish spinal cord edema from hemorrhage (T1 low, T2 high versus T1 high, T2 high, respectively)
 - Short tau inversion recovery (STIR) sequence important in identifying intramedullary cord lesions

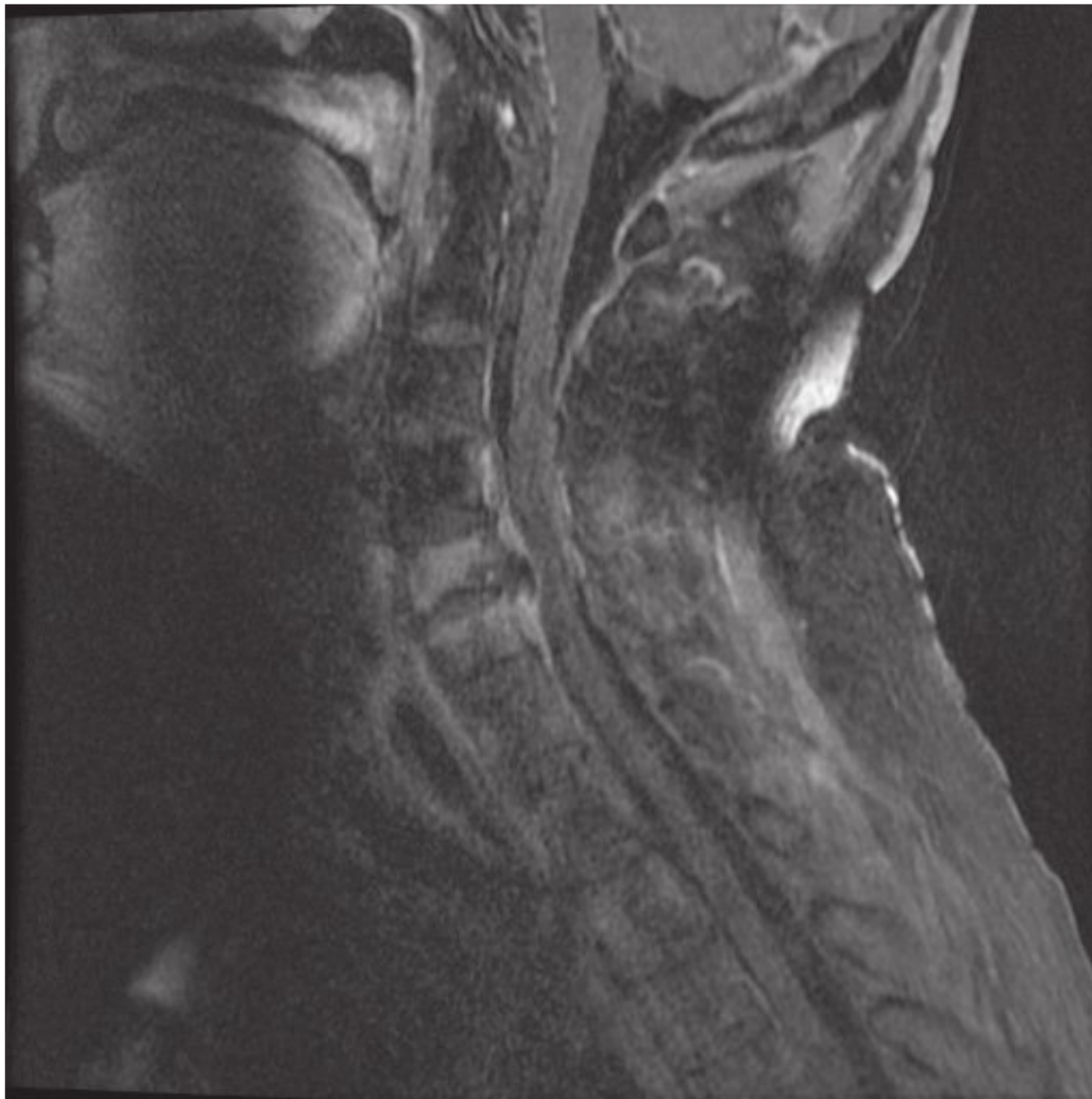


Fig. 8.2 Sagittal T1-weighted fluid-attenuated inversion recovery (FLAIR) MRI image with gadolinium contrast enhancement demonstrating C5–C6 diskitis, vertebral osteomyelitis, and a large ventral epidural abscess with significant spinal cord compression.

- 3-Tesla magnets can image individual spinal cord tracts, providing more accurate posttraumatic neurological assessment
- Disk degeneration:
 - MRI findings must be strictly correlated with the clinical presentation.
 - In general, degenerative changes are age-related and present in both asymptomatic and symptomatic individuals.
 - Advanced imaging should be reserved for patients who have objective evidence of root irritation on examination, patients who have true radicular symptoms, and patients who have failed conservative nonoperative treatment.
 - MRI may show end plate and hypertrophic ligamentous changes, disk pathology, and evidence of instability.²
 - High-intensity zone (HIZ) (radial tear of the posterior annulus, fissure extending from nucleus to the periphery), dark disks with or without height loss, and Modic end plate changes have been extensively studied and have unknown clinical significance.
 - Modic end plate changes:³
 - Type 1

- * T1 low, T2 high
- * Associated with pain and instability
- Type 2
 - * T1 high, T2 normal
 - * Presence of yellow marrow accounts for shortening of T1 signal
 - * More common than type 1
- Type 3
 - * T1 low, T2 low
 - * Less segmental instability secondary to advanced degeneration and sclerosis
- Disk herniation nomenclature:⁴
 - *Protrusion*: Herniation that maintains contact with the disk of origin via a bridge as wide as or wider than the diameter of the displaced material.
 - *Extrusion*: Diameter of disk material beyond interspace is greater than width of bridge that may or may not connect to the disk of origin.
 - *Sequestration*: Extrusion that is no longer contiguous with the disk of origin (**Fig. 8.3**).

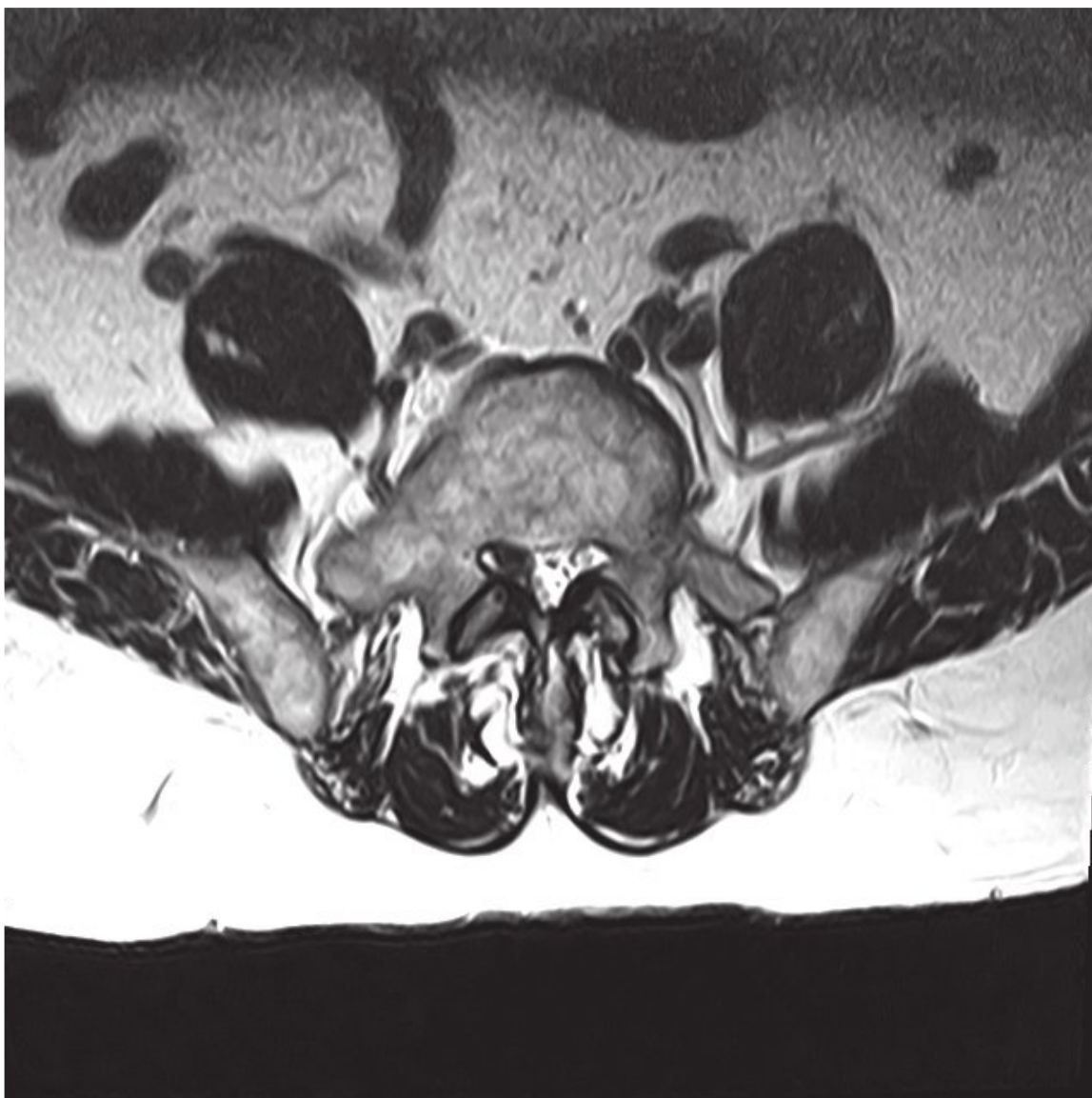


Fig. 8.3 Axial T2-weighted MRI image of the lumbar spine demonstrating an L5–S1 sequestered herniated nucleus pulposus in the right subarticular recess.

- Location (axial plane): Central, subarticular or lateral recess, foraminal or pedicle zone, extraforaminal or far lateral zone, anterior zone.
- Artifact:
 - Spinal implants can result in metal artifact and poor image quality unless specific techniques are used (polymer < titanium < tantalum < stainless steel < cobalt chrome).
 - Strategies to decrease artifact include larger fields of view, smaller voxel sizes, higher readout bandwidths, and geographic orientation of the frequency-encoded direction in relation to the metal objects.
- Contraindications: Metal debris in the eye, cardiac pacemaker, ferrous metal implant in the brain, inner ear implant.
- Bone scintigraphy
 - Superior sensitivity
 - Poor specificity
 - Can help detect lesions in symptomatic patients with negative radiographs
 - Standard part of the work-up of metastatic bone disease
 - Can help distinguish acute versus chronic pars interarticularis fractures

8.3 Surgical Pearls

- Before obtaining advanced imaging of the spine, a plan must be in place for how the results will be used to determine the best treatment regimen.
- A differential diagnosis should be established based on the history and physical examination and refined based on imaging findings before implementation of a nonoperative versus operative treatment plan.
- Advanced imaging can be extremely helpful in establishing a diagnosis and implementing treatment resulting in a high rate of clinical success.

Common Clinical Questions

1. Which specific combination of signal intensities and MRI sequences denote spinal cord hemorrhage?
 - A. High T1, high T2
 - B. Low T1, high T2
 - C. High T1, low T2
 - D. Low T1, low T2
2. A 35-year-old male who has previously undergone a lumbar microdiscectomy for an L4–L5 herniated nucleus pulposus presents to you with an acute onset of recurrent lower extremity radiculopathy 8 months after his previous surgery. Which MRI sequence of choice

will best help distinguish a recurrent disk herniation from epidural fibrosis?

- A. Gadolinium-enhanced T1 sequence
 - B. Gadolinium-enhanced T2 sequence
 - C. STIR sequence
 - D. T1 sequence alone
 - E. T2 sequence alone
3. Which imaging modality is most often used to assess for dynamic instability in the setting of spondylolisthesis?

Answers to Common Clinical Questions

- 1. A.
- 2. A.
- 3. Dynamic (flexion–extension) lateral radiographs.

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9 Radiation Exposure in Spine Surgery

Alexander Tuchman and Patrick C. Hsieh

9.1 Key Points

- Fluoroscopy is an important tool during many spine procedures, but its use carries a variety of radiation exposure–related risks to the patient, surgeon, and operating room (OR) staff.^{1,2}
- Minimizing radiation exposure and using appropriate protective gear is a priority during procedures using fluoroscopy.²

9.2 Description

- Ionizing radiation units of measure
 - Gray (Gy) is an SI unit representing 1 J of radiation energy absorbed by 1 kg of mass.
 - Sievert (Sv) is an SI unit of equivalent dose or effective dose related to the biologic effect of the deposit of a joule of radiation energy in a kilogram of human tissue.
 - It corresponds to the product of the absorbed dose of radiation and a dimensionless quality factor (Q).
 - Gy and Sv share units (joules per kilogram), but by convention Gy is used to refer to dose and deterministic effects of radiation seen at high doses, whereas Sv refers to the stochastic effects of radiation seen at lower doses.
- For medical purposes, X-rays are produced in a vacuum tube that uses high voltage to accelerate electrons from a cathode to a metal anode releasing photons.
- A patient can then be placed between the X-ray tube and X-ray detector (image intensifier for fluoroscopy), and images are generated based on the differential X-ray absorption of tissues.
- X-rays are a type of ionizing radiation having sufficient energy to remove electrons from atoms, so they are defined as a carcinogen by the World Health Organization.²
- Being directly in the X-ray beam places a patient at risk for radiation effects:
 - Radiation exposure to the patient should be minimized within the confines of appropriate medical management.^{1,3}

- Radiation scatter poses a cumulative occupational risk to clinicians performing spine surgery over their career.^{2,3}
- During spine surgery, X-rays are commonly used for a variety of purposes, including the following:
 - Level confirmation
 - Fracture–dislocation reduction
 - Deformity correction
 - Implant guidance for anatomic localization, orientation, or fixation
 - Biopsy
 - Vertebroplasty or kyphoplasty
 - Percutaneous or minimally invasive hardware placement
- Health effects of ionizing radiation include the following:³
 - Tissue effects
 - Testes (temporary sterility)
 - 0.1 Gray (Gy) single exposure or 0.4 Gy annual exposure
 - Testes (permanent sterility)
 - 6.0 Gy single exposure or 2.0 Gy annual exposure
 - Ovaries (sterility)
 - 3.0 Gy single exposure or 0.2 Gy annual exposure
 - Lens (cataract)
 - 0.5 Gy single exposure or 0.5 Gy lifetime exposure
 - Bone marrow (depression of hematopoiesis)
 - 0.5 Gy single exposure or 0.4 Gy annual exposure
 - Heart or brain (circulatory disease)
 - 0.1 Gy single exposure or 0.5 Gy annual exposure
 - Stochastic effects
 - Detriment-adjusted nominal risk coefficient after exposure to radiation at a low dose rate:³
 - 5.5%/Sv for cancer
 - 0.2%/Sv for genetic effects
 - Other factors may increase an individual's risk for cancer:
 - Younger age
 - Female gender
 - Autoimmune, connective tissue, or DNA repair disorders
- Certain steps can minimize radiation exposure to spine surgeon and OR staff:¹
 - All OR personnel should wear lead or lead equivalent shielding during fluoroscopy.
 - Eyes, thyroid, gonads, and femurs require protection.
 - Shielding should not be folded or creased and should be tested annually for defects.
 - Minimize shots of fluoroscopy used per procedure.
 - Use pulsed setting rather than continuous fluoroscopy.
 - Nonessential personnel should leave the room during fluoroscopy.

- Essential personnel should position themselves as far from the machine as the room allows (preferably behind the image intensifier) and behind a lead shield.
- When operating under fluoroscopic guidance, maintain a distance of 2 to 3 feet from the X-ray beam source.
- Stand opposite the source (near the image intensifier) to reduce exposure to scatter radiation during lateral fluoroscopy, and position the source below the patient for anteroposterior (AP) images.
- Turn your head 90° from the patient to reduce scatter to the eyes.
- Use collimation and automatic brightness control when appropriate.
- Place the image intensifier as close to the patient as possible to reduce scatter.

9.3 Surgical Pearls

- Always wear appropriate protection and stand as far from the X-ray beam as possible, for radiation dose to the operator decreases exponentially with distance from the source.^{1,2,3}
- Use a “hands-off” technique when appropriate to minimize exposure during² procedures.^{1,2,3}
- Consider using intraoperative stereotactic navigation when many levels are involved.

Common Clinical Questions

1. All the following organ damage has attenuated effects with fractionation *except* which?
 - A. Testes (sterility)
 - B. Eyes (cataracts)
 - C. Bone marrow (depression of hematopoiesis)
 - D. Ovaries (sterility)
 2. All of the following are effective measures to decrease operator radiation exposure *except* which?
 - A. Wearing leaded goggles
 - B. Wearing leaded apron and thyroid shield
 - C. Collimating the X-ray beam
 - D. Standing opposite the image intensifier during lateral fluoroscopy
 - E. Positioning the source below the patient
 3. True, or false? Low-dose exposure to radiation has deterministic effects with a threshold dose where the risk for cancer clearly increases.
-

Answers to Common Clinical Questions

1. B.
 2. D.
 3. False.
-

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10 Electrodiagnostic Testing in Spine Surgery

Kamakshi Patel and Holli A. Horak

10.1 Key Points

- Electromyogram/ nerve conduction studies (EMG/NCS) are performed for patients who show evidence of peripheral nervous system damage/injury.
- EMG/NCS data can be used, along with the history, physical examination, and neuroimaging, to guide the timing and localization of surgery.
- EMG/NCS cannot assess central nervous system problems such as myelopathy.
- EMG/NCS should be considered if there may be other neurologic reasons for the patient's symptoms or if the history does not match clinical examination findings. This includes evaluation for myasthenia gravis, Guillain-Barré syndrome, diabetic neuropathy, amyotrophic lateral sclerosis (ALS), and myopathies.
- EMG/NCS of the affected limb(s) can distinguish between nerve root avulsion, radiculopathy/multiple radiculopathies, plexopathy, and neuropathy.¹
- EMG/NCS can also assist in assessing severity and time course of nerve root compression/ damage.
- It is often preferable to wait 4 to 6 weeks after symptom onset before performing EMG/NCS unless the patient is critically ill or has rapid-onset paralysis. This is because of a delay in EMG manifestations owing to the time it takes for the affected axons to degenerate.

10.2 Description

10.2.1 Nerve Conduction Study

- Nerve conduction study (NCS) is a separate test from EMG, but the two are performed in conjunction > 90% of the time.
- NCS evaluates the integrity of the nerve itself and can assess for axonal or demyelinating damage.

- NCS parameters include the amplitude (related to the number of functional nerve fibers), latency (marker of speed of conduction from point of stimulation), and conduction velocity of individual peripheral nerves.²
- Reduced amplitude is associated with axonal loss. This is a feature of idiopathic peripheral neuropathies.
- Prolonged latency and significantly slow conduction velocity are seen in demyelinating processes. An example of this is the prolonged latency seen in carpal tunnel syndrome (CTS).
- Both motor and sensory nerve fibers can be evaluated.
- In a radiculopathy, the sensory nerve conduction study may remain normal even when the patient has clear sensory loss.
 - This is because the cell nucleus of the sensory nerve is located in the dorsal root ganglion, external to the spinal canal (and often distal to a disk herniation) (see **Fig. 10.1**).

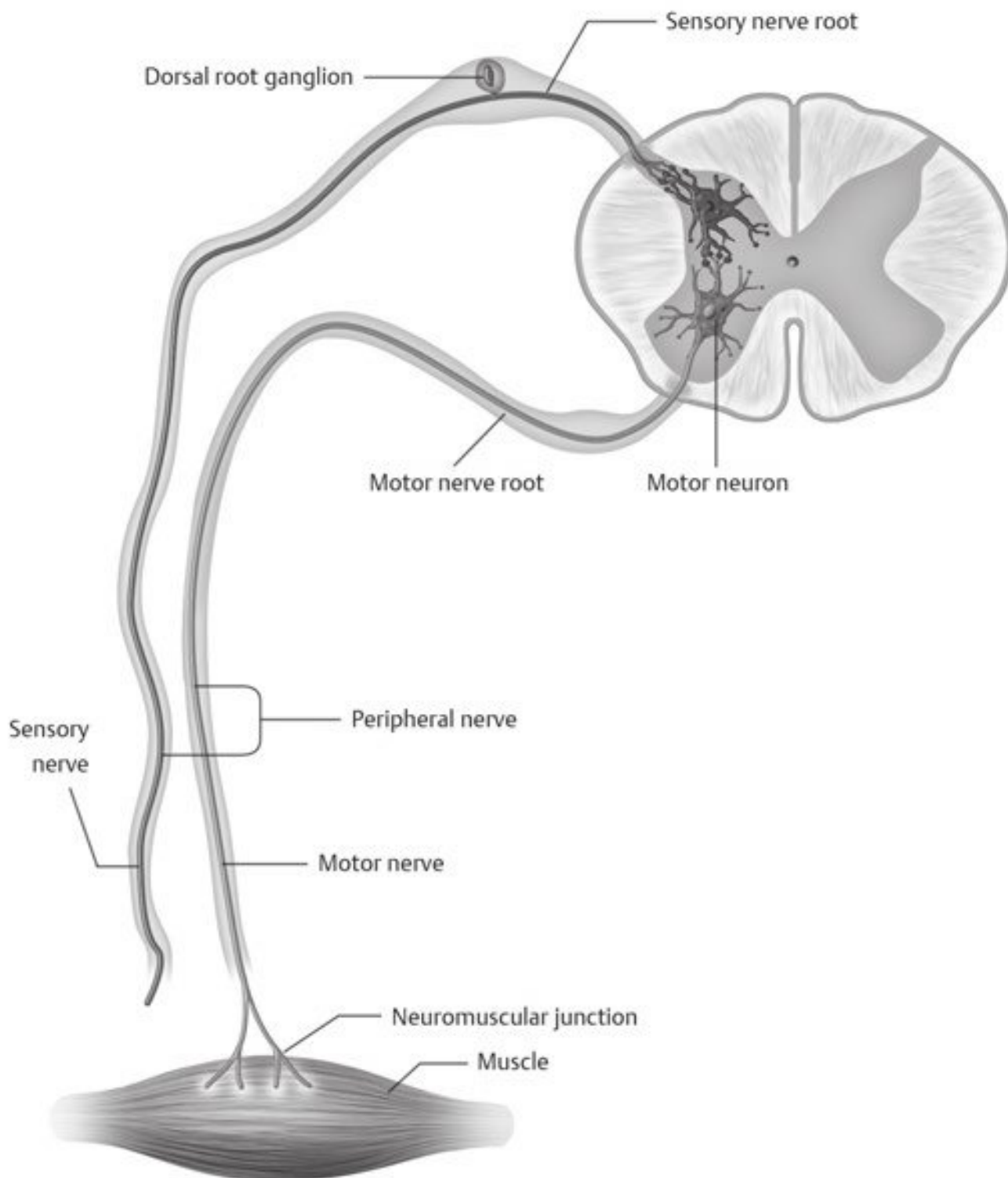


Fig. 10.1 Axial image of spinal cord showing relative location of dorsal root ganglion.

- Because the nucleus and distal axon are unaffected, the sensory NCS will be normal even if there are sensory symptoms (because the proximal portion of the axon is affected).

10.2.2 Electromyogram (EMG)

- EMG is the evaluation of motor unit potentials (MUPs) in the muscle.
- EMG is performed in the affected limb(s) either to evaluate the muscle fibers or to evaluate motor neuron innervation to the muscles of that specific myotome.
- EMG assesses two parameters: Spontaneous activity and motor unit potentials (MUPs):³
 - Spontaneous activity is the presence or absence of abnormal discharges from individual muscle fibers, the most common being fibrillations and positive waves.
- MUPs are the electrical potentials generated by the contraction of all muscle fibers innervated by one individual motor neuron.
- EMG can potentially distinguish between hyperacute (less than 3-wk), acute (3–12-wk), and chronic (> 12-wk) lesions.
- Sensitivity and specificity depend on the degree of damage/injury and presence of other coexisting peripheral nerve diseases, such as peripheral neuropathy.

10.2.3 Pathophysiology of Radiculopathy

- After 3 to 4 weeks (acute stage), fibrillations and positive waves will develop.
- Fibrillations and positive waves are strongly associated with acute axonal injury, but because they can be seen in other conditions as well, clinical correlation is needed.
- After 6 weeks or so (subacute), the surrounding remaining axons sprout new branches to the denervated muscle fibers; this is called reinnervation (see **Fig. 10.2**).³
- With early reinnervation, the MUPs demonstrate immature potentials (satellite potentials) and irregular firing rates (also called instability).
- During reinnervation, the MUPs become polyphasic and larger as the newly reinnervated muscle fibers become integrated into the motor unit.
- After 12 weeks or so (chronic), muscle fibers are reinnervated and the potentials become more mature, giving rise to triphasic, long-duration, large-amplitude stable motor units on EMG.
- See **Table 10.1** for the more commonly affected muscles associated with specific radiculopathy levels.

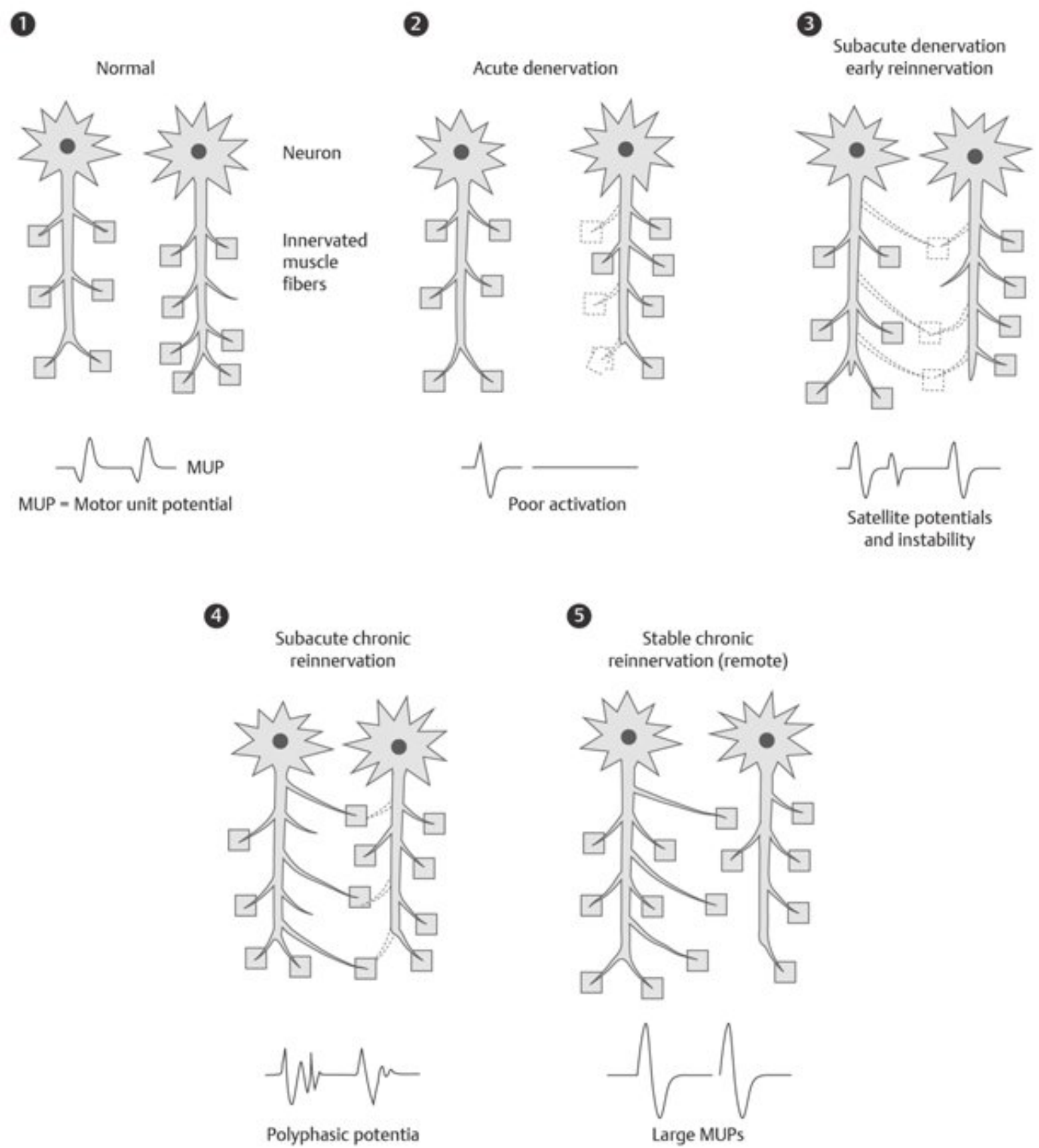


Fig. 10.2 Graphic demonstrating acute denervation and reinnervation of a motor unit.

10.3 EMG/NCS Pearls

- EMG does not show denervation changes initially; it may take 3 to 4 weeks for denervation (axonal injury) to be evident on EMG, in the form of abnormal insertion with fibrillations.⁴
- Accordingly, waiting 3 to 4 weeks, if possible, provides greater sensitivity.

Table 10.1 Table of radiculopathy findings

Nerve root level	Example of commonly affected muscles
C5–C6	Deltoid, biceps
C6–C7	Triceps, pronator teres
C8	First dorsal interosseous, abductor pollicis brevis
L3–L4	Vastus lateralis, vastus medialis
L4–L5	Tibialis anterior, gluteus medius
S1	Gastrocnemius, gluteus maximus, short head of biceps femoris

- The earliest changes (less than 4 wk) on EMG may simply represent difficulty activating the muscles in the affected nerve root distribution.
- Sensory nerve conduction studies may remain normal in the setting of severe nerve injury (avulsion, severe radiculopathy), but motor fibers will show low or absent amplitude

Common Clinical Questions

1. A patient who has diabetes with low amplitudes on NCS and chronic denervation changes on EMG seen only in the distal muscles of the bilateral L5 to S1 myotomes likely has what?
 - A. Acute bilateral L5 and S1 radiculopathies
 - B. Acute inflammatory demyelinating polyneuropathy (Guillain-Barré syndrome)
 - C. Polymyositis
 - D. Peripheral neuropathy
2. A patient who has a suspected L5 radiculopathy has an EMG done within 1 week. What is it most likely to show?
 - A. +4 fibrillations and positive waves, polyphasic, rapid-firing, large motor units
 - B. Prolonged latencies on both motor and sensory nerve conduction studies
 - C. Normal insertion but very long-duration, rapid-firing motor units
 - D. Normal insertion, but poor activation of muscles in the right L5 myotome

-
3. A patient who has fibrillations, positive waves and polyphasic motor units in the deltoids, biceps, and midcervical paraspinal muscles, but in whom all other muscles are normal, likely has what?
- A. Axillary neuropathy
 - B. Remote, inactive C5 radiculopathy
 - C. Subacute C5–C6 radiculopathies
 - D. Amyotrophic lateral sclerosis
-

Answers to Common Clinical Questions

- 1. D.
 - 2. D.
 - 3. C.
-

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11 Intraoperative Neurophysiological Monitoring in Spine Surgery

Colin R. Bamford

The purpose of intraoperative monitoring is to provide the surgeon and anesthesiologist with immediate or close to immediate warning that neural structures are at risk of being injured or have been injured as a result of the surgery or as a result of anesthesia. Surgeons and anesthesiologists are often able to intervene when provided timely warning.^{1,2}

11.1 Key Points

- To the extent possible, the precise nature of the operative procedure and the associated risks to structures should be known to the monitoring team in advance of the operative procedure so that the monitoring can be planned to cover the appropriate anatomical distribution and neurophysiological functions.
- The monitoring should incorporate segmental and radicular functions corresponding to the operative site, long tract functions caudal to the operative site, and long tract functions rostral to the operative site. Long tract functions rostral to the operative site should include, if possible, pathways that do not pass through the operative site. This should be done in a thoughtful way to avoid overwhelming the system and the personnel with excessive data collection.
- Significant monitoring abnormalities arise as a result of dysfunction remote from the operative site as often as they localize to the operative site. These remote monitoring abnormalities include the effects of anesthesia, hypotension, anoxia, hypothermia, neuromuscular blocking, hemorrhage, and ischemic stroke.
- Recognition of real-time or close to real-time change is critical in recognizing the offending cause and bringing about its reversal.
- Effective, reliable, comfortable communication between the monitoring team, the surgeon, and the anesthesiologist is paramount. An uneasy relationship results in delayed or incomplete communication.

11.2 Methods

- A typical setup for a patient presenting with a T12 burst fracture would include the following:

- Median somatosensory evoked potential (SSEP) recording from EP, CII, C4', and C3'
- Tibial SSEP recording from Pop fos, CII, Cz'
- Transcranial electrical motor evoked potential (TCeMEP) with recordings from abductor pollicis brevis, rectus abdominis, vastus lateralis, tibialis anterior, medial gastrocnemius, abductor hallucis, flexor digiti minimi brevis pedis, and anal sphincter
- Free-run EMG with recordings from the same muscles
- Pedicle screw stimulation (triggered EMG) with recordings from the same muscles
- Train of Four recorded at the abductor pollicis brevis
- EEG recorded at C4' and C3'

11.3 Somatosensory Evoked Potentials

Any large or small nerve in the body can be used to generate an SSEP as long as it is in part composed of afferent fibers. Large peripheral nerves in the extremities are typically used for intraoperative evoked potential monitoring (IOM), because they produce a larger amplitude response with a smaller number of averaged sweeps. Typically, in the upper extremities, the median and ulnar nerves are used, and in the lower extremities, the tibial and fibular nerves are used. Because these nerves are made up of multiple roots, they are insensitive to single root dysfunction. If information about single root dysfunction is desired, dermatomal SSEPs would be required. If extremely caudal information is desired, pudendal SSEPs would be required.

- Technique: Upper- and lower-extremity SSEPs
 - Stimulating electrodes are placed over the median or ulnar nerves at the wrists or at the tibial nerve at the ankle or the fibular nerve at the fibular neck.
 - For the upper extremities, the recording electrodes are placed at Erb's point (EP), over the C2 spinous process (CII), and over the contralateral cortex (C3', C4'). For the lower extremities, the recording electrode is placed at Cz'. A variety of reference and ground electrode positions can be thoughtfully used; usually Fz is the common reference and FPz the ground position.
 - The peripheral nerve can be stimulated at a frequency between 1.9 and 6.1 Hz, avoiding harmonics of 60 Hz, using a constant current stimulator. The faster the stimulating frequency, the more rapid the acquisition—up to a point, beyond which attenuated amplitude and skin blistering may result. The stimulus intensity should be sufficient to produce a muscle twitch if a mixed nerve is being stimulated (or $3 \times$ sensory threshold if a pure sensory nerve is being stimulated). A square wave stimulus of 0.2 millisecond duration is given using a constant current stimulator.

- Sweeps consisting of the desired waveform, EEG, EMG, EKG, and extraneous artifacts are averaged until the time-locked desired waveform is identified. The number of sweeps averaged is variable and depends on the signal:noise ratio. The sweep duration should be long enough to fully encompass the desired waveform, which could be normal or pathological. The bandwidth should be limited to the frequencies represented by the desired waveforms.
- Typical recording and stimulating parameters for SSEPs:
 - Sensitivity: 0.5 to 2.0 $\mu\text{V}/\text{division}$.
 - Bandwidth: 30 Hz to 3000 Hz.
 - Sweep duration: 50 millisecondsec for UE, 100 millisecondsec for LE.
 - Pulse width: 0.2 millisecondsec.
 - Stimulation rate: 1.9 to 6.1 Hz.
 - # of sweeps: 50 to 500.
 - Alarm criteria: $> 50\%$ attenuation of amplitude or $> 10\%$ increase in latency.³
 - Pros: Minimal disturbance to surgical procedure; can be constantly monitored; neuromuscular blocking does not impact monitoring.
 - Cons: Provides information about the posterior third of the spinal cord only, N_2O impacts monitoring.

11.4 Transcranial Electrical Motor Evoked Potentials

The motor cortex is stimulated with a proprietary stimulator that provides three to seven positive electrical shocks to the motor cortex. By selecting the precise location of stimulation and adjusting the stimulus intensity, TCeMEPs may be set to result in focal or generalized muscle contractions, recorded from the muscles as polyphasic compound motor action potentials.

- Technique: TCeMEP
 - Stimulate the brain with anode over appropriate motor cortex.
 - Cathode over opposite hemisphere.
 - Use a bite block.
 - Shock intensity: Up to 600V.
 - Shock duration: 50 microseconds.
 - Three to seven pulses 2 to 3 millisecondsec apart.
 - Alarm criteria: Disappearance or severe attenuation of MEPs or persistent increase in the stimulation threshold by 100V.⁴
 - Pros: Monitors motor functions of the anterior two-thirds of the spinal cord.^{5,6}

- Cons: Inhalation anesthetics and neuromuscular blocking agents impact monitoring, cannot be continuously monitored, and are disturbing to the operative procedure.
- Must reverse neuromuscular (NM) blocking and must use total intravenous anesthesia (TIVA), no N₂O and no more than 0.5 minimum alveolar concentration (MAC) of inhalation anesthetic.

11.5 Free Run EMG

Needle electrodes are placed subcutaneously or intramuscularly in the selected muscles. The purpose of this monitoring is to detect irritation or injury to peripheral nerve, which elicits solitary discharges, brief bursts, and/or prolonged trains of motor unit potential (MUP) activity.

- Technique: Free run EMG (sEMG)
 - EMG needles are placed subcutaneously or intramuscularly in a muscle above the surgical site as well as in several muscles supplied by nerve roots traversing the surgical site.
 - Sensitivity 200uV/div.
 - Sweep duration 100msec/div.
 - Bandwidth 10 to 10,000 Hz.
 - Alarm criteria: Trains of MUP activity arising from a muscle supplied by a nerve that is being physically manipulated.
 - Pros: Monitors specific nerve roots within the surgical site.
 - Cons: May fail to detect acute total nerve root transection.
 - Must reverse NM blocking.

11.6 Triggered EMG

Nerve root or peripheral nerve may be stimulated with a sterile handheld stimulator being positioned by the surgeon in the operative field. The purpose of this technique is to identify excitable nerve tissue. The excitable nerve tissue could be hidden within tumor or a placode. The nerve could be normal but may be in closer proximity to a pedicle screw than is desirable, or there could be breach in a pedicle with a low nerve stimulation threshold.

- Technique: Triggered EMG (tEMG)
 - The sterile stimulating electrodes are held or positioned by the surgeon to explore a tumor or placode for nerve or to explore a pedicle screw hole or to stimulate a pedicle screw to identify a breach.
 - The stimulator is triggered by the technologist.
 - The recording electrodes are the same electrodes that have been used for sEMG and TCeMEP.

- Threshold for identifying breach is decided by the surgeon. Higher values represent a careful, conservative approach. The lumbar thresholds used would be 7.6 to 11 mA, thoracic thresholds used would be 6 to 8 mA, and cervical thresholds would be 7.6 to 10 mA. A threshold in the 20 mA range would strongly suggest that a breach has not occurred. A threshold in the 3 mA range would strongly suggest that a breach has occurred.
 - Pros: Identifies viable nerve tissue in tumors or placodes. Identifies pedicle breach.
 - Cons: May fail to identify a breach if the adjacent nerve has an elevated threshold for stimulation based upon the presence of a comorbidity of nerve disease or injury.

11.7 H Reflex

- H reflexes can be recorded during surgery and under anesthesia with the same limitations as TCeMEPs: TIVA will be required, and neuromuscular block must be reversed.
- Disappearance of an H reflex unilaterally would suggest that the S1 to S2 roots have been compromised on that side.
- Disappearance of H reflexes bilaterally would suggest that the spinal cord has been severely compromised rostrally, and there should simultaneously be loss of TCeMEPs in the lower extremities.

Common Clinical Questions

1. The latencies to P23 of the median SSEP are longer than the latencies to P37 of the tibial SSEP. What is the likely explanation for this problem?
 - A. The patient has a cervical central cord syndrome.
 - B. The technologist has inadvertently switched the electrode connections from the arms and legs.
 - C. This is a normal physiological variant.
2. A patient is undergoing monitoring for a C5–C6 anterior cervical discectomy and fusion. TCeMEP recorded at the left deltoid shows a 55% amplitude decrease and a 12% increase in latency. These changes are sufficient to raise an alarm.
 - A. True
 - B. False

-
3. All the following would represent real-time recording techniques *except* for which?
- A. sEMG
 - B. tEMG
 - C. TCeMEP
 - D. Tibial SSEP
 - E. EEG
-

Answers to Common Clinical Questions

- 1. B.
 - 2. B.
 - 3. D.
-

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12 Bedside Procedures

Daniel C. Lu and Praveen V. Mummaneni

12.1 Key Points

- Halo orthosis and traction: Skull fracture or severe skull osteoporosis is a contraindication for halo placement. Scalp abrasion or infection overlying the intended pin sites is also a contraindication for the procedure.
- Lumbar puncture (LP) or lumbar drain: Known or suspected intracranial mass, infection, tethered cord, or coagulopathy are potential contraindications to the procedure.

12.2 Indications

- Halo orthosis and traction: Halo orthosis is effective at controlling abnormal motion at the C1–C2 articulation resulting from fracture or ligamentous injury. The purpose of the halo is to maintain normal alignment and/or immobilize the cervical spine to prevent further spinal injury and to allow for bony fusion in cases of fractures. Halo traction is used to limit fracture–dislocations and maintain normal alignment.^{1,2}
- Lumbar puncture or lumbar drain: A lumbar puncture is indicated for collection and analysis of cerebrospinal fluid (CSF) for infection, subarachnoid hemorrhage, or elevated intracranial pressure. Additionally, intrathecal administration of medication or contrast (for myelography) can be performed via a lumbar puncture. A lumbar drain is placed if temporary CSF diversion is indicated for hydrocephalus (communicating) or wound management (pseudomeningocele, CSF leak, etc.).

12.3 Technique

12.3.1 Halo Orthosis and Traction

- Patients should be positioned either in a sitting head-neutral position or a supine head-neutral position at the end of the bed so that the head slightly overhangs the bed. A semirigid collar may be used to immobilize the neck during halo application.³
- The appropriate-sized halo ring is selected. The halo ring should accommodate the entire head circumference with clearance of ~ 1 cm.

- Halo pin sites are selected at this time, with two anterior and two posterior sites.
 - The anterior sites are centered in the groove between the supraciliary ridge and frontal prominences. Pins should be placed just superior to the lateral half of the eyebrows to avoid the supraorbital nerve and vessels. This location avoids muscular structures to diminish discomfort.
 - A posterior pin should be placed 1 cm above the apex of the pinna of each ear. A line connecting the posterior pin site with its contralateral anterior pin site should roughly bisect a line drawn between the remaining two pin sites at a right angle. This provides distribution of force for stability.
- The planned pin sites are sterilely prepared and injected with 1% lidocaine. The two pins—one front and the diagonally opposite back pin—are then finger-tightened to just touch the skin; this is repeated for the other pins. In children, multiple pins (> 4) are sometimes used to distribute the pressure more evenly.
- A torque screwdriver (set to 6 to 8 lbs. pressure) is then used to tighten the pins in diagonal pairs. The pins are now stabilized and locked down with appropriate locking nuts to the halo frame. The halo vest is then placed on the patient, the semirigid collar is removed, and halo and vest are stabilized with halo rods.
 - The halo vest should be adjusted so that the straps make contact with the patient's trapezius and shoulder area. There is a tendency for the vest to ride high and not touch the shoulders unless care is taken during vest application.
- For traction placement, a variety of devices are available. Gardner-Wells tongs or halo rings are the most common (**Fig. 12.1**). Pin sites for Gardner-Wells tongs are 2 to 3 fingerbreadths (3 to 4 cm) above the ear pinnae. The Gardner-Wells pins are spring-loaded with a force indicator; these pins are tightened until the indicator protrudes 1 mm beyond the flat surface. Pins are retightened daily until the indicator remains at this location for 3 days. If used, halo rings have the advantage of a compatible vest orthosis to secure the tractioned position.
- After tong or halo ring placement, the patient is transferred to a bed with a headboard attached to a pulley system with weights. With the pulley placed above the patient's head, flexion and traction can be accomplished. If the pulley is placed at the level of the pins, then straight traction forces can be applied. If the pulley is placed below the level of the patient's head, extension and traction are possible. Lateral X-rays should be obtained immediately after application of traction and after each weight adjustment. Typically, evaluation begins with 5 pounds of traction for upper C-spine injuries and 10 pounds for lower C-spine injuries.
- For upper cervical injuries, evaluation of the atlantooccipital joints is important to rule out atlantooccipital dislocations. Such injuries should

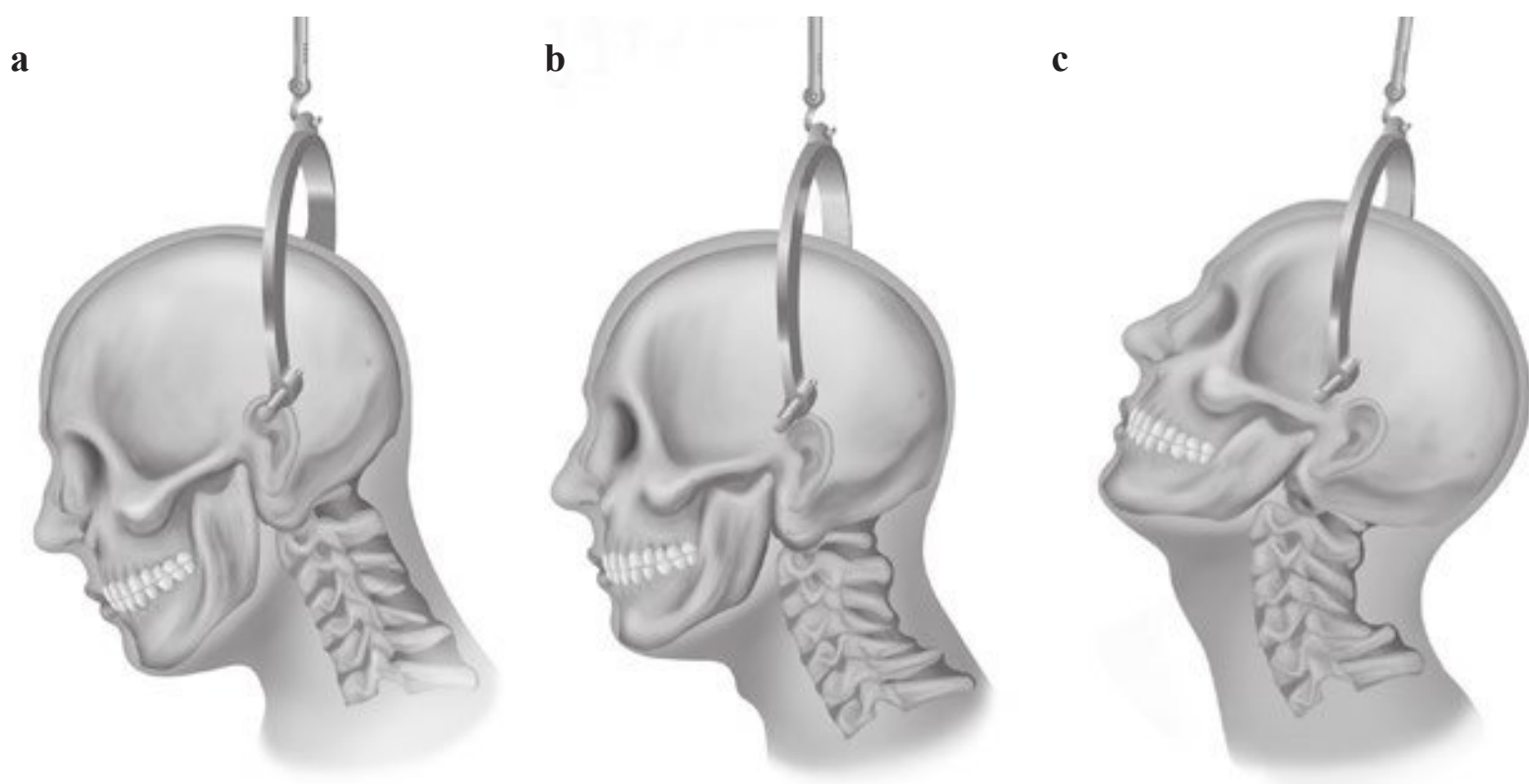


Fig. 12.1 Proper fixation points for Gardner-Wells tongs application. **(a)** Posterior placement of tongs to produce flexion of head. **(b)** Normal placement of tongs to produce straight traction. **(c)** Anterior placement of tongs to produce hyperextension of head. (Adapted from Vaccaro, AR and Albert TJ, eds. *Spine Surgery: Tricks of the Trade*, 2nd ed., Thieme; 2009.).

not use traction. For midcervical locked facets, 5 pounds per level of traction weight should be applied to the injury (e.g., slowly work up to 25 lbs. for a C5-level facet subluxation). Before applying traction for cervical facet subluxation, consider MRI imaging to rule out a coincidental anterior herniated disk with cord compression. If a herniated disk is present, consider anterior operative correction instead of a trial of traction.

12.3.2 Lumbar Puncture or Lumbar Drain

- This bedside procedure can be performed with the patient sitting or lying down.
 - For the recumbent position, the patient is placed in a lateral decubitus posture, with neck flexed and knees brought up to the chest. This distracts the space between the spinous processes, facilitating passage of a spinal needle into the thecal sac.
 - For the sitting position, the patient should be sitting with head and arms resting on a pillow placed on a bedside stand. The back is sterilely prepared and draped.
- LP can be safely attempted at the L3 to S1 interspaces in the anatomically normal patient. The intercrestal line is identified and palpated in the midline for the L4 spinous process.
 - Initially, 1% lidocaine is infiltrated subcutaneously. Subsequently, the lumbodorsal fascia is injected.

- The spinal needle with stylet is aimed slightly rostrally to the umbilicus to approximately parallel the spinous process, and the bevel should be turned parallel to the length of the spinal column to reduce the chance of post-LP headaches.
- The needle is advanced with a midline trajectory, and a “pop” should be felt as the needle penetrates the ligamentum flavum and passes into the dura (**Fig. 12.2**).
- The stylet is then withdrawn to check for CSF flow; if none is seen, reinsert the stylet and advance the needle further; if no CSF flow is present, attempt another trajectory.
- If blood is seen, wait for the blood to drain and clear, as this may represent a traumatic tap. If it does not clear, advance the needle or attempt another trajectory.
- If a lumbar drain is selected, a lumbar drain needle (14-gauge Tuohy) should be used. After entering the thecal sac with needle bevel facing laterally, the bevel is turned superiorly, and a lumbar drainage catheter with wire stylet is inserted (20–40 cm).
- The needle and stylet are sequentially removed, and cerebrospinal fluid (CSF) flow is confirmed by dropping the catheter below the patient.

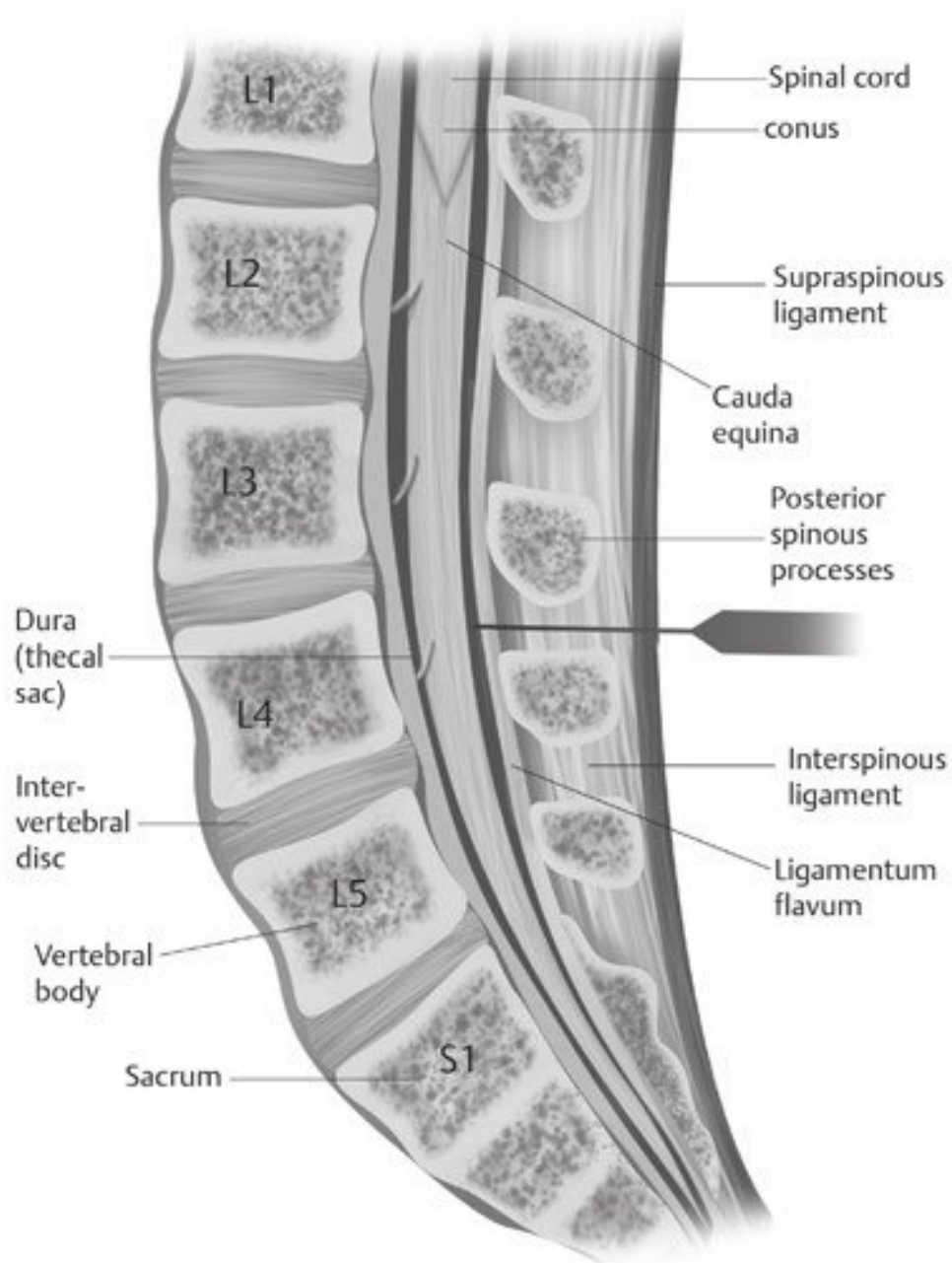


Fig. 12.2 Lumbar puncture technique, with pertinent spaces and ligaments labeled.

- A 2×2 gauze section is placed around the insertion site of the catheter and a Tegaderm (3M; St. Paul, Minnesota) pad is placed on top to secure the catheter.
- Several more Tegaderm pads are placed along the flank of the patient to secure the catheter to the patient's body.

12.4 Complications

12.4.1 Halo Orthosis and Traction

- Patients who have osteoporosis may have a thin skull and suffer from skull fracture with halo pin placement or after falls while wearing their halo. Consider a head CT before placing a halo in patients who have osteoporosis.
- Pin loosening occurs in 60% of patients over a 3-month period. Pins may require retightening.
- Pin site infection (10 to 20%). Treat by placing a pin at a new, adjacent site and giving the patient oral antibiotics.
- Neurologic deterioration after traction may occur secondary to retropulsed disk. Consider obtaining a preprocedure MRI to rule out this condition before traction.
- Overdistraction is another potential complication of halo/traction. This could manifest in deficits or pain and can typically be identified on the lateral X-ray.

12.4.2 Lumbar Puncture or Lumbar Drain

- Infection can occur in certain cases, especially those involving prolonged use of lumbar drains.
 - Superficial infection can be treated with drain removal and antibiotic treatment.
 - Epidural abscess (depending on size and neurologic compromise) may require surgical intervention (laminectomy and evacuation).
- Radicular pain can occur secondary to nerve root irritation. If persistent, consider repositioning of drain.
- Post-LP headache:
 - Options include bed rest (24 h), abdominal binder, desoxycortisone acetate, caffeine sodium benzoate, high-dose steroids, and blood patch.
 - If related to lumbar drain, consider decreasing output.
- Spinal epidural hematoma (usually in setting of coagulopathy or anticoagulation).
- Tonsillar herniation (in the presence of mass-occupying lesion).

- Intracranial subdural hygroma or hematoma.
- Epidermoid tumor (increased likelihood with needle introduction without stylet). This can occur in a delayed fashion, but incidence is very low.
- Abducens palsy (often delayed 5–14 d post-LP and resolving without intervention in 4–6 wk).

12.5 Postprocedure Care

12.5.1 Halo Orthosis and Traction

- The pins should be retightened once a day for ~ 3 days at the same pressure and then retightened every week for 3 weeks.
 - A persistently loose pin may indicate migration into the inner table and should be removed and a new one placed at different site.
 - Postprocedure radiographs are taken to verify proper head positioning with halo orthosis and traction placement.

12.5.2 Lumbar Puncture or Lumbar Drain

- For lumbar drain care, prophylactic antibiotics may be continued while the drain is in place, with dressings changed every three days. Drains should be removed or changed after a week.

12.6 Outcomes

- Halo orthosis: Fusion rates are as high as 84% in nonelderly patients who have type II odontoid fracture treated with a halo; risk factors for nonunion include advanced age and displaced odontoid fractures.
- Halo traction: Reduction of bilateral locked facets is typically easier to achieve than reduction of unilateral locked facet.
- LP: Risk of persistent or disabling complication is rated at 0.1 to 0.5%

12.7 Surgical Pearls

12.7.1 Halo Orthosis and Traction

- Pin tension should be uniform. Unequal pin tension will lead to migration of halo as pins migrate in the direction of the pin with the least tension.
- Adjustments during follow-up should not be limited to the halo pins. Inspection of alignment with the vest should be performed to ensure that

shoulder straps are making contact with the trapezius and shoulder area. X-ray radiographs should accompany follow-up visits to ensure proper alignment.

12.7.2 Lumbar Puncture or Lumbar Drain

- Care must be taken in removing the Tuohy needle from the lumbar catheter to avoid shear of the catheter by the sharp bevel of the needle. The trajectory and rotation of the needle must not be altered during removal.
- Evaluation of anatomy with preoperative radiograph is essential, especially in patients who have degeneration and osteophyte formation.
- If attempts at LP or drain placement are unsuccessful, placement of lumbar drain under fluoroscopic guidance may be necessary.

Common Clinical Questions

1. Frontal halo pins may compromise which nerve?
 2. During retightening of halo pins during a follow-up visit, it is found that the pins can no longer be torqued to 6 pounds after one complete turn. What has happened? What should be done?
 3. Patient develops nausea, vomiting, and headaches 2 weeks after a work-up for meningitis. What is the likely diagnosis? What is the treatment?
-

Answers to Common Clinical Questions

1. Supraciliary nerve.
 2. The pin has likely breached the cortical inner table. The pin should be removed and a new pin placed at a new site.
 3. The patient has likely developed post-LP headache. Treatments include bed rest, abdominal binder, hydration, and medication. If these measures fail, a blood patch is indicated.
-

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13 Pharmacology

Mark S. Greenberg

13.1 Key Points

- Treat pain early with effective doses: Results in overall less consumption of pain meds.
- Adjuncts to opioids for pain: Nonsteroidal anti-inflammatory drugs (NSAIDs), muscle relaxants, acetaminophen, Tramadol (not a conventional opioid), and centrally acting pain meds (e.g., gabapentin) for neuropathic pain.
- Use of steroids for spinal cord injury remains controversial, but benefits probably do not outweigh risks.
- Deep venous thrombosis (DVT) prophylaxis in spinal cord injury is critical. If prophylactic anticoagulation is contraindicated, then vena cava interruption filter should be considered.

13.2 Pain Medication^{1,2}

- Nonopioid analgesics
 - Acetaminophen (APAP)
 - An effective pain medication. Does not inhibit peripheral cyclooxygenase activity and is thus not associated with altered platelet function, bronchospasm, or gastric ulceration.
 - Potentiates narcotic pain medication and NSAIDs.
 - Main hazard is hepatic toxicity. Use with caution with active liver disease, with chronic heavy alcohol consumption, and with glucose-6 dehydrogenase deficiency.
- Nonsteroidal anti-inflammatory drugs (NSAIDs)
 - Anti-inflammatory and antipyretic.
 - Single as-needed doses are effective against pain even without “anti-inflammatory dosing.”
 - Adverse effects include reduction of renal blood flow, platelet function inhibition (permanent with aspirin, temporary with other NSAIDs), peptic ulcers. Deleterious effect on bone healing is controversial—many

surgeons hold NSAIDs for 2 weeks after fusion (a longer hiatus is not appropriate).

- Examples of NSAIDs include naproxen (Naprosyn), diclofenac (Voltaren), and ketorolac tromethamine (Toradol). Toradol can be given parenterally (parenteral use should not exceed 3 to 5 days). Oral dosing only as continuation of parenteral dosing (not for routine use as an NSAID).
- Opioid analgesics
 - No single agent has been shown to be more effective or better tolerated as a rule, although individual differences in patients may make certain opioids more effective in a given patient.
 - All produce dose-related respiratory depression. Some lower the seizure threshold. Diversion of prescribed narcotics to sale on the street for recreational use is a burgeoning problem.
 - With chronic use, tolerance develops. All may be habit-forming.
 - Dosing depends more on age and prior narcotic use than on body weight.
 - Use of meperidine (Demerol) is not recommended owing to the lack of advantages and numerous drawbacks, including seizures and drug interactions (serotonin syndrome) with SSRIs or monoamine oxidase (MAO) inhibitors. Because of its anticholinergic properties, it is sometimes still used to treat postanesthetic shivering.
- Weak opioids: For mild to moderate pain.
 - Codeine: Typically prescribed in combination with APAP. Significant incidence of nausea and vomiting.
 - Hydrocodone: In the United States, available only as a combination drug—e.g., with APAP (Vicodin, Lortab, etc.) or ibuprofen (Vicoprofen).
- Opioids for moderate to severe pain.
 - Oral: Oxycodone + acetaminophen (Percocet).
 - Parenteral (IM or IV): For example, morphine, hydromorphone (Dilaudid). Monitor for respiratory depression. May be used for patient-controlled analgesia (PCA).

13.3 Muscle Relaxants

- Oral and IV agents used for low back pain have no activity at the neuromuscular junction. They do exhibit some centrally acting analgesic affect that appears to be independent of muscle spasms. The most consistent effect of these drugs is drowsiness/sedation, which may help the patient rest. Tolerance develops.
- Commonly employed agents include cyclobenzaprine (Flexeril), diazepam (Valium), tizanidine (Zanaflex), and carisoprodol (Soma).
- Use beyond 2 to 3 weeks is not recommended for most of these.

13.4 Anticoagulation (e.g., heparin, LMWH)

- Prophylactic anticoagulation
 - For patients without risk factors for blood clots, prophylactic anticoagulation for elective spine surgery is not recommended.³
 - For spinal cord injuries,⁴ prophylaxis with either of the following:
 - Low molecular weight heparin (LMWH), a rotating bed, adjusted dose heparin, or some combination of these
 - Low-dose (minidose) heparin + pneumatic compression stockings or electrical stimulation
- Treatment for documented deep vein thrombosis/pulmonary embolism (DVT/PE)
 - Therapeutic anticoagulation with heparin transitioned to warfarin or dabigatran (Pradaxa)
 - Post-op: In the first 1 to 2 weeks after spinal surgery, because of the risk of spinal hematoma, a vena cava interruption filter is preferred for DVT/PE. But for acute myocardial infarction (MI) or cardiac ischemia, therapeutic heparin may have to be used; in this case, monitor patient's neuro exam frequently.

13.5 Antifibrinolytic Agents

- May be used off label to reduce blood loss in complex spine operations.
- These agents inhibit the conversion of plasminogen to plasmin within a clot. Plasmin promotes the breakdown of fibrin.⁵
- Thromboembolic complications have been minimal with the two drugs currently in use (aprotinin was withdrawn from the market because of these complications).
- Available agents:
 - Tranexamic acid: 10 mg/kg loading dose followed by infusion of 1 mg/kg/hr during surgery.⁶
 - Epsilon-aminocaproic acid (EACA): A common regimen is 100 mg/kg bolus, followed by 10 mg/kg/hr infusion during surgery.⁶

13.6 Steroids (e.g., Decadron, methylprednisolone)

- Acute nerve injury.
- Spinal cord injury protocol still controversial:
 - The assertion: Administration of methylprednisolone according to protocol within 8 hours of a spinal cord injury (SCI) (complete or incomplete) benefits sensory and motor function at 6 weeks, 6 months, and 1 year.^{7,8}

- The controversy: Results could not be replicated,⁹ steroid-induced myopathy might have produced a transient initial worsening that was misinterpreted as improvement when it subsided,¹⁰ and the risk of side effects (infection including sepsis, gastrointestinal [GI] bleeding diabetogenic complications) is substantial.¹¹
- Use of the protocol among spine surgeons continues to decrease,¹² but it is still used by as many as 56% of respondents to one survey.¹²
- Protocol: Within 8 hours of SCI, bolus with methylprednisolone 30 mg/kg IV over 15 minutes, wait 45 minutes, then start a maintenance infusion of 5.4 mg/kg/hr typically maintained for 23 hours. Do not start the protocol > 8 hours postinjury.
 - Spine tumors: For acute symptoms of spinal cord compression from metastatic tumor, use Decadron 10 mg IV or by mouth every 6 hours for 72 hours, followed by 4 to 6 mg every 6 hours.
 - Epidural steroids.
- Perioperative epidural steroids after routine surgery for lumbar degenerative disease may result in a small reduction of postop pain, length of stay, and the risk of not returning to work at 1 year,¹³ but most of the evidence originates from studies not using validated outcome assessment that favor positive results, and further study is recommended (various agents, dosages, and delivery methods were reported).
- As part of pain management:
 - Chronic low back pain: Not recommended¹⁴ (may be used to provide temporary relief in select cases).
 - Acute radiculopathy: Prospective studies show varying efficacy.¹⁵
- Low back pain: Oral steroids (e.g., steroid dose pack) may provide temporary improvement in symptoms, but no difference from placebo is found at 1-week or 1-year follow-up. Use caution when combining with NSAIDs because of GI complications.

13.7 Treatment of Osteoporosis

- Treatment for documented osteoporosis is usually provided under the direction of a primary care physician. Improvement is usually slow, and documentable changes on dual-energy X-ray absorptiometry (DEXA) scan may take up to 2 years (radiographic improvement lags behind actual reduction of fracture risk).
- Forteo (teriparatide), a recombinant human parathyroid hormone, is the fastest-acting treatment for osteoporosis. When the risk of failure as a result of osteoporosis is high and the downside of waiting for improvement in bone density is significant, off-label use of Forteo (teriparatide) has been described for 2 to 3 months preop, as well as postop in osteoporotic women, in whom it produced a slight decrease in

time to fusion.¹⁶ Studies are ongoing to evaluate further. It is administered as a 20 mcg daily subcutaneous injection. Because of risk of osteosarcoma, Forteo should not be used in patients who are at increased risk for osteosarcoma (including patients who have Paget's disease) and should not exceed the recommended total duration of treatment of 24 months in a patient's lifetime.

13.8 Surgical Pearls

- Pain medication: Early treatment with effective doses before pain becomes severe reduces the total quantity of pain medication needed to control the pain.
- Use of Demerol (meperidine) as a first-line analgesic should be avoided.
- The maximum lifetime use of Forteo (teriparatide) for osteoporosis should not exceed 2 years.

Common Clinical Questions

1. The following are options for DVT prophylaxis in spinal cord injury:
 - I. Low-dose (minidose) heparin alone.
 - II. Low molecular weight heparin alone.
 - III. Oral anticoagulation alone.
 - IV. Low-dose heparin + pneumatic compression device

What are recommended treatments?

 - A. I and III
 - B. II and IV
 - C. I, II, and III
 - D. IV
2. Which of the following drugs/classes of drugs should be avoided initially after spine fusions because of interference with osteoblasts?
 - A. Opioids
 - B. NSAIDs
 - C. Heparin
 - D. Acetaminophen
3. Which of the following statements about the high-dose methylprednisolone protocol for use in spinal cord injury is false?
 - A. Administration should not be undertaken > 8 hours after the injury.
 - B. Some of the apparent benefit of methylprednisolone may have been

the result of patients' recovering from steroid-induced myopathy.

- C. An almost equal number of studies have shown a benefit from employing the protocol as have shown lack of benefit.
 - D. Risks of high-dose methylprednisolone include sepsis, pneumonia, GI bleeding, and deleterious effects of elevated blood glucose.
4. What is the total lifetime recommended dose of Forteo?
- A. 24 mcg
 - B. 240 mcg
 - C. 24 months
 - D. 24 IU
-

Answers to Common Clinical questions

- 1. B.
 - 2. B.
 - 3. C. The benefit initially demonstrated could not be found to be replicated in any other study encountered in a meta-analysis of the literature.
 - 4. C.
-

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14 Radiation Therapy in Spine Surgery

Alexander Tuchman and Patrick C. Hsieh

14.1 Key Points

- Radiotherapy is an important primary or complementary treatment for spinal oncology.^{1,2,3}
- Patient status, type of pathology, goals of treatment, and radiation risks must all be taken into account when formulating a treatment plan.^{1,4,5}

14.2 Description

- Conventional external beam radiation therapy (EBRT)¹
 - Ionizing radiation typically delivered via a photon beam from a linear accelerator is collimated and directed at a target lesion with the goal of DNA damage causing cell death or preventing cell replication.
 - Intensity modulated radiation therapy (IMRT) reduces normal tissue toxicity but increases the volume of normal tissue receiving a low dose.
- Stereotactic body radiotherapy (SBRT) or stereotactic radiosurgery (SRS)^{1,4,5}
 - Radiation is delivered via the convergence of multiple beam orientations.
 - Image guidance helps precisely deliver high-dose radiation to a target, allowing for a higher treatment dose in fewer fractions while sparing surrounding tissues.
- Proton beam therapy^{1,4}
 - Higher atomic weight of protons compared with photons leads to a reduction in radiation dose to normal tissues along entry path and no exit path dose.
 - Improved therapeutic ratio allows for dose escalation.
 - Is often the radiation treatment of choice for residual chordoma.
- Radiation-induced spinal cord injury¹
 - Reversible myelopathy
 - 2 to 6 months after radiation therapy.
 - Typically presents with L'Hermite's sign.
 - Thought to be caused by transient demyelination.
 - No treatment is necessary; it is not associated with chronic progressive myelopathy.

- Chronic progressive radiation-induced myelopathy
 - Bimodal peak at 13 and 29 months after treatment.
 - Diagnosis requires neurologic deficit localized to the area of irradiation and exclusion of other causes.
 - Progressive motor and sensory disturbance culminates in complete spinal cord injury.
 - EBRT risk of myelopathy at total dose of 50, 60, and 69 gray is 0.2%, 6%, and 50% respectively.
 - SBRT maximal spinal cord dose < 13 gray in one fraction or < 20 gray in three fractions associated with < 1% risk of myelopathy.
- Metastatic disease to the spine¹
 - Radiotherapy (RT) for spinal cord compression
 - Typical EBRT dose is 30 gray in 10 fractions.
 - Typical SRS doses include 16 to 24 gray for single fraction, 24 to 27 gray in three fractions, or 30 to 35 gray in five fractions.
 - Local tumor control seen in 78 to 93% of patients in large series
 - “A score predicting posttreatment ambulatory status in patients irradiated for metastatic spinal cord compression,” Rades D, Int J Radiat Oncol Biol Phys 2008 Apr 22:
 - **Tumor histology:** Myeloma/lymphoma (9 points), breast (8), prostate (7), colorectal/small cell lung cancer (SCLC)/renal/other tumors (6), non-small cell lung cancer (NSCLC)/unknown primary (5)
 - **Interval since tumor diagnosis:** > 15 months (8), ≤ 15 months (6)
 - **Visceral metastases:** No (8), yes (6)
 - **Motor function:** Ambulatory without aid (10), ambulatory with aid (9), nonambulatory (3), paraplegic (1)
 - **Time to motor deficit:** > 14 days (9), 7 to 14 days (7), ≤ 7 days (4)
 - * Score ≥ 38 (post-RT ambulatory rate 98.5%): Recommend treating with radiotherapy (RT) alone
 - * Score 29 to 37 (post-RT ambulatory rate 70.9%): Consider adding circumferential decompression and stabilization to RT
 - * Score ≤ 28 (post-RT ambulatory rate 10.6%): Short course RT for palliation of pain
 - Surgery and RT versus RT alone
 - NOMS criteria:³
 - A framework for decisions regarding surgery or radiation taking into account **N**eurologic, **O**ncologic, **M**echanical stability, and **S**ystemic disease patient factors.
 - Surgery used for patients with neurologic deficits from radioresistant tumors or mechanical instability.
 - External beam radiation used for radiosensitive tumors.

- Radioresistant tumors without cord compression or instability can be controlled with SBRT or IMRT.
- “Direct decompressive surgical resection in the treatment of spinal cord compression caused by metastatic cancer: A randomized trial.”²
- Direct decompressive surgery plus postoperative radiotherapy is superior to treatment with radiotherapy alone for patients with spinal cord compression.
- Primary spinal canal lesions
 - Completely resected low-grade astrocytomas or ependymomas^{1,5}
 - Local failure rate is < 10% without further treatment.
 - Typically do not require adjuvant radiation therapy, but long-term follow-up is required.
 - Incomplete or piecemeal resection of low-grade astrocytomas or ependymomas
 - Adjuvant radiation therapy improves progression-free survival.
 - Dose is 50.4 gray in 1.8 gray daily fractions.
 - High-grade spinal astrocytomas and ependymomas radiation therapy routinely recommended as part of multimodal treatment approach^{1,5}
 - High-grade astrocytoma total dose is 54 gray in 1.8 gray daily fractions.
 - High-grade ependymoma and multifocal low-grade astrocytoma total dose is 50.4 to 54 gray.
 - If cerebrospinal fluid (CSF) dissemination, consider craniospinal irradiation.
 - Typical dose is 36 to 45 gray with boost of 50.4 to 54 gray to site of tumor.
 - SRS for benign intradural extramedullary tumors (meningioma, schwannoma, neurofibroma)^{1,5}
 - Dose is 16 to 30 gray in one to five fractions.
 - Radiographic response seen in 28 to 39% of lesions.
 - Large series report local control in 83 to 100% but no study features median follow-up beyond 37 months.

14.3 Surgical Pearls

- For metastatic disease to the spine, tailoring radiation and surgical treatment to the individual patient’s needs is key.
- Radiation plays an important adjuvant role in the management of intradural lesions.
- Radiation dosing to the spinal cord must be kept to a minimum to avoid radiation-induced neurologic injury.

Common Clinical Questions

1. For single-fraction SBRT, to keep the risk of radiation-induced myelopathy below 1% the maximal radiation dose to the spinal cord delivered should not exceed what?
 - A. 2 gray
 - B. 7 gray
 - C. 13 gray
 - D. 20 gray
2. A previously ambulatory patient having no comorbidities beyond breast cancer diagnosed 5 years ago presents with a single metastasis at T6 causing radiographic spinal cord compression, mechanical back pain, and 3 days of leg weakness requiring a walker for ambulation. Which treatment would best preserve ambulatory status in this patient?
 - A. Surgical decompression and stabilization
 - B. Surgical decompression and stabilization followed by radiation therapy
 - C. EBRT alone
 - D. SBRT alone
3. True, or false? Adjuvant radiation therapy for incompletely resected ependymomas has no effect on progression-free survival.

Answers to Common Clinical Questions

1. C.
2. B.
3. False.

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15 Spinal Radiosurgery Therapy

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and Peter Carlos Gerszten

15.1 Key Points

- Conventional fractionated radiotherapy, defined as radiation delivered in one or two radiation beams without high-precision or highly conformal techniques, results in local tumor control, pain palliation, and maintenance or recovery of neurological function in patients with radiosensitive spinal tumors.
- Stereotactic spinal radiosurgery precisely delivers tumoricidal radiation to a spinal tumor and limits toxicity to surrounding normal structures.
- Spinal radiosurgery is a safe, effective, and evolving treatment strategy for spinal tumors.

15.2 Description

- Epidemiology
 - In North America, the annual incidence of spinal tumors is 200,000 cases. The vast majority are metastatic lesions.
 - Up to 40% of patients who have cancer will develop vertebral metastatic disease.
 - In ~20% of patients, spinal metastases will progress to neural element compression.
 - With improved multimodality approaches for cancer treatment and greater long-term survival, the incidence and prevalence of spinal metastases will likely increase.
- Overview
 - Traditional therapies for spinal tumors, whether primary or metastatic, include open surgical excision, systemic chemotherapy, and conventional fractionated radiation therapy, alone or in combination.
 - The goals of spinal radiotherapy are to deliver lethal radiation to the tumor, avoid damage to normal structures, palliate pain, and maintain or recover neurological function.
 - Conventional external beam radiation therapy (cEBRT) delivers a uniform radiation dose to a spinal tumor and its surrounding normal structures, including the spinal cord. It is typically delivered over several fractionated treatment sessions (to provide normal tissues with opportunity for repair) and results in pain palliation and maintenance

or recovery of neurological function in patients who have radiosensitive tumors, such as hematologic malignancies, seminoma, breast, prostate, and neuroendocrine tumors.

- Because excessive radiation to the spinal cord can cause myelopathy, the radiation dose that cEBRT can deliver to a tumor is limited. For this reason, cEBRT often fails to control radioresistant solid tumors, such as melanoma, renal cell carcinoma, and sarcoma.
- Stereotactic spinal radiosurgery is an image-guided, intensity-modulated, conformal radiation technique that applies multiple radiation beams passing through multileaf collimators with different apertures to deliver targeted tumoricidal radiation dose to a spinal tumor and limit radiation dose to surrounding normal structures. It is typically delivered in one session (**Fig. 15.1**).
- Radiosurgery offers the advantage of applying radiobiologically effective doses to a target while sparing the surrounding structures.
- Conventional radiotherapy
 - Three randomized trials have been published for spine metastases.
 - 30 gray delivered in 10 fractions is commonly used. Variations in dose-fractionation schedule do not significantly affect clinical outcomes.
 - Achieves local tumor control in 60 to 90% of cases.
 - Achieves pain palliation in 50 to 70% of patients.

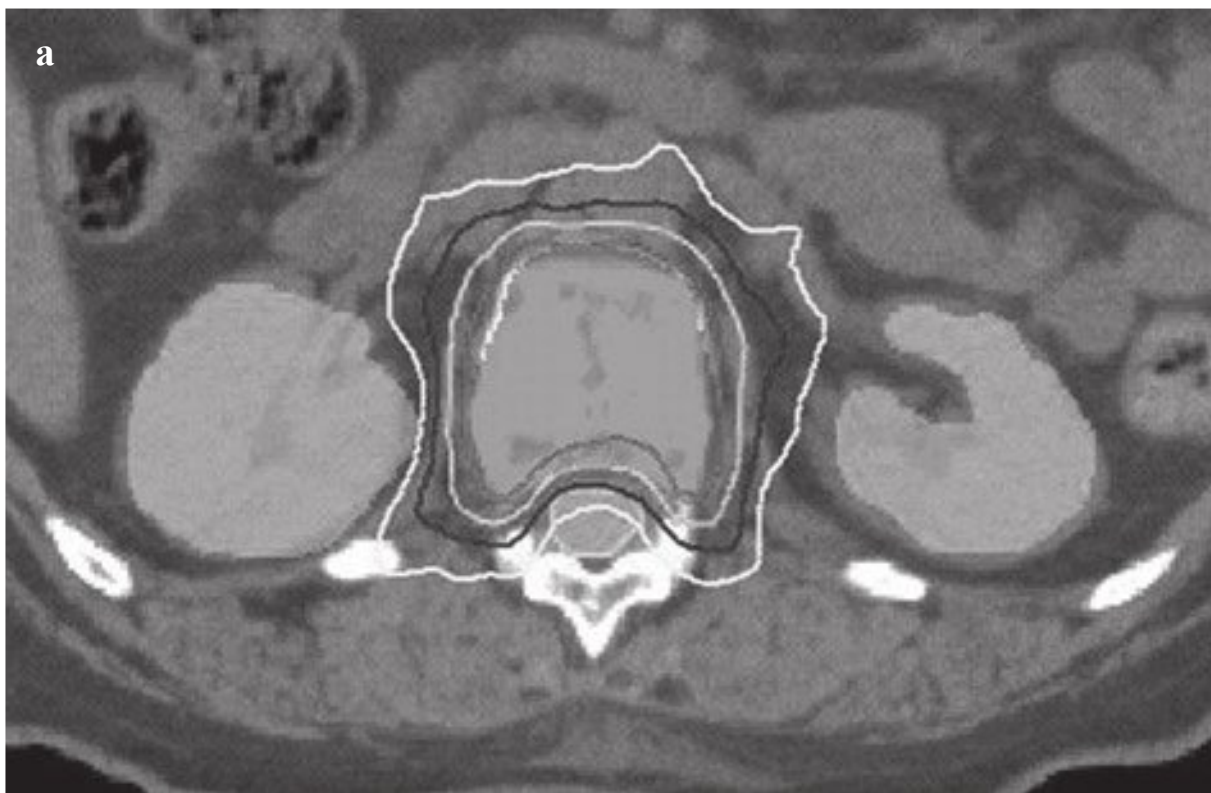


Fig. 15.1 Case example of a 54-year-old woman having symptomatic progression on MRI of L1 and L4 breast metastases after prior conventional radiotherapy. She complained of significant mechanical back pain upon ambulation. She first underwent a percutaneous cement augmentation procedure, followed by radiosurgery. The prescribed dose to the planned tumor volume was 16 gray using nine coplanar beams (Synergy S, Elekta, Atlanta, GA). (**a**) Axial images of the treatment plan are presented. (Continued on page 96)



Fig. 15.1 (Continued) (b) Sagittal images of the treatment plan are presented.

- Achieves maintenance or recovery of neurological function in 70% of patients who have epidural spinal cord compression.
- Radiosurgery
 - No randomized data are available to date.
 - Use of both a single dose (16 to 24 Gy $\times$ 1) and fractionated dosing (4 Gy $\times$ 4, 6 Gy $\times$ 5, 8 Gy $\times$ 3, 9 Gy $\times$ 3) has been reported.
 - Achieves local tumor control in $\sim$ 90% of cases.¹
 - Achieves pain palliation in 85 to 100% of patients.
 - Radiosurgery appears to be a safe and effective treatment option for benign spinal tumors, although open surgical resection remains the mainstay initial treatment. Patients who have benign spinal tumors generally have longer life expectancies than patients who have malignant disease, and the delayed effects of radiation are thus a unique concern in this population.
 - Radiosurgery is a particularly good “salvage” therapy for recurrent or progressive malignant spinal tumors that have failed cEBRT, because these patients are poor open surgical candidates (concern for radiation-induced hypoxia and fibrosis) and poor repeat cEBRT candidates (concern for radiation-induced myelopathy).
 - Radiosurgery may be particularly useful for spinal tumors that require subtotal surgical resection.² Though less aggressive surgical resection may result in decreased morbidity, adjuvant radiosurgery compensates for subtotal resection by providing excellent local tumor control.

Because radiosurgery achieves such a steep falloff gradient, tissues adjacent to the lesion including paraspinal musculature and skin receive only negligible radiation dose. For this reason, adjuvant radiosurgery can be administered in the early postoperative period.

- Limitations: In an effort to protect the spinal cord, the interface between tumor and adjacent spinal cord may be underdosed. Microscopic epidural tumor in this transitional area thus may not receive a tumoricidal radiation dose, resulting in disease progression.³
- Complications: Most common are low-grade skin or esophageal toxicity. Most morbid is myelopathy. Radiosurgery may also be associated with increased risk for vertebral compression fractures at treated levels, especially when a kyphotic deformity or lytic tumor is present.⁴
- Spine radiosurgery: Fundamental principles
 - Treatment planning
 - CT is used for simulation. MRI may be used for tumor and spinal cord visualization and contouring. MRI and CT data are fused for treatment planning. In cases with spinal instrumentation, CT myelography may be used to delineate spinal cord. Functional PET imaging may also be used for tumor delineation.
 - Gross tumor volume (GTV) delineates the spinal tumor. Clinical target volume (CTV) accounts for any microscopic disease outside the GTV. Planning target volume (PTV) accounts for positioning and delivery errors outside the CTV.
 - Target immobilization
 - Early spinal radiosurgery protocols applied an approach similar to that of cranial radiosurgery through the application of an invasive rigid frame directly to the spine.
 - Frameless techniques have become the methodology of choice, being less invasive.
 - Target localization
 - Careful patient positioning and acquisition of pre- and intratreatment volumetric patient imaging is achieved with gantry-mounted imaging sources, and detectors acquire real-time tomographic data during gantry rotation.
 - This data facilitates robust registration, highly accurate radiation delivery, and submillimeter targeting errors.
- Indications for treatment with spinal radiosurgery
 - Primary therapy for spinal tumors not causing severe cord compression or spinal instability
 - “Salvage” therapy for recurrent or progressive spinal tumors that have failed cEBRT
 - Primary therapy for inoperable spinal tumors

- Primary therapy for spinal tumors causing severe compression in nonsurgical candidates
- “Separation” adjuvant therapy for spinal tumors with subtotal surgical resection
- Postoperative adjuvant therapy for spinal tumors with gross-total surgical resection treated with or without instrumentation

15.3 Surgical Pearls

- cEBRT is an established, safe, and effective treatment strategy for radiosensitive spinal tumors, such as hematologic malignancies, seminoma, breast, prostate, and neuroendocrine tumors.
- cEBRT is an appropriate initial treatment option for spinal tumors where no contraindication exists. Contraindications include radioresistant pathology, prior radiation, and severe cord compression or spinal instability.
- Radiosurgery is an evolving, safe, and effective treatment strategy for spinal tumors and should be considered for the treatment of solid-tumor metastases in the setting of oligometastatic disease or radioresistant histology.
- Radiosurgery should be considered over cEBRT for treatment of recurrent or progressive spinal tumors that have failed cEBRT, for postoperative adjuvant therapy for spinal tumors that have been surgically resected, and for inoperable spinal tumors.
- Single-fraction, highly conformal, and large-dose radiosurgery treatments offer excellent tumor control, symptomatic relief, and patient convenience.

Common Clinical Questions

1. What is the major factor limiting the effectiveness of conventional radiotherapy for the treatment of spine tumors?
 - A. The relative resistance of the spinal cord to high-dose radiation
 - B. The intolerance of the spinal cord to high-dose radiation
 - C. The resistance of tumors of the spine to even high doses of radiation
 - D. The instability of the spine that develops after conventional radiotherapy treatment
2. Which of the following statements is true?
 - A. The most commonly used radiation dose prescription for the treatment of spine metastases is 20 gray in 10 fractions.
 - B. Randomized clinical trials have demonstrated the superiority of radiosurgery over conventional fractionated radiotherapy for the treatment of spine tumors.

-
- C. Certain histologies, such as myeloma and lymphoma, are known to be relatively resistant to radiation therapy.
 - D. Reported outcomes demonstrate that 85 to 100% of patients experiencing pain from spine metastases report improvement after radiosurgery.
3. Radiosurgery should be considered as a first-line treatment for a spine tumor over conventional radiotherapy in which setting?
- A. The setting of gross spinal instability
 - B. When the lesion has already undergone treatment using fractionated radiotherapy with spinal cord tolerance doses
 - C. For lymphoma, myeloma, or seminoma
 - D. The setting of widely metastatic spine disease
-

Answers to Common Clinical Questions

- 1. B.
 - 2. D.
 - 3. B.
-

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16 Spinal Navigation

Roger Härtl and José A. Corredor

16.1 Key Points

- Navigation guides surgical interventions in real time through a virtual representation of the surgeon's instruments relative to the patient's anatomy while being displayed on a separate computer screen.
- Routine spinal navigation surgery includes meticulous preoperative planning, registration process, and real-time surgical tracking.
- Navigation is useful in minimally invasive and complex spinal surgeries.
- Improved accuracy in pedicle screw placement and reduction in intraoperative radiation exposure have been described as the main advantages. Some disadvantages, such as the significant learning curve, registration inaccuracy, and costs, are of concern.

16.2 Description

- Spinal navigation is a tool to plan and guide surgical interventions in real time. It generates a virtual surgical reality onto a separate computer screen while guiding the surgeon's instruments relative to the patient's spinal anatomy. Routine spinal navigation includes meticulous preoperative planning, registration of medical imaging onto navigation device, and real-time surgical tracking:
 - Preoperative planning of each case in addition to the training of the surgeon and surgical staff must be completed before using the navigation device. The footprint and interface of various devices in the operating room must be discussed and well understood by the team (**Fig. 16.1**).
 - Registration of the patient's anatomy onto the navigational device involves the use of preoperative or intraoperative images. First generation navigation systems used manual point- and surface-matching techniques for registration in open surgery cases. They were considered cumbersome to use, disruptive to the surgical workflow, and lengthy. The current generation navigation system has shifted to the use of intraoperative cone-beam computed tomography (cbCT), which uses a reference array located on both the cbCT device and the patient's anatomy as an intraoperative image is obtained. Preoperative 3D datasets acquired by a high-resolution CT scanner can also be used.¹

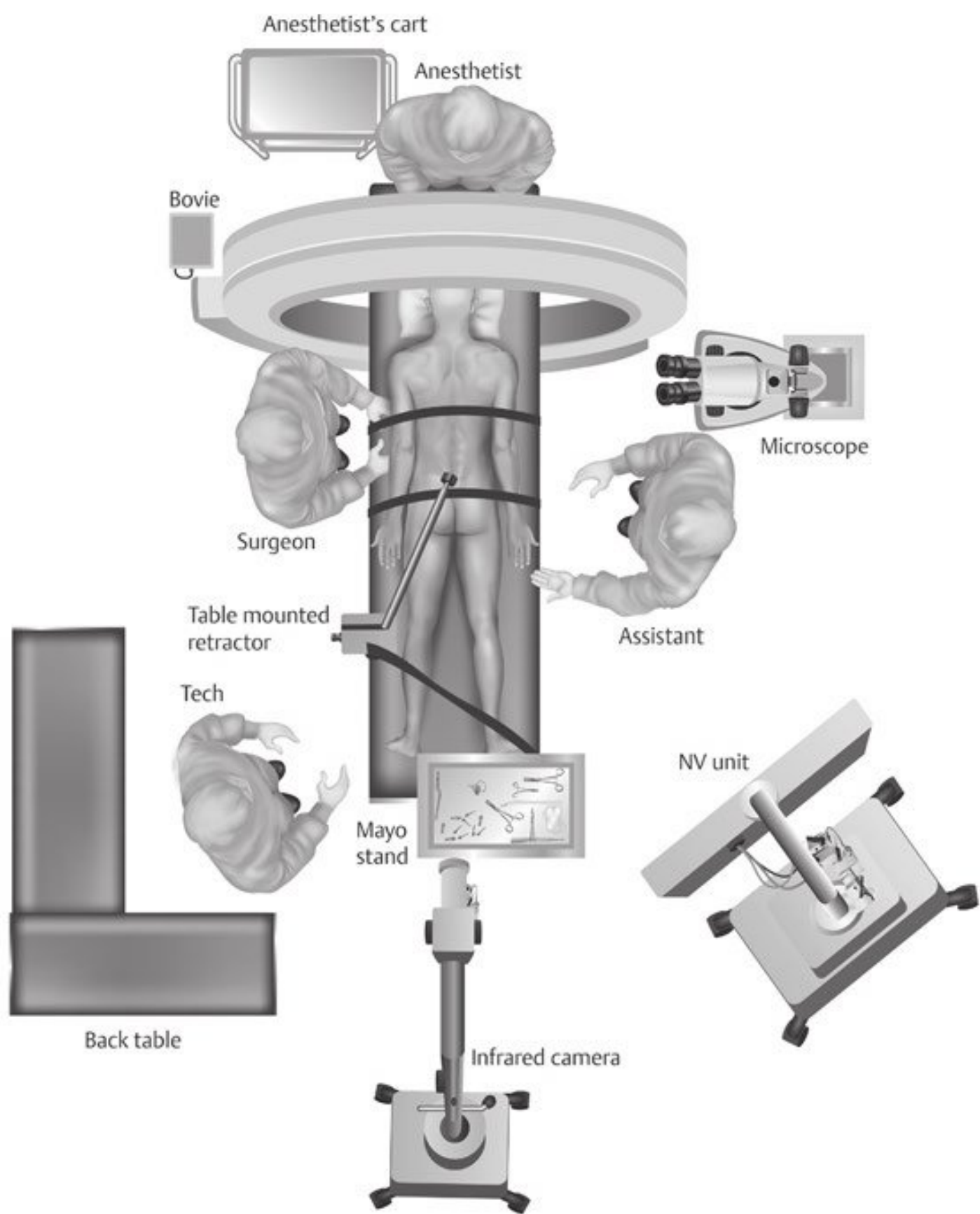


Fig. 16.1 Operating room (OR) layout for MIS TLIF case using Airo Mobile Intraoperative CT navigation system. After screws have been placed using 3D navigation using a navigated guide tube, the microscope is brought in for the decompression, facetectomy, and cage and rod placement. 3D navigation through a 22-mm tubular retractor reduces the need for fluoroscopy, especially in deformity cases.

These images are then automatically imported onto the navigation software to create a 3D representation of the patient's anatomy, thus completing the registration process. This improves the identification of the anatomic target and eliminates the need for manual point- and surface-based registration. The most recent generation of navigation systems (Airo Mobile) includes a large-bore intraoperative CT scanner that provides high-quality CT images and is integrated with the

operating room table and the navigation system. This integration maximizes images quality and streamlines the workflow.

- The 3D representation created is linked to the working position in relation to the patient's anatomy through a tracking system that is presented on a separate computer screen.² Real-time tracking is conducted by two infrared cameras and matched with previously obtained imaging data through the use of a reference point on the patient and on the surgical instruments¹ (**Fig. 16.2**). Optical systems require a direct line of view of the camera and the tracked instruments to link surgical anatomy to the 3D dataset on the computer workstation. Tracking using electromagnetic instead of infrared technology has shown some promising results. The system can then triangulate the instrument's tip location, angle, and trajectory, enabling localization of surgical instruments or implants in real time.²
- Accurate navigation requires vigilant interpretation of information presented on the computer screen in tandem with the surgeon's tactile feedback. Subtle discrepancies may indicate a mismatch between the actual anatomy and information displayed on screen; this requires immediate troubleshooting.



Fig. 16.2 Real-time information is presented on the computer screen during pedicle screw insertion (inset). We use a navigated guide tube that allows drilling, tapping, and placing a pedicle screw through one instrument and without the need of K-wires.

16.3 Indications and Advantages

- Navigation is considered one of the four pillars of minimally invasive spine (MIS) surgery, next to access strategies, implants and microsurgical techniques.¹ Indications and advantages include the following:
 - Required in all MIS procedures, navigation is mainly used to facilitate the placement of screws in all regions of the spine. Numerous published studies have demonstrated the efficacy of 3D computer navigation systems in improving the accuracy of pedicle screw placement as compared with that of conventional or “freehand” placement.³ The navigation system enables optimal screw placement through the use of larger pedicle screws and higher screw-to-pedicle diameter ratio, which are important in patients who have poor bone quality or deformities.¹
 - Can confirm the correct spine level and guide procedures such as those of decompression.
 - Guides the placement of interbody devices for lateral transpsoas approaches.¹
 - Advantageous in patients requiring complex spinal procedures such as significant spondylolisthesis, thoracolumbar deformity, tumor resection, or revision surgery. The use of navigation in complex anatomy cases over conventional anteroposterior/lateral fluoroscopy offers the benefit of accurate and convenient screw application. Screw fit is maximized, and there is no need to “skip” levels in response to small or complex pedicle anatomy, as is frequently seen in multilevel cases using conventional techniques.
 - In the lumbar spine, navigation is used to determine the length of rods and screw alignment during a multilevel fusion, facilitating percutaneous rod placement.
 - Surgery through navigation system is safe because of the reduced intraoperative radiation exposure and neurovascular complications. The comparison of neurological complications with and without navigation favors the use of navigation for pedicle screw insertion, though no statistical significance has been indicated.²
 - Latest-generation intraoperative CT scanners in conjunction with navigated guide tubes allow drilling, tapping and screw placement through one navigated instrument. This allows navigation efficiency;⁴ K-wires are potentially not needed for MIS procedures, the workflow is streamlined, and the use of additional intraoperative fluoroscopy has been eliminated or minimized.
- Some disadvantages to spinal navigation system have been reported, among them a significant learning curve and high cost (**Table 16.1**).

Table 16.1 Advantages and disadvantages of spinal navigation system

Advantages	Disadvantages
Improve accuracy placement and size of instrumentation used	Limited imaging quality and field of view
Reduce radiation exposure to surgeon and staff	Up-front costs of the capital equipment
Potential for less invasive approaches through smaller access corridors	Potential line-of-sight limitations for optical systems
Allow verification of screw accuracy intraoperatively	Additional equipment and footprint in the OR
Lower the risks of wrong-level surgery	Lack of scientific data supporting its clinical benefit
Decrease reoperation rate	The learning curve could be significant
Used to determine the ideal positioning and trajectory of the TLIF or PLIF cage	Concerns about accuracy and interference with metallic instruments
Measure intraoperative distances.	Potential increase in OR time
Perform MIS surgeries without K-wires	Interruption of surgical “flow”

Abbreviations: MIS, minimally invasive spine; OR, operating room; PLIF, posterior lumbar interbody fusion; TLIF, transforaminal lumbar interbody fusion.

16.4 The Future of Spinal Navigation

- Demand of navigation systems will expand globally. Additionally, companies are working to offer better intraoperative image quality through superior software and more compact footprint devices that can be easily adapted into small spaces.
- Spinal navigation will further move away from the use of K-wires, minimizing the need for intraoperative X-ray use.
- User-friendly and cost-efficient navigated-robotic surgery has been introduced and will progressively gain acceptance among surgeons for the placement of pedicle screws. With the continuous technological

advancements being made, spinal navigation systems are likely to become a standard of care in surgical procedures.

16.5 Surgical Pearls

- The reference array should be positioned between the camera and anatomical target region.
- To maximize navigational accuracy and efficiency, pedicle screw placement should constitute the initial step of the procedure.
- The reference array should be positioned into the iliac crest for cases up to L2. If the procedure extends above, it should be repositioned cranially using a spinous process clamp.
- Intraoperative electromyography monitoring may be used to stimulate the screws to double-check the accuracy of screw positioning.

Common Clinical Questions

1. Which feature of second-generation spinal navigation systems spurred its widespread use?
 - A. Easy adaptation to 2D or 3D stereotactic devices
 - B. Accurate registration of the patient's spinal anatomy without the need for matching points
 - C. Use of point- and surface-matching technique for registration
 - D. Possibility for active tracking
2. In the traditional routine of spinal navigation surgery, the registration process constitutes which of the following?
 - A. Converting real-time information to a 3D representation based on the different reflective angles and presented in the screen
 - B. Constantly interpreting what is presented on the computer screen
 - C. Registering patient data into the system
 - D. Placing a reference array on the patient and obtaining intraoperative images through an imaging system
3. Which of the following spinal navigation advantages is not supported by clinical evidence?
 - A. Accuracy in pedicle screw placement
 - B. Less surgical time
 - C. Decrease radiation exposure
 - D. Less neurovascular injuries

Answers to Common Clinical Questions

1. B.
 2. D.
 3. B.
-

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17 Spine Biologics

Zorica Buser, Rahul Basho, and Jeffrey C. Wang

17.1 Key Points

- Successful spine fusion relies on the osteoconductive, osteoinductive, and osteogenic characteristics of the graft.
- The iliac crest autograft is the only standalone graft that provides all three features.
- Each graft carries risks and benefits. The appropriate choice depends on the condition of the patient and the surgical location.

17.2 Description

- The role of the bone graft is to initiate and support healing cascades that will lead to the formation of new bone.
- The ideal graft should be (**Table 17.1**)
 - Osteoconductive: Provides mechanical structure for blood vessels, cell migration and bone formation
 - Osteoinductive: Contains growth factors for cell proliferation and differentiation
 - Osteogenic: Contains osteoblasts and stem cells that form new bone
- The two main graft categories are autografts and allografts.
- Depending on function, allografts can be further divided into graft extenders, enhancers, and substitutes.

17.2.1 Autografts

- Autografts represent an ideal graft, possessing osteoconductive (collagen/bone matrix), osteoinductive (bone morphogenetic proteins [BMP]), and osteogenic (cell) components.
- The bone can be cancellous, cortical, or vascularized cortical.
- Autografts provide immediate and long-term stability, no compatibility issues, and low costs.
- Autografts are divided in two groups: Local grafts and iliac crest bone grafts (ICBG).
- Local graft:

- Advantages: Harvested from the processes, lamina or facets; no need for additional surgery; provides immediate stability.
- Disadvantages: Cortical bones, due to their pore size, lead to slow bone remodeling and, potentially, to long-term instability.
- Iliac crest bone graft (ICBG) (the gold standard):
 - Advantages: Pure cancellous bone; provides immediate and long-term stability owing to blood vessel and cell penetration.
 - Disadvantages: Additional surgery for graft harvest; potential comorbidities (pain at the donor site, fracture, infection, hernia). However, studies have shown similar pain rates in patients with and without ICBG.¹
- Overall drawbacks with the use of autologous grafts are donor age (bone quality decreases) and the large amount of graft material needed for multilevel fusions.

17.2.2 Allografts

- Allografts are harvested from cadaver tissue and possess osteoconductive and minor osteoinductive abilities.
- Suitable bones are cortical or corticocancellous.
- Most commonly, allografts are used for anterior cervical discectomy and fusion and for spine tumor cases requiring spinal reconstruction (because some tumor patients may seed the iliac crest or local bone with tumor cells).
- Depending on the harvest and preparation, grafts can be fresh, fresh frozen, or freeze-dried.
- The main drawbacks are potential immunogenicity, contamination, and graft quality.

17.2.3 Demineralized Bone Matrix

- Cell-free allograft consisting of collagen matrix, growth factors and calcium phosphate.
- Demineralized bone matrix (DBM) particle size plays a crucial role in osteoinductivity.
- DBM is usually mixed with a carrier such as calcium sulfate, glycerol, gelatin, or hyaluronic acid.
- DBM can be in the form of powder, chips, putty, or gel-filled syringes.
- DBM can be used as a slow-release delivery vehicle for growth factors and other active molecules.²
 - Advantages: Obtainable in large amounts.
 - Disadvantages: Lot-to-lot variability is significant; fusion success is influenced by the type of DBM.³

17.2.4 Ceramics

- Ceramics are osteoconductive grafts but lack cells and growth factors.
- Hydroxyapatite and β -tricalcium phosphate (β -TCP) are the most commonly used materials and can produce good fusion rates.
- Silicate substituted calcium phosphate and calcium sulfate have been used, but with inconsistent fusion rates.
 - Advantages: Easily obtainable in large amounts, no contaminations and suitable pore size for cell and blood vessel ingrowth
 - Disadvantages: Brittle scaffolds lacking mechanical stability

17.2.5 Bone Morphogenetic Proteins

- Bone morphogenetic proteins (BMPs) are osteoinductive by stimulating stem cells toward osteogenic differentiation.
- BMPs play an important role in bone ossification and tumorigenesis.
- BMPs are synthesized using recombinant DNA or isolated from donor bone.
- Commonly used carriers: Ceramics, absorbable collagen I sponges (ACS), or allografts.
- rhBMP2 is FDA-approved for anterior lumbar interbody fusion, and rhBMP7 (OP-1) is used under a humanitarian device exception for posteriolateral lumbar fusion.
 - Advantages: Availability; high fusion rates
 - Disadvantages: High costs; potential complications

17.2.6 Bone Marrow Aspirate

- Bone marrow aspirate (BMA) has osteogenic and osteoinductive properties.
- Commonly used carriers are DBM or collagen sponges.
 - Advantages: Contains stem cells, osteoblasts, and growth factors
 - Disadvantages: Harvest site morbidities; quality varies with donor age; low stem cell numbers

17.2.7 Platelet Gels

- Contain platelets and various growth factors.
- Current literature has shown lower fusion and higher nonunion rates.

17.3 Surgical Pearls

- Autograft remains the gold standard; regardless of whether surgeons choose to use it, they must be proficient in autograft harvesting techniques.
- Meticulous decortication of transverse processes, facets, and the pars interarticularis is paramount for a successful fusion.
- Preoperative planning is key. Surgeons should consider the patients biology, number of levels involved, and previous history of autograft harvesting when deciding graft choices.
 - The fusion rates and biology of fusion are much more favorable in adolescent scoliosis versus adult scoliosis patients.
 - Fixation into the pelvis may preclude harvesting of autograft.

Common Clinical Questions

1. What are the two main graft types?
2. What is the only bone graft option with osteogenic potential?
3. What are the disadvantages of DBM?

Answers to Common Clinical Questions

1. Autograft and allograft.
2. Autograft (including bone marrow aspirate) is the only bone graft option with live cells capable of producing bone.
3. Main drawbacks are donor age and bone quality, preparation technique, and lot-to-lot variation.

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III

Spinal Pathology

18 Congenital Anomalies

Jared Fridley, Eric Momin, and Andrew Jea

18.1 Key Points

- Congenital abnormalities of the spine are a heterogenous group of disorders in which there has been compromise of one of the stages of spinal embryological development (**Fig. 18.1**).
- Other systems, including the cardiovascular and genitourinary systems, may be affected.

18.2 Vertebral Developmental Abnormalities

18.2.1 Congenital Scoliosis

- Background
 - Scoliosis that occurs secondary to abnormal vertebral development
 - Classified based on either embryologic failure of formation (hemivertebra) or failure of segmentation (bar or block vertebra) (**Fig. 18.2**)
- Signs, symptoms, and physical exam
 - Associated spinal abnormalities occur in ~20% of patients.
 - Cardiac, genitourinary, and skeletal abnormalities occur in 30 to 60% of patients.
- Neuroimaging
 - CT of the spine or spinal radiographs is used to assess bony abnormalities and to plan any surgical intervention.
 - MRI of the spine is used to look for associated intraspinal abnormalities.
 - Echocardiogram and renal ultrasound should be performed in all patients.
- Treatment
 - Bracing does not prevent curve progression, nor does it correct underlying pathology.
 - Surgery is considered for symptomatic curves $> 45^\circ$ or progressing $> 10^\circ$ per year.

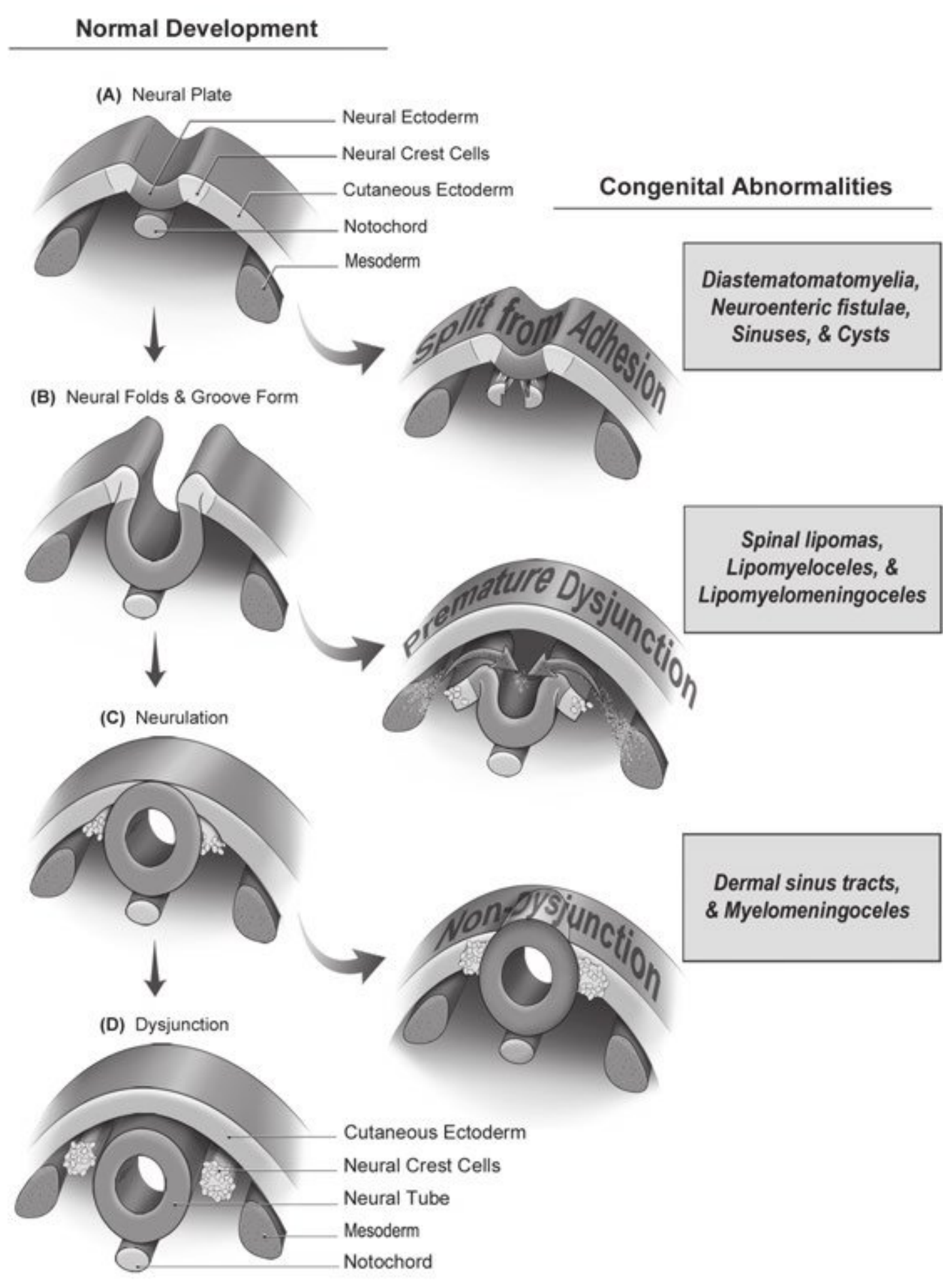
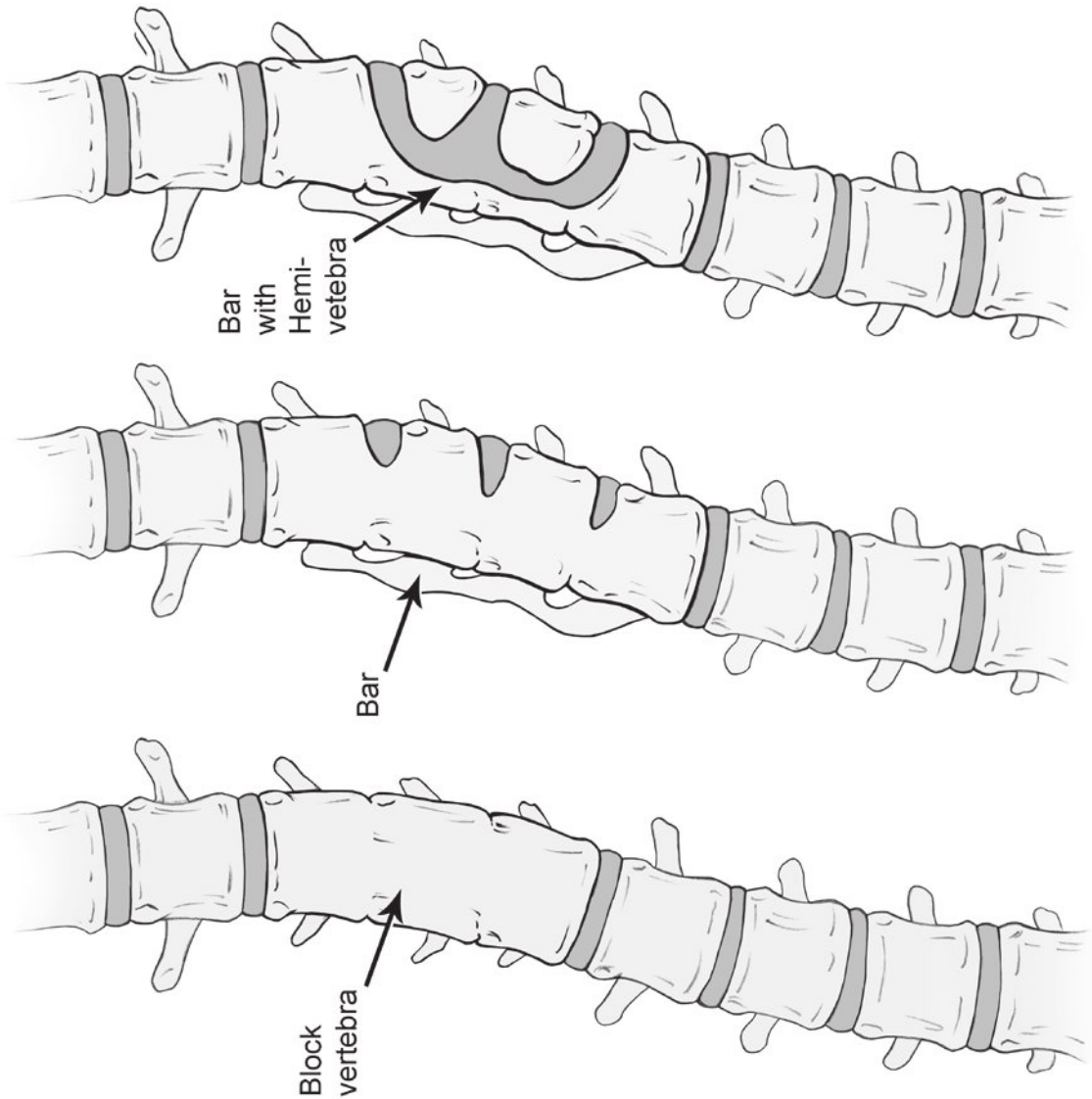
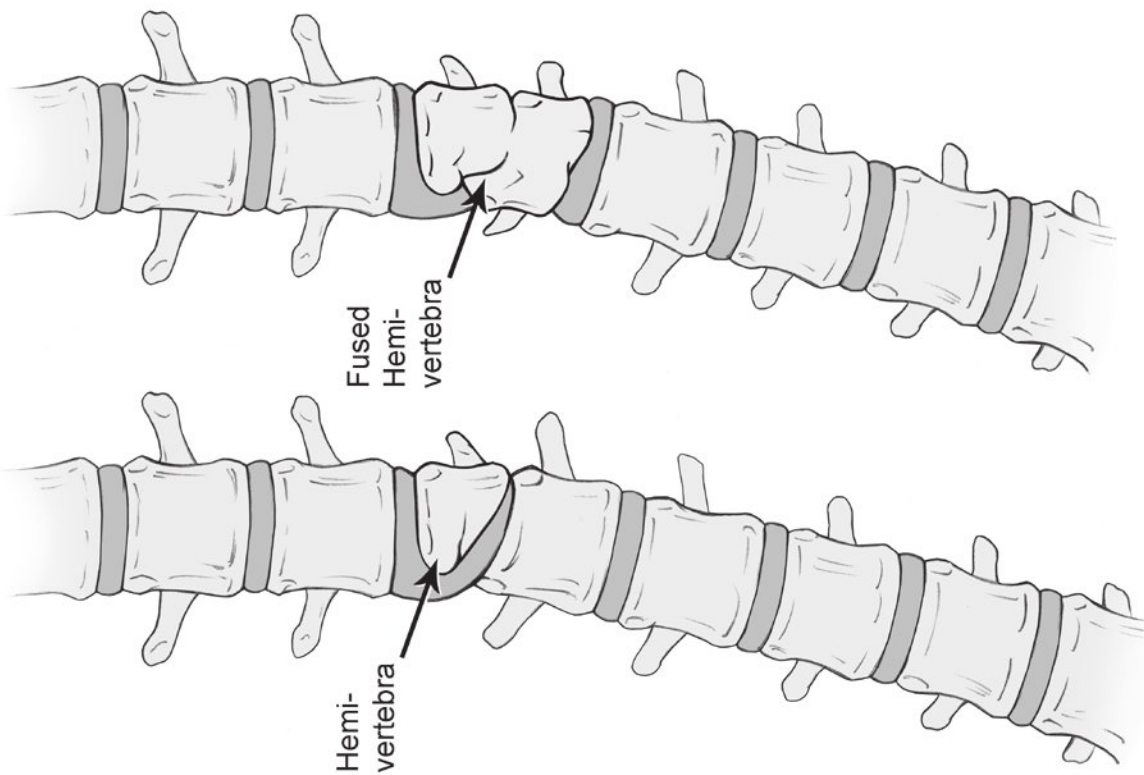


Fig. 18.1 Defects during different stages of spinal cord embryology lead to congenital anomalies of the spine.

Failure of Segmentation



Failure of Formation



18.2.2 Klippel-Feil Syndrome (KFS)

- Background
 - Characterized by multiple fused cervical vertebrae due to failure of segmentation.
 - Three types:¹
 - Type 1: Fusion of many cervical and upper thoracic vertebrae. High risk of scoliosis (~ 30%).
 - Type 2: Fusion at one or two interspaces combined with other congenital spinal abnormalities.
 - Type 3: Fused cervical vertebrae with thoracic or lumbar fused vertebrae.
- Signs, symptoms, and physical exam
 - Classic clinical triad of short neck, low posterior hairline, and decreased neck range of motion is seen in less than 50% of patients.
 - Associated with scoliosis, deafness, genitourinary abnormalities, cervical ribs, and cardiovascular abnormalities.
 - Present with neck pain or neurologic compromise.
- Neuroimaging
 - Cervical radiographs or CT of the cervical spine reveal multiple fused cervical vertebrae.
 - MRI of the cervical spine is performed if neurologic signs/symptoms present.
 - Echocardiogram and renal ultrasound are used to look for associated abnormalities.
- Treatment
 - Surgery is reserved for cases of spinal instability, compression of neurologic elements, or spinal deformity.

18.3 Anomalies of Notochord Formation

18.3.1 Split Cord Malformations (SCMs)

- Background
 - Classified into two types: Type I is characterized by two hemicords, each within its own dural tube, separated by a bony/cartilaginous septum, and Type II is characterized by two hemicords in a single dural tube, separated by a fibrous septum.²
- Signs, symptoms, and physical exam
 - Often associated with open and occult spinal dysraphism manifestations.
 - Common clinical presentations include tethered cord syndrome (TCS), pain, and/or sensorimotor disturbances, with hypertrichosis found in nearly 50% of all patients.

- Neuroimaging
 - High-resolution thin-cut CT scans of the spine are useful for defining the type of split cord malformation.
 - MRI of the spine is useful for identifying other associated intraspinal anomalies.
- Treatment
 - The primary goal of surgery is detethering of the spinal cord by removal of any offending bone, septa, or fibrous bands.

18.4 Anomalies of Dysjunction

18.4.1 Spinal Lipomas³

- Background
 - Includes multiple pathologies: Lipomas of the filum or conus medullaris, lipomyelomeningocele, intradural spinal lipomas, epidural lipomatosis, lipomyelocele, and spinal angiolioma.
- Signs, symptoms, and physical exam
 - Classically associated with TCS as a result of tethering of neural tissue by the lipoma.
 - Presenting symptoms/signs can include lower extremity weakness, orthopedic abnormalities of the lower extremities, numbness, gait dysfunction, spasticity, urinary hesitancy/incontinence, back/leg pain, and cutaneous stigmata of the low back.
- Neuroimaging
 - MRI of the lumbar spine without contrast is the modality of choice for defining the precise nature of the lipoma (T1-weighted sequence) and surrounding neural tissue (T1- and T2-weighted sequences) and is key for planning any surgical intervention.
- Treatment
 - Natural history of these lesions is not clear, so the risk of surgical morbidity (including future retethering and neurological injury) must be weighed against any possible surgical benefit.

18.4.2 Myelomeningocele

- Background
 - Multiple factors are involved in etiology of myelomeningocele, both maternal and fetal, including a strong association with maternal folate deficiency.
 - Nearly all patients have an associated Chiari II malformation.

- Signs, symptoms, and physical exam
 - Neurological deficits are dependent on the location and length of the defect.
 - High risk of hydrocephalus (80–90%).
 - Associated with orthopaedic and genitourinary abnormalities.
 - Risk for developing tethered cord syndrome and scoliosis.
- Neuroimaging
 - Fetal ultrasound has nearly 100% sensitivity for detection of myelomeningocele.
 - Fetal MRI can also be useful, especially when planning surgical treatment.
- Treatment
 - Surgical repair of the myelomeningocele should occur within the first 48 to 72 hours of life because of the risk of infection from exposure of neural elements.
 - With the publication of the encouraging results of the landmark MOMS trial,² in utero repair of myelomeningocele must be considered in eligible patients.

18.5 Surgical Pearls

- The goals of surgery are to arrest progression of the deformity and to correct it, if it can be corrected safely.
- Patients who have Type I KFS should be followed closely in clinic for development of spinal deformity, particularly scoliosis.
- All Type II SCMs should be surgically explored in symptomatic patients, even if imaging is unrevealing of any median septae.
- Surgery carries a risk of not only neurologic injury and cerebrospinal fluid (CSF) leak, but also scar tissue formation and spinal cord tethering, all of which must be considered when deciding whether to operate on an asymptomatic patient.
- Placode is dissected from surrounding epithelium around the malformation, with care taken not to leave any attached epithelial elements because of the risk of developing dermoid tumors (**Fig. 18.3**).
- Lateral edges of the placode are reapproximated, followed by closure of the dura, the dorsal fascia/muscle layers (depending on surgeon preference), and, finally, the skin.

a

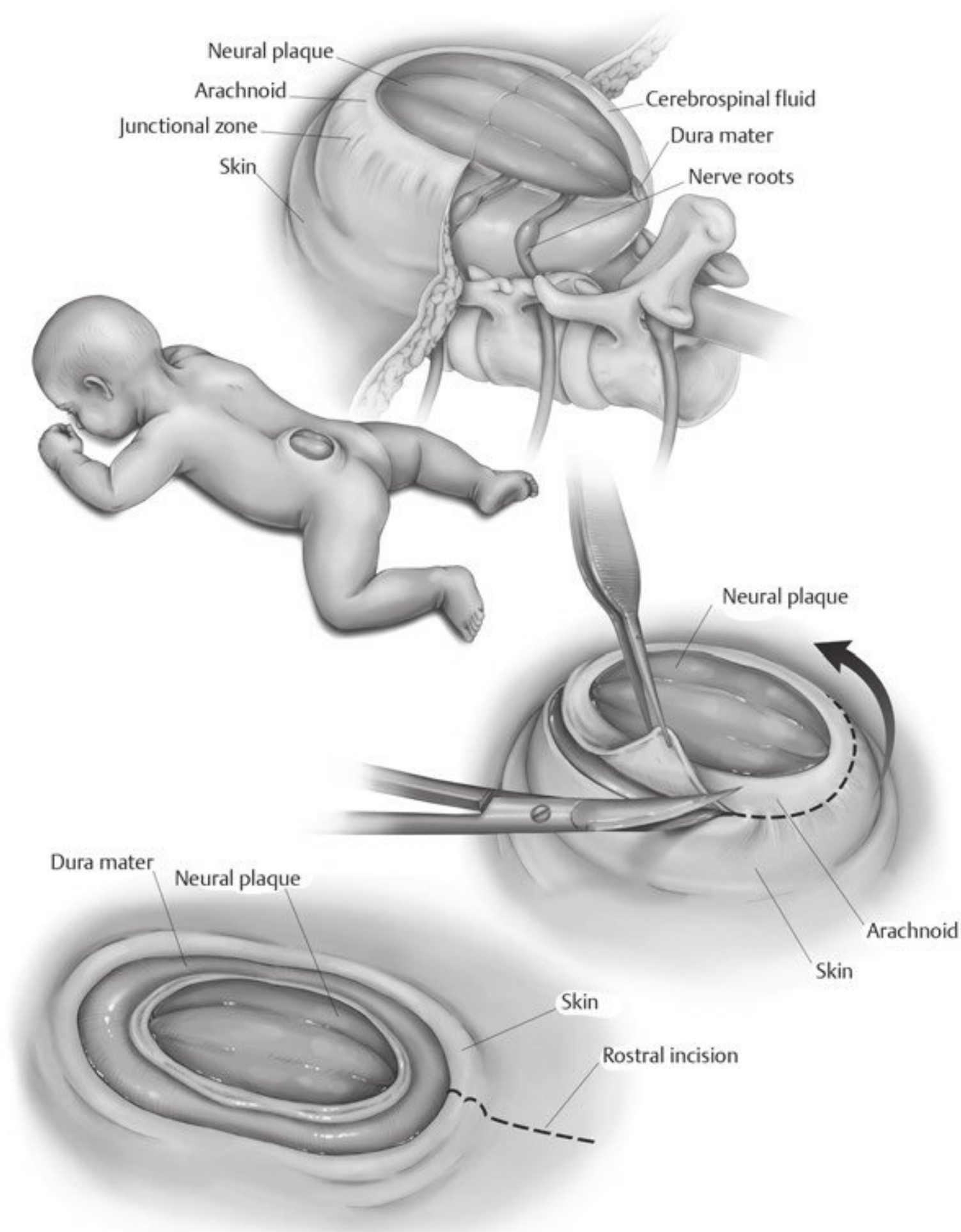


Fig. 18.3 (a) The multilayer closure begins with an initial circumferential incision following the arachnoid–skin junction to isolate the neural plaque and spinal cord. The skin incision is continued rostrally in the midline to observe the caudalmost intact lamina and the dura beneath it. (Continued on page 122)

b

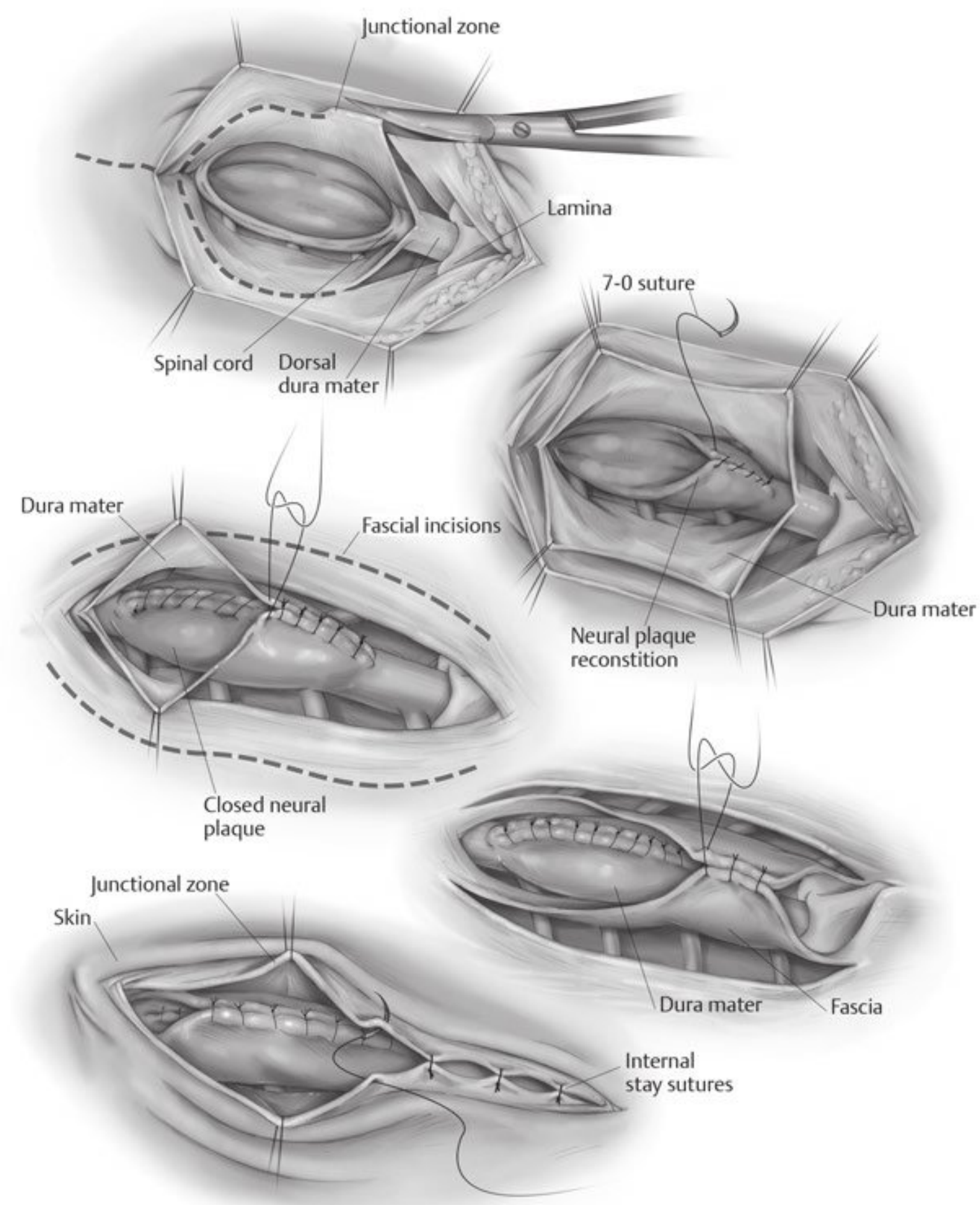


Fig. 18.3 (Continued) (b) An inferior and lateral dissection toward the caudal edge of the defect isolates the dura. The caudal defect is extended and the residual dura dissected. In the midline, the dura is then approximated following that of the lateral arachnoid. The lumbo-sacral fascia is identified and incised bilaterally with dissection from the posterior iliac crest and sacrospinalis muscle, with care taken to not disrupt the sacral fascial attachments. The lateral edges of the fascial flaps are then folded toward the midline and sutured in place over the dorsal surface of the dura. The subcutaneous tissue, if present, is closed. The placement of sutures in the area where the dura joins the dermis will provide for tight internal sutures that will reduce tension during later skin closure.

Common Clinical Questions

- 1. What is the most common cause of congenital scoliosis?
- 2. What is NOT commonly associated with congenital scoliosis?
- 3. Myelomeningocele is highly associated with what maternal factor?

Answers to Common Clinical Questions

- 1. Hemivertebrae are the most common etiology of congenital scoliosis.
- 2. Anorectal abnormalities are not commonly associated with congenital scoliosis.
- 3. Folate deficiency or maternal metabolism of folate is thought to be a major contributor to the development of myelomeningocele. Maternal folate supplementation has led to a significant decline in the incidence of patients who have myelomeningocele.

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19 Cervical Trauma

Gregory D. Schroeder and Alexander R. Vaccaro

19.1 Key Points

- Early decompression increases the chance of neurologic recovery in patients who have an acute spinal cord injury.¹
- The use of high-dose methylprednisolone in patients with an acute spinal cord injury is controversial.^{2,3,4}
- C1 ring fractures rarely require surgery.
- Geriatric patients who have a type II odontoid fracture who are healthy enough to undergo general anesthesia should consider surgical stabilization—preferably a posterior C1–C2 fusion.^{5,6}
- The algorithm for the treatment of subaxial cervical spine fractures is based on bone stability, injury to the diskoligamentous complex, and the neurologic injury of the patient.⁷
- Subaxial cervical facet fractures involving more than 40% of the absolute height of the intact lateral mass or more than 1 cm long from the tip of the superior articular process to the fracture line are at increased risk for failure of nonoperative treatment.⁸

19.2 General Principles

- Field management
 - ABCs: **a**irway, **b**reathing, **c**irculation
 - Spinal immobilization
 - Rigid cervical collar, lateral bolsters, rigid backboard.
 - Pediatric patients require recessed headboard or pediatric board.
- Emergency room management
 - ABCs plus **d**isability and **e**xposure
 - Standard imaging
 - Neurogenic shock
 - Hypotension in the presence of bradycardia
 - Management with vasopressors and modest fluid resuscitation
- Classification
 - American Spinal Injury Association (ASIA) scoring scale (**Table 19.1**):
 - Complete: No motor or sensory function below zone of injury
 - Incomplete: Partial motor or sensory function below zone of injury (**Table 19.2**)

Table 19.1 ASIA Neurological Scoring System

Grade	Description
A	Complete: No motor or sensory function is preserved in the sacral segments.
B	Incomplete: Sensory but not motor function is preserved below the neurologic level and includes the sacral segments.
C	Incomplete: Motor function is preserved below the neurologic level, and more than half of the key muscles below the level have a muscle grade < 3.
D	Incomplete: Motor function is preserved below the neurologic level, and at least half of the key muscle groups below the level have a muscle grade > 3.
E	Normal.

Note: The caudalmost normal level is the neurologic level.

Table 19.2 Incomplete spinal cord injury patterns

Syndrome	Prognosis	Description
Central cord	Fair	Due usually to hyperextension injury with greater upper extremity involvement and more proximal than distal muscle groups
Anterior cord	Poor	Due to injury of anterior spinal artery; loss of pain, temperature, and motor
Brown-Sequard	Best	Due to hemitransection of cord or lateral injury; ipsilateral motor loss, vibration, and position sense and contralateral pain and temperature
Posterior cord	Fair	Loss of vibration and position sense

- Ten percent to 15% of patients have noncontiguous spinal fractures.
- Orthogonal radiographs (anteroposterior [AP]/lateral C, T, and lumbosacral spine).
- CT:
 - Useful for visualizing occipital–cervical and cervicothoracic junctions
 - Must include sagittal and coronal reconstructions if plain X-ray not used

- Not as useful for ligamentous injury assessment
- MRI:
 - Required in all cases with neurologic impairment or deficit
 - Ligamentous structures visualized on T1
 - Edema visualized on T2 with short tau inversion recovery (STIR)
- Steroid management
 - Controversial
 - Congress of Neurological Surgeons recommended against the use of high-dose methylprednisolone in 2013 owing to its limited efficacy and high rate of complications.³
 - 47.4% and 56.4% of Cervical Spine Research Society members reported using steroids in patients who had complete or incomplete spinal cord injury (SCI) in 2014.²
- Surgical timing
 - Surgical decompression within twenty-four hours has been shown to lead to statistically significant improvements in neurologic function six months after injury.¹

19.3 Upper Cervical Spine Trauma

19.3.1 Atlantooccipital Dissociation

- Background
 - Clinical suspicion paramount
 - Mechanism: High-energy rotational or flexion–extension force
- Signs, symptoms, and physical exam
 - Derangements in cardiorespiratory parameters common
 - Survivors⁹
 - 40% quadriplegic
 - 40% hemiplegic
 - 20% neurologically normal
 - Neurologic deterioration can occur in up to 35% of patients who live through the initial injury.
- Neuroimaging
 - Radiographs
 - Multiple radiographic measurements have been described, but none has been universally accepted as superior.
 - Powers ratio:¹⁰ Ratio of distance from basion to C1 lamina divided by distance from opisthion to anterior ring of C1 (**Fig. 19.1**).
 - Identifies anterior subluxation if ratio > 1.
 - Wackenheim line:¹¹
 - Line from posterior surface of clivus—normally its inferior extension should barely touch posterior aspect of the odontoid tip

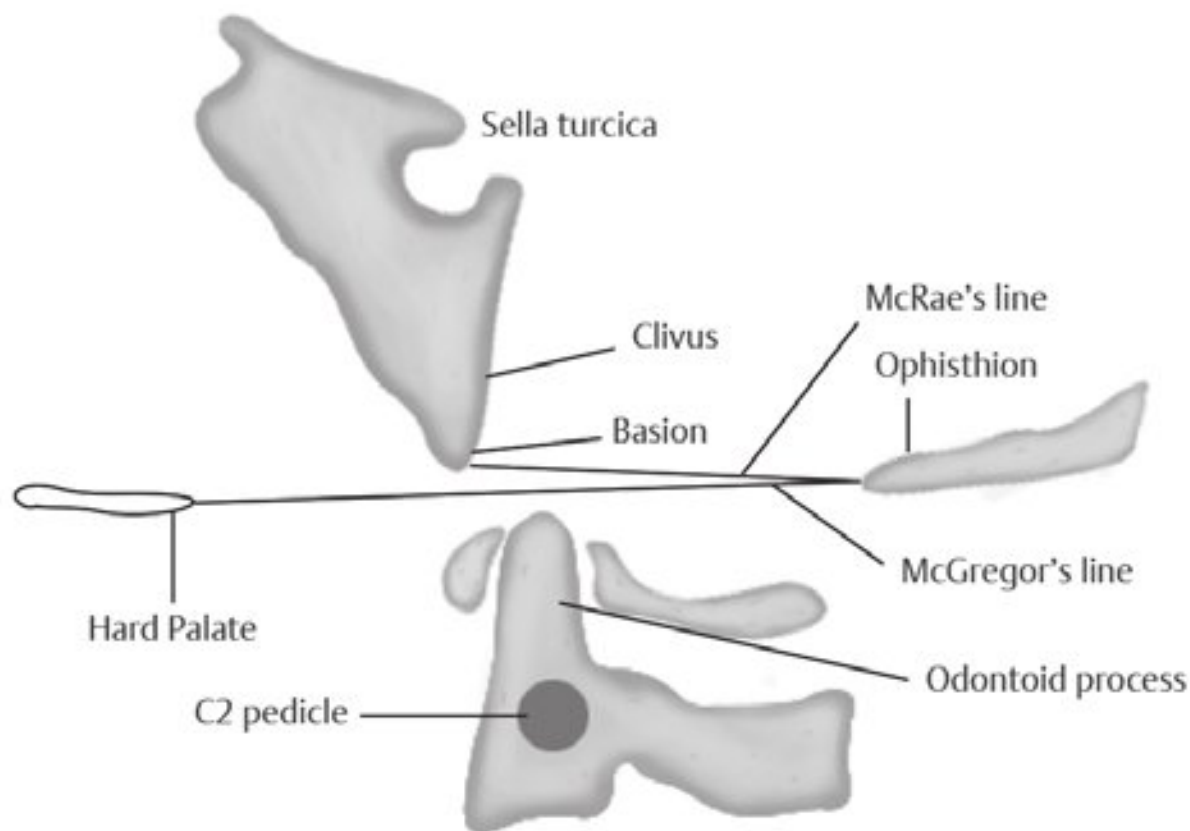


Fig. 19.1 Different lines and parameters to assess for craniovertebral stability.

- If line runs behind the odontoid, there is posterior dissociation.
 - If line runs in front, there is anterior dissociation.
- Harris's Rule of 12:¹²
 - A basion–axial interval or basion–dental interval > 12 is indicative of an atlantooccipital dissociation.
- CT
 - Revised condyle–C1–interval:¹³
 - Distance between the occipital condyle and the C1 lateral mass on sagittal imaging.
 - A value greater than 2.5 mm is consistent with a dissociation.
- Treatment
 - Occipitocervical fusion
- Surgical pearls
 - Obtain a CT angiogram (CTA) or MR angiogram (MRA) to assess vertebral artery integrity as part of the preoperative planning if suspicion exists for vessel injury.
 - Use of MRI for assessing integrity of ligaments may help with decision to fuse occiput to C1, C2, or a more caudal spot.

19.3.2 Atlas (C1) Fracture

- Background
 - Neurologic injury is rare due to large space available for the spinal cord.
 - Can occur anterior, posterior, or combined (Jefferson).
 - Mechanism is axial loading.

- Status of transverse alar ligament (TAL) is critical to surgical decision making.
- Dickman Classification¹⁴ of TAL injuries:
 - Type 1—Midsubstance disruption of TAL
 - Ligament will not heal.
 - Type 2—Avulsion of the tubercle with TAL attached
 - Bony injury that may heal.
- Signs, symptoms, and physical exam
 - Typically does not lead to neurologic deficits.
 - With severe fractures, complete and incomplete injuries are possible, including medullary dysfunction.
- Neuroimaging
 - Radiographs
 - Open-mouth odontoid view to assess the relationship of lateral mass of C1 on C2
 - Spence rule¹⁵
 - Classic rule indicating that > 7 mm composite overhang is indicative of a TAL injury.
 - Newer studies have demonstrated little correlation between bony displacement and transverse ligament integrity.¹⁶
 - CT scan
 - Coronal and sagittal reconstruction can replace open-mouth radiographs
 - Important to look for tubercle avulsion to determine the healing potential of the TAL
 - CTA to rule out vertebral artery dissection/occlusion if suspected
 - MRI
 - Best method to assess the TAL¹⁶
- Treatment
 - Rigid cervical orthosis or halo
 - TAL intact
 - Dickman Type 2 TAL injury
 - C1–C2 fusion
 - Dickman Type 1 injuries
 - Persistent instability after nonoperative treatment
 - Clinical deformity (rotatory subluxation)
- Surgical pearls
 - Obtain a CTA or MRA to assess vertebral artery integrity as part of the preoperative planning if vessel injury suspected.
 - Vast majority of C1 fractures are amenable to nonoperative treatment.

19.3.3 Axis (C2) Fracture and Traumatic Spondylolisthesis of C2 (Hangman's Fracture)

- Background
 - Associated with high-velocity trauma.
 - C2 fractures are among the most common spinal fractures in the elderly.
 - Mechanism is a combination of hyperextension, compression, and rebound flexion (**Fig. 19.2**).

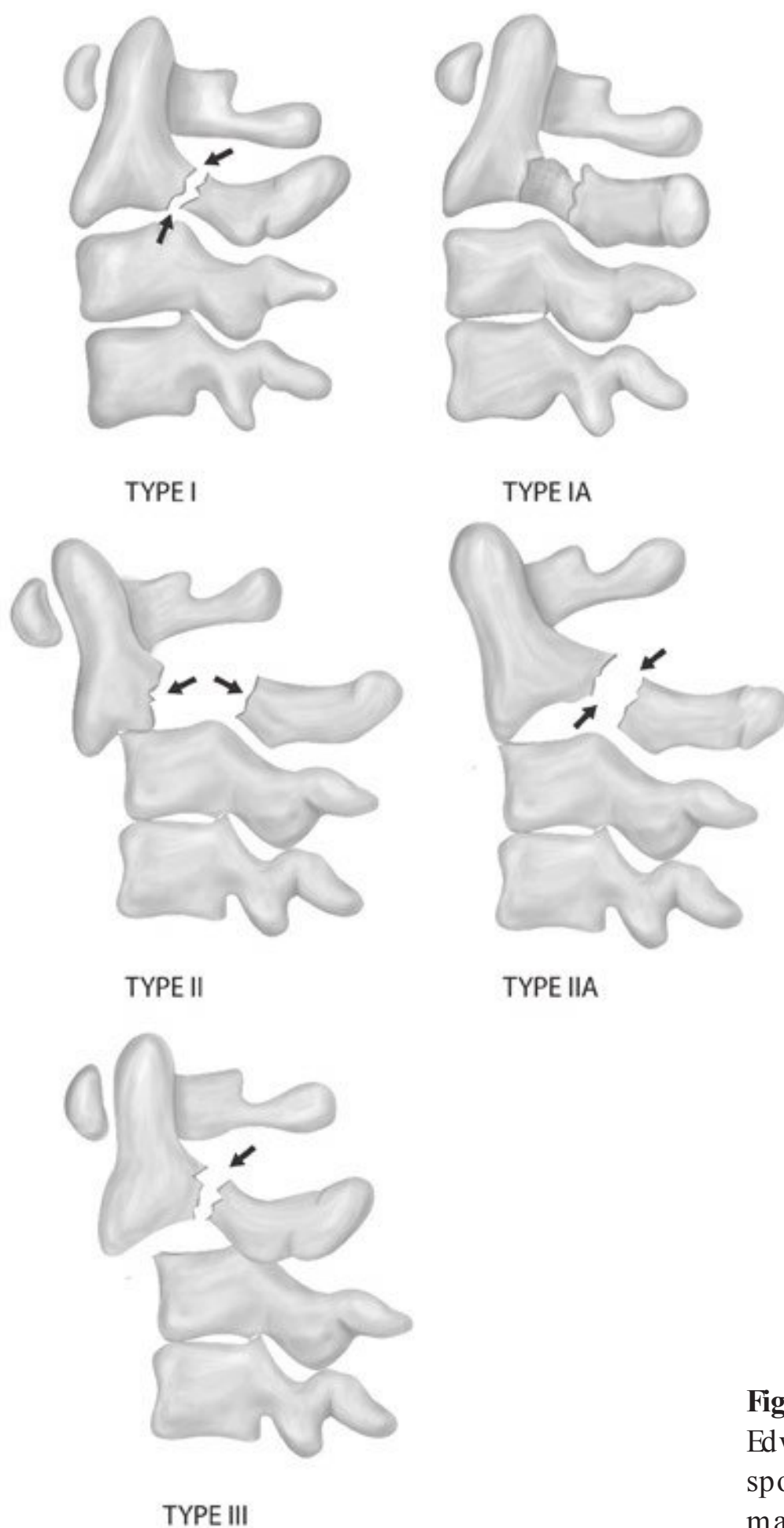


Fig. 19.2 The Effendi/Levine and Edwards classification of traumatic spondylolisthesis of the axis (hangman's fracture).

- Signs, symptoms, and physical exam
 - Patients typically present with neck pain only in nonangulated, nondisplaced fractures.
 - Complete or incomplete injuries are possible with severe fractures.
 - Cerebellar findings (nausea/vomiting, asymmetric exam, ataxia) may suggest vertebral artery injury.
- Neuroimaging
 - CT scan with reconstructions
 - CTA to rule out vertebral artery dissection/occlusion if suspected
- Classification/treatment¹⁷
 - Type I: Minimal displacement (< 3 mm) $\rightarrow$ hard cervical collar
 - Type II: Significant displacement (> 3 mm) and angulation $> 11^\circ$
 - Nonoperative—reduction and halo or collar immobilization
 - Operative
 - Disruption of the C2/C3 disk or an inability to maintain the reduction $\rightarrow$ may consider surgery
 - * If C2 pedicle screws placement is possible, an isolated C2–C3 fusion is preferred.
 - * If C2 pedicle screw is not possible, a C1–C3 posterior fusion or C2/C3 anterior cervical discectomy and fusion (ACDF) can be performed.
 - Type IIa: Minimal displacement (< 3 mm) but angulation $> 11^\circ$
 - Nonoperative—no traction
 - Halo
 - Operative
 - Disruption of the C2/C3 disk or inability to maintain the reduction
 - C2–C3 fusion preferred
 - Type III: Associated facet dislocation
 - All require either anterior C2–C3 or posterior C1–C3 fusion.
 - Atypical fracture¹⁸
 - Fracture extends into the vertebral body.
 - Displacement of the portion of the posterior vertebral body attached to pedicle often significantly narrows the spinal canal.
 - Increased risk of neurologic injury, specifically Brown-Sequard injury.
- Surgical pearls
 - Obtain a CTA or MRA to assess vertebral artery integrity as part of the preoperative planning if vessel injury is suspected.
 - Often a C2 pedicle screw can be placed, allowing for surgical treatment with only a C2–C3 fusion.

19.3.4 Dens Fracture (C2)

- Background
 - Among the most common spinal fracture injuries in the elderly after falls
 - Sometimes associated with simultaneous C1 fractures
 - Mechanism: Hyperflexion or hyperextension (**Fig. 19.3**)
- Signs, symptoms, and physical exam
 - Neck pain and tenderness on palpation are typical.
 - Usually does not cause neurologic deficits (due to the width of the canal in this region).
 - Severely angulated fractures may cause complete or incomplete injuries.

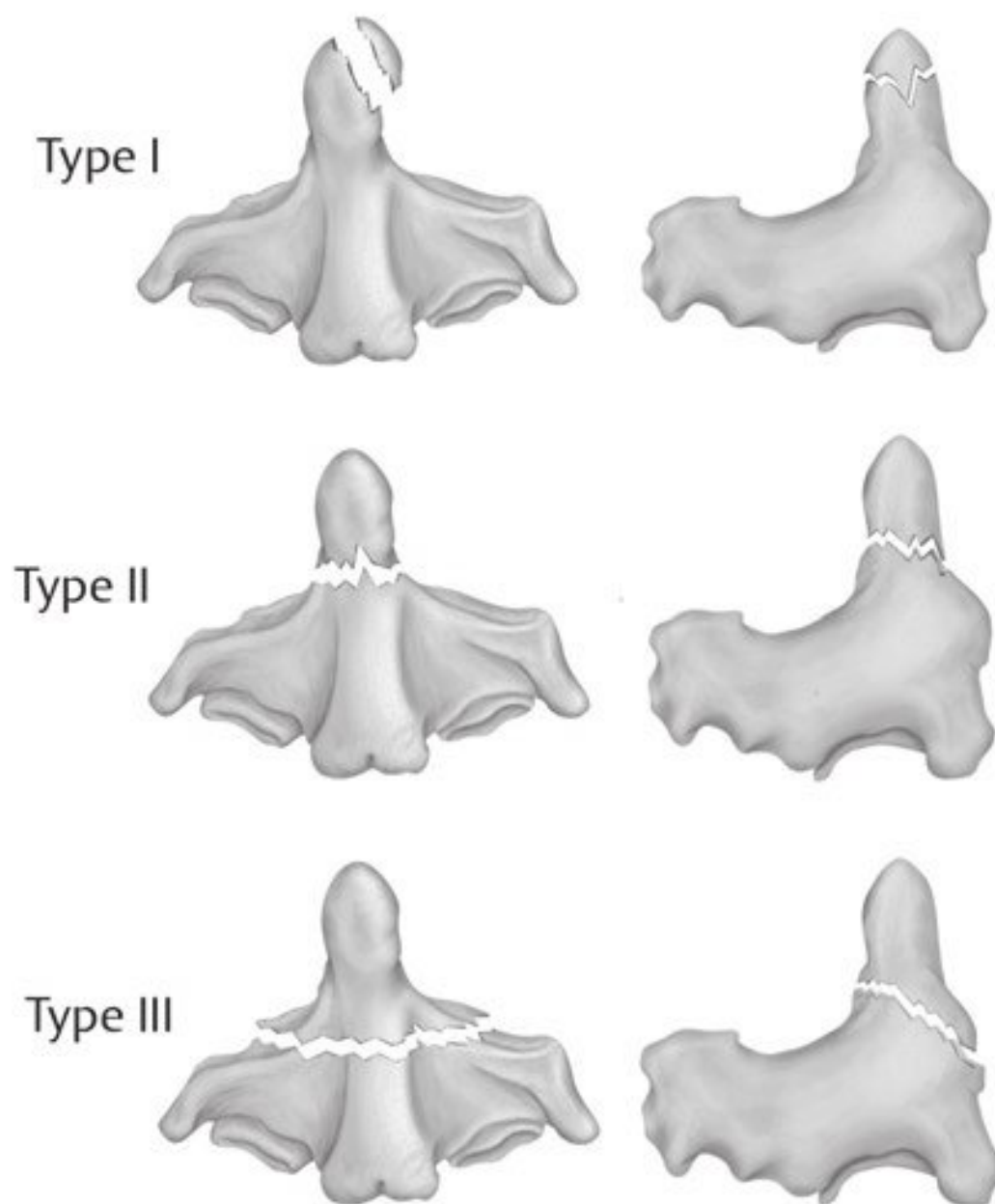


Fig. 19.3 Anderson D'Alonzo classification of dens fractures.

- Neuroimaging
 - CT scan with reconstructions is recommended even if fracture is evident on plain radiograph.
 - MRI may be helpful for assessing the integrity of cruciate ligament.
- Classification
 - Anderson and D'Alonzo¹⁹
 - Type I: Avulsion fracture at tip.
 - Type II: At the waist/base.
 - Type II: Extends into the C2 vertebral body.
- Treatment
 - Type I and type III
 - Hard collar or halo.
 - If significant ligament disruption associated with a type I injury, make sure there was not a spontaneously reduced atlantoaxial dissociation.
 - Type II
 - Controversial.
 - Risk factors for nonunion:²⁰
 - 5 mm of displacement, angulation $> 10^\circ$, age > 50 , and posterior displacement
 - A young patient having no nonunion risk factors can be treated in a halo or a hard collar.
 - Young patient with concern for nonunion:
 - Anterior odontoid screw or C1–C2 posterior fusion
 - Geriatric odontoid fracture:
 - If healthy enough to undergo general anesthesia, should undergo a C1–C2 fusion.^{5,6}
 - If possible, do not treat in a halo.
 - Increases the risk of dysphagia and death.
- Surgical pearls
 - Ensure that the anterior cruciate ligament is intact before attempting an anterior odontoid screw fixation.
 - Multiple posterior fusion options for a C1–C2 fusion have been described, and the authors routinely use a C1 lateral mass and a C2 pedicle screw.

19.3.5 Subaxial Cervical Spine Trauma (C3–C7)

- Background
 - Particularly common in high-speed motor crashes and diving accidents
 - Wide variety of injuries that can range from benign spinous process fracture to fractures associated with complete spinal cord injuries
- Classifications
 - Subaxial cervical spine injury classification system⁷

- Assigns values based on injury morphology, integrity of the diskoligamentous complex, and neurologic status of the patient.
- Patients with a score > 4 should undergo surgical treatment.
- Patients with a score < 4 should undergo a trial of nonoperative treatment.
 - Morphology
 - * Compression = 1
 - * Burst = 2
 - * Distraction = 3
 - * Rotation = 4
 - Diskoligamentous complex
 - * Intact = 0
 - * Indeterminate = 1
 - * Disrupted = 2
 - Neurologic status
 - * Intact = 0
 - * Root injury = 1
 - * Complete spinal cord injury = 2
 - * Incomplete spinal cord injury = 3
 - * Persistent spinal cord compression with a neurologic deficit = +1
- AOSpine Subaxial Cervical Spine Injury Classification System²¹
 - Three types of fractures:
 - A: Compression (A0—Minor fractures such as an isolated lamina fracture or spinous process fracture. A1—Compression fracture involving a single end plate and not the posterior vertebral wall. A2—Coronal split of the vertebral body involving both end plates, but not the posterior wall. A3—Burst fracture involving a single end plate and the posterior wall. A4—Burst fracture involving both end plates and the posterior wall.)
 - B: Tension band injury (B1—Bony injury only. B2—Posterior tension band injury including the ligamentous structures. B3—Anterior tension band injury.)
 - C: Translation injury (Any translation or displacement of one vertebral body in relationship to the adjacent level.)
 - Additional descriptors:
 - Facet fractures (F1—Nondisplaced fracture measuring $< 40\%$ of the height of the lateral mass and < 1 cm. F2—Displaced fracture, or a fracture measuring $> 40\%$ of the height of the lateral mass or > 1 cm. F3—Floating lateral mass. F4—Subluxed, perched, or dislocated facet. BL—Bilateral facet involvement.)
 - Neurology (N0—No neurologic injury. N1—Transient neurologic deficit. N2—Persistent radiculopathy. N3—Incomplete spinal cord injury. N4—Complete spinal cord injury. NX—Unable to evaluate neurologic status. +—Continued spinal cord compression.)

- Patient specific modifiers (M1—Injury to the posterior ligamentous complex. M2—Significant disk herniation. M3—Stiffening bone disease, such as ankylosing spondylitis or diffuse idiopathic skeletal hyperostosis. M4—Vertebral artery abnormality.)
- A spine injury score will eventually be established to allow the AO classification to guide treatment.
- Signs, symptoms, and physical exam
 - Wide variability in presentation
 - Mild compression fractures (A1 and A2) are only associated with axial neck pain.
 - Translational injuries are often associated with significant neurologic injuries.
- Neuroimaging
 - CT and MRI are typically necessary to appreciate the extent of injury.
 - CTA is recommended if any fracture involves the transverse foramen.
- Treatment
 - Based on stability and neurologic injury
 - Considerations
 - Is there mechanical instability?
 - Is there neurologic compromise necessitating direct or indirect decompression?
 - Are there patient factors (e.g., multitrauma)?
 - Stable: Nonoperative, rigid cervical orthosis
 - Unstable: Surgical fusion via anterior versus posterior versus combined approach
- Surgical pearls
 - Ongoing compression of the spinal cord is a surgical emergency.
 - Early decompression has been shown to increase the chance of neurologic recovery.¹
 - Facet fractures measuring $> 40\%$ of the height of the lateral mass or more than 1 cm in length from the tip of the superior articular process to the fracture line have a high incidence of failing nonoperative treatment.⁸
 - C7 burst fractures have a higher likelihood of falling into significant kyphosis than do burst fractures at other levels.
 - Ensure the posterior elements are intact when an anterior cervical vertebral body fracture is present. Posterior element involvement may indicate a highly unstable injury.

Common Clinical Questions

1. List these incomplete spinal cord injuries in the order of best prognosis to worst.
 - A. Brown-Sequard, anterior cord, central cord
 - B. Central cord, Brown-Sequard, anterior cord
 - C. Brown-Sequard, central cord, anterior cord
 - D. Anterior cord, central cord, Brown-Sequard
2. A 70-year-old female presents with significant neck pain after a fall. Imaging demonstrates a type II odontoid fracture with 3 mm of displacement. What treatment will lead to the lowest mortality and highest health-related quality of life?
 - A. Hard cervical collar
 - B. Halo
 - C. Posterior C1–C2 fusion
 - D. Traction and bed rest
3. A polytrauma patient is intubated with an unknown neurologic injury. What is the best modality for assessing acute spinal cord injury?
 - A. CT scan
 - B. T1-weighted MRI
 - C. T2-weighted MRI
 - D. Flexion and extension cervical radiographs

Answers to Common Clinical Questions

1. C.
2. C.
3. C.

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20 Thoracolumbar Trauma

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20.1 Key Points

- The thoracolumbar junction (T11–L2) is the most common site of traumatic spine injury in the thoracic and lumbar spine.
- Injury recognition is key in clinical decision-making and prevention of secondary neurologic injury.
- Surgery is indicated for spinal cord injury, acute or delayed mechanical instability, and intractable pain.

20.2 General Principles

- Field management
 - ABCs: **a**irway, **b**reathing, **c**irculation
 - Spinal immobilization on a rigid backboard with “log-roll” precautions
- Emergency room management
 - Repeat ABCs and perform secondary trauma assessment.
 - Standard imaging.
 - Treat neurogenic shock (not to be confused with spinal shock) if present.
 - Most common with cervical and high thoracic injuries.
 - Acute loss of background sympathetic tone, resulting in hypotension in the presence of bradycardia.
 - Manage with vasopressors and moderate fluid resuscitation.
- Physical exam
 - Sensorimotor exam of all dermatomes/myotomes
 - Reflexes (including bulbocavernosus)
 - Rectal tone and saddle anesthesia
 - Assess bladder function
- Imaging
 - Orthogonal plain films should be avoided for injury screening unless CT is unavailable.¹
 - CT:
 - Primary screening tool for suspicion of thoracolumbar injury
 - Must include axial, coronal, and sagittal reconstructions
 - Not useful for assessing ligamentous injury

- MRI:
 - Best for evaluation of spinal cord injury and diskoligamentous injury (such as with suspected chance injury of the disk)
- Degree of injury
 - Determination of complete versus incomplete injury is imperative.
 - Spinal shock must be excluded if injury appears complete initially.
 - American Spinal Injury Association (ASIA) scoring scale (**Table 20.1**).²
- Thoracolumbar Injury Classification and Severity Score (TLIC-SS)^{3,4} is a scoring system that maintains that surgical intervention is mainly based on three factors:
 - Morphology of the fracture
 - Posterior ligamentous complex
 - Neurologic status
- Based on the scoring system outlined in **Table 20.2**, a TLIC-SS ≤ 3 is usually a nonoperative candidate, whereas a score of 4 is considered the “gray zone” in which patients may be managed surgically or conservatively. On the other hand, a TLIC-SS ≥ 5 generally warrants surgical intervention.

20.3 Thoracolumbar Fractures

- General features:
 - The most common site of injury is at the thoracolumbar junction (T11–L2).

Table 20.1 ASIA Neurological Scoring System

Grade	Description
A	Complete: No motor or sensory function is preserved in the sacral segments.
B	Incomplete: Sensory but not motor function is preserved below the neurologic level and includes the sacral segments.
C	Incomplete: Motor function is preserved below the neurologic level, and more than half of the key muscles below the level have a muscle grade < 3 .
D	Incomplete: Motor function is preserved below the neurologic level, and at least half of the key muscle groups below the level have a muscle grade > 3 .
E	Normal.

Note: The caudalmost normal level is the single neurologic level.

Table 20.2 Thoracolumbar Injury Classification and Severity Scoring System

Category	Finding	Points
Radiographic findings	Compression fracture	1
	Burst component/lateral angulation > 15°	1
	Distraction injury	2
	Translational/rotational injury	3
Neurologic status	Intact	0
	Nerve root injury	2
	Complete SCI	2
	Incomplete SCI (including cauda equina)	3
Posterior ligamentous complex	Intact	0
	Unknown/cannot determine	2
	Disruption	3

Abbreviation: SCI, spinal cord injury.

- Attributed to the transition from the rigid thoracic spine to the mobile lumbar spine
- Thoracolumbar fractures can generally be classified as stable versus unstable based on the Denis 3 column theory:⁵
 - Anterior column includes the anterior half of vertebral body and disk, as well as the anterior longitudinal ligament.
 - Middle column comprises the posterior half of vertebral body and disk, the posterior longitudinal ligament, and the pedicles.
 - Posterior column includes the posterior bony arch as well as the posterior ligamentous complex (supraspinous and interspinous ligaments, ligamentum flavum, and facet joints).
- Most isolated anterior column fractures should be considered stable, except in the following cases:^{5,6,7}
 - Loss of > 50% vertebral body height or excessive kyphotic angulation at one segment
 - Three or more continuous compression fractures
 - Neurologic deficit
 - Progressive kyphotic deformity on serial imaging studies
- Middle column fractures are considered unstable except in the following cases:

- Fracture above T8 with intact ribs and sternum
- Below L4 if posterior elements intact
- Neurologic deficit
- Posterior column fractures are typically stable unless accompanied by middle column failure and/or bilateral facet fractures.
- There are six types of thoracolumbar fractures, according to the McAfee classification system, which Denis further simplifies into four categories (**Fig. 20.1**):⁵
 - (1) Compression fractures
 - Due to a compressive flexion injury.
 - Factors of instability: $> 50\%$ loss of vertebral body height, $> 30^\circ$ of anterior to posterior angulation, $> 30^\circ$ of focal kyphosis.
 - If any of the foregoing factors is present, posterior fusion or percutaneous fixation at least one level above and below the affected level(s) should be performed.
 - (2) Burst fractures
 - From axial compression.
 - Make up roughly 10 to 20% of all thoracolumbar fractures.⁸
 - Surgery is usually indicated, especially when presenting with neurologic deficit, angular deformity $> 20^\circ$, $> 50\%$ spinal canal compromise, anterior body height $< 50\%$ of posterior height, and/or progressive kyphosis.⁹

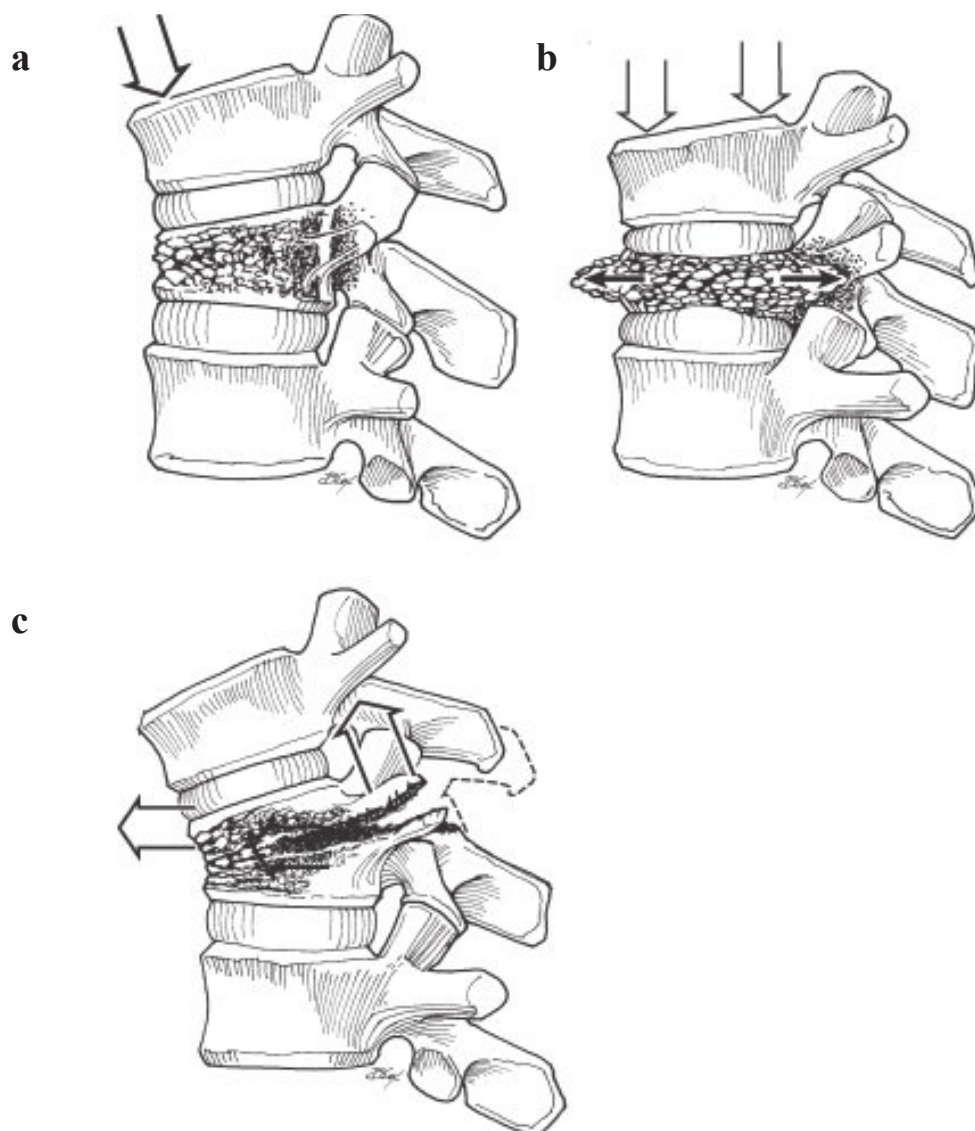


Fig. 20.1 (a–c) Compression: (a) compression fracture, (b) burst fracture, (c) chance-type fracture.

- However, in patients presenting with a burst fracture and no neurologic deficits, several series have shown little or no neurologic decline with conservative measures (i.e., orthotics), with ultimately no significant difference between operative versus conservative treatment arms.^{10,11}
- An anterior approach allows adequate decompression of any retropulsion, whereas a posterior approach allows reduction and restoration of sagittal balance. A combined approach could be warranted in setting of significant instability.
- Direct comparison of anterior/posterior approaches shows similar results regarding functional outcome and pain. Perioperatively, the anterior approach has been associated with greater intraoperative blood loss¹² but may result in fewer long-term complications and subsequent surgeries.¹³
- (3) Chance fractures (seatbelt injury)
 - Flexion–distraction injury that results in a horizontal fracture extending posteriorly through the vertebral body and into the posterior column (most often involving the pedicles).
 - A bony Chance fracture without ligamentous injury, with absence of disk injury, or with no dislocation may be treated with a hyperextension brace, such as a thoracolumbar sacral orthosis (TLSO).
 - Posterior ligamentous disruption or extension through the disk usually requires posterior fusion due to instability.
 - Rule out intra-abdominal or pelvic injury.
- (4) Fracture–dislocation
 - Due to rotation, shear, compression, or tension causing three-column failure.
 - May result in severe subluxation or dislocation.
 - There are three main subtypes:⁵
 - Flexion rotation: Anterior compression and total rupture of the middle and posterior columns under rotational and tension forces. Imaging finding include subluxation or dislocation, increased interspinous distance, decreased canal diameter, jumped facets, and an intact posterior vertebral body.
 - Shear: All three columns, including anterior longitudinal ligament, disrupted; force directed most commonly in posteroanterior direction fracturing the posterior arch, resulting in “free-floating lamina” and the superior facet of the inferior vertebra. Often results in complete spinal cord injury (SCI).
 - Flexion distraction: Radiographically similar to a Chance-type fracture ± subluxation due to torn annulus fibrosus.
 - Treatment with urgent decompression and stabilization is indicated in cases with incomplete neurologic deficit, > 50% loss of height with angulation and kyphotic angulation > 40%

- Other classification systems for thoracolumbar injury include the following:
 - AO-Magerl classification system¹⁴
 - Separated into types A (compression injuries), B (distraction injuries), and C (rotational injuries) based on pathomorphological characteristics
 - McAfee classification system¹⁵
 - Uses CT primarily to group fractures according to the type of failure seen within the middle column, which includes axial compression, axial distraction, and translation within the transverse plane
 - Load-sharing classification¹⁶
 - Based on amount of comminution within the vertebral body, degree of fracture fragment displacement, and amount of kyphotic correction on later plain films

20.4 Bracing

- High thoracic fractures
 - Stable fractures between T1 to approximately T5 may be treated with a cervicothoracic orthosis (CTO), for a TLSO does not extend far enough superiorly to provide adequate support.
- Midthoracic to lumbosacral fractures
 - Stable fractures anywhere between the midthoracic spine to L5 may be treated with a hard TLSO.
 - In most cases of acute lumbosacral trauma, a soft lumbosacral orthosis (LSO) brace does not provide adequate stabilization. A TLSO remains the best option for external stabilization.
- Always assess stability of fracture in brace with upright X-rays in brace.

20.5 Conus Medullaris and Cauda Equina Syndromes

- Conus medullaris syndrome
 - Often presents suddenly and bilaterally
 - Is associated more often with axial back pain than with radicular pain
 - Features perianal numbness
 - Features overflow incontinence
- Cauda equina syndrome
 - Results from injury to the cauda equina below the L2 level
 - Is associated with acute weakness either unilaterally or bilaterally, perianal or saddle anesthesia, and urinary retention
- Both syndromes require urgent decompression ± fusion.

20.6 Spinal and Neurogenic Shock

- Speedy recognition and differentiation is crucial for acute stabilization and surgical decision-making
- Spinal shock
 - A temporary physiologic loss of all sensorimotor function caudal to the level of injury occurring virtually at the time of insult. Autonomic dysfunctions (i.e., loss of sympathetic tone) may occur with gradual return.
 - Mechanism is yet to be fully elucidated but may be due to displacement of K⁺ into the extracellular space, thus reducing axonal transmission. As K⁺ levels normalize, the manifestations of spinal shock slowly improve.
 - Separated into four clinical phases spanning from immediately postinjury up to months after:¹⁷
 - Phase I (days 0–1): Areflexia, flaccid paralysis, anesthesia, and loss of bowel/bladder function. The bulbocavernosus reflex is typically one of the first reflexes to return and should be tested in every patient for distinguishing complete versus incomplete SCI.*
 - Phase II (days 1–3): Initial reflex return; gradual recovery of DTRs.
 - Phase III (day 4–1 mo): Continued return of DTRs with some early hyperreflexia.
 - Phase IV (1–12 mo): Spasticity and hyperreflexia with overactivity of most cutaneous and deep tendon reflexes. Detrusor paralysis is estimated to return between 4 and 6 weeks.
 - Absence of the bulbocavernosus reflex acutely after injury does not rule out incomplete SCI due to the potential presence of spinal shock.
 - Manage conservatively with observation of return of function.
- Neurogenic shock
 - A neurological emergency resulting in loss of sympathetic tone producing unopposed vagal activity after SCI not caused by any other traumatic source.
 - Results in hypotension, bradycardia, and occasionally priapism in males.
 - Estimated to occur in up to 20% of isolated cervical injuries and 10% of isolated thoracolumbar injuries with an Abbreviated Injury Score > 2.¹⁸
 - Should be treated similar to other forms of shock. IV fluids, vasopressors (e.g., phenylephrine) are warranted to achieve hemodynamic stability and prevent end-organ failure.

Common Clinical Questions

1. A 9-year-old boy was a restrained passenger in a high-speed motor vehicle accident. He presents without obvious neurological deficits, but a CT scan demonstrates a significant flexion–distraction injury at the thoracolumbar junction. Which of the following should be assessed radiographically before spinal surgical intervention is considered?

- A. The adjacent vasculature looking for potential aortic injury
 - B. The adjacent visceral organs including pancreas, spleen and bowel
 - C. The chest cavity looking for pneumo- or hemothorax
 - D. The occipital cervical junction looking for potential atlantooccipital dislocation
 - E. All of the above
2. A 60-year-old female with a remote history of breast cancer presents with back pain and L1 vertebral body fracture on CT. She has no neurological deficits. She denies trauma. Which of the following is an appropriate management strategy?
 - A. Reassurance, discharge, and follow-up as needed
 - B. Application of TLSO brace and follow-up in 6 to 8 weeks in clinic
 - C. Scheduling an elective L1 biopsy and cement augmentation of the vertebral body
 - D. Obtaining an MRI to rule out pathologic fracture

Answers to Common Clinical Questions

1. E.
2. D.

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21 Sacropelvic Trauma

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21.1 Key Points

- Sacral fractures often are associated with high-energy trauma and can cause pelvic ring instability.
- Highly displaced and multiplanar fractures can result in spinopelvic instability or dissociation.
- A thin cut CT scan of the sacrum with coronal and sagittal reconstructions is the imaging modality of choice for the diagnosis of high-energy traumatic sacral fractures.
- Treatment can range from observation to a variety of surgical approaches aimed at stabilizing the pelvis and sacrum.

21.2 Sacral Fractures

- Background
 - The sacrum connects the pelvis to the lumbar spine and helps dissipate the weight of the axial skeleton to the ground via the pelvis and the lower limbs.
 - Sacral fractures typically are the result of high-energy trauma but can also be observed in elderly patients who have osteoporotic bone after a low-energy mechanism such as a fall.
 - Fewer than 10% of sacral fractures occur in isolation, and they often are associated with other injuries to the pelvic ring.
 - Denis et al¹ classified sacral fractures into three types based on the relationship of the fracture line to the sacral neuroforamen (**Fig. 21.1**).
 - Zone 1: Fractures are lateral to the neural foramen; Zone 2: Fractures pass through the neural foramen; Zone 3: Fractures are medial to the neural foramen and involve the spinal canal.
 - Roy-Camille² further classified Denis zone 3 injuries based on the mechanism of injury and degree of displacement.
 - Furthermore, vertical stresses cause bilateral intraforaminal fractures, which can cause the sacrum to become dissociated from the pelvic ring. A horizontal fracture then develops at the weakest portion of the sacrum, the S1 to S2 junction. This results in spinopelvic dissociation.³
- Signs, symptoms, and physical examination
 - Sacral injury should be suspected in any patient who has sacrococcygeal pain.

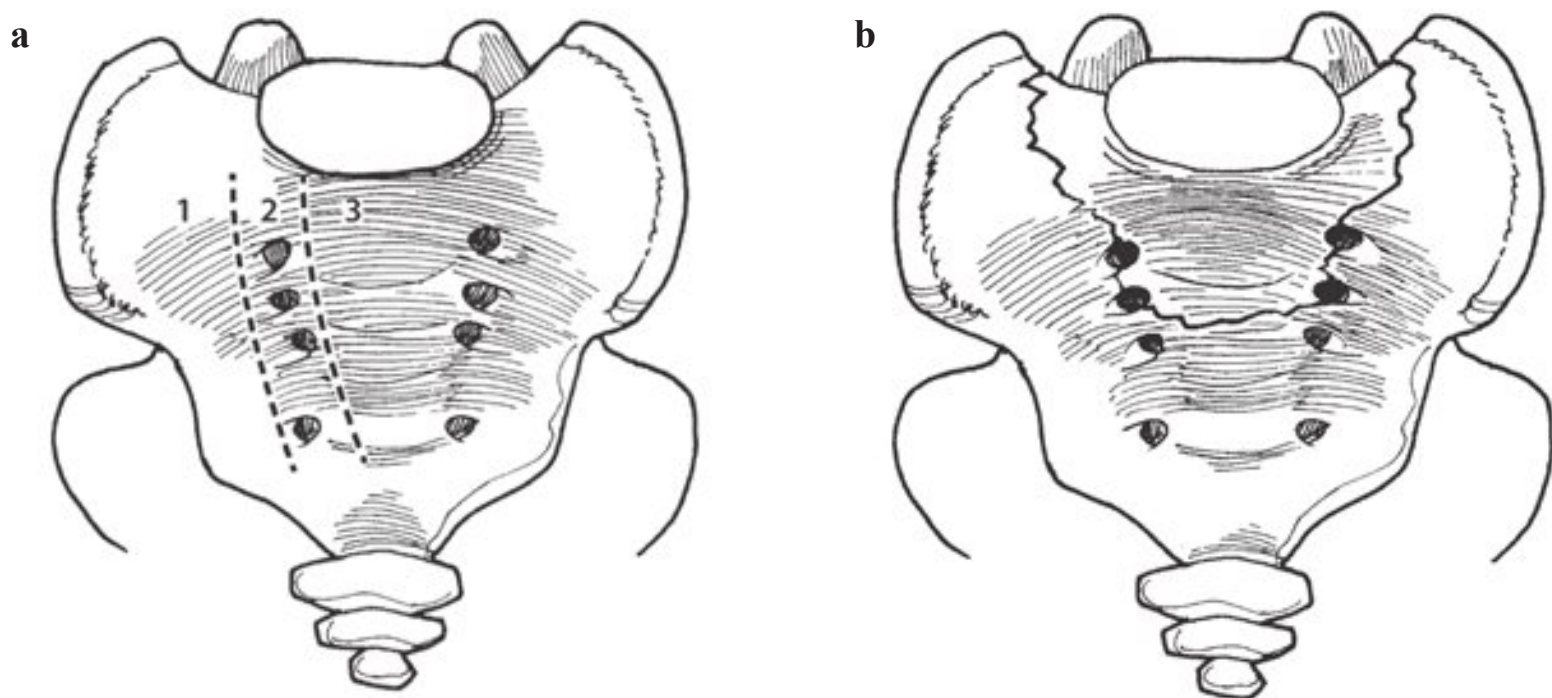


Fig. 21.1 Sacral fractures according to the Denis classification. **(a)** Zones 1, 2, and 3. **(b)** U-shaped fracture.

- Thorough evaluation of the pelvis should be conducted to assess stability of the pelvic ring. This can be done by applying gentle rotational pressure to the iliac wings.
- In patients who have lacerations, ecchymoses, tenderness, swelling, or crepitus around the pelvic ring, the presence of sacropelvic injury should be suspected.
- Evaluation for a sacropelvic degloving, or Morel-Lavelle lesion, should be conducted. This usually manifests as a palpable subcutaneous fluid mass.
- Digital rectal examination should be performed to evaluate for occult open fractures that might have involved the rectum. Likewise, lacerations in the perineal region should be investigated to ensure that an open fracture is not missed.
- Evaluation of the remainder of the spine should be conducted to assess for noncontiguous injuries.
- A detailed and thorough neurological examination, including motor and sensory assessments, should be performed.
- Work-up
 - Standard Advanced Trauma Life Support (ATLS) protocol should be adhered to at the time of initial presentation.
 - All life-threatening injuries should be addressed and the patient adequately resuscitated.
 - If provisional stabilization of the pelvic ring is necessary, placement of a pelvic binder, a pelvic sheet wrapping, or an external fixator can be performed. The principles of damage control orthopaedics—immediate and temporary stabilization with later definitive treatment—should be adhered to.
- Neuroimaging

- The ATLS protocol for imaging a suspected sacral fracture includes obtaining an anteroposterior (AP) view of the pelvis.
- The AP view, however, is not always optimal for visualizing the fractures secondary to the inclination of the pelvis and overlying bowel gas. Approximately 30% of all sacral fractures can be identified from a single AP view of the pelvis.
- Inlet and outlet views of the pelvis are recommended to improve visualization and constitute a complete radiographic series in patients who have suspected pelvic injuries.
- Treatment
 - Nonoperative treatment is an option for patients who are neurologically intact and who have minimally displaced or angulated fractures. These usually are zone 1 or 2 injuries in conjunction with a stable pelvic fracture.⁴
 - Operative techniques include sacral screws, lumboiliac instrumentation, and a combination of the two, known as *triangular osteosynthesis*.
 - Iliosacral screws can be placed by open or percutaneous techniques.
 - Lumboiliac fixation offers advantages such as familiar anatomy, prevention of kyphotic deformity, and possibility of concomitant neural decompression, if necessary.
 - Lumbopelvic fixation can be performed through a percutaneous or mini-open approach for fractures that do not require open reduction. These approaches have the benefit of minimal blood loss and minimal soft tissue dissection.⁵
 - Sacropelvic fixation using screws placed through the S2–alar–iliac pathway has been described. The pathway is defined using intraoperative fluoroscopy and carries the benefit of a minimally invasive approach and the ease of attaching the screws to the spinal portion of the construct.⁶
- Surgical pearls
 - Open surgical reduction and fixation continues to be recommended for sacral fractures involving the lumbosacral junction and for sacral fractures that require urgent neural decompression.
 - In cases of sacral fracture involving vertical displacement, sacroiliac screws are prone to loosening and can result in loss of reduction.
 - Appropriate intraoperative visualization of the necessary anatomic landmarks is crucial to performing percutaneous fixation.

21.3 Transverse Sacral Fractures

- Background
 - 45% of traumatic fractures of the pelvic ring transverse the sacrum.¹
 - However, transverse sacral fractures represent only 3 to 5% of all sacral fractures.⁷

- This injury was initially termed the *suicidal jumper's fracture* because the mechanism of injury includes fall from height. Other mechanisms of injury include motor vehicle collisions, blast injuries, and rush injuries.
- As a result of potential injury to the sacral nerve roots, transverse sacral fractures often are associated with devastating complications such as bowel and bladder dysfunction, sexual dysfunction, sensory deficits, and lower motor weakness.
- Signs, symptoms, and physical examination
 - The presence of root transection with a transverse sacral fracture is 35% so a thorough neurological examination, rectal examination, and light touch and pinprick sensory examination must be performed.
- Work-up
 - Adherence to standard ATLS protocols and treatment
 - Thorough and complete neurological examination, including motor testing, sensation to light touch and pinprick, rectal examination, and careful evaluation of the skin for lacerations, ecchymosis, and evidence of open fractures
- Neuroimaging
 - Radiographs, including standard pelvic views, and CT scans of pelvis and spine with reformatted coronal and sagittal view images
 - MRI when clinically indicated
- Treatment
 - Most stable sacral fractures can be treated nonoperatively, but those involving a pelvic ring disruption or spinopelvic instability should be stabilized.
 - Historically, surgical treatment consisted of only sacral laminectomy in an effort to address and treat sacral nerve root injuries.
 - New techniques have been developed to internally stabilize the fracture.
 - Goal of treatment is pain relief, stabilization, and earlier mobilization. Depending on the injury, neurological recovery might not be possible, although a partial return of function often is achievable.
 - The rare nature of the injury has limited published treatment strategies to case reports and case series.
 - Posterior pelvic ring fixation in the form of percutaneous iliosacral screws, sacral plating, or transiliac bars is an effective treatment strategy for transverse sacral fractures. However, if associated spinopelvic instability is present, spinopelvic fixation should be applied that stabilizes the lumbar spine to the ilium.
- Surgical pearls
 - Appropriate preoperative imaging is crucial in establishing and understanding the fracture pattern. Furthermore, adequate intraoperative radiographic visualization of the bony landmarks is essential for safe placement of iliosacral screws and spinopelvic fixation.

21.4 U-Shaped Sacral Fractures

- Background
 - U-shaped sacral fractures can be thought of as a subdivision of transverse sacral fracture and are even rarer.^{8,9}
 - The injury is characterized by a transverse sacral fracture in conjunction with bilateral sacral fracture–dislocations. This results in a mechanical dissociation of the upper sacrum and spine from the pelvis.
 - U-shaped sacral fractures were classified into three types by Roy-Camille: Type 1, flexion fracture with upper fracture fragment dislocated anteriorly; type 2, flexion fracture with upper fragment dislocated posteriorly; and type 3, extension fracture with upper fragment vertical and displaced anteroinferiorly.
 - Strange-Vognsen and Lebech¹⁰ added a fourth subdivision: With complete comminution of the upper sacrum but without a distinct transverse fracture or displacement.
 - The injury results from a high-energy force with substantial axial load (e.g., fall or jump from height, motor vehicle collision, and crush mechanism).
 - A substantial amount of associated soft tissue damage, hemorrhage, and other orthopaedic and systemic injuries often is present; 96.8% of patients who have U-shaped sacral fractures have associated injuries.
 - A high mortality rate is associated with these injuries, and the injury can be missed or underdiagnosed because of the severity of the patient's other injuries.
 - Sacral nerve root injuries caused by comminuted fracture fragments or displacement can present as incomplete monoradiculopathy to a complete cauda equina syndrome.
- Signs, symptoms, and physical examination
 - The diagnosis might be missed or delayed because of other serious distracting injuries or difficulty visualizing the sacrum.
 - Sacral injury should be suspected in any patient who has sacrococcygeal pain.
 - The standard ATLS protocol should be followed at the time of initial presentation.
 - Considering that the patient might be critically ill at the time of initial presentation, secondary surveys should be obtained as the patient's condition stabilizes and improves.
 - The most common neurological deficit associated with U-shaped fractures is cauda equina dysfunction secondary to injury to the lower nerve roots. This will cause impaired bladder function, decreased rectal tone, and saddle anesthesia, sometimes in the absence of any lower extremity dysfunction.

- The L5 and S1 nerve roots can also be damaged, with the damage manifesting as foot drop.
- Detailed neurological examination, including rectal examination and assessment of light touch and pinprick sensation, must also be performed.
- Work-up
 - Radiographic clues that suggest a U-shaped sacral fracture include bilateral transforaminal sacral fractures, irregularity of the superior sacral foraminal lines, and L5 transverse process fractures.
 - Angulation of the fractured segment can also produce a pseudo inlet view on a standard AP pelvis view.
- Neuroimaging
 - A lateral-view sacral radiograph is useful when evaluating U-shaped sacral fractures, because the transverse fracture line often is difficult to appreciate.
 - CT of the spine and pelvis is necessary to further define the injury and for operative planning. Coronal and sagittal reformatting are used to assess fracture angulation, translation, canal compromise, and the fracture pattern.
 - MRI is obtained when patients have a known sacral fracture and neurological deficit.
- Treatment
 - Nonoperative treatment options previously included bed rest with skeletal traction or immobilization using a cast or orthoses.
 - U-shaped sacral fractures typically result in spinopelvic instability and are thus not typically treated nonoperatively.
 - Definitive treatment of this complex injury occurs only after all other life-threatening injuries have been addressed and the patient adequately resuscitated.
 - The principles of damage control orthopaedics should be adhered to; temporary stabilization of the pelvic ring might be necessary.
 - If signs or symptoms of neurological compromise are present, decompressive maneuvers, either indirectly via fracture reduction or directly with a laminectomy or focal foraminotomy, should be performed.
 - Percutaneous iliosacral screws are best used to treat a U-shaped fracture that is not comminuted and that is minimally displaced.
 - Lumbopelvic fixation is best suited to treat highly unstable fractures having a substantial rotational component. The technique can be combined with iliosacral screws or a transiliac plate to form triangular osteosynthesis. This is the biomechanically superior method.

- Surgical pearls
 - If a preoperative CT scan reveals bone fragments in the foramen, sacral nerve root decompression should be performed before fracture reduction so that the fragments do not injure the roots during a reduction maneuver.
 - The sagittal relationship of the spine to the sacrum and the ilium should be reviewed preoperatively, considering that this relationship can be difficult to interpret intraoperatively because of the lack of intact bone at the sacral ala.
 - A true lateral view of the pelvis, with overlap of the sciatic notches, is necessary for the safe placement of iliac screws. The obturator–inlet and obturator–outlet views are used to determine the screw starting point and final position.
 - A combined obturator oblique outlet view, with $\sim 45^\circ$ caudal and 45° anterolateral angulation of the radiographic beam, produces overlap of the anterior–inferior iliac spine and the posterior–superior iliac spine and is also known as the *teardrop view*.
 - If adequate intraoperative visualization is not feasible, direct palpation of the sciatic notch might be necessary to ensure safe screw placement.

21.5 Traumatic Spinopelvic Dissociations

- Background
 - Separate and distinct entity from lumbosacral fracture–dislocations or bilateral sacroiliac joint dislocations.
 - Spinopelvic dissociation occurs only in **H**-, **U**-, **II**-, **T**-, and **Y**-shaped sacral fracture patterns.^{8,9}
- Signs, symptoms, and physical examination
 - Initial care of the patient adheres to the ATLS protocols; a pelvic compression device is sometimes necessary.
 - If the patient is suffering from an active pelvic hemorrhage as a result of the injury, and if the hemorrhage cannot be adequately controlled with pelvic compression, fixation by a pelvic clamp, laparotomy, and pelvic packing should be performed, as has been described in damage control algorithms. Radiographic coiling in interventional radiology might also be necessary.
- Work-up
 - Considering that spinopelvic dissociations usually are associated with other life-threatening injuries, the majority of patients are intubated and sedated. This makes a thorough and adequate neurological examination extremely difficult.

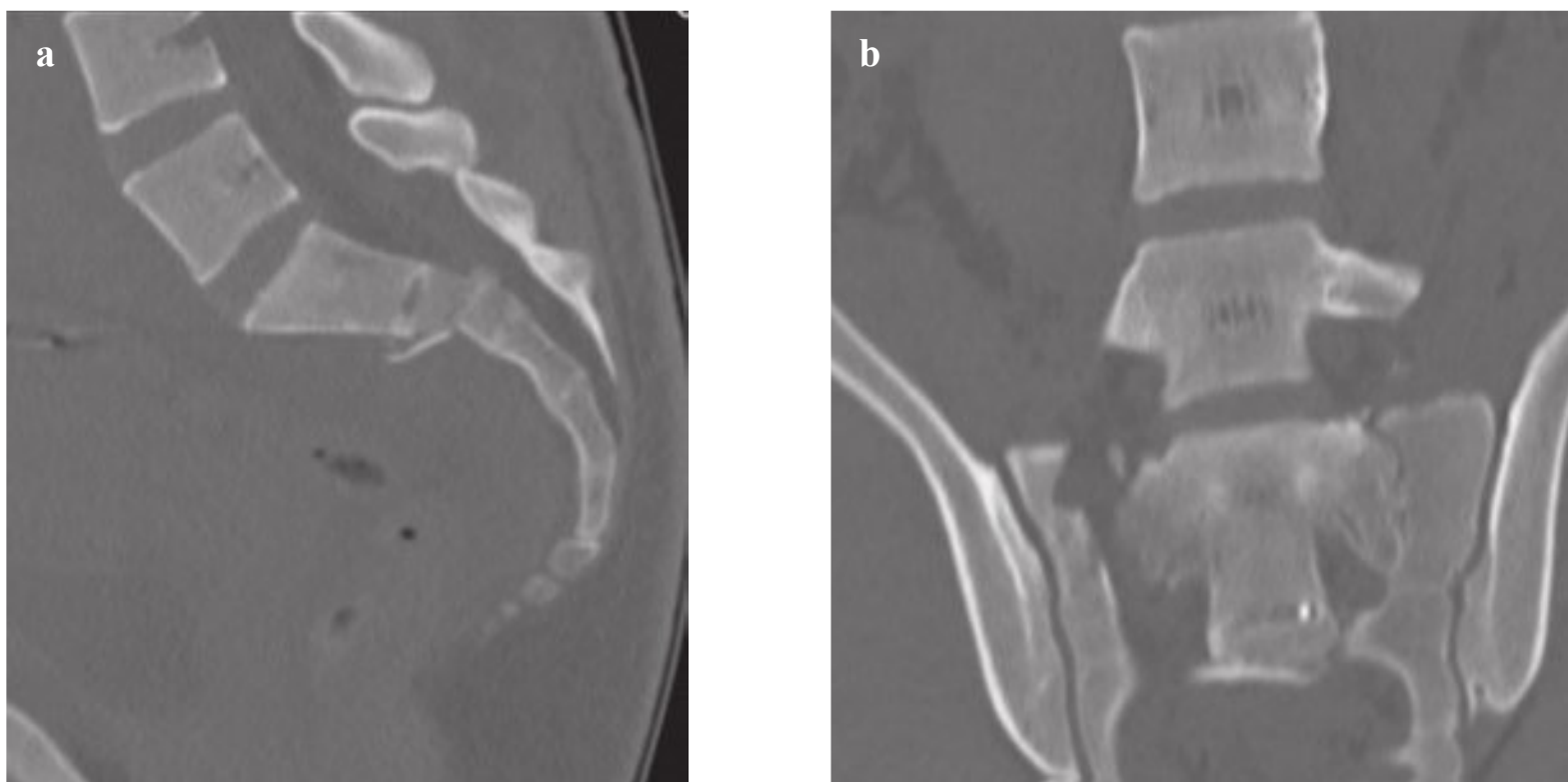


Fig. 21.2 (a) Computed tomographic sagittal reconstruction shows H-type sacral fracture. (b) Computed tomographic coronal reconstruction shows H-type sacral fracture.

- Evaluation for ecchymosis, lacerations, and Morel-Lavelle lesions are essential components of the workup.
- Neuroimaging
 - A CT scan, often a whole-body scan, is almost universally warranted and is mandatory for these patients.
- Treatment
 - Triangular stabilization with spinopelvic fixation and iliosacral screws is the most stable method of fixation and is used for **U**- and **H**-shaped fractures (**Fig. 21.2**).
 - For **II**-shaped fractures, triangular osteosynthesis can be used, but because this fracture is not as rotationally unstable, bilateral iliosacral screws or anterior tension plating might be sufficient.
 - **T**- and **Y**-shaped fractures can also be addressed by anterior stabilization using plates and screws.
- Surgical pearls
 - Approach is based on type of fracture and method of stabilization.
 - For direct manipulation of the sacrum to achieve reduction, a Schanz pin can be inserted into S1. Unilateral or bilateral femoral traction can also be helpful.

Common Clinical Questions

1. According to the Denis classification of sacral fractures, which type is most likely to be associated with neural injury?
2. Indications for surgical stabilization of sacral fractures include which of the following?
 - A. Zone 1 fracture with > 2 cm displacement
 - B. Minimally displaced transverse sacral fracture at S4
 - C. Lumbosacral dissociation
 - D. Nondisplaced Zone 2 fracture without neurological symptoms
 - E. A and C

Answers to Common Clinical Questions

1. Zone 3 (central sacral canal fractures).
2. E.

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22 Infection

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22.1 Key Points

- Swift and accurate diagnosis of spinal infections is necessary to prevent structural instability or neurologic compromise.
- Hematogenous spinal infections develop from the seeding of the cartilaginous end plates, which allows pathogens to propagate within the avascular disk space before spreading to the adjacent vertebral bodies.
- Granulomatous infections may be caused by one of several fungal, bacterial, and spirochete pathogens that generally give rise to an indolent clinical course.
- Postoperative infections most often develop secondary to direct inoculation of the wound with skin flora and are characterized by pain and tenderness at the surgical site.

22.2 Pyogenic Vertebral Osteomyelitis and Diskitis^{1,2,3,4}

- Background
 - Vertebral osteomyelitis accounts for ~ 1% of all skeletal infections.
 - Diskitis typically arises as a result of hematogenous spread such that the pathogens emanate from the vascular end plates into the avascular disk space before disseminating to the adjacent vertebral bodies.
 - Pyogenic infections most frequently involve the lumbar spine (58%), followed by the thoracic (30%) and cervical (11%) regions.
 - Gram-positive organisms such as *Staphylococcus aureus* and *Streptococcus* species are the most commonly isolated organisms (67 and 24% of cases, respectively).
- Signs, symptoms, and physical examination
 - The most prevalent signs and symptoms are axial pain (86%) and fever (60%).
 - Neurologic changes such as radicular numbness and muscle weakness may be present in as many as a third of patients.
 - Patients should be questioned regarding any ongoing constitutional symptoms, travel history, or recent procedures that may be suggestive of a diagnosis of infection.

- Work-up
 - White blood cell count (WBC) may be normal in up to 50% of patients.
 - Erythrocyte sedimentation rate (ESR) is more sensitive but is relatively nonspecific for infection.
 - C-reactive protein (CRP) is elevated in at least 90% of patients with spinal infections and is a reliable indicator of the disease course because it tends to rise acutely with the onset of an infection but returns to baseline rapidly as it resolves.
 - Blood cultures may be useful because they have been shown to reveal the causative pathogen in up to 58% of hematogenous spinal infections.
 - Urinalysis and culture with sensitivities should also be obtained to rule out an infection of the urinary tract, which may spread to the spine.
- Neuroimaging
 - Plain radiographs are the standard form of initial imaging study for suspected cases of spinal infections and will frequently exhibit abnormalities (89% of cases).
 - CT may display early pathologic changes within the spinal column as well as any fluid collections in the surrounding soft tissues.
 - MRI is the ideal diagnostic modality for identifying pyogenic diskitis.
 - Edema and fluid may be evident within the disks and adjacent tissues on T2-weighted images.
 - Addition of gadolinium contrast can enhance the visualization of paraspinal and epidural enhancement suggestive of active infection.
 - Technetium-99m/gallium-76 citrate bone scans or indium-111 tagged WBC studies are extremely sensitive for diagnosing spinal infection but are less specific.
- Treatment
 - First-line treatment for pyogenic infections is administration of broad-spectrum intravenous (IV) antibiotics for at least 6 to 8 weeks, until culture-specific regimens may be initiated.
 - Identify the pathogen with biopsy, blood, or tissue cultures before initiating treatment.
 - Immobilization may be beneficial for reducing pain and stabilizing the spine.
 - Surgery may be warranted if appropriate medical management fails or if the patient develops neurologic deterioration or spinal instability/deformity.
 - The goals of surgery include debridement of the infected tissue, decompression of the neural structures, and stabilization of the spine.
- Surgical pearls
 - Vertebral osteomyelitis and diskitis frequently affect the spinal column and may require an anterior procedure (**Fig. 22.1**).



Fig. 22.1 Cervical diskitis/epidural abscess with end plate involvement, relative sparing of the vertebral body, and epidural collection compromising the cervical canal.

- Posterior stabilization may also be necessary in instances in which there is significant instability or deformity.
 - Avoid stainless steel implants in these cases, for there is a tendency for bacteria to form a biofilm on them.
 - Titanium implants are preferred because they don't promote bacterial biofilm colonization.
- Autogenous bone is an excellent graft material for fusion in an infected surgical field, although allograft and metal may also be reasonable options for select cases.

22.3 Granulomatous Infections^{1,2,4,5}

- Background
 - Granulomatous infections caused by certain bacteria, fungi, and spirochetes are far less common than pyogenic infections in the United States.

-
- Spinal inoculation generally is done within the peridiskal metaphysis of the vertebral body adjacent to the end plate.
 - An inflammatory response produces a granuloma with a caseous abscess.
 - The infection propagates along the anterior longitudinal ligament to encompass contiguous levels.
 - The most common pathogen is *Mycobacterium tuberculosis*, an atypical bacteria that is more prevalent in underdeveloped nations but that is becoming more common in the United States.
 - Pott described the first surgical drainage of a tuberculous abscess in 1779, and his name has become synonymous with spinal disease.
 - Neurologic deficits may be observed in up to 47% of affected individuals.
 - The spine is the most common site of skeletal involvement (1% of all patients and nearly 50% of those who have musculoskeletal manifestations).
 - Fungal species (e.g., *Aspergillus*, *Blastomyces*, *Coccidioides*, *Cryptococcus*, and *Histoplasma*), spirochetes (e.g., *Actinomyces israelii* and *Treponema pallidum*), and parasites are more unusual causes of granulomatous spinal infections.
 - Signs, symptoms, and physical examination
 - Granulomatous spinal infections are characterized by a prolonged duration (i.e., months to years) of symptoms, including back pain and constitutional complaints.
 - The thoracic spine is the most common region of the spine for infection, which may give rise to significant kyphotic deformities.
 - Neurologic compromise and paraplegia may develop more frequently with tuberculous infections than with pyogenic osteomyelitis owing to their predilection for the posterior elements.
 - Work-up
 - All patients should be assessed with a tuberculin purified protein derivative (PPD) skin test.
 - ESR, CRP, WBC, and blood cultures may be less informative in these cases.
 - Sputum should be acquired for acidfast bacilli (AFB) and fungal cultures.
 - Neuroimaging
 - Individuals suspected of having tuberculous infections should be assessed using a chest X-ray, which may demonstrate pulmonary disease as well as any extensive bony lesions or focal kyphosis.
 - MRI is the imaging modality of choice and will often show destruction of the vertebral bodies, with relative sparing of the intervertebral disks.
 - Treatment
 - Pathogen-directed antimicrobial therapy:

- Six to 12 months of a multidrug treatment, including isoniazid, rifampin, pyrazinamide, and streptomycin or ethambutol is the standard regimen for *Mycobacterium tuberculosis*.
- Antifungals such as amphotericin B or ketoconazole are employed for culture-documented fungal disease.
- Surgical intervention is not routinely performed for these infections unless there is a failure of pharmacotherapy, progression of deformity, instability, or neurologic decline.
- Surgical pearls
 - These types of infections may bring about considerable destruction of the vertebral bodies, which may require reconstruction of the anterior column.
 - Supplementary posterior fixation may also be indicated to minimize the risk of developing a postoperative deformity.
 - Colonization of metallic implants rarely occurs with granulomatous infections.
 - Avoid stainless steel implants.
 - Postoperative imaging with CT and MRI is easier with titanium implants (decreased artifact compared with stainless steel implants).

22.4 Epidural Infections^{6,7}

- Background
 - Abscesses most often arise from adjacent vertebral osteomyelitis/diskitis but may also develop from hematogenous extension or direct inoculation from spinal procedures.
 - *Staphylococcus aureus* is the most common pathologic organism.
 - Most studies highlight the importance of early diagnosis and surgical intervention in the presence of evolving neurologic decline.
 - Preoperative neurologic status may strongly influence treatment method and subsequent outcomes.
 - Epidural infections usually affect the thoracic and lumbar spines, where they typically exist in the posterior epidural space; in the cervical spine, they are normally located anterior to the thecal sac.
- Signs, symptoms, and physical examination
 - Patients regularly complain of axial pain, but symptoms may be more subtle with less virulent organisms.
 - Larger lesions may compress the neural elements and give rise to neurologic deficits.
- Work-up
 - Standard infection laboratories (e.g., WBC, ESR, and CRP) with blood cultures.

- Identification of a specific pathogen may require a needle or open biopsy to acquire tissue for culture.
- Neuroimaging
 - MRI is the most sensitive and specific imaging study, for it clearly demonstrates any fluid collection in the epidural space.
- Treatment
 - Patients who are neurologically intact may be candidates for nonsurgical treatment consisting of long-term antibiotic therapy.
 - Surgical decompression ± stabilization is generally indicated for individuals who have failed medical management or who present with neurologic deterioration.
 - Medical management has a high risk of failure in immunocompromised populations (e.g., elderly, diabetes), virulent organisms, or patients with neurologic compromise.
- Surgical pearls
 - Operative approach is largely influenced by the location of the infection.
 - Though a laminectomy may be sufficient to address posterior abscesses, an anterior procedure may also need to be performed in the setting of vertebral osteomyelitis or ventral abscess.
 - Concomitant arthrodesis may be warranted in cases in which spinal instability has been caused by the infection or any subsequent decompression.

22.5 Postoperative Infections^{1,2,4,8,9,10}

- Background
 - Surgical site infection (SSI) has been reported in up to 12% of adults undergoing spinal operations.
 - SSI is associated with longer hospital stays, higher complication rates, and increased mortality.
 - Risk factors for SSI include increased age, obesity, diabetes, tobacco use, poor nutritional status, greater intraoperative blood loss, prolonged surgical time, complete neurologic injuries, revision surgery, placement of instrumentation, disseminated cancer, and a posterior operative approach.
 - Postoperative SSI ordinarily arises following direct inoculation of the wound with normal skin flora.
- Signs, symptoms, and physical examination
 - The typical SSI is clinically evident and associated with obvious erythema, edema, and drainage from the incision, although subfascial lesions may yield few external signs of infection.
 - Patients may or may not complain of pain or exhibit constitutional symptoms.

- Work-up
 - WBC, ESR, and CRP are frequently elevated and may be used as serial markers to confirm the resolution of a SSI.
 - Wound cultures are essential for directing antimicrobial therapy.
 - Sampling of deep wound collections is preferable since superficial drainage is prone to contamination with skin flora.
 - Intraoperative cultures represent the ideal method for identifying the causative organism.
- Neuroimaging
 - CT may reveal pathologic changes such as an abscess or hematoma, but these findings may be difficult to differentiate from nonspecific postoperative changes.
 - MRI is ideal for identifying fluid collections, but in many cases there will be an increased signal on T2-weighted images and contrast enhancement in the surgical field regardless of whether an SSI is present.
- Treatment
 - Prophylactic antibiotics administered 60 minutes before a spinal procedure have been shown to reduce the incidence of SSI by up to 60%.
 - Additional doses of intraoperative antibiotics should be dispensed for prolonged surgical procedures with significant blood loss or gross contamination.
 - Multiple studies have demonstrated that the use of vancomycin powder in surgical wounds decreases the postoperative infection rate.
 - Definitive therapy for established SSI is open irrigation and debridement.
 - IV antibiotics are typically continued for a minimum of 6 weeks, at which time patients may be switched over to oral medications depending on clinical course and laboratory profile.
- Surgical pearls
 - Deep wound cultures should be obtained intraoperatively before delivery of antibiotics and irrigation of the tissues.
 - Metallic hardware is frequently left in place to maintain stability, but any loose bone graft should be removed to prevent the formation of a nidus of bacteria.
 - Following open irrigation and debridement, the wound may be closed primarily over drains, covered with a vacuum dressing, or left open.

Common Clinical Questions

1. Which laboratory test is best for following the resolution of a spinal infection?
 - A. White blood cell count
 - B. Erythrocyte sedimentation rate
 - C. C-reactive protein
 - D. Platelet count
 2. What are the most common organisms that are isolated in cases of vertebral osteomyelitis and diskitis?
 - A. Gram-positive cocci
 - B. Gram-negative rods
 - C. *Mycobacterium tuberculosis*
 - D. Fungi
 3. Surgical intervention is most clearly indicated for which of these conditions?
 - A. L3–L4 diskitis associated with low back pain
 - B. Incisional drainage following a recent discectomy
 - C. Granulomatous infection involving the T6–T7 vertebral bodies with no obvious deformity or compression of the neural elements
 - D. C5–C6 epidural fluid collection resulting in progressive weakness and numbness
-

Answers to Common Clinical Questions

1. C.
 2. A.
 3. D.
-

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23 Spinal Column and Spinal Cord Tumors

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23.1 Key Points

- Spine tumors are broadly organized into three general categories depending on the spine compartment invaded by the neoplasm: Extradural, intradural extramedullary, and intradural intramedullary (**Fig. 23.1**).
- Overall, MRI is the best modality for determining the localizing compartment of a neoplasm.
- Determining which compartment is affected is essential to yield the primary differential diagnosis (**Fig. 23.1**), the pathophysiologic mechanism connecting the presence of the tumor to the clinical syndrome, and the appropriate surgical strategy.
- Tumor histopathology can influence the surgical approach and treatment paradigm that gives the best clinical outcome.

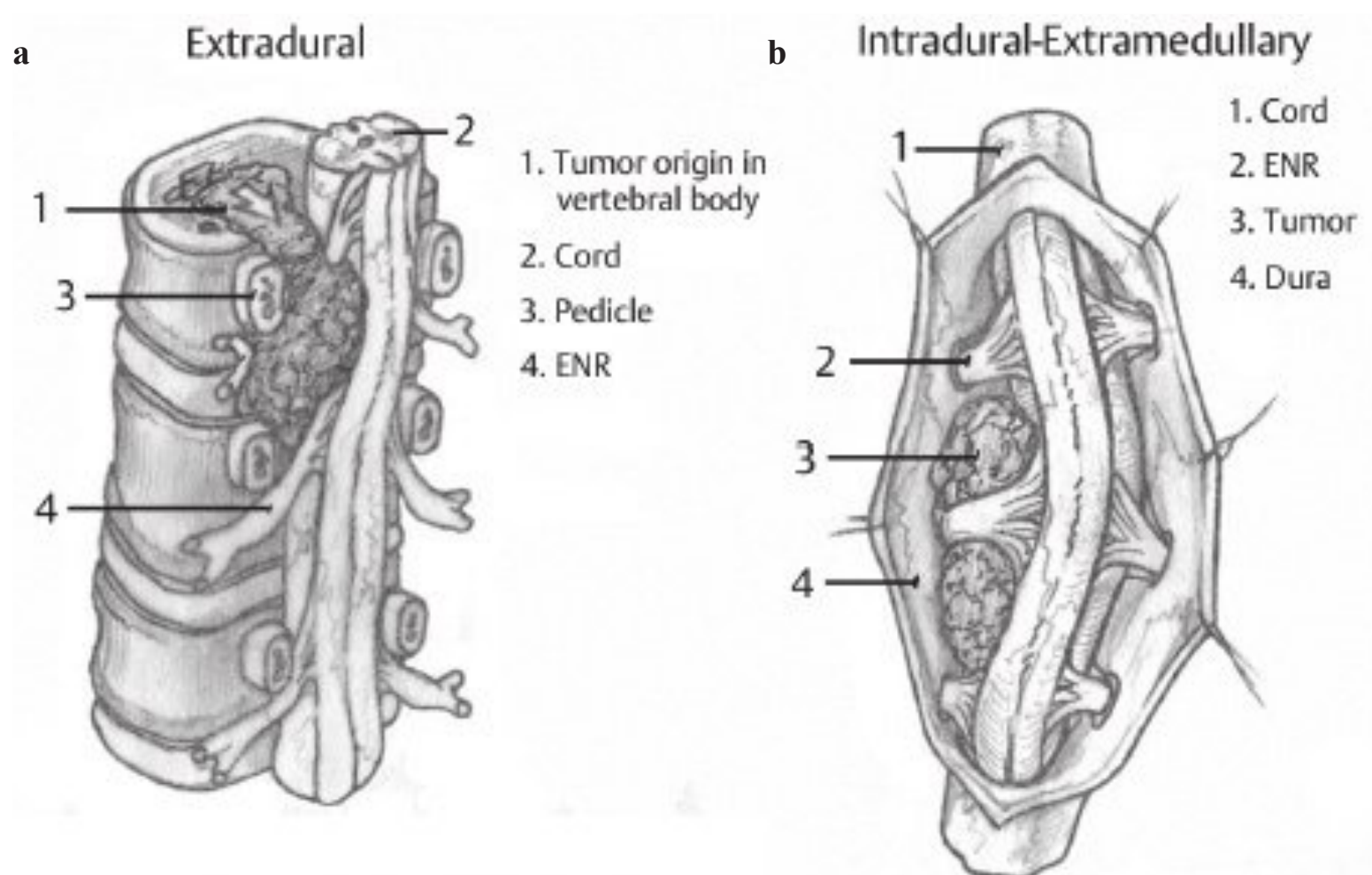


Fig. 23.1 (a,b) Artistic rendition demonstrating the structural relationship of differently compartmentalized neoplasms to the spinal cord and adjacent structures. Differential diagnosis based on compartmentalization of the neoplasm. (From Khanna AJ, ed. MRI for Orthopaedic Surgeons, Thieme; 2010.) (Continued on page 166)

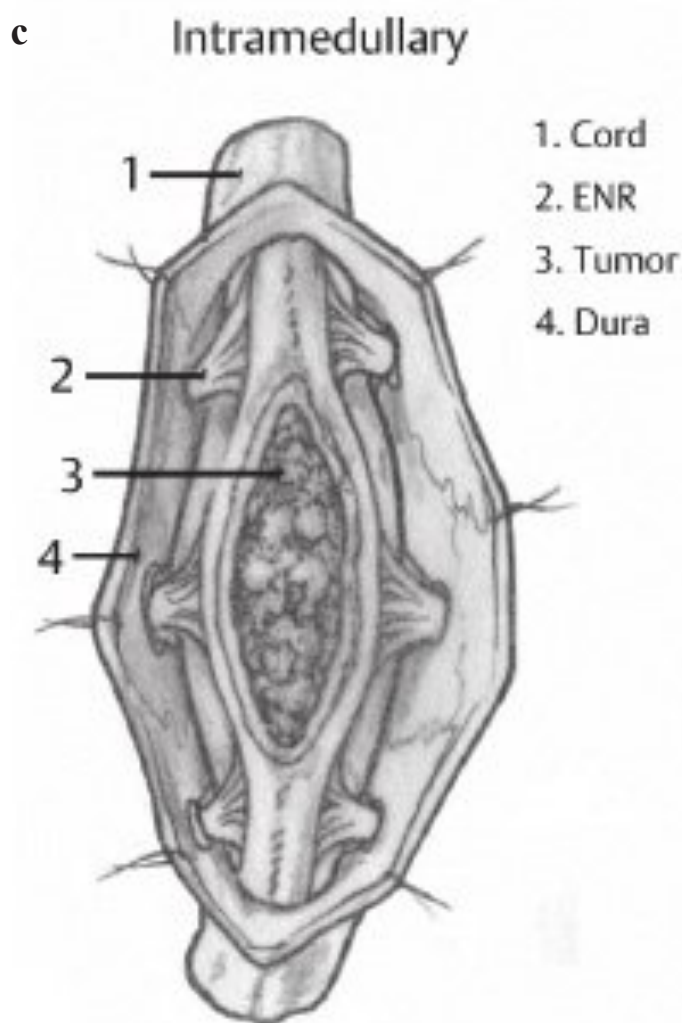


Fig. 23.1 (Continued) Fig. 23.1 (c) Artistic rendition demonstrating the structural relationship of differently compartmentalized neoplasms to the spinal cord and adjacent structures. Differential diagnosis based on compartmentalization of the neoplasm. (From Khanna AJ, ed. *MRI for Orthopaedic Surgeons*, Thieme; 2010.)

23.2 Extradural: Metastatic Spine Tumors

- Background
 - Metastatic spine tumors are more prevalent than primary spine tumors.^{1,2} Of the 1.5 million cancer cases diagnosed annually, 10% result in symptomatic secondary metastases. The bony spine is the third most common site of metastasis (commonly thoracic).¹ The most likely origins of spinal metastasis are breast, lung, and prostate cancer.¹ Peak incidence: Fourth to sixth decade. Men are more likely to be afflicted, possibly because of their higher prevalence of lung and prostate cancers compared with breast cancer.¹ The two most common mechanisms of spread to the spine are hematogenous spread and direct extension.¹
- Signs, symptoms, and physical exam
 - Presentation is a factor of systemic tumor spread, amount of bony destruction, extent of neural compression, and tumor growth rate.^{1,3} Physical examination can elicit appropriate signs of neurologic dysfunction, pain, and palpable masses. A thorough history is essential to elicit risk factors (e.g., cigarette smoking). Pain is the most common initial complaint and can be categorized as radicular

(radicular compression or foraminal stenosis), mechanical (spinal instability resulting from compromised vertebral bodies and adjacent structures, axial-loading pain), or local pain. Motor and autonomic dysfunctions are the second most common signs of metastatic epidural spinal cord compression (MESCC).¹ Sensory abnormalities are the third most common sign and can include anesthesia, hyperesthesia, and paresthesias. Signs and symptoms of myelopathic origin can be described as being distributed in a bandlike fashion. Compromise of bowel or bladder function and loss of the ability to ambulate are crucial prognostic factors. Other important signs include those of systemic neoplastic disease, such as marked weight loss and clinical syndromes such as the sequelae of functional metabolic tumors.

- Work-up and neuroimaging
 - Diagnostic blood work should include prostate-specific antigen assays, blood chemistry, and blood cell counts. The gold standard for imaging of metastatic spine disease is MRI with and without contrast because the quality of bone–soft tissue interface visualization allows the clinician to elucidate compression and/or invasion of osseous, neural, and paraspinal structures (**Fig. 23.2**). T2-weighted images and T1 contrast-enhanced images have the highest yield diagnostically.^{1,3} Other imaging modalities include CT (detailed rendition of osseous anatomy), myelography studies (in cases of MRI contraindication), CT angiography, magnetic resonance angiography or angiogram (evaluation of vascular anatomy), and whole-body scans (allow detection of the primary tumor of origin). Plain radiographs are relatively insensitive for tumor diagnosis but can be useful to screen for pathologic fractures, spinal deformities, sclerotic lesions, lytic lesions, and large masses.
- Treatment
 - Common surgical indications include preservation of neurologic function, mechanical stabilization, and pain relief with the goal of achieving optimal resection, decompression, and/or stabilization (i.e., reconstruction and instrumentation). Curative treatment is possible only in select cases (i.e., solitary renal cell carcinoma). Surgical candidacy factors include functional status (the most prognostic factor of postoperative neurologic function),¹ age, life expectancy (with 3 mo considered a minimum), and systemic disease burden. Objective scales are available for patient selection.³ Nonsurgical candidates can be managed via minimally invasive means such as vertebroplasty or kyphoplasty. Adjuvant treatments include pharmacotherapy (tumoricidal and palliative) as well as radiation therapy (conventional radiotherapy or stereotactic radiosurgery).¹



Fig. 23.2 (a) Sagittal T1 MRI with contrast, (b) sagittal T2 MRI, (c) axial T1 MRI with contrast, and (d) axial T2 MRI demonstrating a T10 vertebral body metastatic lesion with epidural spinal cord compression. Postoperative (e) lateral and (f) anteroposterior plain radiographs demonstrating T8 to T12 pedicle screw fixation with T10 titanium mesh cage.

- Surgical pearls
 - Although surgical exposure is always of utmost importance, these patients often have poor healing capabilities as a result of systemic cancer spread, previous use of corticosteroids, and radiation exposure to the surgical field. For this reason, surgical exposure and wound closure should be performed with a strategy of maximal avoidance of postoperative wound complications (some advocate for plastic surgery involvement).

23.3 Extradural: Primary Spine Tumors

- Background
 - Ten percent of the neoplasms that affect the bony spine are primary spine tumors, occurring more frequently in men than women.² The most frequent primary spine tumors are chordomas, chondrosarcomas, osteosarcomas, and Ewing sarcoma.^{2,4} Chordomas are slow-growing tumors and represent 1% of primary spine tumors.⁵ Chondrosarcomas are responsible for 7 to 12% of primary spine tumors.² Osteosarcomas are less common but are the most common malignant tumor of *osseous* origin. Predisposing factors include adolescence, a family history of retinoblastoma, and previous exposure to ionizing radiation.² Among children, eosinophilic granulomas and Ewing sarcoma are the most common benign and malignant tumors, respectively.² Hemangiomas are the most common benign tumor and plasmacytomas are the most common malignant primary spine tumors in adults.² Rare tumors include giant cell tumors, aneurysmal bone cysts, osteoid osteomas, and osteoblastomas.^{2,4}
- Signs, symptoms, and physical exam
 - Chordomas most common symptoms are back and neck pain. Nearly a third of patients will demonstrate signs of neurologic deficit, and physical exam may identify a palpable mass.^{2,5} Chondrosarcomas' common signs and symptoms are radiculopathy, myelopathy, cauda equina syndrome, and nocturnally exacerbated pain.^{2,4} Osteosarcomas most commonly present with an insidious development of nocturnally exacerbated back pain.^{2,4} Ewing sarcoma presents with symptoms of pain and local inflammation (frequently misdiagnosed as infection). Signs and symptoms of systemic illness, such as weight loss and fever, are also common.^{2,4} Plasmacytomas additionally present with diffuse osteoporosis, bone fractures, and osteolytic bone lesions.^{2,4}

- Work-up and neuroimaging
 - Histopathological analysis of the lesion is crucial for selecting appropriate intervention with respect to each tumor origin (i.e., prognosis and sensitivity to pharmacotherapy or radiotherapy).³ Chordomas and chondrosarcomas are both hyperintense under T2-weighted MRI.^{2,5} However, chondrosarcomas can be differentiated from chordomas and other tumors via gadolinium enhancement, which yields a characteristic ring-and-arc pattern.⁴ The gold standard for imaging osteosarcomas is positron emission tomography (PET) scanning owing to its ability to measure bone turnover. Highly mineralized tumors are hyperintense on T1-weighted MRI, whereas low mineralized tumors are hyperintense on T2-weighted imaging.² Ewing sarcoma can be detected via plain radiography, because the bone has a mottled, moth-eaten appearance. The possibility of distant metastases should be ruled out via a whole-body CT scan.² When multiple myeloma is suspected, blood cell counts, blood chemistries, and serum/urine electrophoresis proteins should be obtained. These studies may show evidence of renal failure, infections, hypercalcemia, anemia, or Bence-Jones proteins. Diagnostic bone marrow biopsies can also be considered. Radiologic diagnosis can be supported via CT or MRI (T2-weighted) and are usually not hot (i.e., no increased uptake) on a bone scan.^{2,4}
- Treatment of primary extradural spine tumors
 - The most important prognostic factors for primary spine tumors include tumor histopathology, location or extent of invasion or bony destruction, size, and histologic grade. Ideally, primary spinal column neoplasms should be completely excised via en bloc resection with wide surgical margins and avoidance of tumor capsule breach. Capsule breach is correlated with higher rates of local recurrence and decreased survival.² Concomitant adjuvant therapy such as chemotherapy and radiotherapy can reduce rates of local recurrence (note, however, that osteosarcomas and chondrosarcomas are relatively radioresistant). Ewing sarcoma can be managed with chemotherapy (typically the combination of doxorubicin, cyclophosphamide, vincristine, and dactinomycin). Ewing sarcoma is radiosensitive and can be treated via conventional radiotherapy.² Plasmacytomas are most commonly managed with pharmacotherapy (chemotherapy and bisphosphonates) and radiotherapy. Surgical intervention is reserved for cases of marked spinal instability.²
- Surgical pearls
 - Tumors: En bloc spondylectomy is ideal for radioinsensitive primary osseous tumors (chordoma, chondrosarcoma, etc.). Such techniques should be attempted by experienced surgeons, often in conjunction with a multidisciplinary team including thoracic surgery, orthopedic surgery, general surgery, vascular surgery, and plastic surgery.

23.4 Intradural Extramedullary Spinal Cord Tumors

- Background
 - Intradural extramedullary tumors are the second most common tumor type in the spine.⁶ Primary lesions in this compartment arise from perineural coverings of nerve roots or from the meninges; thus meningiomas, schwannomas, neurofibromas, and paragangliomas account for the majority of tumors in this compartment.⁶ Although the majority of tumors are benign, they can result in significant neurologic dysfunction, most commonly as a result of compression of neural structures.
- Signs, symptoms, and physical exam
 - Symptomatology is usually of insidious onset. The most common complaint is localized or radicular pain. Other signs and symptoms include gait impairment, weakness, paresthesia, impotence, and autonomic dysfunction.⁶ Physical examination findings include Brown-Sequard syndrome and long tract signs, such as Babinski sign, clonus, and hyperreflexia. Intradural extramedullary tumors can be distinguished from intramedullary tumors by the observation that intramedullary tumors can spare dorsal tracts, whereas extramedullary tumors commonly affect all sensory modalities.⁶
- Work-up and neuroimaging
 - MRI can identify the tumor compartment (i.e., intramedullary versus extramedullary), but when contraindicated, CT myelography is the imaging of choice. Meningiomas, schwannomas, neurofibromas, and paragangliomas are all hyperintense on MRI T2-weighted images. They are all iso- or hypointense in T1-weighted images.⁶ Schwannomas can be distinguished from meningiomas, because schwannomas may demonstrate cystic changes within the tumor and appear as focal areas of increased signal in T2-weighted images. In contrast, meningiomas rarely develop cystic changes. Schwannomas are also often dumbbell-shaped.⁶ Paragangliomas demonstrate strong enhancement upon administration of gadolinium contrast. Metastatic lesions should be suspected in a patient with a previous history of cancer, and in such a case imaging should include a whole-body scan to assess systemic spread of the metastatic neoplasm.¹
- Treatment
 - Complete microsurgical excision of intradural extramedullary tumors is optimal but not always possible owing to factors that dictate surgical approach (i.e., anterior or posterior cord location) and the degree to which neural structures are involved by the neoplasm. For example, neurofibromas normally grow from the central root as an enlargement of the nerve itself, making complete surgical resection

very challenging without incurring a neurologic deficit. This is in contrast to schwannomas, which typically involve only one fascicle of a nerve root, making complete dissection of the mass and preservation of the remainder of the nerve root function possible.⁶ When complete resection is not possible without neurologic compromise, partial resection avoiding neurologic compromise is at times appropriate. This decision is individualized to patient preference, and factors to consider are patient age, neurologic status at presentation, tumor histopathology and size, and factors predisposing to local recurrence, such as a medical history of neurofibromatosis. The risk of local recurrence can be reduced postoperatively with adjuvant chemotherapy and radiotherapy for sensitive tumors. Radiosurgery can be considered in cases of recurrence, multiple lesions, and an absence of compressive myelopathy.

- Surgical pearls
 - Tumors should be removed from the spinal cord rather than attempting to initially dissect the tumor-spinal cord interface. Often, this requires initial internal tumor debulking to avoid manipulating the spinal cord in its most compressed state.

23.5 Intramedullary Spinal Cord Tumors

- Background
 - Intramedullary spinal cord tumors make up roughly 6 to 8% of all tumors of the central nervous system.⁶ The two most common intramedullary spinal cord tumors are low- or high-grade astrocytomas and ependymomas.⁶ Low-grade astrocytomas are more common in children, whereas ependymomas are more prevalent in adults.⁶ High-grade astrocytomas often have a poor prognosis, because they are highly infiltrative and have a high rate of recurrence.⁶
- Signs, symptoms, and physical exam
 - Intramedullary spine tumors have a nonspecific presentation, with insidious onset or following a trivial injury. Signs and symptoms include radicular pain, localized pain, dysesthesia, paresthesia, spasticity, torticollis, extremity weakness, Brown-Sequard syndrome, and autonomic dysfunction.⁶ In children, these tumors can present as a failure to achieve developmental milestones.⁶ Intramedullary tumors that localize to the cervical spine may also be accompanied by hydrocephalus.²
- Work-up and neuroimaging
 - T1-weighted MRI with contrast reveals the solid tumor component. T2-weighted images allow visualization of cystic elements and the cerebrospinal fluid. When viewed axially, astrocytomas are located eccentrically in the spinal cord and may display heterogeneous

enhancement under T1 weighting. In contrast, ependymomas are most often in the center of the spinal cord when imaged axially. Ependymomas can exhibit homogeneous enhancement.⁶ Plain radiographs may be useful preoperatively as a baseline reference for managing spinal deformity, as in the case of patients who have associated scoliosis.

- Treatment
 - Histologic grade and preoperative neurologic function are the most significant prognostic factors for surgical management of intramedullary spinal cord tumors. High-grade astrocytomas have approximately an 80% mortality rate within the first 6 months of diagnosis. Conversely, it is possible to completely surgically resect and cure ependymomas.⁶ Astrocytomas have a grayish yellow, glassy appearance and should be resected starting at the midpoint of the neoplasm and moving in an inside-to-outside fashion. The tumor should be debulked until the border between the spinal cord and tumor can be reasonably demarcated.⁶ Ependymomas have a red or dark gray appearance and display a characteristic visible boundary in relation to the spinal cord. Ependymomas can be resected en bloc, and separation at the tumor–spinal cord boundary can be achieved by applying a microsurgical laser or plated bayonet in the axial direction. Electrophysiologic monitoring can be used to assess the patient’s neurologic status throughout the procedure. This includes somatosensory evoked potentials (SSEPs) and motor evoked potentials (MEPs) as well as epidural recordings.⁶
- Surgical pearls
 - Tumors can often be easily suctioned off of the normal parenchyma of the spinal cord when they are necrotic (high-grade astrocytoma) or when a good tissue plane is noted between the tumor and spinal cord (ependymoma or low-grade astrocytoma). Extreme care should be taken to avoid dissecting the plane between tumor and cord with instruments as these maneuvers contuse and stretch the cord; rather, suction should be used for debulking and sharp dissection in locations where the tumor is focally tethered to the cord.

Common Clinical Questions

1. Which is the *least* common histology of metastatic tumor arising in the spine?
 - A. Breast adenocarcinoma
 - B. Prostate adenocarcinoma
 - C. Nonsmall cell lung adenocarcinoma
 - D. Transitional cell carcinoma of the bladder

2. Classic initial presentations of metastatic epidural spinal cord compression include all of the following *except* which?
 - A. Abdominal pain
 - B. Bladder retention
 - C. Localized back pain
 - D. Gait imbalance
 3. Which is the most important predictor of survival in patients who have spine tumor?
 - A. Presence of metastases in the liver
 - B. Histopathology of the lesion
 - C. Number of spinal levels involved
 - D. Local invasion into paraspinal tissues
-

Answers to Common Clinical Questions

1. D.
 2. A.
 3. B.
-

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24 Cervical and Thoracic Spine Degenerative Disease

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24.1 Key Points

- Cervical and thoracic degenerative disease is a chronic condition but can present acutely.
- MRI is usually the imaging modality of choice for diagnosis.
- Radicular and axial pain may be treated conservatively, but myelopathy and/or worsening neurologic function generally requires surgical intervention.

24.2 Cervical Disk Herniation

- Background
 - Dehydration, fissuring, and fragmentation of the nucleus pulposus of cervical disks with age are natural processes.¹
 - The annulus and often the posterior longitudinal ligament may tear, allowing the nucleus to herniate into the spinal canal, where it may compress the cord and/or the adjacent root at its foramen.²
 - Acute disk rupture occurs more often laterally in the spinal canal due to the relative weakness of the posterior longitudinal ligament in that area; as a result, root compression occurs more often than cord compression.²
 - Rarely, infarction of the cord and root may occur if compression and ischemia are severe.
- Signs, symptoms, and physical exam
 - Lateral disk herniations cause pain that radiates from the neck to the shoulder/arm and into the hand; disk usually impinges on nerve exiting from the neural foramen at the level of the herniation (e.g., a C6/C7 disk causes C7 radiculopathy).
 - C5 symptoms: Shoulder abduction (deltoid) weakness; shoulder paresthesias; deltoid and pectoralis reflexes diminished.³
 - C6 symptoms: Forearm flexion (biceps) weakness; upper arm, thumb, and radial forearm sensory alteration; biceps and brachioradialis reflexes diminished.³
 - C7 symptoms: Elbow extension (triceps) weakness; second and third digit sensory alteration; triceps reflex diminished.
 - C8 symptoms: Hand weakness; fourth and fifth digit sensory alteration; finger-jerk reflex diminished (rarely used).³
 - Central disk herniation can cause myelopathy and central cord syndrome.

- Work-up
 - Complete history and physical
 - Basic laboratory studies
 - Neuroimaging studies based on localization of signs and symptoms (**Fig. 24.1**)
 - MRI is the imaging modality of choice for visualizing soft tissue and the spinal cord/nerve roots.
 - Use CT myelogram when MRI cannot be done.
 - Plain CT is good for bone detail.
 - Plain X-rays are good for bone imaging, anteroposterior/lateral views are useful for visualizing alignment, and flexion–extension views are useful for assessing subluxation/dynamic instability (instability with movement).
- Treatment
 - More than 90% of patients who have acute cervical radiculopathy as a result of cervical disk herniation can improve without surgery.³
 - Surgery is indicated for those who fail to improve or those who have progressive neurologic deficit while undergoing nonsurgical management.
 - Anterior surgical options: Anterior cervical discectomy with or without fusion, plating, or artificial disk (arthroplasty), anterior cervical foraminotomy.
 - Posterior surgical options: Posterior cervical laminectomy/foraminotomy with or without fusion, keyhole laminectomy (for lateral “soft disk” herniation).
- Surgical pearls
 - Partial or complete corpectomy may be required if herniated disk is sequestered posterior to vertebral body and is not accessible by discectomy.



Fig. 24.1 MRI is the imaging modality of choice for visualizing soft tissue and the spinal cord/nerve roots but cannot distinguish between disk and bone.

24.3 Cervical Spondylotic Myelopathy

- Background
 - Caused by the reduction of the sagittal diameter of the cervical spinal canal as a result of congenital and degenerative changes.³
 - Often caused by degeneration of the intervertebral disk producing a focal stenosis as a result of a “cervical bar” that is usually a combination of osteophytic spurs and/or protrusion of disk material.²
 - Most common type of spinal cord dysfunction in patients older than 55 years.²
 - Cord injury likely occurs as the result of several interrelated factors: Direct compression of the cord, microtrauma associated with neck flexion and extension, and vascular injury.⁴
 - Risk factors include cigarette smoking, frequent lifting, and diving.
 - Signs and symptoms may overlap with those of amyotrophic lateral sclerosis (ALS; also known as motor neuron disease).² Both conditions may present with upper motor neuron findings in the LEs (hyperreflexia, spasticity) and lower motor neuron findings in the UEs (weakness, hyporeflexia). Sensory disturbance is usually more prominent in cervical disk disease. Dysarthria and tongue fasciculations may occur in ALS.
- Signs, symptoms, and physical exam
 - Gait disturbance, often with lower extremity weakness or stiffness, is a common early finding.⁴
 - Cervical pain and mechanical signs are uncommon in cases of pure myelopathy.
 - Earliest motor findings are typically weakness in the triceps and hand intrinsics.
 - Clumsiness with fine motor skills (writing, buttoning buttons).⁴
 - Glove-distribution sensory loss in the hands or sensory level several levels below the area of cord compression.
 - Reflexes are hyperactive at varying distance below the level of stenosis, pathologic reflexes may be present (Hoffmann’s, Babinski, clonus, etc.).
 - Central cord syndrome, in which motor and sensory deficits affect the upper extremities more than the lower extremities, may occur acutely after trauma with hyperextension in those who have cervical stenosis.
- Work-up
 - Complete history and physical
 - Basic laboratory studies
 - Neuroimaging studies based on localization of signs and symptoms
 - MRI is the imaging modality of choice for visualizing soft tissue and the spinal cord/nerve roots but cannot distinguish between disk and bone.
 - CT myelogram: Use when MRI cannot be done or when better bone imaging is needed. Can still visualize spinal cord/nerve roots, but does not provide information about changes within the spinal cord

parenchyma; risks of lumbar puncture and/or intrathecal contrast injection need to be considered.

- Plain CT: Good for bone imaging. May demonstrate narrow canal, but does not provide adequate information regarding soft tissues.
- Plain X-Ray: Good for bone imaging. May demonstrate narrow canal (posterior vertebral line to spinolaminar line < 12 mm). Flexion–extension view may demonstrate dynamic instability.
- Treatment
 - Nonoperative management (prolonged immobilization with rigid cervical bracing, eliminating “high-risk” activities, a short course of bed rest, and anti-inflammatory medications) may be considered for mild myelopathy.
 - More severe myelopathy should be treated with surgical decompression.
 - Surgical approaches:
 - Posterior: Not ideal for correction of kyphotic deformity. Preferred for most cases of myelopathy due to OPLL (ossification of the posterior longitudinal ligament).
 - Laminectomy alone (higher incidence of late kyphotic deformity).
 - Laminectomy + instrumentation/fusion (lateral mass screws, etc.).
 - Laminoplasty (if patient has myelopathic symptoms without axial neck pain).
 - Anterior: Ideal for correction of kyphotic deformities.
 - Anterior cervical discectomy and fusion (ACDF).
 - Corpectomy and fusion: When compression extends beyond region of disk space.
 - Combinations of ACDF + corpectomy and fusion.
 - Anterior procedures that include more than three disk levels will additionally need posterior instrumentation/fusion for stability.
- Surgical pearls
 - Bone imaging (CT or X-ray) is important to detect ossified posterior longitudinal ligament (OPLL) when suspected on MRI. If present, it may influence approach (posterior instead of anterior) or procedure (corpectomy instead of ACDF) to prevent intraoperative durotomy.

24.4 Thoracic Disk Herniation

- Background
 - The incidence of clinically significant herniation is 1 patient per 1 million people.⁵
 - 0.25% of all herniated disks.²
 - $< 4\%$ of operations for all herniated disks.⁵
 - 75% occur below T8, most commonly T11/T12.²
- Signs, symptoms, and physical exam
 - Axial pain may be mechanical. Can sometimes be confused with cardiac, pulmonary, or abdominal pathology.

- Lateral or centrolateral herniations can present with radicular pain around chest wall along the path of an intercostal nerve in a dermatomal pattern.
- Central herniations are associated with a high incidence of spinal cord compression and long tract signs (lower extremity hyperreflexia, Romberg sign, Babinski reflex, clonus, ataxic gait, loss of rectal tone, and decreased perianal sensation).
- In severe cases, the lesion may cause loss of bowel or bladder function, progressing rapidly to incomplete or total flaccid paraplegia.
- Work-up
 - Complete history and physical
 - Basic laboratory studies
 - Neuroimaging studies based on localization of signs and symptoms
 - MRI is the imaging modality of choice for visualizing soft tissue and the spinal cord/nerve roots but cannot distinguish between disk and bone; scout film with MRI of entire spine is needed to localize level precisely.
 - CT myelogram: Use when MRI cannot be done, when better bone imaging is needed, and when spinal cord/nerve roots can still be visualized. Does not provide information about changes within the spinal cord parenchyma. Can show calcification in disk (occurs in 30–70% of symptomatic thoracic disks). Risks of intrathecal contrast injection need to be considered.
 - Plain CT: Good for bone imaging, but does not provide adequate information regarding soft tissues. Should be done in almost all cases to differentiate “soft disk” from calcified disk herniation, which affects surgical decisions.
 - Plain X-ray: Good for bone imaging. Essential as an intraoperative reference to determine correct level.
 - Intraoperative fluoroscopy: It is often easier to count vertebrae upward from sacrum. Be sure to correlate carefully with MRI (which usually counts from top down), for not all patients have 12 rib-bearing thoracic vertebrae.
- Treatment
 - Asymptomatic thoracic disk herniations without evidence of spinal cord compression require no treatment.
 - Symptomatic thoracic disk herniations without evidence of spinal cord compression should initially be treated nonoperatively (at least 4–6 wk).
 - Acute herniations resulting in axial pain should be treated with
 - Activity modification
 - NSAIDs
 - Physical therapy
 - Radicular pain/paresthesias can be treated with
 - Oral corticosteroids
 - Epidural steroid injections
 - Surgery may be considered for unrelenting symptoms despite nonoperative treatment.

- Symptomatic thoracic disk herniations causing spinal cord compression should be treated surgically.
 - Approaches
 - Anterior: Good for midline or broad-based herniations or densely calcified disk herniations
 - * Transsternal or via resection of medial aspect of clavicle (for upper thoracic lesions)
 - Anterolateral: Good for midline or broad-based herniations or densely calcified disk herniations
 - * Transthoracic transpleural via thoracotomy (usually right-sided to avoid heart)
 - * Video-assisted thoracoscopic (VATS) not widely used
 - * Minimally invasive transthoracic transpleural or retropleural
 - Posterolateral: Good for lateral or soft disk herniation
 - * Transpedicular
 - * Costotransversectomy
 - * Lateral extracavitary
 - Posterior (laminectomy): Not recommended due to high incidence of neurologic injury
- Surgical pearls
 - The anatomy involved with surgery for thoracic herniated disks is less often encountered by spine surgeons than the cervical or lumbar regions, so be sure to understand this anatomy thoroughly before taking a patient to the operating room.
 - Calcified disks are difficult to treat via posterolateral approaches, so consider an anterior or anterolateral approach for these lesions.
 - Ensure that the exact disk level is localized by intraoperative fluoroscopy before proceeding with bone removal and discectomy.
 - Posterolateral approaches may require instrumentation, especially if performed bilaterally.
 - Consider SSEPs and MEPs intraoperatively, especially if myelopathic.

Common Clinical Questions

1. A C6–C7 lateral disk herniation will compress which nerve root and cause what physical exam findings?
 - A. C6 nerve root with deltoid weakness, paresthesias over shoulder, and reduction of biceps reflex
 - B. C6 nerve root with triceps weakness, paresthesias in thumb, and upgoing toes
 - C. C7 nerve root with triceps weakness, diminished triceps reflex, and sensory loss in digit 3
 - D. C8 nerve root with weakness of hand grasp, reduced pinprick sensation in digit 5, and reduced finger flexor reflex

2. On lateral X-ray C-spine, what AP spinal canal diameter is considered to be stenotic?
 - A. 12 mm measured from posterior vertebral body to the spinolaminar line
 - B. 12 mm measured from posterior vertebral body to base of spinous process
 - C. 18 mm measured from posterior vertebral body to base of spinous process
 - D. 20 mm measured from posterior vertebral body to base of spinous process
3. True, or false? Laminectomy is the best approach for excising a herniated thoracic disk.

Answers to Common Clinical Questions

1. C. C7 nerve root is compressed, causing triceps weakness, second- and third-digit paresthesias and/or sensory loss, and diminished triceps reflex.
 2. A.
 3. False. Laminectomy is not recommended because of the high rate of neurological injury associated with this approach.
-

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25 Degenerative Lumbar Spine Disease

Michael Y. Wang

25.1 Key Points

- Degenerative disease of the spine is a ubiquitous problem and part of the natural aging process. Treatment, surgical or otherwise, is directed at specific symptoms, pathologies, and syndromes.
- Skeetoligamentous disorders are typically the result of intervertebral disk disease or facet disease or both. Sacroiliac and hip joint pain can also be contributors to back pain.
- Determining the “pain generator” is not always straightforward and requires a synthesis of data from the medical history, physical exam, provocative testing, and imaging.

25.2 Background

- Degeneration of the lumbar spine is the result of natural aging, environmental insults, and genetic predisposition.
- Degenerative changes may be asymptomatic.¹
- Specific disease states, such as rheumatoid arthritis and ankylosing spondylitis, can accelerate or alter the pathology and clinical presentations.
- Changes that develop over time include loss of water volume within the nucleus pulposus with loss of disk height, disk bulging, facet joint degeneration, and osteophyte formation across the joints (both the intervertebral disks and facet joints). Eventual loss of motion occurs.

25.3 Specific Conditions

25.3.1 Radicular Pain

- Signs, symptoms, and physical exam
 - Diagnosis requires a correlation between the pain distribution and the compressed nerve root identified on imaging.
 - Sharp, shooting pain in a given dermatome, but it sometimes may be dull or aching.

- Pain, paresthesias, weakness, and diminished reflexes can be found on exam.
- Work-up and neuroimaging
 - Diagnostic testing includes MRI, CT scan (if the offending pathology is suspected to be osseous), or, in some cases, CT myelogram.
 - If there is doubt about the level or location of the pathology, a test nerve root injection with anesthetic may be diagnostically helpful. Electromyography (EMG) is also frequently used to confirm the radiculopathy or to distinguish it from a peripheral nerve problem.
 - Classic pathology is a paracentral disk herniation. Other possibilities include compression as a result of foraminal collapse, facet joint overgrowth, or a far lateral disk herniation.
- Treatment
 - Treatment of radiculopathy is directed at nerve decompression. This can be done with a laminectomy, laminoforaminotomy, or microdisectomy.
 - In cases in which compression is caused by foraminal collapse resulting from loss of spinal alignment (e.g., spondylolisthesis or degenerative scoliosis), a fusion with intervertebral height restoration may be indicated.
- Surgical pearls
 - During a laminectomy, laminoforaminotomy, or microdisectomy, care must be taken to avoid excessive facet joint removal. These joints are critical to the stability of the spine, and no more than half of a joint should be removed in a unilateral approach.
 - After surgical decompression, scar formation can result in radiculitis or, rarely, arachnoiditis. Minimizing manipulation and nerve root retraction may be helpful.

25.3.2 Neurogenic Claudication

- Signs, symptoms, and physical exam
 - Diagnosis is highly dependent on the history. Patients typically complain of unilateral or bilateral leg pain, numbness, and weakness that can be precipitated by standing or walking.
 - The physical examination is frequently completely normal.
 - The ability to walk farther when bending forward (as when pushing a shopping cart) is classic.
 - Vascular claudication is suggested when relief occurs with rest without the need to flex at the waist, and must be ruled out by evaluating peripheral pulses (for suspected cases, an ankle–brachial index [ABI] may be useful).

- Work-up
 - Diagnostic testing includes an MRI or CT myelogram to evaluate the size of the lumbar spinal canal and neuroforamina.
- Neuroimaging
 - Compression may result from congenital stenosis, disk bulging, ligamentum flavum hypertrophy, facet joint overgrowth, or spondylolisthesis or any combination of these.
- Treatment
 - Surgical intervention is directed at patients who have failed conservative measures, such as epidural injections, and includes decompression with the possibility of an adjunct fusion.
 - Standard treatment involves the use of a midline laminectomy with foraminotomies.
 - Indirect decompression can be achieved with an interspinous spacer device, which causes focal flexion and stretching of the ligamentum flavum and facet joints.
 - Unilateral hemilaminotomy for bilateral decompression is another less invasive option that preserves more of the skeletoligamentous structures.
- Surgical pearls
 - Curved Kerrison rongeurs can be useful to reach out to the distal foramina and should be placed on the caudal and dorsal aspect of the foramen, this being the least likely location for the exiting nerve root.
 - In performing a hemilaminotomy for bilateral decompression, care should be taken to preserve the ipsilateral ligamentum flavum, which will displace the dura ventrally. This reduces the risk of dural injury while work is being performed on the contralateral nerve roots. Ipsilateral decompression can be performed after the opposite side has been decompressed.

25.3.3 Axial Back Pain from Intervertebral Disk Disease (without Deformity)

- Signs, symptoms, and physical exam
 - The most classic disk-related mechanical pain syndromes present as pain that worsens with activity and lessens with rest.
 - Anterior thigh pain can also be associated with low back pain, as the symptoms may follow a somatotopic pattern.
 - Physical examination may reveal pain relief with extension maneuvers and provocation with flexion, although these findings are not universal.
 - It is important to eliminate other insidious causes of axial pain, such as infection or malignancies.

- Work-up
 - MRI to examine the disk and neighboring structures
 - CT myelogram when MRI is contraindicated
 - Flexion–extension films to rule out spondylolisthesis
- Neuroimaging
 - The disk nucleus progressively loses the high signal on T2 (so-called black disk), and loss of disk height will also occur.
 - Adjacent bony end plates may exhibit high T2 signal, classified as Modic changes.
 - Plain X-rays may also reveal degeneration, but dynamic flexion and extension views may be more effective in revealing any instability.
 - CT scans may demonstrate osteophytes and are less useful. In marginal cases, a provocative diskogram can be used to assess the disk's internal morphology and ability to tolerate the injectate's pressure, as well as any symptoms associated with injection.²
- Treatment
 - Treatment of disk-related pain remains controversial and requires that patients have failed conservative treatment measures.
 - For patients who have intractable symptoms and demonstrable pathology, treatment may be directed at disk removal or the elimination of motion.
 - This can be performed with a posterolateral instrumented fusion; anterior, posterior, or transforaminal lumbar interbody fusion (ALIF, PLIF, TLIF); lateral interbody fusion; extreme lateral interbody fusion (XLIF); or total disk arthroplasty.³
- Surgical pearls
 - It is generally believed that if surgical treatment of disk-related pain is warranted, then an interbody fusion is preferable to posterolateral fusion.
 - For patients who will undergo disk arthroplasty, care must be taken to exclude those who have facet disease, as the posterior joints must continue to function after the operation.

25.3.4 Axial Back Pain from Facet Joint Disease

- Signs, symptoms, and physical exam
 - The zygapophyseal joint (facet joint) is a synovial joint that is prone to arthritic changes and is a possible pain generator (**Fig. 25.1**).
 - Pain that worsens with provocative maneuvers such as back extension may be a simple diagnostic clue.
 - Facet pain can radiate into the lower extremity, mimicking a painful radiculopathy.

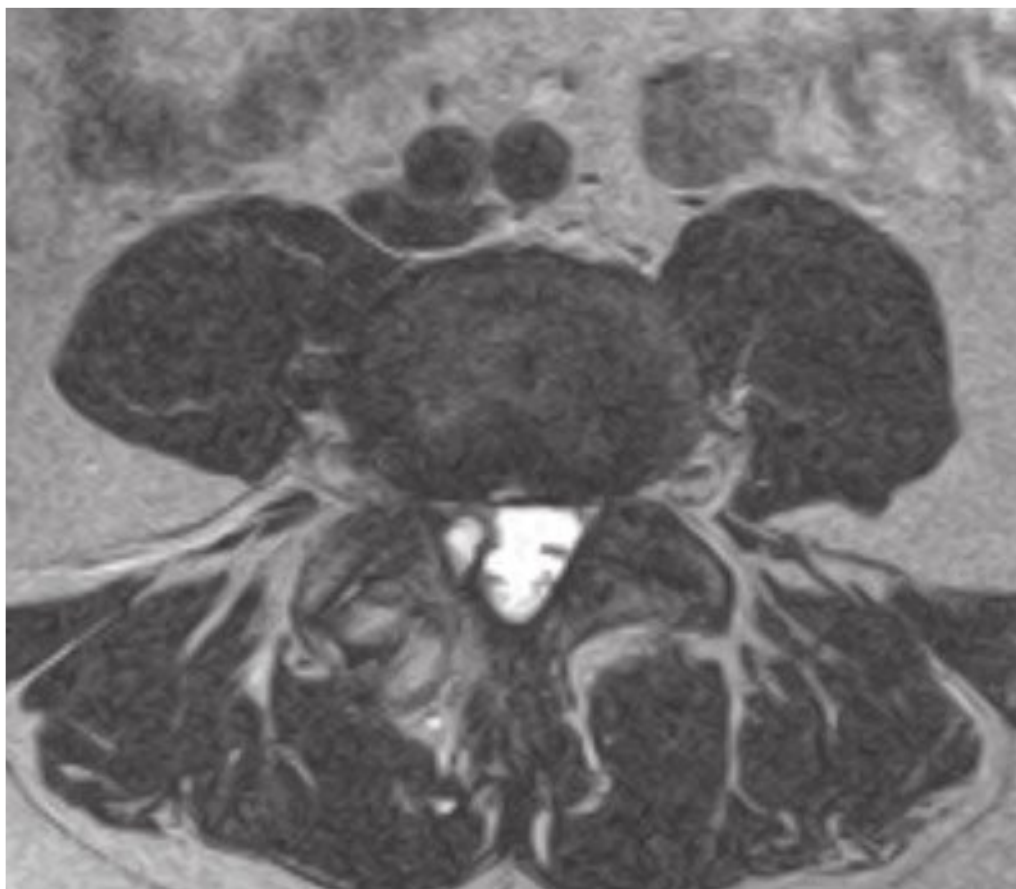


Fig. 25.1 Right synovial cyst causing radiculopathy at the L5/S1 level.

- Work-up
 - MRI, flexion–extension films, and single-photon emission computed tomography (SPECT) bone scan are helpful.
 - The definitive diagnostic test is an anesthetic or steroid injection (controversy surrounds the relative efficacy of intra-articular versus periarticular joint injections).
 - Hip or sacroiliac joint pathology must also be considered as a possible source of “back pain.”
- Neuroimaging
 - High T2 signal in the joint as well as increased focal uptake on SPECT bone scans can be valuable predictors.⁴
- Treatment
 - Interventional treatment of isolated facet joint disease is with anesthetic/steroid injections or dorsal ramus rhizolysis.
 - In addition, physical therapy and anti-inflammatory medications are often helpful.
- Surgical pearls
 - Obtain dynamic films to rule out spondylolisthesis.
 - Fusion/fixation of the facet joint remains controversial.

25.3.5 Spondylolisthesis

- Signs, symptoms, and physical exam
 - Spondylolisthesis may be the result of congenital, traumatic, degenerative, or iatrogenic etiologies.

- A degenerative spondylolisthesis typically affects the L4–L5 level, although any level may become involved.
- A fatigue fracture of the pars followed by progression of slippage may also be classified as a form of degeneration caused by chronic mechanical load.
- The clinical syndrome may cause leg pain or back pain or both.
- Work-up
 - MRI and flexion–extension films are mandatory.
- Neuroimaging
 - Diagnostic testing can include MRI imaging to evaluate neural entrapment. Flexion–extension X-rays are mandatory to assess the gross stability of the level, which affects treatment decisions (**Fig. 25.2**).
- Treatment
 - Typically involves decompression and fusion. A common approach is laminectomy for decompression followed by instrumented fusion.
 - Controversy exists over the utility of correcting the slippage.
 - The use of interbody fusion increases the rate of arthrodesis but poses the threat of higher complication rates.
- Surgical pearls
 - High-grade slips where realignment is desired can often be treated more effectively by instrumenting the level above (L4–S1 in a L5/S1 slip) to bring the middle intermediary screw up to a rod connecting the two end screws.
 - The exiting nerve root at the slip level is displaced ventrally. Aggressive correction of a high-grade slip can result in foot drop. Close electromyographic (EMG) monitoring for nerve root irritation may allow assessment of the stretch/tension on this structure.

25.3.6 Degenerative Scoliosis and Kyphosis

- Signs, symptoms, and physical exam
 - Evaluation of patients with spinal deformity requires not only a standard physical examination, but also an assessment of the patient's standing and lying postures.
 - Any progression of deformity should also be evaluated with serial imaging. The frequent coincidence of hip and sacroiliac joint arthritis and leg length discrepancies needs to be assessed.
 - Because this disease entity typically affects older females, an evaluation of bone density may be warranted and appropriate measures to augment bone density undertaken.
 - In addition, assessment of any contributing hip flexion contractures may be necessary.

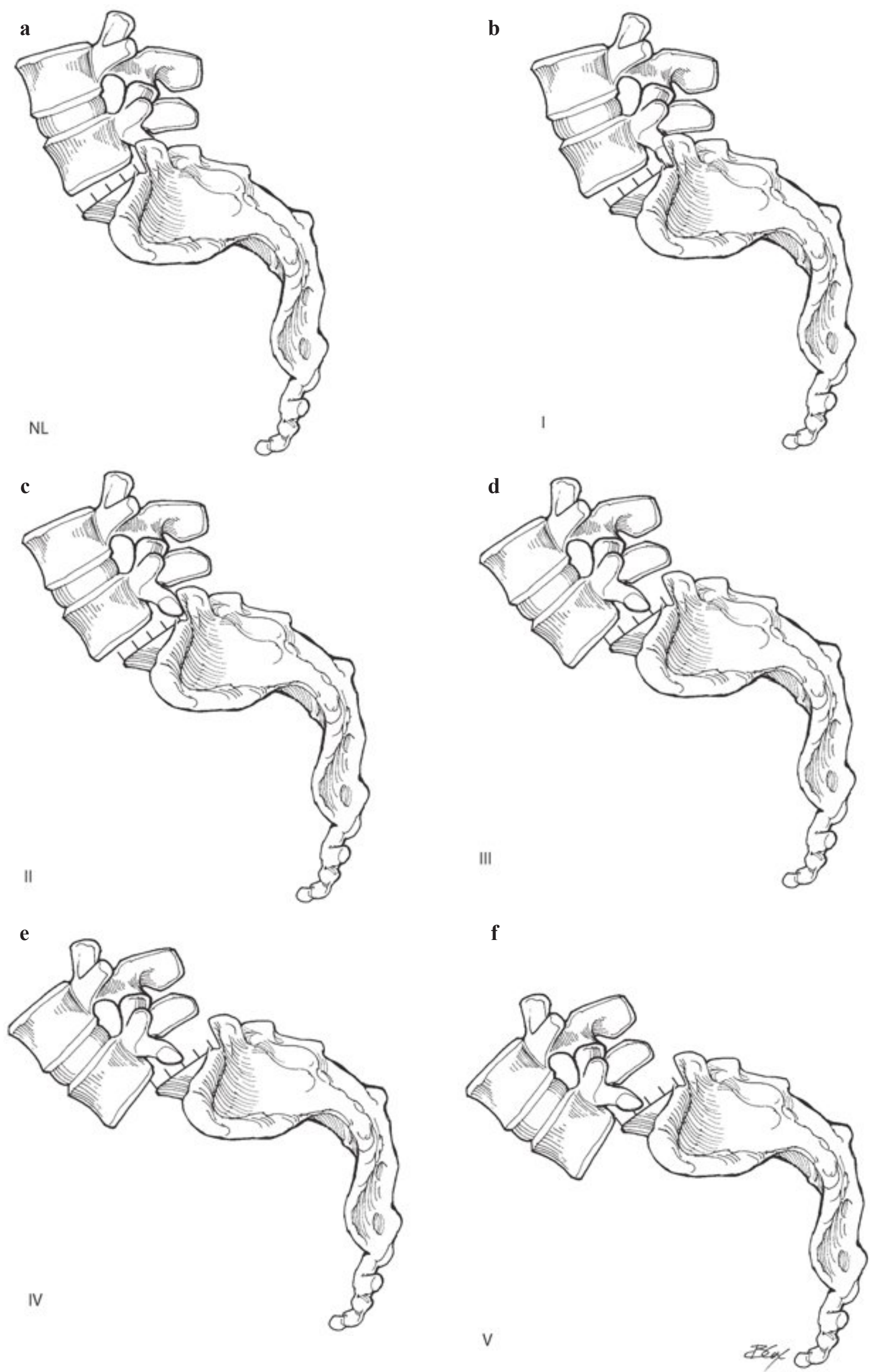


Fig. 25.2 (a–f) Meyerding classification of spondylolisthesis: Normal, Grade I, 0 to 25%; Grade II, 26 to 50%; Grade III, 51 to 75%; Grade IV, 76 to 100%; Grade V, > 100%(spondyloptosis).

- Work-up
 - Diagnostic imaging requires an assessment of neural element integrity (MRI or myelography) and disk and facet joint disease, as well as local and global spinal alignment.
- Neuroimaging
 - MRI, dynamic X-rays, and CT scanning are essential for preoperative planning.
 - Thirty-six-inch standing X-rays are mandatory to determine coronal alignment (Cobb angle) and loss of sagittal balance.
- Treatment
 - The treatment of this patient population is complicated, requiring significant investment in preoperative planning, surgical intervention, and postoperative care. Critical decision-making factors include the following:
 - Source and nature of complaints (pain, reduced activities of daily living (ADLs), neurologic symptoms, abnormal postures, cosmetic concerns)
 - The degree of deformity and its relative contribution to the patient's symptoms
 - The extent (levels and amount of correction) of the deformity that needs treating
 - The amount (quality and number of levels) of fixation needed
 - The surgical technique necessary to achieve correction
 - Need for fusion adjuncts (osteobiologics, bracing, and bone stimulation)
 - Anesthetic considerations and risks
 - Nutrition and metabolic concerns
 - The extent to which rehabilitation will be necessary after surgery
- Surgical pearls
 - Careful preoperative planning, including measurements of preop sagittal/coronal imbalance, is essential to achieving good clinical and surgical outcomes.

Common Clinical Questions

1. Lumbar facet joint pain is best treated with what?
 - A. Massage therapy
 - B. Electrostimulation
 - C. Joint injections
 - D. Surgical fusion

2. Causes of spondylolisthesis include all of the following *except* what?
 - A. Neoplastic
 - B. Degenerative
 - C. Traumatic
 - D. Iatrogenic
 3. Studies critical to the evaluation of a spinal deformity include what?
 - A. Blood tests for rheumatoid arthritis
 - B. Genetic assessment of predilections for deformity
 - C. SPECT nuclear medicine bone scan
 - D. 36-inch standing X-rays
-

Answers to Common Clinical Questions

1. C.
 2. A.
 3. D.
-

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26 Congenital and Neuromuscular Scoliosis

Marie Roguski, Amer F. Samdani, and Steven W. Hwang

26.1 Key Points

- Congenital and neuromuscular scoliosis differ from adolescent idiopathic scoliosis by the presence of vertebral anomalies and of an underlying neurologic disorder, respectively.
- Both conditions occur in patients who may have significant medical comorbidities and decreased pulmonary function as a result of associated anomalies or progressive neuromuscular disease, necessitating a thorough preoperative evaluation.
- Treatment is often initially conservative and involves bracing. However, surgical correction is indicated for severe or progressive disease, especially when considering curve magnitude in relation to age.

26.2 Congenital Scoliosis

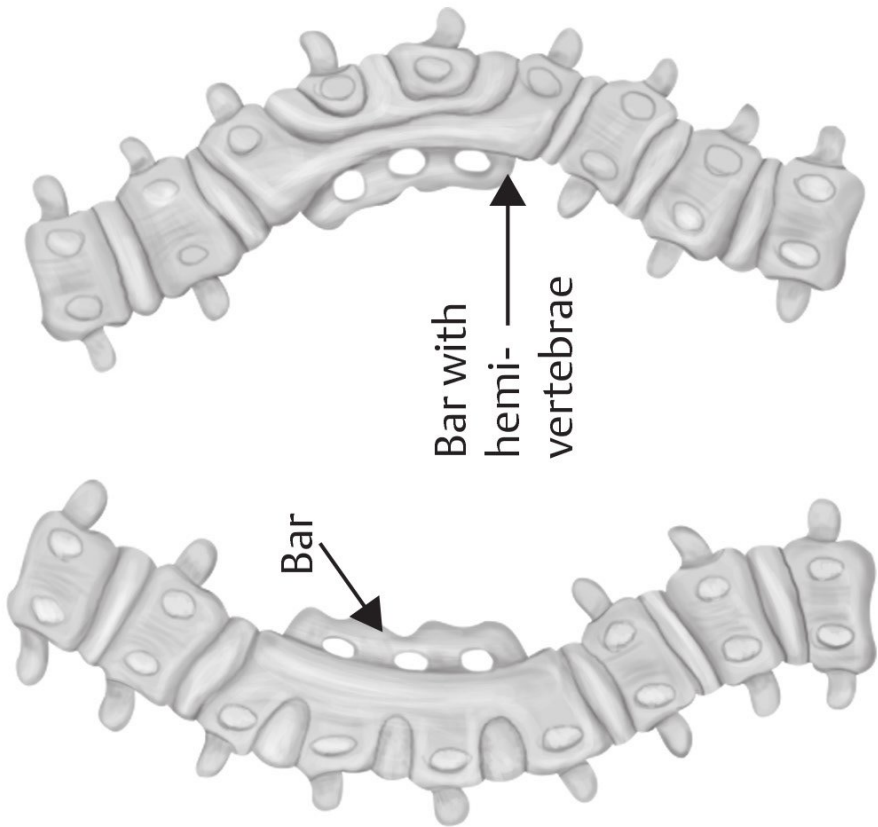
- Background
 - Scoliosis is an abnormal curvature of the spine; congenital scoliosis is distinguished from other types of scoliosis by the presence of vertebral anomalies at birth.
 - Congenital vertebral anomalies are classified according to the error in embryological development: (1) Failure of formation, (2) failure of segmentation, or (3) both failure of formation and segmentation (**Table 26.1**).¹
 - Incomplete and complete failure of formation results in a mildly wedged or completely absent vertebra, respectively. Asymmetric failure of formation results in a hemivertebra (a wedged vertebral body with a single pedicle and hemilamina), which is the most common cause of congenital scoliosis.
 - Failure of segmentation results in fusion of vertebrae. This can occur unilaterally or bilaterally resulting in an unsegmented bar of bone or block vertebrae, respectively (**Fig. 26.1**).
 - Although the vertebral anomaly is present at birth, scoliosis generally does not develop until growth phases of childhood or adolescence. Whether a congenital scoliosis develops depends on whether growth of the spine is balanced bilaterally. For example, the lack of growth plate in a unilateral unsegmented bar coupled with normal growth in the

Table 26.1 Etiologies of congenital and neuromuscular scoliosis

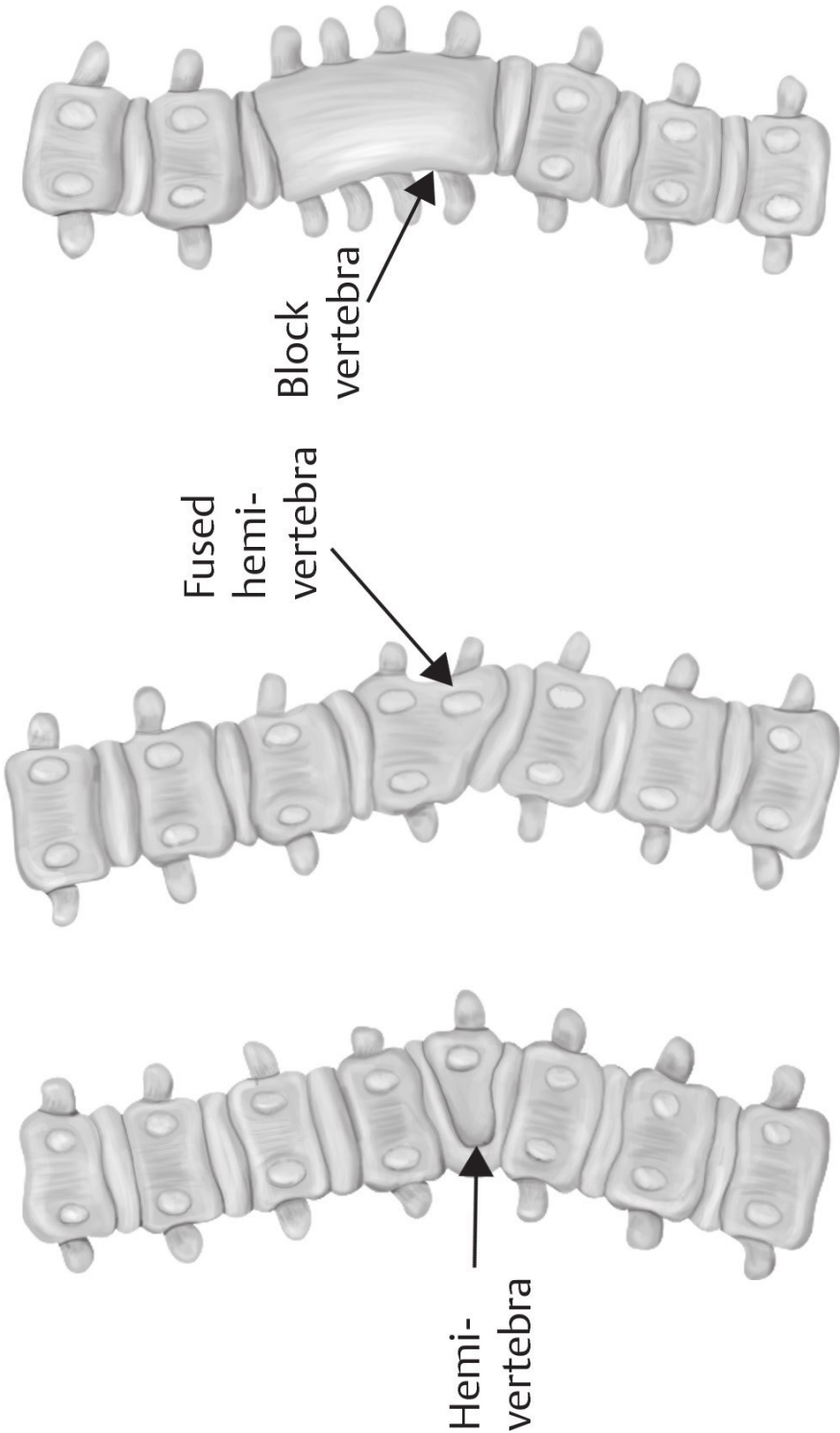
Congenital scoliosis	Neuromuscular scoliosis ⁶
Failure of formation	Primary neuropathies
Fully segmented hemivertebra	Upper motor neuron pathology
Semisegmented hemivertebra	Cerebral palsy
Incarcerated hemivertebra	Spinocerebellar degeneration
Nonsegmented hemivertebra	Syringomyelia
Failure of segmentation	Spinal cord tumor
Unilateral bar	Spinal cord trauma
Block vertebrae	Lower motor neuron pathology
Mixed anomalies	Poliomyelitis
Congenital scoliosis	Other viral myelitides
Dysplastic spondylolisthesis	Traumatic
	Charcot-Marie-Tooth disease
	Spinal muscular atrophy
	Combined upper and lower pathologies
	Myelomeningocele
	Primary myopathies
	Muscular dystrophy
	Duchenne’s muscular dystrophy
	Limb-girdle dystrophy
	Facioscapulohumeral dystrophy
	Arthrogryposis
	Congenital hypotonia
	Myotonic dystrophy

Note: Congenital scoliosis is classified by the underlying error in development. In contrast, the Scoliosis Research Society classifies neuromuscular scoliosis according to whether the underlying cause is a primary neuropathy and myopathy.

FAILURE OF SEGMENTATION



FAILURE OF FORMATION



contralateral side from a hemivertebra has the highest risk of progressive deformity owing to a tethered growth on the side of the anomaly.

- Other types of congenital deformity include congenital kyphosis, congenital lordosis, congenital stenosis, and dysplastic spondylolisthesis.
- Signs, symptoms, and physical exam
 - Spinal examination assesses for pelvic and truncal imbalance. Head tilt, shoulder height asymmetry, apparent leg length discrepancy, and abnormal pelvic tilt are indicators of imbalance.
 - Adam's forward bend test assesses for thoracic rotation and is quantified though a scoliometer.²
 - Neurological signs and symptoms may be present and necessitate a thorough neurological examination.
 - Signs may include neurocutaneous stigmata of spinal dysraphism, such as skin tags, hair tufts, dermal sinus tracts, or abnormal pigmentation, as well as musculoskeletal abnormalities, such as asymmetric calves or clubfeet.
- Work-up
 - Evaluation includes a complete history and physical examination. History includes prenatal and birth history and achievement of developmental motor and cognitive milestones.
 - Attention to medical comorbidities is warranted due to an association of renal, respiratory, and cardiac anomalies with congenital spinal disorders.³
 - Congenital vertebral anomalies can also be associated with congenital rib fusions. When present simultaneously with scoliosis, patients can develop severe restrictive lung disease and thoracic insufficiency. Pulmonary function tests may be needed as part of a thorough preoperative work-up.
- Neuroimaging
 - Full-length upright posteroanterior (PA) and lateral spine X-rays are often the first radiographic studies obtained to evaluate for suspected congenital scoliosis and to monitor for curve progression over time.
 - CT provides additional bony anatomy and is useful in detailing underlying vertebral anomalies. Three-dimensional reconstruction of complex deformities and measurement of pedicle and vertebral body size may also aid in operative planning (**Fig. 26.2**).
 - MRI depicts relationships of the deformity to neural structures and also detects coexisting abnormalities, such as syringomyelia or diastematomyelia.
- Treatment
 - Treatment is often initially conservative or involves bracing. However, nearly 75% of 202 patients who had congenital scoliosis treated conservatively had significant progression.⁴

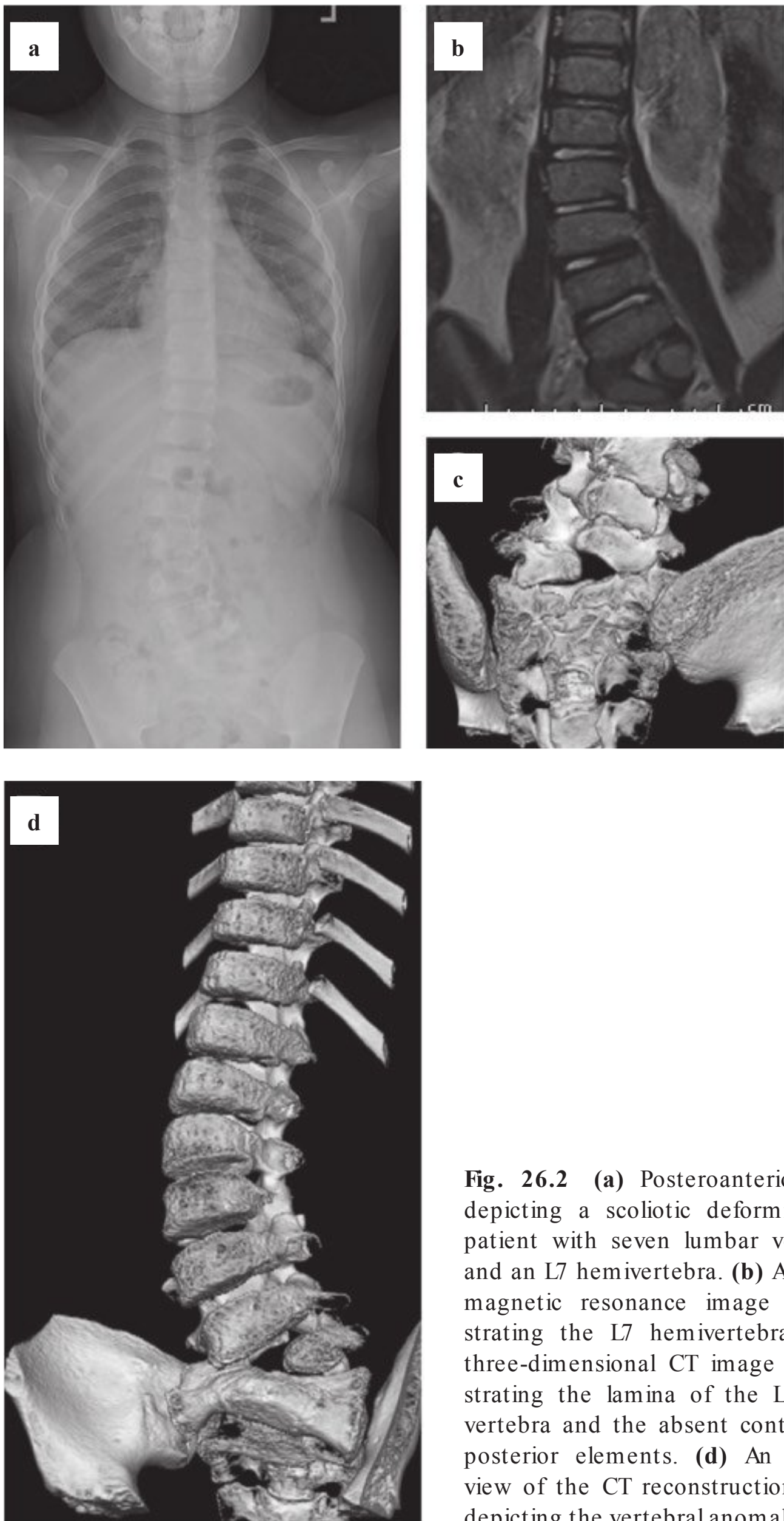


Fig. 26.2 (a) Posteroanterior X-ray depicting a scoliotic deformity in a patient with seven lumbar vertebrae and an L7 hemivertebra. (b) A coronal magnetic resonance image demonstrating the L7 hemivertebra. (c) A three-dimensional CT image demonstrating the lamina of the L7 hemivertebra and the absent contralateral posterior elements. (d) An anterior view of the CT reconstruction image depicting the vertebral anomaly.

- Early surgery is often the best option for severe or progressive congenital scoliosis. Surgical options include spinal fusion in situ, convex hemiepiphysiodesis, excision of a hemivertebrae, spinal instrumentation without fusion, growing systems, and thoracoplasty with a vertical, expandable prosthetic titanium rib (VEPTR).
- Surgery is accompanied by high surgical morbidity and mortality. Reames et al reported a morbidity of greater than 10% and mortality of 0.3% in a series of more than 2,000 patients undergoing surgical treatment for congenital scoliosis.⁵
- Surgical pearls
 - Special attention is needed to screen for associated renal, cardiac, respiratory, and congenital anomalies.
 - Surgical treatment is indicated for severe or progressive congenital scoliosis.

26.3 Neuromuscular Scoliosis

- Background
 - Scoliosis is a common occurrence in patients with neuromuscular disease and is often more severe and progressive than in patients without an underlying disorder.
 - The Scoliosis Research Society classifies neuromuscular scoliosis into categories of upper motor neuron disease, lower motor neuron disease, mixed pathology, and primary muscle disease (**Table 26.1**).⁶
 - Patient factors that contribute to the development of spinal deformity include asymmetric paraplegia and muscle tone, mechanical force imbalance across the spine, related congenital spine and neural anomalies, altered sensory feedback, and abnormal posture.
 - The severity of the underlying disorder often determines whether a deformity will develop. Scoliosis develops in up to 70% of patients who have cerebral palsy, in 85 to 90% of males who have Duchenne's muscular dystrophy, and in nearly 100% of patients who have traumatic tetraplegia with onset before skeletal maturity.
- Signs, symptoms, and physical exam
 - The signs and symptoms of neuromuscular scoliosis are similar to those found in congenital scoliosis. The spinal deformity is more likely to involve the pelvis with an associated “windswept” hip deformity, and joint contractures and spasticity are more common. These can result in significant postural difficulty and functional impairment.
 - Patients with neuromuscular disorders are often nonambulatory at presentation and may also have significant medical comorbidities necessitating special attention to pulmonary and nutritional status.

Pressure sores may be present at scoliosis apices and other pressure-bearing sites.

- Work-up
 - Evaluation includes a complete history, physical examination, and neurological examination. Static conditions such as cerebral palsy or myelomeningocele should be separated from progressive conditions such as Duchenne’s muscular dystrophy.
- Neuroimaging
 - The role of X-ray, CT, and MRI in the evaluation of neuromuscular scoliosis is similar to the aforementioned role in the work-up of congenital scoliosis.
- Treatment
 - Nonoperative treatment is generally the initial treatment and includes bracing and customized wheelchair seating systems to provide support for the head, back, and pelvic girdle.
 - The role of bracing in prevention of curve progression is controversial; however, bracing may aid in delaying surgery in very young children who have neuromuscular scoliosis.
 - The goals of surgical treatment are often improvement in the quality of life by pain relief, improvement of seated posture and pulmonary mechanics, prevention of further deformity, and alleviation of caregiver burden.
 - Whether surgical treatment is appropriate is determined on a case-by-case basis, and an exact threshold for curve severity has not been established. However, typically a curve of greater than 50° is often considered to be surgical indication. Surgical options typically include growing systems in skeletally immature patients or long fusions.
 - Morbidity and mortality is high. Reames et al reported a morbidity of greater than 18% and mortality of 0.3% in a series of 4,657 patients undergoing surgical treatment for neuromuscular scoliosis.⁵
- Surgical pearls
 - Development and progression of scoliosis in patients who have neuromuscular disease depends on the severity of the underlying illness.
 - Surgical correction is indicated for severe or progressive scoliosis, but the morbidity of surgery is high.

Common Clinical Questions

1. A hemivertebra develops as a result of what?
 - A. Bilateral failure of formation
 - B. Unilateral failure of formation
 - C. Failure of segmentation
 - D. Both failure of formation and segmentation

2. Which of the following pathologic entities has the highest risk of developing scoliosis?
 - A. Cerebral palsy
 - B. Duchenne's muscular dystrophy
 - C. Skeletally immature tetraplegic spinal cord injury
 - D. Rhett's syndrome
 3. Which of the following is associated with the highest risk of scoliotic progression?
 - A. Growth spurts
 - B. Skeletal maturity
 - C. Ambulatory status
 - D. Vitamin deficiency
-

Answers to Common Clinical Questions

1. B. A hemivertebra results from unilateral failure of formation.
 2. C. Skeletally immature cervical spine injuries are associated with a 100% risk of developing scoliosis in some series.
 3. A. Growth spurts are associated with the highest risk of scoliotic progression.
-

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27 Scheuermann's Kyphosis

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27.1 Key Points

- The etiology of Scheuermann's disease remains unknown. Genetics is thought to have a prevalent role.
- Adolescents with deformity greater than 75° may have self-esteem or postural concerns. Pain is a prevalent complaint. The incidence of pain was higher (78%) when the kyphosis involved the first or second lumbar vertebrae.
- Patients should be examined carefully to exclude spastic paraparesis manifested by ataxia and hyperreflexia, which may rarely be present and should be secondary to cord compression by the angular deformity, herniated thoracic disk, or a spinal epidural cyst.
- The successful surgical management of Scheuermann's disease requires addressing this sagittal imbalance and choosing the correct upper (UIV) and lower (LIV) instrumented levels.

27.2 Background

- Scheuermann's disease (juvenile kyphosis dorsalis) is a structural kyphosis of the thoracic spine initially described by Scheuermann in 1921. It occurs commonly in adolescents (0.4–8.3% of the general population) and in most cases is characterized by minimal deformity and few clinical symptoms. Scheuermann's disease is rarely associated with neurological complications. The natural history of Scheuermann's disease is poorly understood.¹ The treatment is based on the age of the patient, the severity of the deformity, the curve pattern, the progression and presence or absence of pain, and neurological deficits. Operative intervention for Scheuermann's disease is indicated for patients who have progressive deformity, painful curves, unacceptable cosmetic appearances, restrictive pulmonary disease, and neurological deficits. The main goals of surgery are to relieve pain and improve sagittal balance and aesthetic and health-related quality of life (HRQoL) issues.²

27.3 Signs, Symptoms, and Physical Exam

- Adolescents who have deformity greater than 75° often have self-esteem or postural concerns.
- The deformity may be associated with pain and fatigue. Damborg et al found that Danish adults who had Scheuermann's disease had statistically significant worse health-related quality of life Short Form Physical Component Summary (SF-12) than controls. Albanese and Guntz noted pain in only 20% of their patients, whereas Nathan and Kuhns reported an incidence greater than 60%. Pain is more commonly the chief complaint of adults.
- Spondylolysis or spondylolisthesis has been identified in up to 50% of adolescents with Scheuermann's disease and must be excluded in any Scheuermann's patient who presents with back pain.
- The incidence of pain was also much higher (78%) when the kyphosis involved the first or second lumbar vertebrae. The pain of Scheuermann's disease is usually intermittent, aching, and nonradiating in nature, localized over the kyphotic region.
- Scheuermann's disease is rarely accompanied by a neurological deficit.
- On physical exam, the patient who has Scheuermann's kyphosis typically has a well-demarcated angular deformity of the thoracic or thoracolumbar spine, with compensatory hyperlordosis of the cervical or lumbar spine, characterized by a prominent head and neck thrust. The kyphosis is normally rigid and not fully correctible when the patient attempts extension of the thoracic spine in the prone position. The lordosis of the lumbar spine is usually not structural and readily corrects using the Adams forward bending test.
- The thoracic deformity may be increased with forward bending, when viewed from the side. Thoracolumbar or lumbar kyphosis is not easily visible but may be detected by prominent spinous processes of the thoracolumbar or lumbar spine. A mild scoliosis of 10 to 20° may be noted on forward bending in 20 to 30% of patients.
- Muscle tightness and apparent contractures, particularly of the hamstrings, are common findings.
- Patients should be examined carefully to exclude spastic paraparesis manifested by ataxia and hyperreflexia, which may rarely be present and be secondary to cord compression by the angular deformity, herniated thoracic disk, or a spinal epidural cyst.

27.4 Work-up

- The criteria for the diagnosis of Scheuermann's disease are (1) more than 5° of wedging of at least three adjacent vertebrae at the apex of the

kyphosis, (2) end plate irregularities, and (3) a thoracic kyphosis of more than 45° on lateral standing radiograph (14×36 cassette).

- Radiographic evaluation of Scheuermann's disease involves acquiring orthogonal standing (posteroanterior and lateral) views of the spine.
- In our institution, EOS imaging is preferred, for this unique digital imaging system enables long-length PA and lateral images to be acquired simultaneously and at a 9.2 times lower radiation dose, with a quality that is 97.2% equivalent to or better than conventional radiology.
- Standing PA radiographs are required to exclude an associated scoliosis and may also be used to determine interpedicular distance, which is increased in patients who have spastic paraparesis secondary to a spinal epidural cyst.
- The successful surgical management of Scheuermann's disease requires addressing this sagittal imbalance and choosing the correct upper (UIV) and lower (LIV) instrumented levels.
- Any Scheuermann's patient who has a neurological deficit or significant back pain requires further evaluation with an MRI scan of the whole spine.

27.5 Neuroimaging

- The sine qua non for radiologic diagnosis of Scheuermann's disease is irregularity of end plates.
- No constitutional effects are found in adolescent kyphosis, and vertebral defects are bounded by sclerotic rim.
- MR imaging reflects changes seen on plain films more clearly and earlier.
 - The disk material is clearly seen to herniate into the end plate defect beneath the nonfused ring apophysis forming Schmorl's nodes.
 - Sometimes prolapse of large foci of intervertebral disk tissue may be observed anteriorly and appear submarginally behind the anterior longitudinal ligament; subsequently, a portion of the apophyseal ossification center may be separated from the vertebral body and produce a limbus vertebra.
- The adolescent patient's combination of cartilaginous nodes, irregular vertebral outlines, and kyphosis in area of dorsal spine is virtually pathognomonic of Scheuermann's disease.
- Recent reports at the Scoliosis Research Society and in the literature have reported posterior disk herniations after cantilever correction of the kyphotic deformity and have recommended MRI for all patients undergoing deformity correction.³ In the authors' opinion, MR imaging is indicated for atypical deformity, neurologic symptoms, or significant pain.

27.6 Treatment

27.6.1 Nonoperative Treatment

- The treatment is based on the age of the patient, the severity of the deformity, the curve pattern, the progression and presence or absence of pain and neurological deficits. Nonoperative treatment using a thoracolumbosacral orthosis (TLSO) with corrective sternal pad is usually the mainstay of initial treatment if the curve measures between 50 and 75°, and is at least 40% passively correctible.
- Riddle et al, however, have found that patients who have curves greater than 75° will respond less favorably to bracing; surgical intervention should be considered in these patients.⁴
- If nonoperative treatment is prescribed, then the brace should be worn for more than 18 hours per day and should be accompanied by an exercise program.

27.6.2 Operative Treatment

- Operative intervention for Scheuermann's disease is indicated for patients who have progressive deformity, painful curves, unacceptable cosmetic appearances, restrictive pulmonary disease, and neurological deficits.
- The main goals of surgery are to relieve pain and improve sagittal balance and cosmesis and HRQoL issues.
- Current surgical management of Scheuermann's disease trends toward a posterior-only approach with posterior column osteotomies, in the form of either Ponte osteotomies in the thoracic spine or wide posterior releases in the lumbar spine.^{5,6}
 - However, some advocate anterior release for more rigid curves that do not reduce to less than 50°. Horton et al have shown that radical disectomy in vitro is the single most effective technique for increasing the range of motion of the thoracic spine.
- Appropriate level selection is important to minimize risk of adjacent segment deformity (e.g., proximal and distal junctional kyphosis).⁶
 - Current concepts and literature supports symmetry, with the number of levels above the apex being equal to the number of levels fused distal. UIV will typically be typically T2 or T3 for high and midthoracic apical deformities.
 - LIV is typically the vertebra that is most closely bisected by the sagittal sacral vertical plumb line, and this vertebra should be positioned distal to the first lordotic disk and rostral to a lordotic disk (**Fig. 27.1**).

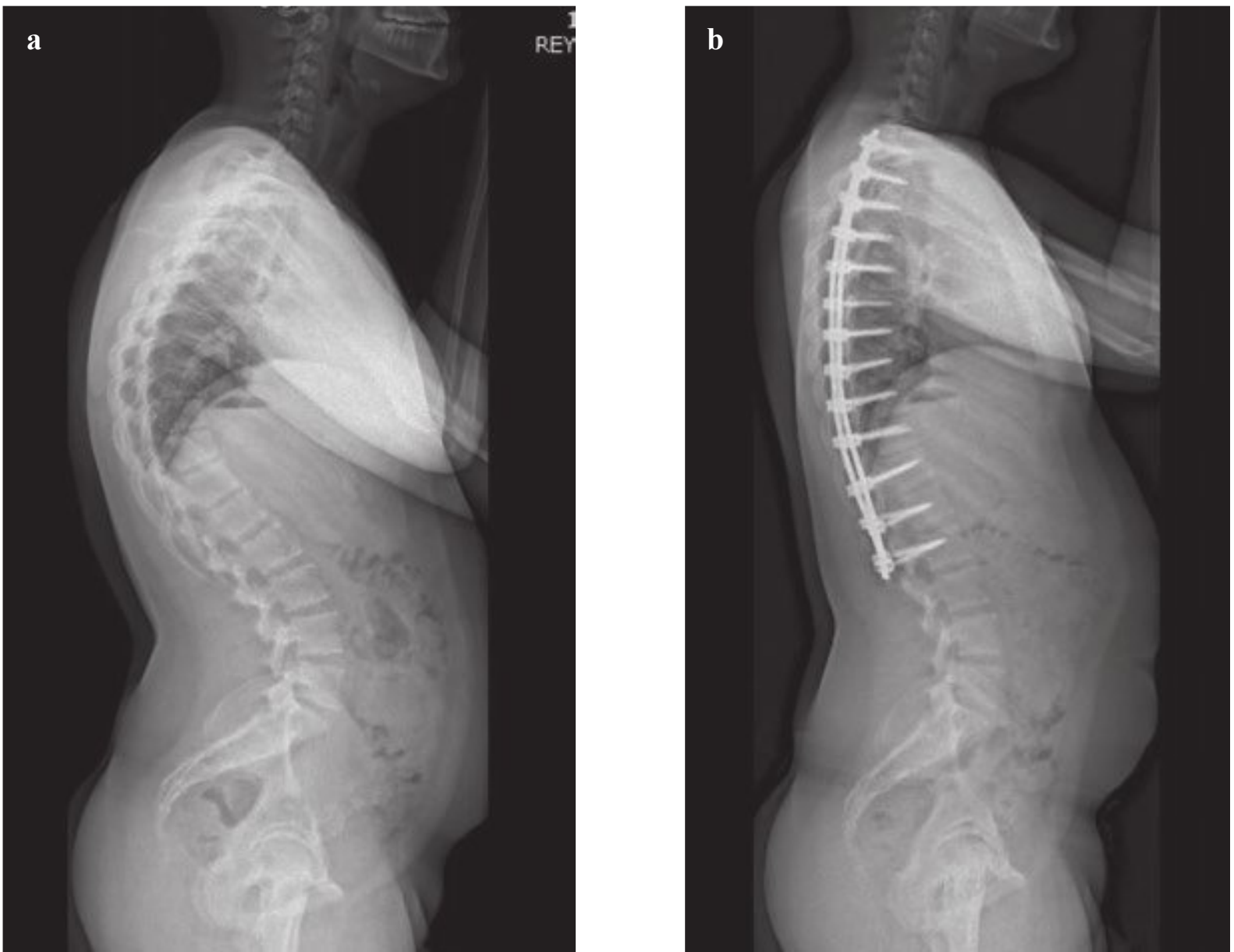


Fig. 27.1 (a,b) 17-year-old adolescent male with 85° kyphosis who underwent posterior spinal fusion an Ponte-type posterior column osteotomies from T2 to L2 with 35° residual thoracic kyphosis.

- Correction is usually obtained by a combination compression and cantilever rod reduction.
 - The rods are placed in the proximal anchors, and these anchors are compressed. The rest of the reduction is achieved by using a cantilever maneuver with the rod while inserting it into the distal anchors, thus engaging and reducing the rods in a sequential and bilateral manner.
 - This is facilitated by the use of reduction screws below the apex of the deformity.

27.7 Surgical Pearls

- LIV is typically the vertebra that is most closely bisected by the sagittal sacral vertical plumb line, and this vertebra should be positioned distal to the first lordotic disk and rostral to a lordotic disk.
- Generous Ponte osteotomies and posterior releases around apex and into lumbar spine.

- Use of reduction screws to facilitate correction and rod reduction.
- Maintain appropriate thoracic kyphosis; do not overcorrect.
- Use a modified Wiltse approach at cephalad end of construct to protect soft tissue sleeve and lessen risk of proximal junctional kyphosis.

Common Clinical Questions

1. A 13-year-old skeletally immature boy is brought to your office because his mother is concerned about his poor posture. A lateral radiograph shows thoracic kyphosis of 68° . It is partially correctable on hyperextension. What is the initial course of management?
 2. What are the radiographic criteria for the diagnosis of Scheuermann's kyphosis?
 3. Based on current literature, what is the best way to prevent distal junctional kyphosis in a patient who undergoes fusion for Scheuermann's kyphosis?
-

Answers to Common Clinical Questions

1. A corrective TLSO worn 18+ hours to prevent progression of kyphosis.
 2. The radiographic criteria for the diagnosis of Scheuermann's disease are (1) more than 5° of wedging of at least three adjacent vertebrae at the apex of the kyphosis, (2) end plate irregularities, and (3) a thoracic kyphosis of more than 45° on lateral standing radiograph (14×36 cassette).
 3. LIV is typically the vertebra that is most closely bisected by the sagittal sacral vertical plumb line (SVPL), and this vertebra should be positioned distal to the first lordotic disk and rostral to a lordotic disk. When fused short of the SVPL, the rate of distal junctional kyphosis has been found to be dramatically higher.
-

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28 Adolescent Idiopathic Scoliosis

Lawrence G. Lenke and Todd M. Chapman Jr.

28.1 Key Points

- The natural history of adolescent idiopathic scoliosis (AIS) is predictable based upon the skeletal maturity and degree of scoliosis present.
- All patients should be examined thoroughly for neurologic findings, which are rare.
- The goals of surgical management include halting progression of scoliosis while leaving the patient as many mobile segments as possible and a well-balanced spinal alignment.

28.2 Background

- Adolescent idiopathic scoliosis is defined as a coronal Cobb measurement of 10° or more in children 10 to 18 years old.¹
- One percent to 3% of children in this age group will have curvatures of 10° or more, but only 0.15 to 0.30% will have a curve requiring treatment.¹
- Natural history of adolescent idiopathic scoliosis (AIS) varies depending on skeletal maturity and severity of curvature.²
 - Skeletal maturity is measured using bone age radiographs, Risser sign, menses, and chronologic age.
 - Peak height velocity is a time of maximal growth rate in adolescents that occurs 6 to 12 months before onset of menses in females.
 - Risser score is also an important variable and can predict risk of curve progression based on curve severity.
 - Curves of the same magnitude are less likely to progress in children who are more skeletally mature.¹
- Long-term studies confirm that at 50-year follow-up, 61% of AIS patients who were not treated will have back pain, but 70% of these will not be limited physically by this pain.³

28.3 Signs, Symptoms, and Physical Exam

- Children may present with a variety of physical symptoms or cosmetic signs or none at all. True neurologic findings are rare in AIS.

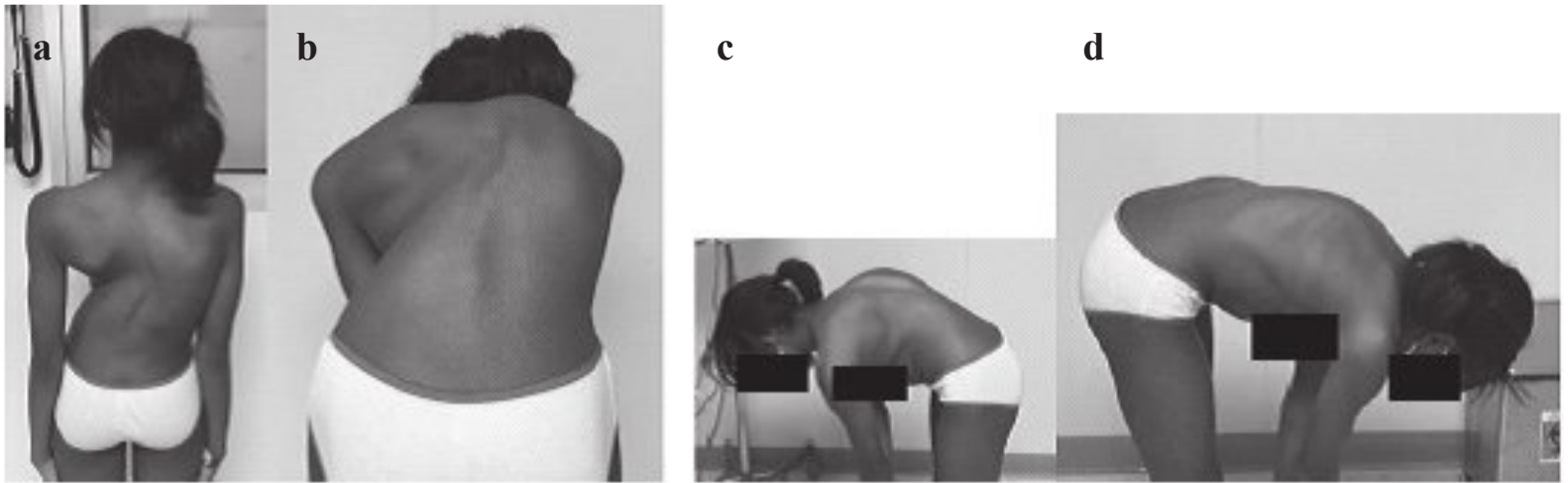


Fig. 28.1 (a–d) Representative clinical pictures of a patient who has prominent rib hump, waist crease, and shoulder height difference.

- Some patients will notice the physical appearance of their scoliosis. This could be waist asymmetry, shoulder height difference, or prominent rib hump, as seen in representative clinical pictures (**Fig. 28.1**).
- Other children who have greater curve magnitudes may present with back pain. This pain can be worsened with activity.
- A comprehensive history and physical exam must be undertaken, including the child's family medical history, personal birth history, and surgical and medical history.
- Physical exam includes a detailed neurologic exam of the entire upper and lower extremities including motor and sensory evaluations, abdominal reflexes, clonus, Babinski reflex, and extremity reflexes.

28.4 Work-up

- Standard work-up includes upright posteroanterior and lateral 36-inch scoliosis radiographs (**Fig. 28.2**).
- Preoperative work-up for AIS includes supine bending films to appropriately classify the curve according to the Lenke Classification System (**Fig. 28.3**).
- Additional work-up, such as total spine MRI, is indicated in patients who have presumed AIS and who have an atypical curve at presentation or abnormal curve progression.
- MRI is also indicated if any signs of upper or lower neurologic abnormalities are detected on physical exam or if pain is out of proportion to the deformity.

28.5 Treatment

- Total spine MRI, as discussed, may be indicated in certain cases.
- CT scan is avoided, if possible, in children. It can be helpful if there is any question of congenital etiology, such as failure of formation or segmentation.

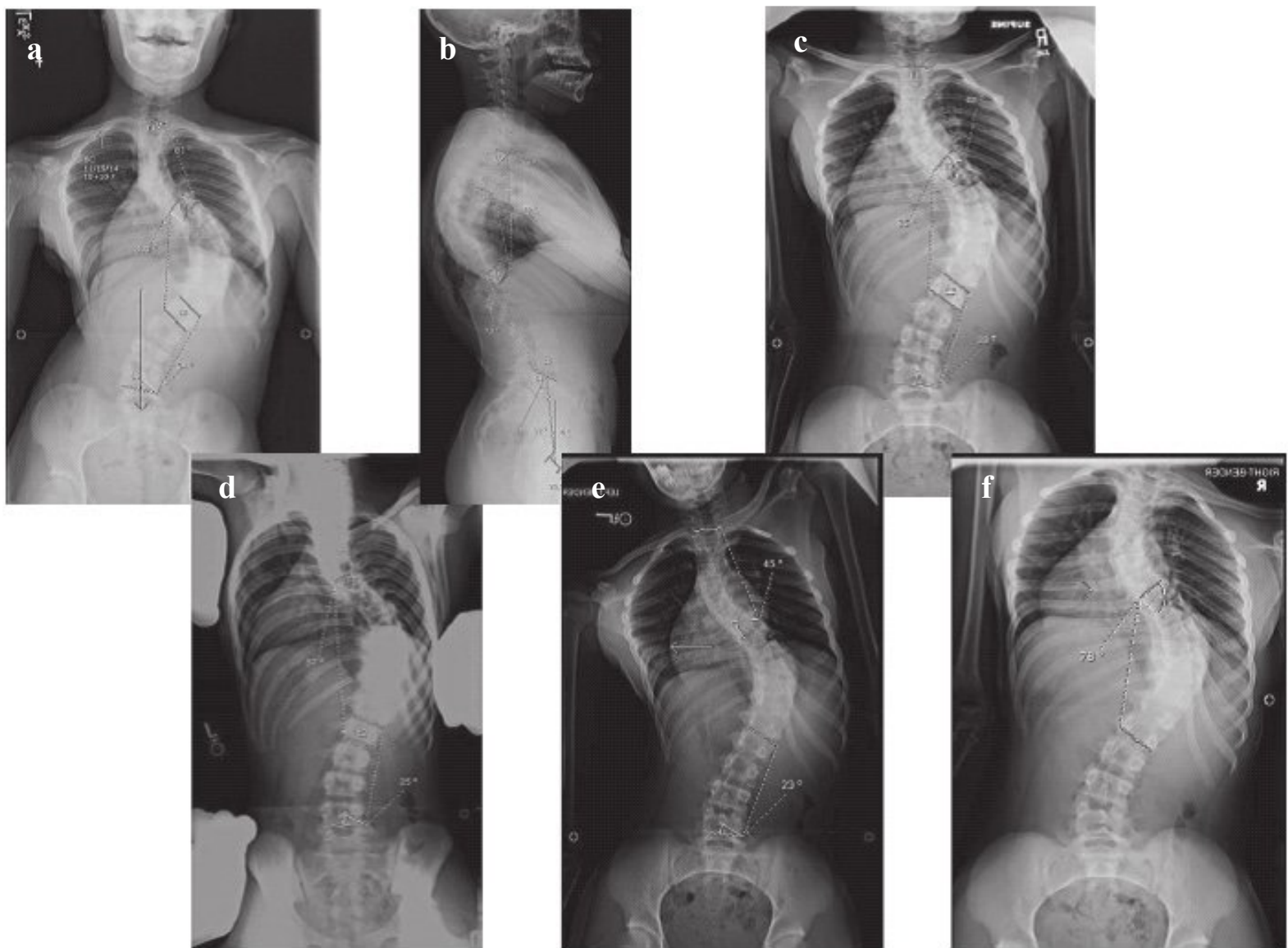


Fig. 28.2 Radiographic images. (a) PA, (b) lateral, and (c) supine, (d) push-prone, and (e) left bending and (f) right bending films of an 11-year-old female who has AIS.

28.5.1 Observation

- Observation⁴
 - Observation is indicated for skeletally immature children who have curves measuring less than 25°.
- In these children, close radiographic and clinical follow-up is recommended. As the typical AIS major curve progresses on average 1° per month, 6-month follow-up is appropriate.
- Skeletally mature patients who have little to no growth potential remaining and who have asymptomatic curves less than 50° may also be observed.

28.5.2 Bracing

- Bracing
 - Bracing remains a challenging issue for patients, parents, and physicians.
 - Bracing is recommended for skeletally immature patients who have curves measuring between 25 and 40 to 45°.
 - The functional goal of bracing is not to reverse the scoliosis but rather to prevent or slow the progression of the deformity by applying a force to the flexible, deformed spine.

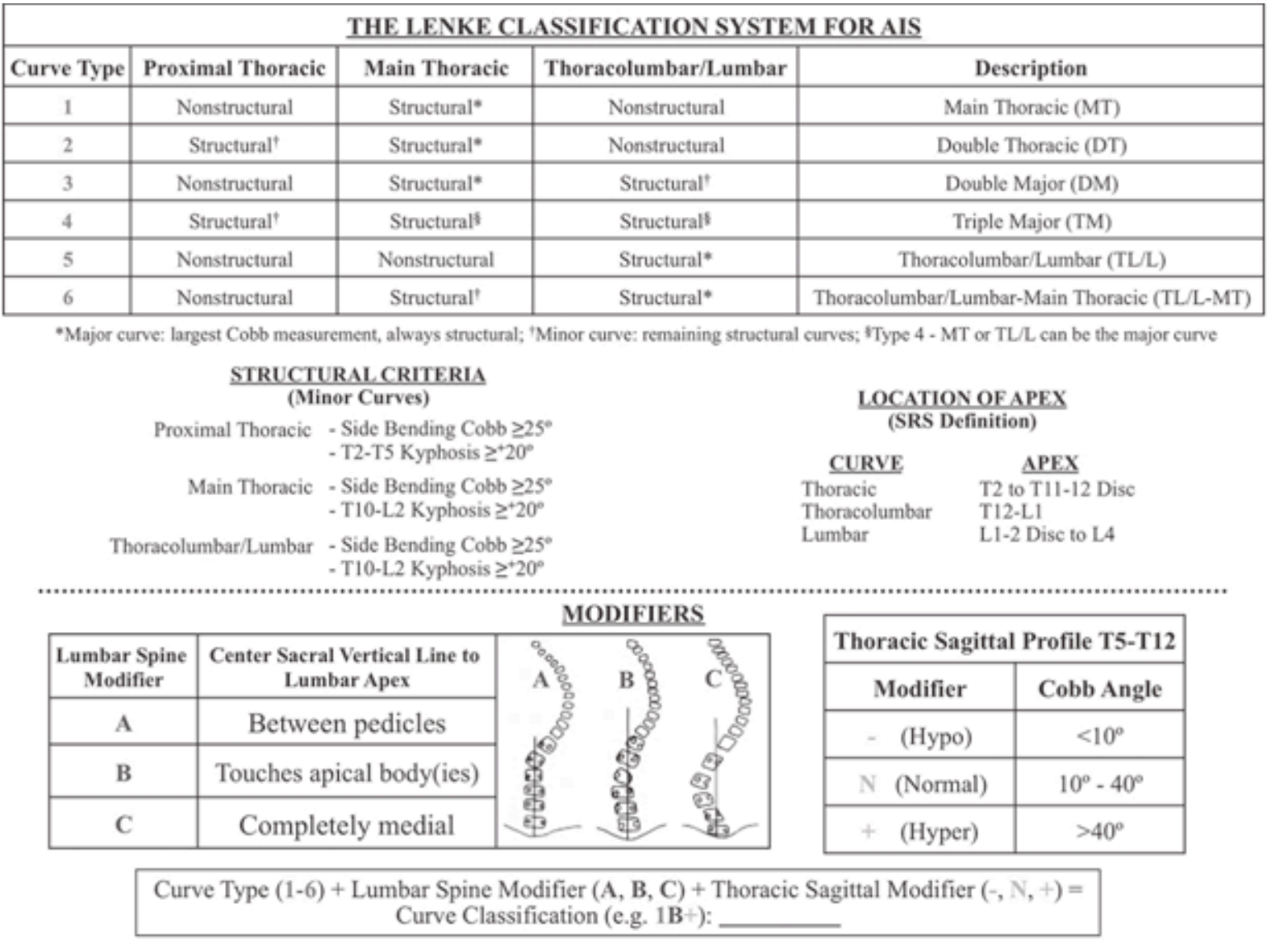


Fig. 28.3 Summary of the Lenke Classification.⁴ (Used with permission from Dr. Lenke. From Lenke IG, Betz RR, Harms J, et al. Adolescent idiopathic scoliosis: a new classification to determine extent of spinal arthrodesis. J Bone Joint Surg Am. 2001;83–A(8):1172.

- Bracing prescriptions vary based on curve type, but in skeletally immature patients younger than 15 years old, with curves ranging from 25 to 45°, there were similar rates of progression to surgery for braced and observed patients.⁵
- Compliance with brace wear is a significant driver of outcomes of bracing. High compliance with brace wear has, in a small prospective study, correlated with an 11% rate of curve progression. Low compliance with brace wear in this study, on the other hand, was correlated with a 56% rate of progression.⁶

28.5.3 Surgery

- The tenet of deformity correction in AIS is to provide reduction in the magnitude of the curvature without neurologic deficit and to achieve a fusion that will prevent further progression of the curvature regardless of whether anterior or posterior approaches are used.
- Anterior thoracotomy for fusion is associated with decreased pulmonary function immediately postoperatively, persisting for 2 years. No change

in pulmonary function is seen in thoracolumbar anterior fusion. There is significant increase postoperatively using all posterior instrumentation and fusion.⁷

- Posterior spinal instrumentation and fusion is the most commonly performed procedure for AIS. Three-column fixation of the spine with pedicle screws allows direct control of the vertebrae for correction of not only the coronal and sagittal deformities, but also the rotatory component of scoliosis.
- Upper instrumented vertebra (UIV) and lower instrumented vertebra (LIV) are determined based on the Lenke curve type, shoulder balance, flexibility, and remaining growth potential.
- Center sacral vertebral line (CSVL) touch vertebra (TV) rule can be used to determine the LIV and can preserve motion at some segments based on the most cephalad thoracolumbar/lumbar vertebral body touched by the CSVL.
- The addition of osteotomies of the spine, posterior column, pedicle subtraction, and vertebral column resection are safe adjuncts to posterior instrumentation to allow maximal correction in AIS cases.

28.6 Surgical Pearls

- The goal of surgery is to leave the patient with as many mobile, unfused segments as possible while obtaining a balanced, stable spine.
- Posterior pedicle screw constructs have been found to be a neurologically safe treatment for all AIS classifications and achieve significant reduction of spinal curvature with improved pulmonary function and clinical outcomes. Direct vertebral rotation with the use of pedicle screw constructs allows the surgeon to address the apical three-dimensional deformity of AIS.
- Osteotomies can be safely added to the surgical plan to dramatically correct AIS.
- The CSVL TV rule can be used to help determine the LIV and preserve motion segments in children who are past their peak height velocity stage of growth.

Common Clinical Questions

1. A 10-year-old premenarchal female returns to your clinic after she was found to have scoliosis on school screening. She presents with a 15° increase in curve magnitude since her last radiograph, 6 months ago. What is the most appropriate next step in her care?
 - A. Observation with follow-up in 6 months
 - B. Bracing
 - C. Surgical intervention with an all posterior instrumentation and fusion
 - D. MRI

2. Which of the following is the most predictive of risk for progression in AIS curves?
 - A. Age
 - B. Risser sign
 - C. Peak height velocity
 - D. Tanner stage

Answers to Common Clinical Questions

1. D. MRI is indicated in this patient, for the patient presents with a curve progressing at a rate faster than expected.
2. C. Peak height velocity is the time of maximal growth and also the time of highest risk of curve progression.

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29 Adult Degenerative Deformity

Salman Abbasifard and Ali A. Baaj

29.1 Key Points

- Adult degenerative deformity is characterized by abnormal coronal and sagittal spinal curvature in the adult spine.
- Asymmetric disk and facet joint degeneration are thought to be the most common causes of degenerative deformity.
- The most commonly presenting symptom is intractable back pain, but other symptoms such as radiculopathy and neurogenic claudication may also be present.
- The primary goals of surgery are to restore sagittal balance, decompress the neural elements and achieve stabilization and arthrodesis.
- The absolute degree of coronal curve correction in an adult deformity patient is less important than maintaining or restoring sagittal balance.
- Despite the high risk of complications in patients undergoing reconstructive surgery for adult degenerative deformity, most studies suggest a significant improvement in quality of life and high rate of patient satisfaction if the surgical objectives are achieved.

29.2 Background

- Spinal deformity is a dynamic process involving coronal, sagittal, and rotational elements. Any coronal plane deformity is called scoliosis, whereas sagittal plane deformity can result in kyphosis or hyperlordosis.
- Adult spinal deformity refers to abnormal curvatures of the spine in patients who have completed their growth.
- There are several terms used to describe the adult degenerative deformity; adult degenerative scoliosis, degenerative lumbar scoliosis, de novo scoliosis, collapsing scoliosis, senescent lumbar scoliosis, and primary degenerative scoliosis.
- The pathophysiology of adult degenerative deformity is associated with degenerative change.
- The asymmetrical degeneration of disk and facet joints puts an asymmetric load on the spine that lead to degenerative deformity.

- Degenerative changes in the vertebral body and disks are more obvious in adult degenerative deformity, commonly resulting in spinal stenosis.
- Degenerative deformity is characterized by a curve in the lumbar area and no compensatory curve in the thoracic vertebra, while adult idiopathic scoliosis has a major curve as well as compensatory curve similar in size to the major curve.
- Rotatory deformity is limited at the apex of the curve in adult degenerative deformity accompanied with lateral subluxation, while it is observed in the entire lumbar spine in adult idiopathic scoliosis.
- Sagittal imbalance is more common in adult degenerative deformity than adult idiopathic scoliosis.
- The risk factors of curve progression may be a curve with Cobb angles over 30° , an asymmetric disk above and below the apical vertebra, a lateral listhesis over 6 mm, an apical rotation greater than Grade II, osteopenia, and L5 vertebra being located above rather than below the intercrest line.
- Patient's age or sex is not associated with curve progression in adult degenerative deformity.
- Adult degenerative deformity curves tend to progress 1 to 6° per year (average 3° per year).^{1,2,3,4}

29.3 Signs, Symptoms, and Physical Exam

- The most common symptom of adult deformity is back pain.
- Back pain results from degenerative changes in the lumbar disk and facet joint, as well as from muscle fatigue.
- Spinal stenosis is present in only 31% of adult idiopathic scoliosis, whereas it is observed in 90% of adult degenerative deformity.^{1,2,3,4}
- Leg pain presents as intermittent claudication or radiating pain caused by spinal stenosis.
- Spinal stenosis is more frequently presents as foraminal stenosis than central stenosis, and more common on the concave side than on the convex side.
- In addition to scoliosis, sagittal and coronal imbalance is frequently observed in adult degenerative deformity, which causes improper posture leading to low back pain.
- Pain at the convexity is caused by fatigue of the paraspinal muscles or arises from the facet joints.
- Pain at the concavity of the curve is thought to be caused by destruction of the facet joints and degenerative changes in disk spaces.
- Radicular pain at the concavity can arise from narrowed foramina or ruptured disks causing radiculopathy.

29.4 Evaluation of Adult Deformity

- A detailed neurologic examination, including muscle power, reflexes, sensory testing, and gait testing, is necessary for patients with spinal deformity.
- Any degree of shoulder or pelvic asymmetry as well as pelvic obliquity and leg length discrepancy should be evaluated.
- The range of motion in hip and knee joints is evaluated to identify any contracture.
- Full-length (36-inch) standing posterior–anterior and lateral X-ray images should be obtained to assess global sagittal and coronal balance.
- All images should be compared with any previous films to assess curve progression.
- Lateral flexion and extension views may suggest curve flexibility and can reveal dynamic instability. However, adult degenerative lumbar deformity is typically a fixed deformity.
- If neurological symptoms are present, magnetic resonance imaging is usually recommended.
- In patients with significant risk factors for osteoporosis, dual-energy X-ray absorptiometry (DEXA) may be necessary.
- Assessment of previous treatments is essential. A trial of conservative treatment is mandatory before surgical intervention is considered.
- In surgical candidates, work-up may include cardiac echo and pulmonary function tests to ensure patients are fit for extensive spinal surgical reconstruction surgery.

29.5 Treatment

- Initial intervention might include weight loss, physical therapy, exercise, and oral nonsteroidal anti-inflammatory drugs (NSAID).
- Surgical intervention is considered if there is neurologic deterioration or high risk for curve progression, or if back pain becomes refractory to nonsurgical treatment.
- Surgical intervention requires careful patient selection, preoperative assessment and preoptimization to reduce the incidence of complications.
- The goal of surgery is to relieve back pain, improve radiating pain and claudication and correct deformity.
- Surgical options include decompression alone, decompression and limited short fusion, and decompression and long fusion with correction of deformity.
- Decompression alone may induce curve progression and worsening of subluxation.
- In large scoliosis curve and severe subluxation of the apical vertebra, correction of deformity with decompression and long segment fusion is required.

- Restoration of sagittal imbalance is achieved by anterior column release or additional posterior techniques such as osteotomies.
- Typical surgical constructs include thoracic to sacral or to iliac stabilization with pedicle screw and rod systems, coupled with osteotomy techniques to provide decompression and correction.
- Anterior interbody techniques can provide additional corrective capabilities and enhance arthrodesis capability.
- Surgery should be selected with a full understanding of the cause of symptoms while considering the advantages, disadvantages, indications, and complications of each surgical option.

28.6 Complications

- There are substantial surgical risks for patients with adult degenerative deformity because of their increased age and frequently associated medical comorbidities.
- The reported complication rates of adult deformity surgery are as high as 40% and the rates are even higher in the population older than 75 years and in revision setting.^{1,2,3,4}
- The most common complications are excessive blood loss, deep wound infection, and pulmonary embolism.
- Increase fusion level and longer operation time are associated with more blood loss. This is particularly the case where osteotomy techniques such as pedicle subtraction osteotomy or vertebral column resection techniques are employed.
- Late complications include adjacent segment disease, pseudarthrosis, and instrumentation failure.

Common Clinical Questions

1. What are the indications for surgery in adult degenerative deformity?
2. What are the most common complications of surgical intervention in adult degenerative deformity?

Answers to Common Clinical Questions

1. Surgical intervention is considered if there is neurologic deterioration or curve progression, or if back pain becomes refractory to nonsurgical treatment.
2. The most common complications are excessive blood loss, deep wound infection, and pulmonary embolus.

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30 Radiographic Parameters of Spinal Deformity

Khoi D. Than, Andrei Fernandes Joaquim, and Praveen V. Mummaneni

30.1 Key Points

- Sagittal balance, or a sagittal vertical axis less than 5 cm, is the most important radiographic parameter to accomplish in surgery for adult degenerative kyphoscoliosis.
- The ideal lumbar lordosis is within 10° of the pelvic incidence.

30.2 Background

- Normal or near normal sagittal balance has been directly correlated with clinical outcomes regarding pain and functional status.¹
- This sagittal balance is composed by the harmonious relation of the cervical, thoracic, and lumbar curvatures but also is influenced by the relationship between the pelvis and the spine. This relationship is better known as “spinopelvic balance.”²
- Assuming a normal distribution and negative values for kyphosis and positive values for lordosis, there is a wide variation of neutral upright curvatures.
 - The C2–C7 angle (the greatest variation in regional curves) ranges from -11° of kyphosis to $+45^\circ$ of lordosis.
 - The thoracic kyphosis (T1–T12) ranges from -25 to -65° .
 - The lumbosacral lordosis (T12–S1) ranges from $+40$ to $+84^\circ$. Of note, $\sim 70\%$ of the total lumbosacral lordosis occurs between the superior end plate of L4 to the superior end plate of S1.

30.3 Signs, Symptoms, and Physical Exam

- Initial clinical evaluation of patients includes assessment of their main complaints and comorbidities.
- Physical examination includes assessment of neurological deficits, body habitus, posture, muscle contractures (hip flexion, knee flexion, etc.), pelvic obliquity, shoulder imbalance, and limb deformities.

30.3.1 Neuroimaging

- Radiographic assessment includes static standing full-length 36-inch X-rays with anteroposterior and lateral views of the whole spine to evaluate sagittal balance and measure the lumbo–sacral–pelvic parameters.
- Additional radiographic evaluation may include a lumbar MRI or CT myelogram (for patients with previous lumbar instrumentation), dynamic lumbar radiographs to help identify mobile listhesis, and side-bending X-rays in scoliosis cases to evaluate curve flexibility. Assessment of bone density is also of importance in patients who will undergo surgical treatment.
- Several forms of measurements have been proposed to assess a patient's global sagittal balance (**Fig. 30.1**).

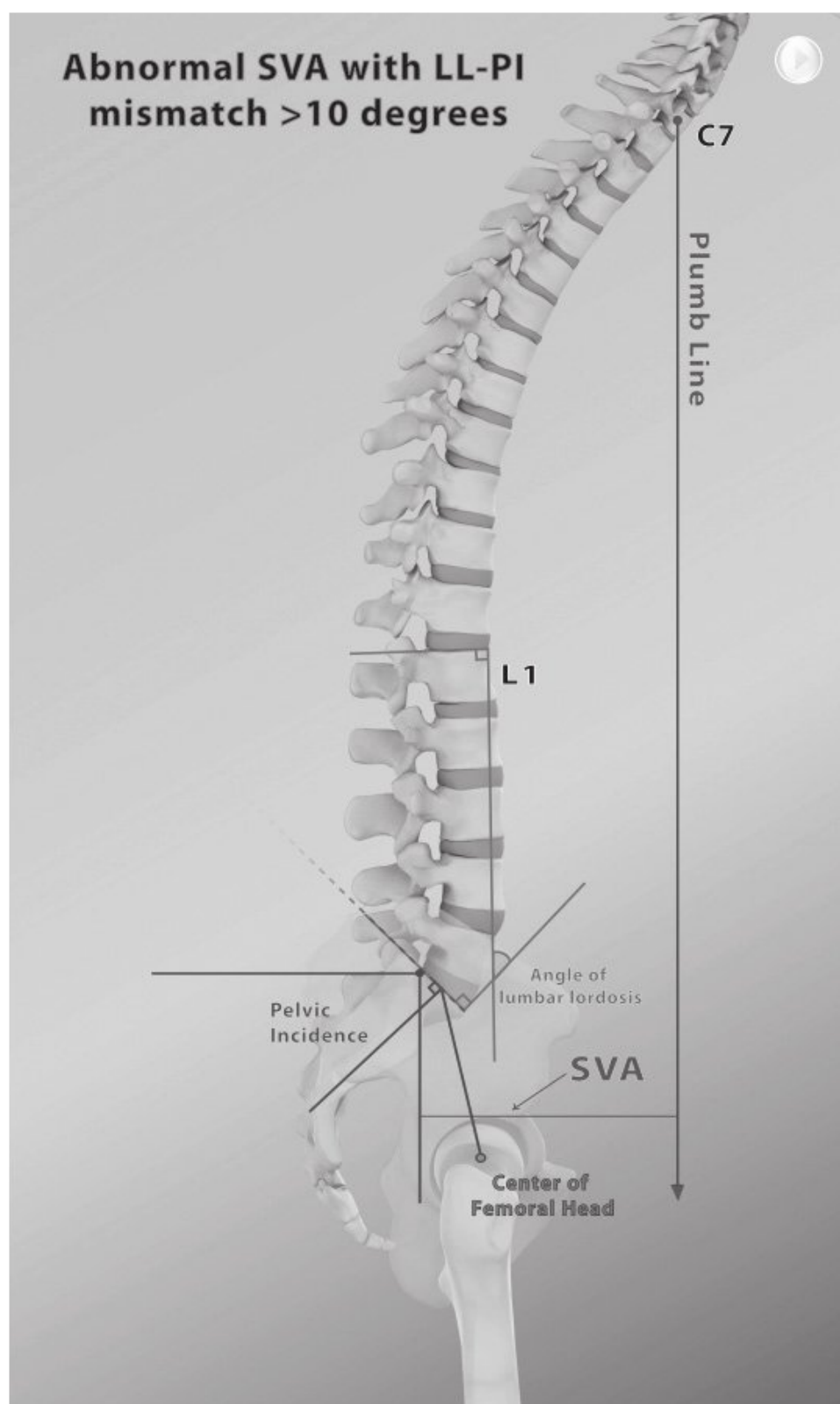


Fig. 30.1 Calculation of sagittal vertical axis (SVA), pelvic incidence (PI), and long lordosis (LL).

- One important parameter is a vertical line drawn from the center of the C7 vertebral body with respect to the posterior superior corner of the S1 end plate (the “sacral reference point”). This “C7 plumb line” should be no further than 5 cm from the sacral reference point in normally aligned individuals. The C7–SVA (sagittal vertical axis) has a positive value when the vertical plumb line is anterior to the sacral reference point and a negative value when posterior to it.
- Pelvic incidence (PI) is defined as the angle formed by a line drawn from the midpoint of the approximated center of the femoral heads to a line perpendicular to the center of the sacral end plate. The PI is established at the end of adolescence and is a constant value unaffected by body posture. Normal values range from 48 to 55°.
- Pelvic tilt (PT) is defined as an angle formed between a vertical reference line at the midpoint of the sacral end plate and a line drawn from that point to the midpoint of the approximated center of the femoral heads. Normal values range from 12 to 18°.³
- Sacral slope (SS) is defined as an angle formed between a horizontal reference line and the sacral end plate. Normal values range from 36 to 42°.
- From a geometrical point of view, $PI = SS + PT$. Spinopelvic alignment is normal when there is a direct relationship between the PI, SS, and long lordosis (LL). An elevated PT is a sign of compensated sagittal imbalance.
- Considering that PI cannot be changed, surgeons must pursue an adequate LL when planning surgeries to achieve a normal sagittal balance and improve patients’ final outcome. To determine the ideal LL, we can use the equation $LL = PI \pm 10^\circ$.

30.4 Treatment

- Nonoperative management of sagittal and coronal imbalance can be performed in patients who have good functional status and who have mild pain. Physical therapy, stretching, and nonsteroidal anti-inflammatory drugs are the primary conservative treatment options. Spinal epidural steroid injections may benefit some patients who have radicular pain.
- Operative treatment is recommended in patients who have moderate to severe symptoms, patients who fail conservative measures for pain control, and patients with neurological deficits. Surgical planning is highly dependent on the radiographic parameters discussed above. Additional factors that may influence surgical planning are patients’ comorbidities, previous surgeries, bone quality, and body habitus.
 - Decompression without fusion can be considered in patients without evident instability, in patients without severe sagittal imbalance (mild curves), and when symptoms are secondary to nerve root compression

(i.e., in patients with minimal axial low back pain). Minimally invasive decompressive techniques can be used to decrease blood loss and hospital stay. The main limitations of decompressive techniques are the inability to restore sagittal balance and improve axial back pain. There is also a small risk of later instability secondary to decompression.

- Anterior approaches include the ALIF (anterior lumbar interbody fusion) and LLIF (lateral lumbar interbody fusion) techniques. Potential advantages of these approaches include indirect decompression via expanded foraminal height, as well as avoidance of posterior lumbar muscle trauma. The LLIF technique can also improve coronal spine alignment, especially from L1 to L4 levels (the L5–S1 level is limited by the iliac crest). ALIF may have approach-related complications, such as retrograde ejaculation or vascular injuries. LLIF has some risk of lumbar plexus injuries as the psoas muscle is traversed. In general, anterior spine techniques are useful to restore lumbar lordosis and to provide interbody fusion with indirect nerve root decompression but most of the time will require a combined posterior approach to achieve normal or near normal restoration of the spinopelvic measurements.
- The majority of spinal deformities are treated by a posterior approach with instrumentation and fusion, with or without an anterior approach. Posterior approaches allow direct decompression of a stenotic spinal canal or foramina. Advantages of the posterior approach include direct decompression, the possibility to perform interbody fusion using posterior or transforaminal techniques (posterior lumbar interbody fusion or transforaminal lumbar interbody fusion, respectively), and access to perform spinal osteotomies that can correct coronal and sagittal imbalance.
- Osteotomies are used to attain coronal and sagittal deformity correction, especially in rigid deformities. The goal of spinal osteotomies is to restore the SVA to under 5 cm and to match LL within 10° of the PI. The chosen osteotomy technique may vary according to the degree of correction required to restore sagittal balance. Smith-Petersen osteotomy (SPO) includes resection of the adjacent facet joints, lamina, and ligaments of the involved level and may correct up to 10° of lordosis per level.
- The pedicle subtraction osteotomy (PSO) consists of resection of the facet joints, lamina, pedicles, and wedge-shaped portion of the vertebral body. Closure of the PSO can result in up to 30° of lordosis correction and significant improvement of the SVA. The complication potential of PSO compared with SPO is higher, including blood loss and neurological deficits.⁴
- In cases in which there is a rigid coronal and sagittal deformity, surgeons can perform a vertebral column resection, which consists

of resection of the vertebral body, the adjacent disks, and all the posterior elements of the affected level, resulting in the most significant correction rate of all osteotomies but having a significant risk of new neurological deficits and blood loss.

- After surgery, patients must receive adequate pain management. Early ambulation is recommended, in addition to a rigorous physical therapy program. Nutritional support is also started for adequate bone healing, as well as for stabilization of any concomitant comorbidity. Antibiotics are prescribed for the duration of drain placement, although this is practitioner-dependent.

30.5 Surgical Pearls

- Because of patient medical comorbidities, clinical status, or even unrealistic expectations, less extensive surgical procedures may sometimes be preferred over total correction of spinopelvic parameters and sagittal balance. The choice of the surgical plan depends on many factors, but a surgeon's judgment is of paramount importance to consider a realistic procedure in light of the patient's general condition.
- Iliac screw fixation should be considered when there is a high risk of lumbosacral junction pseudarthrosis, such as in patients who have poor bone quality, long-segment fusion to the sacrum (L2 or above to S1), treatment of L5–S1 pseudarthrosis, high-grade spondylolisthesis, and inadequate sacral fixation with screws. Unilateral iliac screw fixation has demonstrated similar results to bilateral iliac screw fixation, avoiding added muscular and fascial dissection, decreasing surgical operative time, and possibly decreasing the risk of infection.⁵
- Minimally invasive deformity surgery can be used for select patients with mild adult deformity, such as those with flexible curves, LL–PI mismatch of less than 10°, SVA of less than 6 cm, PT less than 25°, minimal lateral listhesis (< 6 mm), coronal Cobb angle of less than 20°, and no thoracic hyperkyphosis.⁶

Common Clinical Questions

1. Above what value is the sagittal vertical axis considered abnormal?
2. How does one measure pelvic incidence?
3. What is the ideal relationship between lumbar lordosis and pelvic incidence?

Answers to Common Clinical Questions

1. 5 cm.
2. Pelvic incidence (PI) is defined as the angle formed by a line drawn from the midpoint of the approximated center of the femoral heads to a line perpendicular to the center of the sacral end plate.
3. They should be within 10° of one another.

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31 Vascular Pathology of the Spine

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31.1 Key Points

- The Spetzler et al nomenclature for spinal vascular lesions classifies arteriovenous malformations (AVMs) and arteriovenous fistulas (AVFs) according to anatomic location. Lesion classification does not rigidly dictate optimal treatment strategy but may provide a useful framework to guide decision making.¹
- Spinal angiography is the gold standard and is warranted for all arteriovenous lesions.
- Thorough understanding of spinal angiographic anatomy, surgical vascular anatomy, and segmental variability is essential for decision making.
- A high index of suspicion should be maintained in the presence of spinal vascular lesions. Protean clinical findings, imaging appearance, and low incidence often result in diagnostic delay.

31.2 Cavernous Malformations

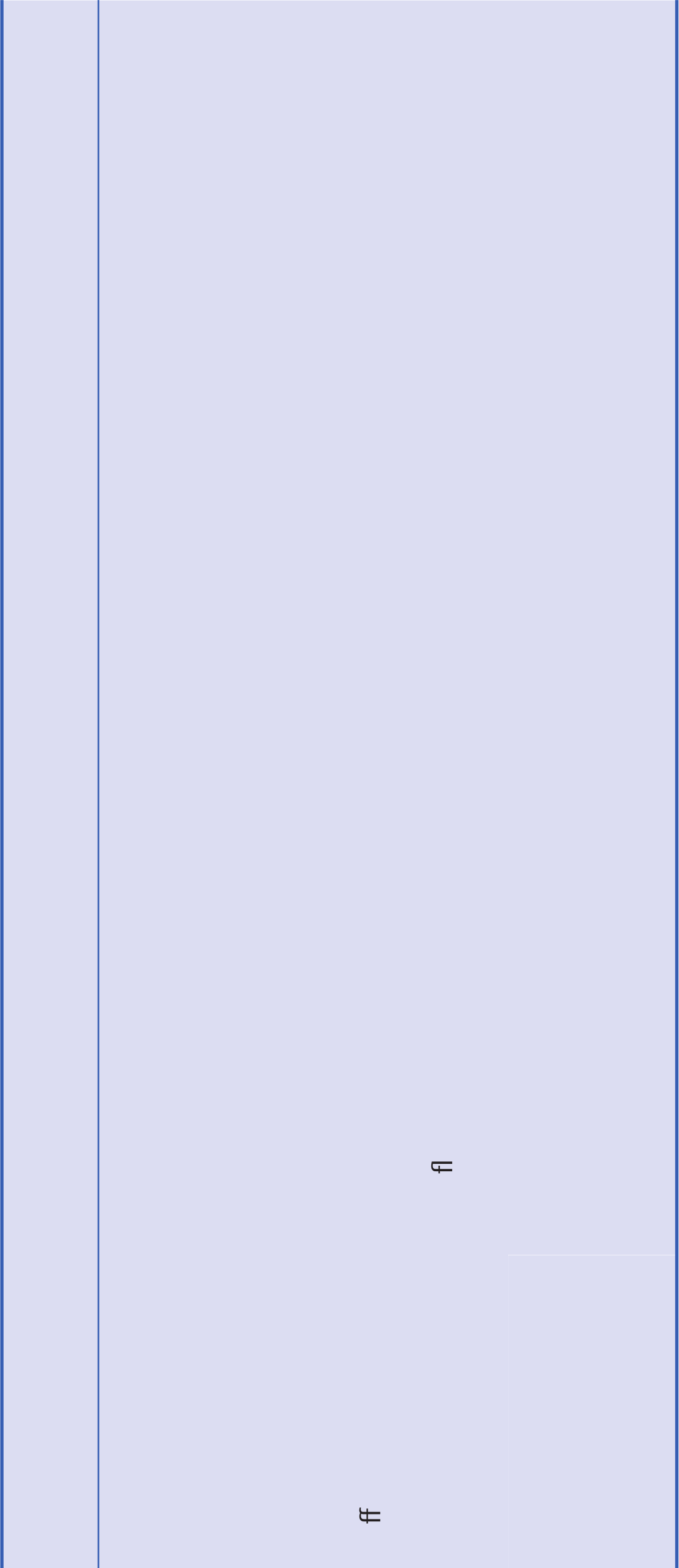
- Background
 - Cavernous malformations are benign vascular neoplasms that may occur in sporadic or familial forms. Spinal cavernomas favor thoracic over cervical locations, with lumbar next in frequency.
 - Intramedullary is the most common location, although intramedullary exophytic, intradural extramedullary, and extradural locations have been reported.
 - Peak incidence of symptomatic hemorrhage is reported in the fourth decade.
- Signs, symptoms, and physical examination
 - Acute: Due to large hemorrhage, may result in long-tract dysfunction or radicular symptoms (including pain) depending on location; acute meningeal signs are rare.
 - Progressive decline or stepwise deterioration: Due to repeated hemorrhages and/or hemosiderin toxicity; improvement between events is usually incomplete.
 - Estimated bleed rates have been reported at 1.4 to 4.5% per patient-year. The rate of subsequent hemorrhage may approach 66% per patient-year.

- Work-up
 - MRI with gradient-recalled echo (GRE) sequences.
 - Detailed family history, MRI of the brain, and possible familial screening should be considered. As many as 50% of patients may harbor intracranial cavernomas as well.²
- Neuroimaging
 - MRI is the modality of choice. Cavernomas are angiographically occult.
 - MRI appearance: T1 and T2 “popcorn” heterogeneity reflects vascular sinusoids containing blood products of different ages. Surrounding T1 and T2 hypointensity reflects the rim of hemosiderin-stained parenchyma. Hypointense “blooming” notable on GRE sequences, with absence of flow voids.³
- Treatment
 - Gross total excision is the treatment goal and is protective against future hemorrhage.
 - Conservative management is a consideration for small asymptomatic lesions, especially deep-seated lesions that fail to reach the pial surface on axial images.^{2,4}
- Surgical pearls
 - Appropriate zones of entry from a posterior or posterolateral approach include midline myelotomy, dorsal lateral sulcus, or laterally between the dentate ligaments.
 - Whenever possible, sharp dissection is preferred. Piecemeal excision is common.
 - Care is necessary to preserve hemosiderin-stained parenchyma surrounding the cavernoma. The developmental venous anomaly should be preserved whenever possible.^{2,4}

31.3 Arteriovenous Lesion

- Background
 - Intradural–dorsal AVF
 - Considered an acquired lesion, attributed to venous outflow dysfunction.
 - Low-flow fistula involves radiculomedullary artery as it pierces the dural root sleeve, resulting in arterialized coronal venous plexus. Type B lesions recruit feeders from adjacent levels, but a single fistulous point is always present.
 - Accounts for 60 to 80% of spinal vascular lesions. Typically affects males more than females, ages 40 to 60, and has predilection for the thoracolumbar spine.

- Intradural ventral
 - High-flow anastomosis between anterior spinal artery (ASA) and ventral venous plexus. Varix formation, flow rate, complexity, and multiplicity of feeding pedicles increase with types A to C.
 - Occurs in younger patients (20–60 y) and favors the thoracolumbar spine.
- Extradural
 - High-flow direct anastomosis between epidural artery and vein may receive multisegmental arterial contributions.¹
 - Sporadic de novo formation, congenital, syndromic associations (e.g., neurofibrosis (NF)-1), and traumatic etiologies have all been reported (**Table 30.1**).
- Extradural–intradural AVM
 - High-flow AVM, irrespective of tissue boundaries. Commonly involves multiple or entire spinal segments. May interdigitate with functional cord tissue. Commonly fed by ASA and posterior spinal artery (PSA). Rare.
- Intramedullary AVM
 - High-flow AVM with diffuse and compact forms. High-risk features, including varix formation and associated aneurysms, are common.
 - Typically symptomatic early in life (10–30 y), with predilection for the cervical spine depending on report; represents 15 to 20% of all spinal vascular lesions.
- Conus AVM: High-flow, complex shunting pattern found at conus. Multiple shunts and nidi may be pial or intramedullary. Rare (**Table 30.2**).¹
- Signs, symptoms, and physical examination
 - Intradural–dorsal AVF: Protean history of progressive myelopathy, dominated by gait and sphincter dysfunction. Pathophysiology relates to venous vascular congestion.
 - Intradural ventral: Similarly protean, progressive, and variable history dominated by myelopathic findings. Symptomatic progression resulting from steal, compression, and hemorrhage increases with grade and flow rate.
 - Extradural: Myeloradiculopathy is common, with symptoms due to compression, steal, and hemorrhage. Venous congestion is typically rare.
 - Extradural–intradural AVM: Malignant natural history characterized by progressive myelopathy. Radicular pain referable to the involved segment is typical. Pathophysiology involves mass effect, steal, hemorrhage, and compression.



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- Intramedullary AVM: Malignant natural history characterized by progressive myeloradiculopathy. Stepwise deterioration or acute decline attributable to repeated hemorrhage, steal, and compression.
- Conus AVM: Progressive myeloradiculopathy, symptoms referable to conus.¹
- Work-up
 - Spinal MRI/magnetic resonance angiography (MRA) has proven especially useful for intradural dorsal lesions. MRA may be sufficiently sensitive to identify the fistula type, to pinpoint the level(s) of the shunt, and to confirm treatment at follow-up.
 - Spinal angiography, however, remains the gold standard imaging modality (for all spinal AVFs and AVMs) and may be directed more precisely after careful inspection of prior MRA.
- Neuroimaging
 - AVMs
 - MRI is most useful to delineate the size and configuration of the nidus (compact versus diffuse), to evaluate for hemorrhage, and to assess the angioarchitecture.
 - Intradural dorsal AVF
 - MRI: In the appropriate clinical setting, extensive intramedullary T2 hyperintensity along with intradural flow voids along the dorsal pial surface is nearly pathognomonic. Enhancement may be variable.
 - Angiography: Venous outflow is sluggish, and long venous phase may be necessary. Suspected but occult intradural–dorsal fistulas require surgical exploration.
 - Intradural ventral
 - MRI: Reveals variable T2 hyperintensity and/or enhancement with dilated flow voids along the ventral surface of the spinal cord. Identifies varix formation.
 - Angiography: Reveals a direct fistula between the radiculomedullary artery and ASA. The ASA is identified by the classic hairpin loop and may be displaced from midline. The ventral arterialized vein often harbors a prominent venous varix.
 - Extradural: MRI/MRA: Prominent epidural flow voids and enhancement, typically with significant mass effect. Variable T2 intramedullary signal change.³
- Treatment
 - Intradural–dorsal AVF: Microsurgical clip occlusion at intradural fistulous point remains the gold standard. Embolization with liquid agent is now an acceptable first-line treatment but is associated with greater risk of recurrence.⁵
 - Intradural–ventral AVF: Microsurgical fistula obliteration remains the treatment of choice, but embolic strategies are often used as well.

Vessel caliber (particularly type A and B) and proximity to ASA present challenges to endovascular management.

- Extradural AVF: Large-caliber feeding vessels favor coil embolization.
- Extradural–intradural AVM: Curative resection is atypical. Multimodal palliation strategies ameliorate symptoms as a result of steal, compression, and/or hemorrhage.
- Intramedullary AVM: Surgical resection, typically via a posterior or posterolateral approach, remains the gold standard and is typically preceded by attempts at embolization in appropriately selected patients.⁶
- Conus AVM: Multimodal strategies include aggressive embolization and resection.¹
- Surgical pearls
 - Indocyanine green angiography (ICG) is ideal for intraoperative confirmation of location and essential for the identification of angiographically occult lesions.
 - Intramedullary AVMs are commonly fed by at least one branch of the ASA. Steal phenomena may obscure ASA involvement on preoperative angiography, but serial runs midresection may reveal the ASA as shunting is progressively eliminated.
 - Arterial supply is circumferentially addressed first for intramedullary AVMs. Venous outflow is spared until the lesion is sufficiently devascularized.

Common Clinical Questions

1. The Spetzler et al nomenclature divides arteriovenous lesions solely on the basis of what characteristic?
2. What imaging modality is the gold standard and is mandatory for all spinal arteriovenous lesions?
3. Describe the typical imaging findings of an intradural–dorsal fistula on MRI.

Answers to Common Clinical Questions

1. Anatomic location; all other characteristics follow suit.
2. Angiography.
3. Dorsal flow voids, extensive T2 signal change, variable and patchy enhancement.

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32 Spondyloarthropathies

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32.1 Key Points

- Spondyloarthropathies are subdivided into seronegative and seropositive arthropathies (positive versus negative antinuclear antibody [ANA] or rheumatoid factor [RF]).
- A family of described disorders/syndromes with overlapping symptomology with varying human leukocyte antigen (HLA) associations, including ankylosing spondylitis (AS), psoriatic arthritis, enteropathic arthritis, Reiter syndrome, ossification of the posterior longitudinal ligament (OPLL), and rheumatoid arthritis (RA).
- RA with high incidence of C-spine involvement ($> 85\%$).¹
- In RA, atlantoaxial subluxation (AAS) classified by anterior atlantodental interval (ADI) ≤ 3 mm (stable) and posterior atlantodental interval (PADI) ≤ 14 mm (increased risk of injury).
- OPLL: Asians, mostly asymptomatic, surgical approach controversial. CT for evaluation of posterior longitudinal ligament (PLL) calcification/ossification.
- Diffuse idiopathic skeletal hyperostosis (DISH): Sacroiliac sparing, osteophytic change, associated with globus and dysphagia.

32.2 Ankylosing Spondylitis Seronegative Arthropathy

- Background
 - Also called Marie-Strümpell disease.¹ Affects 0.1 to 0.2% of the population, with a male: Female ratio of 3:1. Peak age of onset: Teens to fourth decade of life.²
 - A chronic systemic inflammatory disease (strongest correlation with HLA-B27). CD4, CD8, and cytokine (tumor necrosis factor [TNF]– α and TNF- β) mediated.^{3,4,5}
 - AS specifically involves the sacroiliac joints and progresses to involve the entire spine. It may also variably involve peripheral joints, eyes, skin, and the cardiac and intestinal systems.
 - Increased risk (up to 20%) if there is a first-degree relative with HLA-B27 and AS.⁶

- Enthesitis: Chronic inflammation at the insertion point of tendons that leads to ossification.⁵
- Enthesopathy leads to osteoporosis of vertebral bodies and disk with sparing of nucleus pulposus (bridging osteophytes), producing the so-called bamboo spine.¹
- Signs, symptoms, and physical exam^{1,5}
 - Initial nonradiating back pain and morning stiffness (> 45 min) that improves with exercise/activity
 - Stiffness of the gluteal and lumbosacral junction (sacroilitis)
 - Progression of symptoms to entire spinal axis
 - Eventual decrease in range of motion (autofusion)
 - Tendon/ligament involvement (plantar fasciitis/Achilles tendonitis)
 - Differential diagnosis: Rheumatoid arthritis (RF+), DISH (also called Forestier disease) (spares facet and sacroiliac [SI] joint, later age of onset than AS), psoriatic arthritis/Reiter syndrome (milder with asymmetric sacroiliitis)
 - The Patrick or FABERE (flexion, abduction, external rotation, extension) test: Flex hip, flex knee, and place the lateral malleolus on contralateral knee; then press the ipsilateral knee downward (by stressing the hip joint); will cause pain in bursitis, sacroiliitis, and other hip joint pathology
 - Cauda equina syndrome: Usually no obvious etiology or compression
 - Unstable rotatory subluxation (occipitoatlantal or atlantoaxial joint)
 - Myelopathy: Due to bow stringing of cord
 - Significant fractures with minimal trauma
- Work-up
 - Modified New York criteria (radiographic sacroiliitis and back pain > 3 months, limited spine motion in sagittal/frontal planes, or limited chest expansion)
- Neuroimaging
 - Plain X-rays (crucial for diagnosis): Obtain X-rays of entire spine (bamboo spine) and pelvis (SI joint) (**Fig. 32.1**)
 - MRI: For further evaluation of disk spaces and ligamentous changes
 - Bone scan: Increased uptake at SI joint
- Treatment
 - Medical management with nonsteroidal anti-inflammatory drugs (NSAIDs), sulfasalazine, TNF- α antagonists, and corticosteroids
 - Stable fracture: Treatment with rigid brace
 - Surgical decompression for inpatients with unstable fractures and/or progressive neurologic deficit
- Surgical pearls^{1,5}
 - Special care is needed with routine neck immobilization after trauma intraoperatively in patients with AS. These patients tend to have the

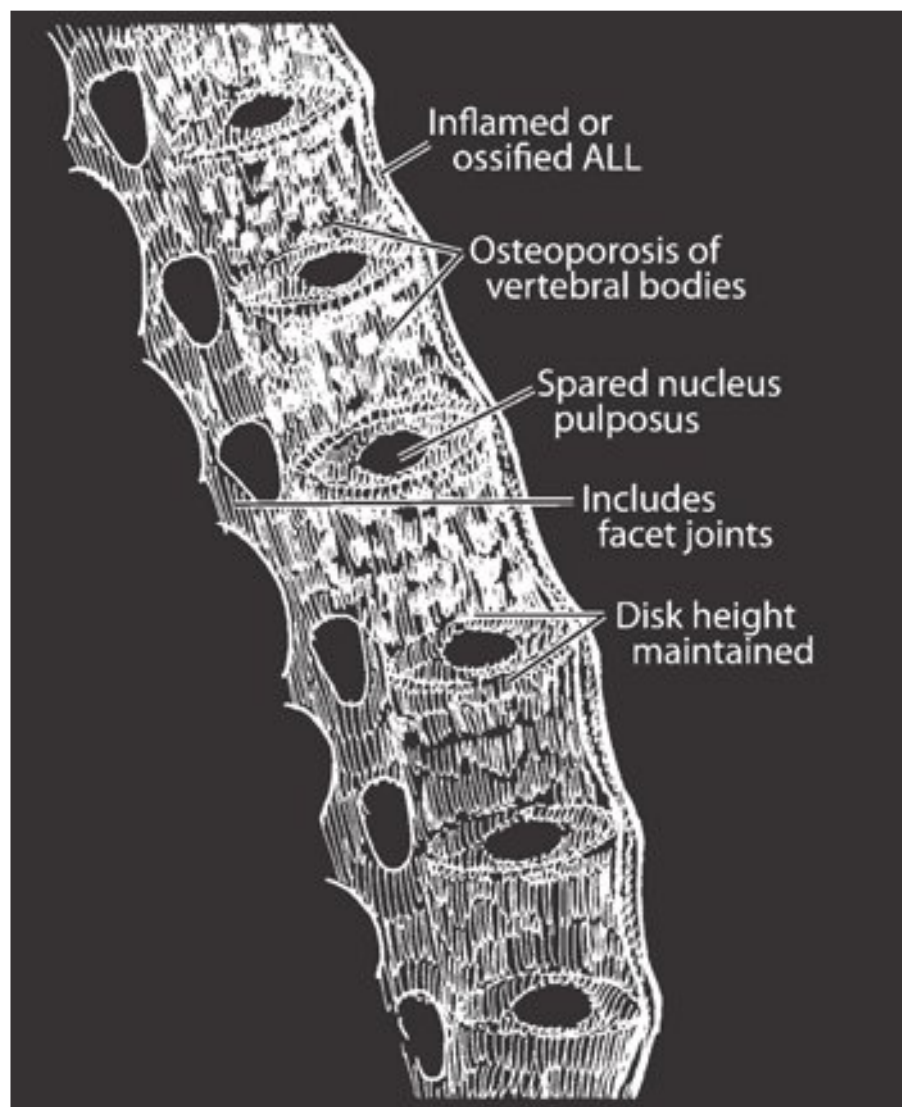


Fig. 32.1 “Bamboo spine” in ankylosing spondylitis.

neck fixed in flexed position. Forced inline fixation may be deleterious, so fix neck in natural position.

- If patient is placed in traction, special attention to degree of kyphosis is necessary. Traction must be in line with patient’s natural kyphosis and not directly horizontal.
- No consensus on treatment for cord injury in AS patients (halo versus internal fixation) without obvious compression.
- Root/cord compression: Laminectomy and fusion recommended.
- Posterior approach should be strongly considered secondary to anterior bridging osteophytes and concerns with fixation of anterior plate for osteoporotic vertebral body.
- Consider posterior osteotomy and fusion for correction of severe kyphotic deformity.

32.3 Rheumatoid Arthritis

- Background
 - Very high incidence of C-spine involvement (atlantoaxial subluxation [anterior > posterior], basilar impression, pannus granulation of odontoid)^{5,7}
 - 2:1 female:male ratio; peak incidence in fourth to fifth decade of life

- Serum RF+: 1 to 2% prevalence
- AAS: Erosion at C1–C2 joint and at transverse ligament insertion
- Signs, symptoms, and physical exam
 - Morning stiffness, symmetric multijoint arthritis (particularly the proximal interphalangeal [PIP], metacarpophalangeal [MCP], and metatarsophalangeal [MTP]), rheumatoid nodule (extensor surface)
 - Radiographic decalcification at joints (X-ray hand)
 - Neck pain with possible C2 radiculopathy
 - Headache, paresthesias, difficulty with ambulation, and signs of cervicomedullary junction compression (basilar impression)
- Work-up
 - ADI: ≤ 3 mm for evaluation of integrity of the transverse ligament; does *not* correlate with risk of injury.⁸
 - PADI: Essentially the amount of space for the cord at C1–C2; *does* correlate with risk of injury if ≤ 14 mm.^{5,8}
 - Look for basilar impression and cervicomedullary junction compression.
- Neuroimaging
 - Lateral C-spine X-ray: ADI (anterior aspect of odontoid to arch of C1)/ PADI (posterior aspect of vertebral body to spinolaminar line)
 - MRI for evaluation of degree of stenosis (pannus)
- Treatment
 - Surgical treatment of asymptomatic AAS can be considered when ADI > 6 to 10 mm.⁹
 - AAS in RA will progress with time, so treatment is recommended, especially if myelopathy exists.
 - Rigid collar does not support C1–C2 and is thus a poor option.
 - Immobilization of the C1–C2 joint via halo or posterior fusion alone may reduce the size of pannus over time.
 - May use halo traction to align the odontoid and return it to its neutral position (start with 5 lbs.).
- Surgical pearls⁵
 - Anterior, posterior, or vertical subluxation: Most cases will require 360° of fusion.
 - Rotational or lateral subluxation: Posterior-only approach is adequate.
 - Posterior fusion with or without laminectomy (C1): C1–C2 fusion, C2–occiput
 - Anterior approach: Odontoidectomy. For transoral approach, the patient's mouth must be able to open at least 25 mm.¹ Patient is to remain in halo traction until fusion.
 - RA with concomitant basilar impression should first be reduced and then fused.
 - Transoral approach is associated with higher morbidity and usually reserved as a second option to posterior fusion.

32.4 Diffuse Idiopathic Skeletal Hyperostosis

- Background
 - Significant osteophyte formation in absence of significant degeneration. Distinct from degenerative disease, OPLL, and AS (**Fig. 32.2**).¹⁰
 - Males in seventh decade of life.
 - SI joint spared.¹¹
 - Osteophytes do not stabilize, and unfused they are unstable. Minor trauma can lead to significant injury.
- Signs, symptoms, and physical exam
 - Morning stiffness (milder than with AS)
 - Globus: Sensation of lump in the throat, secondary to large anterior vertebral body osteophyte adjacent to esophagus
 - Dysphagia with or without weight loss¹
- Neuroimaging/work-up/treatment
 - Dysphagia: Speech therapy consult to rule out primary esophageal pathology, diet modification, barium swallow (to localize obstruction). Progressive dysphagia with weight loss may benefit from surgical debulking.⁵
 - CT scan is superior to X-ray for evaluation of osteophytic structures.

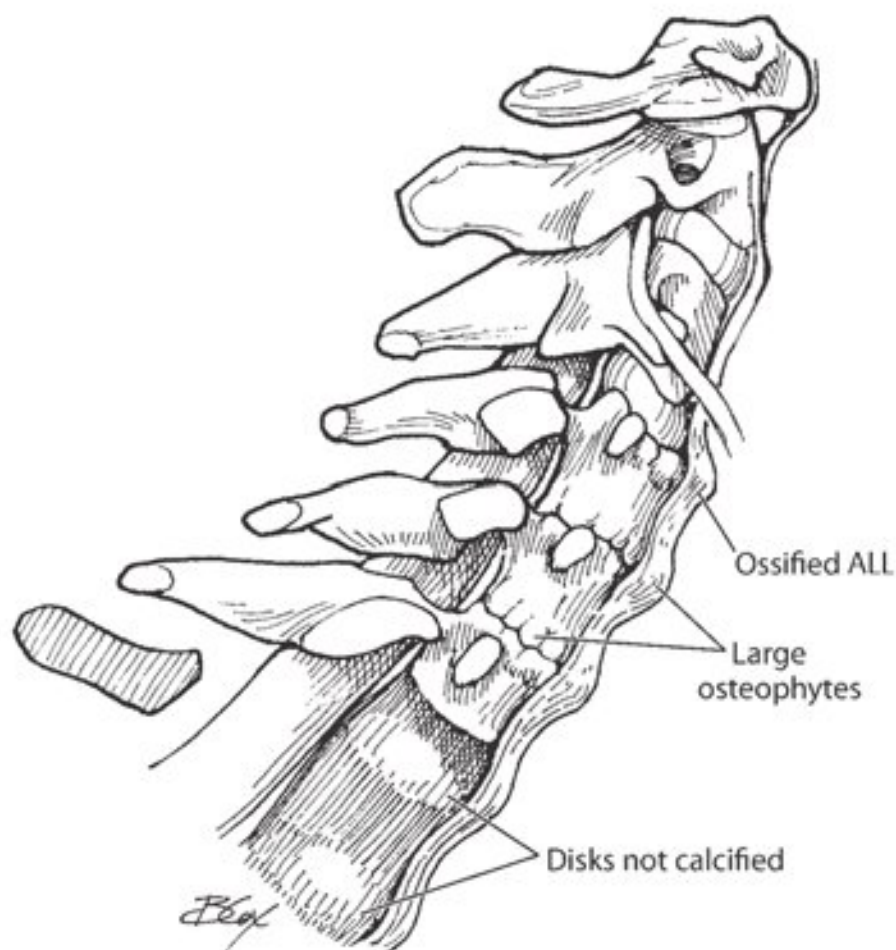


Fig. 32.2 Diffuse idiopathic skeletal hyperostosis in the cervical spine. Note the continuous osteophyte formation along the anterior border of the cervical spine.

- Conservative treatment unless mass effect on esophagus or surrounding structures causes significant health risk (i.e., weight loss from dysphagia, pneumonia, and respiratory difficulty).
- Initially, a change in diet to soft mechanical may be beneficial.
- Surgical pearls
 - Anterior approach, drill down osteophyte. Special attention needed for protection of surrounding structures.
 - No instrumentation needed. Debulking only. Do not violate disk space.⁷
 - Initially, postop patients may have increased dysphagia. Risk of gastrostomy tube requirement.¹

32.5 Ossification of the Posterior Longitudinal Ligament

- Background
 - Calcification with subsequent ossification of PLL. Can occur in any part of the spinal column and can extend into dura. Most commonly cervical (C3–C6).
 - Classified as segmental when ossification skips area behind disk space and is present only behind each vertebral body. (Mixed type and focal form exist.)
 - Increased incidence in Asian (Japanese) population (prevalence around 2%).^{5,12} Prevalence increases with age (average age at time of diagnosis is mid-50s).
- Signs, symptoms, and physical exam
 - Most are asymptomatic but can progress to myelopathy over time.¹³
 - Symptoms can range from subjective neck pain to severe myelopathy.
- Neuroimaging/work-up
 - Plain X-rays will miss the ossification, so CT is suggested when OPLL is suspected.
 - MRI or CT with intrathecal contrast for evaluation of degree of stenosis.
 - MRI: Ossified PLL is dark on T1 and T2.
 - Consider glucose level check given the higher frequency of OPLL patients with diabetes mellitus (DM).
- Treatment
 - Surgical treatment required to decompress the spinal cord. Patients who have myelopathy can benefit from early decompression.
 - Patients who have mild subjective complaints can be treated conservatively.
- Surgical pearls
 - Nasotracheal/fiberoptic intubation should be strongly considered to prevent hyperextension.

- Surgeon’s prerogative whether to leave a thin rim of ossified PLL attached to the dura during decompression. Note that the ossification usually extends into the dura and is inseparable.¹
- Anterior approach (corpectomy) as opposed to posterior (laminectomy/laminotomy) decompression is controversial because of the significant risk associated with resecting all of the OPLL.¹⁴ Leaving a thin layer of bone that is adherent to the dura is recommended. Nerve root decompression is required.⁷
- Somatosensory evoked potential (SSEP) monitoring is highly recommended.

Common Clinical Questions

1. A 65-year-old male comes to the hospital after a motor vehicle accident complaining of neck and back pain. Further questioning reveals that the patient has had a history of chronic neck ache with morning “back stiffness.” He also admits to having mild dysphagia with the sensation of a lump in his throat. Initial X-rays show significant vertebral osteophyte formation but are otherwise negative. Hip X-rays show normal hip and pelvic joints with no fractures. What is the patient’s most likely diagnosis?
 - A. OPLL
 - B. DISH
 - C. Ankylosing spondylitis
 - D. Acute cervical fracture
2. A 65-year-old Japanese male comes to your office complaining of not being able to keep objects in his hands and “dropping things.” Your exam is positive for bilateral Hoffman 3+ reflexes and 4/5 weakness of intrinsic hand muscles. Cervical spine MRI with central canal stenosis and cervical dynamic X-rays are normal. Careful analysis of MRI indicates a hypointense signal on T1 and T2 lining the posterior side of the vertebral body at C3–C5 with sparing of disk space. What is the most likely diagnosis?
 - A. Primary bone tumor
 - B. Metastatic disease
 - C. DISH
 - D. Segmental OPLL
3. True, or false? An ADI > 3 mm correlates with increased risk of cervical injury.

4. Dynamic imaging indicates C1–C2 instability. Which of the following is *not* a recommended treatment?
 - A. Cervical fusion secondary to ADI > 10 mm
 - B. Rigid cervical collar with close follow-up
 - C. Halo fixation with subsequent close follow-up imaging in an effort to decrease pannus size
 - D. Cervical fusion and decompression secondary to symptoms of myelopathy
 5. True, or false? Enthesitis is a chronic inflammation at the insertion point of the tendons into the spine that can become ossified.
-

Answers to Common Clinical Questions

1. B. Diffuse idiopathic hyperostosis (DISH) is the most likely diagnosis. The patient admits to morning stiffness, chronic axial spine pain, sensation of throat lump (“globus”) with dysphagia. The formation of osteophytes and the sparing of the sacroiliac joints also point to DISH as the most likely diagnosis. These bridging osteophytes seen with DISH do not provide any added stability.
 2. D. The patient’s diagnosis is most consistent with segmental OPLL. Calcified lesions are hypointense on T1 and T2. In addition, OPLL has a high incidence in the Japanese population. Finally, because the area behind the disk was spared, the OPLL is classified as segmental.
 3. False. A PADI > 14 mm does, not an ADI.
 4. B. A rigid collar is a poor choice here, for it does not provide stability at C1–C2.
 5. True. It is seen particularly in patients with ankylosing spondylitis, but it can also be seen in a variety of spondyloarthropathies.
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33 Spinal Emergencies

Puya Alikhani, Andreas K. Filis, and Frank D. Vrionis

33.1 Key Points

- Prompt and accurate diagnosis of spinal emergencies is critical, for return of function is highly dependent on early intervention.
- Multidetector CT with coronal and sagittal reconstruction is sufficient in evaluating spine injury in patients who have neck pain or have altered mental status and remains the imaging of choice in acute settings.¹
- MRI is the imaging modality of choice in diagnosing ligamentous injury, spinal hematomas, acute herniated nucleus pulposus, spinal epidural abscesses, and neoplastic conditions.
- Trauma and tumors may also present as “spinal emergencies,” as discussed elsewhere.

33.2 Spinal Hematomas

- Background
 - As in the cranium, these include subdural, epidural, and subarachnoid hematomas.
 - In up to a third of cases, no etiologic factor can be identified.
 - Anticoagulant therapy and vascular malformations represent the second and third most common causes.
 - Spinal hematomas are typically localized dorsally to the spinal cord at the cervicothoracic and thoracolumbar regions.^{2,3}
 - Subarachnoid hematomas can extend along the entire length of the subarachnoid space.
 - Intramedullary hemorrhages, caused by cavernomas or arteriovenous malformations (AVMs), typically produce devastating neurologic symptoms but are often not managed using emergent surgical decompression.
- Signs, symptoms, and physical exam
 - Epidural and subdural spinal hematomas present with intense, knifelike pain at the location of the hemorrhage (“coup de poignard”).
 - This may be followed by a pain-free interval of minutes to days.
 - Subarachnoid hematoma can be associated with meningitis-like symptoms, disturbances of consciousness, and epileptic seizures (often misdiagnosed as cerebral hemorrhage based on these symptoms).

- Symptoms depend on the location and extent of hemorrhage and may include motor weakness, sensory and reflex deficits, and acute bowel/bladder dysfunction.²
- Acute traumatic central cord syndrome (ATCCS) is a very common acute incomplete cervical cord injury presenting with severe weakness of upper than lower extremities.⁴
- Work-up
 - Hematology (including platelets), electrolytes, and partial thromboplastin time (PTT)/prothrombin time (PT)/international normalized ratio (INR).
 - Disseminated intravascular coagulation (DIC) panel and specific hematology factors may need to be assessed.
 - Appropriate neuroimaging.
- Neuroimaging
 - The imaging modality of choice is MRI with or without gadolinium (**Fig. 33.1**).
 - The appearance of hematomas in MRI is highly dependent on the age of the clot.
 - Hyperacute bleeding (< 24 h): T1 isointense, T2 slightly hyperintense.
 - Acute bleeding (1–3 d): T1 slightly hyperintense, T2 hypointense.

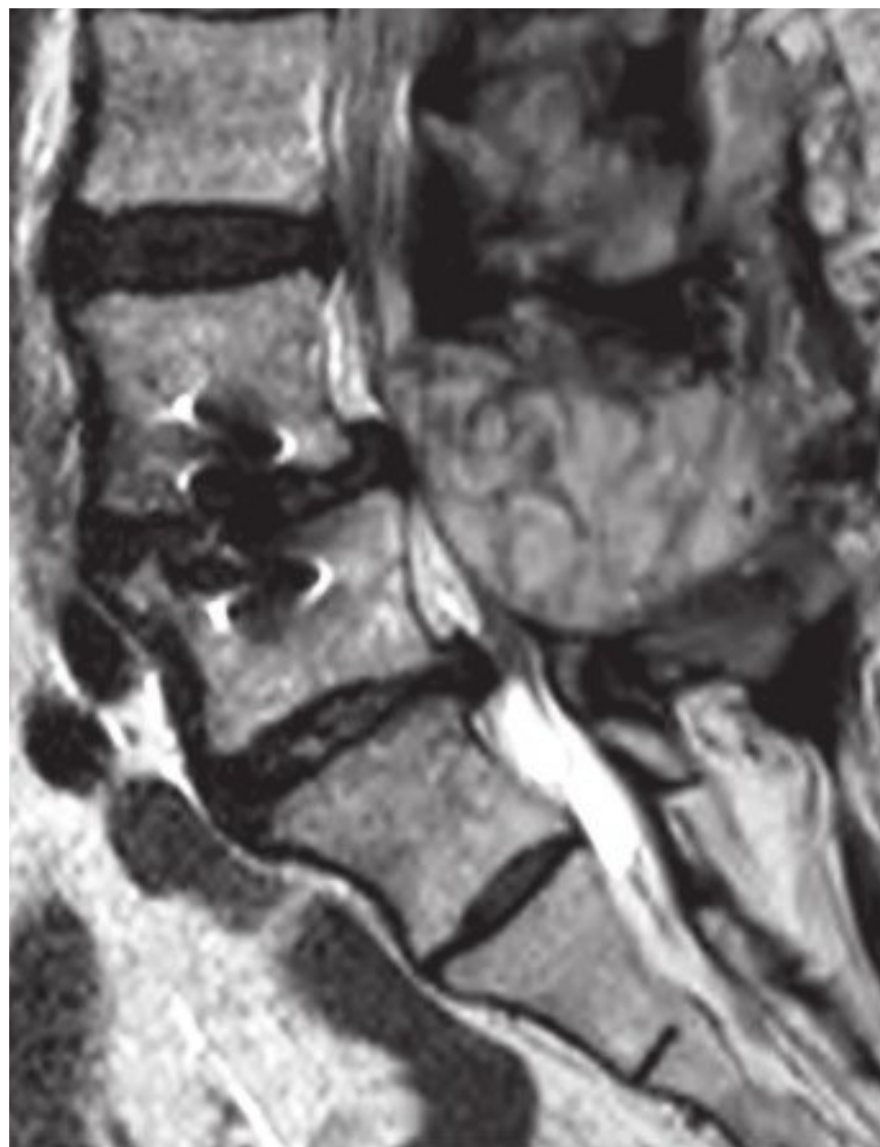


Fig. 33.1 T2-weighted MRI of the lumbar spine showing postoperative mixed signal fluid collection (consistent with epidural hematoma) in the epidural space compressing the dural sac.

- Subacute bleeding (> 3 d): T1 hyperintense, T2 hypointense (T2 may be hyperintense for late subacute).
- Treatment
 - The treatment of choice is correction of coagulopathy, if present, and emergent surgical decompression.
 - Benefit of surgical intervention is debatable if the only symptom is pain.
 - Surgery typically involves laminectomy without the need for fusion.
 - For cervicothoracic and thoracolumbar junction multilevel laminectomies, consider instrumentation and fusion.
 - For the ATCCS, in the past, early surgical decompression was considered as contraindicated. Newer studies advocate early decompression. Nonetheless, the timing of surgery remains controversial.⁴
- Surgical pearls
 - Subfascial drains may diminish the incidence of symptomatic epidural blood collections after multilevel laminectomy procedures.

33.3 Cauda Equina and Conus Syndromes

- Background
 - Cauda equina syndrome (CES) refers to the clinical condition that results from compressive, ischemic, and/or inflammatory neuropathy of multiple lumbar and sacral nerve roots in the lumbar spinal canal.^{5,6}
 - Conus syndrome has features similar to those of CES but involves compression at the level of the conus medullaris (T12–L1 typically).
 - The most common cause is disk herniation in the lumbar region.
 - It may also be caused by traumatic injury, lumbar spinal stenosis, primary or metastatic tumors, epidural abscess, ankylosing spondylitis, spinal subdural or epidural hematoma, spinal manipulation, or vascular malformation.^{5,6}
- Signs, symptoms, and physical exam
 - Urinary retention is the most consistent finding, occurring in 90% of patients presenting with CES.
 - Anal sphincter tone diminished in 80% of patients.
 - Saddle anesthesia is the most common sensory deficit (75% of patients).
 - After total perineal anesthesia develops, patient will likely have permanent bladder dysfunction.
 - Low back pain and radicular symptoms.
 - Complete CES entails painless urinary retention and overflow incontinence.
 - Conus lesions have the same features except that motor and sensory loss is typically asymmetric.
- Work-up
 - Basic laboratory studies and appropriate neuroimaging.

- If imaging demonstrates pathology other than herniated nucleus pulposus (HNP), further work-up is indicated (e.g., tumor or infection work-up).
- Neuroimaging
 - MRI is the best initial study if CES or conus syndrome is suspected.
 - MRI assesses soft tissue compression as well as signal changes within the spinal cord, especially the T2-weighted images.
 - CT scan: Unenhanced CT scan demonstrates bony anatomy, which may be helpful when the structural integrity of the spine is compromised and for surgical planning. CT scan with intrathecal contrast (CT/myelo) is the test of choice when MRI is contraindicated.
- Treatment
 - Prompt surgical decompression. Timing is controversial but should ideally be < 24 hours.⁷ In cases of spinal cord compression, the literature suggests better outcomes if surgery is performed within 48 hours.⁸
 - Surgical strategy is usually focused on the underlying causes.
 - Typically involves laminectomy and discectomy (for HNP).
 - More extensive surgery (e.g., vertebrectomy, tumor removal) may be necessary for other pathologies.
 - Return of function is dependent on the extent and duration of preoperative deficits.
 - Moderate to high doses of steroids for a few days can be helpful. Methylprednisolone has been shown to be more beneficial than dexamethasone in animal studies.^{9,10}
- Surgical pearls
 - Complete hemilaminotomy or laminectomy may be required for removal of a large central disk fragment causing conus or cauda equina syndrome.

33.4 Spinal Epidural Abscess

- Background
 - Spinal epidural abscess (SEA) is responsible for 0.2 to 2 cases per 10,000 hospitalizations.
 - Thoracic level is the most common site (50%), followed by lumbar (35%).
 - Often associated with vertebral osteomyelitis/diskitis.
 - Risk factors include diabetes mellitus, sepsis, trauma, intravenous drug abuse, alcoholism, and epidural anesthesia or analgesia.
 - Skin abscesses and furuncles are the most common sources of infection.
 - Gram-positive *Staphylococcus aureus* is the most common causative agent.
- Signs, symptoms, and physical exam

- Diagnosis is often achieved in delayed fashion due to the vagueness of presenting signs and symptoms.
- The most common presenting symptoms include excruciating pain localized over the spine, radicular pain, weakness, and sensory deficits.
- Average time from back pain to root symptoms is 3 days, and 4.5 days from root pain to weakness.
- Leukocytosis and fever may be absent.
- Work-up
 - Hematology (complete blood count [CBC] with differential), electrolytes (comprehensive metabolic panel [CMP]), acute phase reactants (erythrocyte sedimentation rate [ESR], C-reactive protein [CRP]), blood cultures. Cardiac echo to rule out endocarditis may be indicated.
- Neuroimaging
 - MRI with gadolinium is the modality of choice in diagnosing SEA.
 - Typical finding: T1 shows hypo- or isointense epidural mass.^{11,12}
 - Plain radiographs are often helpful for suspected diskitis and will show chronic, erosive changes in the end plates. CT is helpful in diagnosing osteomyelitis.
- Treatment
 - For SEAs that show clear involvement of the spinal canal and cause dural compression, surgical decompression and intravenous antibiotic therapy for 2 weeks, then a switch to oral antibiotics, is the treatment of choice.
 - Surgical therapy is associated with improved motor function in comparison to delayed surgery after failure of medical management. Diabetes mellitus, leukocytosis > 12.5k, CRP > 115, and positive blood cultures have been identified as predictors of failure of medical management alone.¹³
 - Another consideration in favor of early surgery is to obtain material for cultures so that antibiotic treatment is not delayed and is pathogen-specific.
 - SEA is often seen in association with diskitis/osteomyelitis. In these cases, typically only a thin film of epidural enhancement is seen. Surgical intervention is not necessarily indicated in these situations.
 - SEA is fatal in up to a third of elderly patients, and mortality is usually due to the original focus of infection or as a complication of neurologic compromise.
- Surgical pearls
 - Most spinal instrumentation is safe, effective, and at times necessary in the treatment of epidural abscess or diskitis/osteomyelitis of the spine.
 - Instrumentation in patients who have spinal infection after decompressive surgery does not increase the recurrent rates of infection.¹⁴

Common Clinical Questions

1. Cauda equina syndrome describes the clinical condition that results from what neuropathy involving multiple lumbar and sacral nerve roots?
 - A. Compressive
 - B. Ischemic
 - C. Inflammatory
 - D. All of the above
2. Conus lesions have the same features as cauda equina *except* which?
 - A. Urinary retention
 - B. Anal sphincter tone diminished in 80% of patients
 - C. Saddle anesthesia
 - D. Motor and sensory loss typically asymmetric

Answers to Common Clinical Questions

1. D.
2. D.

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IV

Surgical Techniques

34 Occipitocervical Fusion

Gisela Murray, Edwin Ramos, and Juan S. Uribe

34.1 Key Points

- Main goal is solid arthrodesis. The keys to achieve it are adequate decortication, bone graft placement, and precise alignment of craniocervical structures.
- Traction is not applied in cases of occipitocervical dislocation or significant ligamentous injury on MRI.
- CT angiography of the neck may be performed to evaluate the anatomical course of the vertebral artery.
- As part of preoperative planning, make sure to review the depth of the midline suboccipital keel and the thickness of the paramedian cranium.
- It is important to verify with fluoroscopy that the occipitocervical junction is in a neutral anatomical position before application of instrumentation.

34.2 Indications

- Occipitocervical instability is most commonly a result of inflammatory diseases like rheumatoid arthritis, trauma, infection, tumor, iatrogenic injury (after transoral odontoidectomy), congenital anomalies, or cranial settling with brainstem or cord compression.¹

34.3 Technique

- Awake fiber-optic¹ intubation is often performed, and baseline somatosensory evoked potentials (SSEPs) and motor evoked potentials (MEPs) are obtained with the patient initially in the supine position.
- Maintaining cervical alignment, rotate the patient to the prone position onto chest rolls or a Jackson frame.
- The head is secured with a Mayfield cranial fixation system (Schaerer Mayfield; Randolph, Massachusetts).
- Cervical alignment is maintained and checked with fluoroscopy to ensure a neutral occipit relationship.
- The suboccipital area is shaved and cleansed.
- A midline incision is extended from the inion to the lowest level to be incorporated in the construct.
- Subperiosteal dissection is performed to expose the suboccipital bone and the dorsal elements of the cervical spine. Special care is taken to leave a

cuff of fascia near the inion for subsequent closure. This ensures that the occipital plate will be fully covered by muscle, reducing the chances of hardware's eroding through the skin.

- When exposing the arch of C1, we prefer to do blunt dissection with a Penfield 1 to avoid injury to the vertebral artery.
- Any decompression required (suboccipital or cervical) is now performed, with the bone saved for autograft.
- Occipital fixation can be performed using a variety of methods: Occipital wiring, occipital screw fixation (occipital plate; **Fig. 34.1**) or occipital condyle screw fixation.
- For the placement of the occipital plate, first place the plate in the appropriate position and mark the midline keel with a marking pen. The plate is removed, and a handheld power drill is used to make a bicortical hole. To avoid injury to the suboccipital neural structures, drill in a progressive fashion—first to a depth of 6 mm, then slowly increasing until bicortical penetration is felt, usually around 10 to 14 mm. Then the hole is tapped. The plate is placed in the correct location and fixed with a 4.5- to 5-mm-diameter screw.
- Most occipital plates provide for at least two to three screw placements in the midline and other paramedian holes. The same steps should be followed for all screws (drilling, tapping, screw placement), with the difference that the paramedian bone is usually less thick and probably will require a shorter screw.
- The plate should be placed high in the suboccipital bone (closer to the inion than to the foramen magnum) to leave a small area of bone caudal to the plate for fusion surface.
- An alternative to plate systems is occipital condyle screw fixation.² It is particularly useful in situations in which a suboccipital decompression

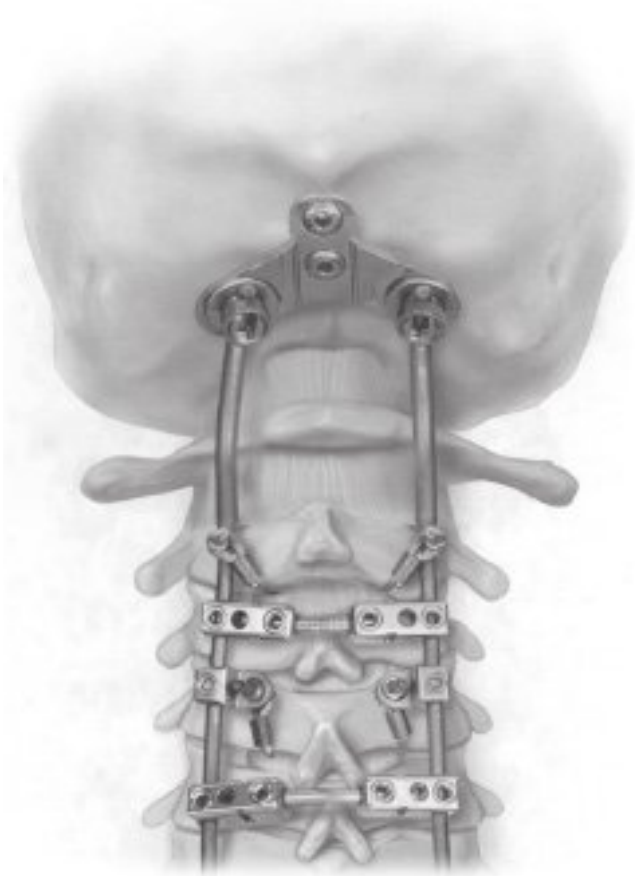


Fig. 34.1 Occipital plate and rod system. Illustration of the Summit occipital plating and rod system (DePuy Spine, Raynham, MA). Note that the occipital anchor plate is extended to the cervical spine via rods anchored by sublaminar wires that pass through custom cable connectors. (Reprinted with permission from DePuy Acromed; SCST pg. 434, Fig. 26-14.)

is required. With this technique, a 3.5×20 - to 22-mm effective length and 10- to 12-mm lag shank screw (30–34 mm) is placed in the center of the condyle, inferior to the hypoglossal canal. Although different techniques for its placement have been described, in general the screw has a medial trajectory (10 – 25°) from an entry point ~ 5 mm lateral to the foramen magnum on the condyle itself. Image guidance and free-running electromyography (EMG) monitoring of the hypoglossal nerve are highly recommended with this technique.

- The occipital fixation is then anchored to cervical instrumentation that could be a combination of rod/sublaminar wires, rod/screws, or screws/plate.
- In terms of fixation points, cervical fixation may or may not include the atlas.
- Fusion is often taken down to C2.³

34.4 Complications^{4,5}

- In contemporary series, complication rates (minor and major) range from 12 to 30%
 - Wound infection, cerebrospinal fluid (CSF) leak, intracranial injury (sub-/epidural hematoma), spinal cord injury (instrumentation into spinal canal), vascular injury (hardware into vertebral artery)
 - Hardware failure (loosening, pullout, breakage)
 - Nonunion requiring reoperation
 - Fixation of patient's neck in exaggerated flexion or extension

34.5 Postoperative Care

- Cervical collar
- Upright X-rays
- Prophylactic antibiotics for 24 hours
- Prompt mobilization

34.6 Outcomes⁶

- Overall all technique, rod/screw, screw/plate, and rod/wiring constructs have overall reported fusion rate greater than 90%.⁶
- Rod/screw construct (occipital plate/condyle screws) have highest fusion rates (94–100%) and wiring/onlay graft the lowest (88.6%).
- Rod/screw constructs have highest rate of neurologic improvement and wiring/rod the lowest (81.6 versus 51.8%).⁶
- Early biomechanical studies suggest that condyle screw fixation is biomechanically equivalent to suboccipital plate systems in terms of providing craniocervical stability.²
- Instrumentation failure is lowest using the rod/screw technique.

34.7 Surgical Pearls

- Cervical traction is not applied in cases of occipitocervical dislocation.
- Before placement of condyle screws, the morphology of the occipital condyles should be reviewed on the preoperative CT scan.
- Before locking in the construct, verify that a neutral occipitocervical relationship has been achieved.

Common Clinical Questions

1. Cervical traction is contraindicated in which occipitocervical instrumentation cases?
2. Before locking the construct down, why are anteroposterior and lateral fluoroscopy views obtained?
3. Which construct has the lowest pseudarthrosis rate?

Answers to Common Clinical Questions

1. Those performed for occipitocervical dislocation or when significant ligamentous injury is suspected.
2. To ensure adequate alignment and avoid exaggerated flexion or extension.
3. Screw-and-rod–based constructs.

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35 Chiari I Decompression

Mark S. Greenberg

35.1 Key Points

- Consistent feature of Chiari I: Disruption of normal cerebrospinal fluid (CSF) flow through the foramen magnum. Most symptomatic cases have descent of cerebellar tonsils ≥ 5 mm below the margins of the foramen magnum best seen on MRI. Radiographic criteria should not be used alone in the diagnosis; clinical correlation is crucial.
- Surgical treatment for symptomatic patients consists of enlarging the foramen magnum (suboccipital decompression), usually combined with C1 laminectomy.
- Syringomyelia, if present, will usually respond to suboccipital decompression alone.

35.2 Indications

- Symptoms include
 - Pain (the most common symptom; mostly suboccipital H/A that is exacerbated by neck extension), neck pain, arm pain, weakness in ≥ 1 limb, numbness in ≥ 1 limb, loss of temperature sensation (dissociated sensory loss),¹ balance difficulties. Fifteen percent to 30% of patients meeting radiographic diagnostic criteria are asymptomatic.²
- Signs³ include
 - Hyperactive lower extremities reflexes, downbeat nystagmus, gait disturbance, hand muscle atrophy, cerebellar signs, Babinski sign

35.3 Technique

- Position: Patient is prone on chest rolls with the neck flexed and the head in a Mayfield head holder or on a horseshoe head rest.
- Skin incision: Midline incision from the inion down to the C2 spinous process.
- The fascia is opened in a Y or T, leaving a cuff of tissue attached to the occipital bone for use during closure.

- The occipital bone is exposed down to the foramen magnum. The posterior C1 arch is exposed (with caution regarding the vertebral arteries, which pass over the superior aspect of the posterior arch of C1).
- At minimum, the surgery consists of enlargement of the foramen magnum (FM) (suboccipital decompression) often combined with C1 laminectomy. The removal of occipital bone should be as wide as the FM, but should be limited to no more than ~2.5 to 3 cm above the FM (to avoid cerebellar herniation, referred to as cerebellar ptosis). Techniques include thinning the bone with a high speed drill and removing the residual bone with a Kerrison rongeur (**Fig. 35.1**). Additional adjuncts to the suboccipital decompression include the following:
 - C2 laminectomy: Reserved for cases with severe tonsillar descent below the superior margin of C2.
 - Duraplasty: A Y-shaped incision (**Fig. 35.1**) is made in the dura (some authors preserve the arachnoid⁴) and a patch graft sewn in watertight closure with 4–0 Nurodon (Ethicon, Cincinnati, Ohio). Options for sources of graft: Pericranium, fascia lata, and allograft dural substitutes. Pericranium can be harvested through the same incision by subcutaneous dissection.⁵
 - Instead of opening the dura in all cases, some surgeons simply lyse extradural constricting bands. Then intraoperative ultrasound may be used to determine whether there is adequate room for CSF circulation. If not, a duraplasty is performed.
 - An alternative to duraplasty (primarily in pediatrics): Partial thickness scoring of the dura with several parallel passes of a scalpel.
- Closure: A multilayered watertight closure is performed. Skin approximation with sutures is preferred over staples to reduce risk of CSF leak. A wound drain is not used. A lumbar drain is occasionally used for 2 to 3 days.

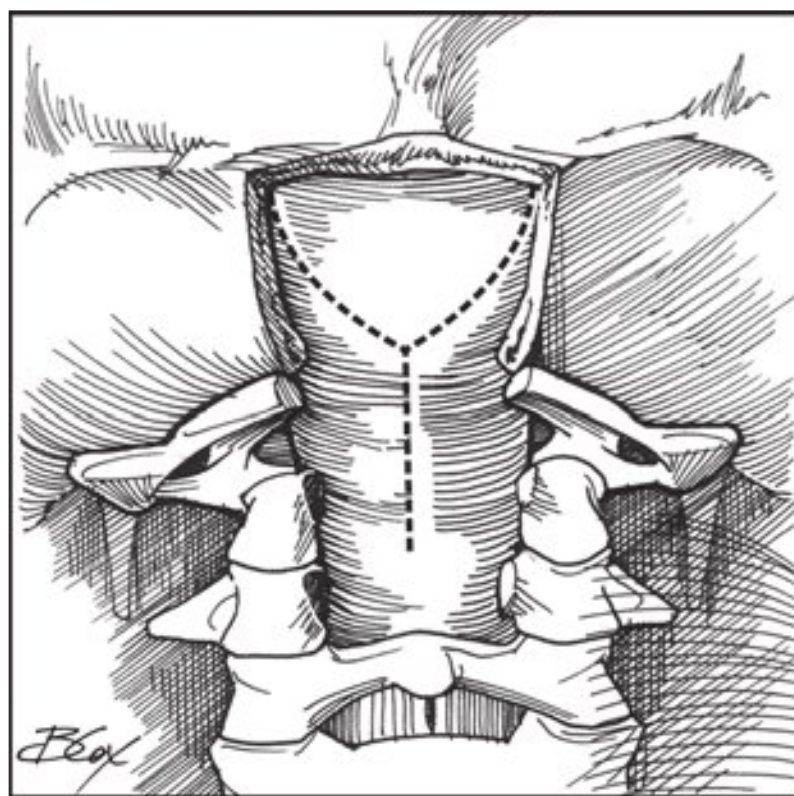


Fig. 35.1 Posterior exposure of the dura after bony removal, showing an outline of the Y-shaped dural incision. (From Vaccaro AR and Albert TJ, eds. *Spine Surgery: Tricks of the Trade*, 2nd ed., Thieme; 2009.)

35.4 Complications

- The major complications from surgery is CSF leak. This may be external (may be initially treated by oversewing the site of leak and temporary lumbar drainage) or subcutaneous (pseudomeningocele).
- Overaggressive removal of occipital bone can lead to cerebellar ptosis (sagging of the tonsils).
- Injury to brainstem or posterior inferior cerebellar arteries (PICAs). Avoid aggressive treatment of tonsillar adhesions. Use an operating microscope if needed.
- Postop apnea or respiratory depression. Tends to occur within the first few days postop. More common in pediatrics. Monitor for apnea and increasing arterial $p\text{CO}_2$.

35.5 Outcomes

- Preop symptoms of headache or pain respond in 82% with 4 years' follow-up.¹ Weakness is less responsive to surgery, especially if atrophy has occurred. Symptoms of greater than 2 years' duration have a worse prognosis.

35.6 Surgical Pearls

- Treating the Chiari malformation via suboccipital decompression corrects syringomyelia in the majority of cases without the need for any other procedure.
- Fifteen percent to 30% of patients with radiographic criteria of Chiari I malformation are asymptomatic.²

Common Clinical Questions

1. Three weeks after a suboccipital decompression with duraplasty for a symptomatic Chiari 1 malformation, a patient develops a tense, very painful fluid collection under the incision, and an MRI shows that it has the appearance of CSF without any other significant abnormalities. What is the best management option?
 - A. Percutaneous tapping of the fluid after careful skin prep, then tightly wrapping the head with bandages to prevent reaccumulation.
 - B. Placement of a lumbar drain and having the patient lie flat in bed for 3 days.
 - C. Surgical exploration of the wound in the OR with repair of dural defect, and placement of lumbar drain for 3 days with the head of bed elevated more than 30°.

- D. Placement of a external ventricular drain to divert the CSF from the wound and conversion to a ventriculoperitoneal shunt if the patient is drain-dependent after 5 days.
2. A 48-year-old female elementary school teacher presents with a 20-year history of headaches that occur almost every day at the same time. They vary from left to right side and are often associated with neck pain. No medications or change in position has provided any relief. Her primary care physician ordered a brain MRI, on which the only abnormality identified is that the inferior pole of the cerebellar tonsils are 4 mm below the foramen magnum. She is neurologically intact. What should you do?
- A. Order a cine flow MRI.
- B. Have her see a neurologist to rule out other causes of chronic headache.
- C. Order a cervical MRI.
- D. All of the above.
-

Answers to Common Clinical Questions

1. C. The patient probably has a “ball–valve” effect through a dural flap. Tapping the fluid will not prevent reaccumulation. Placing the patient flat will actually put more pressure on the incision. An external ventricular drain and/or shunt has no role in the absence of hydrocephalus.
2. D. The headaches sound atypical for Chiari malformation, and she is borderline for the tonsillar descent (which is at 2 standard deviations below normal for her age—see **Table 35.1**). The cine may MRI give additional helpful data, but before recommending surgery, it is critical to rule out other explanations for her headache. A cervical MRI will rule out cervical spondylosis, which may cause neck pain and headache, and it is also necessary to rule out a syrinx associated with the tonsillar descent. Because she is neurologically intact, there should be little risk in proceeding cautiously.
-

Table 35.1 Variation with age of the location of the inferior tonsillar pole relative to the foramen magnum⁶

Age (years)	2 standard deviations below the FM (mm)
0–9	6
10–29	5
30–79	4
80–89	3

Abbreviation: FM, foramen magnum.

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36 Transoral Odontoidectomy

Steven M. Presciutti, Martin Quirno, Colin B. Harris,
and Frank M. Phillips

36.1 Key Points

- The transoral approach allows the surgeon to gain access from the upper clivus rostrally to the top of the C3 vertebral body caudally.¹
- The authors prefer this route for extradural pathology as dural closure can be quite difficult with this approach. In this case, a lateral approach may be preferable.
- A thorough preoperative assessment is necessary to ensure that the patient is free from any oral or dental pathology, which may be contraindications to this approach.
- A minimum of 2.5 to 3 cm of dental clearance should be present to allow adequate exposure for odontoid resection.²
- This approach is best avoided if the compressive pathology lies above the hard palate, as this may require a more extensile approach.
- The endoscopic endonasal approach is a more recently described alternative for craniocervical junction pathology that may decrease the risk of infection, swallowing, and pulmonary issues.³

36.2 Indications

- Extradural ventral pathology causing compression of the brainstem or upper cervical spinal cord from the clivus to the C2 vertebral body
 - Possible etiologies include rheumatoid or other retro-odontoid pannus, os odontoides, tumor, infection, or fracture.
- Continued symptoms of cord compression after C1–C2 posterior decompression and fusion

36.3 Technique

- General endotracheal (ET) anesthesia with a reinforced endotracheal tube is the authors' preferred method of intubation.
 - Tends to allow for easier positioning of the tube away from the midline as opposed to when a nasotracheal intubation is performed.
 - Preoperative tracheostomy should also be considered for cases that require more extensive exposure as well as when ventilator dependence for several days postoperatively is anticipated.

- The authors recommend the routine use of somatosensory evoked potentials (SSEPs) and, often, motor evoked potentials (MEPs) as well.
 - Baseline recordings may be indicated before final positioning of the patient.
- The patient is positioned supine on a standard operating room (OR) table with the head secured by either a Mayfield headrest (Schaerer Mayfield; Randolph, Massachusetts) or with skull tongs/pins for greater control.
- A Spetzler-Sonntag transoral retractor (Aesculap; San Francisco, California) is positioned over the mouth with the retractors used to form a rectangular exposure: The palate is elevated cephalad and the tongue and endotracheal tube are retracted caudally; the tonsils and lateral oropharyngeal walls are covered in moist gauze and retracted laterally (**Fig. 36.1**).
- The field is prepared by filling the oral cavity with a povidone–iodine (Betadine) solution after placing the patient into a Trendelenburg position to keep the solution from entering the lungs should the ET cuff deflate.

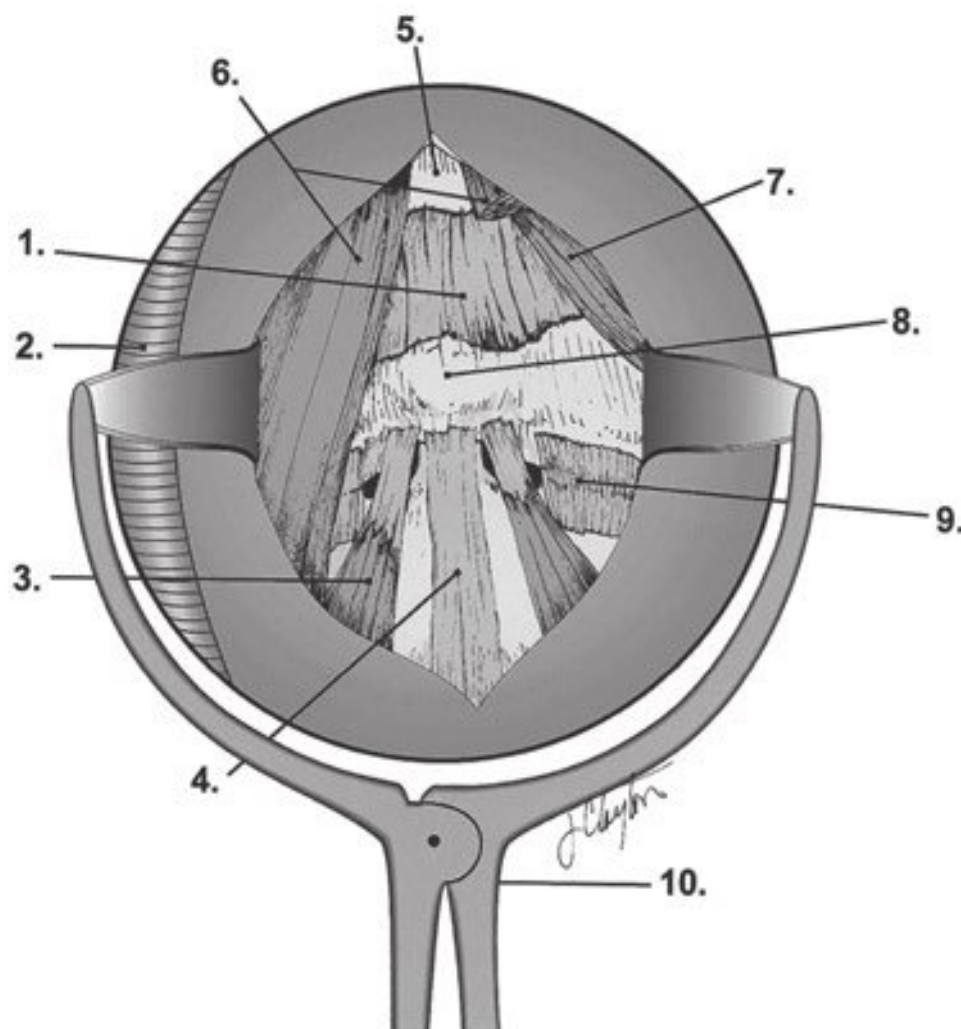


Fig. 36.1 Retropharyngeal anatomy and retractor. (1) Anterior atlantooccipital membrane; (2) armored nasotracheal tube; (3) longus colli muscle; (4) anterior longitudinal ligament; (5) clivus; (6) longus capitis muscle; (7) rectus capitis anterior muscle; (8) anterior tubercle of C1; (9) lateral atlantoaxial joint capsule; (10) retropharyngeal soft tissue retractor. (From Maher T, Merola A, eds. *Surgical Technique for the Spine*, Thieme, 2003.)

- The authors soak the mouth for 10 minutes before removing the solution.
- A preoperative dose of antibiotics is given that includes coverage for oral flora.
 - The authors prefer a combination of cefepime and metronidazole.
- The surgeon sits at the patient's head, and a surgical microscope is used for visualization.
- The midline is identified by palpating the anterior tubercle of C1 as well as the C2–C3 disk space, which is usually prominent.
- A 3- to 5-cm full-thickness vertical incision is made in the midline with the use of the cutting current of an electrocautery, dividing the mucosa, pharyngeal constrictor musculature, and the anterior longitudinal ligament.
 - The amount of bleeding is significantly reduced by staying in the midline.
- Flaps are elevated to either side of the midline out laterally to the edge of the C2 body.
- Self-retaining retractor blades are then placed to retract the flaps laterally.
- The central portion of the anterior arch of C1 is removed using either a high-speed bur or a sharp rongeur.
 - Typically, 1 cm of bone in each direction from the midline is removed, corresponding to roughly two-thirds of the entire anterior C1 arch.
 - Because the anterior C1 arch is the load-bearing portion of the bone, an effort should be made to preserve a portion of the arch to prevent lateral mass spread: Typically, a small cephalad ridge of the anterior C1 arch can be left in place.
- Next, the odontoid is resected, which is usually performed one of two ways: (1) Using a high-speed side-cutting matchstick bur or (2) amputating the odontoid at its base and removing it en bloc by severing the soft tissue connections and working in a caudal-to-cephalad direction.
 - A variation of removing the dens with the bur is to hollow out the odontoid, leaving thin lateral and posterior cortical walls of bone. The remaining bone can then be removed using either a fine curette or micro-Kerrison rongeurs.
- To fully remove the tip of the odontoid, it is necessary to cut the remaining soft tissue attachments. Angled curettes or micro-Kerrison rongeurs can be used to detach the apical, alar, and transverse ligaments.
- In patients with rheumatoid arthritis, the retrodental soft tissue pannus is now exposed. Only loose fragments should be debulked, as complete pannus removal is usually unnecessary and risks a cerebrospinal fluid (CSF) leak.²
- The incision is irrigated thoroughly with antibiotic solution, followed by closure of pharynx, typically by 3–0 chromic buried sutures.

- A two-layered closure may also be performed if layered flaps were created during the initial dissection.
- The retractors are removed, and 1% cortisone cream may be applied to the mucosa and tongue to decrease postoperative edema.

36.4 Complications

- Neurologic injury
 - SSEPs and MEPs can provide intraoperative warning of spinal cord compromise.
 - Keep mean arterial pressure (MAP) above 90 mm Hg to prevent ischemic spinal cord injury.
- Dural tear and CSF leak—increased risk of meningitis and encephalitis
 - Direct dural repair (if possible), followed by fascial/fat graft or fibrin glue.
 - Placement of subarachnoid lumbar drain postoperatively can be helpful.
 - Broad-spectrum antibiotics should be given postoperatively for at least 2 weeks.
- Airway obstruction
 - Avoid premature extubation as reintubation can be very difficult and require emergency tracheostomy. Before extubation, evaluate for
 - Resolution of retropharyngeal swelling on lateral radiograph (usually 24–48 h postoperatively)¹
 - Ability to breathe around orotracheal airway²
- Other pulmonary complications
 - Purulent secretions, atelectasis, Adult Respiratory Distress Syndrome (ARDS).
 - Risk factors include American Society of Anesthesiologists (ASA) Grades III and greater, preoperative lower cranial nerve palsies, preoperative pulmonary involvement, and longer operative times.⁴
- Infection—dissection directly through a “contaminated” field
 - Can be minimized with meticulous wound closure and 24 to 48 hours of IV antibiotics (especially helpful when based on preoperative oropharyngeal cultures)
 - Postoperative infection or retropharyngeal abscess considered the origins of wound dehiscence until proven otherwise
- Vertebral or carotid artery injury
 - Preoperative CT or MRI scans should be reviewed to determine the location of these vessels (i.e., an aberrant medial course of the vertebral artery).
 - Dissection of the atlas should stay within 1.5 cm of the midline, and dissection at the level of the C2–C3 disk should stay within 1 cm of the midline.¹

36.5 Postoperative Care

- The authors typically perform immediate posterior stabilization after odontoid resection. A hard cervical collar and manual stabilization can be used to transfer the patient from the supine to the prone position.
 - When possible, the authors recommend performing the posterior fusion before the transoral resection.
- For most transoral cases, immediate postoperative extubation is possible given that the proximal location of the surgical field within the pharynx makes airway compromise less likely.
 - Importantly, however, the tongue and soft tissues should be carefully inspected for severe swelling before extubation of the patient. If there is concern, the patient should be left intubated until the swelling resolves.
- Oral feeding can resume immediately for most patients who have short, simple procedures; however, many require tube feeding for several days.
- Antibiotics should be continued for 24 to 48 hours postoperatively.
- Preoperatively, many patients requiring ventral decompression may be severely debilitated and malnourished. Postoperatively, there may be a need for a course of parental nutrition.

36.6 Outcomes

- No level I studies have been performed to evaluate outcomes.
- Based on a review of smaller case series, improvement of preoperative neurologic deficits can be expected but is dependent on chronicity of spinal cord compression.
- In one 10-year review of 72 cases, there were two postoperative deaths and one pharyngeal infection requiring a repeat operation, with 100% neurological improvement seen in all patients.⁵

36.7 Surgical Pearls

- Care should be taken to avoid entrapment of the tongue against the teeth with the retractor system.
- As an alternative to palatal retractors, a red rubber catheter can be passed through the nares and sutured to the soft palate; it can then be used as a retractor.²
- The soft palate and mandible can be split to provide extensile exposure in the proximal and distal directions, respectively, although the increased morbidity should be considered.

Common Clinical Questions

1. When performing exposure of the odontoid, the surgeon decided to remove 1 cm each side of the midline of the atlas to create a window access. What technique can help in preventing lateral mass spread and destabilization?
 - A. Bur less than 0.8 cm wide from midline.
 - B. Leave a cephalad narrow bony bridge.
 - C. Remove more than anterior arch of C1, because it does not have any effect of C1 stabilization.
 - D. The anterior odontoid exposure does not require removal of any anterior C1 arch.
 2. When removing an os odontoideum, the surgeon notices CSF leakage. Exploring the dura, he finds a small vertical dural tear. What is the best technique to treat this tear during surgery and postoperatively?
 - A. Repair the tear followed, then place subarachnoid lumbar drain postoperatively.
 - B. Place a layer of fibrin glue over the tear and postoperatively place the patient on 2 weeks of broad-spectrum antibiotics.
 - C. Repair the tear, following it with a fascial/fat graft or fibrin glue and postoperatively placing the patient on 2 days of broad-spectrum antibiotics.
 - D. Repair the tear, following it with a fascial/fat graft or fibrin glue, placing a subarachnoid lumbar drain postoperatively and placing the patient on 2 weeks of broad-spectrum antibiotics.
 3. What is the preferred method for intubating a patient about to undergo transoral odontoidectomy?
 - A. Nasotracheal intubation.
 - B. Preoperative tracheostomy should be performed on all patients undergoing transoral odontoidectomy.
 - C. General endotracheal intubation can be performed as long as the tube is retracted laterally.
 - D. General endotracheal intubation can be performed as long as the tube is retracted inferiorly with the tongue.
-

Answers to Common Clinical Questions

1. B. The surgeon can remove up to 1 cm on each side of the midline to create the access window to the odontoid.
 2. D. Patients who have a dural tear are at a higher risk of developing meningitis, and broad-spectrum antibiotics should be given for at least 2 weeks.
 3. D. Patients should undergo general endotracheal intubation, and the tube should be retracted inferiorly with the tongue, allowing direct visualization of operating window without the tube blocking it.
-

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37 C1–C2 Techniques

Jau-Ching Wu, Khoi D. Than, and Praveen V. Mummaneni

37.1 Key Points

- Fixation choices are (1) C1 lateral mass screws in combination with C2 pars screws, pedicle screws, or translaminar screws; (2) C1–C2 transarticular screw (Magerl’s technique); or (3) wiring techniques.
- Posterior C1–C2 fusion techniques are technically demanding, and caution should be used to avoid vascular (vertebral and carotid arteries) injury during screw placement.
- Preoperative CT scan (with or without CT angiography) is helpful to assess the position of the foramen transversarium of C1 and C2.

37.2 Indications

- Most C1–C2 ligamentous instabilities (> 3 mm atlantodental interval [ADI] on flexion–extension X-rays in an adult without rheumatoid arthritis).
- Traumatic fractures are among the most frequent indications for posterior C1–C2 fixation.
- Certain subsets of type 2 and type 3 odontoid fractures are amenable to posterior C1–C2 fixation.
 - Type 2 odontoid fracture associated with
 - Fracture of the atlantoaxial joint
 - Anterior–inferior oblique fracture in the coronal plane
 - Oblique fracture in the frontal plane
 - Jefferson fracture
 - Ruptured transverse ligament
 - Old, unhealed type 2 odontoid fracture
 - Type 3 odontoid fracture associated with
 - Fracture of the atlantoaxial joint
 - Jefferson fracture
 - Chronic unhealed odontoid fracture after immobilization; rotatory subluxation of C1–C2
- Congenital malformations of C2 (e.g., os odontoideum and odontoid agenesis with C1–C2 dynamic instability).
- Inflammatory diseases (e.g., rheumatoid arthritis with abnormal *PADI*).
- Degenerative diseases (with instability and abnormal ADI).

- Infections (with instability and abnormal ADI).
- Neoplasms (with instability and abnormal ADI).

37.3 Techniques

The procedure is technically demanding, and an exact 3D understanding of the anatomy of the region and of the vertebral artery is mandatory (**Fig. 37.1**).

The patient is positioned prone with head fixed by a Mayfield head holder (Schaerer Mayfield; Randolph, Massachusetts). The neck should be in the neutral position and the head kept in the military chin tuck position. A midline posterior neck incision is then made from the suboccipital area to the spinous process of C3, allowing exposure of C2–C3 facet joints and the posterior C1 arch.

37.3.1 C1 Lateral Mass Screw with C2 Pars, Pedicle, or Translaminar Screws^{1,2}

- C1 lateral mass screw^{3,4}
 - Control of hemorrhage from the venous plexus between C1 and C2 must be achieved by bipolar coagulation or hemostatic agents. It is

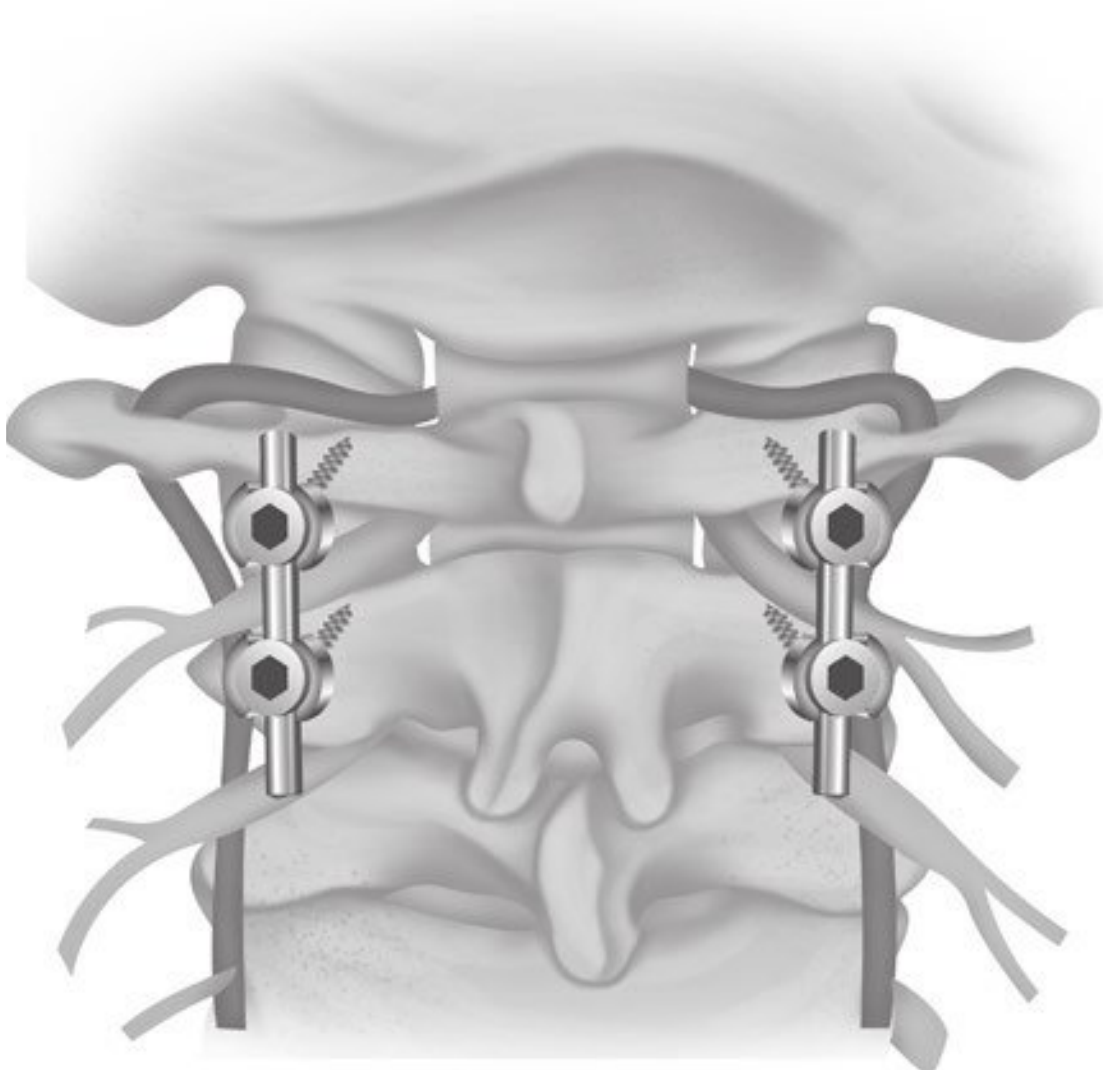


Fig. 37.1 Posterior C1–C2 screw and rod construct. Note the course of the vertebral artery lateral to the pars of C2 and then looping up into the sulcus arteriosus on the superior lateral margin of the C1 lamina.

not necessary to expose the vertebral artery on the superior aspect of the C1 arch (sulcus arteriosus). Usually, the C2 nerve root is mobilized caudally for exposure of the C1 lateral mass inferior to the C1 arch. The medial border of the C1 lateral mass is palpated. A pilot hole can be made with a 3 mm drill bit at the center of C1 lateral mass. The screw trajectory is 10° medial angulation in the axial plane. On lateral fluoroscopic imaging, the drill is aimed toward the anterior tubercle of C1. Stop the drill at the “back side” of the anterior C1 tubercle to prevent plunging the bit into the retropharynx. After tapping of the hole, the C1 lateral mass screw is placed (usually 34–36 mm long).

- C2 pars screws (**Fig. 37.2**)^{3,4}

- The C2 pars is defined as the portion of C2 vertebra connecting the superior and inferior articular surfaces. A C2 pars screw is placed in a trajectory similar to that of a C1–C2 transarticular screw, except that it is shorter. Its entry point is ~3 mm rostral and 3 mm lateral to the inferior medial aspect of the inferior articular surface of C2. The screw should follow a steep trajectory, 45 to 60°, with 10 to 15° of medial

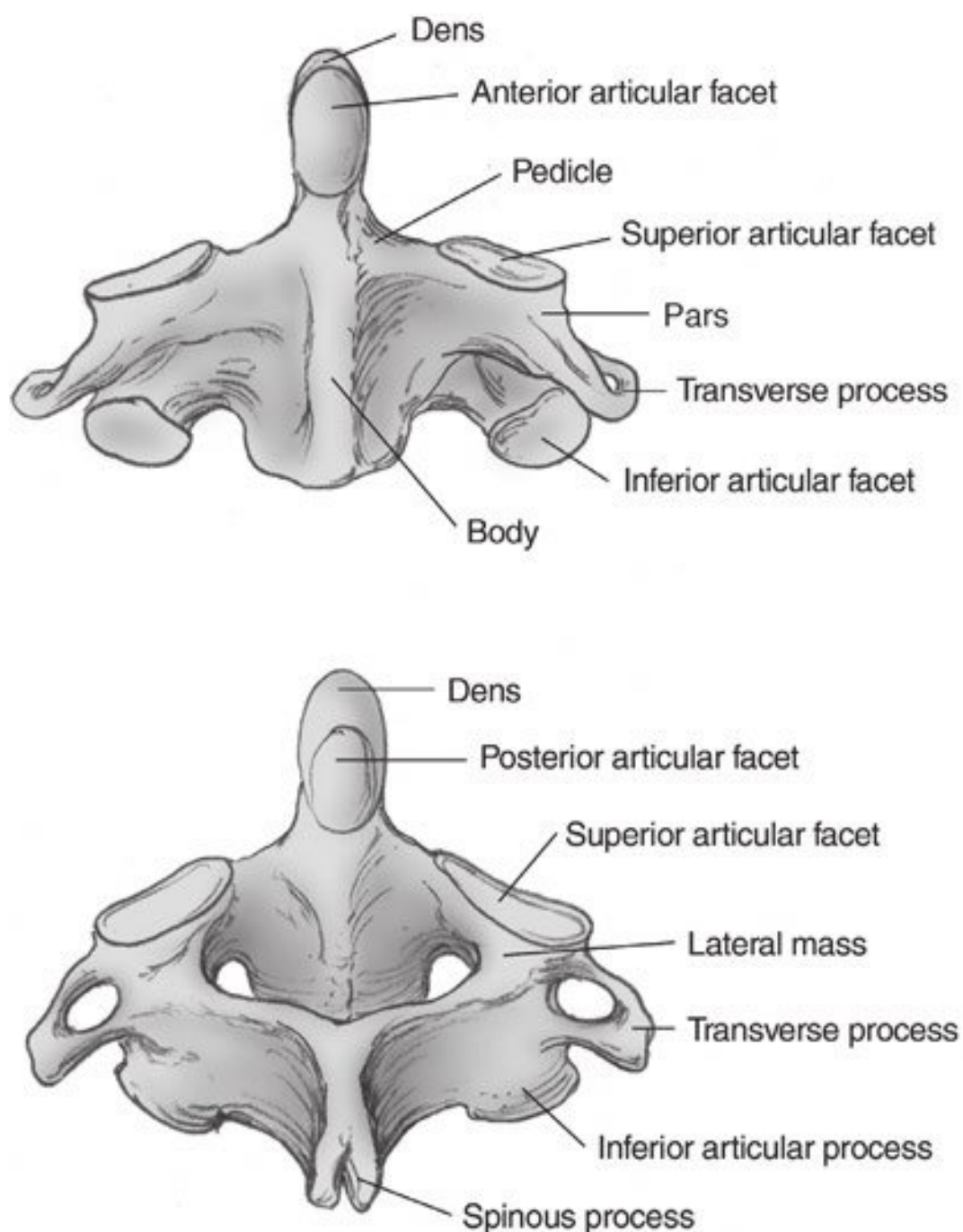


Fig. 37.2 Anatomy of the axis (C2). (From Fessler RG and Sekhar L, eds. *Atlas of Surgical Techniques: Spine and Peripheral Nerves*, Thieme; 2006.)

angulation. Typical screw length is 16 mm, but the screw must stop short of the transverse foramen (check with preoperative CT scan). The risk of vertebral artery injury for C2 pars screws is lower than that for C1–C2 transarticular screws.

- C2 pedicle screws
 - The C2 pedicle is the portion of the C2 vertebra that connects its posterior elements to the vertebral body. The entry point of the C2 pedicle screw is in the pars of C2, lateral to the superior margin of the C2 lamina. This is usually 2 mm lateral and 2 mm superior to the C2 pars screw entry point described above. The pedicle screw requires a medial angulation of 15 to 25° with 20° upward trajectory. For those who have very narrow C2 pedicles, risk of breach into the neural canal or transverse foramen is high (check with preoperative CT scans).
- C2 translaminar screws
 - Translaminar screws serve as a salvage technique for C2 pars screws or pedicle screws in cases of the anomalous high-riding vertebral artery or very thin pedicle. The entry point is at the junction of the spinous process and lamina. The trajectory has to meet the slope of the lamina while aiming mildly dorsally to avoid canal breach. If bilateral translaminar screws are used, offset the entry points craniocaudal to keep the two screw paths from intersecting.
 - Using polyaxial screws, the C1 lateral mass screws can be easily connected with either of the three kinds of C2 screws described above.
 - Bone graft may be placed into the C1–C2 facet joint. This requires sacrifice of the C2 exiting nerve roots, which can cause scalp numbness. Alternatively, C1–C2 wiring can be performed with bone graft placed between the C1–C2 laminae.

37.3.2 C1–C2 Transarticular Screws

- The advantage of the C1–C2 transarticular screw technique is the complete obliteration of the rotational motion of the atlantoaxial joint. The drawbacks of this procedure are the need for anatomical reduction of C1–C2 and the potential for vertebral artery injury. If placement of the first screw likely caused vertebral artery injury, then screw insertion into the contralateral side should not be attempted, for bilateral vertebral artery laceration could result in brainstem infarction and death. The preoperative CT scan must be carefully examined to exclude a high-riding vertebral artery or destruction of bone at the site of intended screw placement.

- The screw entry point is ~ 3 to 4 mm rostral and 3 to 4 mm lateral to the inferior medial portion of the C2-C3 facet joint. The K-wire trajectory is typically 15° medial with the superior angle visualized by fluoroscopy aiming at the C1 anterior tubercle (often 60°). While the K-wire is drilling, subtle changes in resistance may be perceived as the K-wire traverses the four cortical surfaces along the path into the C1 lateral mass: (1) the posterior C2 entry point, (2) the superior articular surface of C2, (3) the inferior articular surface of C1, and (4) the anterior cortex of the C1 ring. After the K-wire is placed, a cannulated drill bit, tap, and screw can be placed. Typical screw length is 36 to 46 mm.
- Wiring techniques:
 - Posterior wiring techniques require an intact posterior arch in both C1 and C2. Thus in cases of Jefferson fracture or hangman's fracture, wiring is of no use. Double-braided titanium cables are preferred (over steel wires), for they are more flexible and have less chance of causing cord or dural injury during the sublaminar passage. Several techniques of sublaminar wiring and bone graft placement have been reported, including the Brooks wiring, the Sonntag wiring, and the Gallie wiring. The Brooks technique places two sublaminar C1 and C2 cables with two interlaminar bone grafts (one on each side). In the Sonntag technique, a single midline iliac autograft is placed between the C1 lamina and the spinous process of C2. A single sublaminar cable is passed under C1, and the cable is wrapped around the spinous process of C2. This may be safer than the Brooks technique, a sublaminar pass under C2 not being needed.

37.4 Complications

- Vertebral artery laceration (unilateral vertebral artery occlusion could be asymptomatic, but bilateral vertebral artery injury could result in a large posterior circulation infarction and death). If there is a unilateral vertebral artery injury, consider placing the screw into the bone to tamponade the bleeding. Consider not placing the contralateral screw. Consider taking the patient for a postoperative angiogram to ensure that there is no artery dissection.
- Dural injury or cord injury.

37.5 Postoperative Care

- A cervical collar is used for 6 weeks postoperatively.

37.6 Outcomes

- Transarticular screws together with the Sonntag wiring technique and bone graft essentially create a three-point fixation that completely obliterates the rotational and flexion–extension motion of the C1–C2 joint. Apfelbaum reports that fusion was achieved in 99% of 198 patients undergoing transarticular screw fixation. However, Apfelbaum also reported a 16.7% complication rate, including five patients who had vertebral artery injuries, one of which was bilateral and fatal.⁵
- C1 lateral mass screws connected to C2 pars/pedicle, or translamina screws, provide biomechanical strength and actually facilitate anatomic reduction with a fusion rate higher than 95%⁶
- Biomechanical analysis in cadaveric specimens showed crossed translamina fixation to be superior to pars screws in strength.

37.7 Surgical Pearls

- Preoperative planning is crucial. The CT scan must be evaluated to rule out an anomalous vertebral artery path and to assess the bony anatomy.
- A thorough understanding of the 3D anatomy of the axis and atlas is mandatory.
- Be aware of the robust venous plexuses around the C1–C2 region, and use a hemostatic agent to diminish bleeding in this area.

Common Clinical Questions

1. Where is the entry point for the C2 pars screw?
2. Where is the entry point for the C1–C2 transarticular screw?
3. Should you place a contralateral C2 pars screw if you have an ipsilateral vertebral artery laceration during insertion of the first C2 pars/pedicle or transarticular screw?

Answers to Common Clinical Questions

1. The C2 pars screw entry point is ~3 mm rostral and 3 mm lateral to the inferior medial aspect of the inferior articular surface of C2.
 2. The same as for the C2 pars screw.
 3. No. Patients may tolerate a one-sided vertebral artery injury, but a bilateral vertebral injury could be fatal.
-

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38 Direct Fixation of Odontoid Fractures

Rajiv Saigal and Dean Chou

38.1 Key Points

- Odontoid fractures constitute 10 to 20% of cervical spine fractures.¹
- The Anderson and D'Alonzo system² divides odontoid fractures into three categories: Type I, fracture of the odontoid tip; type II, fracture of the base of odontoid neck; type III, fracture through the odontoid base and involving the C2 vertebral body.
- Isolated type II fractures in adults should be considered for surgery (**Fig. 38.1**).³
- Odontoid screw fixation is less mobility-limiting than posterior C1–C2 fixation. Approximately half of axial head rotation is generated from the C1–C2 complex.

38.2 Indications and Contraindications

- Indications
 - Type II odontoid fracture
 - Fracture geometry suitable to odontoid screw reduction
 - Acute to subacute fractures

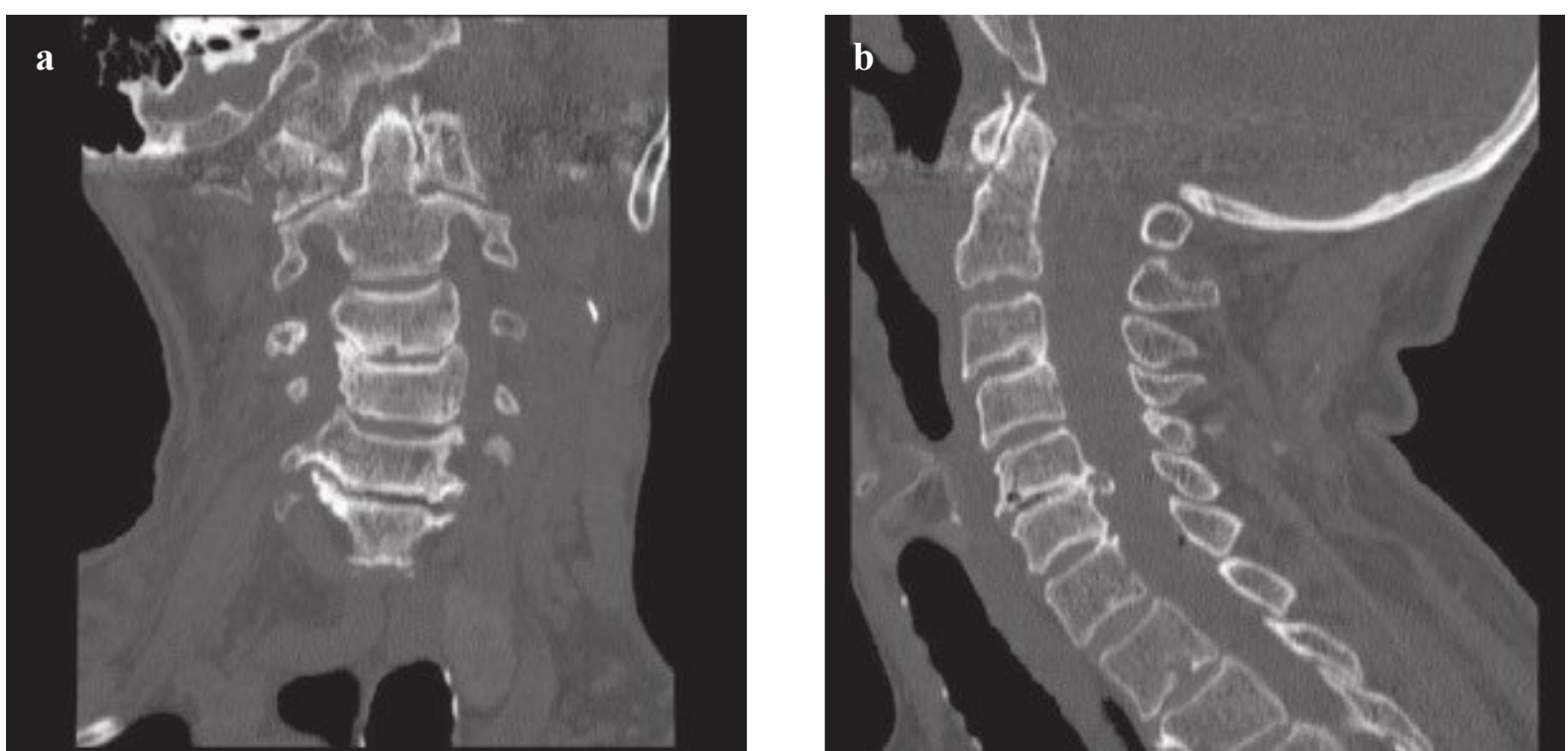


Fig. 38.1 Preoperative (a) coronal and (b) midsagittal CT images show a type II dens fracture with geometry suitable for anterior odontoid fixation.

- Contraindications
 - Disrupted transverse ligament.
 - Barrel chest, or short neck, or other anatomic variant that compromises desired screw trajectory.
 - Fracture angulation unsuitable to anterior screw placement. If the fracture line traverses from anteroinferior to posterosuperior in the sagittal plane, the lag screw will not be able to capture the distal fragment for reduction.
 - Fractures older than 6 months are associated with lower fusion rates, although the exact fracture age cutoff for surgical consideration is controversial. Most surgeons advocate odontoid screw placement within a week of injury if possible.
 - Os odontoideum.
 - Pathologic odontoid fractures from infection or tumor.
 - Osteoporosis.
 - Elderly patients who have osteoporosis and a severely ankylosed spine.

38.3 Technique

- The patient is placed in the supine position.
- Biplanar fluoroscopy is used with two C-arms.
- A large wad of cotton or a radiolucent item is placed to keep the mouth open as wide as possible.
- A lateral preoperative X-ray is taken to align the intended trajectory of the odontoid screw. The cervical spine is prepped and draped the usual sterile fashion.
- A transverse skin incision is made using a 10-blade knife, usually at the C5–C6 level. The skin incision should be centered several segments below the desired entry point.
- Metzenbaum scissors and bipolar cautery are used to dissect the platysma. Dissection is continued down to the spine in a similar manner as for the anterior cervical discectomy and fusion approach. The trachea and the esophagus are mobilized medially.
- The spine is identified and dissected toward C2–C3. The level is confirmed under X-ray guidance. To properly angulate the screw, part of the C2–C3 disk needs to be removed. Many times, if this is not removed, the screw will be too anteriorly placed, possibly compromising the anterior cortex of C2.
- A starting point must be localized in both the anteroposterior and lateral planes for a correct trajectory of the screw. Since this starting point dictates the entire operation, it is critical that this point be precisely placed before placing the K-wire.

- A small pilot hole is drilled at the base of the C2 body after identifying the correct starting point with AP and lateral fluoroscopy.
- If using a cannulated screw system, a Kirschner wire is applied under biplanar fluoroscopy up to the tip of the odontoid, engaging the cortex of the odontoid tip. Extreme care must be taken not to pass the point of the K-wire too deep. The brainstem and/or basilar artery could be critically injured with a deeply placed K-wire.
- The trajectory is tapped over the K-wire into the odontoid but not past the Kirschner wire, lest the K-wire become loose.
- After tapping, a lag screw is inserted under biplanar X-ray guidance to engage the tip of the odontoid. Cannulated and noncannulated screws are both options.
- The lag screw draws the tip of the odontoid toward the body of the odontoid. This is confirmed under fluoroscopic guidance. The screw is then tightened. The ideal screw length should be measured preoperatively on the CT accounting for the fracture gap.
- After meticulous hemostasis is obtained, the wound is irrigated. A surgical drain may be left in place but is usually not necessary. The platysma is closed with interrupted sutures. Skin is closed.

38.4 Complications

- Dysphagia
- Hoarseness
- Nonunion
- Improper implant placement
- Screw failure
- Need for posterior C1–C2 fusion

38.5 Postoperative Care

- After recovery in a postanesthesia care unit, patients are usually transferred to an acute care unit. Early mobilization is recommended.
- The authors obtain an anteroposterior and lateral cervical X-ray (shown in **Fig. 38.2**) to ensure proper implant placement before discharge.
- Use of a supportive cervical collar postoperatively is generally recommended for 1 to 3 months.
- As with other anterior cervical approaches, postoperative dysphagia is common. Diet should be advanced slowly to prevent the risk of aspiration.
- After discharge, the wound should be checked in an early postoperative visit.

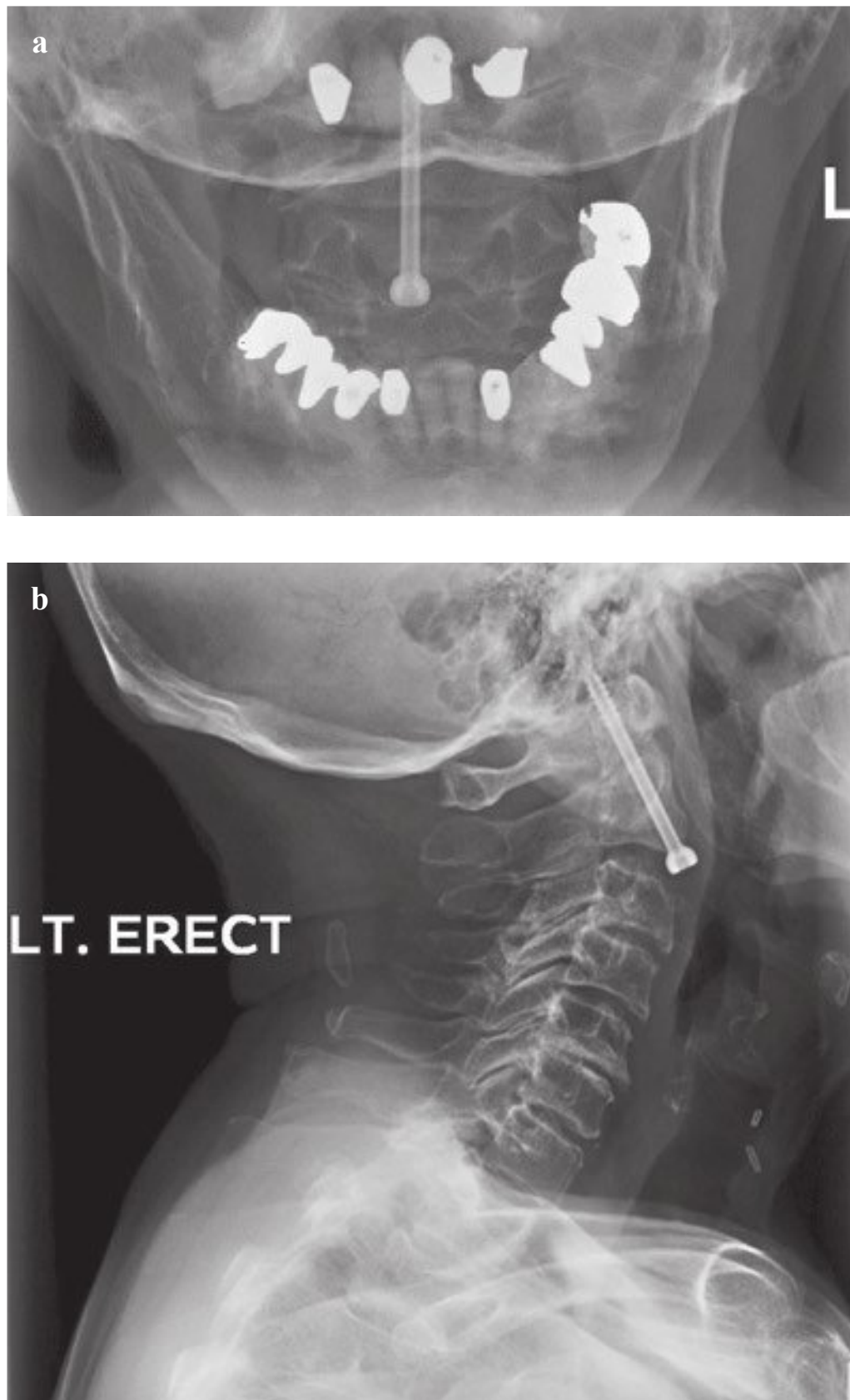


Fig. 38.2 Postoperative anteroposterior (a) open-mouth and (b) lateral X-rays show placement of odontoid screw with fracture reduction.

38.6 Outcomes

- Fractures less than 6 months of age have demonstrated a 90% fusion rate with anterior odontoid screw fixation, compared with 25% fusion rate for those over 18 months.¹

38.7 Surgical Pearls

- Great care must be taken to avoid the tracheoesophageal groove, the anatomic location of the recurrent laryngeal nerve.

- When using a Kirschner wire and cannulated screw system, use X-ray frequently to ensure that the wire is not inadvertently driven deeper than desired.
- Sometimes, the tip of the odontoid is very difficult to see on lateral fluoroscopy. Measure the distance from the top of the C1 tubercle to the top of the odontoid fragment by CT preoperatively. This will at least provide some idea of where the K-wire should stop if the tip of the odontoid is simply not well-visualized.

Common Clinical Questions

1. In general, should an odontoid screw be placed in an elderly patient with severe osteoporosis?
 2. How long after an odontoid fracture occurs is it too late to put in an odontoid screw?
 - A. 3 weeks
 - B. 3 months
 - C. 6 months
 - D. Never
-

Answers to Common Clinical Questions

1. As a general rule, no. Most elderly patients should undergo a posterior C1–C2 fusion.
 2. A. Although some literature says that an odontoid screw can be placed up to 6 months postinjury, most surgeons advocate placement before 3 weeks.
-

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39 Cervical Arthroplasty

Jau-Ching Wu, Ali A. Baaj, and Praveen V. Mummaneni

39.1 Key Points

- Anterior cervical discectomy and fusion (ACDF) remains the gold standard in the surgical management of symptomatic cervical spondylosis and central disk herniation.
- The emerging option of cervical arthroplasty is aimed at preservation of segmental motion with maintenance of adequate stability.¹
- Key elements to achieving a good result and avoiding complications with cervical arthroplasty:
 - Appropriate patient selection (competent posterior elements, without spondylolisthesis)
 - Correct neck positioning during surgery (neutral to slightly lordotic cervical curvature)
 - Generous decompression with resection of the bilateral uncovertebral joints and posterior longitudinal ligament
 - Precise end plate preparation (parallel or domed according to device, with preservation of cortical end plates)
 - Accurate midline acquisition
 - Proper implant sizing (footprint coverage of disk space and avoidance of overdistraction)

39.2 Indications

- The current indications for cervical arthroplasty in the United States include symptomatic one- or two-level cervical radiculopathy or myelopathy in patients who have failed nonsurgical management.
- Relative contraindications include (1) cervical kyphosis, (2) cervical spondylosis with incompetent or significantly degenerated facets ($> 2\text{--}3\text{-mm}$ subluxation on flexion–extension X-rays), (3) cervical ankylosis, (4) osteoporosis, and (5) cervical trauma with ligamentous or facet injury.

39.3 Technique

- Prophylactic antibiotics should be given; perioperative dexamethasone and intraoperative neuromonitoring are optional.

- The patient is positioned supine with the neck in neutral or slight extension, with shoulder retraction to allow adequate fluoroscopic visualization of the target level.
- Create a transverse skin incision along preexisting skin crease near the index level.
- Perform dissection between the carotid sheath and strap muscles (anteromedial to sternocleidomastoid muscle) to expose the retropharyngeal space.
- The trachea and esophagus are retracted and protected medially by self-retaining retractors placed under the elevated longus colli muscles for exposure of the anterior cervical spine.
- After confirmation of the index level by lateral fluoroscopic X-ray, anterior cervical discectomy is performed with curettes, rongeurs, or drill.
- Full bilateral nerve canal decompression must be achieved, including removal of the uncovertebral joints.
- End plate preparation is crucial in cervical arthroplasty and may differ between prostheses. Caution must be taken not to violate the cortical surface, which would increase the risk of implant subsidence or migration.
- Midline acquisition, implant sizing (including footprint size and disk height), and insertion trajectory should be controlled to allow physiological range of motion in the cervical spine after implantation of the prosthesis.
- There are several cervical artificial disks currently on the market, and each has its unique design and fixation system (**Fig. 39.1**). Thus there are

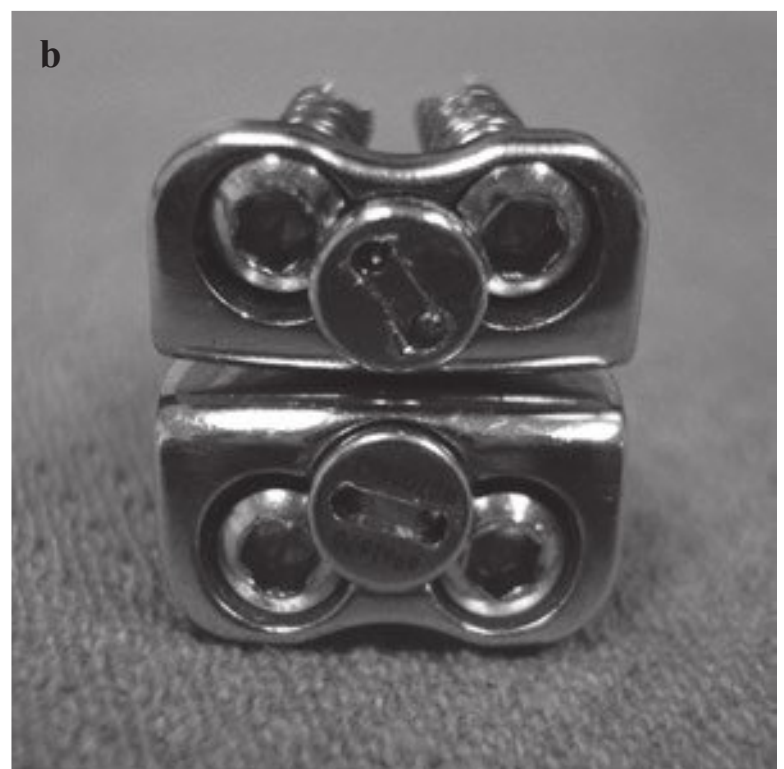


Fig. 39.1 Photographs of three arthroplasty devices currently approved by the U.S. Food and Drug Administration. **(a)** BRYAN device. **(b)** PRESTIGE device.



Fig. 39.1 (Continued) (c) Pro-Disc C device. (From Baaj AA, Uribe JS, Vale FL, et al. History of cervical disc arthroplasty. *Neurosurg Focus*. 2009; 27(3):E10. Reproduced with permission.)

differences in the technique of final implantation, but they all have the common feature of absolutely requiring adequate decompression of the index level. Some differences in material and biomechanics may provide advantages in certain scenarios. In choosing one, the surgeon must have a thorough understanding of the prosthesis’s biomechanical design and familiarity with its implantation (**Table 39.1**).

Table 39.1 FDA-approved cervical arthroplasty devices in the United States

Device	Manufacturer	Classification	Biomaterials
ProDisc-C	Synthes Spine, West Chester, Pennsylvania	Semiconstrained	CCM end plate with UHMWPE inlay
BRYAN	Medtronic Ltd., Memphis, Tennessee	Unconstrained	Titanium alloy shells with polyurethane nucleus
PRESTIGE ST	Medtronic Ltd., Memphis, Tennessee	Unconstrained	Metal on metal

Abbreviations: CCM, cobalt chrome molybdenum; UHMWPE, ultra-high-molecular-weight polyethylene.

39.4 Complications

Complications are similar to those for ACDF, but there are four primary reasons to consider revision of cervical arthroplasty:

- Radiculopathy (or other new-onset neurologic deficit) after cervical arthroplasty
- Subsidence
- Implant migration
- Ankylosed joint (formation of significant heterotopic bone around the implant)

39.5 Postoperative Care

Same as for ACDF, except for the following:

- Perioperative oral nonsteroidal antiinflammatory drugs (NSAIDs) for 2 weeks² might reduce the incidence of heterotopic bone formation.
- Avoid neck collar (encourage normal motion).

39.6 Outcomes

To date, there have been several prospective randomized multicenter U.S. studies comparing cervical artificial disk implantation with ACDF in patients treated for single-level cervical disk disease with radiculopathy or myelopathy.

- The largest of these studies compared the PRESTIGE ST Cervical Disc System (Medtronic; Memphis, Tennessee) with ACDF in a total of 541 patients. At 24 months' follow-up, the arthroplasty group demonstrated maintenance of physiologic segmental motion in association with improved neurologic success, improved clinical outcomes, and a reduced rate of secondary surgeries compared with conventional ACDF.²
- Heller et al compared the BRYAN Cervical Disc (Medtronic; Memphis, Tennessee) with ACDF in 463 patients. At 24 months after surgery, the arthroplasty group showed a statistically greater improvement in the primary outcome variables: Neck disability index score and overall success. No statistical difference was found between the arthroplasty and ACDF groups with regard to secondary surgical procedures or implant-related adverse events. The arthroplasty patients returned to work nearly 2 weeks earlier than the ACDF patients.³
- Murrey et al compared single-level ProDisc-C (Synthes Spine, L.P.; West Chester, Pennsylvania) arthroplasty with ACDF in 209 patients. At 24 months after surgery, visual analog scale (VAS), neck disability index (NDI), and neurologic success rate demonstrated no statistical difference between

the arthroplasty and ACDF groups. Statistically fewer reoperations and less pain medication usage were noted in the ProDisc-C cohort.⁴

- Davis et al published a two-level arthroplasty study with the Mobi-C device (LDR; Austin, Texas). The study compared 225 patients randomized to ACDF versus two-level arthroplasty with 4 years of follow-up. The arthroplasty patients maintained cervical range of motion, had improved outcomes, and had less secondary surgical events compared with the ACDF patients.⁵

39.7 Surgical Pearls

- Generous decompression of the neuroforamen bilaterally (including asymptomatic side) is essential.
- Proper end plate preparation, but not disruption, is essential.
- Correct implant selection (footprint size, height) without oversizing is mandatory.
- The following cervical arthroplasty devices are currently FDA-approved:
 - Prestige ST
 - Prodisc-C
 - Bryan Disc
 - Secure-C
 - PCM
 - Mobi-C (1- and 2-level: The only disk approved for two levels)
 - Prestige LP

Common Clinical Questions

1. How many degrees of neck range of motion loss could result from a single-level ACDF?
2. What is the incidence of development of symptomatic adjacent-level disease after single-level ACDF?
3. Why should patients take oral NSAIDs and avoid neck collar use after cervical arthroplasty?

Answers to Common Clinical Questions

1. A fusion of one level of the cervical spine typically results in the loss of 7° of flexion and extension motion and 6° of rotational motion.²
2. Symptomatic adjacent-segment disease occurs at a rate of 2.9% per year during the 10-year period after the operation (Hillibrand et al, 2008).⁶
3. NSAIDs and motion may reduce the occurrence of heterotopic bone formation around the artificial disk.⁷

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40 Anterior Cervical Corpectomy

Colin B. Harris and Frank M. Phillips

40.1 Key Points

- Anterior cervical corpectomy is a safe and effective technique for decompression of the ventral cervical spinal cord.
- The surgical approach and exposure for corpectomy are similar to the case for the more common anterior cervical discectomy.¹

40.2 Indications

- Generally, corpectomy is indicated when there is ventral spinal cord compression and discectomy is inadequate for neural decompression.
 - Degenerative conditions, such as spondylotic myelopathy in which ventral spinal cord compression is not restricted to the disk level and/or is complicated by cervical kyphosis
 - Tumor infiltration of the vertebral body causing vertebral body collapse and spinal cord compression
 - Traumatic cervical spine injury, such as a teardrop fracture with associated spinal cord compression from retropulsion of vertebral body fragments and spinal instability
 - Osteomyelitis with epidural abscess and concomitant ventral spinal cord compression
 - Ossification of the posterior longitudinal ligament

40.3 Technique

- Intubation via fiberoptic technique, whether awake or asleep, is safer in cases of significant cervical spinal cord compression, myelopathic signs, myelomalacia, and/or cervical instability.
- Awake fiberoptic intubation allows for neurologic assessment immediately after intubation to assess for changes in neurologic function.
- Tape the endotracheal tube away from the side of surgical incision.
- Steroid administration preoperatively or intraoperatively lacks robust scientific support.
- Neurologic monitoring including somatosensory evoked potentials (SSEP) and motor evoked potentials (MEP) is prudent during operations on

myelopathic patients. Pre- and postpositioning signals may help to limit any neurologic deterioration from subtle changes encountered during positioning.

- Some authors routinely use 8 to 10 pounds of traction with Gardner-Wells tongs or chin strap craniocervical traction to increase disk space distraction. Alternatively, Caspar distraction pins (Aesculap Implant Systems, Center Valley, Pennsylvania) can be placed in the adjacent vertebral bodies above and below the corpectomy site to provide direct distraction.
- Place patient supine on the operating table.
 - Pressure points should be padded, especially the ulnar nerve.
 - A roll between the scapulae may improve lordosis and access to the level of interest. In patients who have myelopathy, neck hyperextension must be avoided.
 - Shoulders should be taped down to allow for fluoroscopic visualization of the level of interest, but excessive traction can injure the brachial plexus.
- Localize, with fluoroscopy or using external landmarks, the approximate level of incision along the neck.
- The surgeon may approach the anterior cervical spine from the right or left side; the left-sided approach theoretically involves less risk to the recurrent laryngeal nerve.²
- The incision may be made transversely or longitudinally depending on the number of levels involved and surgeon preference.
- After incision, identify the platysma beneath the subcutaneous layer, and divide it sharply in a transverse or longitudinal fashion.
 - Dissecting under the platysma superiorly and inferiorly allows for better mobilization of the tissues.
- Identify the medial border of the sternocleidomastoid (SCM).
- Using Metzenbaum scissors, dissect in the areolar plane through the deep cervical and pretracheal fascial layers, between the SCM and the medial structures.
 - The carotid artery should be palpated lateral to the plane of dissection to ensure that the trajectory of dissection is correct.
- Using a handheld retractor, retract the medial structures from the SCM to expose the prevertebral fascia.
- Incise the fascia to expose the cervical spine; use Kittners to sweep the prevertebral fascia superiorly and inferiorly.
- Use a spinal marker in the vertebral body to confirm the level with X-ray or fluoroscopy image (avoid placing the marker in the disk space to avoid inadvertent iatrogenic injury to an adjacent level disk).
- With electrocautery or periosteal elevator, dissect the longus colli laterally from their midline attachments.

- Place self-retaining retractors deep to the longus colli and periodically deflate the cuff of the endotracheal tube to minimize esophageal/tracheal edema in prolonged cases.
- Caspar pins may be placed into the vertebral bodies above and below the level of interest, with gentle distraction then applied.
- The disks above and below the level of interest should be incised and removed with curettes and pituitary rongeurs.
- When necessary, anterior osteophytes can be removed with a rongeur.
- A bur can be used to thin the vertebral body to the posterior longitudinal ligament (PLL); alternatively, a Leksell rongeur can be used to first harvest the anterior vertebral body for use as autograft if the bone is not pathologic.
- The PLL is carefully pierced with a small hook or curette.
- A small Kerrison rongeur (1 or 2 mm) can then be used to lift the PLL away from the dura and complete the decompression.
- Lateral decompression (foraminotomy) is generally completed at the level of the uncovertebral joint. The uncus can be thinned with a bur, followed by removal with a microcurette or Kerrison rongeur (**Fig. 40.1**).

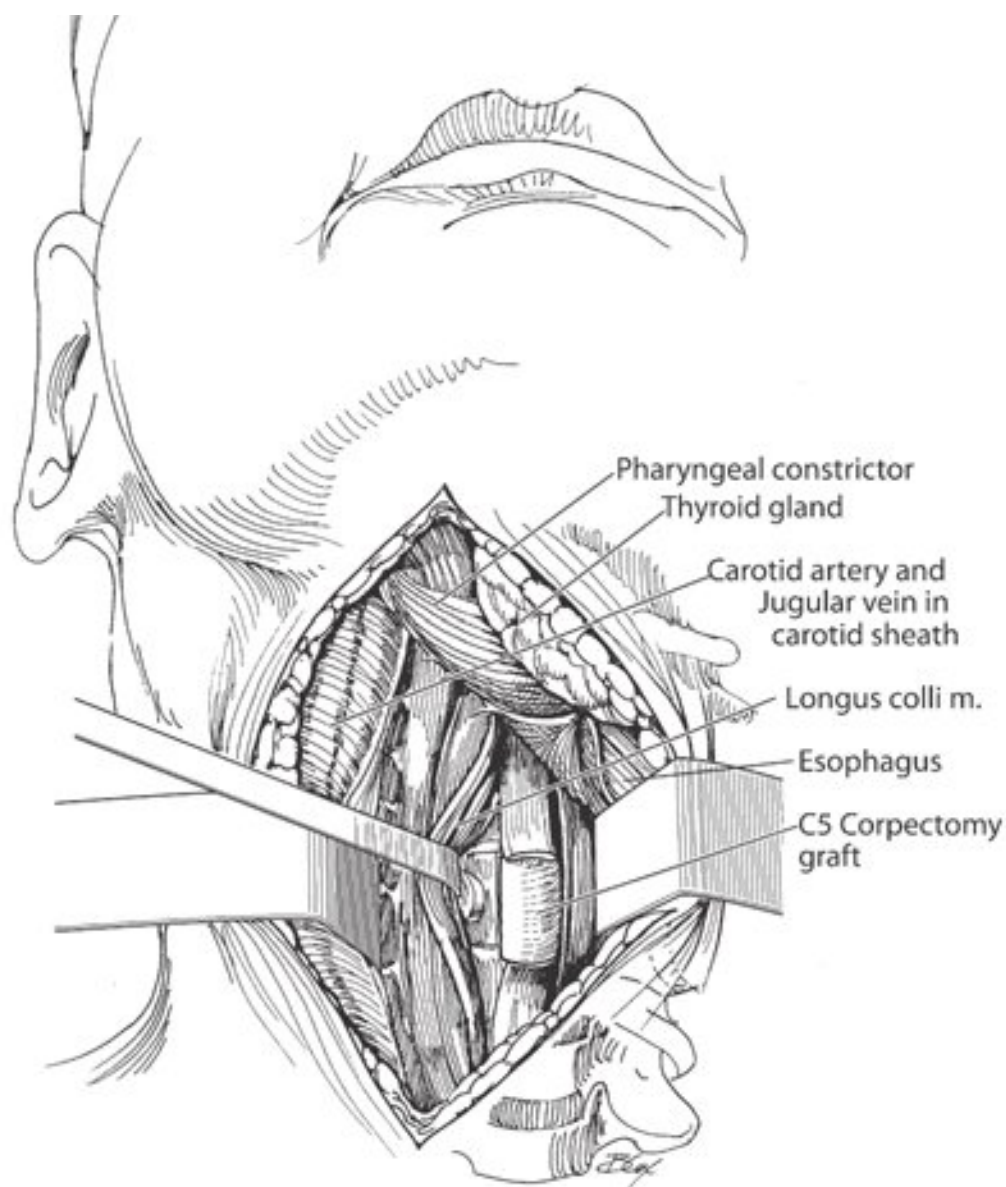


Fig. 40.1 Anterior view of the surgical field after single-level corpectomy in the cervical spine.

- A strut graft or cage filled with autograft, allograft, or synthetic or ceramic allograft substitute can be placed into the defect after contouring of the vertebral body end plates.
- An anterior plate can be used to enhance fusion and prevent graft migration. If possible, multiple points of screw fixation should be established. Posterior instrumentation does, however, provide greater stability and may be required.
- Integrity of the posterior ligamentous structures and facet capsules may allow greater confidence in an anterior only construct with plate fixation.³
- Some authors recommend rigid anterior fixation to enhance rigidity and reduce graft subsidence; others argue that dynamic plates and screws may enhance fusion by loading the graft (Wolff's law). Data in support of either approach are limited to theoretical considerations and retrospective studies.^{4,5}

40.4 Complications

- Wound infection
- Hoarseness
 - Reduce risk by reducing endotracheal cuff pressure and requesting that the anesthetist deflate and then reinflate the cuff once the retractors are set.
 - Avoid dissection into the carotid sheath.
 - Some surgeons advocate a left-sided approach to avoid risk of recurrent laryngeal nerve injury, though doing so is controversial.
- Transient or permanent dysphagia, more common with longer operative times and greater number of operative levels
- Nerve root or spinal cord injury⁶
- Acute airway obstruction from swelling or hematoma
 - Immediately stabilize the airway and intubate.
 - Open the wound at the bedside to allow for decompression of the hematoma and thereby ease intubation in an emergent setting.
 - Reintubation is difficult in this setting and emergency tracheostomy can be required.
- Cerebrospinal fluid (CSF) leak
 - Difficult to close primarily.
 - Consider patching the defect with collagen matrix or tissue autograft and cover with fibrin glue or dural sealant. Be sure that these materials have the potential to cause neural compression, particularly after they swell.
 - Place lumbar drain to divert CSF if there is concern about the risk of persistent drainage.
- Vascular injury

- Vertebral artery
 - The vertebral artery lies lateral to the uncovertebral complex and is often surrounded by a collection of veins.
 - Inadvertent injury can lead to massive bleeding.
 - In the event of injury, the surgeon must immediately administer tamponade to the bleeding, usually with the aid of Gelfoam (Pharmacia & Upjohn, Kalamazoo, Michigan) or another hemostatic agent. The vessel can be repaired primarily or ligated surgically with a reported low risk of neurologic sequelae.^{7,8}
 - An emergent angiogram should be considered to rule out vertebral artery dissection and may require vertebral artery embolization.
 - The artery can be sacrificed with little consequence if it is nondominant; sacrifice of the dominant vertebral artery has a higher incidence of ischemic injury to the brainstem.⁷
 - In cases involving tumor or at higher risk for aberrant vertebral artery anatomy, preoperative CT or MR angiography should be considered to determine vertebral artery dominance.
- Graft dislodgment/instrumentation failure
 - Graft dislodgment may indicate pseudarthrosis, hardware failure, infection, or a combination of factors.
 - Of primary importance is airway stabilization; if there is significant tracheal compression, emergent intubation and reoperation may be required.
 - Revision of the graft may include the use of autograft, such as iliac crest, and posterior cervical instrumentation and fusion to supplement the anterior construct.

40.5 Postoperative Care

- After a prolonged procedure involving concern for anterior soft tissue swelling, consider keeping the patient intubated for 24 to 48 hours (until able to breathe around deflated airway cuff).
- Consider external immobilization of the cervical spine.
- Patients who have myelopathy and gait unsteadiness may benefit from physical and occupational therapy evaluations.
- Ensure that the patient is tolerating oral intake and swallowing normally before discharge.

40.6 Outcomes

- The goals of decompressive corpectomy are neurologic preservation, cervical spine stabilization, and/or reduction of deformity.

- Available outcomes studies are limited in their retrospective nature but indicate excellent technical results (greater than 98% fusion rate) and improvement in radicular and myelopathic symptoms in 86%⁹
- Even in patients with severe myelopathy resulting in a bedridden or wheelchair-bound condition, clinical outcomes showed improvement in more than 60% of patients who underwent surgery.¹⁰
- Outcomes are similar for both anterior and posterior approaches to cervical spondylotic myelopathy,¹¹ though anterior cervical corpectomy and fusion may be preferred in cases of anterior retrovertebral pathology not amenable to discectomy alone or cases of significant kyphosis.
- Restoration of lordosis (cases of more severe preoperative kyphosis) with compression primarily at disk level can be performed more reliably with a multilevel anterior cervical discectomy and fusion or hybrid approach.¹²

40.7 Surgical Pearls

- Identify the location of vertebral artery on preoperative imaging studies.
- Avoid hyperextension of the cervical spine during positioning.
- Generous dissection of the longus colli muscle allows better identification of midline, proper placement of retractor blades underneath the muscle fibers, and decreased pressure on vascular structures laterally or esophagus and trachea medially. However, avoid excessive anterior or lateral dissection to avoid possible vertebral artery injury or sympathetic nerve injury (Horner's syndrome).
- Begin decompression with discectomy cranial and caudal to the intended corpectomy. This will assist in identification of the appropriate depth and width (uncus to uncus) of corpectomy.
- When the PLL is severely adherent, dissociation of the PLL from the adjacent structures, thus eliminating its constrictive effect on the spinal cord and preventing thecal sac manipulation, is the primary goal. Removing the PLL entirely becomes a secondary goal if it can be safely achieved but increases the risk of intraoperative CSF leak and possible neurologic injury.¹³

Common Clinical Questions

1. True, or false? The fusion rate after anterior cervical corpectomy increases with augmented posterior fixation, especially after two- and three-level corpectomies.
 2. True, or false? Anterior corpectomy and fusion can achieve a greater restoration of lordosis than multilevel anterior cervical discectomy and fusion.
-

Answers to Common Clinical Questions

1. True.
2. False.

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41 Anterior Cervical Disectomy

Daniel C. Lu, Khoi D. Than, Kevin T. Foley, and Praveen V. Mummaneni

41.1 Key Points

- Appreciation of the surgical anatomy of vital structures (carotid, esophagus, recurrent laryngeal nerve) during approach is essential.
- To decrease incidence and severity of dysphagia, endotracheal cuff pressure can be decreased during the retraction phase of the surgery. Additionally, intermittent relaxation of the retraction can be used.

41.2 Indications

- Symptomatic herniated nucleus pulposus
 - Radiculopathy (after failed conservative therapy)
 - Myelopathy
- Cervical spondylosis with radiculopathy or myelopathy (avoid anterior approach if the dura is calcified by continuous subtype ossification of the posterior longitudinal ligament [OPLL])
- Ossification of the posterior longitudinal ligament with myelopathy
- Cervical fracture with instability

41.3 Technique

- Place patient in supine position, arms tucked at sides.
 - May use horseshoe headrest with weight strap (7–10 lbs.)
 - May use foam doughnut head holder with shoulder retraction using thick tape
 - Intravenous (IV) bag placed longitudinally between shoulder blades to provide mild head extension
- Localize with fluoroscopy (useful landmark: Cricoid cartilage approximates C5–C6).
 - For a single-level procedure, the incision is marked parallel to the disk space.
 - If a multilevel procedure is performed, the incision is marked at the midpoint (i.e., over the C5 body for a C4–C5, C5–C6 anterior cervical disectomy and fusion [ACDF]).
 - A transverse 3-cm incision is adequate for single-level procedure; a longitudinal incision along the anterior border of the sternocleidomastoid is preferable for a procedure having three or more levels.

- Incision is made and the platysma undermined; the plane medial to the sternocleidomastoid muscle is identified and bluntly dissected with a finger or blunt instrument in a rostral-to-caudal direction (**Fig. 41.1**). The omohyoid muscle overlies the C6 level and can be divided.¹
- The carotid is then palpated and the plane medial to the carotid and lateral to the trachea and esophagus is developed by blunt dissection in the rostral-to-caudal direction.
- The spine is then palpated, the prevertebral fascia is entered, and the disk level is confirmed by X-ray or fluoroscopy.
- Handheld Cloward retractors are used to retract and provide protection medially (the trachea and esophagus) and laterally (the carotid sheath), whereas the longus colli muscles are mobilized with Bovie electrocautery (Bovie Medical Corp., Clearwater, Florida). The midline is marked on the vertebral body to facilitate alignment of the cervical plate.

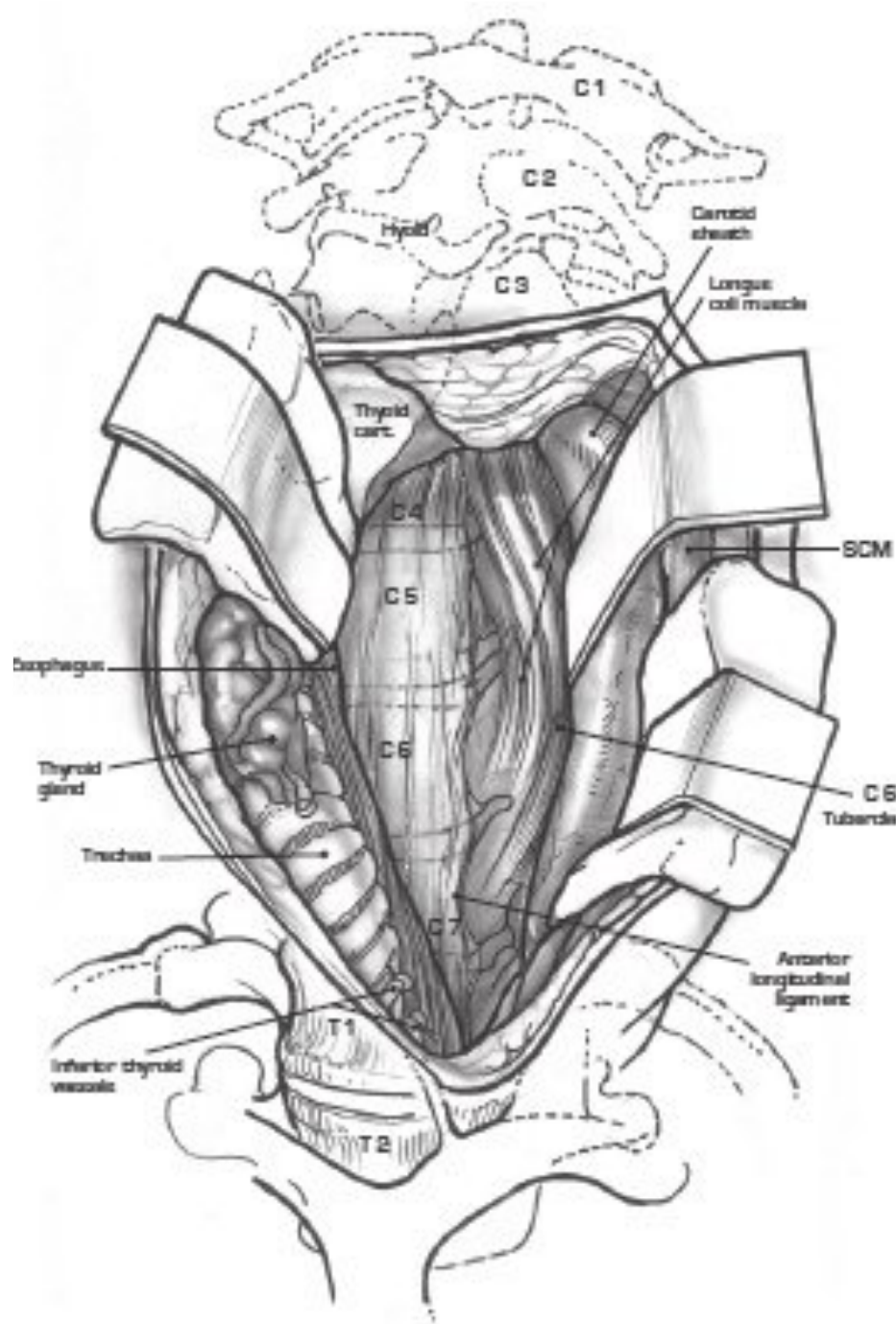


Fig. 41.1 Anterior exposure to cervical spine. (From Hafer R and Merola A, eds. *Surgical Techniques for the Spine*, Thieme; 2003.)

- Self-retaining retractor (Trimline [Medtronic; Memphis, Tennessee] or equivalent) is placed underneath the longus colli. The endotracheal tube cuff may be mildly depressurized at this point.
- Disk anulectomy is performed with a number 15 blade scalpel; additional disk removal is performed with a pituitary rongeur. The cartilaginous end plate is removed from the bony end plate with curettes. Carefully remove the lateral disk demarcated by the uncovertebral joint.
- Osteophytes/calcified disk fragments are drilled down to the posterior longitudinal ligament (PLL), the angle of drilling follows the angle of the disk space as seen with intraoperative radiographs or fluoroscopy. Take care not to drill away the bony end plate.
- The PLL is defined and first entered with a fine-angle nerve hook; it is identified by round longitudinal fibers. It is resected further with fine Kerrison punches. The dural plane is now evident.
- Allograft, autograft, or polyetheretherketone (PEEK)/carbon fiber cages filled with bone may be sized and placed into the disk space.² Temporary sizers are introduced into the intervertebral space with head traction to assess the correct interbody spacer size.
- An anterior cervical plate is placed aligning the midline with previous mark.
- Plates can be fastened to the vertebral bodies with fixed-angle screws or variable-angle screws or a combination of the two.
 - An entirely fixed construct is useful for situations that require rigid fixation (i.e., trauma cases).
 - Variable-angle screws allow for rotational pivoting at the screw–plate interface, which allows for increased loads to be placed upon the bone graft in the disk space. This is called “rotational subsidence” and allows exploitation of Wolff’s law (bone heals when subjected to loading).
 - A hybrid construct uses fixed-angle screws at the most caudal level (acting as a buttress) and variable-angle screws at the more rostral level(s) (allowing for controlled rotational subsidence).
- The wound is irrigated with saline.
- A small subfascial drain may be placed.
- Closure of platysma is done with interrupted 2.0 Vicryl (Ethicon, a Johnson & Johnson company; New York) and the subcutaneous layer with interrupted 3.0 Vicryl. Skin is closed with a subcuticular running 4.0 Monocryl suture.
- Steri-Strips (3M; St. Paul, Minnesota) may be used for skin.

41.4 Complications

- Nerve root injury (C5 nerve root palsy in up to 5%)³
- Spinal cord injury (especially in myelopathic patients; avoid hyperextension)
- Wound hematoma (higher in multilevel cases; may be mitigated by drain placement)
- Hoarseness
- Dysphagia (as a result of edema or recurrent laryngeal nerve palsy)
- Esophageal perforation
- Carotid or vertebral artery injury (0.3%)
- Pseudarthrosis (2–20%)
- Wound infection (< 1%)

41.5 Postoperative Care

- Use soft cervical collar for comfort (one- or two-level) and hard collar for immobilization (three or more levels).
- Use postoperative antibiotics while the drain is in place.
- May be discharged the same day with 6-hour observation in recovery unit.

41.6 Outcomes⁴

- Anterior cervical discectomy and fusion (ACDF) is a successful procedure when performed given the right indications.
- Adjacent-level disease occurs at a rate of around 2 to 3% per year in patients undergoing single-level ACDF.

41.7 Surgical Pearls

- For patient counseling, ACDF is not fully effective in patients who have predominant neck pain.
- The surgeon must know the course of the recurrent laryngeal nerve (RLN) to avoid injury during surgery. On the right, the RLN loops around the right subclavian artery, and the left RLN loops around the arch of the aorta. It runs within the tracheoesophageal groove, where sharp dissection in the paratracheal muscles or prolonged retraction against an inflated endotracheal tube may cause injury. Injury causes hoarseness, cough, aspiration, mass sensation, dysphagia, and vocal cord fatigue. Studies have not demonstrated a difference in RLN injury when comparing right and left anterior cervical exposure.

Common Clinical Questions

1. What is the course of the recurrent laryngeal nerve, and what symptoms would be manifested by a recurrent laryngeal nerve injury?
2. What measures can be undertaken to minimize the risk of dysphagia in ACDF?
3. Which can destruction of vertebral body end plates during discectomy cause?
 - A. Subsidence of the interbody graft is likely
 - B. Foramenal height will increase
 - C. Facets may sublux
 - D. Blood loss for the case will decrease
 - E. All of the above

Answers to Common Clinical Questions

1. The recurrent laryngeal nerve runs within the tracheoesophageal groove after looping around the aortic arch (left) or the subclavian artery (right). Injury would cause hoarseness, cough, aspiration, mass sensation, dysphagia, and vocal cord fatigue.
2. Decreased cuff pressure after placement of self-retaining retractors, intermittent release of retraction during surgery, and intraoperative use of Decadron.
3. A.

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42 Cervical Laminectomy with and without Fusion

Shyam M. Shridharani and F. Andrew Rowan

42.1 Key Points

- In presence of significant central canal stenosis and cervical myelopathy, surgical treatment is appropriate.
- The goal of surgery is to decompress the neural elements and provide stabilization via instrumentation/fusion.
- Focal anterior pathology may be better treated by anterior decompression and fusion.
- Lordotic alignment of the cervical spine is needed to allow the spinal cord to “drift” posteriorly postdecompression.
- Cervical laminectomy and/or fusion are not intended for the treatment of neck pain.
- Preoperative detailed history and physical examination along with radiographs, MRI, and, at times, CT are essential.

42.2 Indications

- Multilevel cervical spondylosis causing cervical stenosis and myelopathy
- Cervical stenosis caused by ossification of the posterior longitudinal ligament (OPLL)
- Ligamentum flavum thickening with resultant stenosis
- Epidural abscess
- Tumor
- Trauma
- Epidural hematoma
- Anterior neck anatomy that would preclude an anterior approach

42.3 Contraindications

- Presence of less than 10° of cervical lordosis on preoperative extension lateral radiograph, in which case an anterior approach should be considered to restore lordosis of the cervical spine
- Anatomy preventing safe placement of instrumentation such as aberrant course of the vertebral artery

42.4 Technique

- Awake, fiber-optic or GlideScope (Verathon Incorporated, Bothell, Washington) intubation may be necessary in cases of severe stenosis or instability.
- Place patient in prone position (requires a team approach).
- Neural monitoring is advised.
 - Important in the setting of severe stenosis or cervical instability.
 - May reduce the rate of neural injury.¹
 - Obtain preflip baselines in the setting of cervical instability or severe stenosis.
- Head positioning options include Mayfield head holder and ProneView (Dupaco, San Diego, California).
- In the setting of an unstable cervical spine or severe cord compression, “sandwiching the patient and spinning prone” using the Jackson table prone positioning technique is advisable.
- Assess head and neck alignment visually and fluoroscopically to ensure appropriate alignment.
- Localize with fluoroscopy and make appropriate-length midline incision.
- Use the midline avascular plane to maintain the integrity of posterior neck muscles and minimize bleeding.
- Using a Cobb and cautery, perform subperiosteal dissection to expose the spinous processes, lamina, and lateral masses.
- Avoid exposing lateral masses and violating the facet joints at levels above or below the operative levels.
- Perform lateral mass fixation.
 - Magerl technique:²
 - The lateral mass is divided into quadrants.
 - The entry point is at the superior medial quadrant and the trajectory is toward the superior lateral quadrant (“up and out”) (**Fig. 42.1**).
 - Used for C3–C6.
 - Roy-Camille technique:³
 - The entry point is at the midpoint of the lateral mass.
 - The trajectory is 10° lateral in the axial plane.
 - Often used at the C7 level.
 - Other described lateral mass instrumentation techniques include Anderson and An techniques.
 - If C7 is included in the fixation, options are pedicle screws versus lateral mass screw placement.
 - Technique for C7 pedicle screw placement requires additional consideration.
 - A laminotomy to palpate the medial edge of the C7 pedicle is useful.
 - The entry point is in line with the lateral mass entry points at superior levels.
 - Medial angulation is necessary to enter and traverse the pedicle (correlate/measure with preoperative cross-sectional imaging).

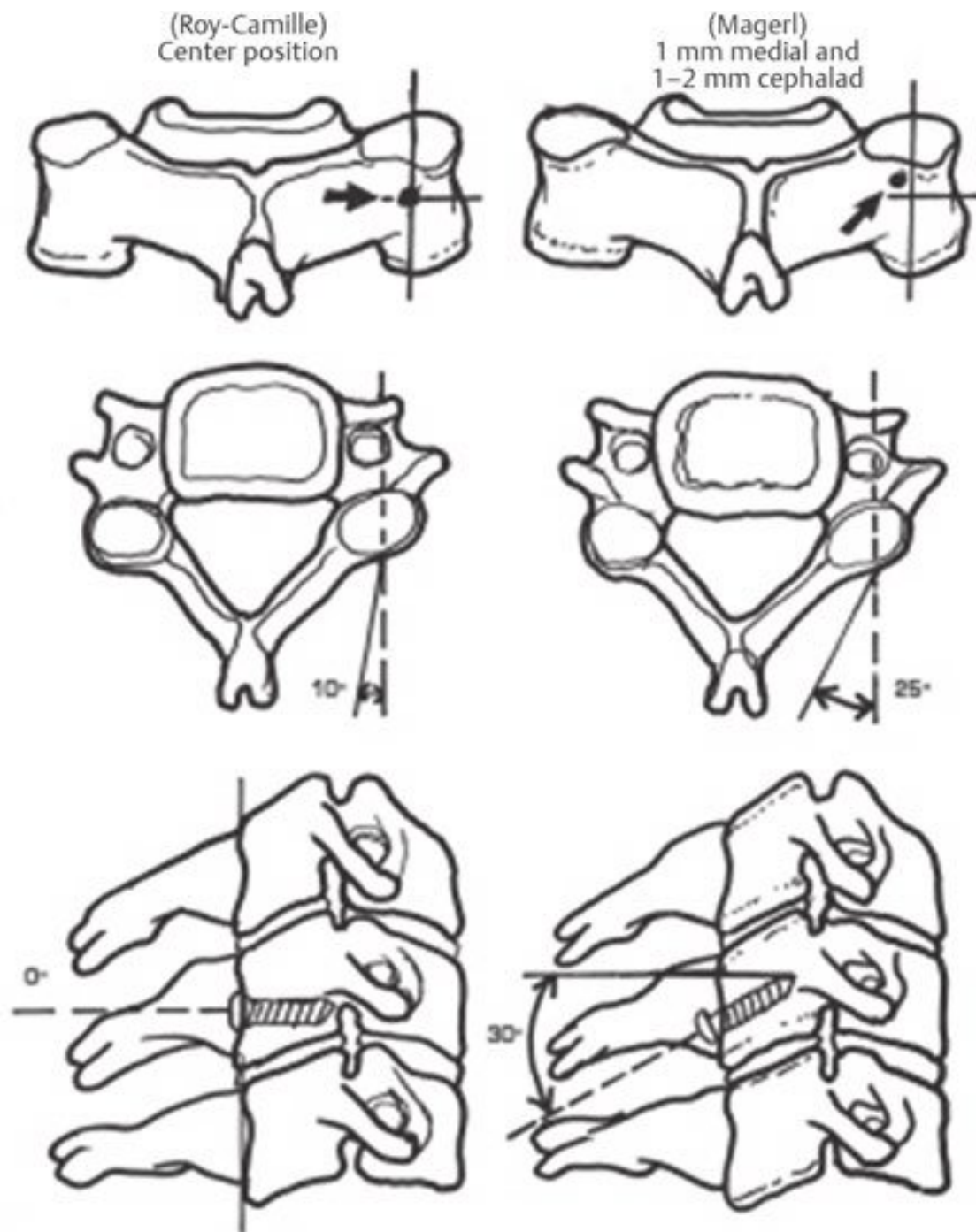


Fig. 42.1 The Roy-Camille and Magerl and An techniques for placement of lateral mass screws in the subaxial cervical spine.

- Lateral mass screw size is typically 3.5 or 4.0 mm by 12 to 16 mm.
- C7 pedicle screw size is typically 3.5 or 4.0 mm by 16 to 24 mm.
- Insert rods of appropriate length and secure with set screws.
 - It is preferred to contour the rods to match the screw heads. (Overusing rod reduction devices such as persuaders may cause loosening or pullout of instrumentation.)
 - Assess sagittal alignment fluoroscopically before final tightening set screws.
 - Final tighten set screws with use of torque limiting screwdriver.
- Performing posterior cervical laminectomy (three options).
 - The senior author's preferred technique is the use an AM-8 drill bit and/or a 2- to 3-mm diamond high-speed bur to create bilateral troughs at the lamina–lateral mass junction and remove the posterior elements in a “lobster tail” or “en bloc” fashion (**Fig. 42.2**).

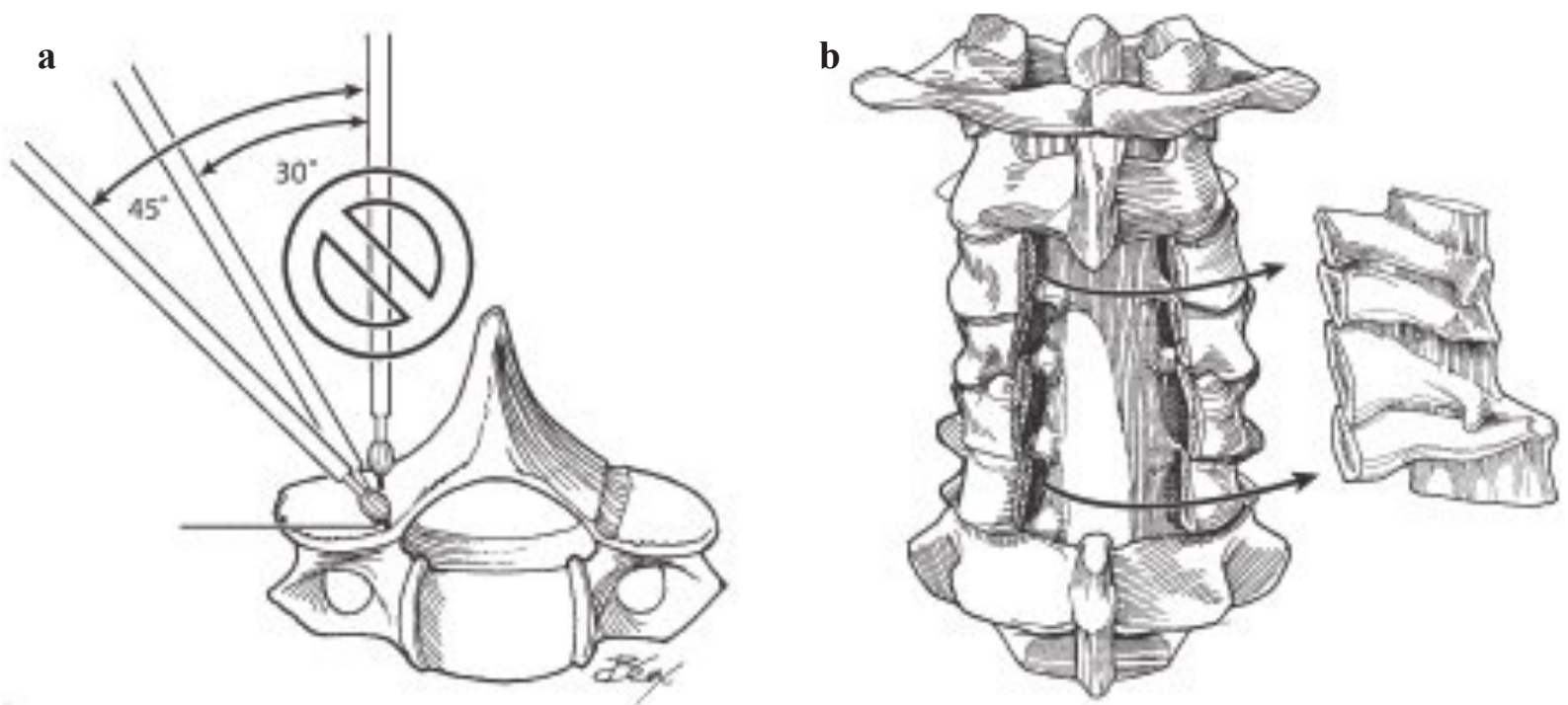


Fig. 42.2 Laminectomy. **(a)** The burr must be directed ~ 30 to 45° in the sagittal plane to avoid burring into the facet joint. Doing so may risk loss of orientation and cause one to bur too deeply, endangering the dural sac or spinal cord. **(b)** En bloc (“lobster tail”) removal of laminae. (From Vaccaro AR and Albert TJ, eds. *Spine Surgery: Tricks of the Trade*, 2nd ed., Thieme; 2009.)

- Apply clamps such as penetrating towel clips to posterior spinous process and apply gentle upward (posteriorly directed) traction while using a fine curette or Woodson to free any adherent ligamentum flavum. (Generally the assistant applies the posteriorly directed traction.)
 - Advantage: No placement of a Kerrison under the lamina and into the spinal canal
 - Disadvantage: Requires proficiency with the drill and places the lateral mass at risk if the troughs are too lateral
- An alternative technique is to use a combination of a rongeur and Kerrison (manual laminectomy), similar to lumbar laminectomy technique.
 - Advantage: More familiar technique, being widely used in the lumbar spine
 - Disadvantage: Potential risk of neural injury with Kerrison repetitively inserted deep/underneath the lamina
- The third technique is to use a footplate craniotome to create bilateral troughs (which may carry higher risk, in the authors’ opinion).
- Decortication of the facets and lateral masses is performed to create a bed for arthrodesis/fusion using a high-speed drill.
- Bone graft material (e.g., local autograft, autogenous morselized iliac crest bone graft, and/or bone graft extender) is placed along the lateral mass edges bilaterally.

- Ensure that no graft material is left in contact of the neural elements or within the spinal canal.
- Ensure hemostasis.
- Place subfascial drain and close muscle, fascia, and skin layers.
- Patients at high risk for infection may benefit from topical application of vancomycin powder to the subfascial space.

42.5 Complications

- Postlaminectomy kyphosis
 - Occurs in 21 to 38% of uninstrumented cases at two years' follow-up.^{4,5,6}
 - More frequent in younger patients and those who have a greater degree of facetectomy.⁷
 - High failure rates have led most authors to recommend instrumented fusion in the setting of cervical laminectomy.
- Cerebrospinal fluid (CSF) leak
 - If possible, primary repair using nonabsorbable suture is preferred.
 - Alternatively, cover with collagen matrix and/or dural sealant (e.g., DuraSeal).
 - Maintain elevated head-of-the-bed (HOB) positioning for 2 to 5 days to decrease pressure on the dural repair.
- Nerve root injury
 - May result either directly during foraminotomy or during screw placement.
 - C5 palsy is the most common occurrence, with an approximate 8% incidence. Etiology is not well understood. May be caused by traction resulting from posterior drifting of the spinal cord, considering that the C5 root is shortest. Noticeable immediately postcervical laminectomy. Generally resolves 6 to 12 months postop.
- Vertebral artery injury
 - Results from misplaced lateral mass screw.
 - Screw should remain in place and postoperative angiogram obtained to assess type of injury (e.g., occlusion, dissection).
- Deep wound infection
- Postop epidural hematoma with neurologic deficit
 - Emergent return to operating room (OR) for evacuation/decompression

42.6 Postoperative Care

- Postop paraspinal spasm may respond well to judicious use of muscle relaxants or low dose diazepam.
- Obtain postoperative upright anteroposterior/lateral C-spine X-rays.

- Remove drain when output becomes less than 30 to 50 mL per 8- to 12-hour shift (typically postoperative day 1 or 2).
- Mobilize the patient with physical therapy.

42.7 Outcomes

- Class III evidence shows that 70 to 95% of patients show postoperative neurologic improvement after cervical laminectomy and fusion for myelopathy.⁸
- However, the primary goal of surgery is to halt neurologic deterioration.

42.8 Surgical Pearls

- Preoperative cervical kyphosis and loss of lordosis are relative contraindications to posterior laminectomy without instrumentation.
- Most authors recommend performing instrumented spinal fusion in the setting of cervical laminectomy to prevent postlaminectomy kyphosis.
- Focal disk herniations and disk–osteophyte complexes should be treated by anterior approaches.
- Preoperative familiarity with the nuances of a particular instrumentation system is helpful.
- When instrumenting to C7, many authors recommend crossing the cervicothoracic junction and extending the instrumented fusion to T1 or T2 to prevent junctional disease, hypermobility/instability, and/or kyphosis. (Fixation at T1 and T2 requires pedicle instrumentation.)
- C2 is not instrumented with lateral mass fixation. Instead, C2 pars, pedicle, or translaminar fixation techniques are used.
- Study preoperative imaging to guide screw size and placement.

Common Clinical Questions

1. When performing lateral mass instrumentation, why is the screw trajectory directed laterally and superiorly (“up and out”)?
 2. Which nerve root is most prone to palsy after posterior cervical laminectomy?
 3. Why is preoperative cervical spine alignment an important consideration?
 4. Why is instrumented spinal fusion recommended in the setting of cervical laminectomy?
-

Answers to Common Clinical Questions

1. To avoid the nerve root and to avoid the vertebral artery.
2. C5 nerve root.
3. Kyphotic alignment prevents posterior “drifting” of the spinal cord postdecompression.
4. To prevent postlaminectomy kyphosis.

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43 Cervical Laminoplasty

Glen R. Manzano and Allan D. Levi

43.1 Key Points

- Laminoplasty is a procedure in which the spinal canal is expanded with the integrity of the posterior elements preserved.
- Laminoplasty was developed in Japan in the 1970s as a “tissue-sparing” alternative to laminectomy to minimize its associated risk of instability and progressive kyphotic deformity and thus improve outcomes.¹

43.2 Indications

- Multilevel spinal cord compression secondary to one or any combination of the following:
 - Ossification of the posterior longitudinal ligament (OPLL)
 - Cervical spondylosis
 - Congenital canal stenosis
- For consideration in children requiring laminectomies
- Not for isolated neck pain or for patients having a significant kyphotic deformity

43.3 Technique

- In cases of severe stenosis, awake fiberoptic intubation should be considered to minimize the risk of hyperextension injury and allow for reexamination of patient prior to positioning.
- Place neurophysiologic monitoring leads (include somatosensory evoked potentials [SSEPs], motor evoked potentials [MEPs], and electromyograms [EMGs]) and obtain baseline recordings before turning the patient.
- Secure patient’s head with or without pins (e.g., Mayfield three-pin headrest [Schaerer Mayfield; Randolph, Massachusetts]) and turn the patient prone with the head secured in a neutral to slightly flexed position.
- Tape superior and dorsolateral aspects of both shoulders and secure to the caudal corner of the operating table to aid in radiographic visualization of the lower cervical vertebrae.

- Use fluoroscopy to plan an incision extending from the spinous process of C2 to that of T1 and infiltrate with 1% lidocaine with epinephrine to minimize skin bleeding.
- Make a midline incision and use a subperiosteal dissection to reflect the paraspinal muscles from the caudal end of C2 to the rostral limit of T1.
- Remove the caudal third of the C2 lamina and the rostral third of the T1 lamina using the combination of a high-speed air drill and a 2 mm Kerrison punch, allowing visualization of the underlying dura at these levels.
- Remove the spinous processes from C3 to C7 with a rongeur and morselize the bone for subsequent autografting.
- Perform laminoplasty of lamina C3 to C7 by creating an “open” side and a “hinged” side. The open side is generally the side with the greatest compression and/or the side most clinically symptomatic.
 - Use the high-speed air drill with a small drill bit to create troughs at the level of the lamina–facet junction from C3 to C7 (**Fig. 43.1**).
 - Drill through the outer and inner cortical margins of the lamina on the open side, but only through the outer cortical margin and cancellous bone (and not the inner cortex) on the hinged side.



Fig. 43.1 Trough is drilled through the inner and outer cortices at the lamina–facet junction on the left (shown from C3 to C6 only). Partial laminectomy of C2 is shown. Crosshatching denotes area of drilling for closed door side.

- Prepare the bone allografts to stabilize the canal expansion.
 - Using rib allografts and the high-speed drill, cut three separate grafts each ~ 8 to 14 mm long.
 - Make transverse grooves along the cut surfaces of the rib grafts, approximating the thickness of the cut laminae.
- Prepare to “open the door.”
 - Use two small curettes placed into the trough on the open side just deep to the outer cortex.
 - Pull the curettes upward, thereby enlarging the lamina–facet gap and creating a greenstick fracture along the trough on the hinged side. Repeat the process on each successive lamina to expand the canal by ~ 4 mm (**Fig. 43.2**).
- Place the rib allografts in the gaps that have been created at the C3, C5, and C7 levels (**Fig. 43.3**).
- Place the morselized spinous process autograft over the decorticated bone surfaces of the facet and lamina on the hinged side at C3, C5, and C7 to provide for an intersegmental fusion. If the patient suffers from radiculopathy as well as myelopathy, one or more foraminotomies can be performed:



Fig. 43.2 With a trough drilled through the outer cortex only on the contralateral side (closed door side), the laminectomy door is “opened,” creating a greenstick-type fracture on the closed side.



Fig. 43.3 Rib allograft is cut and secured into opening in lamina to expand the spinal canal, typically at C3, C5, and C7.

- After the lamina has been elevated and the ligamentum flavum excised, drill the medial third of the facet over the exiting nerve root and widen the opening as needed with 1- or 2-mm angled Kerrison punches.
- Leave the subfascial drain (e.g., Hemovac [Gohar Shafa; Tehran, Iran]) in situ to minimize hematoma formation.
- Variations in the approach:
 - Use spinous processes as autograft instead of the rib allograft.
 - Stabilize the rib allograft with titanium miniplates to the adjacent lamina and facet on the open side or sutures.
 - Use titanium spacers to hold open lamina that is affixed to the lamina–facet joint on the open side.
 - Perform full-thickness splitting of the lamina in the midline and create bilateral troughs, and then spread the lamina and place allograft spacers (double-door laminoplasty).
 - Should rigid stabilization be required, lateral mass screws can be placed. This is best done after drilling and “opening the door,” but before graft insertion.

43.4 Complications

- Early
 - Wound infection (2%)
 - Dural tear/cerebrospinal fluid (CSF) leak (< 1%)
 - Hemorrhage (< 1%)
 - Spinal cord injury (< 1%)
 - Nerve root injury
 - Delayed C5 nerve root injury (2–13.3%)
- Late
 - Postoperative neck pain (40–60%)
 - Reduced range of motion (20–50%)
 - New-onset kyphosis (2–15%)

43.5 Postoperative Care

- Mobilize early with cervical bracing.
- Discharge to home once patient is ambulating and tolerating full diet—usually 2 to 3 days postoperatively.

43.6 Outcomes

- Clinical outcome studies are drawn largely from the Japanese literature on patients with OPLL and are difficult to compare as a result of patient heterogeneity and different outcome measures used.²
- Long-term (5–10 y) outcome studies suggest that 50 to 70% of patients having either OPLL or cervical spondylotic myelopathy show improvement in their neurologic function after laminoplasty.³
- Limited studies comparing this procedure to its corresponding anterior procedure identify similar clinical outcomes, but overall there is a suggestion that outcomes are better than with laminectomy alone.
- A randomized controlled⁴ trial comparing laminoplasty with laminectomy and fusion for cervical spondylotic myelopathy yielded the following results:
 - Both groups showed improvement in postoperative Nurick Grade and Modified JOA Score with low overall surgical complication rates.
 - The laminoplasty group showed statistically significant improvement in postoperative neck, interscapular and arm pain scores as well as in Neck Disability Index and SF-36 scores when compared with the laminectomy and fusion group.
 - Cervical range of motion was reduced by 75% in the laminectomy and fusion group, versus 20% in the laminoplasty group.
 - Although the patient numbers were small, there was a suggestion of increased adjacent level narrowing in the laminectomy and fusion group, noted as early as 1 year postoperatively.

43.7 Surgical Pearls

- Obtain preop X-rays to rule out instability or kyphotic deformity.
- Ensure the provision of drilling gutters at the lamina–facet junction (and not lateral to it) and direct drill medial so as not to disrupt the facet joint
- Lift *gently* when opening the door and ensure that drilling of the closed side (through outer cortex and some cancellous bone) is adequate to avoid creating a true fracture of the lamina. With a true greenstick fracture, some tension should remain in the opening to help maintain compression on the rib graft to hold it in place.
- If multiple foraminotomies are required, plan on making the open side the side where more foraminotomies are required.
- Visually inspect the construct and gently stress it after rib allograft is placed to ensure that you have not compromised the canal and that it remains stable.

Common Clinical Questions

1. True, or false? Patients who have kyphotic deformities and symptoms of myelopathy may benefit from laminoplasty procedures.
 2. List three perioperative surgical complications associated with laminoplasty.
-

Answers to Common Clinical Questions

1. False. Kyphotic deformities are a relative contraindication to performing this procedure.
 2. Delayed C5 nerve root injury (2–13.3%), wound infection (2%), dural tear/CSF leak (< 1%), hemorrhage (< 1%), spinal cord injury < 1%.
-

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44 Posterior Cervical Foraminotomy

Andreas K. Filis and Frank D. Vrionis

44.1 Key Points

- The posterior cervical foraminotomy approach as described by Frykholm (also referred to as laminoforaminotomy) allows nerve root decompression without the need to enter the disk space.
- In contrast to the anterior discectomy approach, the standard posterior cervical foraminotomy is not destabilizing, so there is no need for fusion/implants or harvesting autograft.

44.2 Indications and Contraindications

- Indications
 - Cervical neuroforaminal nerve root compression by a (lateral) soft disk herniation or spondylosis (osteophyte) with progressive or intractable cervical radiculopathy.
 - Cervical nerve root decompression if an anterior approach is not desirable (recurrent surgery, difficult anterior approach, cervicothoracic levels, previous radiation, history of dysphagia).
 - Patients who are at high risk for nonunion might be better candidates for laminoforaminotomy than anterior cervical discectomy and fusion (ACDF).¹
- Contraindications
 - Primary axial neck pain.
 - Cervical instability at the involved level.
 - Cervical myelopathy.
 - Severe kyphosis.
 - Large central disk herniation.
 - Central spinal canal stenosis.
 - Bilateral foraminal stenosis.
 - Osteophytes ventral to the nerve root.
 - Patients with neck pain additional to radiculopathy may benefit more from ACDF (these patients may need some sort of fusion).

44.3 Technique

- Patient positioning
 - After induction of general endotracheal anesthesia, the patient is positioned in a prone position with the head clamped in a Mayfield head holder (Schaerer Mayfield; Randolph, Massachusetts) or in Gardner-Wells tongs with 10 to 15 pounds of traction. The advantages of traction are interspace distraction, stability and alignment.
 - Alternatively, the patient can be positioned in a (semi)sitting position with the head clamped in a Mayfield head holder. The sitting position reduces venous pressure as well as intraoperative blood loss. Additionally, the position of the head in relation to the cervical spine can be better adjusted than in a prone position.
 - The (semi)sitting position carries the risk of air embolism and requires the placement of a precordial or intraesophageal Doppler probe and a central venous line for air aspiration in case of an embolism.
 - The (semi)sitting position is contraindicated in patients who have a patent foramen ovale and the possibility of paradoxical arterial embolism.
- Preparation and draping
 - Confirmation and adequate visualization of the correct level with plain X-ray or fluoroscopy in a lateral position is mandatory.
- Incision, soft tissue dissection, and spine exposure
 - Posterior midline incision ~ 3 to 4 cm long for an open microsurgical approach. Exposure and incision of the nuchal fascia.
 - Unilateral, subperiosteal dissection of the muscle layers from the spinous process and lamina to the facet joint with a Bovie cautery (Bovie Medical Corp.; Clearwater, Florida) and a Cobb elevator. Supra- and interspinal ligaments should be preserved.
 - Placement of a retractor and radiographic confirmation of the correct level.
 - Minimally invasive procedure
 - The proper incision site is marked with a spinal needle and lateral fluoroscopy. The incision is centered slightly rostral to the intended level of decompression.
 - The skin incision is made ~ 2 cm lateral to the midline.
 - If a retractor system is used (small-blade retractor, tubular retractor), only very small incisions are necessary (1.5–2 cm).
 - Incise the fascia with a Bovie cautery.
 - With the use of a bladed retractor, blunt dissection is performed with a finger down to the level of the lamina.

- If a tubular retractor system is used, a K-wire is positioned on the inferior articular facet of the upper vertebral body under fluoroscopic guidance. In this situation, it is imperative that the K-wire engage bone at all times.
- Serial dilators are used to bluntly widen the approach through the muscles, and finally the tubular retractors are placed.
- A final radiographic check is recommended to confirm the correct level.
- The retractor should be placed on the interlaminar space (two-thirds) and on the facet joint (one-third).
- Bring in the draped microscope.
- Foraminotomy and wound closure
 - Expose and mark the cranial and caudal lamina of the appropriate level, and check again with fluoroscopy.
 - Expose the medial parts of the capsule of the facet joint.
 - Perform a partial laminectomy by removing bone from the lateral parts of the caudal and cranial lamina with a high-speed drill or with a small Kerrison rongeur.
 - Open the ligamentum flavum with a nerve hook or a dissector, and resect it with a small Kerrison rongeur.
 - Extend the laminectomy cranially and laterally, and expose the lateral edge of the dura and the nerve root. The nerve root is generally displaced dorsally.
 - Try to remove free disk material under the nerve root with a nerve hook and a pituitary rongeur.
 - If the disk sequestrum is still covered with layers of the posterior longitudinal ligament, it has to be opened with a small dissector or with bipolar forceps.
 - Undercut the superior articular process with 1- and 2-mm Kerrison rongeurs until the entire foramen is patent (but preserve at least 50% of the facet) (**Fig. 44.1**).
 - Epidural bleeding is controlled with Gelfoam (Pharmacia & Upjohn, Kalamazoo, Michigan) and Cottonoids (Saramall; Buenos Aires, Argentina).
 - Placement of a wound drain (Hemovac; Gohar Shafa; Tehran, Iran) is usually not necessary.
 - Close fascia using 0 or 2–0 absorbable sutures.
 - Use inverted, interrupted 3–0 absorbable sutures for closure of the subcutaneous layer.
 - Close the skin with a running, absorbable intracutaneous suture or with staples.

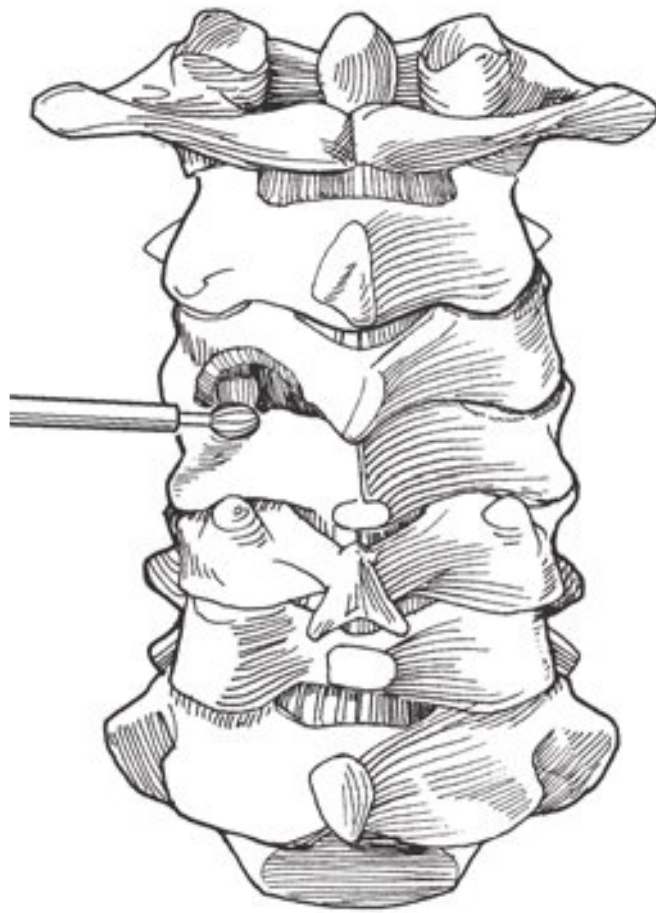


Fig. 44.1 Foraminotomy. The superior articular process can be removed by using a bur to thin it and a 1 mm Kerrison rongeur to remove remaining articular process, leaving the nerve root decompressed. (From Vaccaro AR and Albert TJ, eds. *Spine Surgery: Tricks of the Trade*, 2nd ed., Thieme; 2009.)

44.4 Complications

- Wound dehiscence
- Wound infection and epidural abscess (1.2%)²
- Cerebrospinal fluid (CSF) leak (0.6–2.5%)^{2,3}
- Neurologic deterioration resulting from injury or manipulation of the nerve root or spinal cord or postoperative epidural hematoma (1.2–2.3%)^{2,3}
- Air embolism (sitting position) (1.6–2.3%)^{3,4}
- Persistent radiculopathy indicating residual nerve root compression, epidural scarring, early recurrent disk herniation, or neuropathic pain requiring additional surgery (5.1–8%)^{2,3}
- Recurrent disk herniation (7.6%)³
- Intraoperative hemorrhage caused by epidural veins or as a result of an inadvertent injury to vertebral artery (very rare and usually resulting from a wrong approach)
- Secondary instability due to excessive resection of the facet joint (4.9%)²
- Loss of cervical lordosis (18.5%)²
- Reoperation for adjacent segment disease after posterior foraminotomy estimated at 0.9% per year per level—roughly the same after ACDF (0.8%)⁵

44.5 Postoperative Care

- Mobilize the patient immediately after surgery.
- There is no need for external bracing; patients typically have full range of neck motion.
- Pain control is usually achieved with orally administered opioids. Muscle relaxants help significantly.
- Discharge patients when discharge criteria are met (usually on postop day 1).

44.6 Outcomes

- Large series have reported excellent or good outcomes for posterior cervical foraminotomy in 90 to 96% of patients who have monoradiculopathy.^{2,3,6}
- From a biomechanical standpoint, unilateral foraminotomy does not increase range of motion in adjacent segments.⁷ Nonetheless, annual rate of adjacent segment disease as compared with ACDF does not differ significantly.⁵

44.7 Surgical Pearls

- Removal of more than 50% of the facet significantly compromises torsional stiffness of the cervical spine.⁸
- The entire length of the foramen can be enlarged safely with the use of curettes or a high-speed drill with preservation of more than 50% of the facet by initially working parallel and just inferior to the nerve root and then, as more room is gained, removing the rim just dorsal to the nerve root.
- Minimal bone removal (keyhole exposure) is acceptable, but not at the expense of excessive nerve root retraction or insufficient exposure of the dura with an increased risk of dural laceration.
- In cases of osteophytes, the posterior foraminotomy relies on indirect decompression, which may not be sufficient.
- Avoid placing the Kerrison rongeur between the posterior and anterior rootlets and avoid avulsion.
- Be especially vigilant about the C5 nerve root (high risk of deltoid weakness).

Common Clinical Questions

1. What are the most common indications for a posterior cervical foraminotomy?
2. What are the two options for performing a posterior cervical foraminotomy?
3. How much of the facet joint can safely be removed without creating instability? How much of the facet joint is usually removed during a posterior cervical foraminotomy?

Answers to Common Clinical Questions

1. Cervical neuroforaminal nerve root compression by a (lateral) soft disk herniation or spondylosis (osteophyte) with progressive or intractable cervical radiculopathy, as well as the need for cervical nerve root decompression if an anterior approach is not desirable (recurrent surgery, difficult approach, cervicothoracic levels).
2. The standard procedure with a midline incision and subperiosteal dissection from the midline to the facet joint, and the minimally invasive procedure with a paramedian incision ~ 2 cm from the midline and blunt dissection through the muscles using small-blade or tubular retractors and serial dilators.
3. Removal of more than 50% of the facet significantly compromises torsional stiffness of the cervical spine. During a posterior cervical foraminotomy, typically less than 50% of the facet joint is removed (~ 30%).

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45 Cervical Open Reduction Techniques: Anterior and Posterior Approaches

Harminder Singh, George M. Ghobrial, and James S. Harrop

45.1 Key Points

- Cervical facet dislocations result from high-energy traumatic forces transmitted through *flexion/distraction* vectors.
 - A flexion, rotational, and lateral force can result in a unilateral jumped facet.
 - If the mechanism of injury involves *flexion/compression* forces (e.g., diving headfirst into a shallow pool), then facet and vertebral body fractures may be encountered in the form of jumped and/or perched facets.
 - A severe flexion/distraction injury with a compression component is also called a teardrop fracture.
- Rule of thumb: A unilateral jumped facet results in 25% displacement (*anterolisthesis*) of one vertebral body on another (as seen on lateral plain radiographs or computed tomography [CT] scan of the cervical spine); bilateral jumped facets are associated with a 50%anterolisthesis.
- Acute management usually consists of reduction of the dislocated joints and the realignment of fractured cervical segments, through either open surgical intervention or axial cervical traction.
- Re-creating the force vectors of the mechanism of injury can facilitate reduction of the facets or deformity and realign the spine. For example, the cervical spine is flexed and distracted to realign the facets if the mechanism of injury was flexion/distraction.
- The open anterior approach provides for the removal of extruded disk before reduction of a dislocated facet. This approach may also be effective for reducing fractures and dislocations.
- Posterior instrumentation reestablishes the posterior tension band and provides the optimal environment for fusion.

45.2 Indications

Cervical open reduction can restore spinal alignment, decompress the neural elements, and establish rigid internal fixation.

- Cervical open reduction is indicated when
 - Decompression of neural elements is required

- Closed reduction fails
- Certain contraindications for closed reduction are present:
 - Patient has altered mental status.
 - Skull fracture precludes tong placement.
 - A relative contraindication is a rigid spine such as in ankylosing spondylitis.

45.3 Technique

45.3.1 Anterior Approach

- Neuromonitoring (somatosensory evoked potentials [SSEPs], motor evoked potentials [MEPs]) and electromyography (EMG) allow monitoring of spinal cord function during spinal manipulation.
 - Typically coordinated with a neuroanesthesiologist with the use of total intravenous anesthesia (TIVA). Avoid use of volatile inhaled anesthetics, such as
 - IV Propofol: 50 to 150 $\mu\text{g/kg/min}$ with IV short-acting narcotic agent (e.g., Remifentanyl)
- Localization of the cervical level is possible using fluoroscopy or plain radiographs.
- Anatomic localizing:
 - C4–C5 disk space: Level of the thyroid cartilage
 - C5–C6 disk space: Level of the cricoid cartilage
 - C6 vertebrae: Carotid tubercle (the transverse process of C6)
- If a significant herniated disk is compressing the neural elements, a complete discectomy is performed before anterior reduction.
 - Blunt dissection: Retract carotid sheath laterally and trachea/esophagus medially with retractors to expose anterior spinal column. The anterior longitudinal ligament (ALL) is typically disrupted as a result of trauma.
 - Dissect longus colli laterally with monopolar cautery to expose the cervical vertebrae.
 - In unilateral dislocations, the superior vertebral body will be rotated away from the side of dislocation.
 - The annulus is incised with a scalpel as far laterally as possible near to the uncovertebral joints.
 - The disk and posterior longitudinal ligament (PLL) are removed with curettes and with Kerrison and pituitary rongeurs until the dura is visualized. The PLL is ruptured with a bilateral dislocation.
- Caspar retractors are placed, with pins one level above and below the subluxed and dislocated segments.

- Angulate pins convergently to place the spine in kyphotic (flexed) posture for distraction.
- Distract Caspar pins; the superior body will rotate back and become flush with the inferior body (**Fig. 45.1**).
- After the desired spinal alignment is achieved, a tricortical bone graft harvested from anterior superior iliac crest or allograft may be used for arthrodesis.
- Anterior plate placement: Screws anchoring the plate should terminate by the middle third of the vertebral body to avoid violation of end plates.

45.3.2 Posterior Approach

- More commonly used when there is disruption of the posterior ligamentous complex and associated facet fractures preventing closed reduction.

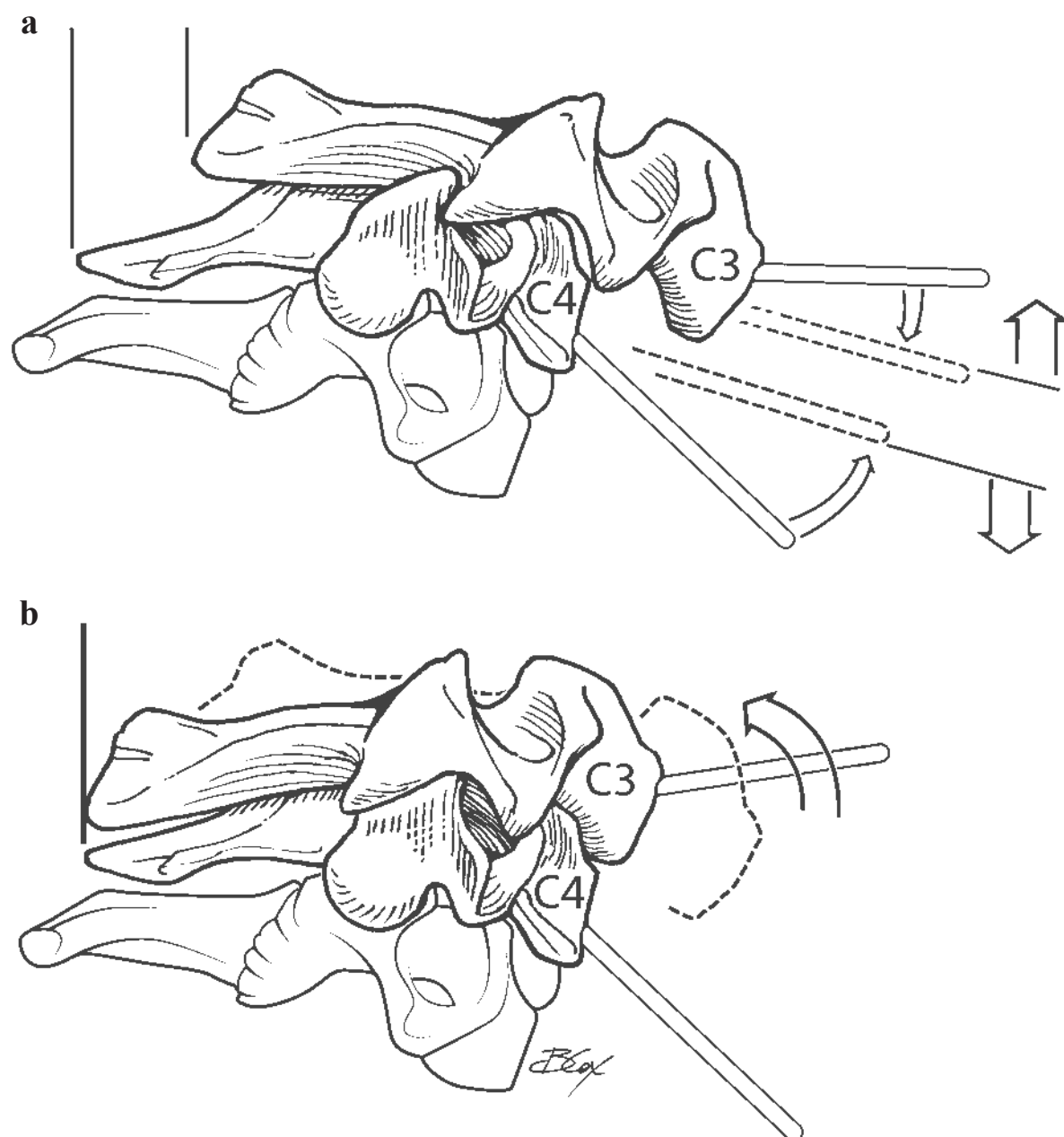


Fig. 45.1 (a,b) Anterior spinal reduction using convergently placed Caspar pins. Distracting the Caspar pins flexes and distracts the spine, unlocking the facets and reducing the spine.

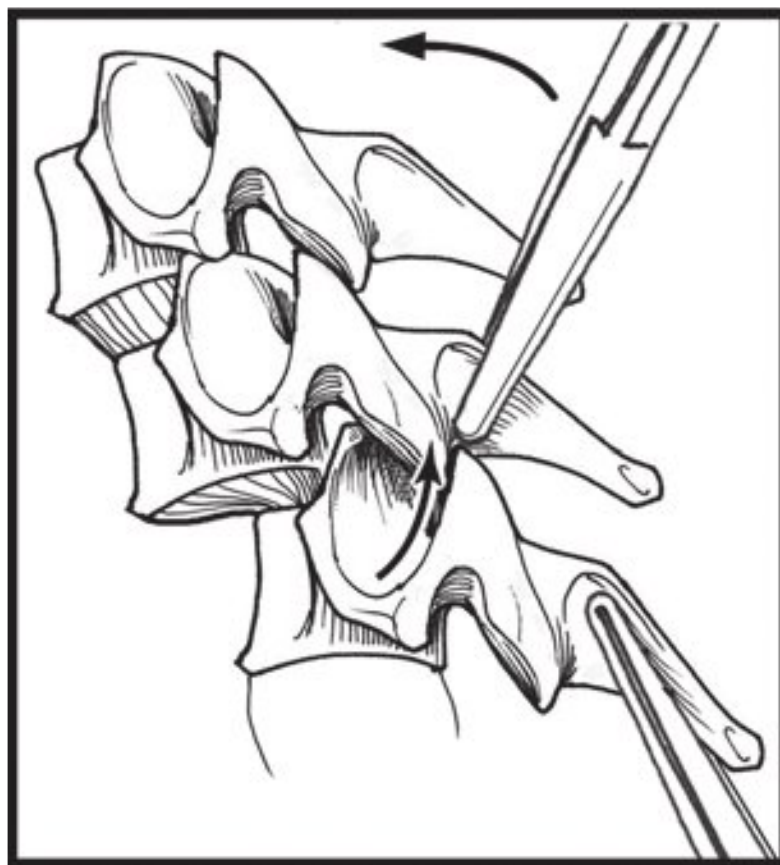


Fig. 45.2 Posterior spinal reduction using two towel clips. The inferior articulating facet of the superior vertebral body is lifted above and over the superior articulating facet of the inferior vertebral body, reducing the spine.

- Prone position in Mayfield head holder (Schaerer Mayfield; Randolph, Massachusetts) or Gardner-Wells tongs, and make a midline incision.
- Perform subperiosteal dissection during exposure of laminae and lateral masses.
- Reduction can be performed with direct distraction using two towel clips in opposite directions on the spinous processes, using a leveraging mechanism (**Fig. 45.2**).
- Alternatively, the superior articulating process of the inferior facet can be drilled away to facilitate reduction.
- Once decompression and reduction are achieved, the spine is locked into place using either lateral mass screws and rods or posterior wiring techniques.
- Decortication of the facets and supplemental bone graft placement in the lateral gutters help with arthrodesis.

45.4 Complications

- Wound infection (increased likelihood in trauma population).
- Consider adjunct intrawound vancomycin powder (mixed evidence)
- Graft donor site pain from autograft. Allograft bone is an alternative but may have a lower fusion rate.
- Anterior exposure (particularly right-sided approach) may result in injury to the recurrent laryngeal nerve.

- Esophageal perforation:
 - Maintain situational awareness of esophagus at all times.
 - Avoid drill use outside of disk space.
- Carotid artery and vagal nerve injury because of violation of the carotid sheath.
- Vertebral artery injury during anterior cervical approach:
 - Evaluate preoperative imaging for anatomical deviations of the vertebral artery.
 - Avoid violation of the superior two-thirds of the uncovertebral joints.
- Vertebral artery injury during posterior fixation:
 - Maintain lateral trajectory while placing lateral mass screws in the subaxial cervical spine.
- Horner syndrome: The sympathetic chain lies adjacent and anterior to the longus colli muscles.
- Cerebrospinal fluid (CSF) leak (which may be present from original trauma).
- Delayed fusion increases the likelihood of screw loosening, plate migration, graft migration, and ultimately construct breakage.

45.5 Postoperative Care

- Immobilization for 6 weeks: Cervical orthosis
- Serial radiographic examination: Anteroposterior, lateral, swimmer's views
- Flexion–extension views for assessment of stability

45.6 Outcomes

- Kwon et al¹ conducted a prospective randomized controlled trial comparing anterior and posterior fixation for unilateral cervical facet injuries and reported equal efficacy between the two techniques.
- Johnson et al² reported a 13% incidence of loss of postoperative radiographic alignment during follow-up of cervical facet fracture–dislocations treated with only anterior decompression and fusion.
- Reindl et al³ reported on 41 consecutive patients with unstable dislocations/subluxations, of whom 8 required anterior reduction after failure of Gardner-Wells traction; in addition, 2 of those 8 failed, requiring posterior surgery as well.
- Fehlings et al reported on 44 consecutive patients treated with posterior cervical fusion for traumatic instability. Long-term follow-up revealed that the cervical spine was successfully stabilized in 93% of cases.⁴

45.7 Surgical Pearls

- Best treated anteriorly: Large disk herniation with cord compression such that spinal cord can be decompressed before distraction.
- Best treated posteriorly: Hyperflexion injury, including PLL injury, unilateral and bilateral facet dislocation without anterior cervical disk herniation.
- Anteriorly, avoid overdistract of the cervical spine during placement of the interbody graft. As a result of the ligamentous laxity prevalent in spinal injury, it is easy to overdistract.
- Posteriorly, be mindful of laminar fractures during exposure. Heat from a Bovie cautery can damage the spinal cord or cause CSF leaks when transmitted through fractured laminae.

Common Clinical Questions

1. When should anterior reduction be used when closed reduction is not possible?
2. When should open reduction be performed?
3. Postoperatively after anterior cervical discectomy with fusion, the patient has a hoarse voice and difficulty swallowing. What should the clinician be concerned about?

Answers to Common Clinical Questions

1. When large anterior disk herniation is present.
2. Decompression of neural elements is required. Closed reduction fails. Closed reduction is contraindicated.
3. Injury to the recurrent laryngeal nerve on exposure.

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46 Anterior Cervical–Thoracic Junction Technique

Muhammad M. Abd-El-Barr, Viren S. Vasudeva,
and Michael W. Groff

46.1 Key Points

- The cervical–thoracic junction (CTJ) is an anatomically unique region of the vertebral column that is particularly susceptible to traumatic injury, intervertebral disk herniation, and iatrogenic (postsurgical) deformity.
- The CTJ is a transitional zone between the mobile, lordotic cervical spine and the more rigid, kyphotic thoracic spine.
- Anterior approaches to the CTJ have distinct advantages over posterior approaches but have important disadvantages and considerations compared with posterior approaches.

46.2 Indications

- Fractures
 - Unstable traumatic fractures/dislocations
 - Osteoporotic fractures causing neurologic deficit, deformity, or persistent pain refractory to less invasive treatments
- Neoplasms
 - Vertebral and epidural metastases
- Infections
 - Osteomyelitis/diskitis meeting criteria for surgery
 - Spinal tuberculoma (Pott disease)
- Deformity (if anterior column load sharing is deficient; often combined with posterior approaches)
 - Postlaminectomy kyphosis
 - Chin-on-chest deformity
 - Degenerative kyphoscoliosis
 - Posttraumatic kyphosis
- Symptomatic herniated nucleus pulposus in the upper thoracic spine (C7–T3)
 - Radiculopathy
 - Neurologic deficit from spinal cord compression
 - Myelopathy

46.3 Technique

- Preoperative imaging:
 - MRI is preferred, for it allows for delineation of important soft tissue components, especially the aortic arch and brachiocephalic vein (BCV), as well as neural elements such as the recurrent laryngeal nerve. The bony anatomy, although better seen by CT, can usually be adequately assessed using MRI.
 - To best decide which anterior approach is best, most advocate making a horizontal line anterior to the most caudal aspect of surgical pathology. If this line is above the sternal notch, then a suprasternal (Smith–Robinson) approach is usually adequate. If line is below, several techniques are available—median sternotomy and additional resection of medial third clavicle, transsternal, transclavicular, and transthoracic (**Fig. 46.1**).¹

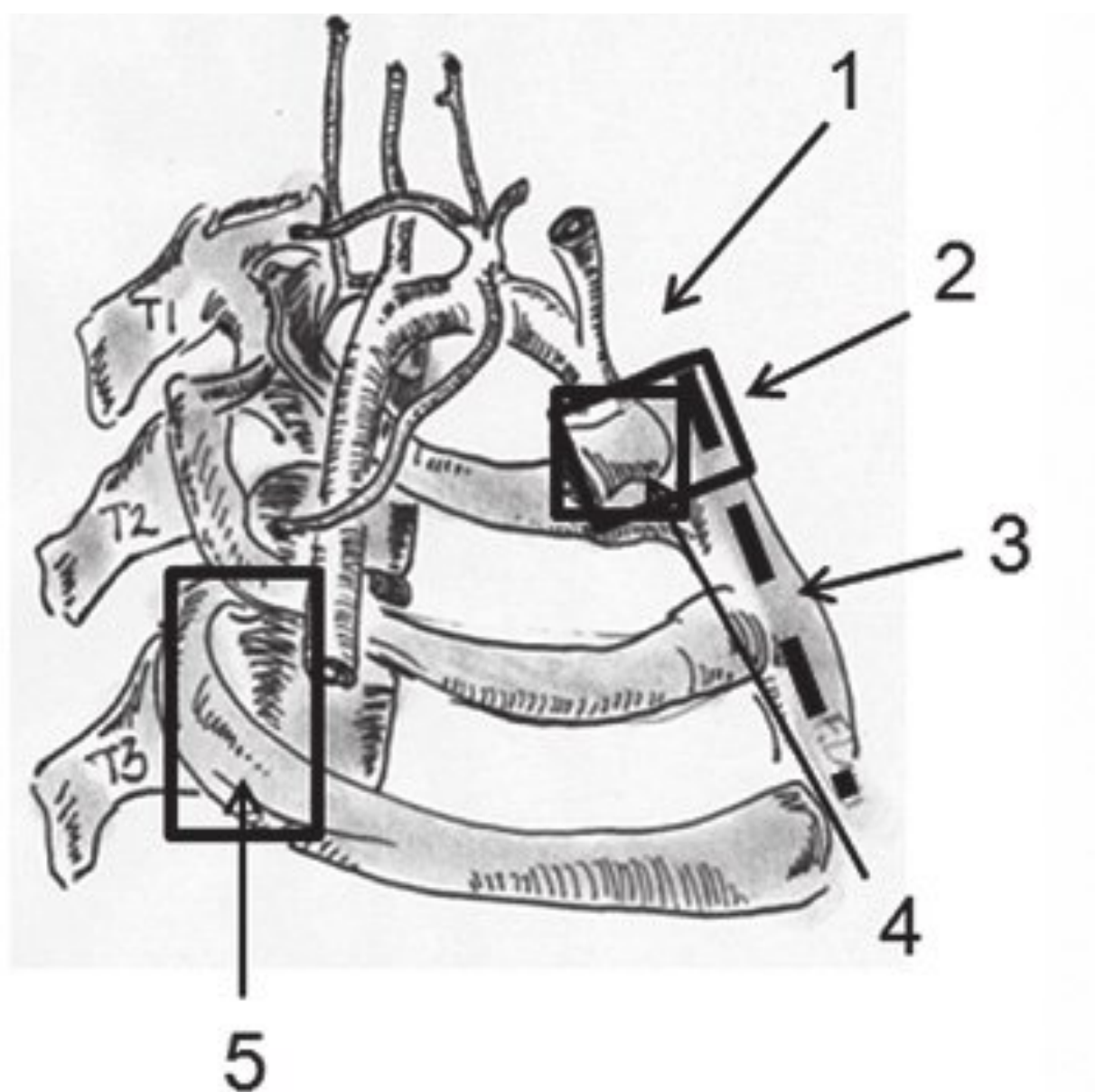


Fig. 46.1 Anterior approaches to upper thoracic spine. (1) Low anterior cervical approach. Approach (Smith–Robinson). (2) Modified anterior cervical approach with median sternotomy and additional resection of the medial third of the clavicle. (3) Sternal splitting technique. (4) Modified low anterior approach leaving the manubrium intact but removing the midclavicle. (5) High anterior transthoracic (transpleural) approach. (From Lam and Groff, reproduced with permission.¹)

- Intraoperative adjuncts:
 - Electrophysiological monitoring in the form of somatosensory evoked potentials (SSEP) and motor evoked potentials (MEP) may be useful, especially in cases of spinal cord compression and/or myelopathy.
 - Severe kyphosis or cases of fracture–dislocations may be helped with preoperative reduction using traction.
 - Corpectomy is more forgiving than disectomy, for it allows for greater maneuverability within tight spaces.
- Smith-Robinson approach to CTJ pathologies:
 - More traditional approach to prevertebral fascia.
 - Involves the retraction of carotid sheath laterally and staying medial to strap muscles.
 - In general, if making a line from the lowest level of pathology to the intended skin incision puts one above the manubrium, this method can be used, and no sternotomy or manubriotomy is needed. In many cases, pathology down to the T1–T2 level can be reached with this method.
- Transmanubriotomy:¹
 - Doesn't destabilize clavicles.
 - Narrow window is acceptable.
 - Intraclavicular line is pivot point for maneuvering instruments and provides a working channel.
 - Visualization of the operative target is via the neck dissection.
- Transsternal approach to CTJ pathologies:
 - Patient is placed supine on radiolucent table and prepped and draped in sterile fashion.
 - Incision:
 - Most use an incision that starts just anterior to the left sternocleidomastoid muscle and curves down over the sternum, with the length depending upon the extent of pathology.
 - A left-sided approach usually avoids the recurrent laryngeal nerve.
 - Divide platysma, followed by sharp dissection to define sternocleidomastoid and strap muscles. Dividing omohyoid may allow for greater visualization.
 - Identify the manubrium and perform subperiosteal dissection posteriorly, being careful to avoid injury to the brachiocephalic veins.
 - Using a high-speed drill or a sternal saw, split the manubrium. Some authors have described using an inverted-T type of manubrial split. It is important to preserve the muscular attachments to help with reconstruction and postoperative pain.
 - Similar to more familiar approaches to anterior cervical disectomies and corpectomies, perform blunt dissection medial to the carotid sheath and lateral to the strap muscles, down to the prevertebral fascia.
 - It is important to avoid the thoracic duct, which is usually on the left at C7–T1 but can be occasionally seen at C6.

- Confirm surgical level using fluoroscopy or plain radiographs. Anteroposterior views that identify the C7/T1 junction may be more practical than lateral views.
- Perform decompression (discectomy and/or corpectomy) at appropriate level(s), with stabilization if needed.
 - Using long-handled, bayoneted instruments allows for manipulation of tissue that is deep and avoids line-of-sight issues.
- Options for anterior column reconstruction:
 - Iliac crest autograft is associated with donor site morbidity.
 - Allograft strut.
 - Polyetheretherketone (PEEK) or titanium cage. Titanium cages may be associated with long-term failure as a result of mismatch of the modulus of elasticity, especially in osteoporotic patients.
 - Reports of using clavicular autograft might make sense in these situations due to proximity of the clavicle in this exposure (see transclavicular approach).² However, thanks to destabilization of the claviculomanubrial junction, this technique is not often used.
 - Most supplement with anterior plate fixation.
- It is important to fill the wound with saline and watch for bubbles with inspiration. The presence of bubbles signifies transgression of the visceral pleura and necessitates the placement of a chest tube.
- Leave a drain and close in layers (wire manubrium).
- Transclavicular approach to CTJ pathologies (originally described by Kurz):³
 - Incision is similar to the transmanubrial approach (**Fig. 46.2**).⁴
 - Manubrial and clavicular heads of the sternocleidomastoid are elevated and retracted laterally. Subperiosteal dissection of the left part of the manubrium and clavicle is done. Using a Gigli saw, the medial third of the clavicle is disarticulated from the manubrium.
 - Care must be taken not to injure the subclavian vein, which is immediately posterior to the middle third of the clavicle.
 - Blunt dissection is made to the prevertebral fascia and decompression and instrumentation is done as above.

46.4 Complications

- Minor
 - Superficial wound dehiscence
 - Injury to strap muscles
 - Nonunion of the clavicle/manubrium
 - Traction injury to the recurrent laryngeal nerve (usually transient)
- Major
 - Injury to the carotid sheath

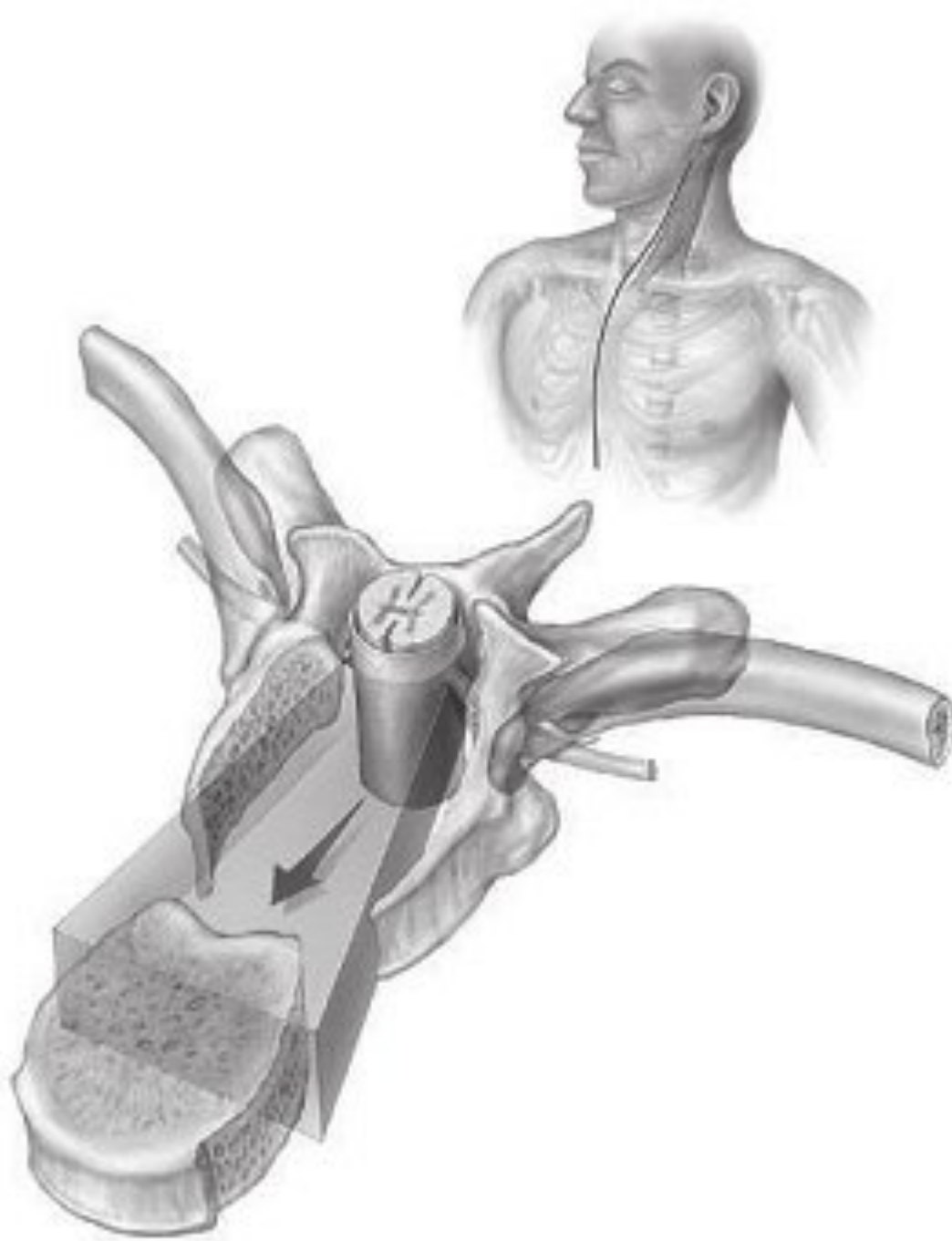


Fig. 46.2 Artist's rendering of anterior approach to thoracic spine using sternotomy. (From Witham et al, reproduced with permission.⁴)

- Injury to thoracic duct (chyle leak)
- Injury to great vessels
- Injury to the sympathetic chain (Horner's syndrome)
- Hemopneumothorax due to injury to the pleura
- Perforation of trachea and/or esophagus
- Cerebrospinal fluid (CSF) leak
- Deep wound infection
- Injury to azygous vein

46.5 Postoperative Care

- Cervicothoracic orthosis may be an option for patients who have poor bone quality or suboptimal fusion. The usual duration of bracing is 8 to 12 weeks in these cases.

- Monitor drain output for chyle leak.
- Postoperative CT is usually helpful to look at instrumentation, for plain X-rays in this region are difficult to interpret.

46.6 Outcomes

- No long-term studies and very few large series have been reported.
- It is difficult to ascertain the rates of morbidity and mortality for anterior approaches thanks to the wide variety of techniques and approaches. Anterior approaches appear to have higher rates of complications than do posterior approaches.
- Anterior approaches are often more desirable to posterior approaches, for they usually give a more direct visualization of most pathologies. Anterior stabilization is also often more desirable thanks to its ability to help with anterior load sharing.

46.7 Surgical Pearls

- Preoperative planning is of vital importance, with special consideration to body habitus and cardiopulmonary comorbidities.
- Soft tissue imaging with MRI is often helpful, for it defines the rostral extent of the aortic arch and defines brachiocephalic vessels.
- If sternotomy and/or manubriotomy is necessary and significant work is to be done in the superior mediastinum, an access surgeon may be very helpful.
- Low cervical approaches (Smith-Robinson) without sternotomy and/or manubriotomy is often possible down to T2–T3 and more desirable thanks to less morbidity and pain to the patient.
- Anterior column reconstruction is essential as the anterior column is responsible for the majority of load bearing.

Common Clinical Questions

1. Under what structure does the recurrent laryngeal nerve loop back in the left chest? On the right?
 2. What are important landmarks in deciding which of the anterior approaches should be used?
 3. What are the advantages of anterior cervicothoracic approaches compared with posterior approaches? What are the disadvantages?
-

Answers to Common Clinical Questions

1. The recurrent laryngeal nerve loops back in the left chest under the aortic arch, close to the ductus arteriosus. On the right side, it loops back under the right subclavian artery.
2. It is important to assess the most caudal pathology and its relationship to the manubrium, as well as the aortic arch and the left brachiocephalic vein.
3. The advantages of the anterior approach include direct visualization of the ventral pathology, anterior release for correction of kyphotic deformity and the ability to reconstruct the anterior column, which is responsible for the majority of load bearing with minimal risk to the neural elements. Disadvantages include the need for an access surgeon for unfamiliar territory, potential injury to the recurrent laryngeal nerve, the great vessels of the mediastinum, and morbidity if manubriotomy and/or sternotomy is needed.

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47 Freehand Thoracic Pedicle Screw Placement Technique

Sheri Palejwala, Jesse Skoch, and Ali A. Baaj

47.1 Key Points

- Thoracic pedicle screw fixation is the principal modality for posterior stabilization of the thoracic spine.
- Small pedicle dimensions and the proximity of the great vessels and neural elements all make placement of thoracic pedicle screws challenging.
- There are established techniques for safely and effectively placing thoracic pedicle screws using freehand techniques.

47.2 Indications

- Stabilization of the thoracic spine following traumatic or pathologic fracture.
- Stabilization after iatrogenic destabilization from decompression of degenerative, neoplastic, infectious, or inflammatory processes.
- Correction of deformity for restoration of sagittal or coronal balance.

47.3 Technique

- Preoperative considerations
 - Overall health of the patient should be optimized as much as reasonably possible. Issues to address include, but are not limited to, smoking, obesity, cardiopulmonary status, life expectancy (especially in the setting of tumor), nutrition, narcotic or substance dependency, and mental health.
 - Imaging:
 - Standing 36-inch X-rays are important when considering long constructs and in instances of deformity correction.
 - CT analysis of pedicle size preoperatively is optional and can assist the surgeon in choosing the appropriate type and size of implant.
- General (**Fig. 47.1**)
 - The authors prefer a freehand technique.¹ Other acceptable techniques include intraoperative fluoroscopy, CT, and image-based navigation.

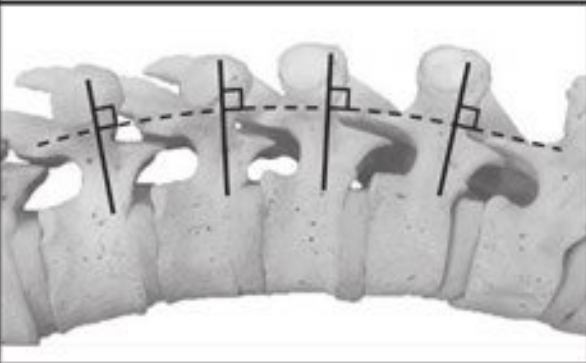
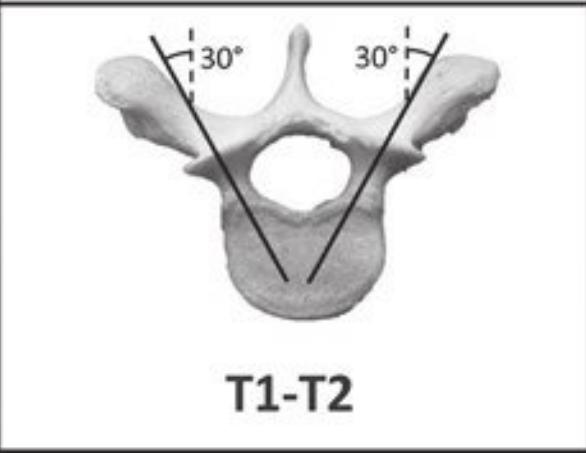
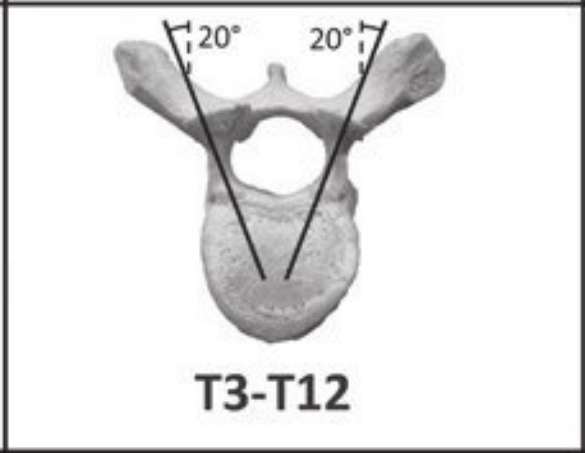
T1-T12 ENTRY POINT		3 mm caudal to the junction of the transverse process – superior articulating process
T1-T12 SAGITTAL TRAJECTORY		Orthogonal to the sagittal curvature of the dorsal spine
AXIAL TRAJECTORY	 T1-T2	 T3-T12

Fig. 47.1 Anatomic landmarks for placement of thoracic pedicle screws using free hand technique. (Copyright Ali A. Baaj, MD, used with permission.)

- Exposure
 - A dorsal midline incision is performed with subperiosteal dissection. The lateral extent of dissection is performed until the lateral facets and transverse processes are visualized, taking care to preserve the facet capsules of the adjacent cranial and caudal levels.
 - Initial fluoroscopic localization is performed to identify the levels of fixation using either the ribs and/or the sacrum as radiographic landmarks. It is essential to identify anomalies on the preoperative imaging—assumptions cannot be made regarding these landmarks.
- Starting point
 - A uniform entry point is used for all levels: Just caudal to the junction of the lateral margin of the superior articulating process and the transverse process. This is created with a small matchstick drill bit or awl.¹

- Trajectory
 - The sagittal trajectory is orthogonal to the curvature of the dorsal spine.
 - The medial angulation is $\sim 30^\circ$ at T1–T2 and 20° at T3–T12.
 - Care must be taken with vertebral rotation in cases of scoliosis.
 - A sharp straight pedicle gearshift is used to create the trajectory through the cancellous bone of the pedicle and vertebral body to a depth determined based on preoperative imaging.
 - A thin ball-ended feeling probe is used to assess all four walls and floor of the pedicle looking for cortical breaches.
 - Tapping is optional but, when performed, uses the same trajectory as the gearshift.
 - An appropriately sized screw is then slowly placed into the pedicle tract based on trajectories heretofore described.¹
- Screw confirmation
 - Final anteroposterior and lateral fluoroscopy is performed to assess for adequate screw placement. We recommend routine intraoperative (if available) or postoperative CT imaging to confirm accurate placement. Although many surgeons confirm adequate screw placement with plain AP and lateral X-rays.
- Salvage techniques
 - A new trajectory can be created as already described. In these instances, a tap may be used to ensure that the screw follows the new trajectory created in place of the previous one.
 - A combination of pedicle, laminar, and transverse process hooks or sublaminar wires may also be used when necessary.
- Other techniques
 - The entry point for T1–3 is in the midpoint to upper third of the thoracic transverse process, whereas at T4–9 it is superior to the transverse process and at T10–12 it is again at the midpoint of the transverse process.

47.4 Complications

- Breach
 - Previous studies describe safe and effective use of the freehand technique with acceptable breach rates.^{1,2,3}
- Neurovascular
 - Rare ($< 1\%$) but can be devastating.
 - The inherently lateral trajectory described here may decrease the risk of cerebrospinal fluid (CSF) leak and significant neural injury from spinal cord impingement.¹

47.5 Postoperative Care

- Imaging
 - Postoperative screw placement can be confirmed via X-ray or CT, the latter for better delineation of trajectory and potential breach.
 - MRI may be used to evaluate neoplastic or infectious processes, though artifact from metallic hardware can significantly reduce the quality of the information received.
 - Standing 36-inch X-rays are important to assess overall sagittal and coronal balance, but these may be delayed to eliminate confounding factors such as postoperative pain and the need for bracing.
- Pain management
 - Patients are generally hospitalized and treated with a combination of oral and intravenous narcotic medications and an aggressive bowel regimen. A strong emphasis is placed on early mobilization.
- Bracing
 - Optional, and depends on underlying pathology, bone stock and other variables.
- Wound care
 - Optimization of blood sugar and nutrition status, mobilization, and avoiding extended periods of dependence on the postoperative site are paramount to adequate incision healing.

47.6 Outcomes

- Multiple clinical and cadaveric biomechanical studies have demonstrated the efficacy of thoracic pedicle screw instrumentation as advantageous for stabilization and deformity correction.

47.7 Surgical Pearls

- The freehand technique for the placement of thoracic pedicle screws is safe, effective, and efficient. It may significantly reduce operative time and radiation exposure.
- The standardized technique already described is easy to teach and implement in cases of standard posterior thoracic instrumentation.
- The extrapedicular (“in–out–in”) technique is a viable option for pedicles that are too small or that are in instances of revision but that are biomechanically inferior to their intrapedicular counterparts.⁴
- Screws may be augmented with polymethylmethacrylate (or other similar agents) in situations of weakened cortical bone.

Common Clinical Questions

1. Which of the following demonstrates the most biomechanically advantageous technique for thoracic fusion?
 - A. Transverse process hooks
 - B. Extrapedicular screw fusion (in–out–in technique)
 - C. Pedicle screw fixation
 - D. Sublaminar wires
2. All of the following are commonly used intraoperatively to confirm adequate pedicle screw trajectory *except* which?
 - A. MRI
 - B. Stereotactic navigation
 - C. Fluoroscopy
 - D. CT
3. Which of the following assists with correct intraoperative localization in the thoracic spine?
 - A. Scout MRI showing C2 down to sacrum
 - B. CT scan of the entire thoracic spine to correctly count the ribs
 - C. Preoperative chest X-ray
 - D. All of the above
 - E. A and B only

Answers to Common Clinical Questions

1. C.
2. A.
3. E.

References

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48 Transpedicular Approach

Vinko Zlomislic and Steven R. Garfin

48.1 Key Points

- Alternative, less extensive, technique for decompression of the neural elements and access to the intervertebral disk, pedicle, and vertebral body through a posterior approach
- Less morbidity than other approaches (transthoracic, thoracoabdominal, extracavitary)
- Particularly effective for soft thoracic disk herniations and intrapedicular lesions
- Utilitarian approach that allows for simultaneous wider decompression and/or posterior instrumentation if needed
- Not indicated for large calcified central thoracic disk herniations or midline anterior spinal cord compression

48.2 Indications

- Soft central or lateral thoracic disk herniations, posterolateral or lateral calcified disk herniations
- Fracture stabilization or reduction
- Infections
- Biopsy or resection of intrapedicular or vertebral body lesions
- Decompression of lateral and anterolateral spinal canal in the setting of tumor
- Posterior release for management of kyphoscoliosis

48.3 Technique

- Patient positioned prone on Jackson table with frame.
- Intraoperative neurophysiologic monitoring with somatosensory evoked potentials and transcranial motor evoked potentials should be considered.
- Fluoroscopy: Correlate levels with preoperative imaging and confirm levels preoperatively via vertebral body counts from sacrum on lateral view, or obtain rib count on anteroposterior (AP) view, having first confirmed the number of rib-bearing segments.¹

- Midline incision: Dissection includes ipsilateral erector spinae muscles to expose lamina, facet capsule complex, transverse process, rib head and costovertebral junction of the desired vertebrae laterally.
- Medial aspect of transverse process overlies the pedicle, which is at the level of the superior edge of the transverse process in the upper thoracic spine, but at the mid to lower half of the transverse process in the lower thoracic spine.
- Intraoperative fluoroscopic localization using radiographic marker at transverse process overlying pedicle of interest is generally more accurate than spinous process marker.
- The pedicle inferior to the level of interest is targeted (i.e., the T8 for a T7–T8 disk herniation).
- A laminotomy is performed adjacent to the pedicle, which allows for direct palpation of the medial pedicle wall, and which may be expanded to a laminectomy if more extensive decompression is indicated.
- After localization of the pedicle via either direct palpation or fluoroscopy, a high-speed drill is used to open the posterior cortex of the pedicle. The cancellous pedicle is then expanded sequentially with combination of curettes, pointed rongeurs, and/or a high-speed drill in “eggshell” fashion, leaving the cortical walls of the pedicle in place.
- Alternatively, a high-speed drill may be used after first developing pedicle channel with pedicle finder and then tapping as for pedicle screw placement.²
- After reaching the posterior vertebral body, a downgoing curette is used to carefully fracture the medial wall out laterally, exposing the lateral margin of the thecal sac. The medial portion of the adjacent facet is resected, and the superior wall of the pedicle is removed to expose the disk space.
- Palpate the ventral epidural space carefully to localize and identify the lesion. For calcified disks, be aware of increased risk of scarring and adhesions to the ventral dura.
- For disk herniations, an annulotomy is created lateral to the dura, without retraction of the spinal cord.³ Disk material is then provisionally removed with pituitary rongeurs. Develop a cavity encompassing the disk space and possibly including the end plates of the adjacent vertebral bodies, and preserve a thin layer of posterior annulus or cortical bone to protect the ventral dura (**Fig. 48.1**). A small downgoing curette is then used to push the disk herniation or mass into the cavity, allowing for removal with pituitary rongeur.
- For retropulsed fracture fragments or tumors, larger amounts of vertebral body may be removed using a high-speed bur into which the epidural mass or bony fragments are gently pushed for subsequent removal.
- Visualization of central lesions is difficult and thus should be carefully undertaken with gentle palpation with a Woodson elevator or small

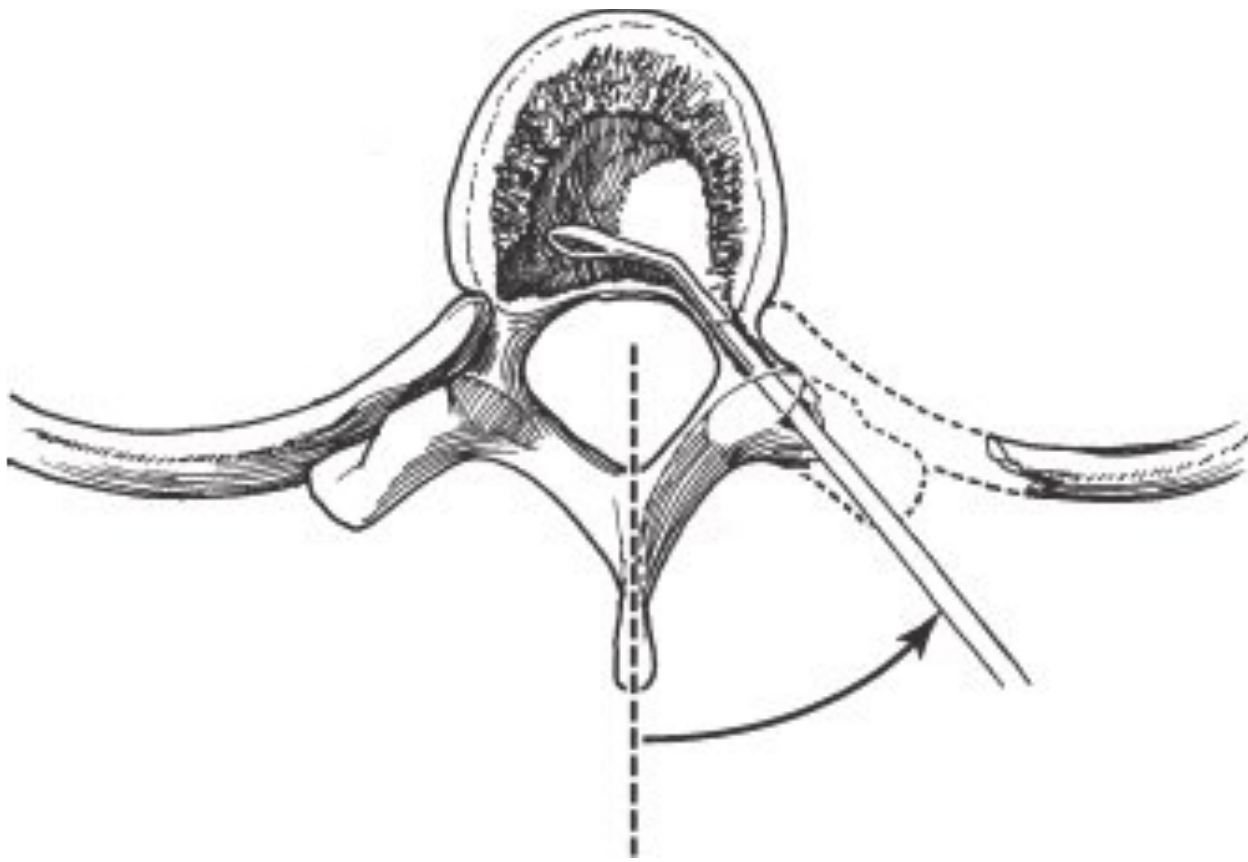


Fig. 48.1 To allow access to the interior of the vertebral body, it may be necessary to resect the rib, rib head, and transverse process. Consider conversion to extrapedicular, costotransversectomy, or anterior approach if needed for adequate decompression. (From Vaccaro AR and Albert TJ, eds. *Spine Surgery: Tricks of the Trade*, 2nd ed., Thieme; 2009.)

downgoing curette. For larger masses, a bilateral transpedicular approach may be considered to include partial resection of the rib head to allow for improved access (**Fig. 48.1**). Intraoperative ultrasound may be used to assess the midline decompression.

- For a bilateral transpedicular approach, or if an unilateral transpedicular approach is expanded to include a facetectomy, consider a posterior instrumented fusion.

48.4 Complications

- Infection
 - Local wound infection most common
 - Diskitis or osteomyelitis
- Durotomy
 - Pseudomeningocele
- Transient or permanent neurologic injury
 - Increased risk of spinal cord injury if excessive retraction placed on dura during decompression
- Inadequate visualization of the ventral epidural space resulting in incomplete decompression
- Iatrogenic instability
 - May occur with wide laminectomy and bilateral transpedicular approach, or if unilateral approach is expanded to include facetectomy

48.5 Postoperative Care

- Frequent neurologic checks for 24 hours postoperatively
- Consider intensive care unit for closer monitoring in patients with preexisting neurologic deficits or significant medical comorbidities
- Urgent MRI in patients with new neurologic deficits
- Routine postoperative imaging not required

48.6 Outcomes

- In properly indicated cases of thoracic disk herniation, neurologic outcomes and capacity for adequate decompression using transpedicular approach affords outcomes similar to a transthoracic approach with less morbidity

48.7 Surgical Pearls

- Review axial and sagittal images on MRI or CT to localize herniated disk and fully understand the relevant anatomy and relationships.
- When anatomy is distorted because of prior surgery or fusion mass fluoroscopy can be used to localize pedicle margins. Consider using a radiographic pedicle marker, as for pedicle screw placement, if additional confirmation is needed.
- If adequate anterior decompression is not achieved transpedicularly, or with a bilateral transpedicular technique, consider performing an extrapedicular or costotransversectomy approach for even greater access (**Fig. 48.1**). May need to convert to anterior approach if lesion remains inaccessible.
- Anterior unintended durotomy or cerebrospinal fluid (CSF) leak can be treated locally with Duragen (Integra Life Sciences Corp.; Plainsboro, New Jersey) or Dura-Guard (Synovis; St. Paul, Minnesota) and a sealant agent followed by lumbar subarachnoid drain.

Common Clinical Questions

1. A large central disk herniation is best addressed via which approach?
 2. Is fusion necessary after routine transpedicular approach?
 3. What is the main disadvantage of the transpedicular approach for anterior decompression?
-

Answers to Common Clinical Questions

1. Transthoracic.
 2. No. Fusion is not typically necessary after a unilateral transpedicular approach, but it may be considered with bilateral transpedicular approach or if facetectomy is performed.
 3. Inadequate visualization and access of ventral midline dura.
-

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49 Costotransversectomy

Hasan A. Zaidi and U. Kumar Kakarla

49.1 Key Points

- Costotransversectomy provides a posterolateral approach for ventral spinal pathology necessitating removal of lateral spine elements.
- The more lateral exposure of the costotransversectomy allows for a progressively improved view of ventral spinal pathology and reduces the need for spinal cord manipulation during surgical dissection; however, the surgeon should be cognizant of critical neurovascular structures.
- Costotransversectomy is indicated for ipsilateral, soft, herniated thoracic disks; it is contraindicated for contralateral spinal cord decompression and large, central, calcified thoracic disks.
- Costotransversectomy is the ideal choice for ventral spinal disease when pulmonary or medical contraindications preclude thoracotomy.

49.2 Indications

- Soft thoracic disk herniation requiring decompression¹
- Paravertebral or epidural spinal abscesses requiring evacuation (originally for tubercular abscess)
- Ventral and ventrolateral space-occupying lesions requiring spinal decompression^{2,3}
- Paraspinal nerve sheath tumors requiring resection²
- Kyphotic deformity requiring correction²

49.3 Technique

- Pre- and postpositioning motor evoked potentials and somatosensory evoked potentials are carefully monitored, particularly in patients who have significant spinal cord compromise evident on preoperative imaging.
- The patient is positioned prone in a neutral or a partially lateral position on a Jackson spine surgery table, with adequate padding at pressure points.
- A straight vertical paramedian or midline incision, with midline convexity, is made.

- For paramedian incisions, surgical exposure is obtained through the skin, trapezius and/or latissimus dorsi (for lower approaches), and the erector spinae, with medial retraction to expose the ribs and transverse process of the caudal vertebrae at the level of interest.
- For midline incisions, a subperiosteal incision down the spinous process, lamina, and transverse process allows the paravertebral musculature to be retracted laterally.
- After the lateral edge of the transverse process is identified, a subperiosteal dissection of the perichondrium of the rib is performed using a semisharp subperiosteal dissector (e.g., Doyen Rib Raspatory [Novo Surgical Inc.; Oak Brook, Illinois]), with care taken to avoid disrupting the parietal pleura and neurovascular bundle at the inferior aspect of the rib (**Fig. 49.1**).

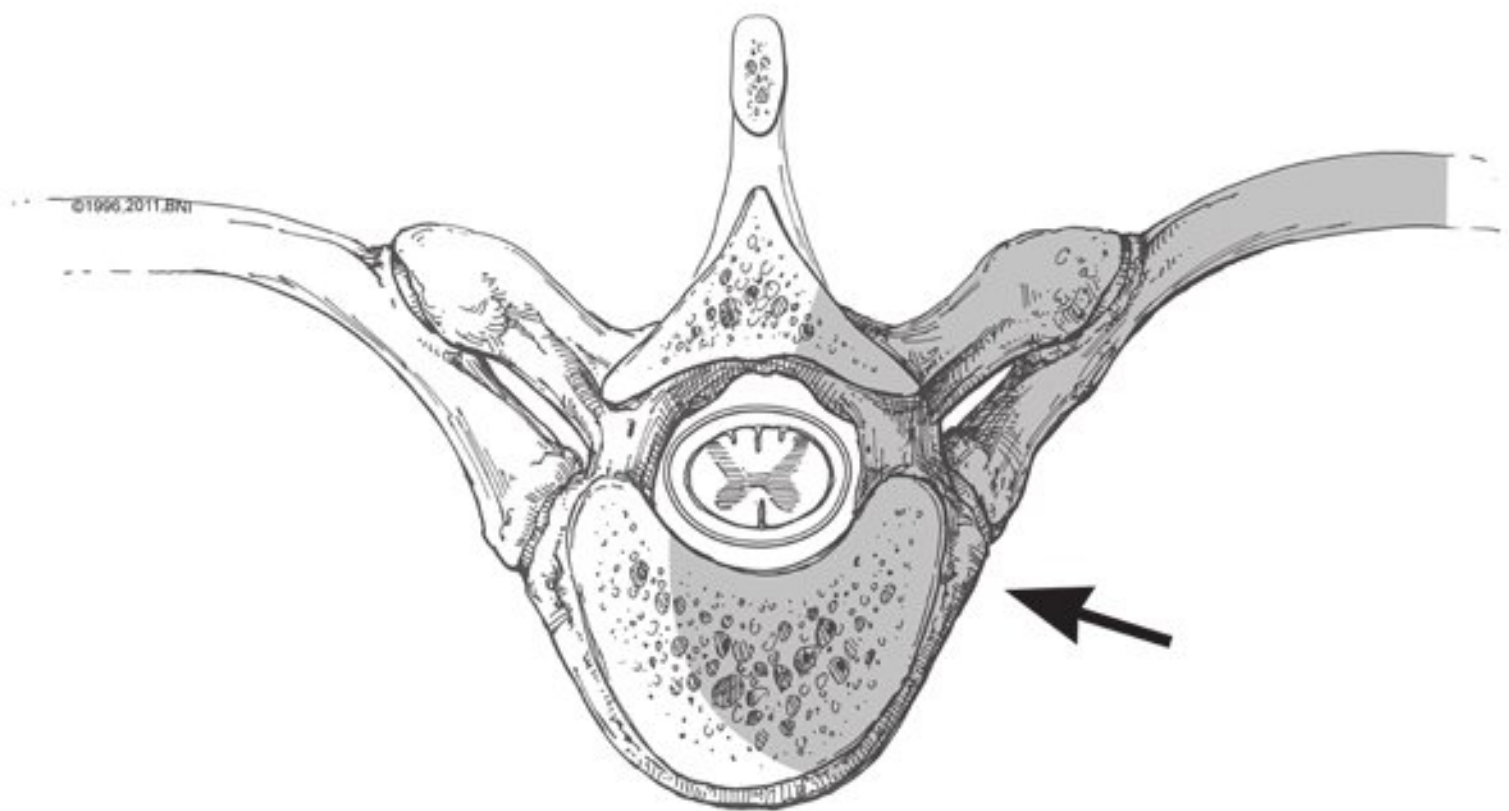


Fig. 49.1 Axial view of the typical thoracic vertebral body shows approach-specific zone (shaded area) of bony decompression for costotransversectomy surgical access. Arrow indicates only the angle of approach (not the skin incision location). Preoperative planning requires correlation with preoperative axial magnetic resonance imaging and computed tomography. With the lateral extracavitary approach, removal of an additional 5 to 7 cm of lateral rib and downward retraction of the pleura provide greater exposure and a more lateral angle of entry, for improved anterior decompression across the midline. (Used with permission from Barrow Neurological Institute.)

- A rongeur, bone punch, or high-speed drill is used to remove the transverse process and 3 to 5 cm of the rib from the costotransverse junction.
- The neural elements must be carefully identified by performing a partial laminectomy and facetectomy, with the lateral margin of the dura identified and the nerve roots followed out of the foramen after identification of the pedicle at the level of interest.
- A malleable brain retractor can be used to protect the pleura and retract it during sharp surgical dissection.
- After removal of the rib and transverse process, sharp dissection is performed on the lateral aspect of the vertebral body and the disk space is identified just superior to the pedicle.
- Manipulation of the spinal cord should be minimized, and no attempt should be made initially to remove disk material herniating within the spinal canal.
- An annulotomy is performed on the lateral aspect of the disk, and a discectomy is performed to create a cavity within which herniated disk material compressing the spinal cord can be pushed inward to facilitate safe removal without spinal cord manipulation; this cavity may be enlarged to include partial vertebral resection, if necessary.
- When vertebral column resection is necessary, a midline approach allows for safe instrumentation and intersegmental fixation.
- If the spine must be destabilized because of vertebral column resection, the first step is posterior instrumentation, with gradual reduction of kyphosis with temporary rods.
- During closure, the operative field should be irrigated and a Valsalva maneuver should be performed to identify any pleural compromise, for which primary repair should be attempted with or without placement of a chest tube.

49.4 Complications

- The type and extent of complications depend on the pathology and the complexity of the procedure performed.
- The overall complication rate for costotransversectomy is 15%⁴
- The most common complications are neurologic decline (5.5%), neurovascular bundle compromise, hemothorax, pneumothorax (up to 25%), pleura injury, lung contusion, atelectasis,^{2,3,4} symptomatic or asymptomatic pseudomeningocele, and infection.

- Inadvertent intraoperative injuries may also occur: Dural tear, injury to great vessels anterior to the vertebral body, and injury to radiculomedullary arterial branches, leading to spinal cord infarct.

49.5 Postoperative Care

- Obtain an immediate postoperative chest radiograph to rule out pneumothorax and hemothorax.
- Conduct continuous pulse oxymetry as part of routine postoperative monitoring.
- Perform incentive spirometry and pulmonary toilet early in the postoperative period.
- Control pain adequately to prevent guarding and shallow breathing.
- Mobilize the patient early as part of standard spinal postoperative care (e.g., to prevent deep vein thrombosis).

49.6 Outcomes

- The duration of spinal cord compression and the age of the patient affect surgical outcomes.
- Improvement may occur in weakness (30–58%), radiculopathy or local pain (42–91%), and spasticity and myelopathy (up to 95%).^{2,3,4}
- Compared with anterior or combined approaches for treatment of oncologic disease, costotransversectomy has similar perioperative complications, but a posterior approach may be better tolerated in patients who have multiple medical comorbidities.

49.7 Surgical Pearls

- The ultimate goal of costotransversectomy is to minimize spinal cord manipulation by removing more bone for better visualization of target pathology.
- Anterior translocation of the vertebral column during destabilization may be prevented by initiating anterior column resection only after intersegmental pedicle screw fixation with a temporary rod.
- The neural foramen can be located by following the neurovascular bundle back, and the disk space of interest can be reached after removal of one or more rib heads.

Common Clinical Questions

1. With central thoracic disk herniation, which side is preferred for the costotransversectomy approach if it is indicated?
 2. What is the origin of the major arterial supply to the spinal cord from T8 to the conus? Where is it located?
 3. During patient positioning, which anatomic landmark is useful in identifying the T7 vertebral level?
-

Answers to Common Clinical Questions

1. The left side is preferred to avoid the great veins. Repair of injury to the thick-walled aorta is technically more feasible than repair of injury to the thin-walled vena cava.
 2. The artery of Adamkiewicz, which is located on the left in most (80%) patients, arises between T9 and T12 in most (75%) patients.
 3. The inferior tip of the scapula.
-

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50 Lateral Extracavitary Approach

Marco Ferrone and Christopher M. Bono

50.1 Key Points

- The lateral extracavitary approach allows for maximal circumferential access to decompress the neural elements (spinal cord and nerve roots) and reconstruct the anterior column through an all-thoracic posterior exposure.
- Anterior column support in the form of a cage or strut graft stabilized with posterior instrumentation can be effected via a single-incision, single-stage surgery.

50.2 Indications

- Degenerative—thoracic disk herniations, spinal cord compression from vertebral body osteophytes (i.e., hard disks)
- Trauma—fractures requiring ventral canal decompression or anterior column support
- Infection—diskitis/osteomyelitis causing substantial bone destruction, anterior epidural abscesses
- Neoplasm—resection, decompression and stabilization for primary or metastatic spinal tumors

50.3 Preoperative Work-up

- Plain radiographs are essential to ensure correct level surgery; intraoperative determination of level is most reliably performed by counting the ribs on an anteroposterior (AP) radiograph and correlating this to the level of pathology on a preoperative advanced imaging study (CT or MRI).
- Advanced imaging, either CT or MRI is especially important. When planning surgery for a neoplasm, it is crucial to evaluate whether the lesion is lytic or blastic and whether epidural soft tissue mass is present.
- Regardless of pathology, CT and MRI allow assessment of pedicle size for screw placement.
- Angiography can be helpful in reducing blood loss and visualizing the artery of Adamkiewicz.

- Pathology-specific work-up—i.e., imaging of entire spine for trauma, culture data, and acute phase reactants for infection; staging; and biopsy for neoplasm—may be necessary.
- Communication with anesthesia and anticipation of the potential for intensive care after surgery.

50.4 Technique

- General endotracheal anesthesia is administered. If there is any anticipation that the pleural cavity will be entered, consider a dual lumen tube.
- Intravenous access and monitoring is established.
- Mechanical anticoagulation and Foley catheter should be placed. Preoperative antibiotics are administered unless held to obtain intraoperative culture data.
- The patient is then positioned prone on a radiolucent table with all bony prominences well-padded.
 - Arms are either tucked at the patient's sides or on arm rests overhead per surgeon preference.
 - The patient must be adequately secured especially if tilting the table is planned.
- Neuromonitoring can be optionally used. In certain instances, steroids can be given before decompression for cord protection.
- Thermoregulatory devices (e.g., Bair hugger) should be used to maintain the patient's temperature.
- Fluoroscopy can be used to ensure good visualization of all levels.
- A sterile operative field is established.
- A midline incision is made from the cranial to the caudal extent of the planned instrumentation levels. Meticulous hemostasis should be obtained whenever possible.
- Standard subperiosteal exposure is undertaken, and levels are confirmed using fluoroscopic imaging.
- It is our preference to place pedicle screws and fashion rods before decompression to maintain the stability of the spine at all times. The rod is placed on the contralateral side to the planned lateral extracavitary approach, if a bilateral extracavitary approach is planned, the temporary rod may need to be switched from side to side to maintain stability.
- Next the dissection is carried laterally. The paraspinal muscles are mobilized and a plane deep to the trapezius, rhomboid, and/or latissimus dorsi (depending on the spinal level) is established to where the posterior angle of the rib is visualized.
- Ribs should be counted and the relationship between the rib and the vertebral body visualized.

- Most ribs (with the exception of 1, 11, and 12) articulate at the level of the disk space with the same numbered vertebral body and the one cranial. For example, rib 4 articulates at the T3/T4 disk space (**Fig. 50.1**).
- Often two ribs need to be removed to gain full access to perform a single-level vertebrectomy.
- Subperiosteal dissection of the rib is performed, taking care not to disrupt the neurovascular bundle running along the caudal surface of the rib.
 - Subperiosteal dissection helps to protect the pleura and avoid a pneumothorax.
 - Visualizing the neurovascular bundle allows the surgeon to trace the root back into the neural foramen.

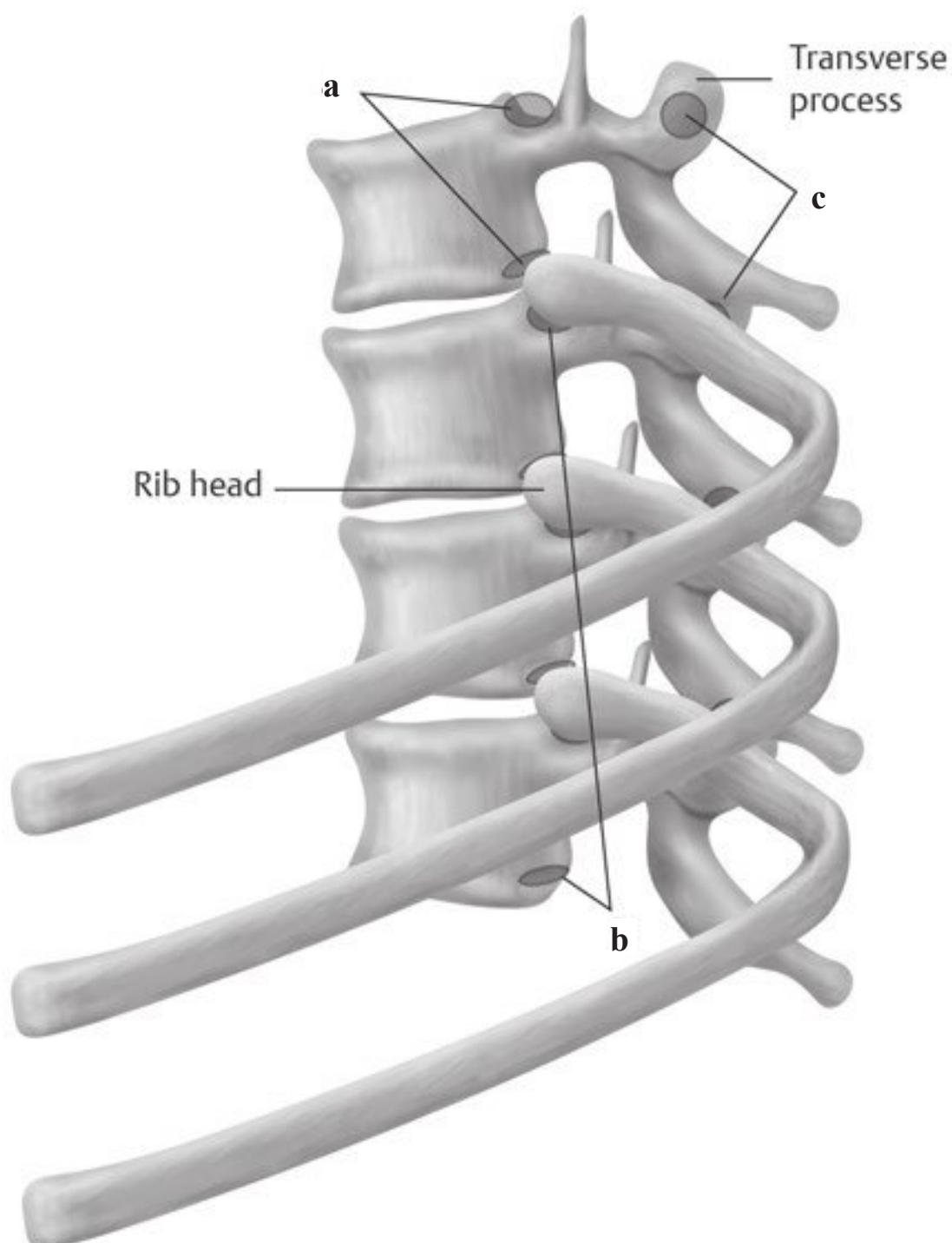


Fig. 50.1 Anatomical relationships between the ribs, vertebral bodies, and disk spaces is critical. Most ribs (with the exception of 1, 11, and 12) articulate at the level of the disk space with the same numbered vertebral body and the one cranial. The rib head articulates with an articular surface (**a**) along the posterolateral aspect of the lower vertebra and another (**b**) along the posterolateral aspect of the upper vertebra. In addition, there is (**c**) an articulation of the posterior rib surface with the anterior aspect of the transverse process that must be considered when removing these structures.

- The rib can then be transected at the posterior angle with a rib cutter (**Fig. 50.2**). A small Cobb elevator can be used to disarticulate the rib from the transverse process, or the transverse process can be removed with a rongeur. Next the rib is disarticulated from the vertebral bodies.
- The lung needs to be gently retracted laterally.
- The vertebral body is now exposed from disk space to disk space.
- Laminectomy and facetectomy can be performed to posteriorly decompress the spinal canal.
- Nerve roots (T2–L1) may need to be sacrificed.
 - Place a 2–0 silk proximal and distal to the dorsal root ganglion.
 - Transect the root distal to the distal tie, leaving the ends long.
 - This tie can then be used to gently roll the dura away from the working area.

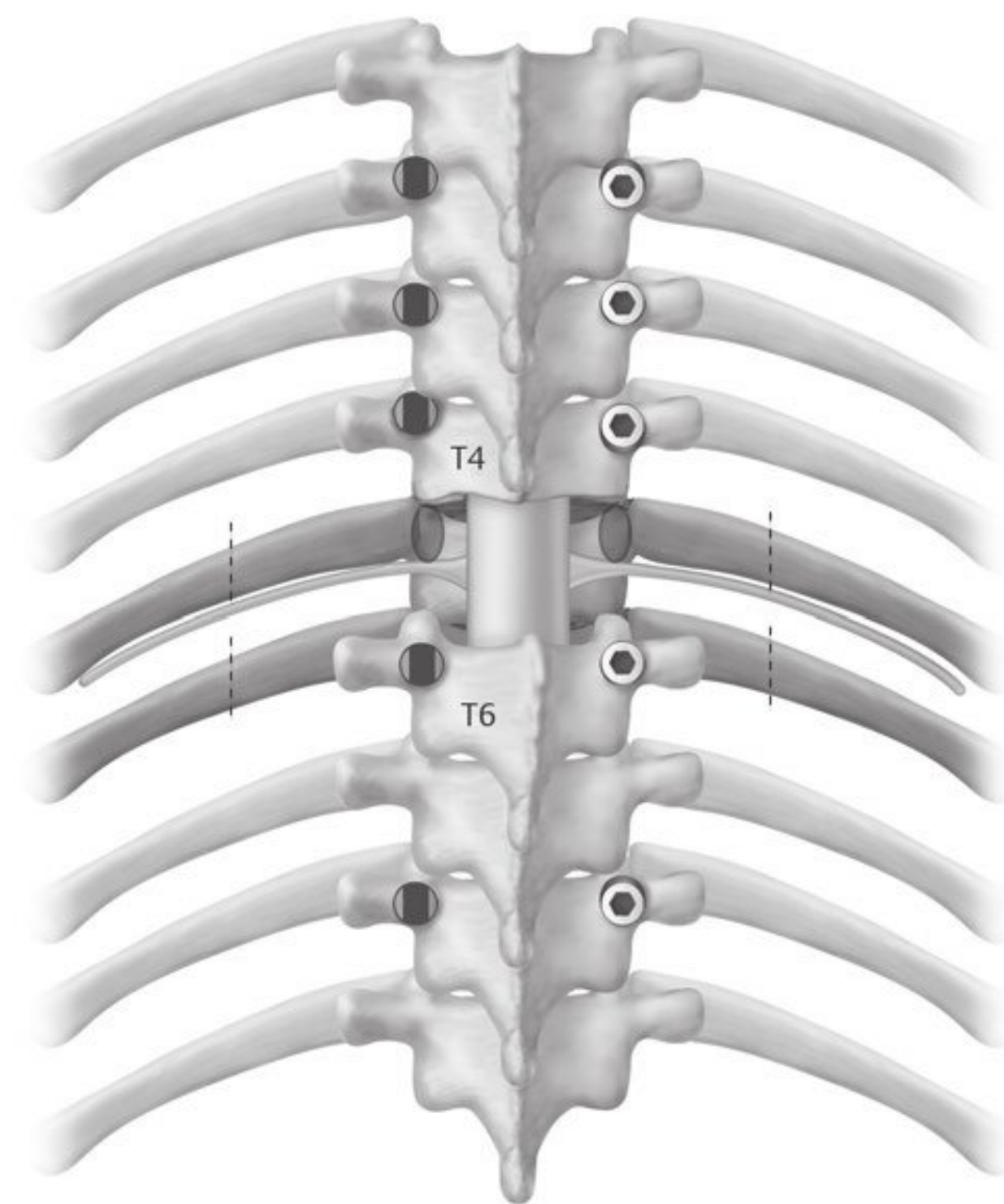


Fig. 50.2 The proposed rib cuts are delineated at the rib angles (dashed lines). After these cuts, the rib head is disarticulated from the vertebral body, and the segment of bone is removed.

- Depending on the goals of the procedure, a discectomy or vertebrectomy can safely be performed using reverse angle curettes and/or a bur, working away from the dura by pushing down into the vertebral body/disk region.
- Care needs to be taken to protect the pleura and great vessels.
- Adequacy of the decompression can be assessed using direct visualization, fluoroscopy, and a dental mirror.
- Anterior column reconstruction can be performed using a variety of techniques:
 - Structural autograft (e.g., rib)
 - Bone cement
 - Metallic mesh cage
 - Expandable cage
- To place the cage, gentle distraction can be applied through the pedicle screws. Fluoroscopy can be used to confirm alignment, followed by gentle compression through the pedicle screws. If an expandable cage is used, this step is usually not needed (**Fig. 50.3**).

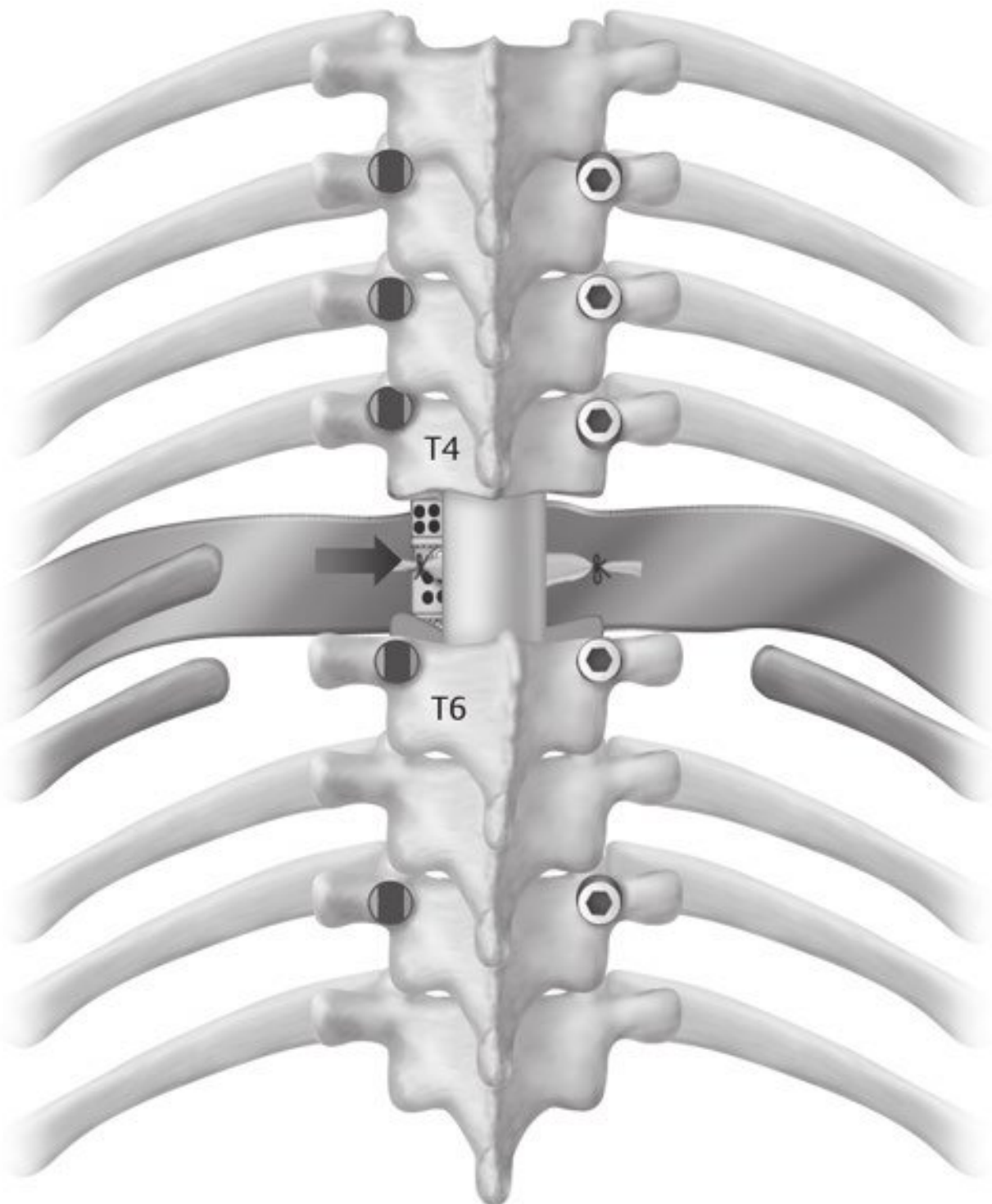


Fig. 50.3 After removal of the vertebral body, the authors prefer to use an expandable cage to reconstruct the anterior column. This is placed lateral and then anterior to the spinal cord.

- The wound is then copiously irrigated and any hematoma evacuated from the chest.
- The lung should be examined for an air leak and the dura for evidence of a cerebrospinal fluid (CSF) leak.
- A red rubber catheter can be left in the chest and the wound closed around it.
- A subfascial drain, such as a 19 Blake, can be left to suction.
- Closure should be watertight and in layers. When the fascia is closed, the red rubber catheter can be withdrawn under negative pressure to evacuate any remaining air or fluid in the chest.
- Skin closure with 3–0 nylon may be preferred when postoperative radiation is required.
- Before leaving the operating room (OR), a chest X-ray should be obtained to rule out pneumothorax.

50.5 Complications

- Large-volume blood loss
- Great vessel injury
- Pneumothorax/hemothorax/chylothorax
- Durotomy
- Neurologic deficit

50.6 Postoperative Care

- The patient should be extubated as soon as possible.
- The patient should be mobilized and participate in physical therapy (PT) and chest PT as soon as possible.
- Drains can be removed when output falls.

50.7 Outcomes

- In a systematic review, Lubelski et al¹ found that the outcomes and risk for complications with a lateral extracavitary approach (or costotransversectomy) are comparable to more traditional approaches such as the transthoracic approach.
- In another article by the same group,² the lateral extracavitary approach was found to be successful in addressing complex pathology such as vertebral body infections but had slightly more blood loss than the costotransversectomy approach.

50.8 Surgical Pearls

- When removing calcified thoracic disk herniations via a lateral extracavitary approach, it is helpful to remove a portion of the vertebral body above and below the disk space before attempting to remove the disk. This can allow the disk region to be more gently pushed into the defect.
- A malleable retractor can be placed anterior to the vertebral body to protect the intrathoracic structures after a subperiosteal plane has been established.

Common Clinical Questions

1. A 45-year-old man is undergoing en bloc resection of a T5 vertebral tumor via a lateral extracavitary approach. After tumor removal, the construct is distracted to allow for easier insertion of a fixed titanium mesh cage. Upon introduction of the cage into the defect, motor evoked and sensory evoked potentials are lost in the lower extremities. The next most appropriate step in management is as follows:
 - A. Full insert the cage.
 - B. Remove the cage and wait.
 - C. Remove the cage and release distraction.
 - D. Wake the patient.
2. During a lateral extracavitary approach for resection of a T9 tumor, you found it necessary to sacrifice the T8 and T9 nerve roots bilaterally. Which of the following postoperative deficits should the patient expect?
 - A. Paraplegia
 - B. Weakness of hip flexion
 - C. Decreased sensation along the inguinal crease
 - D. None of the above
3. Five days following a lateral extracavitary approach you note a milky white substance being emitted from a catheter drain placed into the anterior wound. Which is the most likely diagnosis?
 - A. CSF leak
 - B. Chylothorax
 - C. Wound infection
 - D. Tissue necrosis

Answers to Common Clinical Questions

1. C.
 2. D.
 3. B.
-

References

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51 Pedicle Subtraction Osteotomy/Smith-Petersen Osteotomy

Frank La Marca, Paul Park, and Juan M. Valdivia

51.1 Key Points

- For correction of mainly symptomatic fixed sagittal imbalance (pedicle subtraction osteotomy [PSO]) and nonfixed sagittal imbalance (Smith-Petersen osteotomy [SPO]).¹
- An asymmetric PSO can also be used for correction of coronal imbalance.
- Can aid in the correction of coronal balance via posterior column release (SPO) or three-column release (PSO).
- Can be performed on the thoracic or lumbar spine.

51.2 Indications

- Uncompensated spinal kyphoscoliosis (thoracic and/or lumbar) with or without progression
- Correction of fixed kyphosis requiring more than 30° of correction (PSO) at one spinal segment
- Correction of nonfixed deformity requiring 5 to 10° of correction (SPO) per spinal segment

51.3 Technique

51.3.1 Pedicle Subtraction Osteotomy

- Position on Jackson frame with extended chest bolster.
- Neurophysiologic monitoring should be used (transcranial motor evoked potentials, somatosensory evoked potentials, electromyography) to minimize neurologic injury during correction.
- Place two to three levels of bilateral pedicle screws cephalad and caudal to the target vertebral segment (i.e., commonly L2 or L3 in the lumbar spine).
- Remove transverse process and rib head (if in thoracic spine).

- For 360° extrapedicular vertebral body exposure, dissect soft tissues and vascular structures from vertebral body with placement of malleable retractors to maintain separation.
- Perform Gill laminectomy of the target vertebral body and partial laminectomies of the adjacent segments (**Fig. 51.1a**).
- Decancellate and resect pedicle while contralateral temporary rod is in place.
- Resect (chisel or drill) a posterior wedge out of vertebral body (**Fig. 51.1b**).²
- Repeat steps 5 and 6 on the contralateral side.³
- Resect remaining posterior vertebral body cortex; impact bone into cavity previously created during wedge resection of vertebral body with angled bone tamp or downgoing curette.³
- Use fluoroscopy to confirm adequate resection in addition to visual inspection.²
- Compress across pedicle screws to close down osteotomy (**Fig. 51.1b**).
- Evaluate neurophysiologic monitoring and/or wakeup test to ensure absence of neurologic compromise.

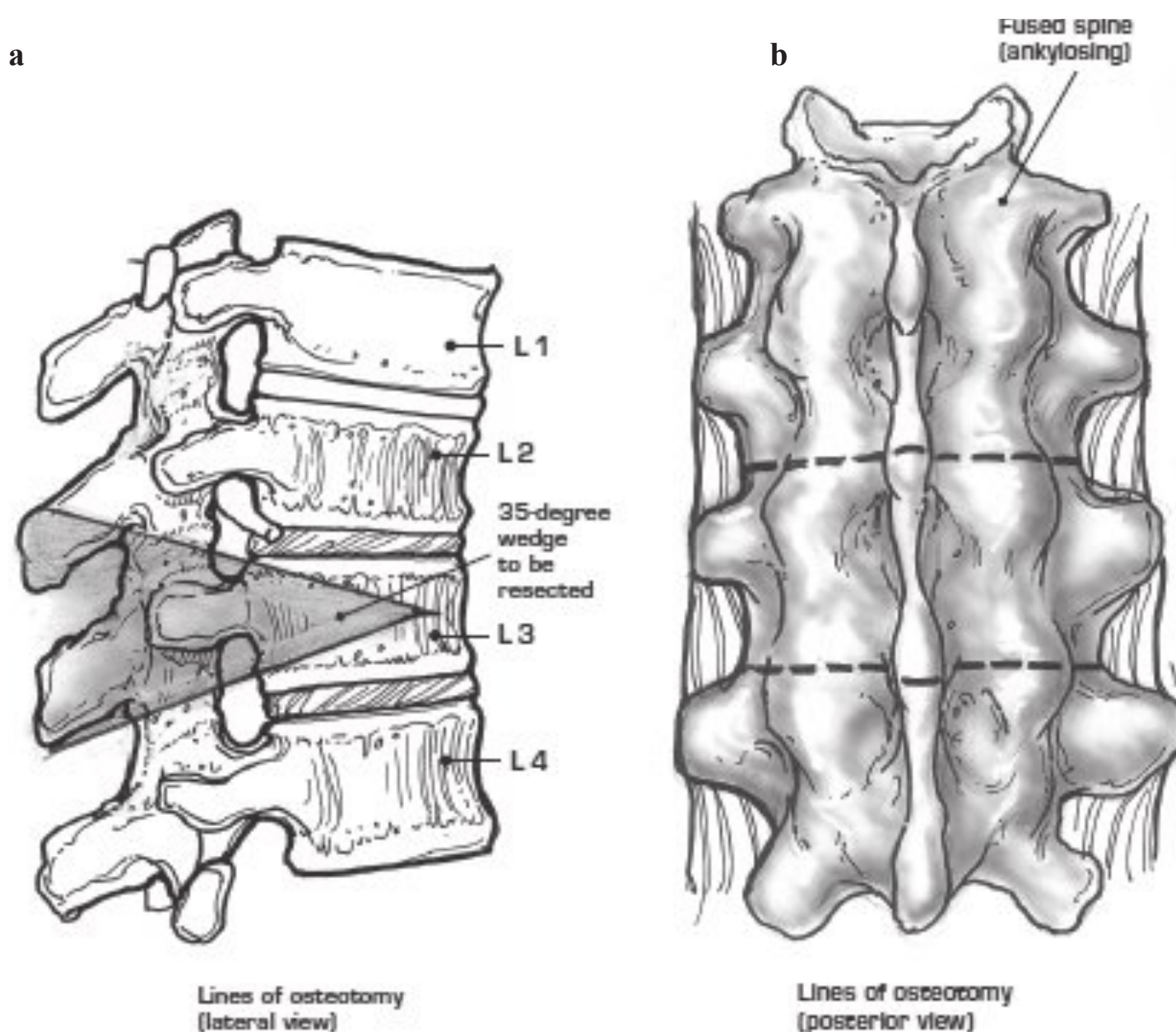


Fig. 51.1 (a,b) Pedicle subtraction osteotomy. (From Haheer R and Merola A, eds. *Surgical Techniques for the Spine*, Thieme; 2003.)

51.3.2 Smith-Petersen Osteotomy

- Position on Jackson frame with extended chest bolster.
- Consider use of multimodal neurophysiologic monitoring to minimize neurologic complications during deformity correction.
- Place pedicle screw instrumentation across targeted segments.
- Remove interspinous ligaments with rongeur.
- Spinous processes are removed at each targeted segment in a 45° angle in addition to removal of the upper and lower lamina edges.
- Bilateral facetectomies are performed with osteotomes and/or a high-speed drill⁴ (**Fig. 51.2a**).
- Remove ligamentum flavum.
- Compress across the screw heads to close the osteotomy⁴ (**Fig. 51.2b**).
- The patient may be placed in extension to assist in closure of the osteotomy.⁴

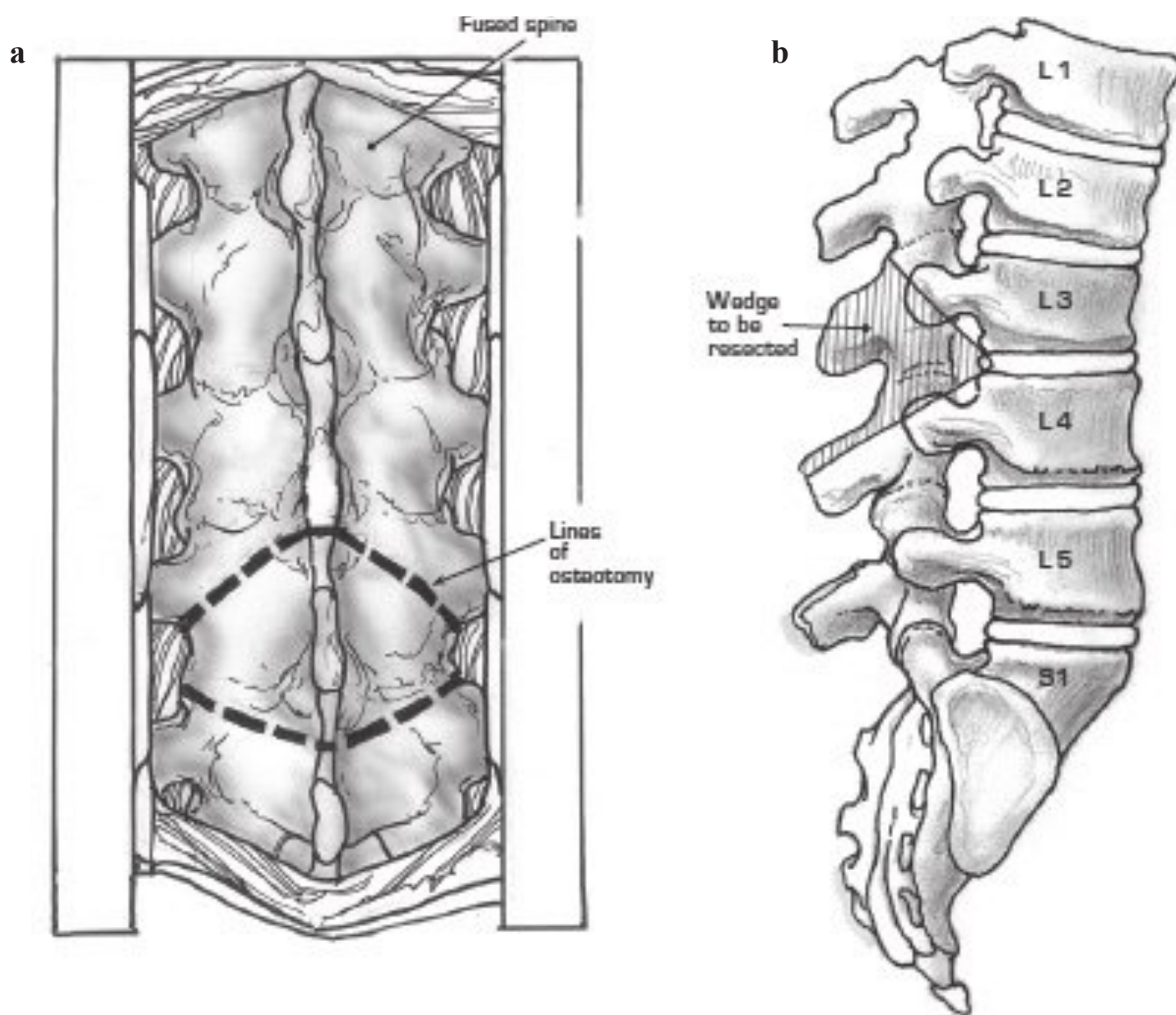


Fig. 51.2 (a,b) Smith-Petersen osteotomy. (From Hafer R and Merola A, eds. *Surgical Techniques for the Spine*, Thieme; 2003.)

51.4 Complications

- Neurologic injury resulting from iatrogenic spinal stenosis upon closure of the osteotomy.
- Spinal subluxation during osteotomy.
- Durotomy.
- Hypotension caused by excessive blood loss during osteotomy. Avoid hypotension, especially during correction maneuvers, lest spinal cord ischemia result.
- Epidural hematoma.
- Vascular injury.
- Pseudarthrosis.

51.5 Postoperative Care

- Immediate neurologic evaluation upon wakeup.
- Admission to neurologic intensive care for close monitoring.
- Maintenance of blood pressure.
- Consider postoperative CT scan to evaluate adequacy of bone resection and juxtaposition of osteotomy margins.

51.6 Outcomes

- With PSO, average curve improvements are 61° in scoliosis cases, 56° in global kyphosis cases, and 51° in angular kyphosis cases.⁴
- Clinical and radiographic outcomes are superior with PSO in the lumbar spine rather than in the thoracic. The lower the PSO is performed, the less angulation is required for sagittal balance correction.¹
- In properly selected patients, sagittal balance can be restored with PSO or multiple SPO safely.

51.7 Surgical Pearls

- Ensure that bone resection is adequate and wide enough laterally in both PSO and SPO to prevent interference upon osteotomy closure or compression of neurologic structures.
- Use thrombin powder or bioabsorbable bone wax to minimize bleeding during PSO without compromising fusion rates. Aprotinin is also an option, although controversy exists over possible associated complications.
- Do not resect the anterior vertebral wall entirely during PSO, for it will serve as a hinge during pedicle screw compression and osteotomy closure and prevent vertebral segment subluxation.

- Posterior laminectomy resection should be complete and include any epidural scar removal to assess dural sac adequately after osteotomy closure and ensure that there is no spinal canal compression.
- During SPO closure, place patient in extension to lengthen the anterior spinal column and facilitate osteotomy closure and to minimize strain on the pedicle screw construct.

Common Clinical Questions

1. How many degrees of correction can be achieved with PSO as opposed to SPO?
2. What is the main advantage of a PSO/SPO procedure?
3. In regard to the anterior and middle columns, what is the main difference between PSO and SPO?

Answers to Common Clinical Questions

1. Up to 30° per level with PSO compared with 10° per level with SPO.
2. Correction of sagittal imbalance and reestablishment of lordosis or correction of kyphosis.
3. PSO is a closing wedge osteotomy involving the three columns and thus shortens the middle and posterior columns. SPO, in contrast, uses the middle column as a pivot, thus lengthening the anterior column and shortening the posterior column.

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52 Thoracoscopic Approaches to the Spine

Meic H. Schmidt and Ricky Kalra

52.1 Key Points

- Thoracoscopic access to the anterior spine should be considered as an alternative minimally invasive access approach to traditional open and mini-open access procedures.
- Thoracoscopy can be used along the entire thoracic spine (T1–T12) and can be extended via transdiaphragmatic incision to the upper third of L2 in the retroperitoneal space. The chest cavity gets smaller at the cranial end and can limit the access for extensive thoracoscopic procedures (e.g., discectomy, corpectomy).
- Good single-lung ventilation with a double lumen endotracheal tube is crucial for thoracoscopy.

52.2 Indications

- Sympathectomy for palmar hyperhidrosis¹
- Thoracic discectomy²
 - Degenerative thoracic disk disease
 - Diskitis
- Neurogenic paraspinal tumors³
- Corpectomy for trauma and tumor^{4,5}
 - Anterior column reconstruction
 - Spinal canal decompression
 - Anterior plate stabilization
- Scoliosis: Release of anterior column and correction of adolescent thoracic deformity⁶

52.3 Techniques

- Single-lung ventilation with a double-lumen endotracheal tube is crucial to achieve complete collapse of the lung. Endotracheal blockers can be used but are less successful in achieving single-lung ventilation.

Neuromonitoring and hand/skin temperature monitoring are used in select cases involving the upper thoracic spine near T1/T2.

- The patient is placed on a rotating, radiolucent table in the lateral decubitus position perpendicular to the operating room table and the floor. The lower extremities are placed on a leg rest, and the upper leg is flexed at the hip joint to relax the psoas muscle. An axillary roll is placed under the dependent arm, and the other arm is supported with an armboard or a pillow.
- See **Fig. 52.1** for setup and intraoperative views.

52.3.1 Sympathectomy

- Typically two, although in rare cases three, ports (camera port, working port, and sometimes a retraction port or suction port) in a triangular arrangement are used for surgical disconnection or resection of the sympathetic chain.
- The second rib is located under the apical fat (first rib is typically not seen), and the sympathetic chain (SC) is located on the rib heads. There are three common methods to disconnect the sympathetic chain:
 - Sympathetic chain can be clipped between the second and third ribs
 - Potentially reversible
 - Sympathectomy: Sympathetic chain is resected
 - Palmar hyperhidrosis: Sympathetic chain from the second to the third rib head (isolate second ganglia)
 - Craniofacial hyperhidrosis: Sympathetic chain from second, third, and fourth rib heads (isolate second and third ganglia)
 - Axillary hyperhidrosis: Sympathetic chain from third, fourth, and fifth rib heads (isolate third and fourth ganglia)
 - Sympathotomy: Disconnecting sympathetic chain with electrocautery or Harmonic scalpel by creating a 5-mm gap
 - Palmar hyperhidrosis: T2 and T3 rib head
 - Axillary hyperhidrosis: T2 to T4 rib head
- Skin temperature increase by 1°C in the ipsilateral hand confirms adequate sympathetic chain disconnection. The temperature increase is due to vasodilation from loss of sympathetic input.
- Chest tubes are typically not required or are only used temporarily while the second sympathectomy is completed.

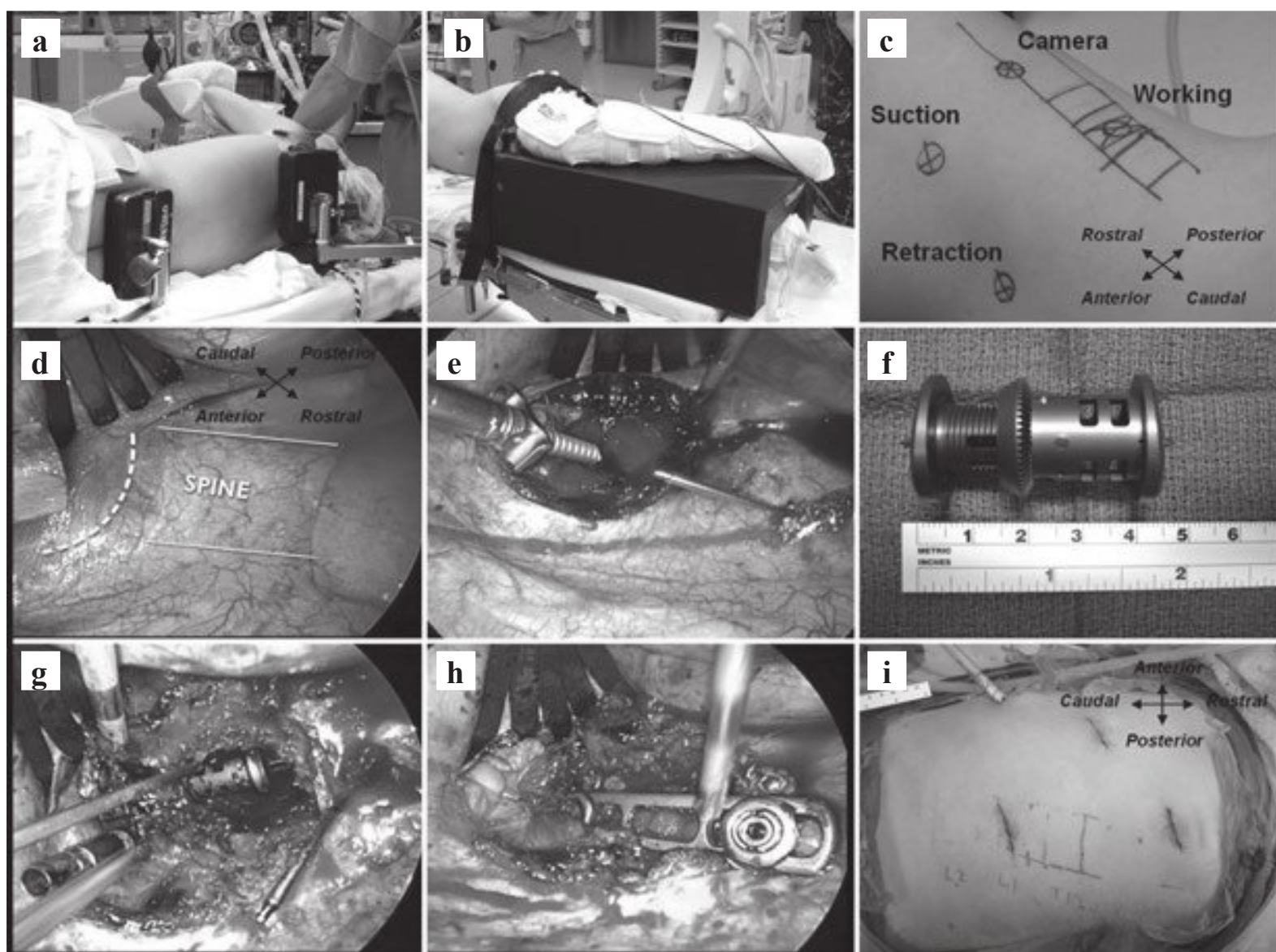


Fig. 52.1 (a) The patient is positioned on a radiolucent table in the right lateral decubitus position for a left-sided thoracoscopic approach to L1. The independent arm is in a Krause frame. Adjustable pads at the pubis, sternum, and lower and upper spine hold the patient in position. (b) The independent leg is slightly flexed at the hip to facilitate iliopsoas relaxation, making it easier to dissect this muscle off the lateral aspect of the vertebral bodies at the thoracolumbar junction. (c) The level of interest is marked, identifying the vertebral body above and below, and the four chest portals are planned. (d) Endoscopic view of the spine (solid lines). The diaphragm is swept inferiorly with a fan retractor and a diaphragmatic incision is planned (dotted lines). (e) A Kirschner wire is placed above the planned corpectomy and a polyaxial screw-clamp combination is placed below it. (f) Lateral view of a fully expanded gear-driven cage. (g) The cage is placed and expanded within the central corpectomy. (h) Final anterolateral plate construct. (i) Closure with chest tube exiting the retraction port. (From Ragel BT, Amini A, Schmidt MH. Thoracoscopic vertebral body replacement with an expandable cage after ventral spinal canal decompression. *Neurosurgery* 2007;61(5 Suppl 2):ONS319.⁷ Reproduced with permission.)

52.3.2 Neurogenic Paraspinal Tumors

- Typically, two to three ports (camera port, working port, and sometimes a retraction port or suction port) in a triangular arrangement are placed around the level of the tumor.

- The tumor is most commonly located under the parietal pleura, which is incised. A plane is identified between the tumor and the parietal pleura.
- Depending on the size of the lesion, debulking with a CUSA (Cavitron Ultrasonic Surgical Aspirator) or pituitary rongeurs is necessary.
- The tumor is then detached from the lateral spine, and the exiting nerve root needs clipping if the mass is a thoracic nerve sheath tumor.
- Chest tubes are not required unless the case takes longer than expected.

52.3.3 Thoracoscopic Microdiskectomy

- Typically uses three to four ports (camera port, working port, suction port, and retraction port for lung or diaphragm).
- For diskectomy, it might be necessary to perform partial corpectomies adjacent to the symptomatic disk space, so the segmental vessels might need to be ligated after the parietal pleura is elevated.
- One centimeter to 2 cm of the rib head is removed with a high-speed drill or rib cutters to expose the corresponding pedicle. Then the disk and adjacent partial corpectomies are performed to create space to decompress the spinal canal. The size, shape, and extent of the space depend on the characteristics of the thoracic disk herniation.
- The spinal canal is entered via partial resection of the pedicle, and the lateral thecal sac is identified. Then the spinal canal floor is thinned until the herniation can be dissected off the dura and delivered anteriorly away from the dural sac into the resection cavity.
- Depending on the defect, bone graft material is placed into the disk space for fusion; it can be supplemented with an anterior lateral plate.
- A chest tube is placed, the lung is reinflated, and the portals are closed.

52.3.4 Thoracoscopic Corpectomy

- Typically requires four ports (camera port, working port, suction port, and retraction port for lung or diaphragm).
- For corpectomy, the parietal pleura is dissected, and the segmental vessels are ligated with a surgical clip.
- The two adjacent disks are incised and completely removed with curettes and rongeurs.
- For anterior column reconstruction (no spinal canal decompression) in spine trauma, the vertebral body is removed with chisels and or a high-speed drill. A cage or structural bone graft is then used as vertebral body replacement.

- For spinal canal decompression (metastatic tumor, burst fracture), the pedicle is identified and resected to expose the lateral spinal canal. Then the canal is decompressed by delivering metastatic tumor or bone fragment into the corpectomy site anteriorly away from the spinal canal.
- The spine can then be stabilized with an anterolateral plate.
- A chest tube is placed, the lung is reinflated, and the incisions are closed.

52.3.5 Scoliosis: Anterior Release and Correction of Deformity

- Multiple ports are placed to perform discectomies at multiple levels.
- Segmental vessels need to be taken if hardware is to be placed.
- If anterior correction is performed, then the rod is reduced to the individual vertebral body screws.

52.4 Complications

- Thoracic access complications: Thoracic neuralgia, pneumothorax, hemothorax, diaphragmatic injury, abdominal organ injury
- Sympathectomy: Horner's syndrome, excessive compensatory sweating
- Paraspinal tumor: Numbness, cerebrospinal fluid leak
- Thoracic discectomy: Cerebrospinal fluid leak, neurological deficit, chyle leak, wrong-level surgery
- Thoracic corpectomy: Cerebrospinal fluid leak, chyle leak, nonunion, hardware complications
- Anterior transthoracic scoliosis surgery: Decreased pulmonary function, limited correction, hardware failure, pseudarthrosis

52.5 Postoperative Care

- A chest tube is rarely needed for sympathectomy and paraspinal tumors unless there was a lung injury. Corpectomy and discectomy have typically longer operating times. When a lung has been atelectatic for a long time, it can result in pulmonary effusion that can be prevented with 24 to 48 hours of chest tube drainage.
- A postoperative chest X-ray and pulmonary therapy are important. Pain control with local bupivacaine or thoracic block can help breathing issues.

- If there was intrathoracic cerebrospinal fluid leak, then a repair (dural-only substitute, suture, or clip) is attempted and supplemented with HemoSeal (Ethicon, Cincinnati, Ohio). Avoid the use of a chest tube or placing the patient on water seal. Most of the time, a lumbar drain should be placed. Sometimes the cerebrospinal fluid leak is not discovered during surgery, and the chest tube or even normal breathing can cause pneumocephalus and postoperative coma.

52.6 Outcomes

- Sympathectomy for palmar hyperhidrosis has a 96 to 100% success rate. Palmar temperature recording during sympathectomy is a prognostic marker for success.
- Thoracoscopic nerve sheath tumor resections are not widely reported. Compared with posterior surgery or open thoracotomy, the thoracoscopy offers the benefits of a minimal access procedure.
- Thoracoscopic microdiscectomy has excellent results: Myelopathy, radiculopathy, and back pain resolved or improved in 91.1%, 97.6%, and 86.5% of patients, respectively.
- Thoracoscopic corpectomy, spinal canal decompression, vertebral body replacement, and plate stabilizations have excellent outcomes for both burst fracture and metastatic spine disease. Fusion rates are > 90%. Spinal canal clearance is 55 to 110%.
- Anterior scoliosis surgery is typically limited to flexible thoracic deformity in adolescence and attains good correction of scoliosis (mean 55–65%) with excellent cosmetic results.

52.7 Surgical Pearls

- Port placement to optimize working distance and avoidance of “fencing” is important. It is better to place a new port than to struggle throughout the entire case.
- Soft and calcified thoracic disks can be removed thoracoscopically, but large, giant calcified, and recurrent disks are better treated with mini-open or open thoracotomy with the microscope.
- Conversion to an open or mini-open procedure is not always a complication. It is good surgical judgment if the procedure is better and safer completed open.
- Consider placing a preoperative marker to avoid wrong-level surgery for thoracic discectomies.

Common Clinical Questions

1. What is the safest and most appropriate course of action if single-lung ventilation cannot be obtained during a thoracoscopic procedure?
2. What is the most common side effect of a thoracoscopic sympathectomy for palmar hyperhidrosis?
3. After a thoracoscopic procedure for a calcified thoracic disk, the patient does not wake up in the intensive care unit. You obtain a computed tomography scan of the head, which shows pneumocephalus. What is the most likely cause of the patient's pneumocephalus?

Answers to Common Clinical Questions

1. Convert the surgery to an open mini-thoracotomy to directly visualize and retract the lung.
2. Reflex or compensatory hyperhidrosis.
3. Durotomy at the thoracic disk level with air entering the central nervous system space.

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53 Lateral Approaches to the Thoracolumbar Spine

Jay Rhee, C. Rory Goodwin, and Jean-Paul Wolinsky

53.1 Key Points

- When deciding the surgical approach, the surgeon must take into account the type and location of the pathology, health status of the patient, surgeon comfort, anatomical knowledge, and goals of surgery. A case requiring extensive anterior exposure and reconstruction or a patient who is unlikely to tolerate prolonged prone positioning may favor a lateral or anterolateral approach to the thoracolumbar spine.
- The laterality of the pathology often dictates the side of the approach. However, in equivocal cases, the thoracolumbar junction may be more easily approached from the left to avoid the inferior vena cava and to avoid retracting the liver.¹
- Lateral access to the thoracolumbar junction can be obtained rostral or caudal to the diaphragm from a retropleural, transpleural, retroperitoneal or combined approach.

53.2 Indications

- Lateral access to the thoracolumbar spine provides wide anterolateral exposure of the vertebral column via a retroperitoneal approach. Decompressions and reconstructions can be performed for a variety of pathologies, including trauma, degenerative disk disease, metastatic or primary spinal tumors, infections, and deformity. The lateral retroperitoneal approach allows the surgeon access to perform single level discectomies and fusions, disk arthroplasties, corpectomies, and lateral plating.

53.3 Technique

- The lateral approach provides access to the thoracolumbar junction (T12–L2), midlumbar region (L2–L5) and the lumbosacral junction (L5–S1). The techniques for each region will be discussed separately.²
- Thoracolumbar junction (T12–L2):
 - All preoperative images should be carefully reviewed with special emphasis on the axial CT slices. The location of the inferior vena

cava and aorta is critical to avoid vascular injury. The laterality of the pathology often dictates the side of the approach. However, a left-sided approach is preferable when appropriate to avoid retracting the liver and injuring the inferior vena cava.

- The patient is placed in the lateral decubitus position with a shoulder roll placed 4 to 5 cm below the inferior axilla to protect the brachial plexus. Preoperative X-rays are obtained to mark the incision at the correct location. Localization of pathology in the lateral position can be difficult, especially in obese or osteoporotic patients, or patients who have variant lumbar anatomy. Accordingly, preoperative imaging is critical to understand how to count. Counting from bottom up is usually more reliable. Preoperative marking with metallic markers or methyl methacrylate can be considered. The incision is typically drawn obliquely along the superior aspect of the 10th or 12th rib medially up to several centimeters onto the abdomen.
- After the incision is made, the subcutaneous tissue, latissimus dorsi, serratus anterior and intercostal muscles are dissected to expose the superior border of the rib. The rib is dissected off of its muscular attachments taking care to avoid injury to the neurovascular bundle on its inferior edge. The diaphragm often attaches to the 11th rib, so its dissection will allow simultaneous entry into the chest and retroperitoneal space. However, reconstruction of the diaphragm may be more complicated after resection of the 11th rib.
- If the 10th rib is dissected, it is resected along the length of the dissection and saved for use as autograft in fusion procedures. Removal of the rib also enhances exposure and may decrease postoperative thoracotomy pain. A self-retaining DeBakey rib spreader can be placed to widen the intercostal exposure. The pleura is usually opened, but care can be taken and a retropleural dissection attempted. The benefit of the retropleural approach is that it may avoid placement of a chest tube and can help keep the lung out of the field. The disadvantage is that it may markedly decrease the exposure. The lung is retracted anteriorly to expose the vertebral column. The caudal thoracic vertebra should be well-visualized. The diaphragm can be incised, taking care to leave a cuff for later repair for access to the upper lumbar segments.
- If the 12th rib is chosen, it is dissected and removed in similar fashion. The internal and external oblique muscles are split to expose the transversalis, which is divided. The peritoneum is bluntly dissected off the inferior surface of the diaphragm and the retroperitoneal space is opened. The diaphragm can be incised with a cuff for a combined retroperitoneal/lateral thoracic exposure of the thoracolumbar junction. There is a risk of lung injury when opening the diaphragm from the

retroperitoneal space, so care must be taken. A table-mounted, self-retaining retractor is placed to aid in retraction.

- The psoas muscle overlies the lumbar spine from the lateral approach and can be dissected from a medial to lateral direction. Dissection from medial to lateral allows the lumbar plexus to rotate posteriorly with the psoas muscle and protects the muscle. A lateral to medial dissection can lead to nerve root injury. Likewise, a transpsoas approach has a higher risk to the lumbar plexus if not carefully monitored. For anterior access to the disk space or in cases requiring vertebral body resection, the segmental vessel can be found at the midposition of the vertebral body coursing in an anteromedial to posterolateral direction. This vessel can be ligated, cauterized, and cut. Care must be taken to fully cauterize these vessels a safe distance distal to the aorta, for they may retract when cut and result in copious, inaccessible bleeding.
- Injury to the artery of Adamkiewicz can lead to postoperative paraplegia. Preoperative spinal angiography can be considered for localization of this vessel. However, this is typically not an issue in conventional spine surgery, for spinal cord perfusion is not based on a single vessel, and there are multiple bilateral arterial feeders to the spinal cord.
- A blunt instrument is used to palpate the disk space(s) of interest, confirmed with an intraoperative X-ray.
- Injury to the cisterna chyli or thoracic duct can lead to a lymphocele or chylothorax. In circumstances where injury is suspected, giving cream through a nasogastric (NG) tube and waiting 30 minutes can help identify a lymphatic chain injury, for the chyle will become milky in color and more identifiable. If the lymphatic chain is injured, it should be ligated using a nonabsorbable suture.
- Once the goals of the surgical procedure are complete, a chest tube can be placed if the pleura has been violated and there is potential for viscera pleural injury. The ribs, latissimus dorsi, and serratus anterior muscles are reapproximated with interrupted reabsorbable or permanent suture.
- Midlumbar region (L2–L5):
 - The patient is placed supine, and a left paramedian incision is marked. The left side is preferred to avoid the more prominent right common iliac vein.
 - An incision is made and the subcutaneous dissection is carried to the external oblique fascia and the anterior rectus sheath, which are incised to expose the left rectus abdominis. The rectus is retracted medially or laterally to expose the semilunar line.

- A space between the posterior rectus sheath and the peritoneum is created with blunt dissection caudal to the semilunar line and the rectus sheath is incised vertically.
- The peritoneum is bluntly dissected in a lateral to medial fashion off of the psoas muscle.
- The left ureter is retracted medially along with the peritoneum until the left iliac artery and vein are identified. Above the L4–L5 disk space, ligation of the segmental vessels will allow for medial mobilization of the aorta if needed. At L4–L5, the iliac vessels are retracted medially, and a table-mounted, self-retaining retractor is inserted to aid in the exposure (**Fig. 53.1**).
- At L4–L5, the iliolumbar vein may obscure the L5 vertebral body and limit the medial retraction of the iliac vein. At this level, it is advisable to find and ligate this vessel to improve exposure and prevent a traction injury resulting in a tear in the inferior vena cava (IVC), which can result in catastrophic hemorrhage.
- When considering instrumentation at L4–L5, a traditional lateral plate should be avoided at this level, for the contour of the L5 vertebral body when compared with the L4 level is dramatically different and will cause screws placed through the plate to preferentially have a trajectory toward the spinal canal. Posterior pedicle screw fixation or lateral screw/rod constructs are acceptable alternatives.

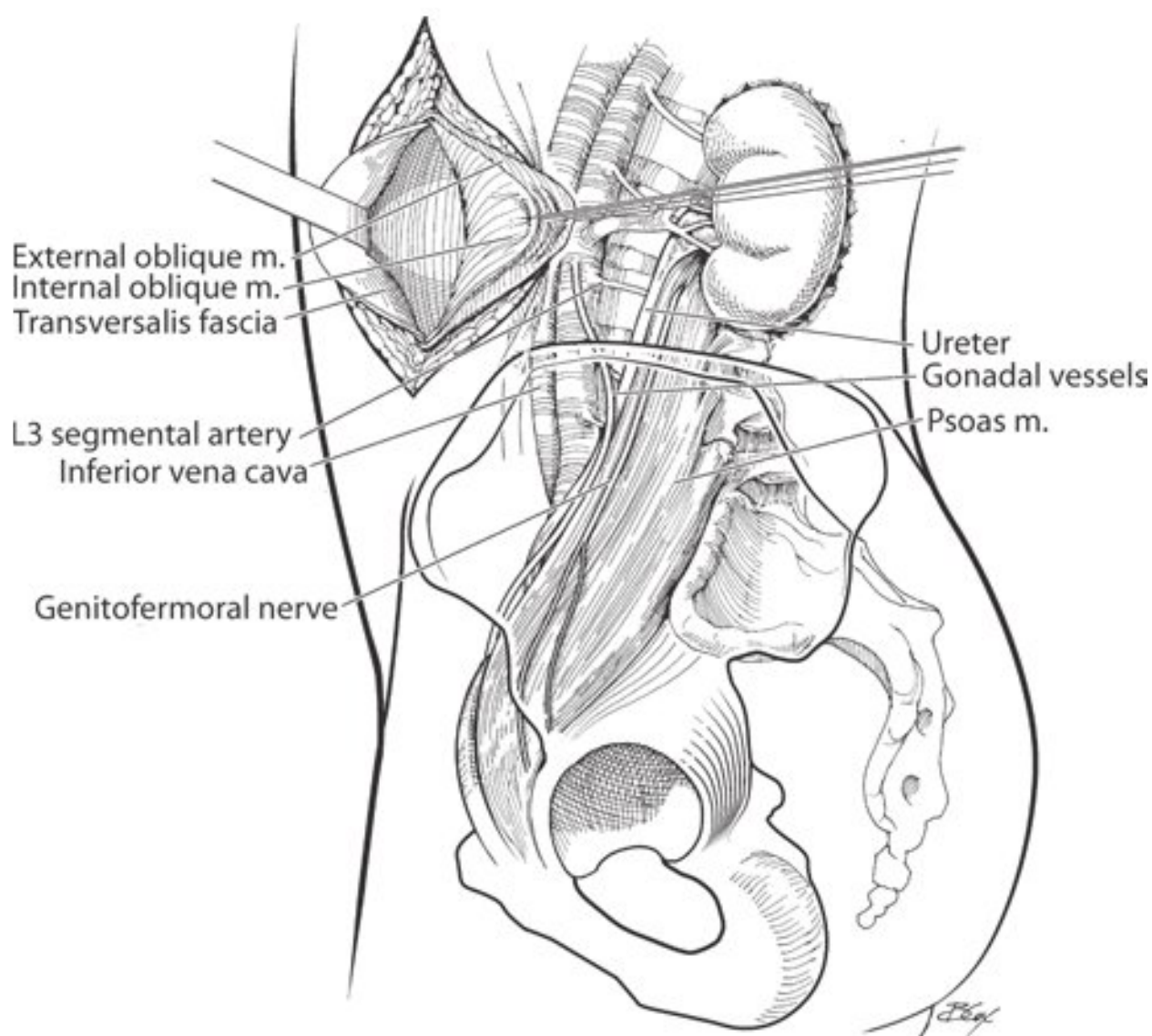


Fig. 53.1 Lateral view of the retroperitoneal approach to the lumbar spine.

- Lumbosacral junction (L5–S1):
 - The L5–S1 level can be approached from a midline exposure to avoid the iliac bifurcation. A paramedian incision is made and the peritoneal contents are retracted medially.
 - The sacral promontory is exposed between the right and left iliac vessels which are retracted laterally. The median sacral vessels are ligated and cut. This allows for safe retraction of the bifurcation of the inferior vena cava superiorly and the iliac vessels laterally, enhancing the exposure of the L5–S1 disk space.
 - Care must be taken to avoid excessive electrocautery of the anterior vertebral margin as injury to the sympathetic plexus may result in retrograde ejaculation in male patients.

53.4 Complications

- Pneumothorax, lymphocele, pseudomeningocele, injury to the lung, bowel, vasculature (aorta, inferior vena cava, iliac vessels), neural elements (nerve roots, genitofemoral nerve, lumbosacral plexus), and retroperitoneal hematomas.^{3,4}
- Pseudohernias may result from incising the abdominal wall musculature and/or disruption of the nervous innervation.
- Retrograde ejaculation may result from bilateral disruption of the sympathetic chain which is unlikely in a lateral exposure to the lumbar spine. This is more likely in midline exposures to L5–S1.

53.5 Postoperative Care

- On the first postoperative day, patients are encouraged to transfer from the bed into a chair and to ambulate as tolerated. Sequential compression devices are in place for all patients. Heparin prophylaxis for prevention of deep vein thrombosis is initiated the day after surgery.
- If surgery involved prolonged retraction of the iliac vein, then a postoperative duplex ultrasound study of the lower extremities is useful to rule out the presence of a deep vein thrombosis.
- Postoperative bowel rest is typically unnecessary after lateral retroperitoneal access. However, patients must be assessed for signs of peritoneal irritation, bowel injury being a potential, albeit infrequent, complication.
- If a chest tube is in place, it is connected to a reservoir on waterseal or -20 cm of H₂O depending on the severity of the injury. Oxygen saturation monitoring and chest X-rays are helpful in cases involving pleural injury.

53.6 Outcomes

- Few high-quality studies compare lateral and anterolateral approaches to the thoracolumbar spine with other techniques. Potential advantages of the lateral approach over a posterior based approach may include less blood loss, less retraction of nerve roots for anterior access, preservation of the posterior tension band, more thorough disk removal, and placement of larger interbody grafts.
- The anterolateral, retroperitoneal approach to the midlumbar region and lumbosacral junction avoids much of the intraperitoneal organ injury associated with an anterior transperitoneal approach, while providing similar spinal exposure. Also, the likelihood of injury to the large vessels and the sympathetic plexus is potentially reduced.

53.7 Surgical Pearls

- A true lateral decubitus position should be confirmed with intraoperative X-ray before incision, for this will aid in visualization of the lateral vertebral column.
- If the pleura is compromised, a chest tube should be placed for postoperative management.
- It is critical to leave a cuff of diaphragm during the exposure for later reapproximation.
- At L4–L5, the iliolumbar vein is ligated and the iliac vessels are retracted medially. At L5–S1, the median sacral vein is ligated and the bilateral iliac vessels are retracted laterally.

Common Clinical Questions

1. Retroperitoneal approaches are preferentially done on the left side of the patient for all the following reasons *except* which?
 - A. It is generally safer to mobilize the aorta than the inferior vena cava.
 - B. The common iliac vein may be more prominent on the right side.
 - C. The presence of the liver may minimize exposure.
 - D. The left ureter is less susceptible to injury than the right ureter is.
2. Complications of retroperitoneal exposure include all of the following *except* which?
 - A. Retroperitoneal fibrosis
 - B. Appendicitis
 - C. Ureteral injury
 - D. Femoral nerve palsy

3. To obtain adequate exposure, which is an essential component of the thoracoabdominal approach?
 - A. Incising the diaphragm along the costal margin and leaving a cuff attached at the level of the spine
 - B. Transecting the psoas to allow full retraction of the muscle
 - C. Dissection of the lumbar plexus from the overlying psoas muscle
 - D. Avoiding sacrifice of segmental arteries

Answers to Common Clinical Questions

1. D.
 2. B.
 3. A.
-

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54 Open and Minimally Invasive Spinal Lumbar Microdisectomy

Ali A. Baaj and Mark S. Greenberg

54.1 Key Points

- Conservative therapy is the initial and primary treatment modality for radiculopathy caused by a herniated nucleus pulposus (HNP) in the lumbar spine.
- Lumbar microdisectomy is a safe and effective procedure when the indication is appropriate.
- Cauda equina and conus syndromes caused by HNP in the lumbar spine constitute neurosurgical emergencies and require immediate evaluation and treatment.

54.2 Indications

- Symptomatic HNP
 - Radiculopathy (after at least 6–8 wk of failed conservative therapy)
 - Cauda equina syndrome
 - Conus syndrome
- Not for isolated axial low back pain
- No instability on preoperative dynamic plain radiographs

54.3 Technique

54.3.1 Open

- The patient is placed in the prone position.
 - Use of a Wilson frame may widen the interlaminar space and assist in access.
- Fluoroscopy is used for localization.
- A 2- to 3-cm midline incision is made with a number 10 scalpel.
- A subperiosteal dissection of tissue from spinous process and lamina on the ipsilateral side is performed.
 - Supraspinous and interspinous ligaments should be preserved.
- The medial facet joint is the lateral limit of the dissection.
- A retractor (e.g., Williams or Taylor) is placed.

- The inferior lamina of the superior level and superior lamina of the inferior level are identified and confirmed with fluoroscopy.
- Using a high-speed drill, a laminotomy (usually) is performed by drilling the inferior part of the superior level.
 - For inferior fragment migration, some of the superior lamina from the inferior level may need removing.
 - A matchstick or diamond drill bit may be used.
- Ligamentous flavum is removed using a curette and Kerrison punch.
- The operative microscope is brought in.
- The nerve sleeve and dura are gently retracted medially (**Fig. 54.1**).
- The posterior longitudinal ligament (PLL) and annulus fibrosus are incised with a number 11 blade, medial to lateral (always incise away from dura).
- Disk material is removed with a pituitary rongeur.
 - The surgeon must have an understanding of where the anterior longitudinal ligament (ALL) is to prevent inadvertent penetration into the retroperitoneal vessels with the pituitary rongeur.
- The disk space is irrigated and loose disk tissue removed.
- An instrument (e.g., Woodson or dental dissector) is passed beneath the dura to ensure that there is no residual fragment.

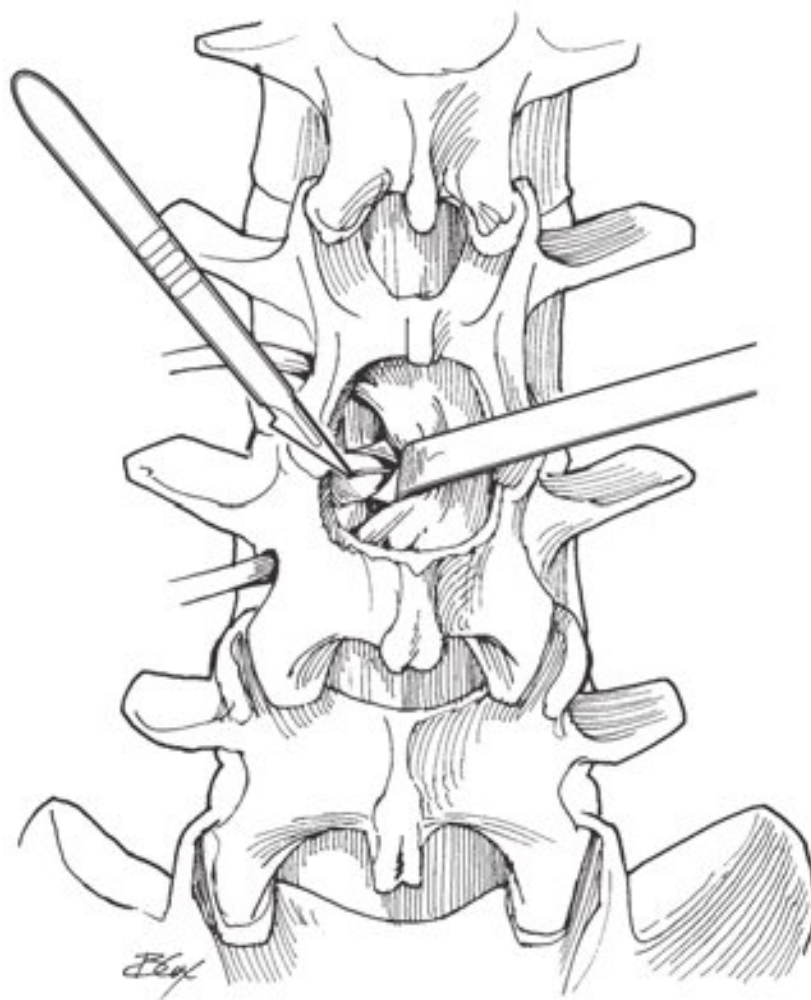


Fig. 54.1 Dorsal view of laminotomy defect showing thecal sac, traversing nerve root (retracted medially), and herniated disk.

- A nerve hook is used to probe under the PLL superior and inferior to the disk space to check for residual.
- The fascia is closed with 0 or 2–0 absorbable suture.
- The subcutaneous layer is closed with inverted, interrupted 3–0 absorbable suture.
- The skin is closed with running suture, staples, or skin adhesive.

54.3.2 Minimally Invasive Technique

- After correct positioning and localization, a 12- to 16-mm incision is made 10 mm lateral to the midline on the appropriate side.
- The thoracodorsal fascia is incised with blade or cautery, and serial dilators are positioned with the use of fluoroscopy.
- The final dilator and working tube should be docked over the interlaminar space with visualization of the inferior lamina of the superior level.
- It is important not to be aggressive with the initial dilators to not penetrate the interlaminar space.
- The operative microscope is brought in.
- Soft tissue is dissected and drilling is performed until ligamentum flavum is reached.
 - If drilling is done too laterally, the pedicle or facet joint is at risk; confirm location with fluoroscopy if needed.
- The ligament is removed and discectomy is performed as in the open approach.
- If a wider decompression is needed, the working tube may be repositioned to allow access to more lamina, and even the contralateral foramen.

54.4 Complications

- Cerebrospinal fluid (CSF) leak (5–8%)¹
 - Attempt primary repair using nonabsorbable suture (typically challenging).
 - Alternatively, cover with Gelfoam (Pharmacia & Upjohn, Kalamazoo, Michigan) and/or fibrin glue.
- Nerve root injury (1%)
- Wound infection (1%)
- Vascular injury (< 1%)
 - Suspected injury to the iliac vessels anteriorly (e.g., bright red blood from disk space or sudden, unexplained drop in blood pressure) requires immediate wound closure, repositioning of patient to supine, and immediate vascular surgery consult.

54.5 Postoperative Care

- Mobilize early; no need for bracing.
- Discharge to home when patient meets discharge criteria (typically, ambulating, tolerating a diet, voiding, and receiving adequate pain control from oral medications—usually same day or on postop day 1).

54.6 Outcomes

- There is an 85% chance of a good (minimal symptoms) or excellent (asymptomatic) outcome.²
- First recurrence is usually treated with repeat microdisectomy. Further recurrence may necessitate fusion.
- Multiple randomized trials have been plagued by crossovers between cohorts, and the most definitive statement that can be made is that decisions for surgery versus conservative treatment based on symptoms, duration, and patient preference result in similar good outcomes in both treatment groups.³

54.7 Surgical Pearls

- Ensure that the HNP is the pain/symptom generator based on exam and imaging.
- Minimal disruption of medial facet (don't confuse the facet for the lamina during the initial approach).
- Overly aggressive drilling of facet may lead to instability.
- Ensure you are not in a nerve root axilla when performing disectomy.
- Preoperative dynamic films may be necessary to rule out instability.
- May need hemilaminectomy or complete laminectomy if treating large central disk (e.g., for cauda equina or conus syndrome).

Common Clinical Questions

1. What is the best way to avoid drilling into the facet joint or pedicle during minimally invasive spine disectomy?
2. When evaluating for recurrent lumbar herniated disk, what radiologic examination is most appropriate?
3. When evaluating a recurrent lumbar herniated disk, what radiologic exam is helpful in ruling out instability?

Answers to Common Clinical Questions

1. Obtain anteroposterior/lateral fluoroscopy images intraoperatively to confirm location.
 2. MRI with and without contrast.
 3. Flexion–extension X-rays.
-

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55 Lumbar Foraminotomy

Ali A. Baaj and Juan S. Uribe

55.1 Key Points

- Removing more than a third of the medial facet joint when performing a foraminotomy could lead to instability.
- A high-speed drill and small-caliber Kerrison rongeur should be used in combination to safely perform a foraminotomy.
- Foraminal stenosis and nerve root compression are the result of hypertrophy/degeneration of the superior articulating facet of the lower vertebra.

55.2 Indications

- Focal lateral recess and/or foraminal stenosis

55.3 Technique

- The patient is placed in the prone position.
 - A Wilson radiolucent frame is adequate if no fixation is planned.
- Fluoroscopy is used for localizing the level of the foramen.
- A paramedian 2- to 3-cm incision is made 1 cm lateral to the midline.
- The fascia is incised with a monopolar cautery.
- Serial dilators and a nonexpandable tube retractor are placed under fluoroscopic visualization.
 - The optimal arrangement is to dock the dilator/tube retractor on the inferior aspect of the lamina of the superior level. For example, for a L4–L5 foraminotomy, the tube is positioned on the inferior aspect of the L4 lamina.
 - This approach is similar to that of minimally invasive surgery (MIS) for microdiscectomy.
- The operative microscope is brought over the field.
 - This is optional, but the microscope provides for better illumination and visualization through the tubular retractor.
- The inferior aspect of the lamina is drilled until the underlying ligamentum flavum is visualized.
- With the use of nerve hooks, curettes, and Kerrison rongeurs, the ligamentum flavum is resected.

- The traversing nerve root is typically visualized at this point.
- The medial facet is undermined until the medial pedicle is palpated with a Woodson or ball-ended probe.
- With a number 2 Kerrison, the foramina above and below the pedicle are widened and hypertrophied ligament is resected until the shoulder of the exiting root is visualized.
- A ball-ended probe should pass easily in the foramina to confirm adequate decompression.
- The space is irrigated and the fascial, subcutaneous, and skin layers are closed in standard fashion.
 - Subfascial drains are not typically used.

55.4 Complications¹

- Cerebrospinal fluid (CSF) leak (< 5%)
 - Technically challenging to repair primarily through the MIS tube
- Nerve root injury (< 1%)
- Wound infection (< 1%)

55.5 Postoperative Care

- Mobilize early without brace.
- Discharge to home when patient meets discharge criteria.
 - Typically same day or postoperative day 1

55.6 Outcomes

- The presumed benefits of MIS foraminotomy include reduced blood loss, less tissue damage, and shorter hospital stays. However, no randomized trial has compared traditional open with MIS foraminotomy.

55.7 Surgical Pearls

- Leaving the ligamentum flavum intact until all the “bony work” is completed protects the dura during drilling and when the Kerrison punches are in use.
- Aggressive drilling of the facet to remove more than a third may lead to facet joint instability.
- Patients whose history and exam findings of radiculopathy correlate with the foraminal stenosis seen on imaging will likely have the best outcomes from this procedure.

Common Clinical Questions

1. What is the roof of the intervertebral foramen composed of?
 2. Which nerve root exits below the L4 pedicle?
 3. How do you ensure that you are not drilling pedicle?
-

Answers to Common Clinical Questions

1. The superior articulating facet of the inferior vertebra.
 2. The L4 nerve root.
 3. Take an anteroposterior fluoroscopy shot intraoperatively or go more medial and expose the ligament/dura medially.
-

Reference

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56 Posterolateral Endoscopic Discectomy

Christopher Yeung and Anthony T. Yeung

56.1 Key Points

- Ideally, the procedure is performed under local anesthetic with Fentanyl and Versed sedation, rather than general anesthesia, so that the aware state patient can alert the surgeon of any potential nerve irritation. This helps prevent iatrogenic nerve injury.
- The patient is positioned to obtain true anteroposterior (AP) and lateral views before needle placement; this will avoid radiographic parallax error and malpositioning of the needle, cannula, and endoscope.
- Initial needle trajectory and placement is essential, because it will ultimately determine the endoscopic field of view. Optimum needle position is determined based on the pathology or region being addressed.
- Fluoroscopy should be used to confirm location if there is any uncertainty about anatomy or location during endoscopy.

56.2 Indications

- All lumbar disk herniations except migrated/sequestered fragments inaccessible through the foramen. Ideal types are foraminal/extraforaminal herniations and recurrent herniations after prior posterior discectomy.
- Foraminal stenosis.
- Symptomatic focal annular tears.
- Diskitis.

56.3 Techniques

- The needle entry point typically starts ~ 10 to 13 cm from the midline and is positioned to allow entry into the disk parallel to the end plates to allow intradiskal positioning without damaging the end plates. The patient's body habitus will determine just how far lateral from the midline to start. After determining the lateral extent of the incision site, check a lateral X-ray and position the incision in line with the disk inclination.
- To optimize the approach trajectory to access the specific herniation type, the entry point can usually be adjusted a little more medial (for foraminal or extraforaminal herniations) or lateral (for central and some paracentral

herniations). Typically, start as lateral as possible to create a shallow entry trajectory ($\sim 15\text{--}20^\circ$ to the floor) to the lateral facet, then slide the needle ventral to the facet, using the facet as a lever arm. The needle will be at the medial wall of the pedicle in the posteroanterior (PA) projection and at the annulus on the lateral view. The trajectory targets the base of the typical paracentral and central herniations. Preoperative measurements can also be performed on the axial MRI images to estimate the most optimal and safe lateral starting point to access the offending pathology.

- A 6-inch-long 18-gauge needle is then advanced from the skin entry point while infiltrating the needle tract and annulus with 0.5 or 1% lidocaine. The needle is advanced to the annulus while monitoring it in the posteroanterior Ferguson view (angled in the disk inclination plane to provide a true PA view of the disk space). You should check a lateral view before advancing the needle tip medial to the medial border of the pedicle to confirm appropriate trajectory and avoid inadvertent dural puncture in the case of an overly shallow trajectory. Ideally, the needle tip will be in the posterior 25% of the disk on the lateral fluoroscopic view when at the midline on the PA Ferguson view to access typical paracentral disk herniations. Fluoroscopically the needle tip can even be located slightly posterior to the posterior border of the vertebral bodies if the herniation is large with extensive posterior migration. For foraminal disk herniations, the needle tip can be in the center of the disk, indicating a 45° trajectory into the disk since the surgeon would not need to access the posterior epidural space as much to remove the foraminal herniation.
- Perform confirmatory contrast diskography at this time. The following contrast mixture is used: 9 mL of Isovue 300 with 1 mL of indigo carmine dye. This combination of contrast ratio gives readily visible radio-opacity on the diskography images, as well as intraoperative light blue chromatization of pathologic nucleus and annular fissures, which help guide the targeted fragmentectomy.
- Insert a guide wire through the 18-gauge needle, remove the needle, and make a small transverse stab incision. Slide the bluntly tapered tissue dilating obturator over the guide wire until the tip of the obturator is firmly engaged on the annulus. Lightly anesthetize the annulus through the eccentric parallel channel in the obturator.
- Remove the guide wire and use a mallet to advance the obturator bluntly through the annulus. Confirm posterior intradiskal placement on the AP/Lat C-arm views.
- Now slide the beveled access cannula over the obturator and into the disk. Remove the obturator and insert the endoscope to get a view of the disk nucleus and annulus.
- The general-purpose access cannula has a bevel hypotenuse of 12 mm and an outside diameter of 7 mm. When the cannula is slightly retracted to the

midstraddle position in relationship to the annular wall, the wide-angle scope visualizes the epidural space, annular wall/posterior longitudinal ligament and the intradiskal space in the same field (**Fig. 56.1**).

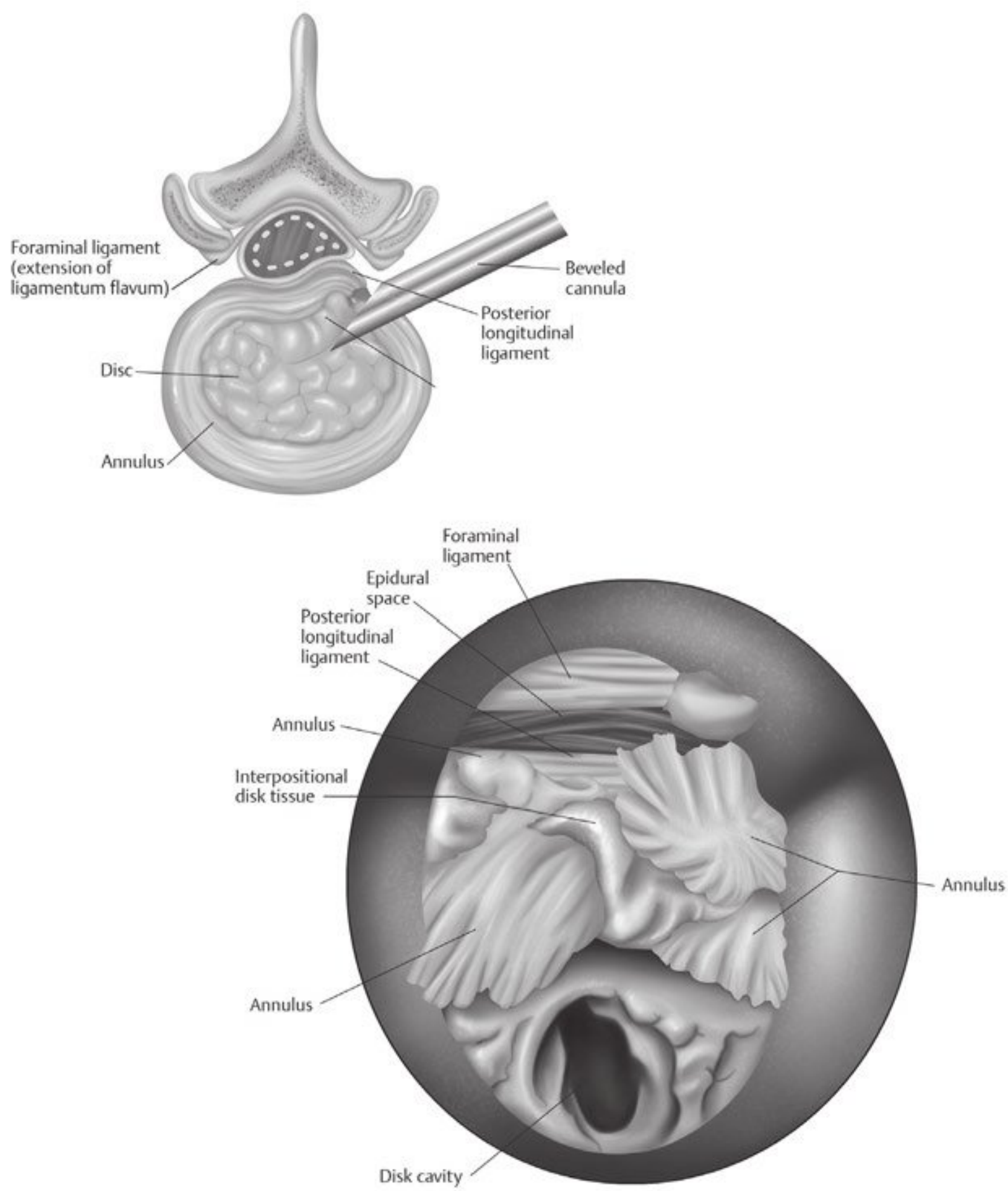


Fig. 56.1 Placement of the cannula in the midstraddle position at the annulus allows visualization of the epidural space, annular wall/posterior longitudinal ligament, and intradiskal space in the same operative field.

- The radiofrequency bipolar probe is used to coagulate any small bleeders to achieve hemostasis and clear visualization. The endoscopic rongeurs can be used immediately to begin removing the blue-stained nucleus pulposus under direct visualization. If there are annular fibers between the opening of the cannula and the herniation, then enlarge the annulotomy medially to release the base of the herniation with a cutting forcep. This connects the surgical annulotomy with the annulotomy from the herniations and allows the extruded fragment to be pulled into the intradiskal cavity and removed. The side-firing Holmium:YAG laser can also be used to enlarge and widen the annulotomy.
- The larger straight and hinged rongeurs are used directly through the cannula after the endoscope is removed if needed. Fluoroscopy and surgeon tactile feedback guides this step.
- The surgeon must be aware that sometimes there are accessory nerves of various diameters within the foramen branching off from the traversing or exiting nerve roots. These nerves are called furcal nerves. They can consist of motor or sensory nerve fibers. The sensory fibers are branches of the dorsal ramus originating ventral to the intertransverse ligament below the foraminal ligament before it exits to the facet joint. Thermal modulation or mechanical trauma to these nerves can produce motor weakness or dysesthetic pain even when the exiting and traversing nerve roots are totally intact and uninjured. Treatment with transformational and sympathetic blocks will provide relief of dysesthesia that is usually temporary.
- After removing the readily visualized herniated nucleus pulposus, one can perform a small debulking decompression by using a straight and flexible suction–irrigation shaver. The debulking process allows better intradiskal visualization, removes the unstable nucleus material to prevent future reherniation, and facilitates the teasing of any residual extruded fragments into the intradiskal working cavity with the endoscopic rongeurs and flexible radiofrequency trigger flex bipolar probe. The flexible radiofrequency bipolar probe is also used to contract and thicken the annular collagen at the herniation site and to maintain hemostasis throughout the case.
- Complete herniation removal is verified by visualizing the decompressed traversing nerve root (**Fig. 56.2**). One should note, however, that if the herniation is a contained paracentral herniation, you would not expect to see the decompressed traversing nerve. You would see the undersurface of the thinned out annulus from within the disk.

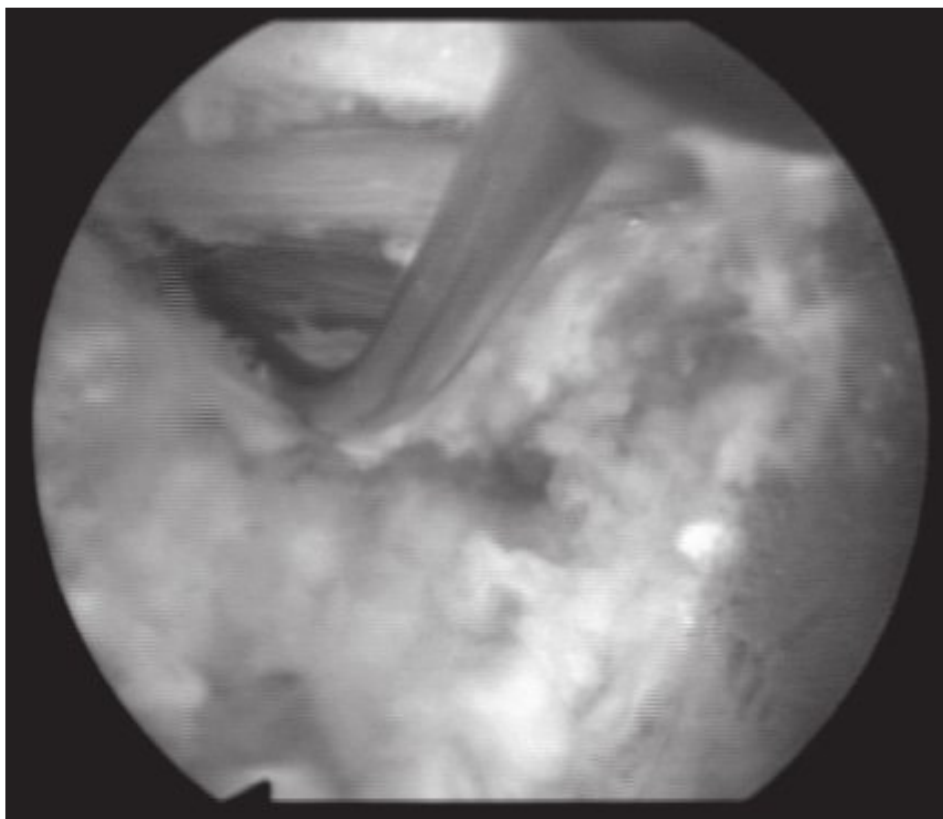


Fig. 56.2 After removal of an extruded herniated nucleus pulposus, complete decompression is verified by visualizing the traversing nerve root from the intradiskal space. Here a probe is pushing the attenuated posterior annular fibers ventrally to get a better view of the traversing nerve root.

- Before removing the endoscope, we also routinely visualize and confirm the exiting nerve root or the perineural fat encasing the nerve root is uninjured and inject 1 mL of 80 mg Depo Medrol in the foramen to help decrease inflammation and reduce the incidence of dysesthesia.
- Treatment of annular tears requires accurate targeting of the tear, removal of any interposed nucleus, and radiofrequency (RF) ablation of the granulation tissue within the tear. Much less disk removal is performed than with treatment for disk herniations.
- Endoscopic high speed burs, lasers, and Kerrison rongeurs can remove bone to decompress foraminal stenosis and allow a more horizontal trajectory into the disk.
- For diskitis, a great tissue sample is obtained for accurate culture, and an aggressive subtotal discectomy with extensive irrigation is performed to remove all the necrotic infected tissue.

56.4 Complications

- Infection, nerve injury, dysesthesia, dural tear, bleeding, scar formation, bowel injury, epidural hematoma

56.5 Postoperative Care

- Postoperative care is the same as a standard posterior microdiscectomy. Avoid excessive bending, twisting, and lifting for the first 4 weeks. Gradually increase physical activity as tolerated after 4 weeks. Neutral spine trunk stabilization exercises can be started 2 to 3 weeks postop if desired.

56.6 Outcomes

- Peer-reviewed literature has consistently shown similar results to those produced by posterior discectomy. There have been three prospective randomized studies comparing posterolateral endoscopic lumbar discectomy to open posterior discectomy and are listed in **Table 56.1**.^{1,2,3} Because of the less invasive nature of the surgical approach, studies have noted less postoperative narcotic usage and quicker return to work.¹ Additional studies focused narrowly on foraminal/far lateral herniations have shown success rates ranging from 85 to 92% figures that compare favorably to literature describing open paramedian discectomy techniques with success rates of 71 to 88%.^{2,4} Treatment of extruded herniations has shown success rates of 88 to 91%. Posterolateral endoscopic lumbar discectomy avoids dissecting through scar tissue when treating recurrent herniations and have shown similar success rates of 82 to 86%.⁵ Complication rates have also been similar to that of open discectomy.

56.7 Surgical Pearls

- Use the “inside-out” technique: The disk is the safest and best starting point, so start the endoscopy intradiskally and then gradually work your way outside and address the pathology accordingly.
- Sometimes (especially at L5/S1) partial lateral facetectomy may be required before entry into the disk. Dock the long or short beveled cannula on the facet, resect the superior articular process (SAP) undersurface from 3 o’clock to 12 o’clock (right-sided approach) or 12 o’clock to 9 o’clock (left-sided approach) until you can gain safe entry into the disk space. Protect the exiting nerve with the cannula, and then use the standard “inside-out” technique.
- Use the specially designed cannulas with a Penfield-like extension to protect the exiting nerve when working in the foramen.
- The patient is awake, so use this to your advantage. If significant leg pain is experienced, stop and reevaluate the patient, asking about the distribution of the pain and reassessing the position, using fluoroscopy to prevent complications.
- When bleeding is encountered, advance the scope back into the disk and slowly pull back the scope, cauterizing the bleeders from inside to out.

Common Clinical Questions

1. What common pathologic condition is not ideally treated with posterolateral endoscopic discectomy?
 - A. Foraminal herniation
 - B. Recurrent paracentral herniation
 - C. Central migrated and sequestered herniations beyond reach from the disk space
 - D. Diskitis
 2. What spinal condition is best treated by the transforaminal endoscopic decompression technique?
 - A. All disk herniations
 - B. Diskitis/spondylodiskitis
 - C. Central spinal stenosis
 - D. Chronic axial back pain
 3. There are endoscopic systems in the market using various disk removal techniques. Which technique is the most flexible and safest?
 - A. Outside-in
 - B. Inside-out
 - C. Targeted
-

Answers to Common Clinical Questions

1. C. Though very experienced surgeons can remove extruded, migrated disk herniations with different trajectories and foraminoplasties to reach an extruded disk herniation, any extruded or sequestered herniation difficult to reach is not ideal for the transforaminal approach and is best removed with the traditional translaminar approach.
 2. B. The literature consistently reports a high success rate with rapid resolution, because the transforaminal approach allows for good tissue excisional biopsy, better tissue sample for gram stain and culture, and more aggressive debridement, irrigation, and direct antibiotic treatment of the infection.
 3. B. The inside-out technique emphasizes visualization of the foramen and intradiskal pathoanatomy for surgical decompression, ablation, and irrigation
-

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57 Lumbar Laminectomy

Benjamin D. Elder and Timothy F. Witham

57.1 Key Points

- The goal of a lumbar laminectomy is to provide decompression of the central canal and the lateral recesses, and it may provide access for further decompression of the central canal through a discectomy and decompression of the neural foramina with foraminotomies.
- Care must be taken to avoid disrupting the bilateral facet joints, which can potentially lead to worsening instability that may require further surgical intervention.
- Additional care must be taken to avoid removal of the pars interarticularis, which may also result in iatrogenic instability.

57.2 Indications

- Focal or multilevel lumbar central and lateral recess stenosis resulting from lumbar degenerative disease, trauma, infectious, or neoplastic conditions.
- Laminectomy without arthrodesis is discouraged if there is instability commonly seen with degenerative spondylolisthesis at the index segment, in a patient with a significant scoliosis at the index level, or if significant facet joint resection is performed.

57.3 Technique

- The patient is placed in the prone position.
 - There are two schools of thought regarding operating room (OR) table selection. If no instrumentation is used, many surgeons prefer using a Wilson Frame, as the patient's lumbar spine is flexed, which opens up the posterior elements and may allow for easier access for the decompression. However, some feel that it may augment the surgeon's sense of adequate decompression intraoperatively owing to the flexed position.
 - Alternatively, some authors prefer using an orthopedic surgical table (e.g., Jackson table) for laminectomies, and this table is generally used if instrumentation will be performed following the decompression. This

table helps maintain the lumbar spine in its usual lordotic position.

Accordingly, if an adequate decompression is performed in this position, then it should be adequate once the patient is mobilized.

- A spinal needle or other radiopaque marker is placed at the desired level and is confirmed with fluoroscopy or a single standard intraoperative X-ray.
- A midline incision is made at the appropriate level, and a subperiosteal dissection is performed to detach the erector spinae muscles from the lamina.
 - It is important to remain subperiosteal during the dissection, for this significantly limits the bleeding that is encountered if the paraspinal muscles are violated.
- The dissection is carried laterally until the medial aspect of the facet joint, but every effort should be made to leave the facet joint capsule intact and avoid disrupting the joint. The bilateral pars are exposed to serve as a lateral marker for the laminectomy.
- A Horsley or Leksell rongeur is used to remove the spinous process(es).
- A high-speed drill with either a cutting or a diamond bur (depending on surgeon preference) or ultrasonic drill is used to thin or remove the lamina. Alternatively, some surgeons perform all of the bone removal with Kerrison and Leksell rongeurs.
 - The ligamentum flavum attaches to the superior edge of the lower lamina, and runs ventral to the inferior half to two-thirds of the upper lamina. Thus you are protected by the ligamentum flavum while drilling the lower half to two-thirds of the lamina.
- Once the drilling is complete, the ligamentum flavum can be released from the lamina, and the remainder of the laminectomy is completed with Kerrison rongeurs.
- Next, lateral recess stenosis is addressed. The dura is dissected from the lateral recess using a Woodson elevator or Penfield 4, and Kerrison rongeurs are used to remove bone (**Fig. 57.1** and **Fig. 57.2**).
 - The facets are often hypertrophied, and the medial third can be drilled or removed with Kerrison rongeurs.
 - The pars may need to be undercut to help decompress the lateral recess, but great care should be exercised, for excessive removal may lead to an iatrogenic pars defect and increase the risk of postoperative spondylolisthesis.
- A ball-tipped probe or Woodson elevator is used to inspect the foramina, as the nerve exits the spinal canal beneath the pedicle.
 - Kerrison rongeurs (generally #2 and #3) are used to complete foraminotomies, if necessary.
- Know your landmarks, and be careful to avoid drilling the facet joint, pedicle, or pars.

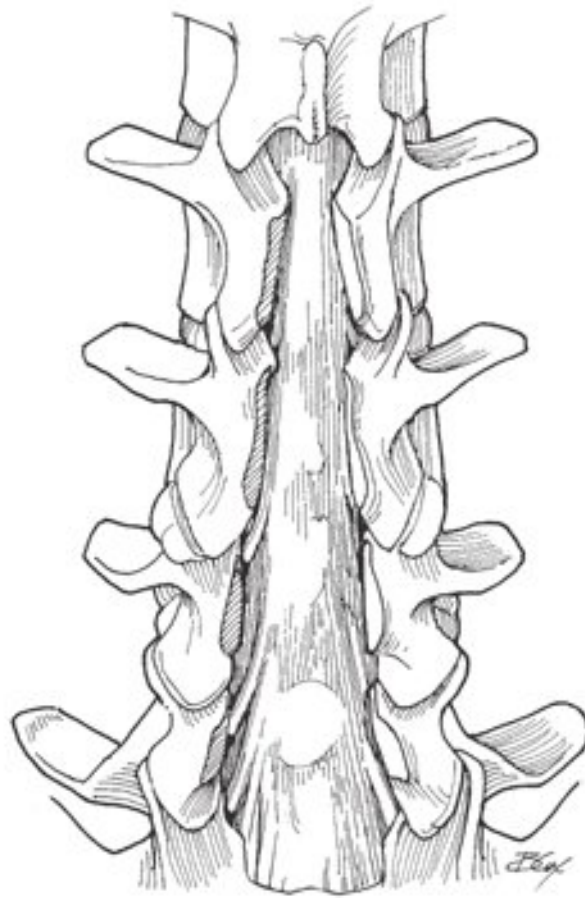


Fig. 57.1 Dorsal view of lumbar spine showing decompressed thecal sac and nerve roots with laminectomy taken to the medial facets laterally.

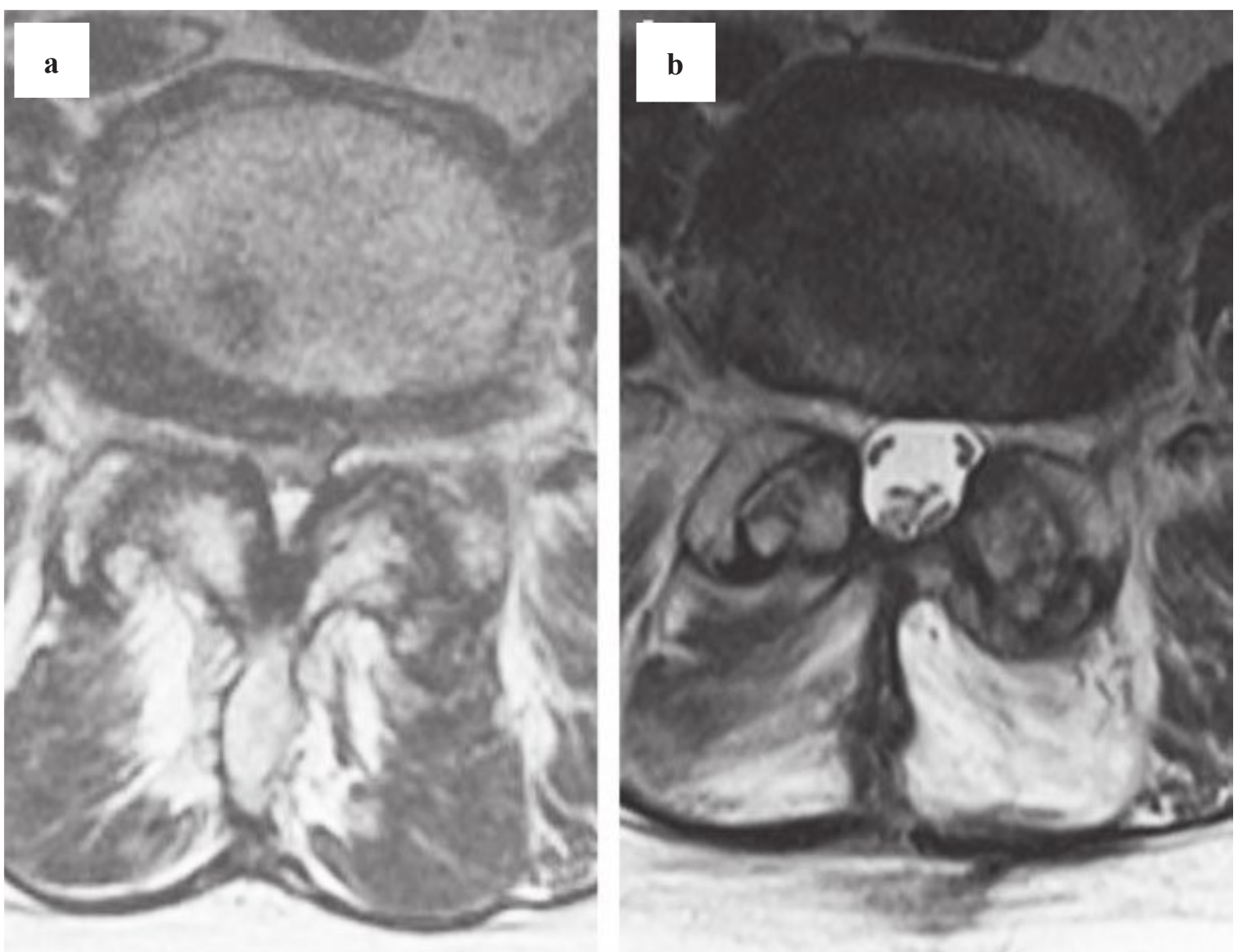


Fig. 57.2 (a) T2-weighted axial MRI demonstrating severe central and lateral recess stenosis from ligamentum flavum hypertrophy, disk protrusion, and facet joint hypertrophy. (b) T2-weighted axial MRI demonstrating thecal sac expansion following lumbar laminectomy.

- A subfascial drain can be used in the wound if warranted, and is often needed if multiple levels are performed, to aid with wound healing as well as help to prevent an epidural hematoma.
- Ensure a tight fascial closure and close the skin in the standard fashion.
- Some alternative procedures have been proposed including laminotomy, spinous process osteotomy, and several variations of minimally invasive approaches, although significant benefits of these alternatives have not been clearly demonstrated in the literature.

57.4 Complications

- Mortality and complications are more common in elderly and patients who have multiple pre-existing comorbidities.
- Durotomy/cerebrospinal fluid (CSF) leak (8–10%)^{1,2}
 - Leads to increased operative time, blood loss, and inpatient stay, but no difference clinically at 1-year follow-up
- Epidural hematoma requiring evacuation (0.18% rate for laminectomy and 0.09% rate for laminotomy in retrospective review of 15,000 patients).³
- Wound infection (1–2%)^{2,4}
- Spondylolisthesis or recurrent stenosis (occurred in 1–4% at 2 y).²
- Inpatient mortality rate (0.2%)⁴
- Neurological deficit (0.7%)⁴

57.5 Postoperative Care

- Operative time is ~2 hours, with an average of 300 mL of blood loss.²
- ~10% of patients required intraoperative transfusion, and ~5% of patients required postoperative transfusion.²
- Average 3-day hospital stay for standard laminectomy.²
- Pain control can generally be managed with oral analgesics alone, depending on preoperative narcotic usage. Use stool softeners with narcotics to prevent a postoperative ileus.
- Antibiotics are typically continued for 24 hours perioperatively.
- The patient can be mobilized on postoperative day 0 and does not require a brace.
- If a durotomy is encountered during the surgery, the patient is usually kept flat for 24 hours to minimize pressure on the lumbar dura.
- Drain removal is dependent on the individual surgeon but often ranges between <30 mL output over 24 hours to <50 mL over an 8-hour shift.
- Discharge home when patient meets discharge criteria.

57.6 Outcomes

- In a recent randomized trial, around 60 to 70% reported improvement in condition from 3 months to 2 years when treated with surgical decompression. Surgery led to greater improvement in Oswestry disability index, stenosis bothersomeness index, and leg pain bothersomeness index.² As reviewed previously,⁴ two other randomized trials showed significant improvement in leg and back pain at 1 year compared with nonoperative treatment, though this difference decreased by 2 years.
- Recurrent/adjacent stenosis or progressive spondylolisthesis reported in 1 to 4% of patients at 1 to 2 years.²
- Several cohort studies demonstrated > 80% improvement in symptoms after decompressive surgery, but at least a third of patients reported recurrent back pain at 7 to 10 years, and 10 to 23% required reoperation by this time.⁴
- Patients with most severe radiculopathy symptoms were most likely to improve.⁴

57.7 Surgical Pearls

- With advanced degenerative disease, obtain preoperative dynamic films to rule out a dynamic spondylolisthesis and spinal instability at the index segment.
- Care should be taken to dissect the dura from the surrounding bone and ligamentum flavum with a curette, Woodson elevator, or Penfield 4 before removing the bone with a rongeur. Never take blind bites with a rongeur without visualizing that the dura is completely dissected.
- Apply bone wax or hemostatic matrix to the bone edges to stop bleeding. Hemostatic matrix or Gelfoam (Pharmacia and Upjohn Company, Kalamazoo, Michigan) can be used to coagulate epidural venous bleeding. If minor bleeding remains or a multilevel laminectomy is performed, consider leaving a subfascial drain to mitigate the risk of postoperative hematoma requiring reoperation and to assist with wound healing.

Common Clinical Questions

1. Why are preoperative dynamic X-rays sometimes obtained before a lumbar laminectomy?
2. During lumbar laminectomy, what location is associated with the highest likelihood of a CSF leak?
3. When should an arthrodesis be considered in combination with lumbar laminectomy?

Answers to Common Clinical Questions

1. To rule out spondylolisthesis or instability, which would indicate that a fusion procedure should be performed in conjunction with the decompression.
 2. The superior aspect of the lamina, as the ligamentum flavum ends at this location, so the dura is not protected. Additionally, this can occur in the lateral recess if there is significant stenosis. Accordingly, care should be taken to dissect the dura from the surrounding bone and ligamentum flavum with a curette, Woodson elevator, or Penfield 4 before removing the bone with a rongeur.
 3. Laminectomy without arthrodesis is discouraged if there is instability or degenerative spondylolisthesis with instability at the index segment, in a patient with a significant scoliosis at the index level, and if a significant amount of the facet joint is removed. Additionally, arthrodesis should be considered if an iatrogenic pars defect is caused by extending the laminectomy too far laterally.
-

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58 Posterior and Transforaminal Lumbar Interbody Fusion

Zachary J. Tempel, Hazem A. Mashaly, and Adam S. Kanter

58.1 Key Points

- Arthrodesis across one or more vertebral disk spaces is an effective surgical treatment for multiple spine disorders that result in instability, severe pain, and/or neural element compression.
- Supplementing posterior instrumentation with an intervertebral graft enhances arthrodesis rates by providing support to and promoting arthrodesis across the anterior and middle columns.¹
- The posterior lumbar interbody fusion (PLIF) procedure involves bilateral interbody graft placement without significant disruption of the medial facet.
- The transforaminal lumbar interbody fusion (TLIF) procedure requires a medial facetectomy (unilateral or bilateral), which permits wider access to the disk space, less retraction of the neural elements, and placement of larger grafts (unilateral or bilateral).²

58.2 Indications

- Spondylolisthesis grade I or II
- Revision surgery for pseudoarthrosis or adjacent segment disease
- Degenerative disk disease causing diskogenic low back pain
- Recurrent disk herniation with mechanical back pain, or recurrent radiculopathy
- Lumbar deformity with coronal or sagittal plane imbalance
- Neural foraminal stenosis from disk space collapse

58.3 Technique

58.3.1 Transforaminal Lumbar Interbody Fusion (Example: L4–L5)

- Exposure of the disk space
 - Expose the vertebral spinous process and lamina of the levels above and below the operative disk space using lateral fluoroscopy to confirm the anatomic level. After confirmation, perform the dissection laterally to

expose the transverse processes. During the dissection, it is important to avoid violating the facet capsule of the superior adjacent level. In cases with dorsal compression, a partial or total laminectomy of the rostral level (L4) is performed.

- A thorough medial facetectomy (inferior articular process of L4) is performed on the symptomatic side over the L4–L5 interspace. If symptoms are bilateral, then bilateral medial facetectomies may be performed. A high-speed drill or osteotome is used to completely transect the pars interarticularis directly over the exiting nerve root (L4) at the caudal aspect of the rostral (L4) pedicle. If a laminectomy was not performed, a sagittal cut is made in the lamina before transecting the pars. A blunt instrument such as a Woodson elevator may be used to identify the neural foramen and protect the exiting nerve root.
- The joint capsule and ligamentum flavum can be detached with monopolar electrocautery or with an upgoing curette. A Leksell rongeur is used to remove the medial facet, which is morselized for fusion autograft. The exiting nerve root (L4) is identified as it courses caudally to the pedicle of the rostral level (L4) and followed laterally using a Kerrison rongeur to decompress the neural foramen. Next, the superior articular process (or lateral facet) of the caudal level (L5) is removed until the superior aspect of the caudal (L5) pedicle is encountered, thus achieving a pedicle-to-pedicle exposure.
- The shoulder of the traversing nerve root is visualized exiting the thecal sac at the level of the disk space. The medial and superior borders of the caudal pedicle limit the amount of the superior facet that can be removed. A high-speed drill can be used to shave bony osteophytes to optimize the trajectory into the disk space and minimize nerve root retraction during interbody graft placement. Inadequate bony exposure may lead to placement of a graft inadequately sized for the disk space.
- Diskectomy
 - The diskectomy is performed by incising the annulus with a scalpel and using curettes and pituitary rongeurs to remove the nucleus pulposus and cartilaginous end plates, thus exposing the bony surface of the vertebral end plates. Preservation of the bony end plates will serve to prevent graft subsidence into the cancellous portion of the vertebral body.
 - To maximize the arthrodesis benefits of interbody grafting, it is important to perform a total diskectomy. The disk space is then packed with morselized autograft.
- Interbody grafting
 - An interbody graft of the appropriate size is packed with autograft or other fusion substrate and carefully impacted into the disk space. A

nerve retractor is placed against the traversing nerve root shoulder to protect it during graft placement. Depending on the anatomy, a second retractor may be used to protect the exiting root as well.

- The interbody graft should not protrude beyond the dorsal aspect of the adjacent vertebrae. Fluoroscopy is then used to evaluate the final graft position and depth.
- If the disk space is significantly collapsed, the pedicle screws may be placed before the interbody graft, and a distraction device is then anchored to the screw heads to open the space, facilitating graft placement.
- Pedicle screw placement
 - The pedicle screw entry point is indicated by the mammillary process: The junction of the transverse process, superior facet, and pars interarticularis. The lateral to medial trajectory varies generally from 5 to 20° from L1 to L5 (i.e., 5° at L1 and increasing at 5° increments per caudal level).
 - The rostral to caudal trajectory should parallel the pedicle as directed by palpation with a Woodson elevator or similar instrument if a decompression has been performed—or, alternatively, using intraoperative fluoroscopy or image guidance.
 - Before rod placement, a high-speed drill is used to decorticate the transverse processes and remaining bony structures at the index level. Autograft or allograft combined with an osteoinductive agent is then packed into gutters bilaterally to promote arthrodesis. Care is taken to ensure that no loose bone chips fall into the laminectomy or facetectomy defects.
 - After the pedicle screws have been placed, a lordotic rod is inserted and the cap screws are tightened with compression across the screw heads. This restores lordosis and subjects the graft to axial loading forces that prevent retropulsion of the graft, maintain disk space height and promote arthrodesis across the end plates.
- Closure
 - A multilayer closure is performed using absorbable sutures in the lumbodorsal fascia and subdermis. Staples or nylon suture is used to close the skin. A surgical drain, such as a Hemovac (Zimmer Biomet, Warsaw, Indiana), may be left in place.

58.3.2 Posterior Lumbar Interbody Fusion

- The PLIF procedure is similar to TLIF with respect to exposure, discectomy, and closure.
- The key difference is that the approach to the disk space is more medial and a portion of the facet remains, necessitating greater neural element retraction to access the disk space (**Fig. 58.1**).

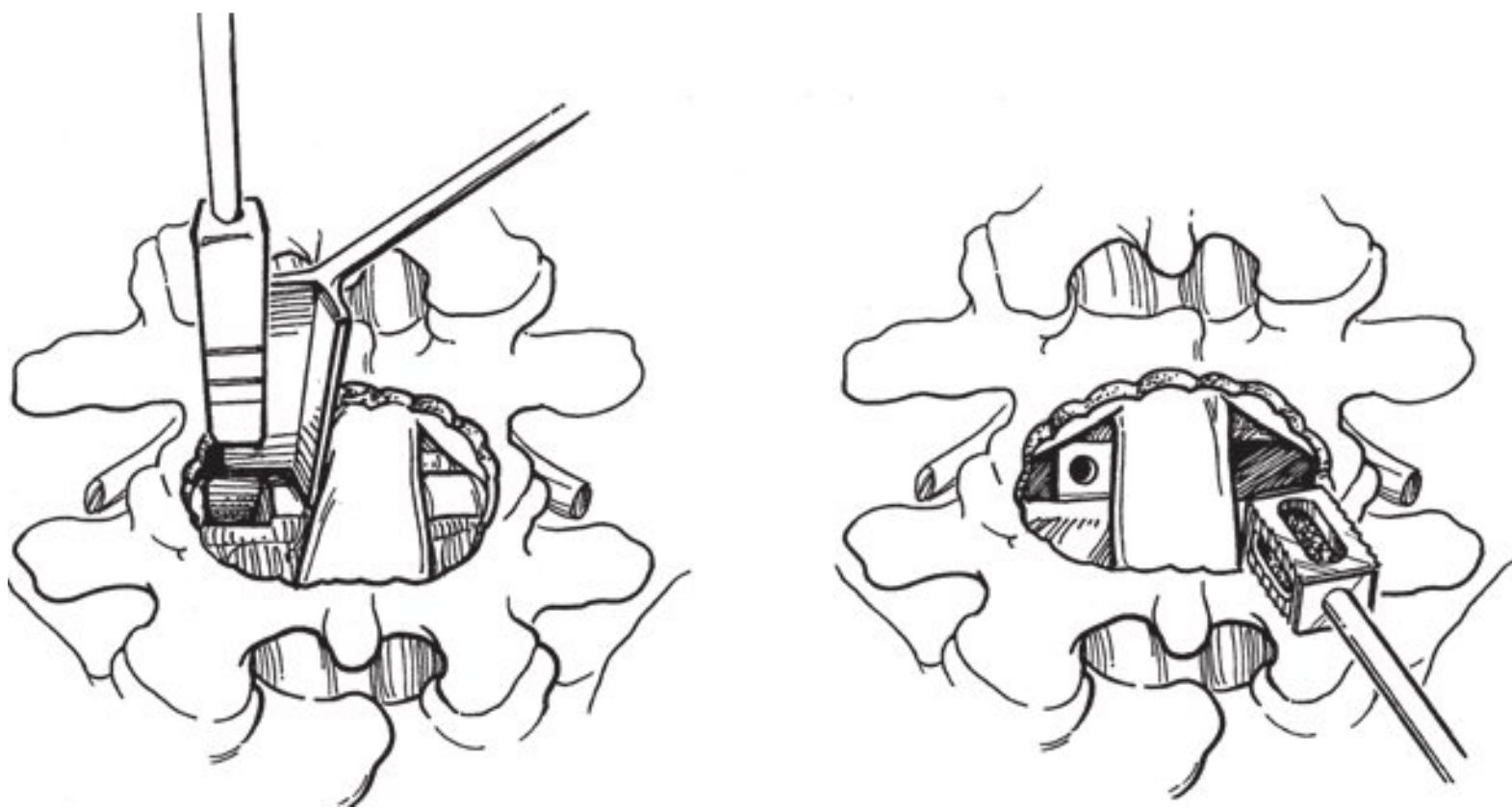


Fig. 58.1 An implant filled with cancellous bone graft is inserted into the rectangular channel with gentle medial retraction of the theca sac during a PLIF procedure. Note the teeth of the cage that engage the vertebral bone. (From Fessler RG and Sekhar L, eds. *Atlas of Neurosurgical Techniques: Spine and Peripheral Nerves*, Thieme; 2006.)

- Laminectomy is routinely performed using a high-speed drill and rongeurs, after which the ligamentum flavum is resected to expose the thecal sac. Bilateral limited foraminotomies are performed with a Kerrison rongeur to ensure adequate nerve root decompression, but the distal aspects of the roots themselves are not typically exposed as in the TLIF procedure.
- After visualization has been accomplished, a dural retractor is used to medialize the neural elements and expose the underlying disk space.
- The discectomy, graft placement, and pedicle screw construct techniques are similar to those for the TLIF technique. Careful attention must be paid during intervertebral work to limit medial retraction of the neural elements.

58.4 Complications

- The incidence of durotomy has been reported to be as high as 17% and most commonly occurs during exposure of the thecal sac and exiting nerve root.^{3,4}
- The incidence of nerve root injury secondary to retraction has been reported as high as 13% with resultant radiculopathy, typically transient.
- TLIF may be associated with less blood loss, shorter operative times and a lower incidence of durotomy and nerve root injury compared with PLIF.^{3,4}
- Vascular injury during the discectomy, typically from breach of the anterior longitudinal ligament, requires packing of the disk space, emergency closure, and vascular consult for laparotomy and/or interventional repair of the injured vessel.

58.5 Postoperative Care

- Mobilize early; consider deep vein thrombosis (DVT) prophylaxis if patient is not ambulating on postop day 1.
- Pain management with patient-controlled administration of narcotics augmented with muscle relaxants, and transition early to oral medications.
- Discontinue wound drain based on output (e.g., less than 100 mL in 24 h).
- A short course of steroids may be beneficial in patients with retraction-related radiculopathy.

58.6 Outcomes

- Approximately 75 to 90% of patients experience significant improvement in low back pain and/or radicular leg pain.³
- Studies have consistently demonstrated an arthrodesis rate of > 90%⁴

58.7 Surgical Pearls

- The PLIF procedure may be more suited to L5–S1 disease because the large dimension of the canal at this level reduces the degree of neural element retraction.
- The conus medullaris limits the use of the PLIF procedure to the upper lumbar spine (i.e., caudal to L2).
- Nerve root retraction can be reduced or eliminated with the more lateral to medial trajectory of the TLIF procedure.
- Using a blunt instrument to identify the neural foramen while drilling the pars is helpful in preventing accidental injury to the rostral pedicle and exiting nerve root.
- Pedicle screw distraction across the disk space must be performed cautiously in patients with poor bone quality to avoid fracturing the pedicle.¹

Common Clinical Questions

1. A patient with known tethered cord would be better suited for what lumbar interbody fusion procedure?
2. During the TLIF operation, complete transection of what structure is required to remove the medial facet?
3. For a TLIF at L3–L4, a laminectomy is performed at what level?

Answers to Common Clinical Questions

1. TLIF. The degree of neural element retraction is less with the TLIF procedure, which reduces the risk of traction on the conus medullaris.
 2. After the lamina is removed, the pars interarticularis must be transected to remove the medial facet and achieve a pedicle-to-pedicle exposure.
 3. For a TLIF at L3–L4, the L3 lamina is removed and care must be taken to preserve the L2–L3 facet capsule when dissecting the L3 transverse process.
-

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59 Minimally Invasive Transforaminal Lumbar Interbody Fusion

Michael Y. Wang

59.1 Key Points

- Minimally invasive transformational lumbar interbody fusion (MIS TLIF) can be used to achieve lumbar spinal decompression, fixation, and fusion.
- The intent behind this approach, which is still unproven, is to reduce soft tissue injury, blood loss, recovery time, and length of hospital stay associated with open TLIF.
- It is important to recognize the anatomic landmarks at the interpedicular space, referred to as a Kambin triangle: Medial (traversing root), lateral (exiting root), and base (inferior pedicle).

59.2 Indications

Harms first described the transforaminal lumbar interbody fusion (TLIF) technique for circumferential fusion in 1982 using an interbody spacer and a supplemental pedicle screw construct.¹ This procedure was a departure from the traditional posterior lumbar interbody fusion (PLIF) in that the TLIF required only a unilateral facetectomy and a single interbody cage. TLIF then became popular because of its lower rates of nerve root injury given the reduced surgical manipulation of neural elements, and it was also found to be effective for revision surgery, because the midline dural sac did not have to be exposed.

A minimally invasive approach for accomplishing the TLIF procedure was subsequently developed.² This technique used the advantages of expandable working tubes and percutaneous pedicle screw technology to reduce the soft tissue injury that has been documented to result from open surgery.³ Although no class I evidence exists to demonstrate the superiority of MIS TLIF over open TLIF, comparative studies suggest a trend toward reduced postoperative pain, hospital length of stay, infection rates, and blood loss.⁴ MIS TLIF can generally be applied in all situations where an open TLIF can be used:

- Single- or two-level degenerative disk disease with correlative clinical symptoms
- Spondylolisthesis less than Meyerding grade III
- Recurrent lumbar disk herniation
- Spondylosis with radiculopathy and back pain

- Focal spinal deformity concentrated at less than three intersegmental levels
- Synovial cysts exhibiting spinal instability

59.3 Technique

There are several variations of the MIS TLIF technique. Here a standard approach is described that has been successfully used by numerous surgeons:

- The patient is positioned prone on a Jackson operating room table, which offers several advantages. The location of the incision is targeted under fluoroscopy, and two paramedian skin incisions are made between 3 and 4 cm lateral to the midline. Sharp incision of the fascia allows for blunt finger dissection between the longissimus and multifidus muscles along a modified Wiltse plane (**Fig. 59.1**).⁵ This not only minimizes trauma to muscular tissues, but also minimizes intraoperative bleeding and creates a natural plane that allows for efficient maintenance of the trajectory of the retractors. The approach, if unilateral, is taken from the side where decompression is most necessary.
- Tubular dilator retractors are then placed through this plane and the level confirmed with fluoroscopy. Monopolar cautery is then used to clear any muscle tissue over the facet joint, making sure to expose the lateral aspect of the joint.



Fig. 59.1 Axial MRI showing the modified Wiltse plane between the longissimus and multifidus muscles. High MRI signal can be seen along this plane between the muscles, giving direct access to the facet joints.

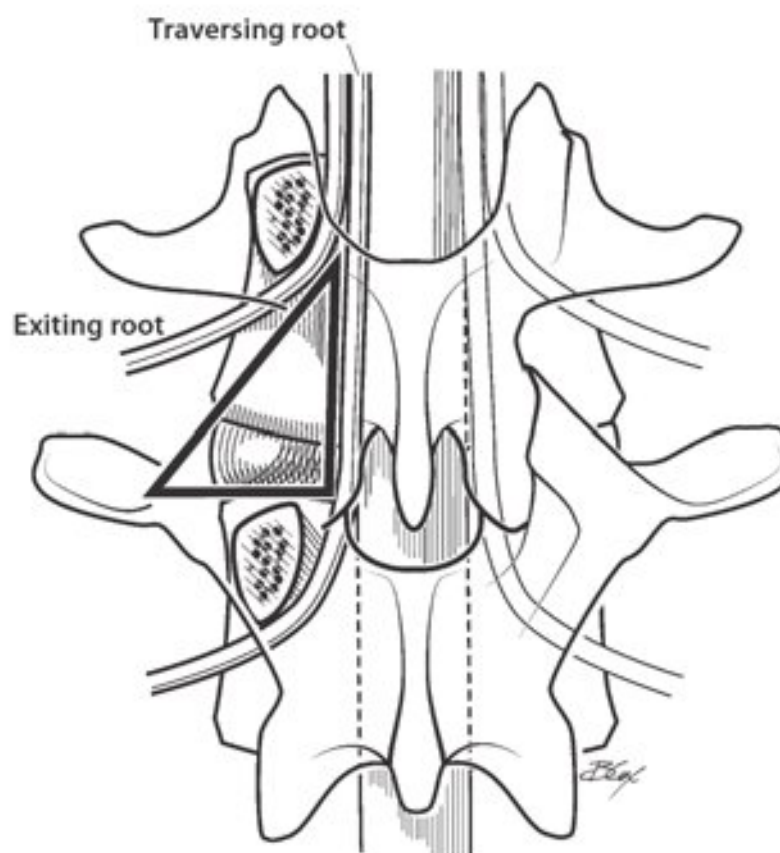


Fig. 59.2 A Kambin's triangle.

- Once the surgeon has confirmed the location of the pars and the lower pedicle, a quarter-inch osteotome or high-speed bur is used to remove the facet joint. Either a partial or complete facetectomy can be performed, with greater bone removal for cases with more medial compression. Care must be taken not to drill into the pedicle. After facet removal, the lateral aspect of the ligamentum flavum is visualized. This is resected until the lateral thecal sac is seen (**Fig. 59.2**). Autograft bone is saved for later grafting purposes.
- Further decompression can then be performed as needed, with removal of ipsilateral ligamentum flavum, decompression under the pars, or contralateral decompression. This is best achieved by angling the retractors more medially. Alternatively, bilateral decompression can be accomplished through dual ports placed on each side.
- After palpation of the caudal pedicle and confirmation of the location of the exiting and traversing nerve roots, the annulus is incised, followed by aggressive intervertebral disk removal and end plate preparation, as with the open technique. This is followed by intervertebral cage placement.
- The neural elements are reinspected. If a posterolateral fusion is desired, the dura is covered with Gelfoam (Pharmacia & Upjohn, Kalamazoo, Michigan), and the lateral facets and transverse processes are decorticated with a drill. This is followed by bone graft application.
- Supplemental pedicle screw placement can then be accomplished either using a mini-open technique (through the retractor with visualization) or percutaneously before or after the decompression. For the mini-open method, the screw placement is very similar to open TLIF. Various methods

are used for percutaneous targeting, including (1) anteroposterior only, (2) the owl's-eye en face approach, (3) biplanar fluoroscopy, and (4) image guidance. All percutaneous methods involve placement of a joshed needle into the pedicle, followed by the use of a cannulated awl and tap to create a screw entry tract. Finally, screw placement and rod tunneling are used to connect the construct components.

- A drain is not typically used.

59.4 Complications

- Cerebrospinal fluid (CSF) leakage is uncommon with these procedures because the dural exposure is limited to those areas that are to be decompressed directly. However, when it occurs, dural repair can be difficult. Some surgeons have advocated leaving the durotomy unrepaired and applying a collagen sponge locally, which suggests that the limited dead space reduces the risk of CSF leakage postoperatively. Use of a Castro-Viejo needle holder can help the surgeon sew through a tubular retractor.
- Incomplete neural decompression is more problematic than in open surgery given the small working corridor. Care must be taken to visualize critical landmarks to ensure that any compression (such as at a far lateral disk) is adequately addressed. The critical landmarks include the caudal pedicle, pars interarticularis, and lateral ligamentum flavum.
- Nerve root injury is best avoided by maintaining an appreciation of the medial–lateral location in which the surgeon is working. Careful identification and control of the exiting and traversing nerve roots prior to discectomy is critical.
- Cage misplacement occurs most frequently when inadequate disk material has been removed or the bony end plate has been violated. Care in end plate preparation is essential to minimize graft settling. In addition, it is desirable that the cage cross the midline, and this can be confirmed with fluoroscopy.
- Screw misplacement is most commonly the result of poor targeting technique. Thus close familiarity with one technique for ensuring that the pedicle is not breached is essential. The anteroposterior (AP) targeting techniques are excellent for preventing neural injuries.
- Pseudarthrosis can be minimized but never totally prevented. Proper end plate preparation, cage sizing, and the use of osteobiologic adjuvants will limit this complication. In addition, attention to postoperative nutrition, bracing, and external electrical bone stimulation are helpful.

59.5 Postoperative Care

Care after surgery is the same as with open TLIF. Patients should be rapidly mobilized with physical/occupational therapists, with external bracing as a treatment option. Muscle spasm occurs more commonly with MIS procedures and is better treated with muscle relaxants and benzodiazepines than with narcotics.

59.6 Outcomes

Proper patient selection is vital for achieving excellent outcomes. Although MIS TLIF has not been conclusively shown to be superior to open TLIF, several reports suggest this pattern. In a well-selected population, radiculopathy will show meaningful improvement in 80 to 95% of cases; axial back pain will improve in 70 to 85% of cases.

59.7 Surgical Pearls

- In cases in which substantial muscle creeps under the edges of the retractor, consider pharmacologic paralysis. Also consider removing the retractor and making a longer fascial opening, followed by placement of a deeper retractor.
- When performing a contralateral neural decompression, leave the ipsilateral ligamentum flavum intact at first. This will push the thecal sac ventrally, keeping it out of the way while the contralateral work is being performed.
- Intervertebral disk can be removed efficiently using an insert-and-rotate scraper. These devices allow entry into collapsed disk spaces and also effectively remove any disk herniations causing neural compression.
- Self-distracting cages, usually with a “bulleted” nose, will allow the placement of larger interbody cages, as it is difficult to use pedicle screw manipulation of a laminar spreader to open the disk space in MIS surgery.

Common Clinical Questions

1. MIS TLIF is used because it is believed to result in all of the following *except* which?
 - A. Decreased soft tissue damage
 - B. Higher fusion rates
 - C. Shorter hospital stays
 - D. Reduced blood loss

2. One of the critical maneuvers to reduce neural injury is which?
 - A. Remove more lateral facet bone so that less nerve retraction is needed.
 - B. Use intraoperative intravenous steroids.
 - C. Use stronger pedicle screw fixation.
 - D. Repair dural tears.
 3. When placing intervertebral cages in a MIS TLIF, it is important to do what?
 - A. Make sure the cage is higher than 13 mm.
 - B. Make sure the cage is longer than 27 mm.
 - C. Use only nondegradable interbody devices.
 - D. Cross the midline.
-

Answers to Common Clinical Questions

1. B.
 2. A.
 3. D.
-

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60 Percutaneous Pedicle Screw Placement

Michael Y. Wang

60.1 Key Points

- Numerous methods can be used for implanting percutaneous pedicle screws, such as image guidance, en face targeting, and biplanar fluoroscopy.
- Percutaneous screws can be safely placed in the thoracolumbar spine using a simple method based primarily on anteroposterior (AP) X-rays.
- Percutaneous screw–rod constructs are a truly minimally invasive surgical (MIS) technique for segmental fixation.

60.2 Indications

Percutaneous placement of transpedicular screws with connecting rods can find application in virtually any setting where open screw fixation might be utilized in the thoracolumbar spine. For a fusion setting, this can be employed in conjunction with interbody fusion for stabilization,¹ such as after an anterior lumbar interbody fusion (ALIF), lateral interbody fusion, transforaminal lumbar interbody fusion (TLIF), or transsacral fusion. In addition, a speculum retractor can be used to expose the lateral masses for posterolateral fusion bed preparation in conjunction with percutaneous fixation. Also, percutaneous fixation can be used for stabilization or “internal bracing” in the absence of a bony arthrodesis.² Thus, regardless of whether a fusion is intended, percutaneous screw–rod fixation can be used to confer spinal stability for degenerative, traumatic, neoplastic, and infectious pathologies.

60.3 Technique

There are various methods for targeting the pedicles of the lumbar or thoracic spine. These include (1) frame-based or frameless image-guided navigation,³ (2) biplanar fluoroscopy in the AP and lateral views, (3) “en face” or “direct down the pedicle” imaging,⁴ (4) a mini-open technique with tactile feedback, and (5) an AP-based targeting method.⁵ This chapter will describe the AP-based targeting method.

1. The patient is positioned prone on a radiolucent operating table. Care is taken to ensure that there are no obstructions to the fluoroscopic image at the level(s) of interest, the technique being heavily dependent on intraoperative imaging.
2. An absolutely precise AP view is registered at the level of screw placement. This is done by first ensuring that the axial rotation of the fluoroscope aligns the spinous process equidistant between the right and left pedicles (**Fig. 60.1**). Either the fluoroscope can be moved or the bed can be rotated to register this image, but the surgeon must compensate for any rotation depending on the method used. The sagittal (Ferguson) angle is then adjusted so that the anterior and posterior cortical rims of the upper end plate of the vertebral body are superimposed into a single line. This ensures that the X-ray beam is completely aligned with the sagittal plane of the pedicles (and thus offers an ideal screw trajectory).
3. The skin is then marked so that a cutaneous entry point can be made 1 to 2 cm lateral to the lateral border of the pedicle. A Jamshidi needle (MedSurge; Chennai, India) is then inserted through the skin and directed slightly medially to contact the bone surface. The needle is then docked at the junction of the transverse process and facet joint near the mammillary process (the attachment of the multifidus muscle).
4. Once the Jamshidi needle has been docked on this ideal pedicle screw starting point, the needle is marked 2 cm above the skin surface with a surgical marker (**Fig. 60.2**). The needle is then hammered into the pedicle to a depth of 2 cm (where the marking meets the skin surface). The trajectory will be medialized to match the patient's anatomy ($10\text{--}30^\circ$). If a proper AP image has been obtained, making the Jamshidi needle parallel to the horizon and parallel to the upper end plate will place the needle in the proper sagittal orientation.

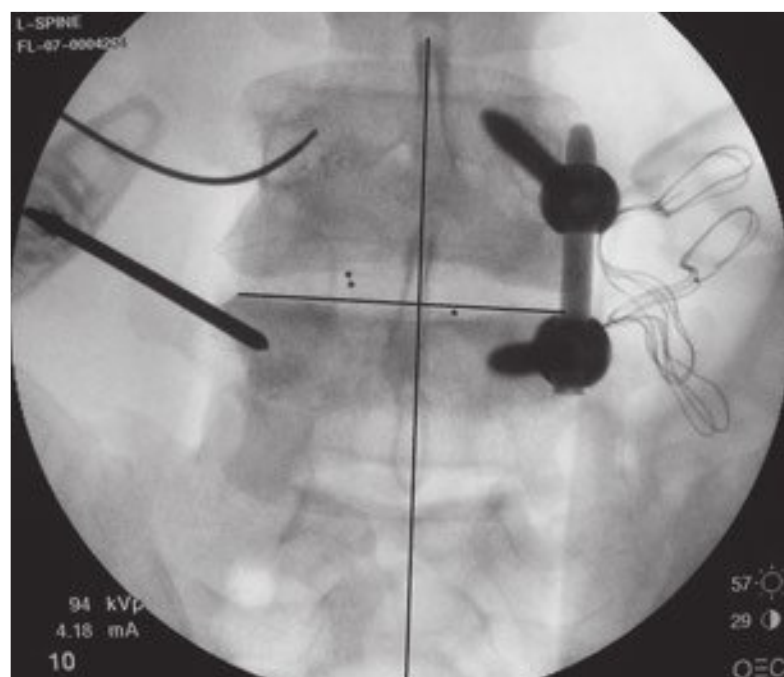


Fig. 60.1 AP projection on the L5 vertebral body. Note that the upper end plate of the L5 body is visualized as a single line and that the spinous process is equidistant between the two pedicles.

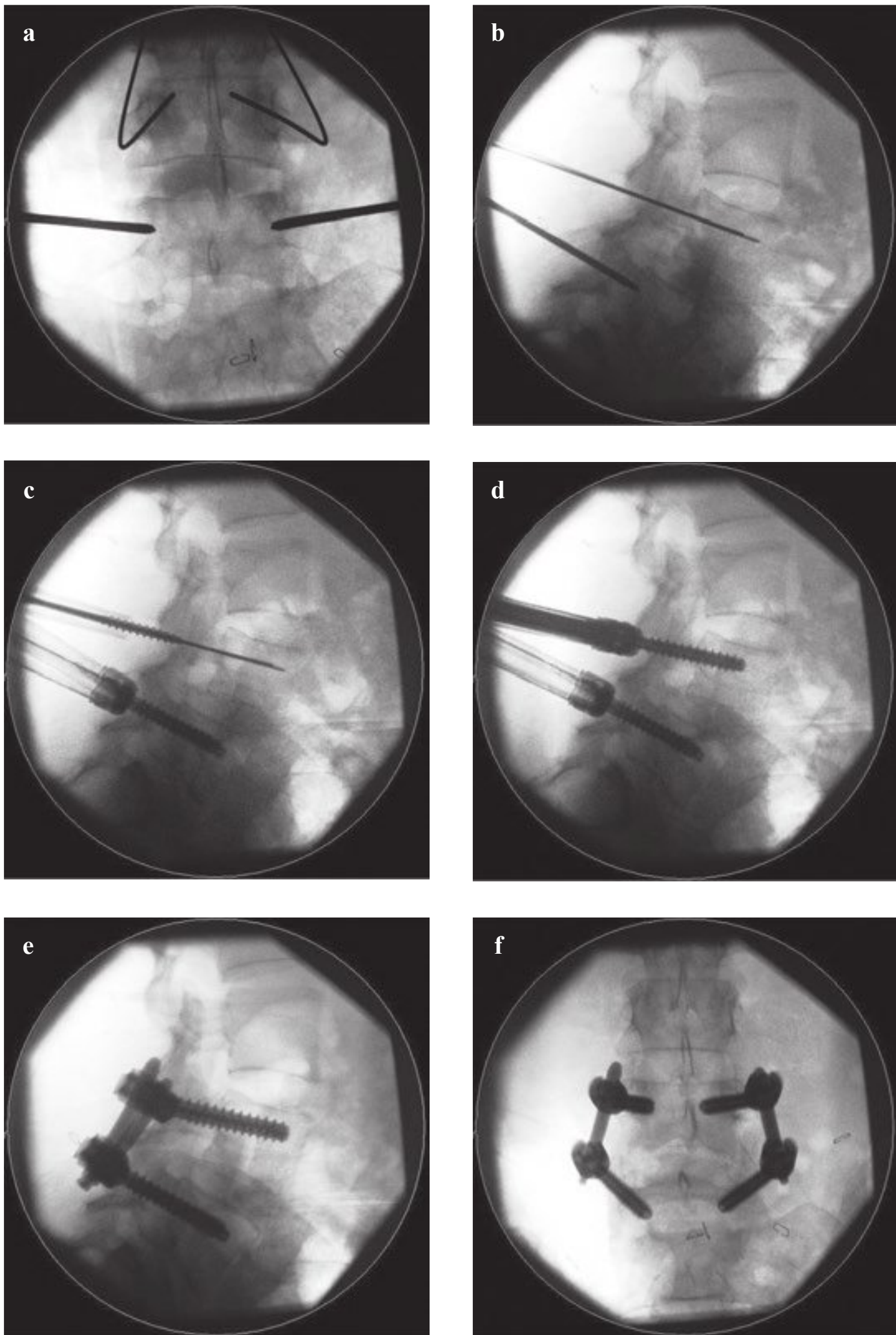


Fig. 60.2 (a) AP view targeting the lower vertebral body. Note the rotation in relation to the body above (where the spinous process is not in the midline). Docking at the junction of the facet joint and transverse process is followed by 2 cm of medial needle advancement. (b) The needle is confirmed on lateral fluoroscopy as having passed the spinal canal, ruling out pedicle wall violation. (c) An insulating sheath is used to minimize soft tissue trauma while an awl and tap are advanced over the K-wire. (d) Final screw positioning with extension tabs attached to guide rod insertion. Final construct is visualized on (e) lateral and (f) AP views.

5. So long as the medial wall of the pedicle is not breached at a depth of 2 cm, there will be no medial pedicle wall violation; the needle will have passed the depth of the pedicle (and spinal canal) following 2 cm of advancement. Once the Jamshidi needle has been placed, the inner stylet is removed, and a Kirschner wire (K-wire) is inserted into the vertebral body.
6. The procedure is then conducted at other vertebral levels with adjustments in fluoroscopic targeting for each level. After all the K-wires have been placed, the fluoroscope is moved to a lateral position. The K-wires are then used to guide an awl and tap for pedicle preparation as with open surgery. Final pedicle screw placement is then performed followed by K-wire removal.
7. During the procedure, care is taken to ensure that the K-wires do not violate the anterior vertebral body and enter the retroperitoneum, which could cause vascular or hollow viscus injury. In addition, care must be taken not to lose control of the wires or have them pull out prematurely.
8. Following screw placement, a percutaneous rod is advanced subfascially through the screw extensions to connect the segmental levels (**Fig. 60.3**). Rod insertion can occur through one of the end screw incisions or through a separate incision distal to the screws.
9. It is helpful to have at least some bend in the rod, for “steering” the rod medially or laterally is made easier by rotating the rod along its long axis to turn the tip medially or laterally. Finally, set screws are placed through the screw extensions to lock the rod to the individual screws.

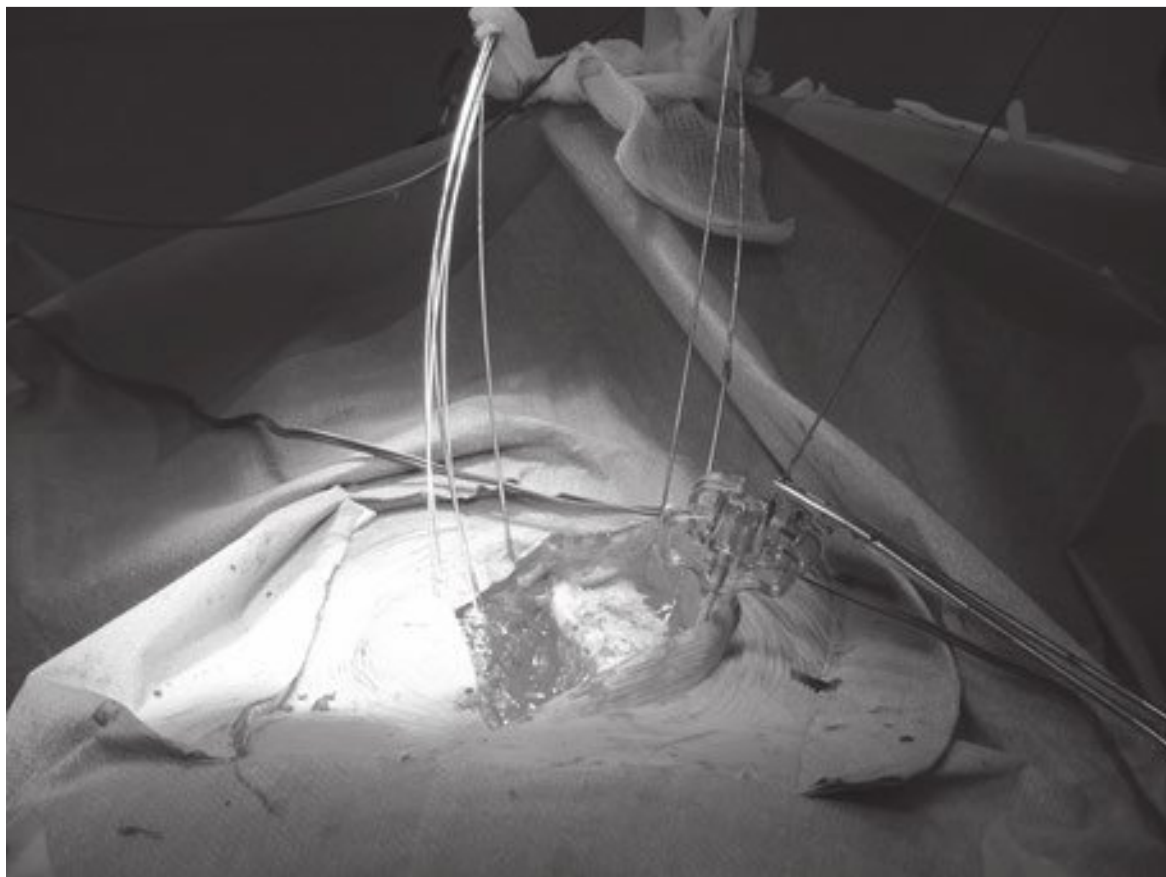


Fig. 60.3 Jamshidi needles and K-wires in place. Note the use of a long skin incision with placement of instrumentation through the exposed fascia and the use of a needle driver to control the K-wire during manipulation.

60.4 Complications

The complications associated with percutaneous screw placement are akin to those encountered with open surgical instrumentation procedures. In addition, risks are associated with loss of control of the K-wires, which can result in inadvertent entry into the retroperitoneum.

Difficulties can be encountered with initial pedicle cannulation or with rod passage and screw–rod connection. Use of the AP-based technique results in a low rate of pedicle violation as long as high-quality fluoroscopic images are used for targeting the Jamshidi needles. Rod passage and connection techniques are often specific to the implant manufacturers and the instruments available for rod and screw manipulation, so they are not described here.

60.5 Postoperative Care

Care after surgery is the same as with open instrumentation. Patients should be rapidly mobilized with physical therapists, with external bracing as a treatment option. Muscle spasm occurs more commonly with MIS procedures and is best treated with muscle relaxants and benzodiazepines as opposed to narcotics.

60.6 Surgical Pearls

- In many instances all of the surgery can be accomplished by a single surgeon standing on the side opposite the fluoroscope. This improves surgical workflow and minimizes the risk of contaminating the sterile field. Because of the highly imaging-dependent nature of the method described here, standing on the same side as the pedicle screw insertion (which is typical for open surgery) is much less important.
- When a rod is passed across the thoracolumbar junction, it is usually preferable to pass the rod from cranial to caudal.
- Because the screw head heights and orientation are not immediately visible, the surgeon should be prepared to pass a prepared rod initially and then to remove it and recontour or resize the implant to achieve the optimal construct. Forcing the screw and rod to mate may result in screw pullout.

Common Clinical Questions

1. Percutaneous pedicle screws can be safely inserted using all of the following techniques *except* which?
 - A. AP-guided imaging
 - B. Pure freehand technique
 - C. Biplanar fluoroscopy
 - D. Image guidance

2. When using the AP-based technique for percutaneous pedicle cannulation, how far is the Jamshidi needle advanced before passing the medial pedicle wall?
 - A. 1 cm
 - B. 2 cm
 - C. 3 cm
 - D. 4 cm
 3. Bending the rod allows what to be accomplished more easily?
 - A. Cutting the rod to size
 - B. Reducing spondylolisthesis
 - C. Achieving a fusion
 - D. Medial and lateral maneuvering
-

Answers to Common Clinical Questions

1. B.
 2. B.
 3. D.
-

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61 Minimally Invasive Lateral Retroperitoneal Transpsoas Interbody Fusion

Gisela Murray, Ali A. Baaj, and Juan S. Uribe

61.1 Key Points

- A successful lateral lumbar interbody fusion (LLIF) technique depends on the following steps:
 - Careful patient positioning
 - Gentle retroperitoneal dissection
 - Directional electromyographic (EMG) monitoring
 - Fusion bed preparation with release of contralateral annulus
 - Appropriate-size interbody implant placement

61.2 Indications¹

- Adjacent level failure
- Axial low back pain associated with degenerative disk disease
- Spinal stenosis (mild to moderate only)
- Spondylolisthesis grade 1 or 2
- Foraminal stenosis (indirect decompression)
- Total disk replacement
- Burst fractures and tumors in the thoracolumbar area (corpectomy)
- Adult spinal deformity (specifically with anterior column release)
 - For correction of lumbar lordosis and sagittal vertical axis (SVA)

61.3 Technique

- Patient positioning
 - The patient is placed on a radiolucent bendable table in true 90° lateral decubitus position with the top of the crest just superior to the table break.
 - Usually the left side is up, unless the crest is higher on that side or there has been previous surgery on that side.
 - Flex the table to increase the distance between the iliac crest and the ribs. This allows access to the disk space (important at L4–L5 to clear the crest and above L3 to clear the lower ribs).

- Localize the skin incision with lateral fluoroscopy by centering crossed K-wires over the disk's midposition. For multilevel approaches, a single longitudinal incision with individual transverse fascial openings for each level is adequate.
- Retroperitoneal access MIS (single skin/fascia incision technique)
 - Work perpendicular to the floor while dissecting through the muscle fibers to avoid entry into the peritoneal cavity, which is anteriorly displaced in the lateral position.
 - Entry into the retroperitoneal cavity is confirmed by the appearance of bright yellow fat and a loss of resistance by the muscle tissues. Finger dissection is then performed so that the surgeon can feel the psoas muscle deep in the cavity and the transverse processes posteriorly.
- Transpsoas approach and retractor placement
 - A series of tubular dilators are placed with EMG monitoring. Directional EMG monitoring (Neurovision: NuVasive; San Diego, California) allows the location of these nerves in relation to the dilator (**Fig. 61.1**). It is essential to guide the dilators with the finger to avoid inadvertent entry into the peritoneal cavity.
 - Lateral fluoroscopy will guide dilator placement into Zone 3 of disk space while bluntly splitting the fibers of the psoas muscle. The exception is at L4–L5 level, where the initial dilator should be placed at the midpoint between Zones 2 and 3. A K-wire is placed through the

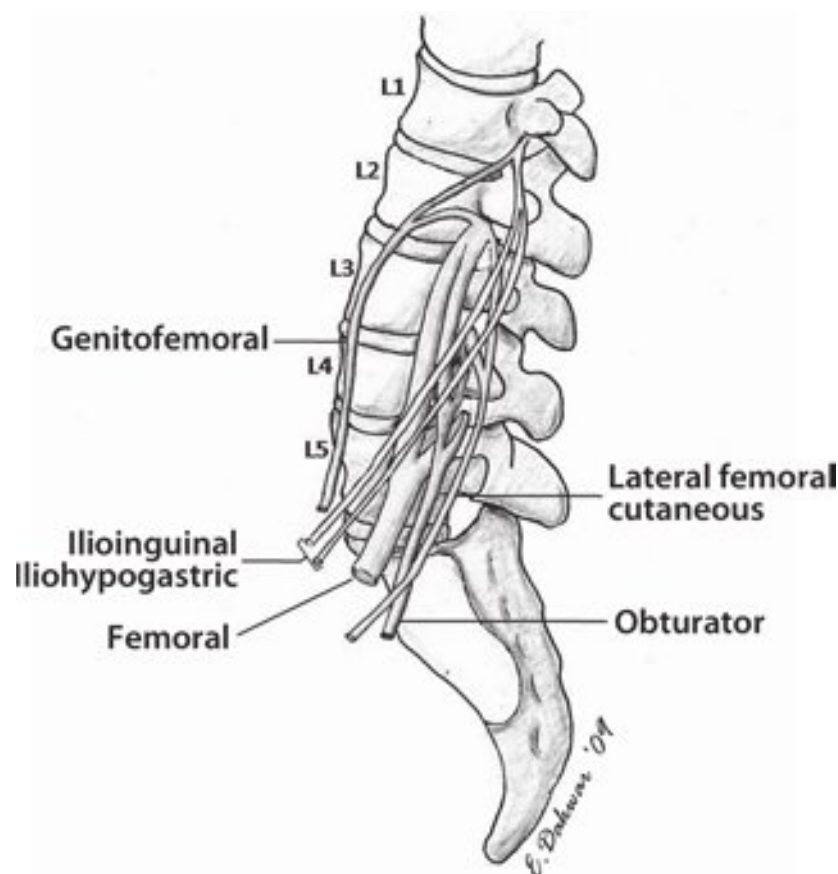


Fig. 61.1 Lateral view of the lumbar plexus. Understanding this anatomy is critical for successful implementation of the lateral transpsoas approach. (From Uribe JS, Arredondo N, Dakwar E, et al. Defining the safe working zones using the minimally invasive lateral retroperitoneal transpsoas approach: an anatomical study. *J Neurosurg Spine*. 2010; 13:260-266. Reproduced with permission.)

initial dilator and into the disk space to hold it in place. EMG monitoring is performed with each dilator before introduction of a larger dilator.

- The retractor is then placed, and a shim needle is introduced to the disk space under anteroposterior (AP) fluoroscopy to secure it.
- Preparation of the disk space
 - The disk space is incised with a knife just anterior to the shim, with a small rim of disk kept just in front of it to keep the retractor from sliding anteriorly.
 - A Cobb elevator is passed along both end plates and through the contralateral annulus under fluoroscopic guidance.
 - A series of instruments (curettes, pituitaries, rasps) are used to clean the space and prepare the end plates.
- For standard interbody fusion:
 - Interbody spacer and instrumentation are then placed.
- For anterior column release (MIS–ACR), additional steps must be performed after discectomy:²
 - A curved custom retractor is inserted gently between the anterior longitudinal ligament (ALL) and great vessels and the sympathetic plexus.
 - A knife is used to cut the ALL in sequential fashion.
 - Hyperlordotic cages (20 or 30°) are inserted into the disk space and fixed to an adjacent vertebral with one or two screws.
- The retractor is removed slowly in the open configuration to allow the psoas muscle to be inspected for bleeding.
- The external oblique fascia is closed with interrupted absorbable suture, and the skin is closed in a subcuticular fashion.
- Other different technique modifications include the following:
 - Transpsoas: Mini-open and open direct retroperitoneal transpsoas dissection
 - Prepsoatic: Mini-open anterolateral retroperitoneal dissection anterior to psoas muscle

61.4 Complications

- Hip flexor weakness¹
 - Self-limited—correlates with trauma to the psoas muscle.
- Thigh numbness (12–30%)³
 - Caused by injury to the anterior femoral cutaneous. Usually resolves without treatment by the time of the 3-month follow-up.
- Femoral nerve injury with associated quadriceps weakness^{1,4}
- Contralateral femoral nerve injury
- Genitofemoral neuralgia³
- Abdominal viscera perforation
- Great vessel injury⁵
- Kidney–ureteral injury
- Graft subsidence¹

- Unintended rupture of anterior longitudinal ligament
- Psoas/retroperitoneal hematoma ipsilateral/contralateral¹
- Abdominal wall paresis or hernia¹
- Rhabdomyolysis³
- Retrograde ejaculation (specific to ACR and prepsoatic approaches)

61.5 Postoperative Care

- For single-level lumbar cases, the patient is mobilized in the immediate postop period without a brace. No drains are placed.
- Expect hip flexion pain on the approach side during the immediate postoperative period.
- The patient is usually discharged on postoperative day 1 for single-level cases.
- Perioperative prophylactic antibiotic for 24 hours.
- In case of significant leg weakness, a CT or MRI scan is indicated to rule out a psoas hematoma.

61.6 Outcomes

- Fusion rates range from 91 to 100%^{3,6}
- For the lateral transpsoas interbody fusion, outcomes scales (Oswestry Disability Index and Visual Analogue Scale leg and back) are significantly improved at follow-up.⁷
- In terms of MIS–ACRs, long-term follow-up is not available yet.

61.7 Surgical Pearls

- Perfect AP and lateral projections of the disk space are imperative before starting the surgery. This is best accomplished by careful patient positioning, not by adjusting the fluoroscopy machines.
- To potentially decrease the amount of tension on the lumbar plexus, minimize the amount of lateral flexion of the patient and flex the ipsilateral hip during positioning.
- Guide the all the dilators with finger dissection through the retroperitoneal space to prevent visceral injury.
- During the directional EMG monitoring, place the nerves *posterior* to the retractor, where it can be safely retracted away from the surgical field without the risk of root avulsion or stretch injury.
- MIS–ACR should be performed once surgeons have mastered the standard lateral transpsoas technique. Careful examination of the location of the great vessels is necessary to prevent vascular complications during the dissection of the ALL.

Common Clinical Questions

1. During the directional EMG monitoring, where the lower numbers should be located?
2. What is the most likely explanation for postop hip flexor weakness?
3. Retraction against which nerve at the L4–L5 level is responsible for postop anterior thigh numbness?

Answers to Common Clinical Questions

1. Posteriorly.
2. Psoas muscle bruising as opposed to damage to femoral nerve branches to the psoas.
3. The femoral nerve (sensory fibers via the anterior femoral cutaneous nerve).

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62 Anterior Lumbar Interbody Fusion

Junyoung Ahn, Krzysztof Siemionow, Dustin H. Massel, Benjamin C. Mayo, William D. Long III, Krishna D. Modi, and Kern Singh

62.1 Key Points

- Anterior lumbar interbody fusion (ALIF) offers several potential advantages over other surgical approaches.
 - Posterior muscle sparing
 - The ability to perform a complete or subtotal discectomy
 - Large surface area for fusion and structural grafting
 - Favorable fusion environment (compression)
 - Effective for anterior releases, particularly in the setting of high-grade deformity
 - Indirect neural decompression

62.2 Indications

- Spondylolisthesis (typically grade I or II)
- Degenerative disk disease causing mechanical low back pain
- Postdiscectomy collapse with neural foraminal stenosis
- Treatment of posterior pseudarthrosis
- Treatment of postlaminectomy kyphosis
- Treatment of coronal and/or sagittal imbalance
- Adult spinal deformity

62.3 Technique

- Positioning and intraoperative setup
 - Place the patient supine on a regular operating table or a flat Jackson table.
 - A bump under the sacrum will allow for increased lumbar lordosis.
 - Slight Trendelenburg position will allow the abdominal contents to displace cephalad away from the surgical field.
 - The operative level is identified with lateral fluoroscopy. This localizes the skin incision (critical for mini-ALIF approaches).
- Surgical exposure
 - A left transverse or longitudinal paramedian incision is made.
 - The anterior rectus sheath is identified and divided in line with the skin incision.

- Blunt dissection is used to mobilize the rectus and to identify its lateral border.
- The lateral edge of the rectus abdominis muscle is retracted toward the midline, exposing the posterior rectus sheath (above the arcuate line) or, less commonly, the arcuate line and the preperitoneal space (below the arcuate line).
- The arcuate line of the posterior rectus sheath is carefully divided in a superior-to-inferior direction and the preperitoneal space identified.
- Using blunt dissection, the peritoneum is mobilized off of the anterior and lateral abdominal wall, and retroperitoneal blunt dissection is used to identify the psoas muscle.
- Handheld retractors are advanced.
- The ureter, peritoneum, and abdominal contents are mobilized across the midline.
- The psoas muscle is released and then retracted laterally, allowing direct visualization of the disk space.
- With blunt dissection medial to the psoas muscle, the index disk space is identified and confirmed with a marker and fluoroscopy.
- Specific procedure steps
 - Retractors are placed around the lateral annulus, first on the right side, retracting the abdominal contents and the iliac vein (**Fig. 62.1**).
 - Once the disk space is identified, an annulotomy is made with either a knife or electrocautery.
 - A Cobb elevator and curette are used to remove the cartilaginous end plates. As much disk as possible should be removed, but the subchondral bone/end plate should be carefully preserved to reduce the risk of subsequent implant migration and subsidence.
 - Pituitary rongeurs are used to remove the disk fragments.
 - An interbody graft (allograft, titanium, or polyetheretherketone [PEEK]) is appropriately sized to gently distract the disk space and contact the ring apophysis.
 - The anterior rectus sheath is closed with nonabsorbable suture. The posterior sheath does not require repair.

62.4 Complications^{1,2,3,4,5}

- Vascular injury (1.7–5.3%).^{1,2}
- Deep vein thrombosis (0.6–1.7%).^{2,3}
- Abdominal hernia (0.6%).²
- Ileus (0.6%).²
- Retrograde ejaculation occurs secondary to injury to the superior hypogastric plexus (sympathetic chain). The superior hypogastric plexus provides innervation to the internal vesical sphincter. The reported

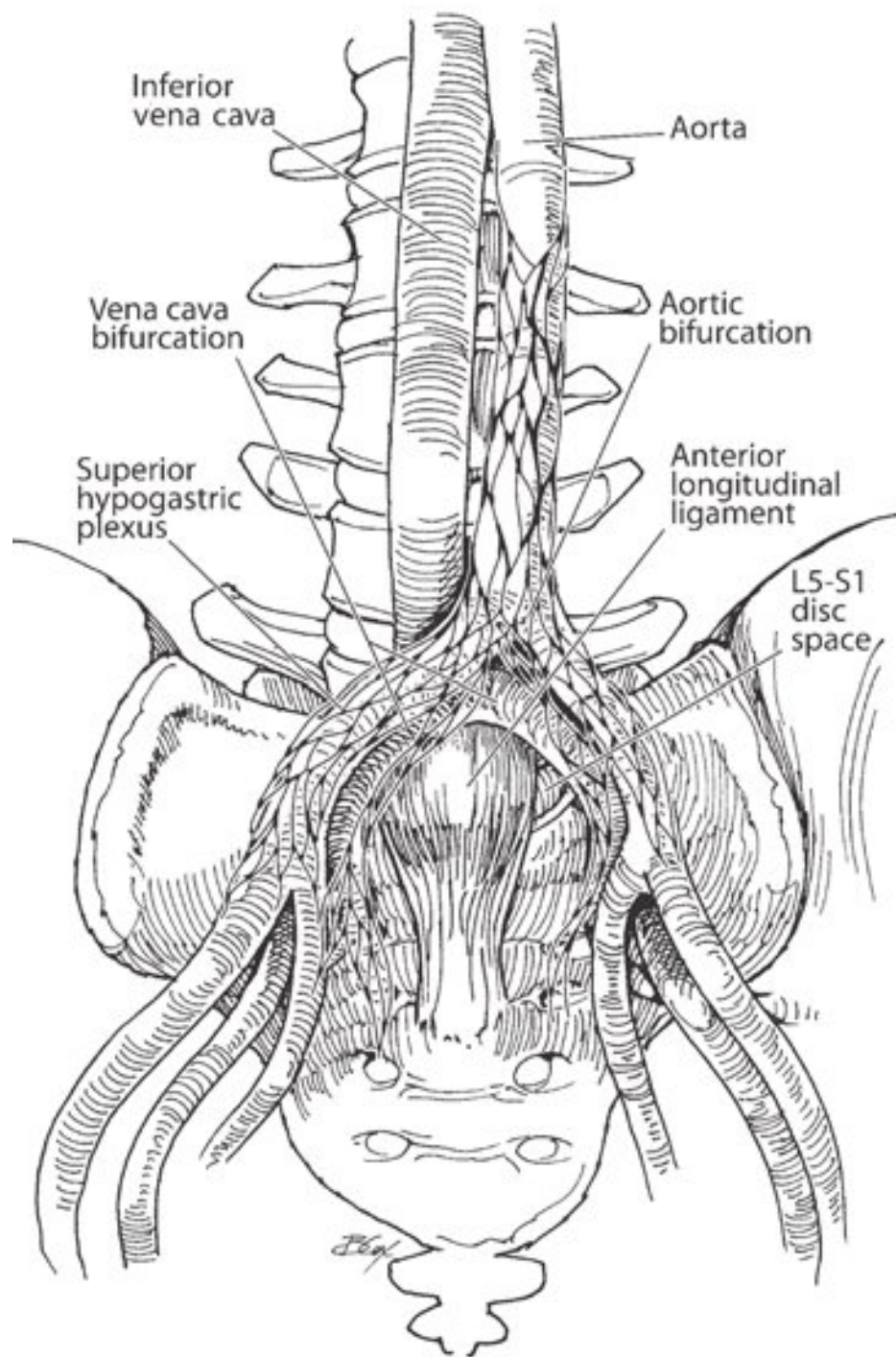


Fig. 62.1 Key anterior retroperitoneal structures.

incidence of retrograde ejaculation after ALIF varies widely in the literature, ranging from 0.4 to 5.9% of male patients.^{4,5}

- Infection (0.4–3.0%).^{1,2,3}
- Ureteral injury (0.3%).¹

62.5 Postoperative Care

- First postoperative day
 - Patients are mobilized.
 - Diet is advanced as tolerated.
- Discharge to home when patient meets discharge criteria (typically, ambulating, tolerating a diet, voiding, and adequate pain control on oral medications).

62.6 Outcomes

- Satisfactory clinical outcomes can be anticipated in 70% of patients.⁶
- Fusion rates vary from 85 to 95% in most series and depend on whether autograft or bone morphogenetic protein (BMP) is used.⁷
- Interbody subsidence is an expected phenomenon and occurs in as many as 85% of cases, particularly with standalone grafts.⁸

62.7 Surgical Pearls

- Localize with fluoroscopy before making a skin incision.
- Trendelenburg position allows the abdominal contents to move cephalad out of the operative field. This position also decreases venous bleeding.
- Avoid the use of monopolar cautery around the sympathetic chain prior to the annulotomy.
- Take care not to violate the subchondral end plates.

Common Clinical Questions

1. What is retrograde ejaculation, and how can the risk be minimized?
2. At what level does the aorta bifurcate, and why is that important for anterior spinal procedures?
3. List four indications for anterior lumbar interbody fusion.

Answers to Common Clinical Questions

1. Retrograde ejaculation occurs secondary to injury to the superior hypogastric plexus, a part of the sympathetic chain, which provides innervation to the internal vesical sphincter. Avoid using monopolar cautery around the sympathetic chain. Consider using a posterior approach in young males. When it occurs, retrograde ejaculation results in a normal sexual climax followed by the lack of ejaculate. Sperm typically is not propelled forward and ends up in the bladder.
2. In two-thirds of cases, the aorta bifurcates at L4–L5. Approaches to the L4–L5 intervertebral disk are more challenging and require mobilization of the iliac vessels.
3. Spondylolisthesis (usually grade I or II), treatment of pseudarthrosis, postdiscectomy collapse with neuroforaminal stenosis, and treatment of lumbar deformity with coronal and/or sagittal imbalance.

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63 Facet Screw Fixation/Fusion

Justin W. Miller and Rick C. Sasso

63.1 Key Points

- Transpedicular fixation has historically been the gold standard for lumbar fusion procedures. Pedicle screw placement, however, is technically demanding, is associated with higher complication rates, requires more extensive soft tissue dissection, and potentially has a negative effect on the adjacent level anatomy (i.e., facet joints).^{1,2} Translaminar facet screws are biomechanically equivalent to pedicle screw constructs when combined with interbody fusion, significantly increasing stiffness and stability of the overall construct.^{2,3}
- Facet screws were first described by King in 1948 as short screws placed horizontally across the facet joint. This technique was later modified by Boucher in 1959, whereby the screw was made longer and directed toward the pedicle to gain greater bone purchase. Magerl in 1984 modified the technique once again, inserting an even longer screw that begins at the base of the spinous process, traverses the contralateral lamina and facet joint, and ends at the base of the transverse process. Magerl's modification is a more effective screw and increases the working length on both sides of the facet joint.⁴
- Magerl's translaminar facet screw (TLFS) technique involves a minimally invasive approach with limited soft tissue dissection. Improved outcomes including decreased complication rates, reduced reoperation rates, shorter operative times, and decreased blood loss have been demonstrated.¹
- The use of image navigation has proven to be a safe, feasible, and efficient method for insertion of translaminar facet screws with low complication rates.¹

63.2 Indications

- In conjunction with interbody fusion for symptomatic lumbar degenerative disease processes unresponsive to at least 6 months of aggressive nonoperative treatment where posterior elements are present and intact
- Primary circumferential lumbar fusion with less than grade II spondylolisthesis
- Repair of pseudarthrosis after standalone anterior lumbar interbody fusion (ALIF)

63.3 Technique

- The patient is placed in the prone position on a radiolucent Jackson table frame under general anesthesia.
- A image navigation reference frame is inserted through a stab incision into the posterior superior iliac spine.
- O-Arm (Medtronic) CT scan is performed to acquire appropriate multiplanar images.
- Image navigation is used to identify and visualize the surgical level.
- A small paramedian stab incision is made to allow insertion of the blunt bullet tip/drill guide. The tip of the guide is placed at the spinolaminar junction with a image projection created across the contralateral lamina and intended facet joint.
- A pilot hole is created with the starting awl. Care is taken when making the first hole to leave adequate room for the contralateral TLFS in the cephalocaudad plane.
- A high-speed drill bit is then advanced under power, following the projection through the base of the spinous process, contralateral lamina, and crossing the intended facet joint. Image navigation is critical during this step of preparation. The use of tactile feel as the drill bit passes through the facet joint cortices is also important feedback. Take care to stay within the lamina and avoid breaching ventrally within the canal. Screw length is determined with the use of a calibrated drill bit and/or length of the image projection (**Fig. 63.1**).
- Without moving the drill guide, the drill bit is carefully removed, and the appropriate length screw is then inserted under power along the prepared trajectory.
- Final tightening of the screw is done by hand screwdriver. Care should be taken not to tighten the screw excessively. A TLFS functions as a threaded bolt and not a lag screw.
- The above steps are repeated in the exact same fashion for the contralateral TLFS.
- After insertion of all instrumentation final imaging is obtained (**Fig. 63.2**).
- For decortication of the facet joints, a tube retractor system can be used to expose and denude the intended facet.
- The wounds are then closed in a standard fashion.

63.4 Complications

- Spinous process fracture
- Ventral trajectory breach with either screw or drill bit
- Neurologic injury³
- Dural tear³

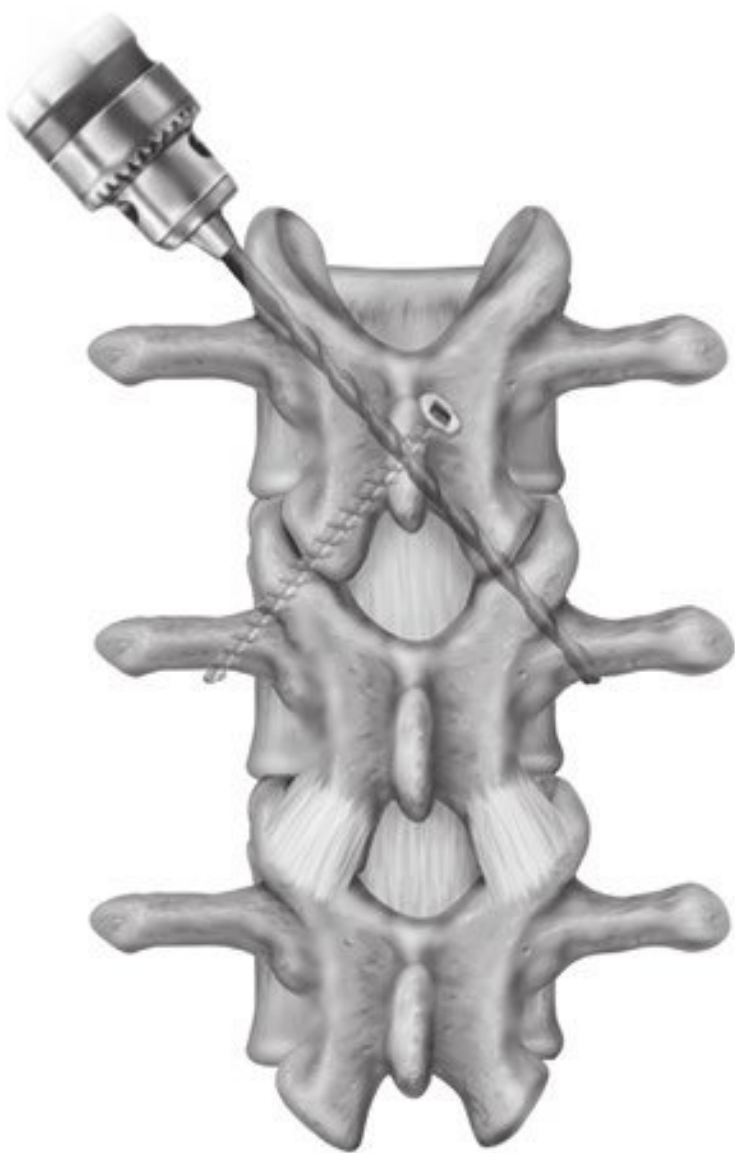


Fig. 63.1 Starting point and trajectory of drill bit.

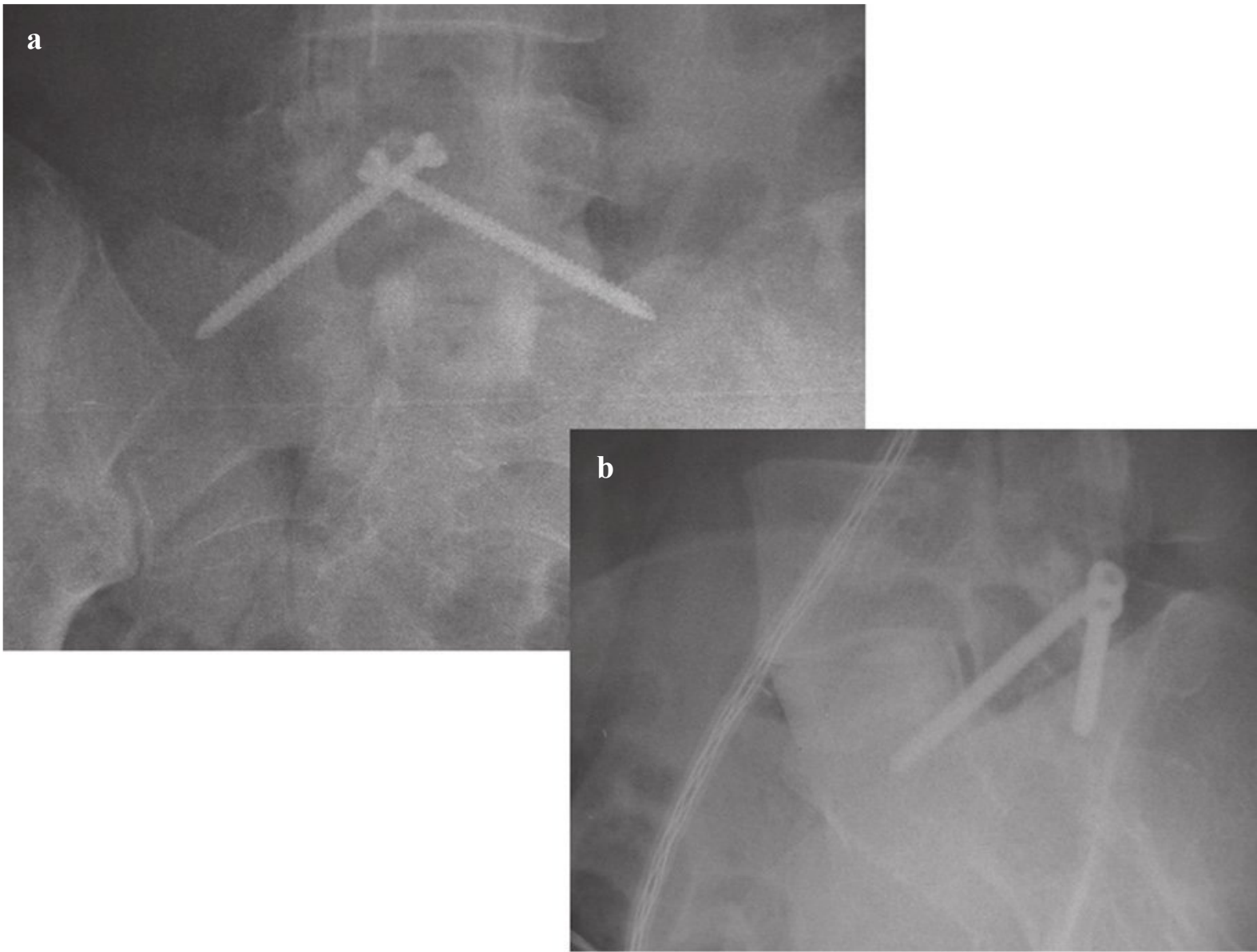


Fig. 63.2 (a) Anteroposterior and (b) lateral fluoroscopy views of final construct.

- Wound infection¹
- Pseudarthrosis¹

63.5 Postoperative Care

- Immediate mobilization following procedure without need for bracing
- Discharge home when patient meets standard discharge criteria

63.6 Outcomes

- Reported reoperation rate after TLFS insertion is 4.7% versus 24 to 37.5% for pedicle screw constructs.¹
- Operative times and blood loss have been shown to be significantly lower in translaminar facet screw cohort versus pedicle screw population.¹
- Patient recorded outcomes showed a significant decrease in postoperative Visual Analogue Scale (VAS) pain scores for both translaminar facet screw and pedicle screw constructs.¹
- Pseudarthrosis rates are low and range from 2 to 9% compared with standalone interbody fusions that can be as high as 25%^{1,3}
- Construct stiffness is significantly increased with TLFS fixation.^{2,3}

63.7 Surgical Pearls

- Obtain adequate visualization of starting point, trajectory, and intended facet joint before screw insertion and preparation of the trajectory.
- Plan for placement of both screws prior to creating the starting points to avoid screw collision.
- Establish appropriate starting point at the confluence of the spinous process and lamina. Avoid starting too high or low to remain within the contralateral lamina.
- Perform final screw tightening by hand, and do not overtighten to avoid screw stripping and/or fracture of the posterior elements. Remember that the TLFS does not function as a lag screw.

Common Clinical Questions

1. Translaminar facet screws function in what capacity?
 - A. Lag screw
 - B. Threaded bolt
 - C. Compression
 - D. Stabilization of anterior column

2. Which of the following is not an advantage of translaminar facet screws in comparison to pedicle screw fixation?
 - A. Less invasive
 - B. Technically simpler
 - C. Potential negative effect on adjacent anatomy
 - D. Lower complication rate
3. Which of the following is true of translaminar facet screws?
 - A. Final tightening should be performed to compress the joint.
 - B. Operative times are equal to pedicle screw placement.
 - C. Blood loss is higher with translaminar facet screw insertion.
 - D. Construct stiffness is significantly increased with translaminar facet screws.

Answers

1. B.
 2. C.
 3. D.
-

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64 Interspinous Process Decompression

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and Peter G. Whang

64.1 Key Points

- Lumbar stenosis refers to compression of the thecal sac, which may result in lower extremity pain/numbness (i.e., neurogenic claudication).
- Interspinous devices (ISDs) are designed to perform an “indirect” decompression by maintaining flexion of a stenotic segment, which increases the dimensions of the spinal canal and foramina.
- Appropriate candidates for ISD should experience clear relief of their claudication with lumbar flexion (i.e., sitting).
- This technique may give rise to reduced surgical morbidity and more rapid rehabilitation compared with laminectomy with or without arthrodesis.
- Elderly patients with significant comorbidities may be excellent candidates for this minimally invasive procedure.

64.2 Indications

- Symptomatic neurogenic claudication secondary to spinal stenosis with or without spondylolisthesis (confirmed by CT or MRI) that has failed to respond to conservative treatments (e.g., physical therapy, medications, or epidural injections).
- Limiting lumbar extension may also target various sources of low back pain by unloading the posterior disk and facet joints.¹
- Contraindications:
 - Significant spinal deformities (spondylolisthesis > grade I, scoliosis > 25°)
 - Bony ankylosis
 - Severe osteoporosis
 - Critical stenosis/cauda equina syndrome
- The X-Stop spacer (Medtronic; Memphis, Tennessee) is FDA-approved for use at one or two levels of the lumbar spine (**Fig. 64.1**).
- Several other spacers are currently undergoing evaluation by the FDA, including the Wallis device (Zimmer Spine, Inc.; Minneapolis, Minnesota), Flexus Interspinous Spacer (Globus Medical, Inc.; Audubon, Pennsylvania), and Superion (VertiFlex, Inc.; San Clemente, California).



Fig. 64.1 Photograph of X-Stop device, Medtronic Spine, LLC, Memphis, Tennessee. (Copyright © Medtronic Spine LLC.)

64.3 Technique

- Preoperative planning:
 - Radiographs: Evaluate for presence of spinal deformity (scoliosis on anteroposterior [AP], spondylolisthesis on lateral) and ankylosis (no segmental motion on flexion–extension views).
 - CT/MRI: Confirm diagnosis of stenosis.
 - Dual X-ray absorptiometry (DEXA): Assess risk for osteoporotic fractures.
- Anesthesia: General, monitored anesthesia care (MAC), or local.
- Positioning: Lateral decubitus or supine on a radiolucent table; make sure that the lumbar spine is maintained in flexion.
- A vertical midline incision centered over the affected segment(s) is made through the skin and fascia, with care taken to avoid attenuating the supraspinous/interspinous ligaments.
- After the levels are confirmed using fluoroscopy or intraoperative X-rays, a subperiosteal exposure of the spinous processes and laminae is performed without disrupting the facet capsules.
- The interspinous space is distracted so that the ISD may be inserted between two adjacent spinous processes (depending on the surgical protocol for each specific implant).
- May also be combined with a microdecompression (e.g., laminotomies).
- The wound is closed in layers, and a drain may or may not be used.

64.4 Complications

- Intraoperative:
 - Spinous process (SP) fracture
 - Malpositioning of implant
 - Inability to safely place the implant (e.g., attenuation of ligaments, excessively small interspinous space secondary to “kissing” SP or facet hypertrophy)
- Postoperative:
 - SP fracture
 - Device migration/dislocation
 - Persistent/recurrent symptoms (same- versus adjacent-segment degeneration)
- In the series of Barbagallo et al, the incidence of complications was 10.1% (primarily spinous process fractures and device dislocations), with a reoperation rate of 7.2%²

64.5 Postoperative Care

- May be performed in ambulatory setting as opposed to inpatient admission
- Immediate postoperative ambulation
- No bracing typically required
- Gradual return to normal activities

64.6 Outcomes

- Richards et al confirmed that the dimensions of the spinal canal and neuroforamina increased after placement of X-stop devices.³
- Kuchta et al published the 2-year results of 175 consecutive X-Stop procedures performed at a single center.⁴
 - X-Stop brought significant improvement in clinical outcome measures.
 - Reoperation rate of 4.6%(removal of implant with posterior decompression).
- Puzzilli et al performed a prospective, multicenter, randomized control trial in 542 patients who had symptomatic lumbar spinal stenosis (422 received X-Stop device, 120 treated nonoperatively).⁵
 - At 1-year follow-up, 83.5%of patients who underwent X-Stop procedure exhibited good clinical results compared with only 50%of the nonoperative group.
 - 4.9%intraoperative complication rate (spinous process fracture, cerebrospinal fluid [CSF] leak) and 11.1%postoperative complication rate (infection, device dislocation)

- 12.3%reoperation rate in the surgical group (5.7%neurological worsening, 4.3%symptomatic dislocation, 2.4%asymptomatic dislocation/spinous process erosion).
- Stromqvist et al compared the outcomes for 100 patients (50 treated with X-stop and 50 who underwent lumbar decompression).⁶
 - Clinical outcomes for both groups were similar at 6, 12, and 24 months.
 - However, only 6%of the decompression group underwent reoperation, compared with 26%of the X-Stop cohort.
- Abrams et al assessed the effect of the X-Stop device on treatment of spinal stenosis in patients who had associated spinal facet cysts.⁷
 - At a mean follow-up of 21 months, patients who had no cysts, cysts < 3 mm, and cysts > 3 mm all had similar improvements in clinical outcomes.
 - The authors concluded that the X-Stop device is an effective treatment for patients with spinal stenosis with or without synovial facet cysts.
- Patel et al reported the results of a prospective, multicenter, randomized, controlled trial comparing patients treated with either the Superior interspinous or the X-Stop.⁸
 - Both spacers gave rise to effective pain relief and improved back function.

64.7 Surgical Pearls

- Appropriate candidates for this technique should experience clear relief of their claudicatory symptoms with lumbar flexion or sitting.
- Flexion–extension lateral X-rays may identify the presence of bony ankylosis at the stenotic level(s), which may preclude the implantation of an ISD.
- The lumbar spine should be maintained in flexion to facilitate intraoperative distraction of the stenotic segments.
- Hypertrophic facet joints may need to be partially excised to allow successful placement of the ISD.
- Care must be taken when inserting the ISD in osteoporotic patients, who may be at greater risk for spinous process fractures.

Common Clinical Questions

1. Which of the following patients is best suited for placement of an ISD?
 - A. Patient with severe low back pain and minimal extremity symptoms
 - B. Patient with claudication secondary to moderate stenosis and a grade II spondylolisthesis at L4–L5
 - C. Patient with moderate stenosis at L3–L4 whose symptoms improve with sitting
 - D. Patient with severe stenosis between L2 and L5 who complains of perianal numbness and bladder dysfunction

2. What is the primary mechanism by which ISD relieves claudicatory symptoms?
 - A. Decreases facet joint forces
 - B. Unloads posterior disk tissue
 - C. Eliminates redundant ligamentum flavum and other compressive lesions
 - D. Increases the dimensions of the spinal canal
 3. Which is not a typical characteristic of neurogenic claudication?
 - A. Pain is worse when walking uphill.
 - B. Symptoms improve with sitting.
 - C. Pain/numbness is worse with lumbar extension.
 - D. Symptoms do not change when riding a bicycle.
-

Answers to Common Clinical Questions

1. C.
 2. D.
 3. A.
-

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65 Lumbar Arthroplasty

Keith Jackson and Joon Y. Lee

65.1 Key Points

- Lumbar arthroplasty has been approved in the United States as an alternative treatment strategy to spinal arthrodesis in the treatment of refractory symptomatic diskogenic back pain.
- Etiology of low back pain remains unclear, with numerous possible pain generators.
- Goals: Maintain segmental spine motion, alleviate back pain, minimize the risk of adjacent level adjacent level degeneration.
- The prosthesis should approximate the size and motion of the physiologic disk, avoid distracting the facet joints, and, ideally, reproduce the normal biomechanics (**Fig. 65.1**).
- Though the majority of lumbar disk arthroplasty procedures are currently performed through an anterior approach, newer devices employing posterior lateral and lateral approaches are under investigation.
- No independent long-term, randomized, prospective study on any artificial disk has been published to date that clearly delineates the safety and efficacy of lumbar arthroplasty.

65.2 Indications

- Patient refractory to a minimum of 6 months of conservative, nonoperative treatment.
- One- or two-level symptomatic degenerative disk disease in patients aged 18 to 60 years.



Fig. 65.1 The Charite artificial disk. (Used with permission from DePuy Spine, Inc.)

- Correlative objective radiographic findings including disk desiccation, vacuum disk, high-intensity zone signal, and Modic signal changes.
- Postdiscectomy axial back pain or juxtafusion disk degeneration.
- Absence of central or lateral recess stenosis that may require posterior decompression to address concomitant radicular leg pain.
- Provocative diskography may demonstrate concordant pain reproduction and confirm diagnosis.

65.3 Contraindications

- Compromised structural integrity of bone
 - Tumor, osteoporosis (T-score < -2.5), osteomalacia, acute fracture
- Potential compromise of stability or alignment of implant
 - Scoliosis, spondylolysis, spondylolisthesis ($>$ grade I), incompetent posterior elements
- Conditions that may compromise clinical outcome of disk replacement
 - Significant facet arthrosis, disk herniation with predominant radicular symptoms, central or lateral recess stenosis, disk height less than 5 mm
- Miscellaneous
 - Obesity (BMI > 40), local or systemic presence of tumor or infection, pregnancy, intraspinal neoplasm

65.4 Technique¹

65.4.1 Anterior Approach

- Positioning (anterior lumbar disk arthroplasty):
 - The patient is placed in the supine position on a radiolucent table with all bony prominences padded after institution of general anesthesia and Foley catheter insertion to decompress the bladder.
 - C-arm fluoroscopy is used to identify the approach angle and location of the disk space and to verify that clear anteroposterior and lateral images can be attained.
- Standard sterile preparation and draping are performed and prophylactic intravenous antibiotics are administered.
- Exposure:
 - The majority of prostheses are implanted by means of an open anterior approach similar to that used for anterior lumbar interbody fusion (ALIF).
 - A general or vascular surgeon can obtain access to the spine through either an anterior retroperitoneal approach (most preferred) or a midline transperitoneal approach.

- A left-sided approach is preferred between L3 and L5 because of the relative resilience of the aorta and ease of mobility compared with the vena cava.
- In accessing L5–S1 in male patients, a right-sided approach is recommended to reduce the rate of injury of the superior hypogastric plexus, which can cause retrograde ejaculation.
- The retroperitoneal space is entered deep to the rectus sheath and the peritoneum, with the ureter retraced medially.
- Blunt dissection reveals the lateral edge of the psoas, and vascular structures are carefully mobilized and elevated off the anterior spine.
- Procedure:
 - Fluoroscopic images are used to confirm the disk level and identify the midline.
 - A subtotal discectomy is performed by excising the anterior longitudinal ligament, annulus, and nucleus pulposus, leaving the lateral portion of the annulus intact.
 - The cartilaginous end plate is debrided from the osseous end plate. The integrity of the end plate is preserved to ensure implant fixation and avoid subsidence.
 - If necessary, the posterior longitudinal ligament (PLL) is released and posterior osteophytes debrided.
 - Special instruments are used to measure the footprint, lordotic angle, and core height.
 - The appropriately sized prosthesis is implanted after device-specific preparation of the end plate.
 - If the device has a polyethylene core, it is trialed and inserted after confirming on lateral fluoroscopy the restoration of desired disk height and lordosis.
- Fluoroscopy confirms the central position of the implant on the anteroposterior view. Ideally, the center of rotation of the device is 2 mm posterior to the sagittal midline of the vertebral body on the lateral view.
- Final confirmatory radiographs are obtained, and the wound is closed in routine fashion.
- The approach varies according to the lumbar level accessed as well as device-specific modifications to the general technique.

65.4.2 Lateral Approach

- Positioning:
 - The patient is placed in a lateral decubitus position with the hip and knee flexed.

- The table can be flexed to create more working space between the iliac wing and the rib cage.
- Obtaining adequate intraoperative fluoroscopy in both the anteroposterior and lateral planes is essential to accurate component placement.
- Exposure
- A retroperitoneal approach is employed down to the level of the psoas muscle.
 - A nerve probe is placed of through the psoas muscle onto the outer layer of the annulus to help maintain a safe distance away from the nerves of the lumbar plexus.
- Procedure:
 - Once the disk is exposed, discectomy and end plate preparation are conducted similar to that described for the anterior procedure, with the exception that the anterior longitudinal ligament (ALL) is maintained.

65.4.3 Posterior Approach

- Positioning
 - The patient is placed in a prone position with all bony prominences padded.
 - Obtaining orthogonal X-rays is essential for accurate implant positioning.
- Exposure
 - A standard posterior midline approach to the lumbar spine or minimally invasive posterior lumbar interbody approach can be used.
 - Once the posterior spinal elements are exposed, a minimal posterior lateral decompression is performed to gain access to the disk space.
- Procedure
 - With the theca sac and the nerve root protected, the disk space is prepared in much the same way as the other two approaches described.

65.5 Complications

65.5.1 Anterior Approach

- Approach-related complications (10–13%): Vascular injury, phlebitis, pulmonary embolism, sexual dysfunction, and retrograde ejaculation.
- Postoperative retroperitoneal scarring makes revision surgery more difficult.
- Failure of the prosthesis primarily involves facet joint degeneration, subsidence, spontaneous ankylosis, device migration, and adjacent-level disease.

- Cases of vertebral body fractures have been reported.
- Heterotopic ossification has been reported in varying degrees in 1.4 to 15.2% of patients.
- A prospective, randomized, multicenter Food and Drug Administration (FDA)–regulated clinical trial reported on complications of 589 patients.²
 - The disk replacement group was found to have an 8.8% reoperation rate, compared with 10.1% in the lumbar fusion control group. Mean time to reoperation was 9.7 months.
 - The primary reason for removal of the implant was device migration (75%).
 - A higher incidence of vascular injury occurred in the reoperation group (16.7%) compared with the primary group (3.4%).
 - Fourteen patients required posterior instrumented fusion for persistent low back pain (2.4%).

65.5.2 Lateral Approach

- Given the early stage of trials of this technique a thorough complication profile has not been reported.
- Approach related complications include injury to the lumbar plexus, vascular injury, or visceral injury.

65.5.3 Posterior Approach

- Given the early stage of trials with this technique, a thorough complication profile has not been reported.
- Approach related risks include durotomy, nerve root injury, and neuritis, as well as epidural fibrosis that may challenge revision surgery.

65.6 Postoperative Care

- Patients bear weight as tolerated and are mobilized on the first postoperative day.
- A brace is typically not needed.
- Standing radiographs are obtained as soon as feasible postoperatively to document the position of the implant in the weight-bearing position.
- A gentle low back and abdominal strengthening program is implemented starting with the first postoperative day.
- The patient is given postoperative restrictions including avoidance of substantial extension, bending, twisting, or heavy lifting for the first 6 weeks.
- Progressive unrestricted activity is allowed after 6 weeks.

65.7 Outcomes

- A large randomized controlled trial comparing patients undergoing one- or two-level disk arthroplasty with those enrolled in an outpatient multidisciplinary nonoperative treatment program demonstrated:³
 - Statistically significant improvements in ODI, low back pain scale and patient satisfaction.
 - Though the differences in Oswestry Disability Index were statistically significant, the magnitude of improvement did not reach the prespecified minimally important clinical difference.
- Clinical and radiographic outcomes after an average of 17 years have raised concerns about the long-term durability of disk arthroplasty.⁴
 - In this retrospective review, spontaneous ankylosis of the operated level was reported in 60% of the population at final follow-up.
 - Perhaps more concerning, patients with functional implants at long-term follow-up (17%) were less satisfied than those who experienced spontaneous ankylosis.

65.8 Surgical Pearls

- The lumbar spine is positioned in the neutral position to minimize tension on retroperitoneal vessels.
- Sympathetic and parasympathetic nerves must be carefully preserved to prevent erectile dysfunction and retrograde ejaculation in male patients.
- The integrity of end plates must be preserved to ensure good implant fixation and avoid subsidence.
- The implant must restore appropriate lordosis, have adequate end plate coverage, and avoid distraction of more than 3 mm to apply proper tension to the posterior ligaments.

Common Clinical Questions

1. All the following are contraindications to total disk arthroplasty *except* which?
 - A. The presence of significant lateral recess stenosis
 - B. Facet arthritis
 - C. Degenerative spondylolisthesis
 - D. Smoking
 - E. Obesity (BMI > 40)

2. Which of the following statements is true for total disk arthroplasty?
 - A. Randomized prospective control trials have illustrated the efficacy of lumbar disk arthroplasty, establishing that it preserves normal biomechanics and reduces the incidence of adjacent-segment degeneration.
 - B. A right-sided approach to the L5/S1 disk space is recommended in male patients undergoing total disk arthroplasty.
 - C. The PLL is essential to the appropriate tensioning of the disk replacement and should not be excised.
 - D. Heterotopic ossification is not a confirmed potential complication of total disk arthroplasty.
 - E. A significant portion of the cranial and caudal end plate must be removed before placement of the implant for proper setting of the prosthesis.
-

Answers to Common Clinical Questions

1. D.
 2. B.
-

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66 Lumbosacroiliac Fixation

Phillip Horne, William D. Long III, and Andrew A. Sama

66.1 Key Points

- Lumbosacroiliac (LSI) fixation provides a strong biomechanical base for arthrodesis that includes the L5–S1 level, improving rates of fusion for long instrumented constructs.
- LSI constructs include fixation into the sacrum and ilium, sharing load from the lumbar spine through the sacrum and sacroiliac joints to the pelvis.
- Understanding pelvic anatomy is critical to ensure proper placement of iliac screws to prevent cortical breach and injury to the pelvic viscera.
- Sacral slope, pelvic tilt, and pelvic incidence are measurement parameters that define the anatomic LSI relationship to assist with surgical planning and instrumentation (**Fig. 66.1**).
- Sacral fracture in osteopenic bone is generally avoided with iliac screw protection of sacral screws.

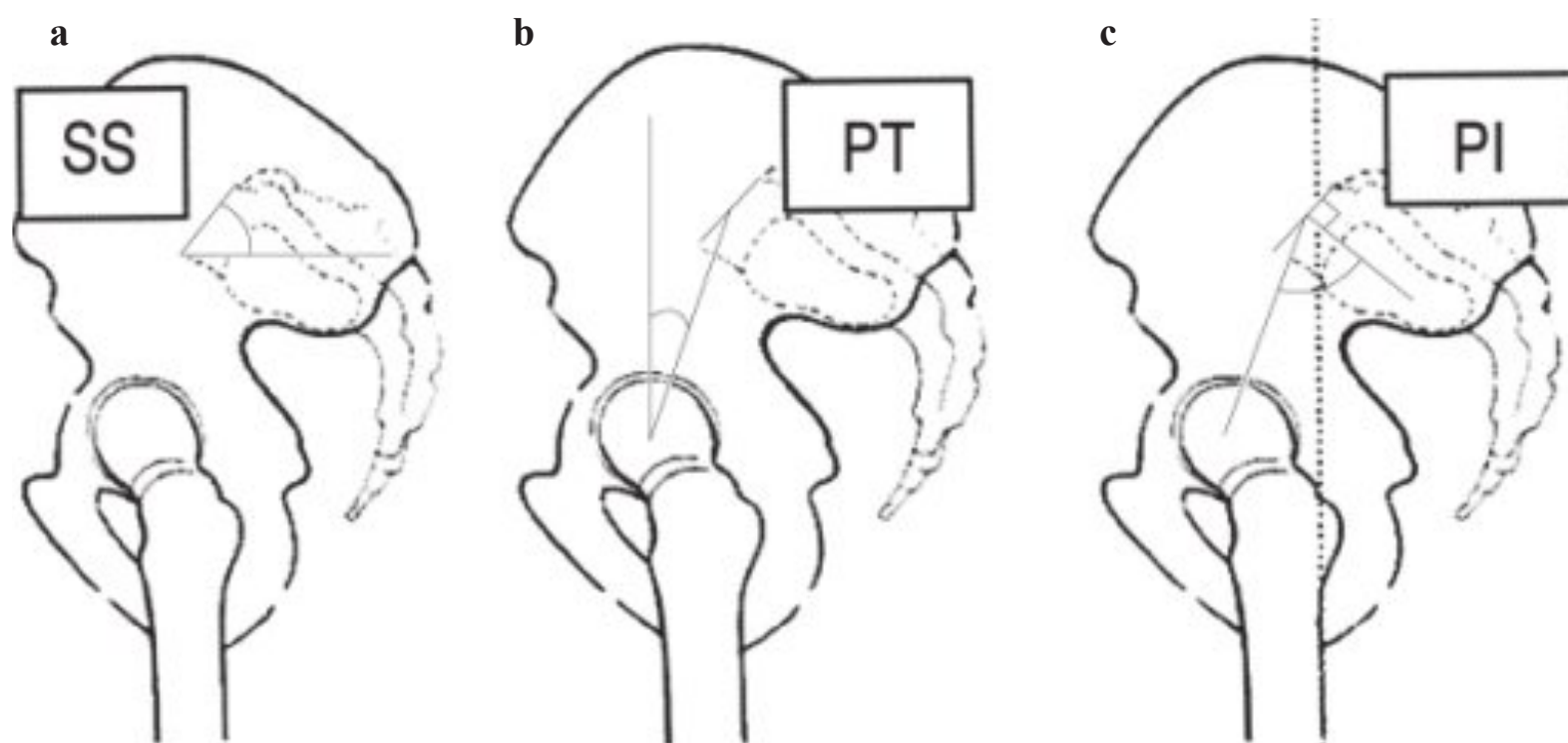


Fig. 66.1 Measurement of the sacropelvic parameters: **(a)** Sacral slope (SS), **(b)** pelvic tilt (PT), **(c)** pelvic incidence (PI). (From Oh SK, Chung SS, Lee CS. Correlation of pelvic parameters with isthmic spondylolisthesis. *Asian Spine J.* 2009;3:21-26. Reproduced with permission.)

66.2 Biomechanics

- The sacrum is comprised largely of cancellous bone with a thin cortical rim, reducing screw fixation strength.
- Large loads are transferred to implants across the lumbosacral junction, including flexion bending moments, cantilever bending on posterior instrumentation, axial, translational, and rotational forces.
- Instrumentation anterior to the lumbosacral pivot point (posterior vertebral line of L5–S1 disk space) best resists flexion moments and enhances fixation strength.¹
- Biomechanical testing has shown increased fixation strength in iliac wings versus S1 and sacral ala.²
- Iliac screws have an oblique trajectory compared with lumbosacral screws and therefore help resist distal pullout.

66.3 Indications

- Long fusions to the sacrum (starting at or above L2)
- Short lumbosacral fusions in osteoporotic bone
- Scoliosis with oblique L5–S1 takeoff
- Correction of pelvic obliquity or significant positive sagittal imbalance
- Rigid structural deformity, paralytic curves, neuromuscular curves, congenital deformities
- Traumatic sacral fractures with or without pelvic discontinuity; insufficiency fractures of the sacrum, or in patients with vestigial S1–S2 disk who require fixation to the sacrum
- Scoliosis patients with L5–S1 spondylolisthesis
- Revision lumbosacral fusion
- Flat back deformity requiring three-column lumbar osteotomy or lumbar vertebral column resection
- High-grade dysplastic spondylolisthesis
- Sacral tumor resection

66.4 Preoperative Planning

- Assess flexibility of the fractional curve (L3/L4–sacrum) and degeneration of the lower lumbar segments with plain radiographs.
- X-rays can also assess sacral anomalies such as a bifid sacrum.
- Cross-sectional imaging can assess any iatrogenic defects from any prior iliac crest bone harvesting.

66.5 Techniques

- A traditional midline posterior approach to the lumbosacral spine is performed.
- Subperiosteal dissection identifies the transverse process and facet joint for pedicle screw start point to instrument the L5 vertebra.³
- Sacral exposure is performed extraperiosteally to avoid injury to the venous structures exiting the dorsal sacral foramen.
- Expose sacral ala and ilium if performing iliac start point screws.
- The S1 pedicle screw start point is 1 cm cephalad and lateral to the S1 foramen, at the junction of the ala and the lateral S1 facet³ (**Fig. 66.2**).
- Place S1 screws directing to the sacral promontory in bicortical fashion.
- Medialize the screw to avoid L5 nerve root laterally on sacral ala.
- S1 alar screws can alternately be placed by directing in an anterolateral direction.

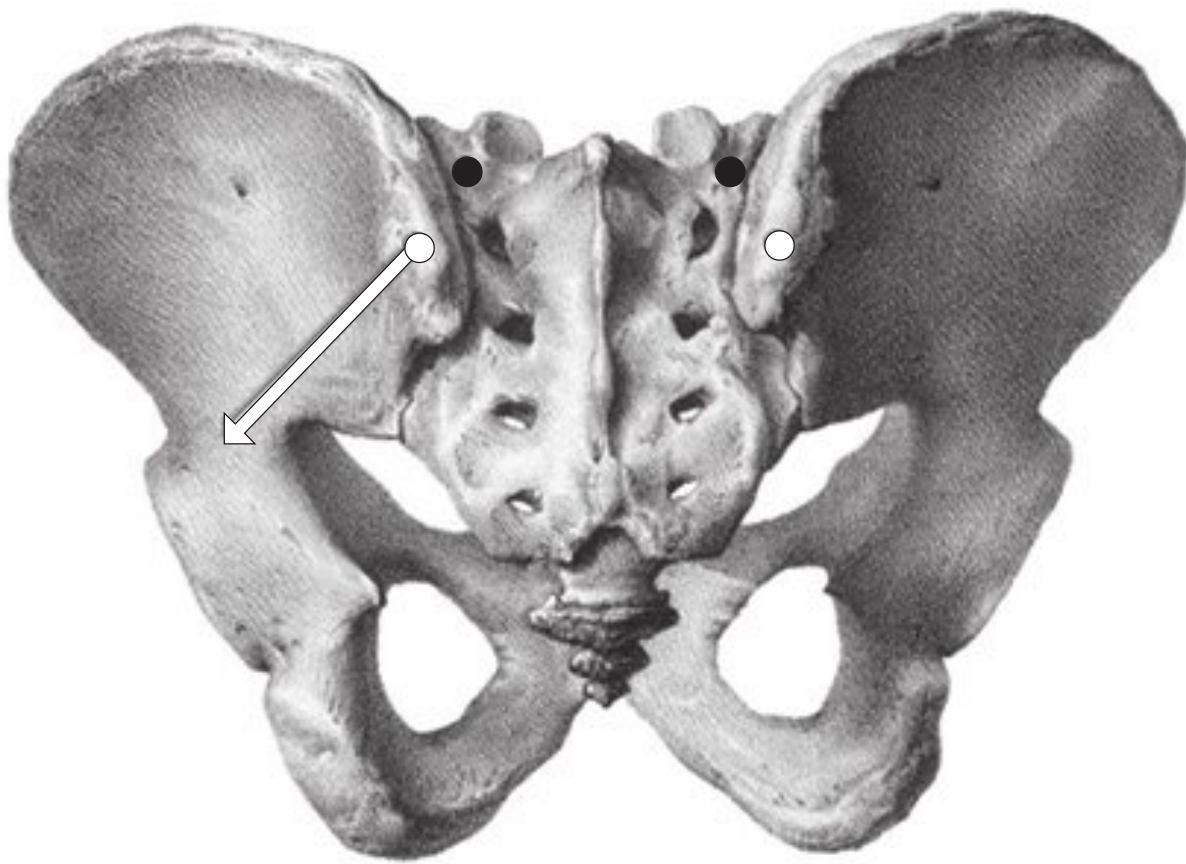


Fig. 66.2 Posterior view of the pelvis demonstrating starting points for the S1 screws (black dots) and iliac screw and trajectory (white dot and arrow).

66.5.1 Iliac Screw Fixation³

- Entry point at distal ilium near the posterior superior ilium spine (PSIS). Place deep to the PSIS along inner table of ilium to reduce prominence of the screw head (**Fig. 66.2**).
- Open cortex with Rongeur or bur. Find cancellous bone and advance probe between inner and out tables of ilium directed $\sim 30^\circ$ caudally and 30° laterally with variation among patients.
- Use teardrop view of iliac wing (obturator oblique) to guide probe advancement and screw placement as needed. Confirm screw does not violate the greater sciatic notch with the iliac oblique view. Confirm that the screw does not violate the hip joint with an anteroposterior pelvis or lateral view. Inlet, outlet, and obturator views should be used to also confirm whether there is any concern for screw position.
- Placement 1 cm above the sciatic notch affords better screw purchase in the supra acetabular region.
- Confirm with sounder that the channel is within bone. Measure depth to be at least 70 to 80 mm.
- Tap halfway and place screw, typically 7.5 to 8.0 mm diameter $\times$ 70 to 80 mm long. Place deep to avoid prominent screw head.
- Connect to S1 screw and lumbar instrumentation with a lateral connector rod. The actual rod will require a substantial lordotic bend to link the iliac screw connector into the lumbosacral pedicle screws.
- Placement of two screws per side is possible if planned appropriately.

66.5.2 S2 Iliac Screw Fixation³

- Typically aligns better with lumbosacral screws obviating connector rod; also avoids the prominent nature typical of an iliac screw.
- Dissection should include S2 dorsal foramen but does not require as far lateral dissection as does an iliac start point.
- Start point is at S2 body, in line with S1 pedicle screw, typically 2 to 4 mm lateral and 4 to 8 mm caudal to the S1 dorsal sacral foramen.⁴
- Bur at start point to access cancellous bone. Advance probe paralleling iliac wing, 40° anterolaterally and 20 to 30° caudally.
- Cortical bone should be encountered at the SI joint 25 to 35 mm deep. Use a ball-tipped probe to confirm location within bone when crossing the SI joint. Advance the gear shift probe to 70 to 80 mm and again confirm position with the sounding probe.
- Tap across SI joint into ilium (~ 40 mm).
- Place screw, typically 7 to 8 mm $\times$ 70 to 80 mm in adults.
- Rods require substantial lordosis to connect from lumbosacral pedicle screws to sacral alar screw.

66.6 Complications

- Pseudarthrosis of the lumbosacral junction is most common complication.
- Infection and wound complications arise because of the superficial nature of the sacroiliac screws and elevation of the erector muscles distally, with an incidence of approximately 4%⁵
- Sciatic notch violation during placement of the iliac screw can cause superior gluteal artery injury.

66.7 Postoperative Management

- Patients should mobilize with progressive ambulation beginning postoperative day 1. Avoid impact and vigorous activity for several months postoperatively, as well as flexion moments of the lumbosacral spine.
- Avoid NSAIDs postoperatively, as well as bisphosphonates, smoking, and poor glycemic control, to promote successful fusion.

66.8 Outcomes

- Fusion rates are greater than 90% when using LSI fixation.
- Prominent hardware can be symptomatic in up to 20% of patients.
- As sacroiliac joint fusion is generally not performed during LSI fixation, iliac screw loosening or breakage can occur from fatigue secondary to cycling of the construct; long term outcomes of implant loosening are not known.

66.9 Surgical Pearls

- A long lumbosacral fusion construct creates large lever arms and generates cantilever forces, which results in high stress at the caudal aspect of the construct.
- Iliac screws protect the S1 screws in lumbosacral fusions, and if iliac screws have inadequate length (< 70 mm) or fixation strength, augment the construct in other ways, including L5–S1 interbody graft, using cross connector between the S1 and iliac or S2 screws, or using a brace postop with thigh extensions postop on side of lesser fixation.
- S1 and S2 pedicles are comprised of cancellous bone, which is often osteopenic in elderly patients, creating increased risk for screw pullout.
- Iliac screws are “out of plane” to the pedicle screws above and anterior to the axis of flexion–extension rotation, creating a biomechanical advantage against distal pull out.

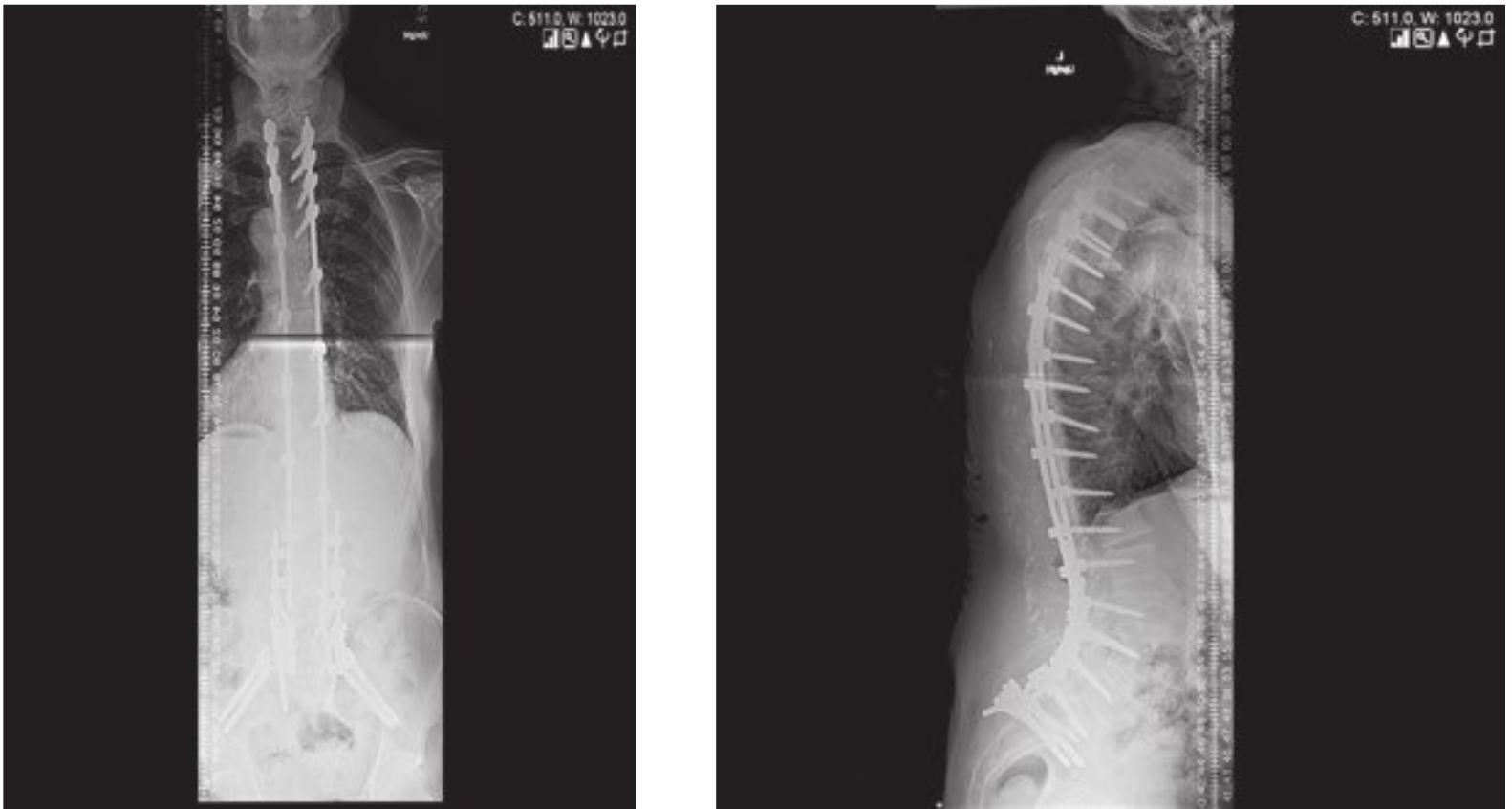


Fig. 66.3 Anteroposterior and lateral X-rays revealing bilateral pelvic fixation with four-screw/four-rod technique in long instrumented fusion in setting of osteopenia. Left side screws are placed in the traditional iliac start point and use a lateral connector to link to the rod. The right screws are S2 alar start point and are in line with the rod.

- One can dissect small abductor window off pelvis to allow advancing a finger to the notch along the outer cortex of the ilium to help with screw trajectory and avoid penetrating the sciatic notch.
- Whenever possible, avoiding fusing to the sacrum, to preserve motion segments, should be attempted; the L5–S1 segment is the site of greatest flexion–extension. However, long fusion segments should not be ended at L5 because of subsequent predictable degeneration of L5–S1 or in setting of risk factors for pseudarthrosis (patient factors: Smoking, diabetes, osteoporosis or metabolic bone disease; pathological factors as above).
- Four rod and/or four iliac screw techniques may be used in the setting of poor bone quality to increase strength of the construct, with reduced rates of implant failure (**Fig. 66.3**).

Common Clinical Questions

1. Violation of the greater sciatic notch during placement of the iliac screws would cause injury to which neurovascular structure?
 - A. Internal pudendal nerve
 - B. Femoral nerve
 - C. Superior gluteal artery
 - D. Lateral femoral cutaneous nerve
 - E. Inferior pudendal vein

2. Which is the S1 pedicle screw start point?
 - A. 1 cm caudal and lateral to the S1 foramen, at the level of the PSIS
 - B. 1 cm cephalad and lateral to the S1 foramen, at the junction of the ala and the lateral S1 facet
 - C. Medial to the alar wing, at the level of the S1 foramen
 - D. 2 to 4 mm lateral and 4 to 8 mm caudal to the S1 dorsal sacral foramen

Answers to Common Clinical Questions

1. C. The greater sciatic notch transmits the superior and inferior gluteal vessels and nerves, the sciatic and posterior femoral cutaneous nerves, the internal pudendal vessels, and the nerves to the obturator internus and quadratus femoris.
2. B.

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67 Iliosacral Percutaneous Fixation (Iliosacral Screws)

Jonathan G. Eastman and Eric O. Klineberg

67.1 Key Points

- Iliosacral (IS) screws can be placed safely and reproducibly with a percutaneous technique.
- Intraoperative anatomy and screw trajectory must be correctly interpreted via osseous fluoroscopic landmarks.
- The posterior pelvic ring may be highly variable and will dictate the available osseous fixation pathways.
- To safely place implants, a clear understanding of the fracture pattern and resultant displacement is required to obtain an adequate reduction.
- Screw length, number and trajectory are dependent upon available osseous fixation pathways, injury pattern and associated instability, bone quality, and patient compliance.

67.2 Indications

- IS screws are advantageous, for they provide posterior stabilization without significant soft tissue dissection, but they are technically demanding and are influenced by the variability of pelvic ring injuries as well as the presence of accompanying orthopaedic, thoracic, abdominal, and genitourinary injuries.
- IS screws may be used as the definitive fixation for many posterior pelvic ring injuries or as an adjunct for more complex pelvic and/or lumbopelvic injuries.
- When treating anteroposterior compression injuries (APC), surgical stabilization is generally indicated for pubic symphysis diastasis greater than 2.5 cm. This injury is typically accompanied by a posterior pelvic ring injury consisting of a posterior ilium fracture, sacroiliac joint disruption, or sacral fracture or a combination of these.¹
- Lateral compression injuries, when there is static or dynamic instability of greater than 1 cm, operative fixation is generally indicated. (Note: Occult instability can be detected with an examination under anesthesia.¹)

67.3 Technique

IS screw placement is technically demanding, and an exact three-dimensional understanding of the posterior pelvic anatomy is critical. The variability in the degree of lumbosacral lordosis and kyphosis must be appreciated. Understanding the unique orientation of each patient's posterior pelvic anatomy will allow the surgeon to obtain and interpret accurate intraoperative fluoroscopic images. The sacrum is unique to each individual and may have an exaggerated L5 sulcus in the anterolateral sacral ala. This has been described as sacral dysmorphism and must be appreciated to safely place IS screws.² The slope of the sacral ala is variable but typically proceeds from posterior to anterior and medial to lateral. In the setting of sacral dysmorphism, there is an osseous void in the superior anterior aspect of the sacral ala, exactly where the L5 nerve root travels (**Fig. 67.1**). The fluoroscopic outlet image shows the cephalad aspect of the sacral ala, which actually correlates with the posterior border of the sacral ala and not the anterior aspect. Failure to recognize this relationship can result in an extraosseous screw and likely injury to the L5 nerve root.² The surgeon should understand the path of the sacral nerve roots as they proceed from the spinal canal out through their respective neuroforaminal tunnels. Each nerve proceeds obliquely from the spinal canal in a cranial to caudal, midline to lateral, and posterior to anterior orientation.³

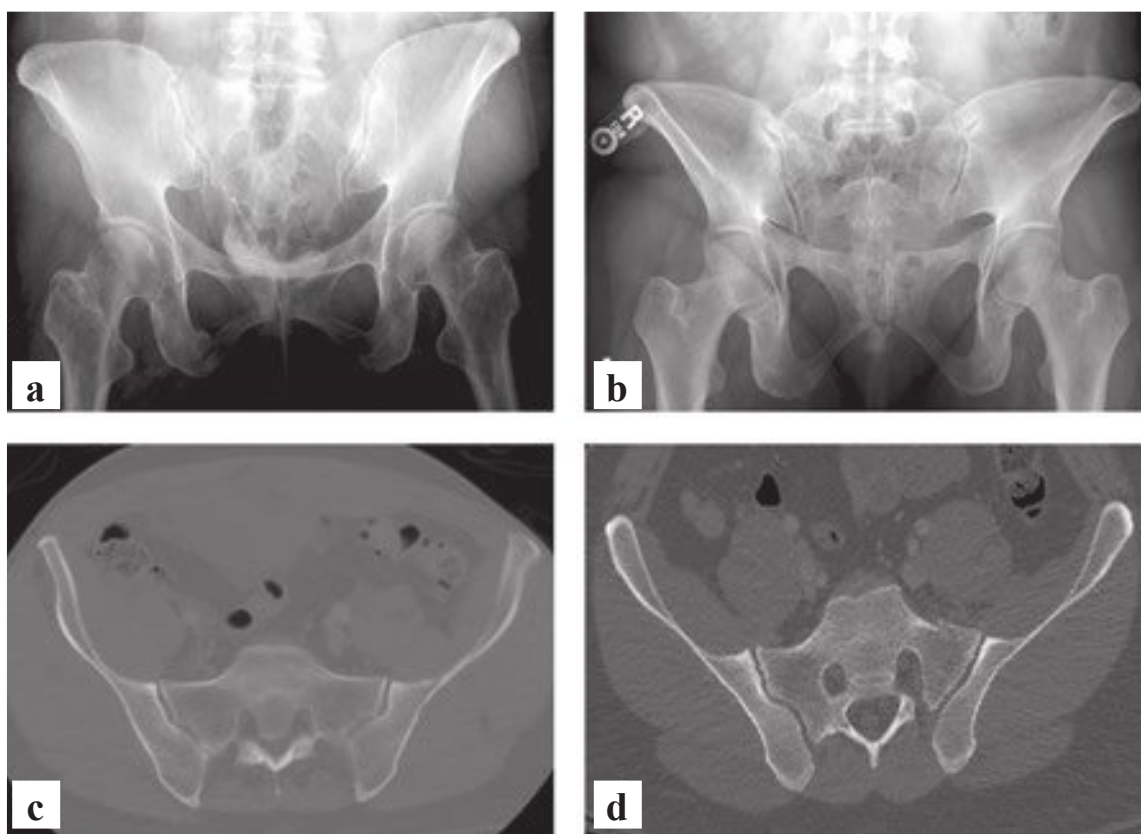


Fig. 67.1 AP pelvis and corresponding axial CT scan of (a,c) normal and (b,d) dysmorphic pelvis demonstrating the variability of the osseous posterior pelvic ring. Note the radiographic features of the dysmorphic pelvis, including collinear relationship of S1 to iliac crest, presence of mammillary processes, residual S1 disk space, large and irregular S1 neural foramen, and the acute downward slope of the sacral ala. The radiographic features correlate to a significantly diminished size of the S1 osseous fixation pathway and a more acute trajectory necessary for an IS screw to be contained within bone.

The following equipment is needed:

- Radiolucent operating table with appropriate positioning equipment/padding.
- Adequate fluoroscopic imaging unit (C-arm).
- Set of appropriately sized Kirschner wires, typically 1.6- or 2.0-mm.
- 4.5- or 5.0-mm cannulated drill.
- A complete set of partially and fully threaded large caliber cancellous screws, washers, and accompanying instrumentation. Specialized equipment may be needed for morbidly obese patients.
- External fixator and universal distractor equipment and tools for fracture manipulation.
- Equipment and table setup for skeletal traction as needed.

The patient may be placed either supine or prone depending on the injury pattern, physiologic status of the patient, need for an open reduction of the osseous injury, and subsequent or simultaneous procedures being performed. Either position requires all osseous prominences to be well padded with the arms positioned and supported appropriately.^{1,4}

If supine, a lumbosacral bump is needed. This typically consists of two operating room blankets that are folded in thirds and are placed directly in the center of the back with the lower end of the blanket ending at the genitals. By centering the blankets in the cranial/caudal and medial/lateral directions, there will be no change in the orientation of the pelvis. Positioning the patient toward the periphery of the bed may be needed for thin patients. This is because the angle required to place a screw can require the surgeon to drop his or her hand, which will be difficult if the patient is in the center of the operating table.⁴

The operative field is defined with adhesive impervious drapes. Care must be taken to prep and drape widely, from the undersurface of the buttock so as to not drape out the necessary skin incision. A large prepped area will allow the surgeon access to the entire pelvis.

- The start site is critical for successful placement of an IS screw, both at the skin as well as on the posterior ilium. The skin start site is generally found by drawing a line extending cranial from the greater trochanter in line with the femur and a line extending posteriorly from the anterior superior iliac spine. The typical start site is in the superior posterior quadrant of this diagram (**Fig. 67.2**). The exact position within this quadrant will depend on the angle of the screw being placed as well as the soft tissue envelope of the patient.
- The appropriate osseous start site and vector for the IS screw is planned by evaluating the preoperative CT scan. A 1.6- or 2-mm Kirschner wire is used first to identify the optimal start site. The smaller-caliber Kirschner wire minimizes the soft tissue damage caused by multiple trajectory corrections. The wire is placed through the skin and onto the posterior

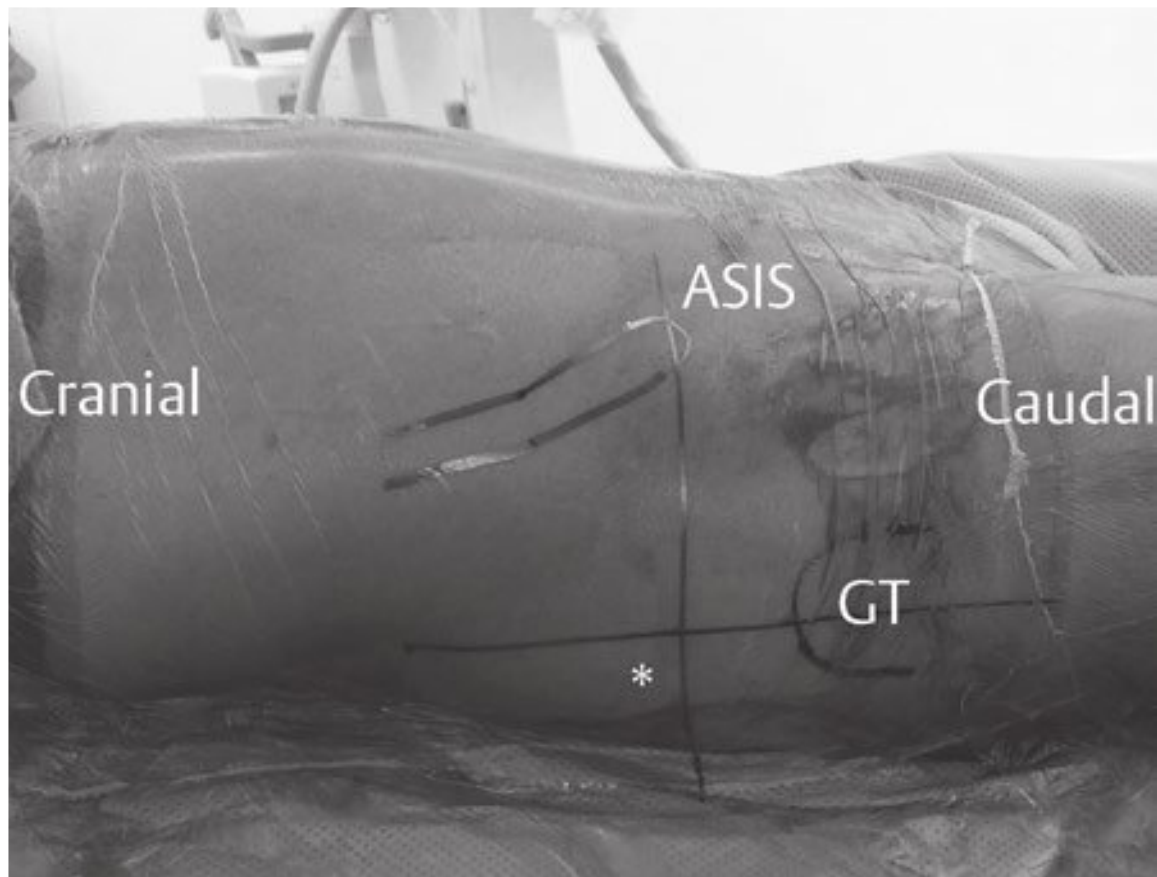


Fig. 67.2 Patient demonstrating supine positioning for IS screw placement. For orientation, the patient's head is to the left. The typical skin level for obtaining an appropriate start site is shown by the asterisk (*). This is located in the superior posterior quadrant of the field comprised of a line extending cranially from the greater trochanter (GT) in line with the femur and a line extending straight posterior from the anterior superior iliac spine (ASIS). Note how posterior draping is required to obtain the appropriate skin start site.

ilium in the anticipated vector. This is altered as needed at both the skin and ilium level and confirmed on inlet and outlet fluoroscopic views.

The appropriate inlet view for an S1 screw is achieved by having the fluoroscopic beam parallel with the anterior border of the S1 body. The appropriate outlet for an S1 screw will have the symphysis pubis overlying the S2 body. Once the defined start site has been confirmed, the wire is oscillated 1 cm into the posterior ilium (**Fig. 67.3**).

- A 1-cm oblique incision is made, and a cannulated drill is passed over the Kirschner wire. The drill is then oscillated into the bone 1 cm to swallow the wire. The more rigid drill then allows the surgeon to more easily make the necessary angular corrections.
- Using the appropriate inlet and outlet views, the drill is then oscillated through the posterior ilium, across the sacroiliac joint, and into the lateral sacral ala in the correct vector. The inlet view verifies the appropriate alignment in the anterior and posterior plane, whereas the outlet view visualizes the appropriate alignment and vector in the cranial and caudal plane. If needed, a correction in the drill trajectory is made by moving the drill into the appropriate plane based upon the fluoroscopic images. The drill is then spun in situ with a quick forward burst to create a new etch in the osseous tunnel and results in a new drill vector (**Fig. 67.4**).

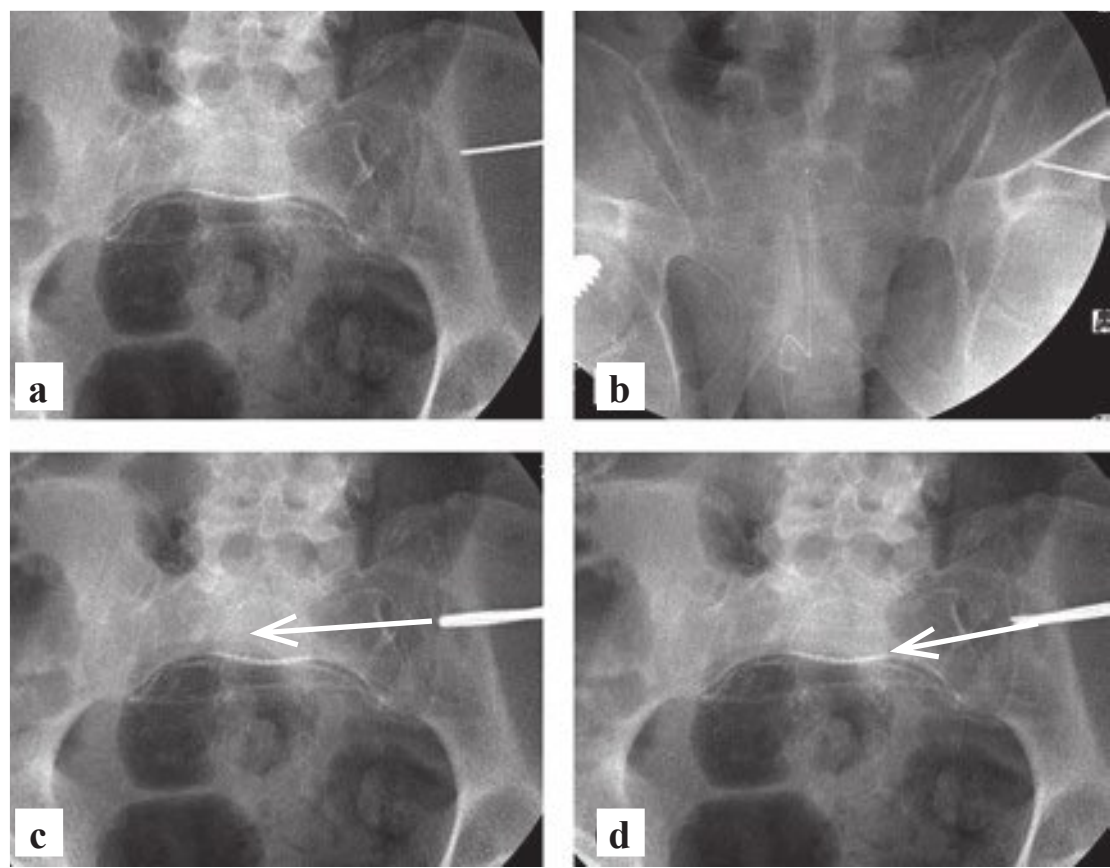


Fig. 67.3 (a) Fluoroscopic inlet view demonstrating wire start site and vector. (b) Fluoroscopic outlet view showing wire start site and vector. (c) Fluoroscopic inlet view after the cannulated drill has been inserted over the wire. Note the trajectory of the drill posteriorly (arrow). (d) Fluoroscopic inlet view after a correction has been made with the drill showing the corrected more anterior trajectory vector (arrow).

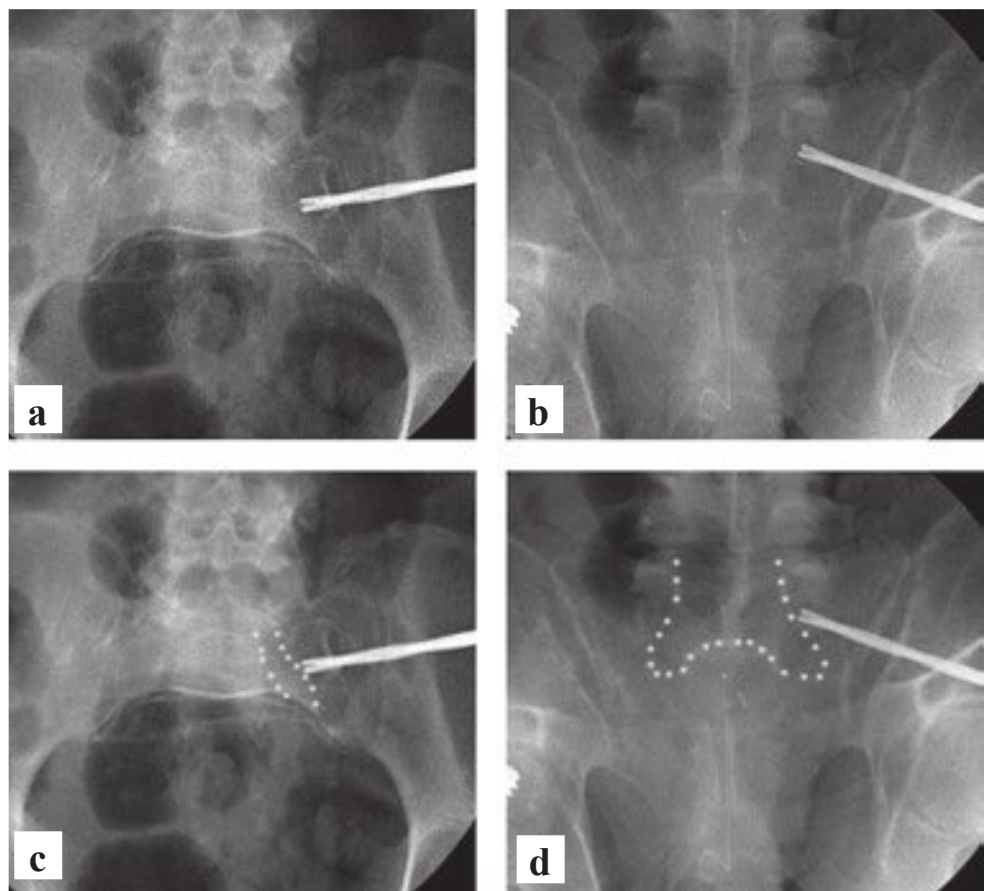


Fig. 67.4 (a,b) Fluoroscopic inlet and outlet views with the drill tip residing at lateral border of left S1 neuroforaminal tunnel. (c,d) Same fluoroscopic views with neuroforaminal tunnel highlighted by dotted lines to demonstrate subtle peripheral cortical densities of the neuroforaminal tunnel, as well as the contained lucency representing the central aspect of the tunnel.

- The drill is advanced using alternating inlet and outlet views until the drill reaches the lateral border of the ipsilateral neuroforaminal tunnel (**Fig. 67.4**). Once there, a lateral fluoroscopic view is obtained to verify drill position. If the drill tip is inferior to the iliac cortical density (ICD), anterior and superior to the neuroforaminal tunnel, and posterior to the anterior border of the sacral body, a safe pathway has been achieved (**Fig. 67.5**).
- Once satisfied, the drill is advanced the remaining 1 to 2 cm and verified on the inlet and outlet views. The drill is removed, and a larger caliber threaded guide wire from the cannulated screw set is then placed into the drill path. The wire is used as a probe to feel the drill path and ensure no osseous breach. Key areas to check are the anterosuperior aspect of the sacral ala lateral to the foramen and the posteroinferior aspect in the neuroforaminal tunnel/spinal canal zone medial to the foramen.
- The guide wire is then advanced into the final position. Care must be taken to not advance the wire or screw to far across midline, for exiting the bone in that location can endanger the common iliac vessels.
- The depth gauge is used to measure length of the desired screw. A guide wire of similar length can also be used with a subtraction measurement technique to verify screw length. A washer should be used, and 1 to 2 mm should be added to determine the final desired screw length.
- The screw and washer are inserted over the guide wire and tightened appropriately. One must not overtighten the screw and intrude the washer through the cortex. An oblique rollover image can be taken to visualize the lateral cortex of the posterior ilium (**Fig. 67.6**).

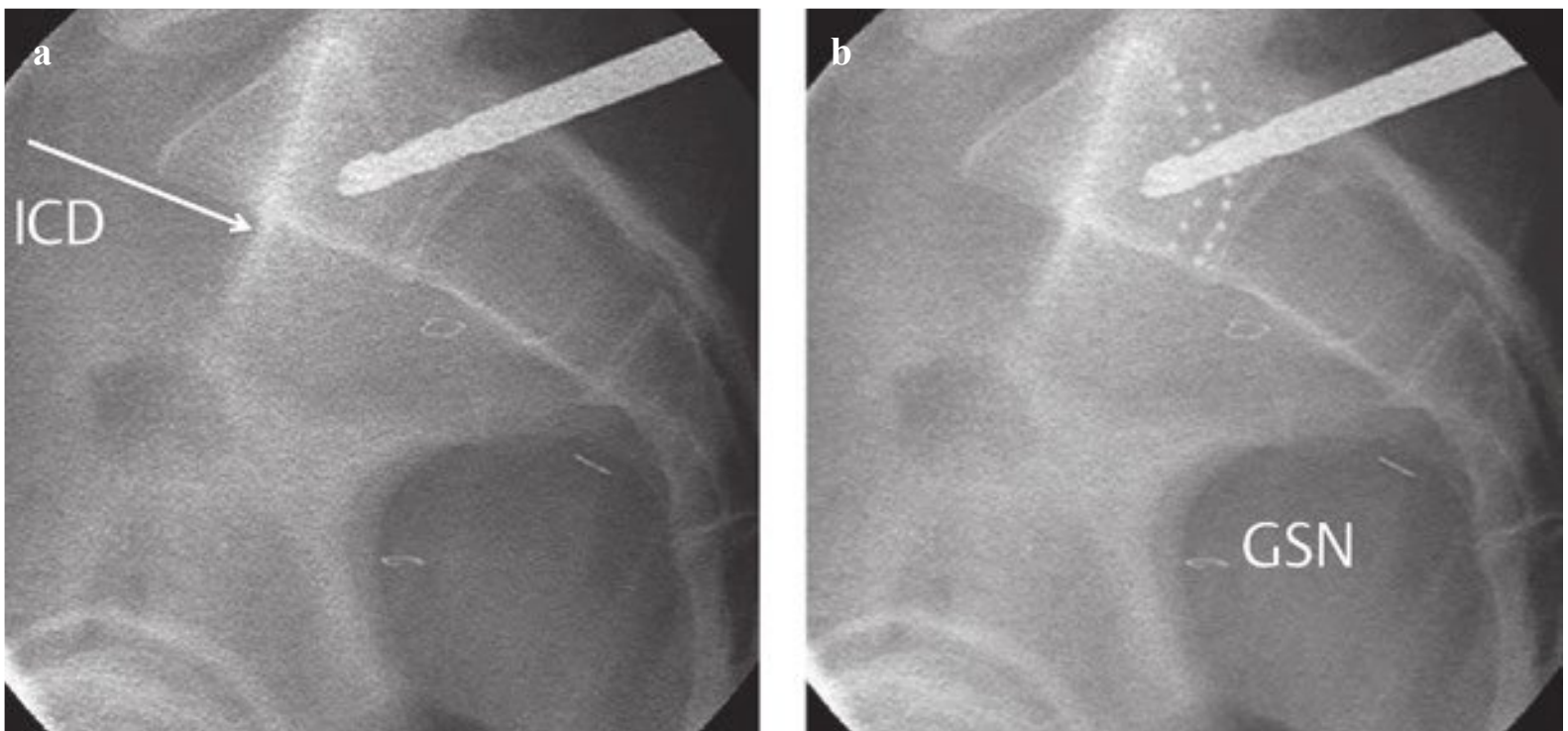


Fig. 67.5 (a) Lateral fluoroscopic view with drill tip residing at lateral border of neuroforaminal tunnel. Note that the drill tip is posterior and inferior to the iliac cortical density (ICD), posterior to the anterior border of the S1 body, and anterior and superior to the S1 neuroforaminal tunnel, confirming a safe position of the drill. (b) Lateral fluoroscopic view showing the oblique course of the S1 nerve root within the neuroforaminal tunnel (dotted lines). Greater sciatic notch, GSN.

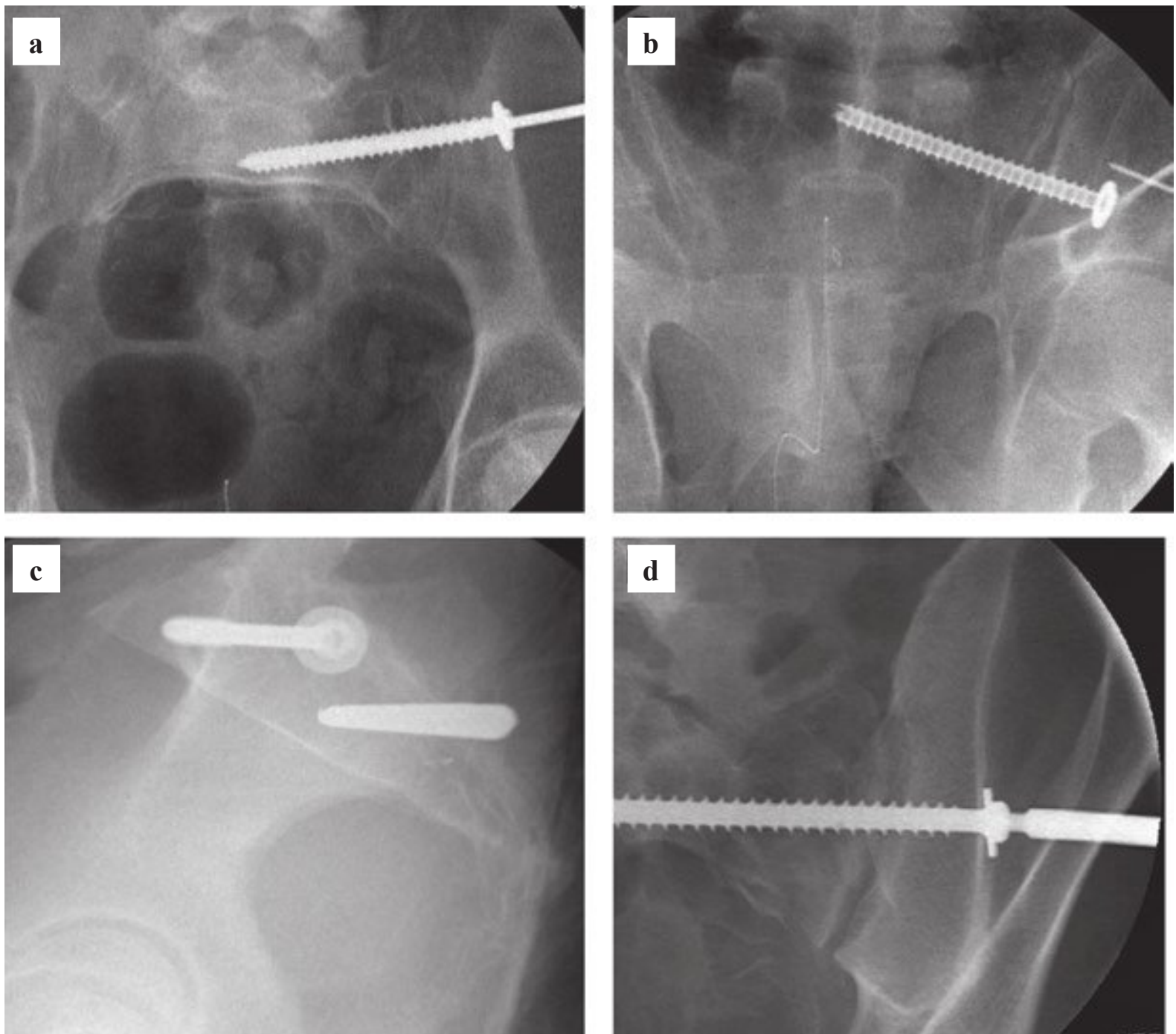


Fig. 67.6 Fluoroscopic (a) inlet, (b) outlet, (c) lateral, and (d) anteroposterior rollover views of S1 iliosacral screw in final position. Note that position of washer against cortex on rollover view is better appreciated as the fluoroscopic beam is parallel to the cortex of the posterior ilium.

- This process is repeated as needed to place screws of desired length and orientation depending on the injury, instability, bone quality, and the available osseous pathways.
- Incisions are irrigated and closed using nylon suture or staples.

67.4 Complications

- Complications related to IS screw placement are usually caused by inadequate appreciation of the pelvic anatomy. Variability in patient anatomy, osseous injury, and displacement are commonplace. Improper patient positioning, poor fluoroscopic imaging, or misinterpretation of good fluoroscopic imaging can also contribute to screw misplacement.

- Several neurovascular structures are in the vicinity of the posterior ilium and sacrum and can be damaged during this procedure. The L5, S1, or S2 nerve roots and the common iliac vessels are in close relationship to the sacrum. Injuring these structures can have catastrophic consequences. If a guide wire or drill is aberrantly placed into the greater sciatic notch, damage to the sciatic nerve, superior and inferior gluteal neurovascular bundles, pudendal nerve, and internal pudendal artery/vein could occur. In addition, branches of the superior gluteal neurovascular bundle are at risk during the procedure while passing through the abductor musculature.

67.5 Postoperative Care

- Prophylactic antibiotics are provided for 24 hours. Mechanical and chemical deep venous thrombosis (DVT) prophylaxis continues provided there are no additional contraindications.
- Postoperative neurological examination is compared with baseline, with particular attention to the L5 and S1 nerve distributions, these being most at risk during this procedure. The wound is monitored for signs or symptoms of infection, and postoperative imaging consisting of radiographs (AP, inlet, outlet) and CT scan (if desired) are obtained to verify reduction and proper implant positioning.
- Postoperative protocols are variable depending severity of the pelvic ring injury, the presence of other lower extremity injuries that limit weightbearing, fixation stability, and bone quality. For a unilateral sacral injury in a patient without other injuries, the patient is usually touchdown weightbearing on the injured side for 6 to 8 weeks followed by 4 to 6 weeks of progressive weightbearing. Full weightbearing should not begin before 8 weeks; some surgeons wait until 12 weeks. Radiographs are obtained at the 6-week, 12-week, 6-month, and 1-year time points to ensure that there is adequate healing and that there has not been any fracture displacement or implant loosening/failure.

67.6 Outcomes

- Because of the varied nature and severity of pelvic ring injuries as well as accompanying injuries, the outcomes associated with these injuries is variable.
- Outcome studies show variable results, but persistent pelvic pain as well as sexual dysfunction are common chronic issues.

67.7 Surgical Pearls

- Preoperative planning is critical for the safe placement of IS screws. The radiographs, axial CT scans, and sagittal and coronal reformats are carefully evaluated to understand the osseous injury, displacement of the hemipelvis, and available osseous fixation pathways.
- Knowledge of the pelvic anatomy, including sacral dysmorphism is vital to minimize complications and place IS screws safely.
- Fracture reduction, either through closed or open, is mandatory for safe IS screw placement.
- Safe placement of percutaneous IS fixation requires obtaining and interpreting intraoperative fluoroscopic images correctly, using the tactile feedback of the oscillating drill, and assessing the drill path with a guide wire.
- Without adequate intraoperative fluoroscopic imaging IS screw placement must be abandoned. Increased soft tissue density from morbid obesity, bowel contrast or bowel gas, and severe osteopenia can obscure intraoperative imaging requiring procedure delay or using alternative imaging techniques if available.

Common Clinical Questions

1. When placing an S1 iliosacral screw, a lateral fluoroscopic image taken intraoperatively with the drill tip medial to the neuroforaminal tunnel can help avoid damage to what structure?
 - A. Ureter
 - B. L5 nerve root
 - C. Sacroiliac joint
 - D. External iliac vessel
 - E. S1 nerve root
2. The path of the S1 nerve root proceeding from the spinal canal out to the ventral neural foramen is what?
 - A. Medial to lateral
 - B. Cranial to caudal
 - C. Posterior to anterior
 - D. A and B
 - E. All of the above

3. It is important to recognize the sacral alar slope when placing an iliosacral screw to avoid damage to what?
 - A. Ureter
 - B. L5 nerve root
 - C. Sacroiliac joint cartilage
 - D. External iliac vessel
 - E. S1 nerve root

Answers to Common Clinical Questions

1. B. The lateral view of the pelvis shows the iliac cortical density, the anterior sacral body cortex, and the neuroforaminal tunnel. The area between these three landmarks contains a safe osseous path for screw placement. If a lateral view is checked when the drill tip is lateral to the foramen and the tip is centered in the safe zone, the surgeon knows the L5 nerve root is safe. The drill tip has not proceeded medially into the zone of the neuroforaminal tunnel and spinal canal, and thus the S1 nerve and spinal canal cannot be in danger. After the drill tip is medial to the foramen, checking a lateral will be able to confirm the location of the drill in space and specifically mark its location in relation to the S1 neuroforaminal tunnel.
2. E. The S1 nerve root proceeds from the spinal canal out to the ventral neural foramen through the corticated neuroforaminal tunnel. The oblique course proceeds from medial to lateral, cranial to caudal, and posterior to anterior.
3. B. The sacral alar slope is a marker of the upper osseous limit of the S1 body. The degree of alar slope is variable but typically proceeds from medial to lateral and posterior to anterior. This is also a marker of the location of the L5 nerve root, as it sits directly on the sacral ala.

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68 Sacroiliac Joint Fusion

Kornelis Poelstra and Jessica Sosio

68.1 Key Points

- Traumatic disruption of the sacroiliac joint (SI joint) used to be the main reason for SI joint stabilization and fusion procedures.
- Fixation choices and fusion technique options have increased recently with the advent of minimally invasive surgery (MIS) techniques to stabilize the SI joint.¹
- Novel MIS techniques are now primarily used to treat painful degenerative conditions of the SI joint.
- Careful work-up with thorough physical examination and positive responses to diagnostic SI joint injections could identify a surgical candidate, but only after conservative approaches are truly exhausted.
- Despite the reduced morbidity of this procedure with the use of MIS techniques, long-term success of SI joint stabilization for degenerative conditions still needs to be determined.

68.2 Indications

- Traumatic SI joint injuries with angular, superior–inferior, or rotational instability of the joint in connection with pelvic ring disruption.
- Degeneration of the SI joint characterized by upper buttock pain just below the level of the posterior superior iliac spine (PSIS) with positional changes (sit-to-stand), climbing or descending stairs or single-leg stance, refractory to nonoperative care, after all other sources are ruled out. Provocative tests can be performed but by themselves (e.g., Patrick’s and Gaenslen’s tests) lack validity.²
- Theories of pain generation include ligamentous or capsular tension, extraneous compression or shear forces, hypomobility or hypermobility, aberrant joint mechanics, and myofascial or kinetic chain imbalances resulting in inflammation and pain.³
- Intra-articular sources of sacroiliac joint pain include osteoarthritis and infection.
- Extra-articular sources of sacroiliac joint pain include enthesitis/ligamentous sprain and osseous sources such as a sacral insufficiency fracture. It may also be caused by primary enthesopathy, hip joint cam impingement,⁴ or (not even just after long constructs) lumbar spinal fusion caused by stress transfer to the SI joint.^{5,6}
- Neoplasm with SI joint instability.

68.3 Techniques

- Direct open reduction of a traumatically disrupted SI joint is occasionally necessary but requires large, bloody exposures. Indirect reduction using intraoperative imaging (fluoroscopy or navigation) and traction is often more appealing, followed by compression lag screw fixation from lateral to medial, aiming for the S1 promontorium to improve screw pullout strength. Bone grafting techniques are rarely used, for scarring and ligamentous healing stabilize the joint over time.
- High-energy sacral fractures involving the SI joint often require sacro- or spinopelvic fixation with medial-to-lateral iliac screw placement and crosslinking, with the possibility for destructive dissection when performed open, especially when associated with degloving injuries of the soft tissues (**Fig. 68.1 a–c**).

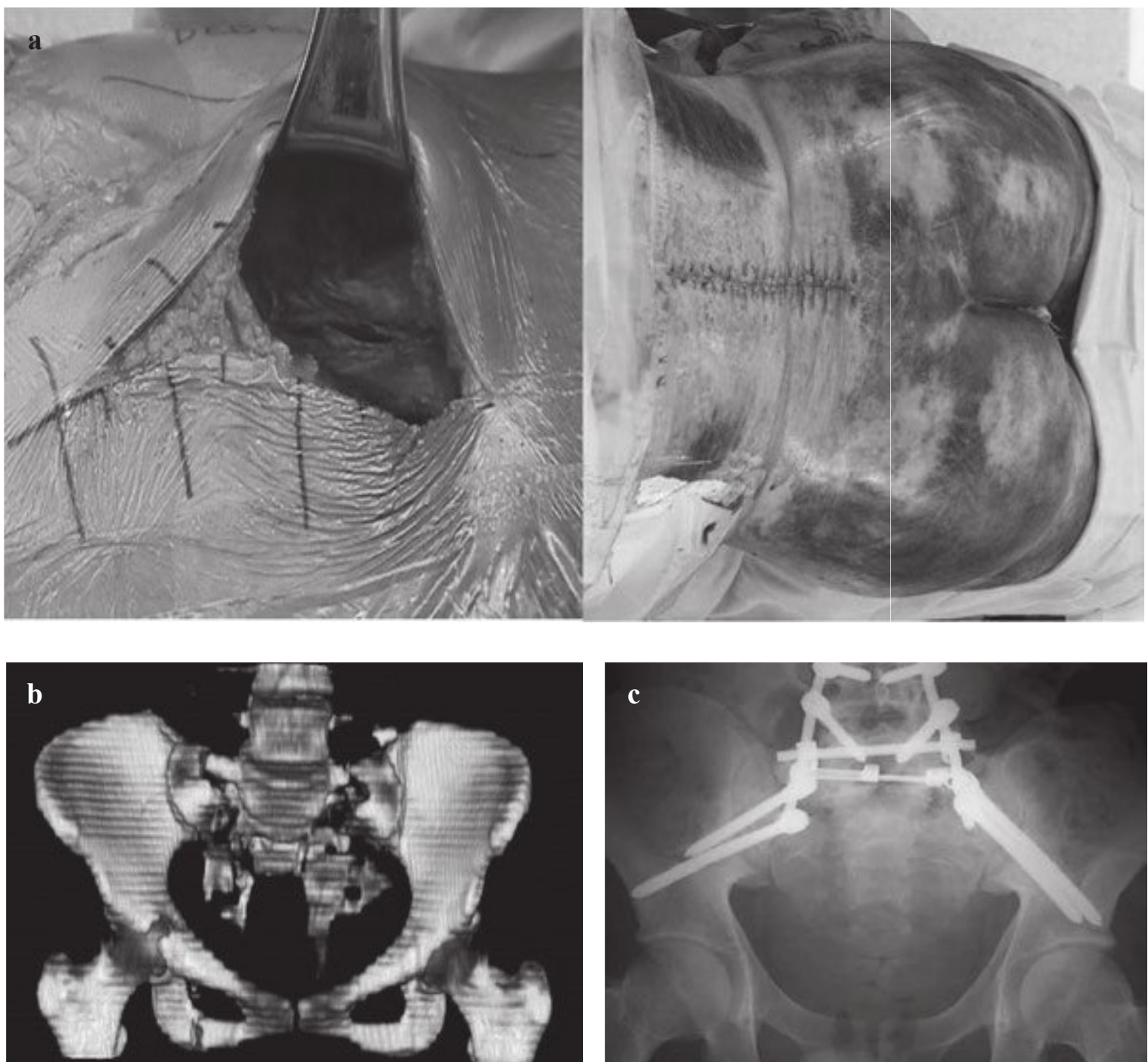


Fig. 68.1 (a) Degloving injury associated with (b) complex bilateral sacral fractures, severely disrupting lumbopelvic stability with (often) neurologic injury. (c) Here after fixation to the lower lumbar levels.

Minimally invasive sacral pedicle screw placement and iliac teardrop targeting has decreased the morbidity of this procedure by making it virtually percutaneous.

- For degenerative conditions, the open approach to SI joint fusion used to be technically demanding, with the creation of a true bony window through the iliac wing into the SI joint from direct lateral. This could also be accomplished via an (often bloody) wide-open anterior or a more limited posterior approach through the posterior fascial reinforcements over the joint, connecting the osseous structures to the gluteus minimus, maximus, iliotibial band, and erector spinae.⁷

After decortication and bone grafting, fixation can be accomplished with lateral-to-medial screws for compression across the joint.

- MIS techniques have recently become increasingly more popular with the use of percutaneous screw placement, often without true bone grafting of the joint itself. Multiple techniques have been described, using direct lateral fluoroscopic or navigational techniques for visualization of the SI joint. The three main MIS surgical techniques for operative treatment of degenerative SI joint pain are as follow:
 - 1. Placement of (ideally three) semiparallel triangular metal bars from lateral to medial across the SI joint (SI Bone iFuse Implant System [San Jose, California]) (**Fig. 68.2a,b**).
 - Motion-free stabilization is anticipated to occur over time via bony ongrowth onto the plasma-sprayed roughened bars on either side of

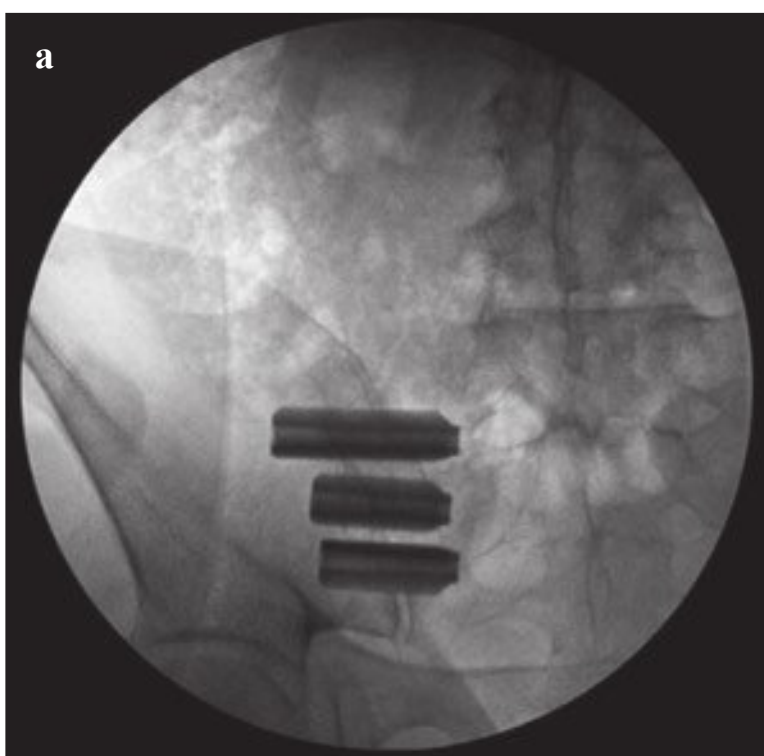


Fig. 68.2 (a) Radiographic image of bar stabilization across the SI joint, (b) with representation of the SiBone coated fusion bars for MIS SI joint stabilization.

the joint within the cortex, inside the trabecular bone of the ilium and sacrum without true decortication or bone grafting of the joint itself. Preliminary data have shown that some bony bridging adjacent to the bars across the joint can occur over time.^{8,9}

- 2. Compression screw placement after percutaneous decortication of the joint via specialized tools to accomplish a solid arthrodesis (SImmetry SI Joint Fusion System: Zyga Technologies, Inc.; Minnetonka, Minnesota). The SI joint is targeted from lateral, and after creation of a window through the ilium, specialized scrapers and expandable 90° reamers are advanced into the joint (**Fig. 68.3a–d**). These allow for a 5 cm² area of decortication and impaction bone grafting around a large compression screw subsequently introduced through the same access portal across the joint into the sacrum. At least one additional screw

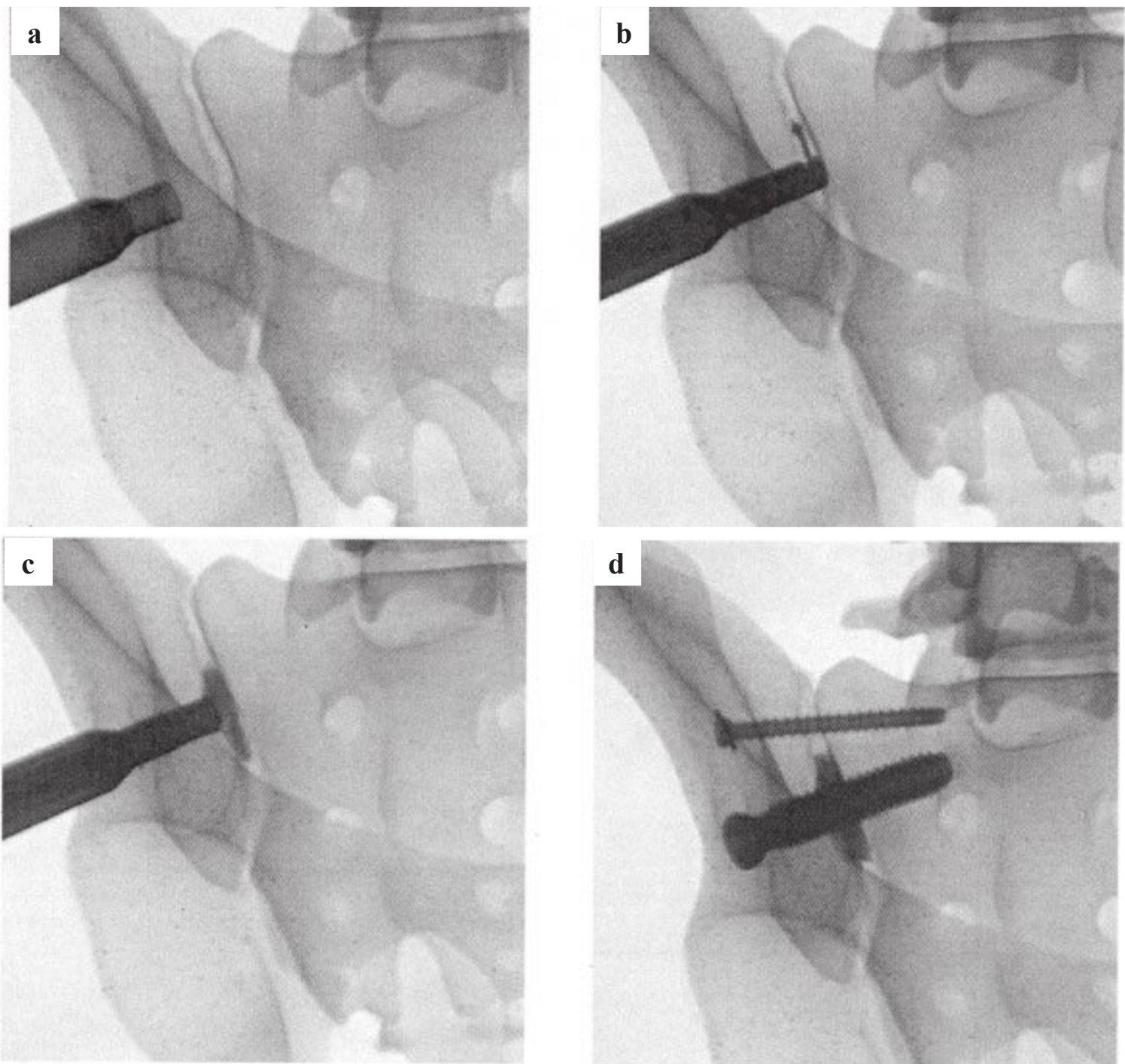


Fig. 68.3 Schematic representation of the Zyga SImmetry system for SI joint fusion stabilization. **(a)** After access perpendicular to the SI joint, **(b)** decortication of the joint surfaces is made possible by special 90° “rasps.” **(c)** Bone graft can be placed in between the decorticated joint surfaces first, after which **(d)** compression lag screw fixation is used for stabilization.

needs to be placed (typically superiorly to this) to provide for rotational stability.¹⁰

- 3. Large-diameter, cannulated compression screw placement across the SI joint from a variety of companies, either from a lateral (e.g., **Fig. 68.4a,b**; slotted and coated screws: Globus Medical Inc., Parsippany, New Jersey) or posterolateral approach (e.g., **Fig. 68.5**; compression

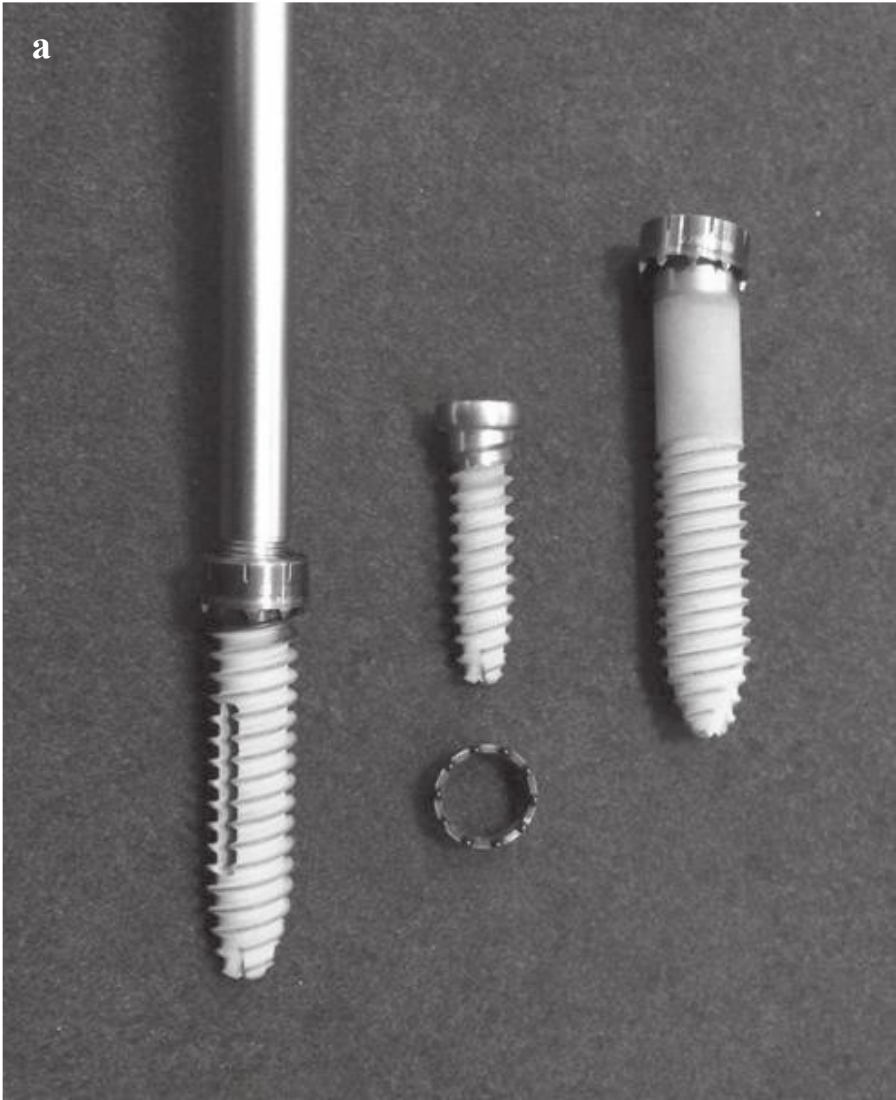


Fig. 68.4 (a,b) Coated, slotted screws for fixation across the SI joint from a lateral approach with the ability to allow for bone grafting and bone growth inside the large lag screw for SI joint fusion.

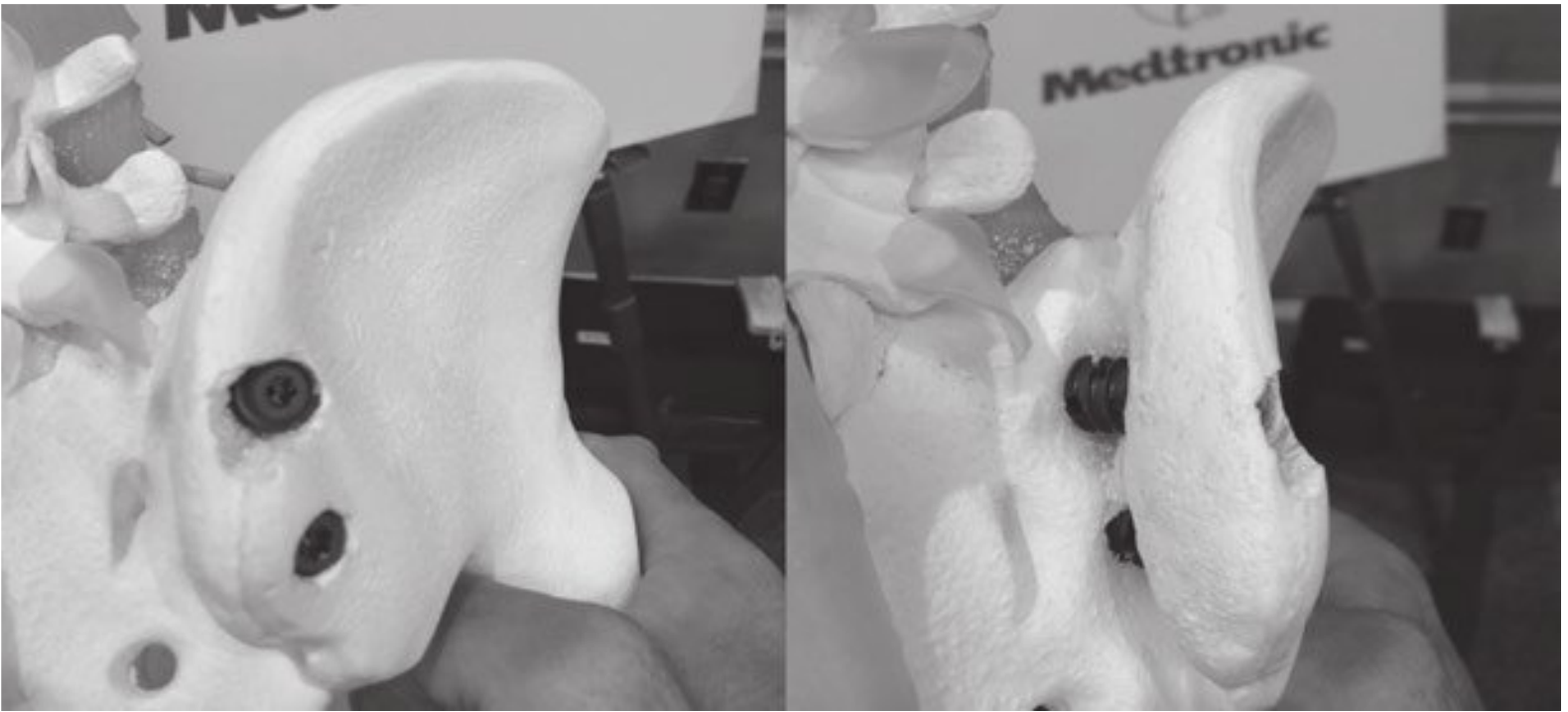


Fig. 68.5 SI joint stabilization with lag screw fixation from a posterolateral angle.

screws: Medtronic, Inc., Memphis, Tennessee), using interference stability by high-pressure lag technique compression to prevent motion of the SI joint.

68.4 Complications

- Vessel or nerve root injury resulting from improper guidewire, drill, or implant positioning, often caused by compromised SI joint visualization
- Wound healing complications after wide dissection for open treatment of sacral fractures with exposed hardware through an area of wound breakdown
- Pseudoarthrosis or implant loosening, as there is often no direct arthrodesis possible, or even attempted, across the joint with the percutaneous techniques (**Fig. 68.6a,b**)
- Failure to improve patient's discomfort, as the SI joint turned out not to be the (sole) pain generator

68.5 Postoperative Care

- If stable fixation is achieved, no external immobilization is required for traumatic SI joint disruptions. Depending on the degree and direction of preoperative instability, weightbearing on the injured side may need to be reduced temporarily.
- Guarded weightbearing is typically advised after operative stabilization of the SI joint for degenerative conditions, for no longer than 6 weeks postoperative.

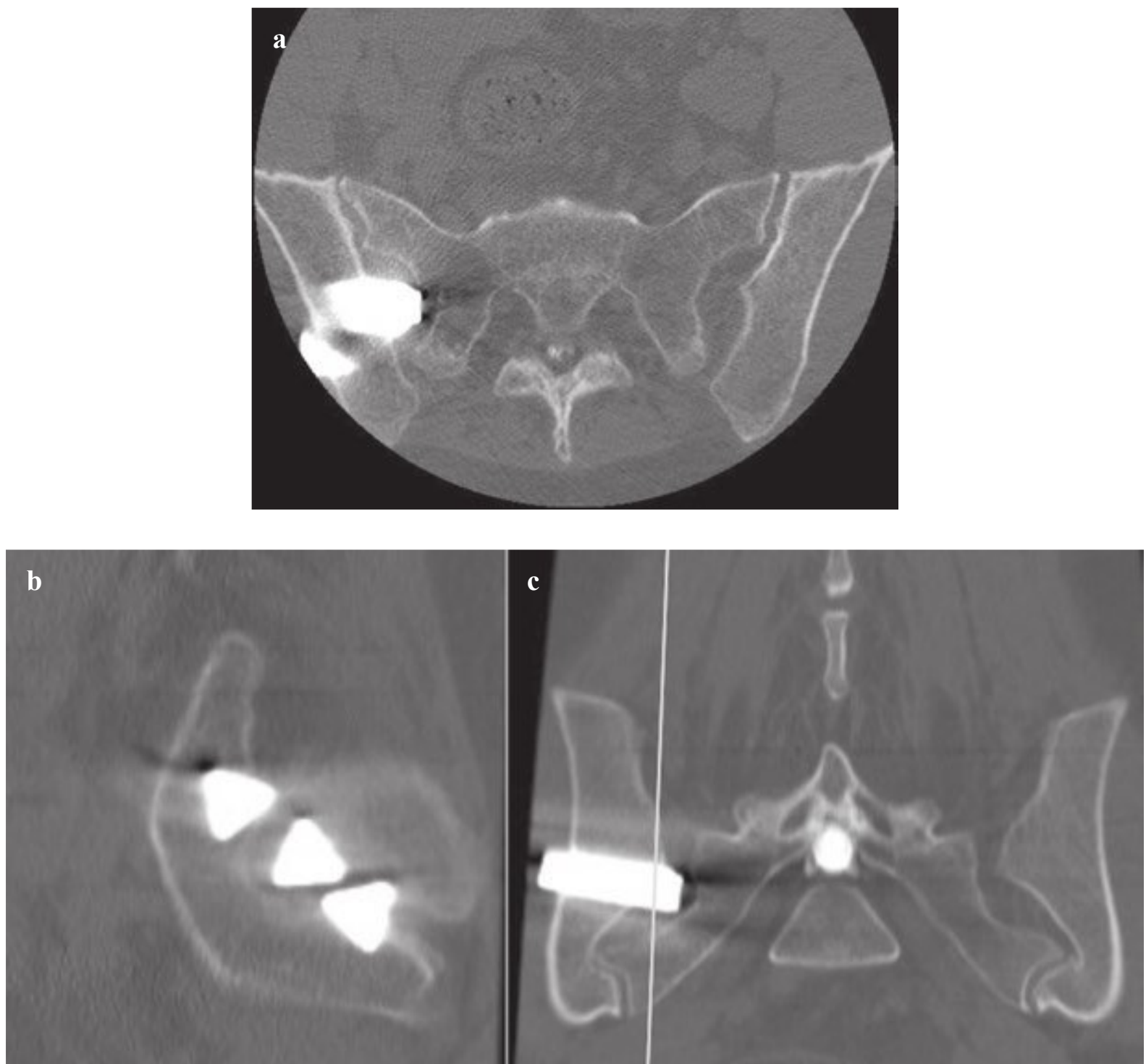


Fig. 68.6 (a) SI joint stabilization attempt with triangular bars without crossing into the sacrum due to location of the bar compared with the joint. This plays a significant role in failure of this MIS “fusion” procedure. MIS fusion on (b) parasagittal and (c) axial CT images in another patient showing inadequate stabilization, leading to ipsilateral lumbar plexus irritation and foot dorsiflexion weakness.

68.6 Outcomes

- Pelvic ring disruptions involving the SI joint(s) are typically high-energy injuries rendering the patient forever “different” from his or her state before the trauma, regardless of the surgical procedure or approach. If the soft tissue injury did not lead to wound breakdown and hardware exposure/failure of fixation and infection with osteomyelitis, the outcomes are fair to good. If wound breakdown occurs, the outcome can be disastrous.
- An abundance of recent literature describes impressively positive outcomes after SI joint “fusion” surgery for degenerative conditions. Most

of these articles, however, have small series, and many of the authors are SI joint fusion company–endorsed.

- Placement of percutaneous nonlag compression bars across the joint should be carefully considered if the anatomy might not allow for adequate stabilization using this technique (**Fig. 68.6a,b**).

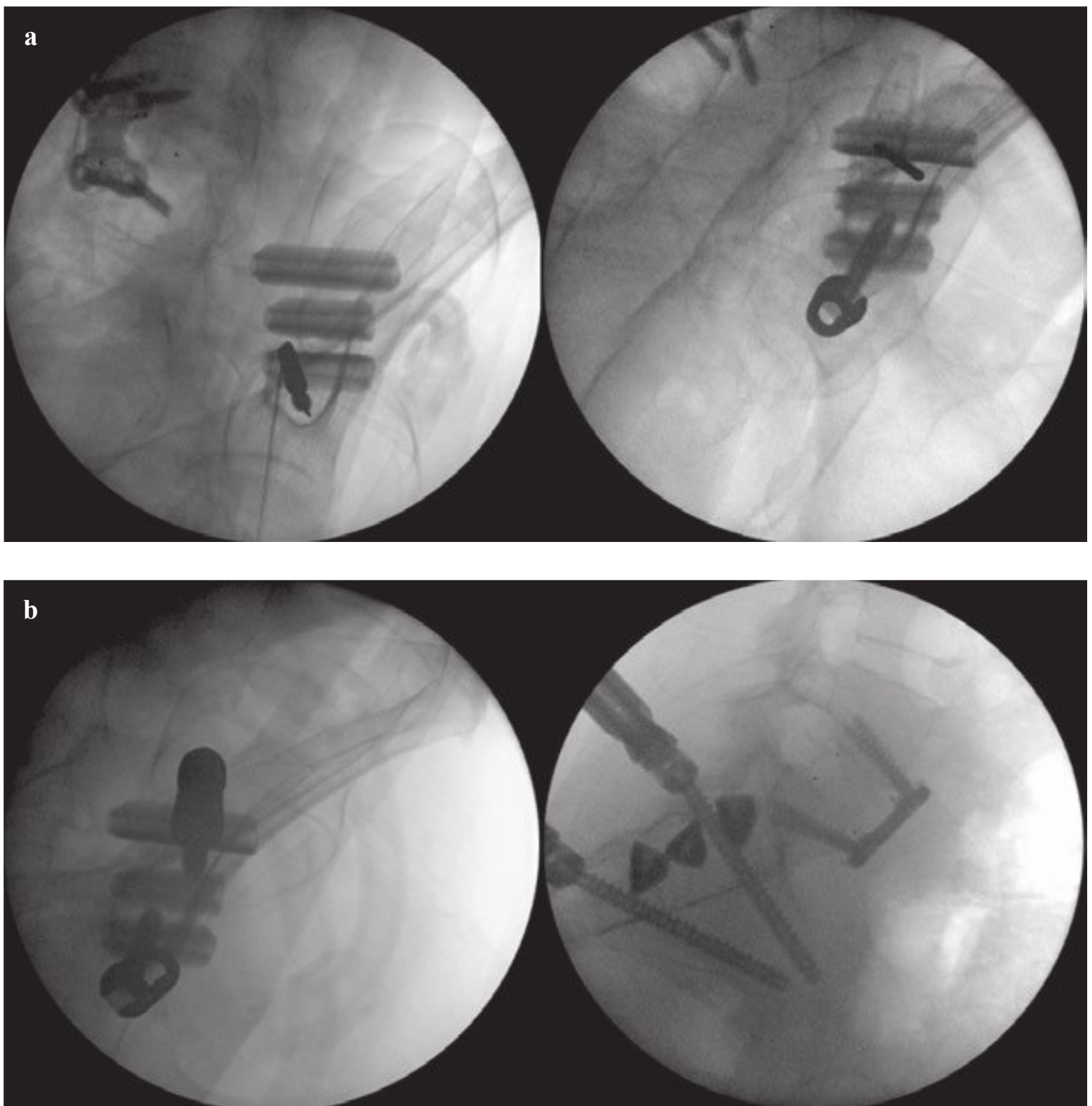


Fig. 68.7 Salvage procedure in a 41-year-old female with prior SI joint stabilization failure and confirmation of the SI joint as her pain generator after lidocaine infiltration of the joint space: **(a)** Modified Ferguson view with fluoroscopy to access a safe corridor between the greater sciatic notch (below) and the most posterior triangular bar across the SI joint using a Jamshidi needle. **(b)** A second screw should be placed into the iliac wing where possible for rotational stability, here performed through another “open” corridor, visualized with Ferguson angle modification. (Continued on page 464)

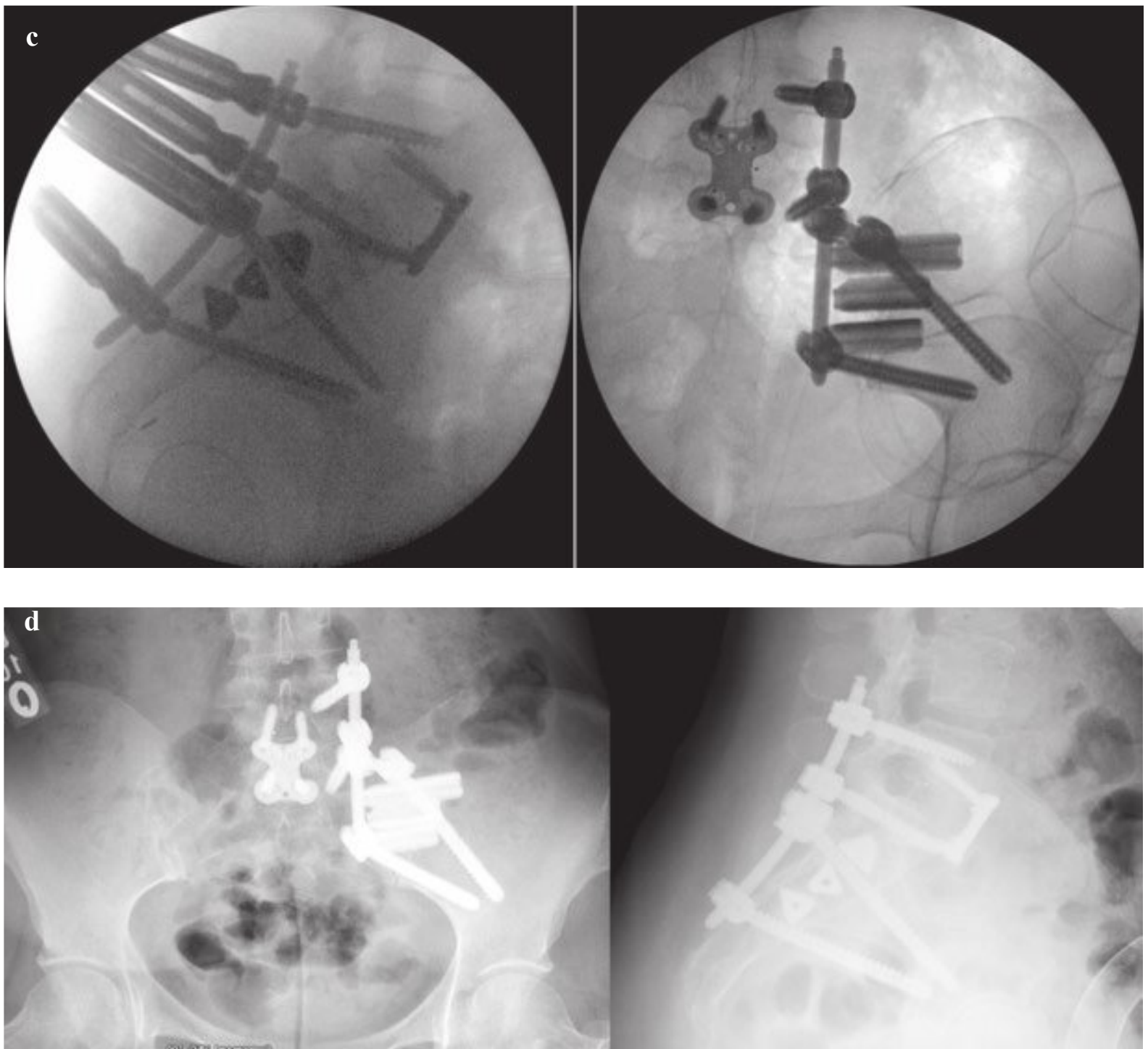


Fig. 68.7 (Continued) **(c)** Ultimately, an MIS posterior construct was created across the stable L5–S1 fusion with two points of fixation above and below the SI joint. **(d)** This was followed by SI joint decortication and bone graft insertion from posterior, leading to excellent stability on standing radiographs. Clinical success is now more likely but is certainly not guaranteed.

68.7 Surgical Pearls

- Fixation of traumatic SI joint disruption is often complex and should be done in specialized centers by surgeons well versed in the technique.
- Preoperative work-up and planning is crucial for operative treatment of degenerative SI joint disease. All other sources of pain need to be excluded, and a thorough understanding of the three-dimensional anatomy is essential when performing the first MIS SI joint “fusion” surgeries.
- Keep the possibility of failure of addressing the pain generator effectively with this technique, and counsel the patient appropriately.

- Salvage of nonhealing MIS SI joint fusion surgery (if confirmed to be the pain generator) is often challenging and typically requires at least *two* iliac wing screws for rotational stabilization of the hemipelvis. Fixation into the lower lumbar spine, superior to the sacrum, is recommended to achieve adequate stability. Placement of iliac bolts between screws or bars crossing the nonfused SI joint can be challenging and requires modification of Ferguson angle fluoroscopy to visualize a safe corridor (**Fig. 68.7a,b**). After stable fixation has been achieved (**Fig. 68.7c**), open posterior bone grafting in between the remaining joint surfaces is the author's preferred method for fusion (**Fig. 68.7d**).

Common Clinical Questions

1. What is the ideal fluoroscopic “view” for percutaneous iliac screw placement to stabilize a traumatically disrupted SI joint?
2. What are pathognomonic signs and symptoms to look for while working up a patient for symptomatic SI joint degeneration?
3. In what position do you position the fluoroscopy machine when placing percutaneous lateral bars or screws across the SI joint?

Answers to Common Clinical Questions

1. Start with placing the fluoroscopy machine parallel to the superior part of S1 (promontorium) and the inner and outer tables of the iliac wing to be instrumented in approximately a “30° Ferguson” and “30° Rainbow” view. The goal is to aim parallel to the Teardrop as visualized in **Fig. 68.8**. The inferior margin of the Teardrop (A) is the sciatic notch (B), and this needs to be in line with the superior lateral part of the acetabulum (C), called the “Sourcele.” The view needs to be parallel to the inner and outer table. The bony column visualized can accept at least one screw—if needed, two screws—measuring 7.5 to 10 mm in diameter by 8 to 10 cm long.
2. Patients complaining of maximal pain below L5 together with sacral sulcus tenderness upon palpation and pain originating lateral to the midline around the posterior superior iliac spine have the highest positive predictive value (60%) of causalgia with SI joint–generated discomfort.
3. Initial SI joint visualization needs to be done with a “perfect” lateral view to the joint, overlapping the sciatic notches and aiming parallel to the superior end plate of the S1 vertebral body. This identifies the starting point for entry into the SI joint from lateral.

A pelvic outlet view (Ferguson view), obtained in straight midline anterior–posterior and parallel position to the S1 superior end plate

allows the visualization of the sacral neuroforaminae. The straight midline anterior–posterior inlet view (90° perpendicular to the outlet view) shows the ventral cortex of the sacrum, identifying the safe direction and depth for lateral-to-medial screw placement.

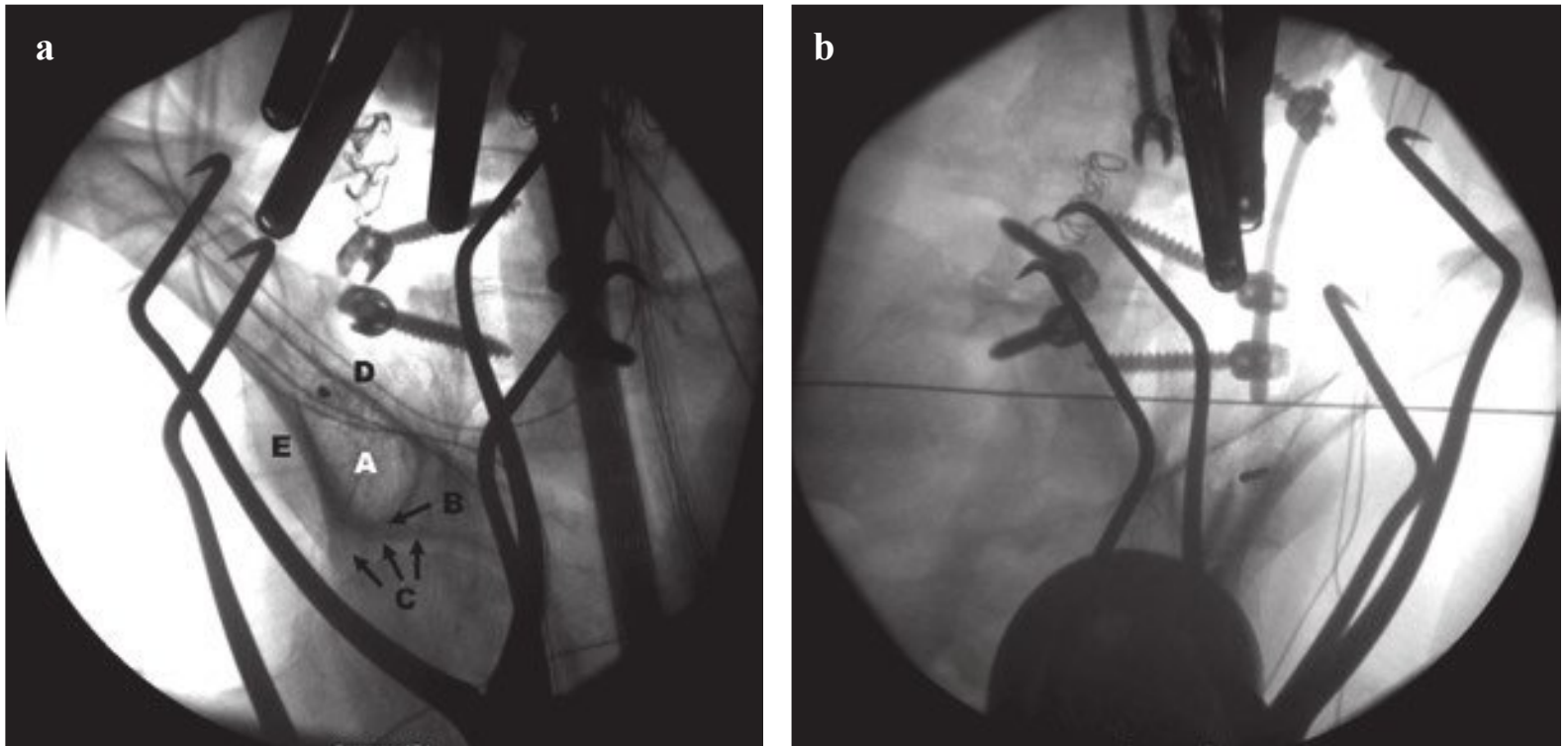


Fig. 68.8 (a) Intraoperative teardrop visualization of the teardrop is possible, even in complex surgery with wires and retractors in the way, and (b) after a total hip arthroplasty. A = teardrop. B = Inferior margin on the immediate teardrop is the greater sciatic notch. C = The sciatic notch needs to be aligned with the superior–lateral margin of the acetabulum, visualized as the sclerotic densification of the sourcele (or “eyebrow”). D = The medial wall of the teardrop is the inner table of the iliac wing. E = The lateral margin is the lateral wall of the iliac wing. Both the medial and lateral walls need to be parallel to the fluoroscopy beam. Choosing a starting point in the teardrop and staying with a Jamshidi needle within the medial and lateral walls and above the greater sciatic notch and the sourcele will ensure a safe trajectory of at least 100 mm and will prevent penetration into the hip joint.

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69 Sacrectomy

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and Ziya L. Gokaslan

69.1 Key Points

- A percutaneous CT-guided biopsy should be done in cases in which diagnosis would influence the decision to proceed with surgery or the extent of surgery.
- It is important to obtain wide negative margins in tumor resection.
- Adjuvant therapy with proton beam or intensity-modulated radiation therapy should be used for contaminated margins or incomplete resections.

69.2 Indications

- Benign tumors of the spine with aggressive invasion of local structures (e.g., osteoblastoma, giant cell tumors, osteochondromas, aneurismal bone cysts, notochordal rests)
- Malignant primary tumors of the sacrum (e.g., chordoma, osteochondroma, osteosarcoma, Ewing's sarcoma, primitive neuroepithelial tumor [PNET])
- Tumors of the pelvis (e.g., osteosarcoma, chondroma, Ewing sarcoma) or pelvic viscera (locally advanced adenocarcinoma of the rectum) invading the sacrum or sacroiliac joint
- Some situations of metastatic tumors of the sacrum causing significant destabilization or pain

69.3 Preoperative Management

- A low-residue diet for several days preop and mechanical cleaning of the bowel by having the patient drink GoLytely will help decrease the risk of inadvertently entering the bowel, especially with anterior approaches.
- Because there can be considerable blood loss, good vascular access with large-bore intravenous catheters and central venous monitoring is necessary, and the surgeon should consider angiographic preoperative embolization if amenable.

- Because the lower extent of posterior sacral incisions is very near to the anus and risk of infection is high, prophylactic antibiotics must be used preoperatively and postoperatively.

69.4 Technique

- Sacrectomy may be performed as a standalone operation or in the context of complex sacropelvic resection (e.g., extended internal or external hemipelvectomy) (**Fig. 69.1**).

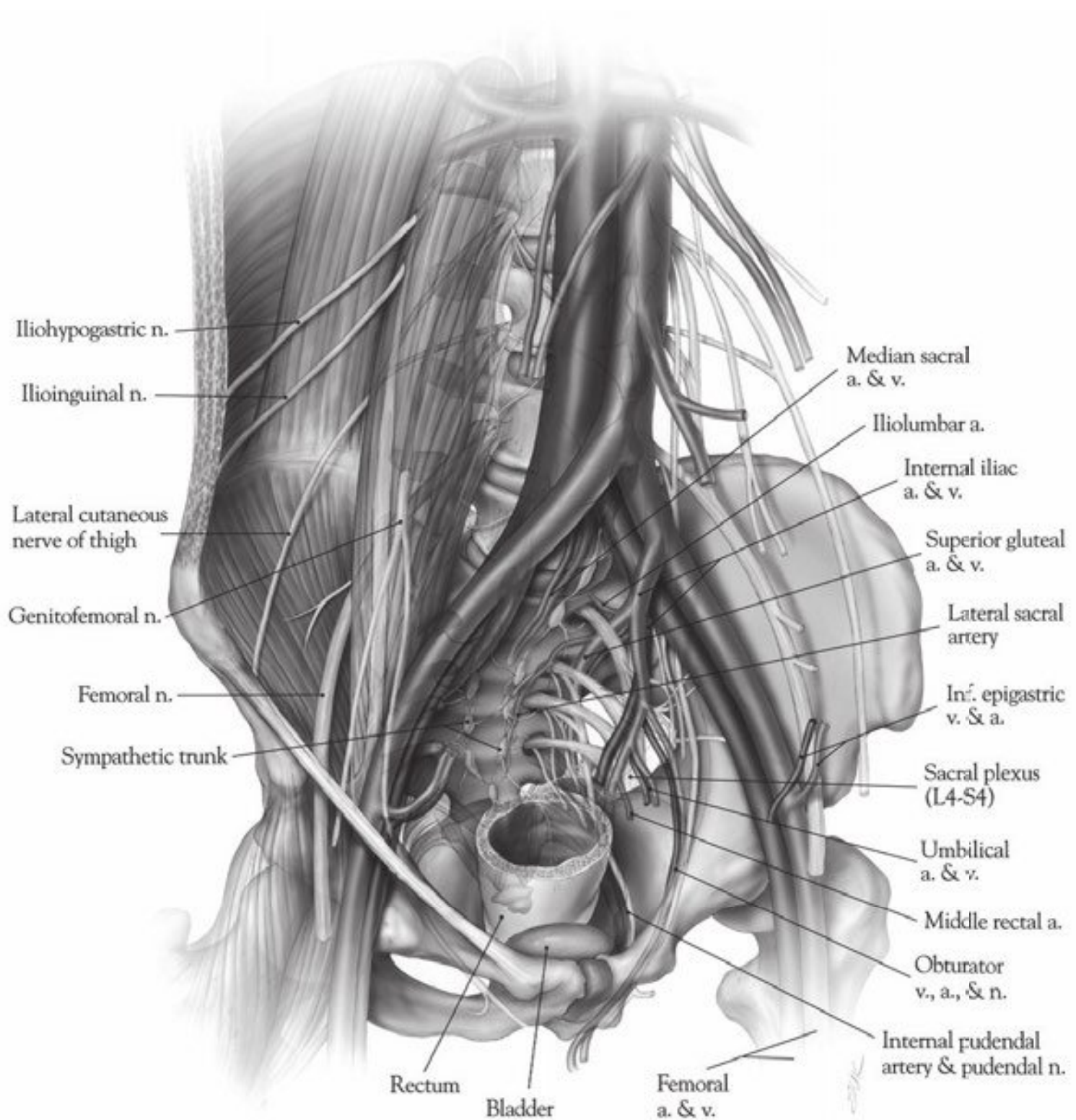


Fig. 69.1 Relationship of the ventral surface of the sacrum to the major pelvic structures, arteries, veins, and neural structures. (From Dickman CA, Fehlings MG, and Gokaslan ZL, eds. *Spinal Cord and Spinal Column Tumors*, Thieme; 2006.)

- Determine whether a partial or total sacrectomy is required and, if a partial sacrectomy is required, whether the osteotomies will be done in the axial or sagittal plane (**Fig. 69.2** and **Fig. 69.3**).¹
- Depending on the tumor location, extent of invasion, and tumor type, a transperineal, posterior, or combined ventral and dorsal approach may need to be used.¹

69.4.1 Perineal Approach

- Allows access to the caudal presacral space and can be used alone for cyst drainage or tumor biopsy but is most frequently used in combination with the posterior midline exposure for en bloc resection of lesions at or below the S3 level.
- The patient is positioned in the Kraske position (flexed prone) with a midline incision extending rostrally from the coccygeal region.

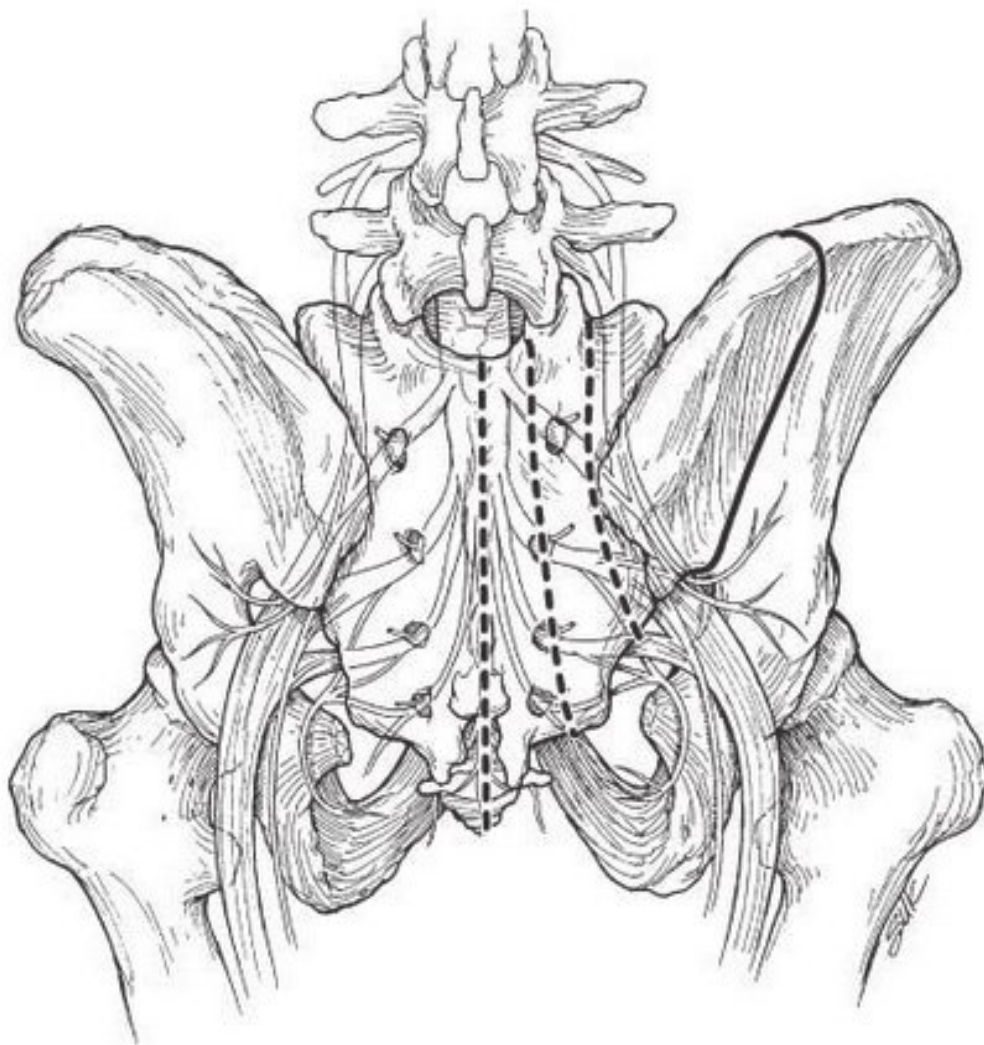


Fig. 69.2 The solid line demarcates the osteotomy for a standard lateral internal hemipelvectomy; Dotted lines demarcate osteotomies for progressively larger resections of the sacroiliac joint and hemisacrectomy. (From Dickman CA, Fehlings MG, and Gokaslan ZL, eds. *Spinal Cord and Spinal Column Tumors*, Thieme; 2006.)

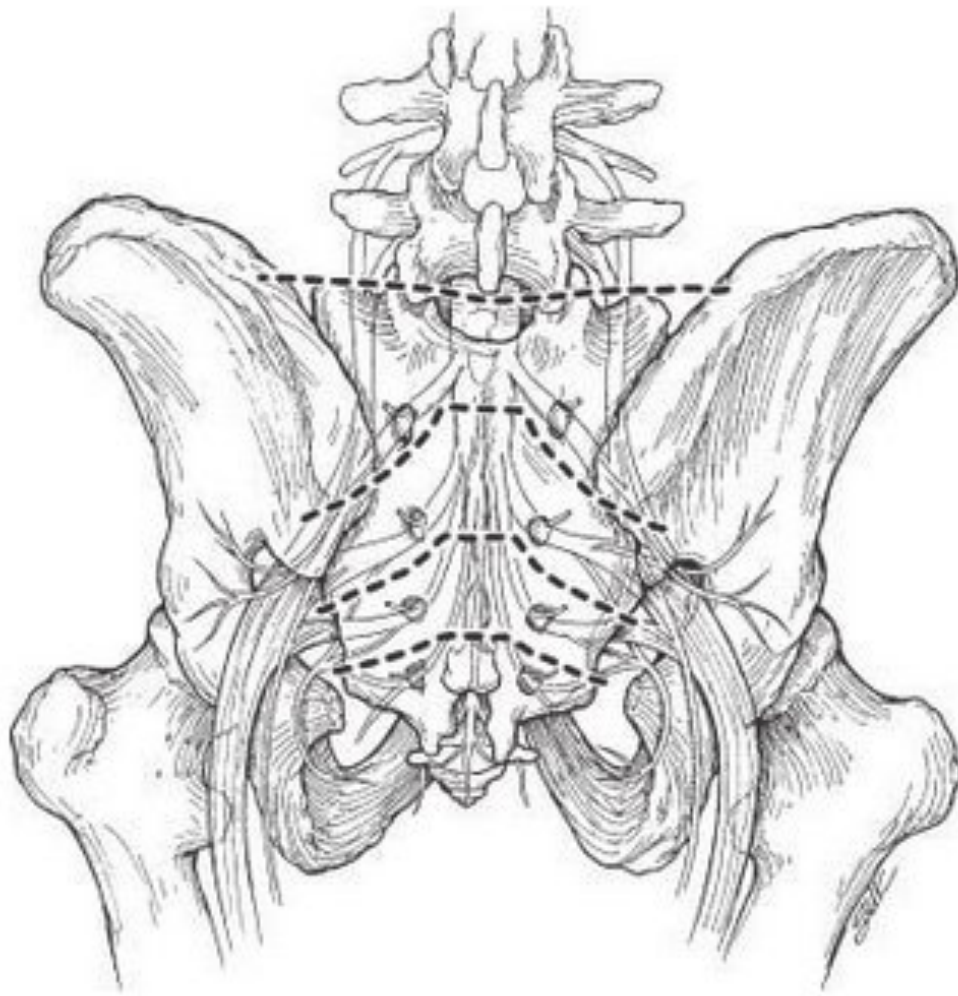


Fig. 69.3 Osteotomies for the low, middle, high, and total sacrectomy. (From Dickman CA, Fehlings MG, and Gokaslan ZL, eds. *Spinal Cord and Spinal Column Tumors*, Thieme; 2006.)

69.4.2 Posterior Approach

- The posterior transsacral approach is the procedure of choice for resection of sacral lesions with little or no presacral extension.
- Incision is midline or transverse depending on the exposure requirements and incorporates the biopsy tract.
- Sacrifice of dorsal rami may be necessary but does not cause serious functional sequelae.
- As long as the sacroiliac joints are preserved laterally, disruption of the sacroiliac ligaments to obtain a wider exposure should not result in mechanical instability.
- Laminectomy is performed with standard technique.
- The cauda equina may need to be ligated with double silk sutures and cut in the axilla of the most caudal nerve root being preserved.
- After the sacrum is dissected and freed from surrounding structures, the sacrectomy can be completed with osteotomies; beware, as significant hemorrhage may occur.

69.4.3 Combined Anterior–Posterior Approach

- Indications include extensive tumor vascularity, primary proximal sacral tumor encroaching into the lumbosacral junction (particularly the S1 end plate), and disease invading the abdominal/pelvic viscera.
- Most often, the anterior approach is performed first, followed by the posterior approach after the patient has had a few days to recover.
- Anterior access is obtained via transabdominal (supine position) or retroperitoneal approach (lateral decubitus position), but the transabdominal approach is favored for high midline sacral tumors and for tumors potentially invading the rectum. It also provides better bilateral exposure of the vessels and tumor.
- Anterior access allows exposure of the upper and middle sacrum facilitating dissection and mobilization of the rectum and other pelvic contents (e.g., iliac vessels) from the tumor surface as well as providing early access to the tumor vascular supply.
- Discectomies and osteotomies are performed at the desired level, and an anterior osteotomy may be performed of the sacroiliac joint if needed.
- The use of rectus abdominis flaps is associated with decreased wound complications.²
- The subsequent posterior stage can then focus on dissecting and transecting the dural contents and completing the sacral amputation with possible sacral reconstruction.

69.4.4 Reconstruction

- The goal is to restore the pelvic ring to facilitate axial load transfer from the spinal column to the pelvis.
- Resection of lesions that are distal to S1 vertebra often do not require subsequent reconstruction. Lesions involving S1 may alter the biomechanics at the lumbosacral junction significantly requiring reconstruction. However, many believe that reconstruction has no role in sacrectomy operations.
- Sacroiliac stability is not significantly affected if more than 50% of the sacroiliac (SI) joint is intact (i.e., at least the S1 segment). The pelvis becomes unstable if more than 50% of the SI joint is compromised.
- Several constructs have been proposed, many involving two to four rod constructs interconnected with sacral bars, pedicle screws placed from the midlumbar region and caudally, and the pelvic/iliac screws anchoring the pelvis.
- In the modified Galveston technique,^{3,4} distally the vertical Galveston L-rods are directed laterally into the ilium to establish a bilateral

connection between the lumbar spine and ilia. Each Galveston rod creates a bridge between the lumbar spine and ilium, and a transiliac rod completes the pelvic ring.

- Another reconstruction involves a triangular frame. Two fibular grafts are placed between L5 and the supra-acetabular region bilaterally along the force transmission lines, increasing resistance to failure.⁵
- Newer techniques are using both anterior and posterior instrumentation, including expandable cages to reconstruct the anterior column (attached inferiorly to the transiliac rod and superiorly to the end plate of intact vertebra) and pedicle and iliac screws as posterior instrumentation.⁶

69.5 Complications

- Injury or bleeding of the internal iliac, iliolumbar, and other sacropelvic vessels.
- Visceral damage to the pelvic contents including ureters or rectum.
- Chances of motor, sensory, bowel/bladder, and sexual dysfunction increase the higher the resection is done (**Table 69.1**). Motor loss occurs if S1 nerve root S1 is ligated. Saddle anesthesia is common. Preservation of at least one S3 nerve root preserves bowel/bladder function in two-thirds of patients.^{7,8} Similar trends have been seen for sexual function where unilateral sacral resection (recall that the pudendal nerve originates from S2–S3) still preserves overall function (albeit with numbness on the side of the resection).⁹

Table 69.1 Bowel and bladder function after sacral resection⁸

Resection and spared level	%normal bowel	%normal bladder
Bilateral S2–S5 resection with both S1 spared	0%	0%
Bilateral S3–S5 resection with both S2 spared	40%	25%
Bilateral S4–S5 resection with both S3 spared	100%	69%
Variable with unilateral S3 root spared	67%	60%
Unilateral S1–S5 resection and contralateral S1–S5 spared	87%	89%

- Wound healing problems and infection (25–46% and even higher after radiation).⁷
- Sacral insufficiency fractures.
- General medical complications (ileus, pneumonia, deep vein thrombosis, etc.).

69.6 Postoperative Care

- Because significant blood loss can often occur during surgery, fluid and blood replacement postoperatively is essential, for these patients are initially managed in the intensive care unit.
- Special air beds or routine position changes help prevent wound breakdown.
- Antibiotics are continued until all suction drains have been removed.
- Considering the close proximity to the anus, special care is taken to keep wounds clean at all times.
- Denervation of the distal GI tract (S2–S4 autonomics) during high sacral resections can result in a significant postoperative ileus, requiring an aggressive bowel regimen in all patients and intravenous nutrition for a prolonged period in some patients. Studies have found higher wound infection rates in patients with postoperative bowel incontinence.⁷
- Bladder training is gradually attempted, and in some circumstances patients need to learn to self-catheterize.
- In patients who had tumor invasion of the rectum requiring resection, colostomy training is necessary.
- Patients should be mobilized as soon as possible, depending on their functional reserve and the stability of the dorsal pelvic ring.

69.7 Outcomes

- Survival depends on tumor type, tumor location, and prior surgeries. Chordoma is the most common tumor treated with sacrectomy, with 5- and 10-year survival rates of 59% and 35% respectively, after surgery.²

69.8 Surgical Pearls

- Tumor vascularity, encroachment of the anterior pelvic fascia, and tumor location relative to S1 must be carefully assessed.
- Wide resection ensuring negative margins throughout is essential to minimizing the chances of recurrence and significantly affects survival.

Common Clinical Questions

1. Which nerve roots should be preserved, if possible, to maintain bladder/bowel function?
2. When is it advisable that a combined anterior–posterior approach be done for sacrectomy?

Answers to Common Clinical Questions

1. Preservation of at least one of the S3 nerve roots has been associated with good bowel/bladder function in about two-thirds of patients.^{2,7}
2. Indications include extensive tumor vascularity, primary proximal sacral tumor encroaching into the lumbosacral junction (particularly the S1 end plate), and disease invading the abdominal/pelvic viscera.

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70 Vertebral Body Augmentation

Andrey Alex Volkov, Ioannis Papanastassiou, and Frank D. Vrionis

70.1 Key Points

- Vertebral augmentation is commonly employed in treating compression fractures in patients who have osteoporotic or pathological compression fractures.
- Two different percutaneous vertebral augmentation methods for cement application into the vertebral body have been documented: Vertebroplasty (VP) and kyphoplasty (KP).
- In vertebroplasty, polymethylmethacrylate (PMMA) cement is injected percutaneously into the collapsed vertebral body.
- Kyphoplasty involves placing inflatable temporary balloons percutaneously into a vertebral body. The inflation of the balloons allows some restoration of vertebral height and creation of a cavity. After deflation, the cavity that has been produced is filled by injection of PMMA.
- Previously, two randomized clinical trials showed that improvement in patients with painful osteoporotic vertebral fractures was similar between those treated with vertebroplasty and those treated with a simulated procedure, at 1-month follow-up.^{1,2} Newer literature review has revealed that balloon kyphoplasty/vertebroplasty provided greater pain relief and less adjacent fractures than nonsurgical treatment.^{3,4}
- One retrospective review study of 80 patients suffering from pathologic vertebral body fractures showed that both techniques are effective in providing pain relief, but KP is associated with lower rate of cement leakage than is VP.⁵
- The CAFÉ study is the first randomized trial showing that in patients who have cancer, KP is an effective and safe treatment for vertebral fractures.⁶
- Compared with nonsurgical management, balloon kyphoplasty resulted in improvements in quality of life and disability measures and reduction of back pain in patients who have acute painful vertebral fractures.^{2,3}
- There is no difference in clinical outcomes between unilateral or bilateral kyphoplasty.^{3,7}

70.2 Indications

- Severe pain or progressive collapse as a result of vertebral body compression fractures in patients with osteoporosis (primary or secondary) or osteopenia.
- Severe pain or progressive collapse as a result of vertebral body metastasis or multiple myeloma. Treatment algorithm for painful thoracic or lumbar vertebral body fractures in cancer patients (**Fig. 70.1**).
- Contraindications to vertebral augmentation include asymptomatic lesions, patients who are improving on conservative care, ongoing local or systemic infection, retropulsed bone fragment or epidural tumor causing myelopathy, neurological deficits, or spinal instability.

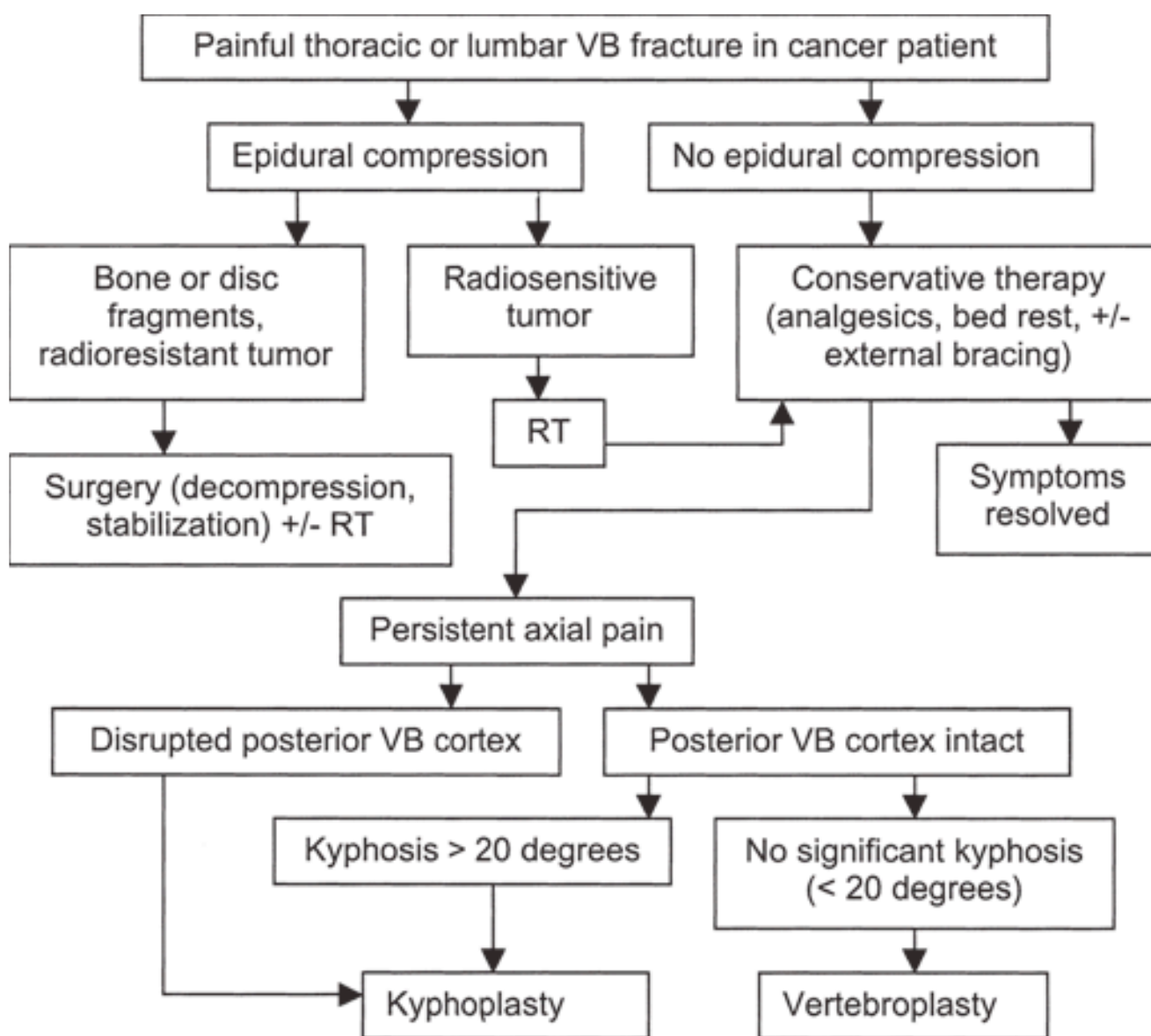


Fig. 70.1 Treatment algorithm for painful thoracic or lumbar vertebral body fractures in cancer patients. See text for details. VB=vertebral body; RT=radiation therapy. (From Fourney DR, Schomer DF, Nader R, et al. Percutaneous vertebroplasty and kyphoplasty for painful vertebral body fractures in cancer patients. *J. Neurosurg.* 2003;98(1 Suppl):21-30. Reproduced with permission.)

70.3 Diagnosis

- Initial diagnosis of vertebral body collapse can be established via X-rays or CT scan of the cervical, thoracic, or lumbar spine.
- To verify and confirm the acuteness of the vertebral fracture, an MRI scan of the spine (T1 and short T1 inversion recovery [STIR] sequences) to detect vertebral body edema and associated impending fractures is used.
- CT scan of the fractured vertebral body with sagittal reconstruction to evaluate posterior vertebral wall integrity.
- Bone scan in some cases to assess acuity of fracture and exclude metastasis in different levels.

70.4 Technique

- Procedures can be performed under general or local anesthesia.
- Patient is placed in prone position on Jackson radiolucent table, with postural reduction of kyphosis if present (using a mobile chest bed or additional pillows).
- Biplanar fluoroscopy is used.
- Both techniques start with the percutaneous insertion of (11 gauge) Jamshidi needle or guide pin into the fractured vertebra via the pedicle of the corresponding vertebra, and end with the injection of PMMA.
- This can be achieved through a transpedicular approach in nearly every case.
- In the thoracic spine the needle can be inserted extrapedicularly, between the rib head and lateral aspect of the pedicle (**Fig. 70.2**).
- Unipedicular or bipedicular approach can be used to inject PMMA, with a goal of filling the center of the vertebral body.

70.4.1 Vertebroplasty

- After correct positioning of the needle, the inner stylet is removed.
- Contrast material is then injected to ensure that the needle is not positioned in the venous flow path (optional).
- Cement, in thin liquid form, is injected into the vertebra using multiple small syringes or continuous injection system.
- The flow of the cement should be followed on the image intensifier.

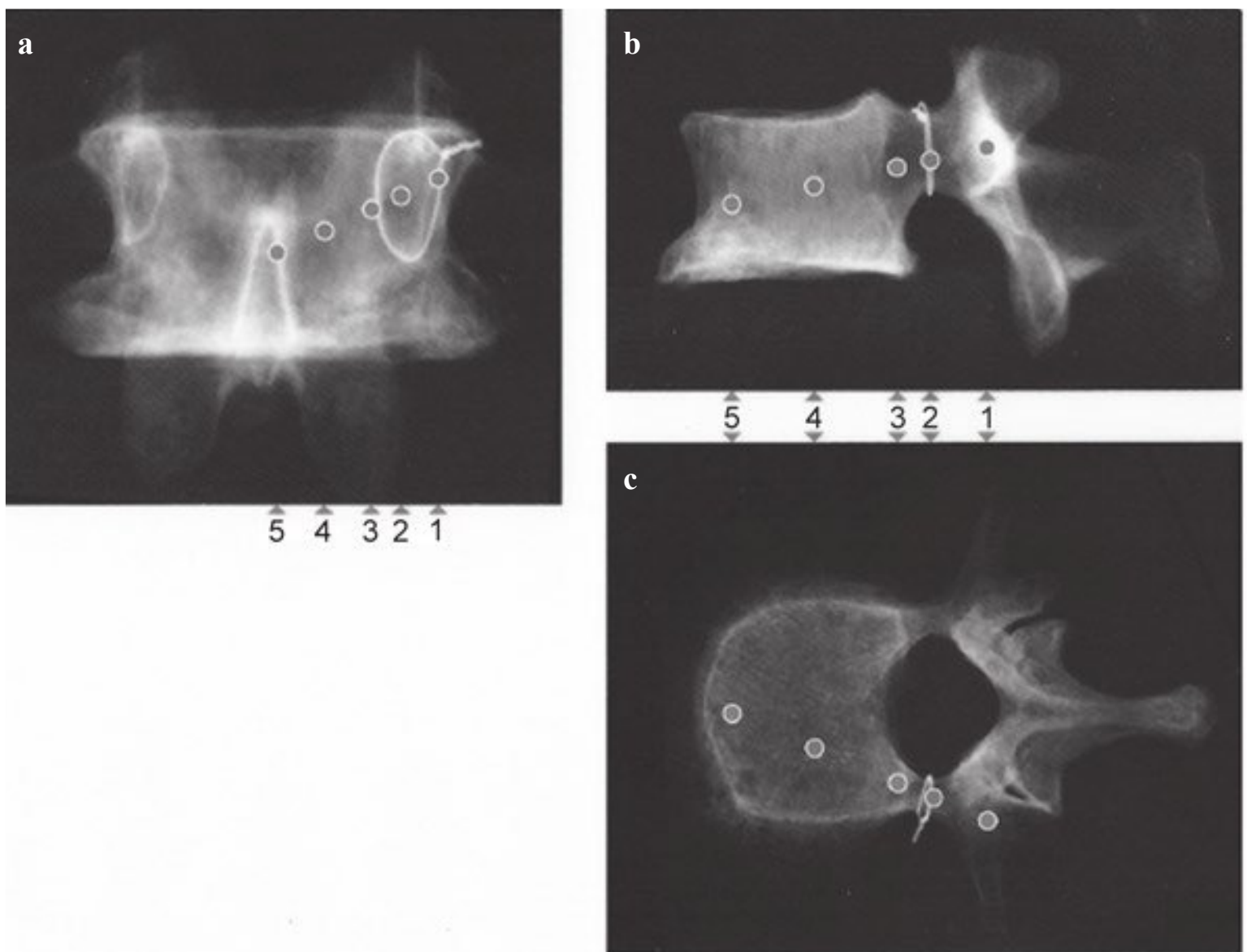


Fig. 70.2 (a–c) Entry points and trajectories for the transpedicular approach during kyphoplasty. (Courtesy of Medtronic.)

70.4.2 Kyphoplasty

- After proper needle positioning, a series of tools (drill, curette) are used to create a working channel/cavity. Once inserted, the balloons are inflated using volume and pressure controls (digital manometer) to create a cavity within the vertebra.
- Once this has been achieved, the balloons are deflated and removed.
- Thick cement can be fed through the cannula under low pressure to fill the cavity created by the balloon tamp.
- In cases involving the upper thoracic spine, the shoulders can interfere with the lateral view, so a stack of pillows of appropriate height should be placed under the chest to lower both shoulders. Alternatively, the arm and the shoulder are left hanging down parallel to the trunk.

70.5 Complications

- Procedure-related: Cement leakage can result from fracture clefts or improper instrument position and can occur in the spinal canal, neuroforamen, or disk space.^{2,8,9}
- Medical: Pulmonary embolism, hemo- or pneumothorax, soft tissue hematomas.^{9,10}
- New adjacent vertebral fracture due to leakage of cement into the disk.¹⁰
- Pedicle fracture.^{7,9,10}

70.6 Postoperative Care

- Conduct an immediate postoperative exam, ensuring that there is no new neurologic deficit compared with preop.
- Mobilize early, with no need for bracing.
- Clinical evaluation and plain spinal radiographs. Use a chest X-ray to rule out pneumothorax in thoracic VP/KP (optional).
- Discharge home when patient meets discharge criteria (usually same day).

70.7 Outcomes

- All studies reported significant improvements in pain score and functional outcome.^{1,8}
- The risk of neurologic sequelae ranges from 0.4 to 4.0% according to various reports.^{2,8,11}
- However, previously, a study by Kallmes et al¹ showed that improvements in pain and pain-related disability associated with osteoporotic compression fractures in patients treated with vertebroplasty were similar to those for the control group, without treatment.
- Results of six randomized, controlled studies (525 treated levels and a total of 424 patients) were evaluated by Bouza et al⁹ to evaluate the safety of KP in treatment of osteoporotic vertebral compression fractures.
 - Cement leakage was detected in 18.3%
 - Only 0.5% of fracture leakages proved to be symptomatic.
 - Clinical complications were recorded in 11.5% with several cases requiring intensive treatment or postoperative surgery.
 - New vertebral fractures were noted in 20.7%

70.8 Surgical Pearls

- The possibility of cement leakage can be overcome through the use of high-quality imaging and slow application of PMMA in a viscous state.
- In cases of focal kyphosis caused by index fracture, the levels above and below the fracture should also be considered in the treatment plan. No more than three to four levels should be treated at one setting. This policy minimizes the risk of microembolization (cement and fat emboli).¹¹ Impending fractures should also be treated.

Common Clinical Questions

1. Contraindications to vertebral augmentation include all of the following *except* which?
 - A. Asymptomatic lesions
 - B. Retropulsed bone fragment
 - C. Epidural tumor causing myelopathy
 - D. Severe pain due to vertebral body compression fractures in patients with osteoporosis
2. Procedure-related complications include all of the following *except* which?
 - A. Cement leakage
 - B. Pulmonary embolism, hemothorax, or pneumothorax
 - C. New adjacent vertebral fracture
 - D. Cement in the pedicle

Answers to Common Clinical Questions

1. D.
2. D.

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71 Spinal Cord Tumor Resection

Mari L. Groves and George I. Jallo

71.1 Key Points

- Intramedullary spinal cord tumors (IMSCTs) are rare lesions accounting for only 5 to 6% of all central nervous system tumors.^{1,2,3,4,5}
- IMSCTs can occur at any age and primarily consist of primary glial tumors, including ependymomas (primarily in the adult population), and astrocytomas (primarily in the pediatric population), as well as gangliogliomas (**Fig. 71.1** and **Fig. 71.2**). Other etiologies include hemangioblastomas or cavernous malformations.^{2,5}
- Prognosis and outcome are dependent on histological diagnosis.²
- Malignant spinal cord tumors account for less than 15% of intramedullary tumors.^{1,2,5}
- Metastases to the spinal cord are rare.

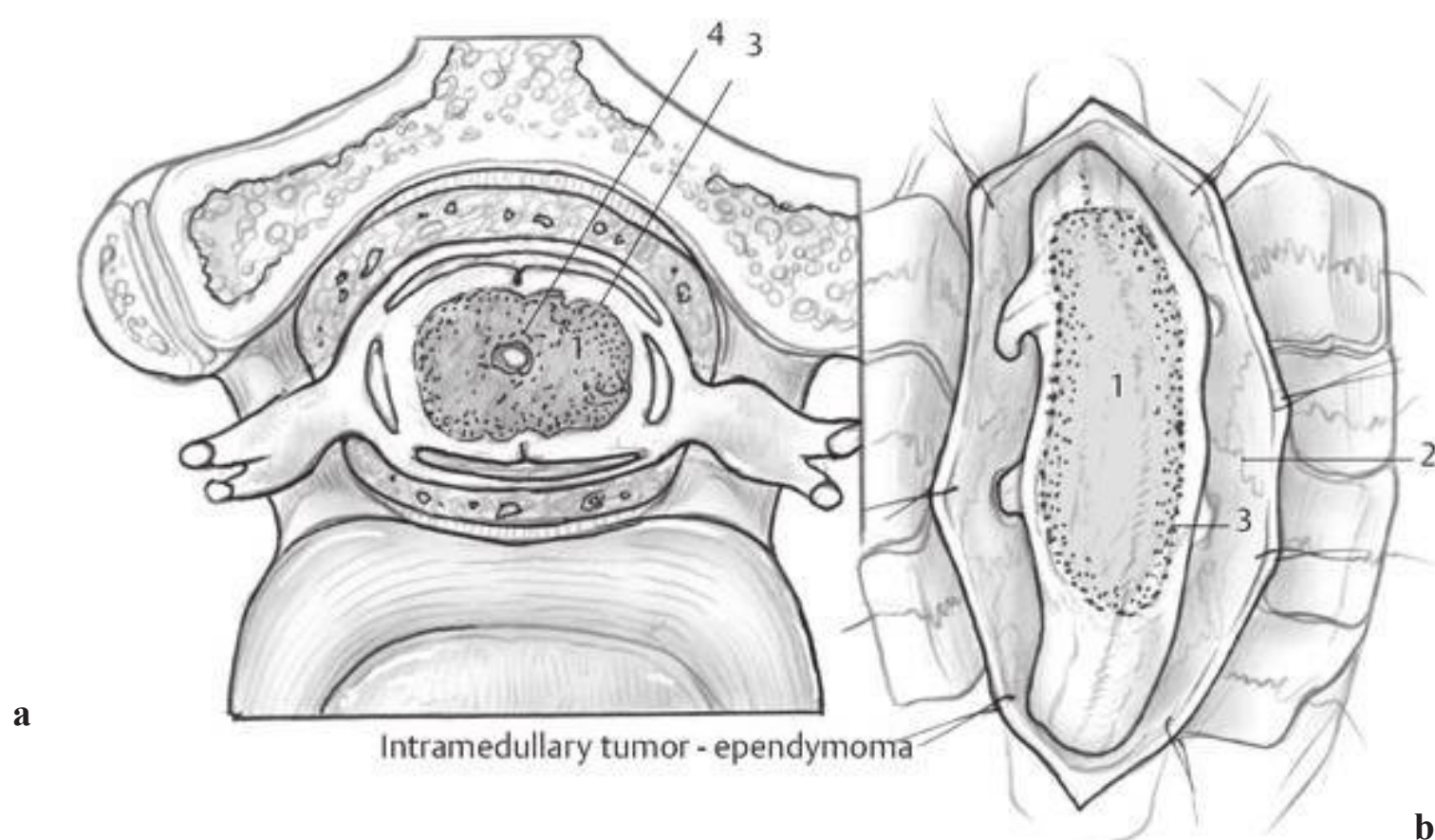


Fig. 71.1 (a) Axial view of an ependymomas within the spinal cord. These are more commonly centrally located. 1. Tumor (stippled); 2. Dura; 3. Cord; 4. Central canal. (b) Intraoperative surgical dorsal view of the tumor. (Adapted from Borges LF, Neurosurgery Tricks of the Trade: Spine and Peripheral Nerves, Thieme; 2014)

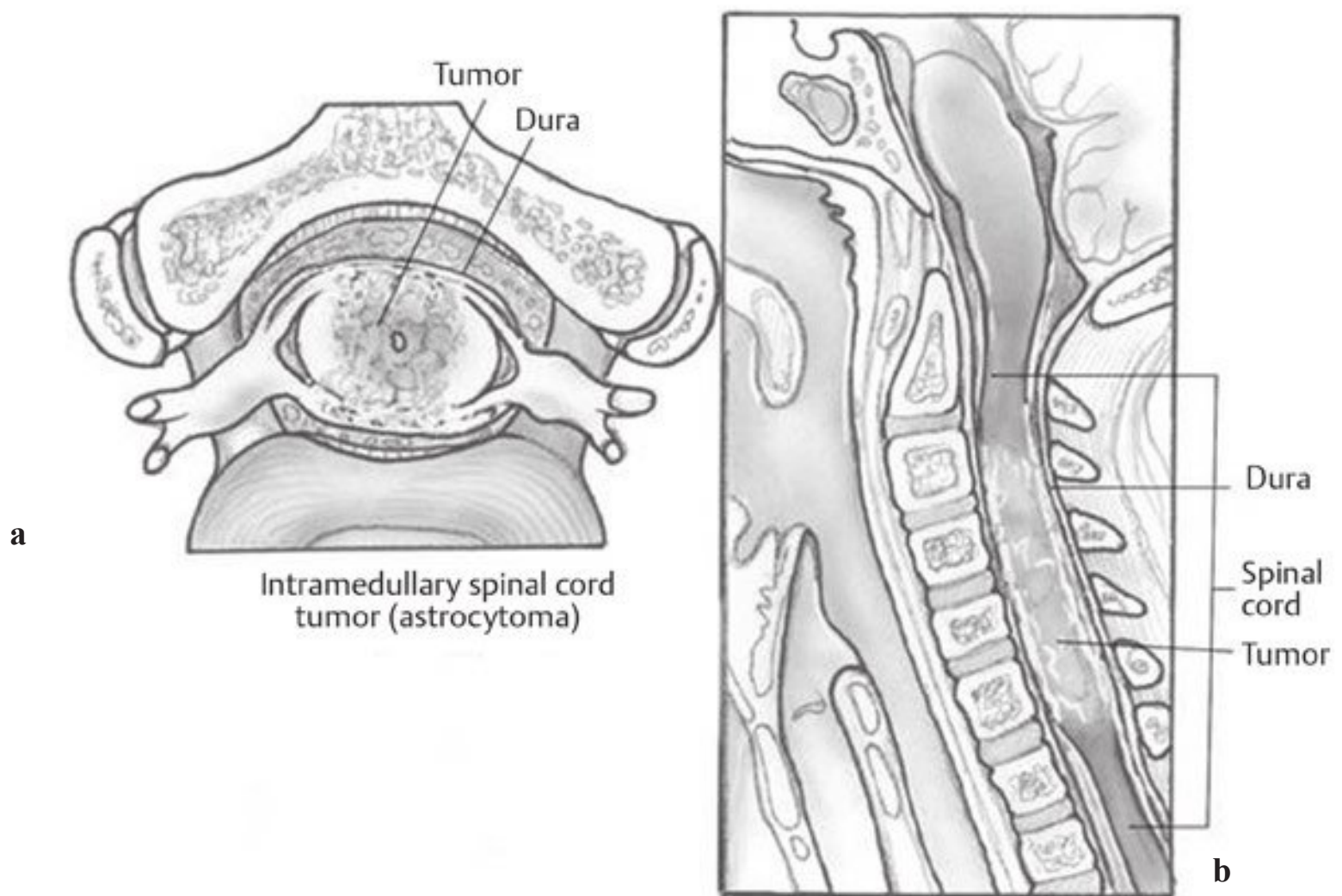


Fig. 71.2 (a) Axial view of astrocytoma within the spinal cord. (b) Midsagittal view illustrating enlargement of the spinal cord at the tumor location. (Adapted from Borges LF, Neurosurgery Tricks of the Trade: Spine and Peripheral Nerves, Thieme; 2014)

- Presenting symptoms can be minimal and insidious. Preoperative deficits typically suggest postoperative deficits.
- Surgical resection is a mainstay in both diagnosis and treatment and should be guided by initial biopsy as well as intraoperative neuromonitoring.
- Adjuvant therapy, including chemotherapy and radiation treatment, can be used in higher-grade lesions but is not a typical first-line therapy.

71.2 Indications

- Diagnosis or treatment for an intramedullary lesion with either clinical deterioration or progression on neuroimaging.²
- Progressive sensory or motor deficits as well as sphincter dysfunction.
- Back pain that localizes to the lesional area that is nonmechanical and exacerbated by lying supine.
- Thorough work-up for immunodeficiency and/or inflammatory processes, including multiple sclerosis or transverse myelitis, should be completed to rule out these lesions and should only be considered if diagnosis still unknown.⁵

- Patients who have drop metastasis or known high-grade lesions should not be considered surgical candidates and are likely more appropriate for adjuvant therapy.

71.3 Technique

- Patients should be placed prone in such a manner as to minimize venous hypertension with neutral alignment of the spine. Patients with cervical and high thoracic lesions should be secured with a three-point Mayfield clamp (Integra LifeSciences; Plainsboro, New Jersey).
- Neuromonitoring, including continuous somatosensory evoked potentials (SSEPs), motor evoked potentials (MEPs), and epidural continuous motor (D wave), should be placed and discussed with anesthesia to minimize any nonsurgical disruption in signals.^{4,6}
- Preoperative corticosteroids, blood pressure parameters, and broad-spectrum antibiotics should be routinely administered.
- Standard midline incision with a subperiosteal dissection of the paraspinal muscles away from the lamina should be performed. The rostral and caudal extent of the incision should be planned based on preoperative neuroimaging studies.
- Bony decompression can be performed through either a laminectomy or laminoplasty with care taken to preserve the lateral ligamentous attachments and facets joints to minimize postoperative instability (**Fig. 71.3**).



Fig. 71.3 The durotomy is completed, and the arachnoid is incised. (From Vaccaro AR and Albert TJ, eds. *Spine Surgery: Tricks of the Trade*, 2nd ed., Thieme; 2009.)

- Hemostasis should be obtained before the durotomy to minimize any rundown of blood into the intradural space.
- Before the durotomy, the tumoral extent may be confirmed with intraoperative ultrasound.
- Epidural leads should be placed, when possible, both rostral and caudal to the lesion. Distal epidural leads will not be possible in the lumbar spine, and rostral epidural leads will not be possible in the high cervical spine. These electrodes monitor the corticospinal tracts.
- Midline durotomy should be performed and the dura tacked up laterally to help with visualization and to minimize epidural rundown. These can either be anchored to the paraspinal muscles or be hung on a stay suture.
- Patients who have had significant localizing pain preoperatively may be under high intraspinal pressure and could be at risk for cord herniation. In the case of cord herniation at the time of the initial durotomy, a timely resection should be performed to help debulk the tumor and decrease intraspinal pressure.
- If the arachnoid is preserved, this may be secured to the dural edges using small vascular clips.
- Myelotomy should be performed to minimize neurologic morbidity. If the tumor is diffuse, a midline approach is preferred to help minimize damage to the dorsal columns. If the tumor comes to the lateral surface, the myelotomy may take place along one of the dorsolateral sulci.
- The posterior median sulcus may be estimated by inspecting the bilateral dorsal root entry zones or identifying the convergence of midline vessels that dive into the median sulcus.
- If there is any concern, dorsal column mapping may be helpful to identify the midline.^{4,6}
- If the tumor is not easily identified on the dorsal surface, the spinal cord may be bluntly spread with a splayed bayonette forcep until the tumor is clearly seen.
- Pial sutures may be used to provide gentle traction, but these typically provide more traction on the dorsal columns and may result in postoperative deficits.
- Pathological analysis via frozen section should be performed immediately. Low grade lesions should be aggressively resected, whereas high-grade tumors should have a minimal debulking only, for they will all require adjuvant therapy.
- The extent of resection is governed by a cleavage plane between the tumor and normal tissue or by a decrement in neuromonitoring.^{4,6}
- The loss of muscle MEPs during surgery with the preservation of the epidural (D wave) will correlate with a temporary motor deficit. Loss of epidural and MEPs greater than 50% of baseline will likely result in a permanent motor deficit.^{4,6}

- Tumoral resection can take place with internal tumor debulking to help minimize collateral damage to the surrounding normal spinal cord. This can take place through either aspiration or an ultrasonic aspirator.^{4,6}
- Copious irrigation should be performed to ensure all bleeding has ceased. This will help identify any persistent bleeders. Coagulation of the anterior spinal artery or its branches should be avoided at all costs.
- Watertight dural closure is necessitated and dural edges can be closed with a running 5–0 Prolene suture (Ethicon Inc., Cincinnati, Ohio).⁶
- Dural patches or grafts may enlarge the diameter of the thecal sac in patients who have prior radiation or expansile lesions that are difficult to close primarily.^{4,6}
- Dural sealant should be considered but may cause mass effect and should be used sparingly in cases of laminoplasty.
- The laminoplasty may be secured with titanium plates and has been shown to decrease the incidence of cerebrospinal fluid (CSF) leak.

71.4 Complications

- Neurologic injury in all capacities may occur including motor, sensory, and sphincter dysfunction.
- Dorsal midline myelotomies may be associated with significant proprioceptive morbidity.
- CSF leak and associated meningitis.
- Wound infection resulting from wound dehiscence.
- Postlaminectomy kyphosis may occur and has a greater incidence in patients who have a preoperative deficit and in patients who require longer laminectomies. The incidence is higher in children than in adults and may be exacerbated by perioperative radiation.
- High cervical lesions may result in diaphragmatic paralysis and difficulty with ventilation and may necessitate placement of a tracheostomy or gastric feeding tube.

71.5 Postoperative Care

- To avoid CSF leaks, precautions should be taken to minimize pressure to the incision. Patients with cervical or cervicothoracic lesions should have an elevated head of bed postoperatively. Patients who have low thoracolumbar or lumbosacral lesions should be kept flat for 24 to 48 hours after surgery.²
- Perioperative steroids can be considered and weaned based on concern for edema and the underlying pathological diagnosis.
- Postoperative MRI with and without contrast within the first 48 hours postoperatively.

- Physical therapy may be critical in patients who have proprioceptive deficits even in the setting of intact motor strength.
- Perioperative deep venous thrombosis (DVT) prophylaxis should be considered with sequential compression devices (SCD) or thromboembolic deterrents (TED) hose as well as subcutaneous heparin outside the immediate perioperative period.
- Sphincter dysfunction should be anticipated and the Foley catheter left in place until the patient is ambulatory. Straight catheterization should be used in cases of retention.²
- Postoperative adjuvant therapy with chemotherapy or radiation therapy may be indicated based on the underlying pathology, and consultation with both medical and radiation oncology should be considered.
- Radiation and/or chemotherapy should be delayed until the wound has healed, typically after 3 to 4 weeks.

71.6 Outcomes

- Overall and progression free survival are dependent on tumor pathology and extent of resection.^{1,2,3}
- Preoperative function correlates with how much neurologic improvement may be seen postoperatively.^{1,2,3}
- Patients who have more severe motor deficits are more likely to have additional deficits postoperatively and are less likely to have significant recovery.^{1,2,3}
- Patients who have sensory deficits will see the most dramatic improvement during the initial 3 postoperative months. However, the full potential for recovery may be seen up to 1 to 2 years postoperatively.

71.7 Surgical Pearls

- Surgery should be offered only in the instance of neurological decline, large associated cysts/syringes, or radiographical progression.
- Adequate discussion with patients regarding expected postoperative deficits is critical, for most patients will have some degree of dorsal column dysfunction postoperatively.
- Neurological monitoring is helpful and should include SSEPs, MEPs, and epidural MEPs.
- Laminoplasty may help decrease the risk of postoperative CSF leak but has not been shown to affect postlaminectomy kyphosis.
- Cystic components of the tumor do not need to be debulked, for these are typically nonneoplastic. In the event of herniation, decompression of the cyst may help with intratumoral pressure.

Common Clinical Questions

1. What are the indications for laminoplasty versus laminectomy?
2. If the tumor has a cystic component, does this need to be resected?
3. What type of neuromonitoring is appropriate?

Answers to Common Clinical Questions

1. Laminoplasty may help reduce the risk of CSF leak by reapproximating the paraspinal musculature. There is no evidence to suggest a decreased risk of spinal deformity, and patients should continue to be monitored for progressive postlaminoplasty kyphosis.
2. The cyst wall is usually nonneoplastic. With removal of the neoplastic component, the cyst will resolve.
3. Patients should have continuous somatosensory evoked potential (SSEPs) as well as motor evoked potentials (MEPs). Continuous epidural (D wave) motors correlate with motor deficit. Loss of epidural and MEPs greater than 50% of baseline will likely result in a permanent motor deficit.

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72 Surgical Resection of Spinal Vascular Lesions

Timothy D. Uschold, Alim P. Mitha, and Steven W. Chang

72.1 Key Points

- The natural history of spinal vascular lesions is usually malignant. Depending on the eloquence of local tissue, early definitive treatment is favored whenever feasible.
- All arteriovenous (AV) lesions require preoperative spinal angiography.
- Preservation of the anterior spinal artery (ASA) is paramount during attempts at surgical treatment or embolization.
- The optimal trajectory for the resection of spinal cavernous malformations is determined by the two-point method. A line connecting the center of the lesion with its most accessible margin determines the approach. Some lesions without pial contact may still be resected.
- Timing of surgical intervention after hemorrhage is controversial, although the presence of acute hemorrhage may aid in early resection of some cavernomas.

72.2 Indications

- Cavernous malformations: Surgery is recommended for all accessible lesions that reach a pial surface. Small, deep-seated asymptomatic lesions can be followed. Deep lesions with evidence of repeated hemorrhage or expansion should be considered for surgery.¹
- Extradural–intradural AV malformations (AVMs): Palliative attempts at flow reduction and/or selective targeting of high-risk features are appropriate in the setting of progressive symptoms related to steal, repeated hemorrhage, mass effect, or congestion.
- Intramedullary and conus AVMs: Surgical resection, typically with preoperative embolization, is recommended if gross total excision can be achieved. Embolization alone for cure has been demonstrated but remains more controversial.
- Intradural–dorsal and intradural–ventral AV fistula (AVF): All require definitive treatment with either surgery or nonparticulate embolization. If clinical history and imaging findings suggest an intradural–dorsal AVF despite a negative angiogram, surgical exploration is warranted.
- Extradural AVF: All require definitive treatment; endovascular intervention is favored.

72.3 Technique

- Posterior or posterolateral approach via laminectomy or laminoplasty:
 - Added costotransversectomy or facetectomy may allow further lateral exposure.
 - The dura is opened in a linear fashion and the arachnoid is tacked to the dura.
 - If additional anterolateral visualization is necessary, the dentate ligaments may be transected and tacked contralaterally with 6–0 Prolene suture (Ethicon Inc., Cincinnati, Ohio) to facilitate gentle rotation of the spinal cord. This may be done across several levels for maximum effect (**Fig. 72.1**).
- Cavernous malformations:
 - When necessary, myelotomy is made sharply with a linear pial opening. Appropriate zones for entry into the cord include the dorsal midline and the lateral dorsal sulcus at the dorsal root entry zone. This trajectory enters the substantia gelatinosa. A third entry option is the lateral trajectory behind the dentate ligaments and between the dorsal and ventral spinocerebellar tracts (**Fig. 72.2**).
 - The lesion is dissected circumferentially. Coagulation of the cavernoma may be necessary to shrink and debulk the lesion's interior. The associated venous anomaly should be preserved. Surrounding hemosiderin-stained spinal cord is not taken.¹

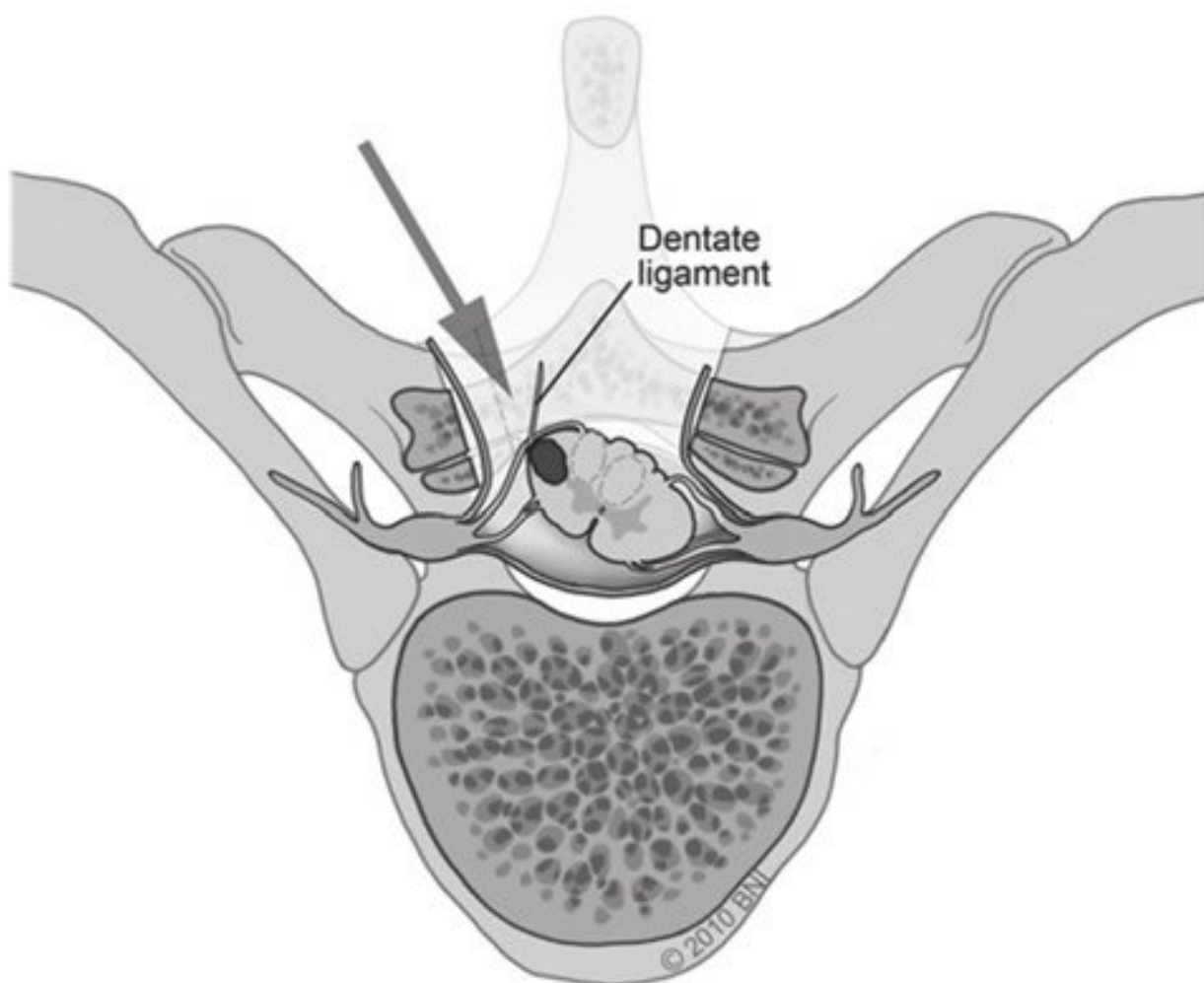


Fig. 72.1 Gentle spinal cord rotation is facilitated by retraction of dentate ligament(s). (Used with permission from Barrow Neurological Institute.)

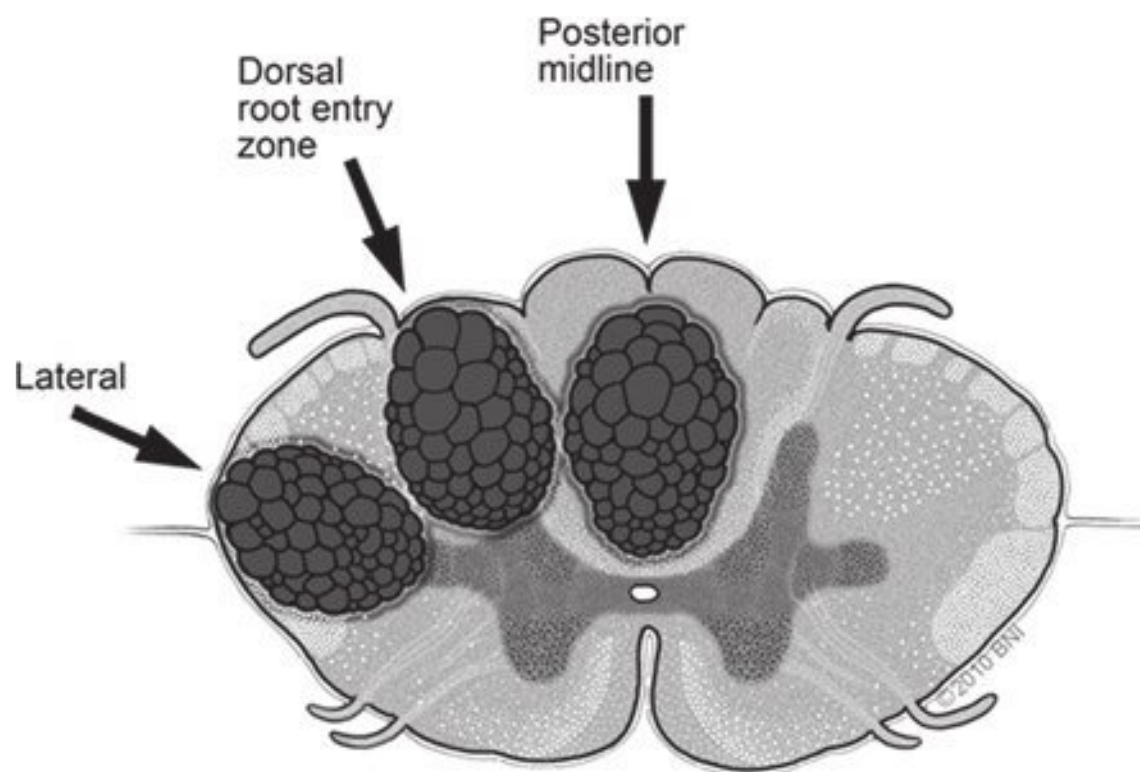


Fig. 72.2 Alternative posterior and posterolateral zones of entry for myelotomy. (Used with permission from Barrow Neurological Institute.)

- **Intradural–dorsal AVFs:** After dural opening, the arterialized vein is followed to its intradural interface along the root sleeve, where the radicular artery pierces the dura. The intradural fistulous connection is clipped, bipolarized, and transected.
- **Extradural AVF:** Transarterial endovascular coil embolization is typically the preferred modality of treatment, although transvenous routes have been described.
- **Intradural–ventral AVFs:** Anterior and posterolateral trajectories have been described. The fistula is bipolarized and transected such that arterial supply to the ASA and normal venous drainage is preserved. Venous varices can be further excised. Nonparticulate embolization may be useful in selected cases, although surgery remains the gold standard.
- **Intramedullary spinal AVMs:** Arterial rather than venous feeders should be sharply dissected circumferentially, cauterized, and transected first. The largest draining vein is typically cauterized and transected last for removal of the specimen.
- **Extradural–intradural AVMs:** Embolization may be the primary palliative modality for the setting of symptomatic steal, repeated hemorrhage, or vascular congestion. Surgery is typically reserved for relief of persistent, symptomatic mass effect.

72.4 Complications

- Cerebrospinal fluid leak, wound complications.
- Delayed or acute neurologic decline: May be attributable to progressive venous thrombosis, especially in the setting of embolization for

intradural–dorsal AVFs. Routine postprocedural heparinization is considered in these patients. Blood pressure control is strict, and judicious use of steroids is advised before intervention.

72.5 Postoperative Care

- Postprocedural angiography and/or MRI is essential to verify treatment effect, especially in the setting of new neurologic deficits.

72.6 Outcomes

- Cavernous malformations: Subtotal resection offers nominal benefit. Initial postoperative deterioration has been reported in 24 to 50% but long-term follow-up data suggest improved neurologic function in as many 46 to 58%.
- Intradural–dorsal AVFs: The primary benefit of surgery over endovascular treatment remains risk of recurrence (as high as 98% initial success for surgery versus 46 to 85% for embolization).² Endovascular abilities are improving. Motor improvements have been reported in ~60% of patients, compared with 40% showing improvements in sphincter dysfunction (Aminoff-Logue [AL] Scale, **Table 72.1**), irrespective of treatment modality. Numerically, improvements in AL scores are typically modest (about 1 point).
- Intradural–ventral AVFs: Available series are mostly small, but neurologic improvement or stabilization after surgery has been reported to range from 87.5 to 95%. Preoperative embolization may be useful in limiting flow rate (and in some cases for cure), but incomplete occlusion has resulted in more modest improvements compared with surgery.³
- Extradural AVFs: Outcomes reported in numerous case reports have been favorable in terms of fistula obliteration and clinical improvement. In the setting of acute hematoma, surgical evacuation within 12 hours is an important prognostic.
- Extradural–intradural AVMs: Most multimodality attempts are purely palliative. Rare cases may be amenable to embolization followed by operative resection.
- Intramedullary AVMs: Clinical improvement rates of 33 to 67% have been reported for surgery, compared with rates of 8 to 20% for clinical decline. Rates of postsurgical angiographic cure have ranged from 59 to 100%. Further endovascular data using Onyx is anticipated. The most appropriate and definitive strategy remains preoperative embolization (when technically appropriate) followed by microsurgical excision.
- Conus AVMs: Little published experience. Aggressive multimodality treatment can result in clinical improvement or stabilization in addition to angiographic cure.

Table 72.1 Aminoff-Logue Scale

Gait dysfunction
1. Leg weakness or gait change without significant activity impairment
2. Restricted walking but no DME assist required
3. Cane for ambulation
4. Crutches or walker for ambulation
5. Bed-bound, does not stand, confined to wheelchair
Urinary dysfunction
1. Hesitancy, urgency, and/or frequency
2. Intermittent incontinence or retention
3. Full incontinence or retention

Abbreviation: DME, durable medical equipment.
Source: Modified from Aminoff MJ and Logue V. The Prognosis of Patients with Spinal Vascular Malformations. Brain 1974;(97):211–218.

72.7 Surgical Pearls

- Intraoperative electrophysiologic monitoring of somatosensory evoked potentials (SSEPs) and motor evoked potentials (MEPs) is essential for all cases. In the setting of AV lesions, serial indocyanine green angiography (ICG) and/or intraoperative spinal angiographic runs before and during resection are often useful.
- Anterior or anterolateral approaches to the resection of spinal vascular lesions are seldom advised due to the risk of ASA compromise and poor dural closure.⁴
- Sharp rather than blunt dissection is favored when possible.
- If required, myelotomy should expose the entire cranial–caudal extent of the lesion to minimize parenchymal retraction and aid in visualizing feeding pedicles.

- Pressure measurements in the arterialized veins of intradural dorsal AVFs can be transduced using a small-gauge needle. Pressure should more closely approximate central venous pressure following fistula obliteration.

Common Clinical Questions

1. Name the primary advantage of surgery (in terms of outcome) over embolization for intradural–dorsal AVFs.
2. Name three zones of entry for myelotomy from a posterior approach.
3. Name three intraoperative tools or strategies useful in guiding and confirming treatment for AVMs and AVFs.

Answers to Common Clinical Questions

1. Decreased risk of recurrence with surgery.
2. Dorsal midline, dorsal lateral sulcus, lateral between the nerve roots
3. Intraoperative monitoring, ICG, intraoperative spinal angiography, intravascular pressure transduction.

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Appendices

I Positioning

Tien V. Le, Juan S. Uribe, and Fernando L. Vale

Topic	Positioning
Prone positioning on Mizuho OSI “Jackson” spinal table	• Patient initially supine on radiolucent imaging top of the Jackson table. • Pad bony prominences appropriately (see below). • Temporarily tuck arms with thin sheet. • Place the Jackson Spinal Surgery Table Top (Fountain Valley, California) over patient and temporarily hold in place using pins for adjustment of pads. • Position rigid plastic head rest directly over ProneView headrest (Mizuho, Union City, California) or other soft head rest. • Position thoracic pad between sternal notch and xyphoid. • Center hip and thigh pads for appropriate support. • Center leg pads over the rest of the legs, leaving room for both feet after pronation. • Place a bedsheet folded to a width of ~ 2 feet centered over the lower half of the arms for later use in tucking. • Attach four seatbelts: One for the upper thorax, one at the hips, one at the thigh, and one at the lower leg. • Remove pin and compress Jackson frame until there is some resistance at caudal end, then replace pin in new slot. • Remove pin and compress Jackson frame until there is firm resistance rostrally, then replace pin in new slot. • Refasten all four seatbelts tightly. • Count for four seatbelts and four pins both rostrally and caudally. • Loosen resistance at head of bed.

Topic	Positioning
	• Discuss to which side to turn patient with partner at caudal end and with anesthesiologist. • Turn 180° into prone position in brisk, smooth fashion. • Remove seatbelts and remove pins securing the posterior bed and remove the posterior bed. • Double-check positioning of all padding and confirm adequacy. • Check foley position and be sure male genitals are free of compression. • Abdomen should be decompressed, allowing for diaphragmatic excursion. • Check for neutral position of cervical spine and reposition ProneView as needed.
Padding and prophylaxis	• Pad all bony prominences to prevent postoperative neuropathies. • May use eggcrate foam padding or any other soft material such as gel pads or Tempur-Pedic foam (Tempur-Pedic North America, Lexington, Kentucky). • Wrists should be padded to decrease risk of median neuropathy. • Elbows should be padded to decrease risk of ulnar neuropathy. • Knees/fibular head should be padded to decrease risk of common peroneal neuropathy. • The pelvis anterior superior iliac spine should be padded to decrease risk of lateral femoral cutaneous neuropathy. • An axillary roll can be placed at the upper chest wall when in a lateral decubitus position to decrease risk of brachial plexopathy. • Eye goggles and slight reverse Trendelenburg position may help decrease undue pressure on eyes when in prone position, which decreases risk of postoperative blindness.

Topic	Positioning
Anterior cervical approach	• Supine position. • Place intrascapular shoulder roll. • Head midline placed in chin strap using 5 to 10 pounds of traction. • Neck in neutral position. • Pad all bony prominences appropriately. • Tape shoulders with 3-inch silk tape and pull to caudal end of operating table to maximize view of caudal cervical spine. • Tuck arms to side in secure fashion, make sure thumbs point up. • May use soft wrist ties using unrolled KERLIX roll (Covidien, Farmington, Connecticut) for later use by circulator for more visualization of caudal levels in extreme cases. • Slight reverse Trendelenburg.
Posterior cervical approach	• Place in prone position using Jackson spinal table, as already described. • Gardner-Wells tongs may be placed in neutral position (3–4 cm directly above the pinna and inferior to superior temporal line) before flipping for additional axial cervical traction and reduction if needed. • Likewise, some may opt to use a Mayfield skull clamp for rigid fixation for certain cases (e.g., severe kyphotic deformities). • Tuck arms using preplaced sheet by wrapping sheet around the arms and onto the thoracolumbar region. • Secure with multiple towel clamps. • Be sure thumbs are facing down. • Tape shoulders with 3-inch silk tape and tape to caudal end of Jackson table. • Tape the skin overlying the trapezius with 3-inch silk tape bilaterally, and tape to caudal end of Jackson table until posterior skin of neck is taught. • Place in slight reverse Trendelenburg.

Topic	Positioning
Posterior cervical approach for occipital cervical fusion	• Place in prone position using Jackson table, as described. • Gardner-Wells tongs may be placed in neutral position (3–4 cm directly above the pinna and inferior to superior temporal line) before flipping for additional axial cervical traction and reduction if needed. • Likewise, some may opt to use a Mayfield skull clamp for rigid fixation before fusion. • Tuck arms using preplaced sheet by wrapping sheet around the arms and onto the thoracolumbar region. Secure with multiple towel clamps. • Be sure thumbs are facing down. • Tape shoulders with 3-inch silk tape and tape to caudal end of Jackson table. • Tape the skin overlying the trapezius with 3-inch silk tape bilaterally and tape to caudal end of Jackson table until posterior skin of neck is taught. • Place in slight reverse Trendelenburg. • Slightly flex the cervical spine and place chin in military tuck position to allow for better exposure of occiput and occipital–cervical junction, keeping the neck neutral.
Posterior lumbar approach	• For unstable fractures, or when manipulation of lumbar spine is to be kept to a minimum, position patient with Jackson table as already described. • For operations without instability (e.g., microdiskectomy), may use a “Wilson frame” or positioning on two large parallel, longitudinally oriented gel rolls with enough room to allow for diaphragmatic excursion and abdominal decompression. • Failure to allow for abdominal decompression leads to increased pressure transmitted to epidural veins, complicating surgery through increased bleeding.

Topic	Positioning
	• For kyphoplasty/vertebroplasty, may use two large gel rolls, with one spanning the width of the chest and one the width of the pelvis. • Regardless of specific procedure, for lumbar operations, arms should be abducted and forearms forward, resting on a soft pillow which is positioned on top of an armrest. • The arms should not be abducted more than 90° and the elbows should not have any direct pressure on them, to minimize ulnar neuropathy. • Axillary padding should be used to minimize brachial plexus injuries. • Keep patient in slight reverse Trendelenburg.
Anterior lumbar approach	• Patient is supine on an operating table with arms tucked to the side. • All bony prominences are appropriately padded. • May use additional roll in lumbosacral region for increased lordosis.
Lateral transpsoas approach	• Cmax table (Steris, Mentor, Ohio) or other table capable of flexion-extension at the hip to be used. • Patient is placed in lateral decubitus position, depending on which side is approached for procedure. • An axillary roll should be placed. • The superior iliac crest should be positioned just past the “break” in the table. • A moderately sized roll may then be placed at the break, especially for an L4–L5 disk space. This is not as important if operating at L2–L3 or higher. • Hips are flexed and knees bent, allowing for more relaxed iliopsoas muscles. • Fluoroscopy is used to establish a “true” AP view with the C-arm in its most neutral position. • Once established, 3-inch silk tape is used to secure the patient at the iliac spines. • More tape is used to secure the thoracic cavity. Be sure to protect the nipples with foam when applying the tape.

Topic	Positioning
	• The legs must be taped down to secure the patient in this fixed position. • An arm board is used to support the lower arm. • An additional arm board is used to support the upper arm, or stacked pillows may be used to support the upper arm.
Lateral thoracic approach	• The same basic positioning is done for the transpsoas approach, but be sure to leave enough room for operating at the appropriate thoracic level(s).

II Selected Spinal Orthoses^{1,2}

Tien V. Le, Juan S. Uribe, and Fernando L. Vale

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III Outcome Scales

Description

Numerous grading scales are available for initial assessment of spine-related disorders as well as for outcome measurement to determine effectiveness of treatment. This appendix presents some of the more widely used scales. For copyright reasons, questionnaires are not reproduced herein.

Modified score

Table A3.1 Modified Japanese Orthopedic Association score for cervical myelopathy¹

Score	Description
Upper extremity (UE) motor dysfunction	
0	unable to feed self
1	unable to use knife & fork; can eat with spoon
2	can use knife & fork with much difficulty
3	can use knife & fork with slight difficulty
4	none (normal)
Lower extremity (LE) motor dysfunction	
0	unable to walk
1	can walk on flat surface with walking aid
2	can walk up and/or down stairs with handrail
3	lack of smooth and stable gait
4	none (normal)

Score	Description	
Sensory deficit		
0	UE	severe sensory loss or pain
1		mild sensory loss
2		none (normal)
0	LE	severe sensory loss or pain
1		mild sensory loss
2		none (normal)
0	trunk	severe sensory loss or pain
1		mild sensory loss
2		none (normal)
Sphincter dysfunction		
0		unable to void
1		marked voiding difficulty (retention)
2		some voiding difficulty (urgency or hesitation)
3		none (normal)

Total score ranges from 0 to 17 (normal)

Functional Independence Measure (FIM)

The Functional Independence Measure (FIM)² was developed to provide uniform evaluation of disability for spinal cord injuries. It rates 18 items shown in **Table A3.2** (13 motor, 5 cognitive) on the seven-level scale shown in **Table A3.3**. The FIM has high internal consistency and is a good indicator of burden of care.

Table A3.2 The Functional Independence Measure™(FIM)

Classif cation	Item
Motor	
Self-care	1. Eating 2. Grooming 3. Bathing 4. Dressing—upper body 5. Dressing—lower body 6. Toileting
Sphincter control	7. Bladder 8. Bowel management
Mobility	9, Bed, chair, wheelchair 10. Toilet 11. Tub, shower
Locomotion	12. Walk or wheelchair 13. Stairs
Cognitive	
Communication	14. Comprehension 15. Expression
Social cognition	16. Social interaction 17. Problem solving 18. Memory

Table A3.3 The FIM™rating levels of disability

Degree of dependency	Level of function	Score
No helper	Complete independence	7
	Modified independence	6
Modified dependence on a helper	Supervision	5
	Minimal assist (≥ 75%independent)	4
	Moderate assist (≥ 50%independent)	3
Complete dependence on a helper	Maximal assist (≥ 25%independent)	2
	Total assist (< 25%independent)	1

Nurick Grade (Table A3.4)³

Table A3.4 Nurick grade of disability from cervical spondylosis

Grade	Description
0	signs or symptoms of root involvement without myelopathy
1	myelopathy, but no difficulty in walking
2	slight difficulty in walking, able to work
3	difficulty in walking but not needin assistance, unable to work full-time
4	able to walk only with assistance or walker
5	chairbound or bedridden

Oswestry Disability Index (ODI)

In this 10-question survey, each response is graded 0 to 5 (0 indicates no disability).⁴ The score is then converted to a percentage, with 0%equaling no disability and 100%the highest level of disability.

Neck Disability Index (NDI)

The Neck Disability Index⁵ is a 10-question survey similar to the Oswestry Disability Index for the lumbar spine. Mild disability is defined as a score of 10 to 28%, moderate as 30 to 48%, severe as 50 to 68%, and complete as 72%or greater.

36-Item Short Form Health Survey (SF-36)

The Medical Outcomes Study Short Form 36⁶ is a 36-item self-administered questionnaire used to measure health status and outcome from the patient’s perspective. Although this questionnaire has been widely used for health policy and in some research, it is not specific enough for many spine-related issues. See reference 5 for the original publication. It produces an 8-scale measure of functional health and well-being. Permission to use the instrument must be obtained from the Medical Outcomes Trust (www.sf-36.com).

Zurich Claudication Questionnaire (ZCQ)

The Zurich Claudication Questionnaire (ZCQ)⁷ is a validated questionnaire that evolved from the Swiss Spinal Stenosis Questionnaire (SSS) and consists of 18 questions that pertain to symptom severity, level of function, and patient satisfaction.⁷ It is oriented toward patients with lumbar spinal stenosis. The final score is calculated as an unweighted mean of all answered questions and ranges from 1 to 5 (5 being more severe). Though not a statistically valid calculation, the questionnaire demonstrated good reproducibility, internal consistency, and validity.

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